

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART C STRUCTURAL DESIGN

SECTION 5 STRUCTURAL CONCRETE

BUREAU OF INDIAN STANDARDS

C O N T E N T S

FOREWORD	... 3
1 SCOPE	... 7
2 TERMINOLOGY	... 7
3 MATERIALS	... 7
4 STRUCTRAL DESIGN USING PLAIN REINFORCED AND PRESTRESSED CONCRETE	... 7
5 STRUCTURAL SAFETY OF TALL REINFORCED CONCRETE (RC) BUILDINGS	... 7
6 EARTHQUAKE RESISTANT DESIGN AND DETAILING OF STRUCTURES	... 7
ANNEX A MATERIALS FOR MAKING CONCRETE	... 8
ANNEX B SELF COMPACTING CONCRETE	... 15
ANNEX C HIGH PERFORMANCE CONCRETE	... 19
ANNEX D DESIGN OF STEEL FIBRE REINFORCED CONCRETE	... 21
ANNEX E CRITERIA FOR STRUCTURAL SAFETY OF TALL CONCRETE BUILDINGS	... 35
LIST OF STANDARDS	... 54

National Building Code Sectional Committee, CED 46

FOREWORD

This NBCS (Part C/Section 5) covers the structural design aspects of plain and reinforced concrete and also the prestressed concrete.

This Section 5 was first published in 1970 having Subsections 5A ‘Plain and reinforced concrete’ and 5B ‘Prestressed concrete’ of SP 7 (Part 6) and were subsequently revised in 1983, 2005 and 2016 as a chapter of National Building Code of India.

This Section is largely based on IS 456 : 2000 ‘Code of practice for plain and reinforced concrete (*fourth revision*)’ and IS 1343 : 2012 ‘Prestressed concrete — Code of practice (*second revision*)’ both of which are currently under revision and are being merged in the fifth revision of IS 456, likely to be titled as ‘Structural concrete — Code of practice’ at the time of the publication of this NBCS.

Major changes have been envisaged in the revision of IS 456 (amalgamating IS 1343). In the absence of availability of finalized version of revised IS 456, at the time of the revision of this publication, the provision of design as per existing IS 456 : 2000 and IS 1343 : 2012 have been continued through appropriate reference to the same.

In the 2016 version of the Code, in addition to the provisions of IS 456, provisions relating to special concrete namely self-compacting concrete, high performance concrete and steel fibre reinforced concrete were introduced.

In this revision, the publication has been renamed as National Building Construction Standards (NBCS).

In this fourth revision of this Section, Subsections 5A and 5B have been merged into a single Section 5, ‘Structural concrete’, and the following additional changes are made:

- a) Special concretes, namely self-compacting concrete, high-performance concrete, and steel fibre reinforced concrete, which were earlier covered in **16A**, **16B** and Annex A of the erstwhile NBC, are now included in [Annex B](#), [Annex C](#), and [Annex D](#) respectively;
- b) Informative Annex ([Annex A](#)) on updated provisions regarding materials for making concrete is introduced;
- c) Criteria for structural safety of tall concrete buildings is introduced in [Annex E](#). This provides prescriptive requirements for design of reinforced concrete tall buildings. The following salient aspects, which are based on the prescriptive approach, are addressed in the [Annex E](#):
 - 1) Structural systems admissible for tall building in different seismic zones;
 - 2) General requirements including: (i) height limitations of different structural systems, (ii) elevation and plan aspect ratios, (iii) lateral drift, (iv) storey stiffness and strength, (v) modes of vibration, (vi) floor systems, (vii) materials, and (viii) progressive collapse mechanism;
 - 3) Wind and seismic effects: (i) load combinations, and (ii) acceptable serviceability criteria for lateral accelerations;
 - 4) Methods of structural analysis to be adopted, and section properties (in cracked and uncracked states) of reinforced concrete member to be considered in analysis;
 - 5) Structural design aspects for various applicable structural systems;
 - 6) Issues to be considered in design of foundations; and
 - 7) Systems needed for structural health monitoring.

The upcoming fifth revision of IS 456 introduces a holistic design philosophy that emphasizes that concrete structures shall not only be strong but also safe, durable, serviceable under expected loads, and resilient in the face of extreme and accidental events. To this end, the standard stipulates that structural design shall meet six design criteria that are strength, serviceability, durability, robustness, integrity, and restorability. Each of these criteria

is associated with performance objectives and corresponding design requirements, ensuring that structural behaviour is assessed comprehensively across its entire lifecycle, including performance during and after any adverse events.

The design philosophy outlined in the above revised standard includes the classification of the design process into the prescriptive process and the closed-loop process. The prescriptive process follows conventional rule-based checks and is applicable to structures where only the basic three criteria (strength, serviceability, and durability) are of concern. In contrast, the closed-loop process is intended for important or critical structures and involves a more rigorous approach, requiring not only quantitative compliance for the first three criteria but also qualitative and performance-based verification for robustness, integrity, and restorability. This two-tier approach introduces flexibility in application while ensuring that safety and resilience are not compromised, especially in vital infrastructure. By integrating modern concerns like post-disaster repairability (restorability), progressive collapse prevention (integrity), and resistance to accidental actions (robustness), the design philosophy ensures that structures are not just built to withstand known demands but also prepared for uncertainties. It promotes reliability, sustainability, and long-term performance.

The revision of IS 456 highlights that structural analysis should correctly capture how structures behave under all types of loads, including static, dynamic, and accidental forces. For regular structures, linear static analysis is generally sufficient. However, for structures affected by dynamic forces like earthquakes, wind, or machine vibrations, dynamic analysis is necessary. In critical cases, the standard also recommends using non-linear dynamic analysis to better reflect the actual behaviour of materials and structures. This approach moves towards more detailed and performance-based analysis to improve safety, serviceability, durability, and resilience in concrete structures.

Apart from drawing from IS 456 : 2000 and IS 1343 : 2012, this revision incorporates provision relating to tall concrete buildings, drawn from IS 16700 : 2023 'Criteria for structural safety of tall concrete buildings (*first revision*)'.

All standards, whether given herein above or cross-referred to in the main text of this Section, are subject to the revision. The parties to agreement based on this Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

As the regulation of land development and buildings is a State subject and is dealt by the States and local bodies such as urban local bodies within their jurisdiction through building regulatory documents like building regulations, building byelaws, and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

For the purpose of deciding whether a particular requirement of this Subsection is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this section.

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Important Explanatory Note for Users of the NBCS

In any Part/Section of this NBCS, where reference is made to '**good practice**' in relation to **design, constructional procedures or other related information**, and where reference is made to '**accepted standard**' in relation to **material specification, testing, or other related information**, the Indian Standards listed at the end of the Part/Section shall be used as a guide to the interpretation.

At the time of publication, the editions indicated in the standards were valid. All standards are subject to revision and parties to agreements based on any Part/Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the list of standards given at the end of a Part/Section, the number appearing within parentheses in the first column indicates the number of the reference of the standard in the Part/Section. For example:

- a) Good practice [C-5(2)] refers to the Indian Standard(s) give at serial number (2) of the list of standards given at the end of this Part C/Section 5, that is, IS 456 : 2000 'Plain and reinforced concrete — Code of practice (*fourth revision*)'.
- b) Accepted standard [C-5(4)] refers to the Indian Standard(s) give at serial number (4) of the list of standards given at the end of this Part C/Section 5, that is, IS 269 : 2015 'Ordinary Portland cement — Specification (*sixth revision*)'.

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART C STRUCTURAL DESIGN

SECTION 5 STRUCTURAL CONCRETE

1 SCOPE

This NBCS (Part C/Section 5) deals with the structural use of plain, reinforced, and prestressed concrete.

2 TERMINOLOGY

For the purpose of this Section, the definitions given in accepted standards [C-5(1)] shall apply.

3 MATERIALS

All materials used in plain, reinforced and prestressed concrete construction shall conform to Part B ‘Building Materials’ of this NBCS as available.

4 STRUCTURAL DESIGN USING PLAIN, REINFORCED AND PRESTRESSED CONCRETE

The provision relating to design and general structural use of plain, reinforced and prestressed concrete, including on:

- a) materials, including special concrete, workmanship, inspection and testing;
- b) general design consideration;
- c) special design requirements for structural members and systems; and
- d) structural design: limit state method.

shall be in accordance with good practices contained in section 2, section 3, section 4 and section 5 of the good practice [C-5(2)] and section 2, section 3 and section 4 of the good practice [C-5(3)].

NOTE — At the time of publication of this section, IS 456 and IS 1343 were under revision, with IS 1343 being amalgamated into IS 456. Once the revised IS 456 is published, it shall replace the provisions given in [4](#).

The informative [Annex A](#) to this Section suggests the updated list of materials and few tests. Users are encouraged to investigate the possibility of applying the provisions given in [Annex A](#) which are listed in consonance with the provisions regarding materials in the good practice [C-5(2)].

5 STRUCTURAL SAFETY OF TALL REINFORCED CONCRETE (RC) BUILDINGS

For criteria for structural safety of tall reinforced concrete (RC) buildings of heights greater than 50 m and less than 250 m, normally intended for use as residential, office and other commercial buildings, reference may be made to [Annex E](#) and accepted standard [C-5(4)].

6 EARTHQUAKE RESISTANT DESIGN AND DETAILING OF STRUCTURES

The provisions on earthquake-resistant design and detailing of concrete buildings are covered in [C-5(5)].

ANNEX A

([Foreword](#) and [Clause 4](#))

MATERIALS FOR MAKING CONCRETE

A-1 AGGREGATES

A-1.1 Normal Weight Aggregates

Aggregates derived from natural sources and other than natural sources (manufactured aggregates such as iron slag, copper slag, bottom ash from thermal power plants, recycled concrete aggregates and recycled aggregates), shall comply with the requirements of the accepted standard [C-5(6)]. These are natural or synthetic aggregates with in-place density (unit weight) between 22.40 kN/m³ to 27.00 kN/m³.

A-1.2 Heavyweight Aggregates

Natural or synthetic aggregates with in-place density (unit weight) more than 27.00 kN/m³ and can range up to 45.00 kN/m³ for aggregates other than steel. Steel aggregates (if used) will have further higher density. Heavy weight aggregate is most commonly used for radiation shielding, counterweights and other applications where a high mass-to-volume ratio is desired. Goethite, limonite, barite, illmenite, magnetite, hematite, ferrophosphorus and steel are some examples of heavy weight aggregate.

A-1.3 Lightweight Aggregates

Natural or synthetic with in-place density (unit weight) between 14.40 kN/m³ to 18.40 kN/m³. The lightweight is due to the cellular or high internal porous structure, which gives this type of aggregate a low specific gravity. The most important aspect of lightweight aggregate is the porosity. They have high absorption values, which requires a modified approach to concrete proportioning. Light weight aggregates namely sintered fly ash aggregates may be used for making light weight structural concrete (a type of special concrete). Sintered flyash coarse aggregates when used shall conform to the accepted standard [C-5(7)].

A-2 CEMENT

A-2.1 For general use in plain, reinforced and prestressed concrete construction, cement used shall be any of the following and the type selected should be appropriate for the intended use:

- a) Ordinary Portland cement conforming to the accepted standard [C-5(8)];
- b) Portland slag cement conforming to the accepted standard [C-5(9)];
- c) Portland pozzolana cement (fly ash based) and Portland pozzolana cement (calcined clay based), conforming to the accepted standard [C-5(10)];
- d) Composite cement conforming to the accepted standard [C-5(11)] and containing a declared clinker content not less than 45 percent, fly ash content not more than 25 percent and a minimum 28-day compressive strength of 43 MPa can be used for reinforced concrete construction; and
- e) Portland calcined clay limestone cement conforming to the accepted standard [C-5(12)]. At locations where temperatures are predominantly below 15 °C for 6 months, the use of Portland calcined clay limestone cement shall not be permitted in underground structures and structural elements in contact with ground water.

A-2.2 Other cements as given below can be used for specific applications:

- a) Sulphate resisting portland cement conforming to the accepted standard [C-5(13)];
- b) Microfine ordinary portland cement conforming to the accepted standard [C-5(14)];
- c) Hydrophobic cement conforming to the accepted standard [C-5(15)];
- d) Low heat Portland cement conforming to the accepted standard [C-5(16)];
- e) Rapid-hardening portland cement conforming to the accepted standard [C-5(17)];

- f) High alumina cement conforming to the accepted standard [C-5(18)];
- g) Supersulphated cement conforming to the accepted standard [C-5(19)]; and
- h) White cement conforming to the accepted standard [C-5(20)].

A-3 WATER

A-3.1 Water used for mixing and curing shall be clean and free from excessive amounts of total dissolved solids (TDS), oils, acids, alkalis, salts, sugar, organic materials or other substances that may be deleterious to steel and setting and strength development of concrete.

A-3.2 Types of Water

In general, the suitability of water for the production of concrete depends upon its origin. Water to be used for mixing and curing of concrete shall be any of the types described below. Any type of water to be used for mixing and curing shall be tested as per stated procedures and shall meet all the requirements given in [A-3.3](#).

A-3.2.1 Potable Water

This water is considered as suitable for use in concrete if meeting the requirements of the accepted standard [C-5(21)] wherein the chloride content shall be limited to 500 PPM and alkalinity is less than or equal to 25 ml ([A-3.3](#)). Such water needs no further testing. In case of any doubt it shall be tested for conformity according to [A-3.3](#).

A-3.2.2 Natural Surface Water

This water, if found suitable, can be used in concrete but shall satisfy the requirements of [A-3.3](#).

A-3.2.3 Treated Water

Appropriately treated water, conforming to [A-3.3](#), from industrial, municipal and other sources can be used in concrete.

A-3.2.4 Water Recovered from Processes in the Concrete Industry

This includes:

- a) water that was part of surplus concrete;
- b) water used to clean the inside of the stationary mixers, mixing drums of truck mixer or agitators and concrete pumps; and
- c) process water from sawing, grinding and water blasting of hardened concrete.

This water may be taken from:

- 1) Basins provided with suitable equipment that distributes the solid matter evenly throughout the water; and
- 2) Sedimentation basins or similar installations, provided the water is left in the basin for a sufficient amount of time to allow the solids to settle properly.

Water recovered from process in the concrete industry contains varying concentrations of very fine particles, the size of which is generally less than 0.25 mm. Water recovered from processes in the concrete industry or combined water can be used provided the following requirements are met:

- i) The additional mass of solid material in the concrete resulting from the use of water recovered from processes in the concrete industry shall be less than 1 percent mass fraction of the total mass of aggregates present in the concrete; and
- ii) The amount of recovered water shall be spread as evenly as possible over a day's production.

A-3.2.5 Water from Other Sources

A-3.2.5.1 Recycled water

This includes water recovered from concrete production operation or water generated from sewage, greywater or storm water systems and treated to a standard that is appropriate for its intended use.

A-3.2.5.2 Desalinated water

Desalinated water conforming to [A-3.3](#) can be used. However normal sea water or brackish water shall not be used for mixing of concrete because of presence of harmful salts.

A-3.3 Requirements for Preliminary Inspection of Mixing and Curing Water

A-3.3.1 Preliminary Inspection

The water shall be examined for oil and fats, detergents, colour, suspended matter, odour and humic matter shall meet the requirements given in [Table 1](#) in accordance with test procedures stated below.

A small subsample shall be assessed as soon as possible after sampling. Bring any material that has settled back into suspension by shaking the sample. Pour 80 ml of the sample into a 100 ml measuring cylinder. Seal with a suitable stopper and shake the cylinder vigorously for 30 s. Smell the sample for any odours other than those of clean water. If in doubt about the odour, test the water for its odour level in accordance with national regulations for potable water. The odour level of the water shall be lower than the maximum level accepted for potable water. Observe the surface for foam. Set the cylinder in a place free from vibration and allow to stand for 30 min. After 2 min, check the sample for the continuing presence of foam and signs of any oils or fats. After 30 min have elapsed, observe the apparent volume of the settled solids and the colour of the water. Measure the pH using indicator paper or a pH metre. Then, add 0.5 ml hydrochloric acid, mix and then smell or test for the presence of hydrogen sulphide.

Put 5 ml of the sample into a test tube. Bring it to a temperature between 15 °C and 25 °C by allowing it to stand indoors. Add 5 ml of 3 percent sodium hydroxide solution, shake and leave for 1 h. Observe the colour.

Table 1 Requirements for Preliminary Inspection of Mixing Water

(Clause [A-3.3.1](#))

Sl No.	Parameter	Requirements
(1)	(2)	(3)
i)	Oils and fats	Not more than visible traces
ii)	Detergents	Any foam should disappear within 2 min
iii)	Colour	Water not from sources classified as potable: the colour shall be assessed qualitatively as pale yellow or pale
iv)	Odour	Water from sources classified as potable: no smell, except the odour allowed for potable water and a slight smell of cement; where blast-furnace slag is present in the water, a slight smell of hydrogen sulphide Water from other sources: no smell, except the odour allowed for potable water; no smell of hydrogen sulphide after addition of hydrochloric acid
v)	pH value	6.0 to 8.5
vi)	Humic matter	The colour shall be assessed qualitatively as yellowish/brown or paler after addition of NaOH

A-3.3.2 Permissible limits for solids shall be as given in [Table 2](#).

Table 2 Maximum Permissible Limit for Solids for Mixing and Curing Water <i>(Clauses A-3.3.2 and A-3.4)</i>				
Sl No.	Parameter	Tested as per Accepted Standard	Maximum Permissible Limit	
			For Mixing Water	For Curing Water
(1)	(2)	(3)	(4)	(5)
i)	Organic	[C-5(22)]	200 mg/l	200 mg/l
ii)	Inorganic	[C-5(22)]	3 000 mg/l	3 000 mg/l
iii)	Sulphate (as SO ₃ ²⁻)	[C-5(23)]	400 mg/l	800 mg/l
iv)	Chloride (as Cl ⁻)	[C-5(24)]	2 000 mg/l for concrete not containing embedded steel and 500 mg/l for reinforced concrete work ¹⁾	500 ¹⁾ mg/l for reinforced concrete or prestressed concrete work and 2 000 mg/l for concrete not containing embedded steel
v)	Suspended matter	[C-5(25)]	2 000 mg/l	2 000 mg/l
¹⁾ 1 000 mg/l chloride in water for reinforced concrete work (but not for prestressed concrete) can be permitted provided the chloride content of hardened concrete is checked at the time of concrete mix design and conform to requirement of total chloride content in the concrete.				

A-3.4 Testing and Requirements of Curing Water

Permissible limits for solids shall be as given in [Table 2](#). Water used for curing should not produce any objectionable stain or unsightly deposition on the concrete surface. The presence of tannic acid or iron compounds is objectionable.

A-3.5 Sampling of Water Concrete Making and Curing

A sample of water not less than 5 litre shall be taken. The sample of water taken for testing shall represent the water proposed to be used for concreting with due account being paid to seasonal variation. The sample shall be stored in a clean container previously rinsed out with similar water.

A-4 ADMIXTURES

A-4.1 Admixtures can be used to:

- Alter the composition of concrete by replacing some of the cement in the concrete by pozzolanic and hydraulic reaction (and are called mineral admixtures or supplementary cementitious materials);
- Enhance a certain function of concrete (and are called chemical admixtures); and
- The following types of admixtures shall be used in production of concrete.

A-4.2 Mineral Admixtures/Supplementary Cementitious Materials (SCMs)

SCMs as given below may be used as part replacement of cement. They can be pozzolanic material or reactive mineral admixture or combination of both.

A-4.2.1 Fly Ash (Pulverized Fuel Ash)

Fly ash conforming to the accepted standard [C-5(26)] may be used as part replacement of ordinary Portland cement. For concrete having steel, the replacement level of fly ash shall be restricted to 35 percent by weight of

cementitious material wherein fly ash shall have a minimum fineness of 280 m²/kg and lime reactivity of 4.5 N/mm² or more.

A-4.2.2 Silica Fume

Silica fume conforming to the accepted standard [C-5(27)] may be used as part replacement of cement. Silica fume is usually used in proportion of 3 percent to 10 percent of the cementitious content of a mix. The total replacement by fly ash and silica fume shall not be more than 35 percent for reinforced cement concrete. The total replacement by GGBF slag and silica fume/Ultrafine ground granulated blast furnace slag (UFGBS) shall not be more than 60 percent for reinforced cement concrete.

The above limits shall also apply where PPC or PSC is used. The fly ash or slag content mentioned on the bag shall be considered.

Only for mass concrete, structures below ground (not exposed to atmosphere) and for plain concrete, the use of more than one type of mineral admixture with ordinary Portland cement or a mineral admixture with blended cements such as PPC and PSC can be permitted after necessary trials (to control heat of hydration). However, silica fume may be used as an additional mineral admixture in reinforced concrete subjected to limits specified above. Uniform blending of the mineral admixtures with the cement should be ensured. In such cases, fly ash and GGBS together may be used within following limits:

Total replacement (<i>Max</i>)	=	45 percent of total cementitious content
Fly ash (<i>Max</i>)	=	20 percent
GGBS (<i>Max</i>)	=	25 percent

A-4.2.3 Metakaolin

Metakaolin conforming to the accepted standard [C-5(28)] may be used as pozzolanic material in concrete as part replacement of ordinary Portland cement only. The replacement level shall be maximum up to 15 percent.

A-4.2.4 Ground Granulated Blast Furnace Slag (GGBS)

GGBS conforming to the accepted standard [C-5(29)] may be used as part replacement of ordinary Portland cement. For concrete having embedded steel, the replacement level of GGBS shall be restricted to 50 percent by weight of cementitious materials. For resistance to sulphate and alkali aggregate reaction, GGBS up to 60 percent can be used.

A-4.2.5 Ultrafine Ground Granulated Blast Furnace Slag (UFGBS)

UFGBS conforming to the accepted standard [C-5(30)] may be used as part replacement of cement. The replacement level shall be up to 20 percent.

A-4.2.6 Ultrafine Fly Ash

Ultrafine fly ash produced through classification route and conforming to the accepted standard [C-5(31)] may be used as part replacement of ordinary Portland cement. The replacement level shall be up to 20 percent.

A-4.3 Chemical Admixtures

A-4.3.1 Chemical admixtures, as mentioned in [Table 3](#) if used, shall comply with the accepted standard [C-5(32)]. Previous experience with and data on such materials should be considered in relation to the likely standards of supervision and workmanship to the work being specified.

Table 3 Types of Chemical Admixtures and their Constituents

(Clause [A-4.3.1](#))

Sl No.	Type of Admixture	Typical Active Constituents
(1)	(2)	(3)
i)	Air-entraining	a) Salts of wood resins b) Some synthetic detergents c) Salts of sulphonated lignin d) Salts of petroleum acids e) Salts of proteinaceous material f) Fatty and resinous acids and their salts g) Tall oils and gum rosin salts h) Alkylbenzene sulfonates j) Salts of sulphonated hydrocarbons
ii)	Retarding	a) Unrefined lignosulphonate containing sugars b) Modified derivatives of unrefined lignosulphonate c) containing sugars d) Hydroxycarboxylic acids and their salts e) Carbohydrates including sugars f) Heptonates related to sugars and starches
iii)	Accelerating	a) Calcium chloride b) Triethanolamine c) Sodium thiocyanate d) Sodium formate or calcium formate e) Sodium nitrite or calcium nitrite f) Calcium nitrate g) Aluminates h) Silicates
iv)	Water-reducing	a) Lignosulfonic acids and their salts b) Hydroxylated carboxylic acids and their salts c) Polysaccharides d) Melamine polycondensation products e) Naphthalene polycondensation products f) Polycarboxylates
v)	High range water-reducing : Normal and retarding	a) Melamine sulfonate polycondensation products b) Naphthalene sulfonate polycondensation products c) Polycarboxylates
vi)	Shrinkage reducing	a) Polyoxyalkylene alkyl ether b) Propylene glycol

Table 3 (Concluded)

(Clause [A-4.3.1](#))

Sl No.	Type of Admixture	Typical Active Constituents
(1)	(2)	(3)
vii)	Corrosion inhibiting	a) Amine carboxylates aminoester organic emulsion b) Calcium nitrite c) Organic alkydicarboxylic d) Chromates e) Phosphates f) Hypophosphites g) Alkalis h) Fluorides
viii)	Permeability-reducing admixture	
	1) <i>Non-hydrostatic conditions (PRAN)</i>	a) Long-chain fatty acid derivatives (stearic, oleic, caprylic capric) b) Soaps and oils (tallows, soya-based) c) Petroleum derivatives (mineral oil, paraffin) d) Fine particle fillers (silicates, talc)
	2) <i>Hydrostatic conditions (PRAH)</i>	Crystalline hydrophilic polymers (latex, water-soluble, or liquid polymer)

A-4.3.2 Chemical admixtures can be achieved by a spectrum of materials and manufacturing processes, but shall comply with accepted standard [C-5(28)]. Further, chemical admixtures can be achieved by a spectrum of materials and manufacturing processes, but shall comply with the accepted standard [C-5(32)]. Further,

- The workability, compressive strength and the slump loss of concrete with and without the use of admixtures shall be established during the trial mixes before use of admixtures;
- The relative density of liquid admixtures shall be checked for each drum containing admixtures and compared with the specified value before acceptance;
- If two or more admixtures are used simultaneously in the same concrete mix, data shall be obtained to assess their interaction and to ensure their compatibility;
- Admixtures shall neither impair durability of concrete nor combine with the constituent to form harmful compounds nor increase the risk of corrosion of reinforcement; and
- The chloride content of admixtures shall be independently tested for each batch before acceptance.

A-4.3.3 The amount of admixture added to a mix shall be recorded in the production record. Additional dose of admixture may be added at project site and mixed adequately in mixer itself to regain the workability of concrete (if necessary) with mutual agreement between the producer/supplier and the purchaser/user of concrete. But, the producer/supplier shall assure the ultimate quality of concrete supplied by him and maintain record of quantity and time of addition.

A-4.3.4 Shrinkage compensating admixtures shall not be used in structural concrete.

A-5 FIBRES

Fibres may be added to concrete for special applications to enhance select properties. But, such fibres shall be approved by the competent authority on the recommendation of a duly constituted competent peer review committee. In any case, the presence of fibres shall not be taken advantage of in the estimation of strength.

A-6 REINFORCEMENT

A-6.1 Reinforcing bars and prestressing strands shall be specified in accordance with the provisions of good practices [C-5(2) and (3)], respectively.

A-6.2 Reinforcement couplers shall comply with the relevant clauses of accepted standard [C-5(33)].

A-6.3 If glass fibre reinforced polymer bars are used, the material shall conform to the requirement of the accepted standard [C-5(34)].

A-6.4 Welded wire reinforcement shall conform to the requirements of the accepted standard [C-5(35)].

A-6.5 Prestressing accessories shall comply with the requirements of good practice [C-5(3)].

ANNEX B

(Foreword)

SELF COMPACTING CONCRETE

B-1 GENERAL

The self compacting concrete (SCC) is highly flowable, non-segregating concrete that fills uniformly and completely every corner of formwork by its own weight without the need for any compaction and encapsulates reinforcement or any other embedment. The provisions relating to its application areas, features, mix proportioning principles, and production and engineering properties are given hereunder. The provisions for concrete making materials and concrete as given in 'materials, workmanship, inspection and testing' section of the good practice [C-5(2)] shall apply to SCC except where different provisions or departures have been indicated hereunder.

B-2 APPLICATION AREAS

SCC may be used in in-situ concrete or for precast concrete applications. It is particularly appropriate for sections with highly congested reinforcement such as, nuclear power plant structures, machine foundations, piers and abutments; and areas having restricted access to concrete placement and compaction like shafts in hydropower structures. SCC is also suitable for the construction of tunnel lining sections and casting of concrete-filled steel tubular columns. SCC may also be used for columns and beams in normal construction with appropriate formwork design.

SCC may be produced in a batching plant on site or in ready-mixed concrete plant and transported to site by transit mixers. It may be placed either by pumping or other placement methods into horizontal or vertical forms.

B-3 FEATURES OF FRESH SELF COMPACTING CONCRETE

The features of fresh SCC are:

- a) Filling ability (flowability);
- b) Passing ability;
- c) Segregation resistance; and
- d) Viscosity.

B-3.1 Filling Ability (Flowability)

Filling ability of SCC determines its ability to flow into and fill all spaces within the formwork, under its own weight. The filling ability is tested using slump-flow test.

B-3.1.1 Procedure

Slump flow test shall be carried out in accordance with the accepted standard [C-5(36)].

B-3.1.2 Classes of Slump-Flow and their Application

There are three classes of slump-flow, as follows:

- a) *SF1* (slump flow 550 mm to 650 mm) – This class of SCC is appropriate for:
 - 1) unreinforced or slightly reinforced concrete structures that are cast from the top with free displacement from the delivery point (such as slabs); and
 - 2) casting by a pump injection system (such as tunnel linings), and 3) sections those are small enough to prevent long horizontal flow (example, piles and some deep foundations).
- b) *SF2* (slump flow 650 mm to 750 mm) is suitable for many normal applications (example, walls, columns); and
- c) *SF3* (slump flow 750 mm to 850 mm) is typically produced with a small maximum size of aggregates (less than 20 mm) and is used for vertical applications in structures having congested reinforcement, structures with complex shapes, etc. *SF3* will often give better surface finish than *SF2* for normal vertical applications but segregation resistance is more difficult to control.

B-3.2 Passing Ability (Free from Blocking at Reinforcement)

Passing ability describes the capacity of the fresh mix to flow through confined spaces and narrow openings such as areas of congested reinforcement without segregation. If there is little or no reinforcement, there may be no need to specify passing ability as a requirement. L-box test is generally carried out to check the passing ability and the procedure shall be in accordance with the accepted standard [C-5 (36)]. In this test, the height of the concrete left in the vertical section (h_1) and at the end of the horizontal section (h_2) is measured. The ratio of h_2/h_1 is calculated as the blocking ratio; which shall be between 0.8 and 1.0.

B-3.3 Segregation Resistance (Stability)

Segregation Resistance is the ability of fresh concrete to remain homogeneous in composition while in its fresh state. Segregation resistance is generally carried out using sieve test to check this property of fresh concrete.

B-3.3.1 Procedure

Segregation resistance (sieve) test shall be carried out in accordance with accepted standard [C-5(36)].

B-3.3.2 Test Results

There are two classes of segregation resistance, namely *SR1* and *SR2*. For *SR1* class, segregation resistance should be above 15 percent and ≤ 20 percent, and for *SR2* it should be ≤ 15 percent.

SR1 is generally applicable for thin slabs and for vertical applications with a flow distance of less than 5 m and a confinement gap greater than 80 mm. *SR2* is preferred in vertical applications if the flow distance is more than 5 m with a confinement gap greater than 80 mm in order to take care of segregation during flow. Segregation resistance becomes an important parameter with higher slump-flow classes and/or the lower viscosity class, or if placing conditions promote segregation. If none of these apply, it is usually not necessary to specify a segregation resistance class.

B-3.4 Viscosity

Viscosity can be assessed by the V-funnel flow time. Concrete with a low viscosity will have a very quick initial flow and then stop. Concrete with a high viscosity may continue to creep forward over an extended time.

B-3.4.1 Procedure

Viscosity by V-funnel test is carried out in accordance with accepted standard [C-5 (36)]. A V-shaped funnel is filled with fresh concrete and the time taken for the concrete to flow out of the funnel is measured and recorded as the V-funnel flow time.

B-3.4.2 Test Results

The viscosity is divided into two classes, namely V_1 and V_2 . For V_1 class, the time taken to pass the concrete from V-funnel shall be ≤ 8 s and for V_2 class, it shall be above 8 s and ≤ 25 s.

V_1 has good filling ability even with congested reinforcement. It is capable of self levelling and generally has excellent surface finish. V_2 class viscosity is more likely to exhibit thixotropic effects, which may be helpful in limiting the formwork pressure or improving segregation resistance. But it may cause negative effects on surface finish and sensitivity to stoppages or delays between successive lifts.

B-4 MIX PROPORTIONING

B-4.1 Mix Proportion Principles

The principles listed below shall be followed:

- a) Lower coarse aggregate content;
- b) Increased paste content;
- c) Low water/powder ratio (see Note);
- d) High range superplasticizer; and
- e) Viscosity modifying admixture, if required.

NOTE — Powder is the material of particle size smaller than 0.125 mm. It includes this size fraction in the cement, mineral admixtures and aggregate.

B-4.2 Mix Design Procedure

For mix design procedure, reference may be made to good practice [C-5(37)].

B-5 PRODUCTION OF SELF COMPACTING CONCRETE

B-5.1 General

SCC is less tolerant to changes in constituent characteristics and batching variances as compared to conventional concrete. Accordingly, it is important that all aspects of the production and placing processes are carefully supervised.

The production of SCC should be carried out in plants where the equipment, operation and materials are suitably controlled under a Quality Assurance Scheme. It is important that all personnel who will be involved in the production and delivery of SCC are adequately trained including practical demonstrations.

B-5.2 Constituents

SCC is more sensitive than conventional concrete to variation in the physical properties of its constituents and especially to changes in aggregate moisture content, grading and shape. So, more frequent production checks are necessary. The moisture content of aggregates should be continuously monitored and the mix adjusted to account for any variation.

Storage of constituent materials for SCC shall be same as that for materials for conventional concrete. However, as the SCC mix is more sensitive to variations, special precautions shall be taken for storage of aggregates, which should be properly stored to avoid cross-contamination between different types and sizes and protected from weather to minimize the fluctuation of surface moisture content and movement of fines. Ground stock should be

stored in purpose-built partitioned bays, which should allow free drainage of excess moisture in the aggregates and rainwater.

B-5.3 Mixing Equipment and Trial Mixes

SCC shall be produced with efficient forced action concrete mixers. The time necessary to achieve complete mixing of SCC may be longer than for conventional concrete due to reduced frictional forces and to fully activate the superplasticizer.

B-5.4 Formwork

Formwork should be watertight and grout tight when placing SCC. The need to design the formwork for water tightness is greater than conventional formwork so as to avoid honeycombs and surface defects. The high fluid nature of SCC may lead to higher formwork pressure than conventional concrete, especially when the casting rate is high. As a result of the highly fluid nature of SCC, a high placement rate is likely and should be anticipated. Formwork designs, that accommodate the expected liquid head formwork pressure, can allow unrestricted placement rates and permit fast casting rate of the SCC. It is recommended to design the formwork for full liquid head.

B-5.5 Curing

Curing is essential for all concrete, and early protection of exposed surfaces is essential to preventing rapid moisture loss that could lead to plastic shrinkage cracking. The top-surface of elements made with SCC can dry quickly because of the increased quantity of paste, the low water/fines ratio and the lack of bleed water at the surface. Initial curing should therefore commence as soon as practicable after placing and finishing in order to minimize the risk of surface crusting and shrinkage cracks caused by early age moisture evaporation.

B-6 ENGINEERING PROPERTIES

B-6.1 General

SCC and conventional concrete of similar compressive strength have comparable properties.

B-6.2 Compressive Strength

SCC with a similar water-cement ratio usually have a slightly higher strength compared to conventional concrete due to the lack of vibration giving an improved interface between the aggregate and hardened paste.

B-6.3 Tensile Strength

SCC may be supplied with any specified concrete grade. For a given concrete grade, the tensile strength may be safely assumed to be the same as the one for a conventional concrete as the volume of paste (cement + fines + water) has no significant effect on tensile strength.

B-6.4 Static Modulus of Elasticity

Increasing the paste volume may decrease the modulus of elasticity, E_c . As SCC has higher paste content than conventional concrete, some difference can be expected and the E_c may be somewhat lower and should be adequately covered by the safe assumptions in design.

If SCC does have a slightly lower E_c than conventional concrete, in prestressing work, this will affect the relationship between the compressive strength and the camber due to pre-tensioning or post-tensioning. For this reason, careful control should be exercised over the strength at the time when the pre-tensioning and post-tensioning strands or wires are released.

B-6.5 Creep

Due to the higher volume of cement paste, the creep coefficient for SCC may be expected to be higher than for conventional concrete of equal strength, but such differences are small and shall be covered by the safe assumptions in design.

B-6.6 Shrinkage

As concrete compressive strength is related to the water-cement ratio, in SCC with a low water-cement ratio drying shrinkage reduces, but with increased cementitious content and increased water content the drying shrinkage and autogenous shrinkage can increase.

B-6.7 Durability

SCC is free from shortcomings due to improper compaction and result in consistently low and uniform permeability, offering less weak points for deleterious actions of the environment.

ANNEX C

(Foreword)

HIGH PERFORMANCE CONCRETE

C-1 GENERAL

High performance concrete is a concrete whose ingredients, proportions and production methods are specifically chosen to meet special performance and uniformity requirements. The higher performance requirements may be:

<i>Sl No.</i>	<i>Higher Performance Requirements</i>	<i>Remarks/Applications</i>
(1)	(2)	(3)
i)	High resistance to deterioration of reinforced concrete due to carbonation	This is to help achieve longer service life. This is particularly applicable in semi-arid regions
ii)	High resistance to chloride attack/ingress	This is for longer service life of reinforced concrete exposed to severe chloride environment/coastal environment
iii)	High resistance to abrasion	This may be required in concrete structures, like parts of hydraulic structures exposed to continuous abrasive action due to silt, such as spillways/glacis; and concrete pavements exposed to heavy traffic
iv)	High resistance to impact	This may be required in structures like blast resistant structures and spillways
v)	Improvement in flexure and shear performance	This is required in bridge girders, joints of RCC framed structure, shear walls, etc
vi)	High resistance to cracking due to shrinkage (plastic and drying shrinkage)	This may be required in concrete/mortar used for repair of distressed concrete structures, etc

High strength concrete generally improves performance of all above parameters. The concrete mix proportioning for high strength concrete may be carried out in accordance with the good practice [C-5(37)]. However, special measures are needed to further improve the specific performance requirements. As a guide, the measures given in [C-2](#) may be adopted. For further details, expert literature may be referred.

C-2 MEASURES FOR IMPROVING SPECIFIC PERFORMANCE REQUIREMENTS

C-2.1 Resistance to Carbonation

Resistance to carbonation of concrete is dependent mainly upon:

- Type of binder, content of binder and pH value of concrete; and

b) Permeability of concrete.

Addition of silica fume, GGBS and fly ash improve denseness and reduce permeability, thus increasing resistance to ingress of carbon dioxide and moisture (needed for carbonation). The addition of fly ash above 35 percent tends to lower the alkalinity, and thus the resistance to carbonation. Lower water-cement ratios and higher grades of concrete can be adopted to improve performance in this case. Minimum OPC content is essential for adequate carbonation resistance for different exposure conditions.

Carbonation resistance can be checked by using accelerated carbonation tests (using 3 percent to 4 percent CO₂ concentrations for minimum 70 days exposure) at mix proportioning stage. For quality assurance purpose, rapid tests like rapid chloride ion penetration test (RCPT), air permeability test or electrical resistivity test may be used. Their values may be established by correlation with accelerated carbonation test values at mix proportioning stage.

C-2.2 Resistance to Chloride Ingress

To improve performance of concrete exposed to chloride environment, two aspects need to be considered:

- a) Total initial chloride content in concrete should be minimized; and
- b) Permeability of concrete should be low.

Whereas chloride content of concrete can be controlled by checking chloride content of mix ingredients, the permeability of concrete can be reduced by careful selection of mix proportions and by use of supplementary cementing materials, either through use of blended cements like PPC and PSC or by direct addition of mineral admixtures, like fly ash, GGBS and silica fume. To control permeability, durability test of resistance to chloride ions shall be carried out by using chloride diffusion test by immersion or ponding method (using 3 percent NaCl exposure for 90 days) at mix proportioning stage. For quality assurance purpose rapid tests corresponding values like rapid chloride ion penetration test (RCPT) or electrical resistivity test shall be carried out. Corresponding values for these tests shall be established by developing a correlation with chloride diffusion test (by immersion/ponding) at mix proportioning stage.

C-2.3 Resistance to Abrasion

To ensure good performance against abrasion, following points should be considered:

- a) Use of higher grades of concrete;
- b) Use of good quality strong aggregate;
- c) Use of silica fume; and
- d) Use of steel fibres (suited for increasing abrasion resistance in spillways/glacis, etc).

Test methods for measuring abrasion resistance are:

- a) Revolving disc type test; and
- b) Under water abrasion resistance test using steel balls.

For further details regarding test methods and criteria, specialist literature may be referred.

C-2.4 Improving Performance in Flexure, Shear, Impact, Ductility and Energy Absorption

Resistance to impact is generally increased when the strength of concrete is high and energy absorption is high. Energy absorption and toughness can be increased by use of steel fibres in concrete. Whereas addition of steel fibres, glass or carbon fibres improves flexural and shear strength also, the use of synthetic fibres like polypropylene fibres do not significantly improve the above properties. Polypropylene fibres do however reduce shrinkage cracking and are suitable for concrete/mortar used for concrete repairs.

For checking impact resistance, falling weight impact test can be used, wherein number of blows for crack initiation in small concrete slab panel are counted.

For further details regarding test methods and criteria, specialist literature may be referred. The use of steel fibre reinforced concrete shall be further governed by [C-2.4.1](#).

C-2.4.1 Use of Steel Fibre Reinforced Concrete (SFRC)

Following properties of concrete can be improved by using SFRC:

- a) Ductility;
- b) Toughness and energy absorption;
- c) Abrasion; and
- d) Flexural strength and shear strength.

Compressive and direct tensile strength gain in flexural and direct tensile strength is very significant when higher percentage of steel fibres are used which give the concrete a strain hardening effect (increase in ultimate load after initial crack). However significant improvements are achieved in strain softening zone also.

Steel fibres having aspect ratio 60 to 80 may be incorporated in concrete of different grades to improve the ductility of concrete. Fibre content varying from 0.5 percent to 2.5 percent volume may be incorporated in to the concrete to improve different mechanical and durability characteristics. However higher addition can cause problems in uniform mixing and placing of concrete which need to be checked at mix proportioning stage before using the mix. Corrosion of steel fibres can also occur particularly in lower grades, so use of coated fibres and higher grades of concrete are recommended. Alternatively, a plain concrete cover of few millimetre can also be provided.

Design of steel fibre reinforced concrete (SFRC) for structural performance shall be in accordance with [Annex D](#).

ANNEX D

([Foreword](#) and Clause [C-2.4.1](#))

DESIGN OF STEEL FIBRE REINFORCED CONCRETE

D-1 GENERAL

This Annex shall apply where steel fibres are used to improve the performance and capacity of reinforced and prestressed concrete structures.

The design of steel fibre reinforced concrete (SFRC) at both the ultimate and serviceability limit states shall be based on the stress (σ) $\neq$ strain (ϵ) relationships for SFRC as specified in [D-3.3](#) (see Note 1).

Steel fibres shall not be relied upon in plastic hinge regions for strength requirements (see Note 2).

Steel fibres shall not be relied upon at constructions joints for either serviceability or strength requirements. Design procedure in this clause is for steel-fibre-reinforced concrete with a softening classification only (see [Fig. 1](#)). Hardening SFRC and the use of synthetic fibres is beyond the scope of this annex.

NOTES

1 When using brittle fibres that rely on a fibre pull-out failure mechanism to obtain member ductility, care is required for cases where a significant number of fibres fracture or where fibres result in local crushing of the concrete due to the local forces imposed on the matrix by the fibres. Fibre fracture may occur where the bond between the fibre and the matrix is high, and this is more likely in a high strength concrete combined with fibres of high bond capacity and of lower strength steels.

2 The anchorage capacity of steel fibres may be lost in areas where crack widths exceed about 3.5 mm. Cracking of this magnitude can be expected in regions where plastic rotations are expected.

D-2 DEFINITIONS

For the purposes of this Annex the following definitions shall apply:

D-2.1 Crack Mouth Opening Displacement (CMOD) — The width of a crack shall be measured at its mouth in a flexural tensile test. To measure the CMOD, a displacement transducers/clip gauge shall be mounted along the longitudinal axis at the mid width of the test specimen (beam with nominal size of 150 mm $\times$ 150 mm and length L so that 550 mm $< L < 700$ mm).

D-2.2 Crack Opening Displacement (COD) — The width of a single localized crack, taken as an average on four sides, for a direct tensile test on a strain softening SFRC dog-bone shaped specimen in accordance with [D-3.3.7](#).

D-2.3 Steel Fibre Reinforced Concrete (SFRC) — A mixture of concrete and steel fibres.

D-2.4 Hardening Behaviour — A material that displays an enhanced strength with increasing crack widths beyond that measured at the point of initial cracking of the cementitious matrix.

D-2.5 Softening Behaviour — A material that exhibits a loss of strength upon cracking.

D-2.6 Target Dosage — The specified quantity of fibres in kilograms per cubic metre of concrete (kg/m^3).

D-3 PROPERTIES OF SFRC

D-3.1 General

SFRC shall be classified in terms of both its characteristic compressive (cylinder) strength (f'_c) (see [D-3.2](#)) and its characteristic residual tensile strength ($f'_{1.5}$) (see [D-3.3.3](#)).

D-3.2 Compressive Strength

The characteristic compressive strength of SFRC at 28 days (f'_c) shall be determined in accordance with the accepted standard [C-5(38)].

In the absence of more accurate data, the mean value of the in situ compressive strength (f_{cmi}) shall be taken as 90 percent of the mean value of the cylinder strength (f_{cm}).

NOTE — The compressive strength of steel fibre reinforced concrete (SFRC) should be determined by means of standard tests on concrete cylinders. The addition of steel fibres to concrete in conventional dosages (less than 100 kg/m^3) does not change the compressive properties of the concrete.

D-3.3 Tensile Properties

D-3.3.1 Classification

SFRC shall be classified as either softening or hardening as shown in [Fig. 1](#). Hardening SFRC is outside the scope of this Annex.

NOTES

1 A hardening material is defined as one with a tensile strength equal to or greater than 1.1 times the strength of the matrix without fibres and taken at a crack opening displacement (COD) of equal to or greater than 0.3 mm.

2 A hardening material is one where multiple cracks can occur prior to crack localization. For material hardening to occur, the fibre dosage is typically such that the strength of the SFRC is at least 10 percent greater than the strength of the same concrete mix without fibres. The value of COD of 0.3 is sufficiently high such that matrix contribution to the strength of the composite is minimal and the tension at this point is substantially taken by the fibres alone.

D-3.3.2 Matrix Tensile Strength

The matrix tensile strength of the softening SFRC (f_{ct}) shall be obtained using flexural tensile testing, in accordance with good practice [C-5(2)].

When only the characteristic compressive (cylinder) strength (f'_c) has been determined, the mean and characteristic flexural tensile strength of SFRC shall be calculated in accordance with the good practice [C-5(2)].

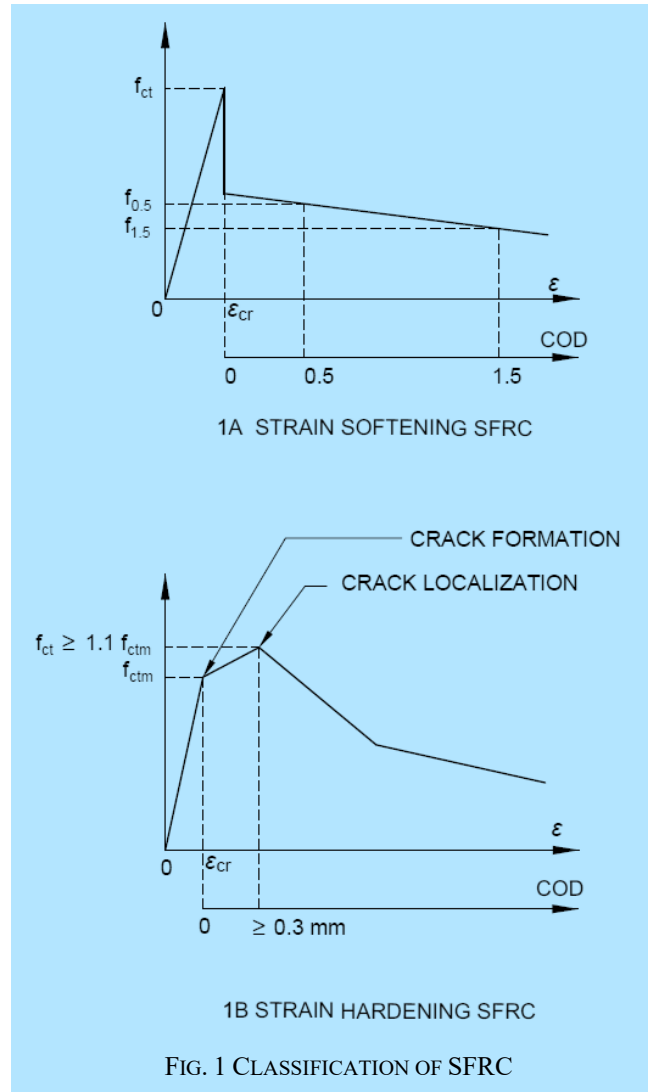
NOTE — For strain softening SFRC, the peak strength of the composite may be approximated as the strength of the concrete mix without fibres.

D-3.3.3 Residual Tensile Strength

The standard characteristic residual tensile strength grades ($f'_{1.5}$) are 0.4 MPa, 0.6 MPa, 0.8 MPa, 1.2 MPa, 1.6 MPa and 2.0 MPa.

The characteristic residual tensile strengths of concrete at 28 days ($f'_{1.5}$) shall be determined statistically from tests carried out in accordance [D-3.3.4](#) or [D-3.3.5](#).

NOTE — The characteristic residual tensile strength depends on the fibre-matrix bond strength, which is usually a function of the compressive strength of the parent concrete, as well as the fibre content. It may be unrealistic to specify a high value of $f'_{1.5}$ when using a relatively low value of f'_c .



D-3.3.4 Determination of Strength by Direct Testing

The characteristic residual tensile strength ($f'_{1.5}$) shall be obtained using direct tensile tests as specified in [D-3.3.7](#).

Alternatively, where matched direct and indirect testing has been undertaken in accordance with [D-3.3.6](#) for similar SFRC mixtures, the characteristic residual tensile may be determined as:

$$f'_{1.5} = k_{R,4} f'_{R,4}$$

Where $f'_{R,4}$ is determined in accordance with [D-3.3.8](#) and calculated statistically, and the factor $k_{R,4}$ determined from [D-3.3.6](#).

For the purposes of this clause, similar SFRC mixtures are defined as having the same:

- Fibre type and content;
- Water to cementitious material ratio;
- Maximum aggregate particle size; and

d) Compressive strength (f'_c).

NOTES

- 1 In calculating the characteristic strength, the population may be treated normally distributed and the strength determined in accordance with accepted standard. A confidence level of 75 percent shall be used that 95 percent of the population exceeds the characteristic value.
- 2 The sample standard deviation shall not exceed 25 percent.

D-3.3.5 Determination of Strength by Indirect Testing

The characteristic residual tensile strength ($f'_{1.5}$) may be obtained using indirect tests as specified in [D-3.3.8](#) and calculated as:

$$f'_{1.5} = 0.4f'_{R,4} - 0.07f'_{R,2}$$

D-3.3.6 Residual Tensile Strength-Residual Flexural Strength Relationship

The relationship between residual tensile strength and the residual flexural strength shall be obtained by matched using the same SFRC mixture. Residual tensile strength specimens shall be prepared and tested in accordance with [D-3.3.7](#). Residual flexural tensile tests shall be in accordance with the procedure as mentioned below:

The average width of the specimen and distance between the tip of the notch and the top of the specimen in the mid-span section shall be determined from two measurements to nearest 0.1 mm width and distance in the notch port of the specimen using calipers.

When the CMOD is measured, the displacement transducer shall be mounted along the longitudinal axis at mid width of test specimen such that the distance 'y' between the bottom of specimen and line of measurement is 5 mm or less as mentioned in [Fig. 2](#).

- a) The concrete mix shall be batched to ensure a uniform distribution of fibres, the SFRC shall be placed in the moulds in a manner that does not interfere with the distribution of the fibres and, the SFRC shall be compacted using lightly applied external vibration; and
- b) A minimum of 12 specimens shall be tested.

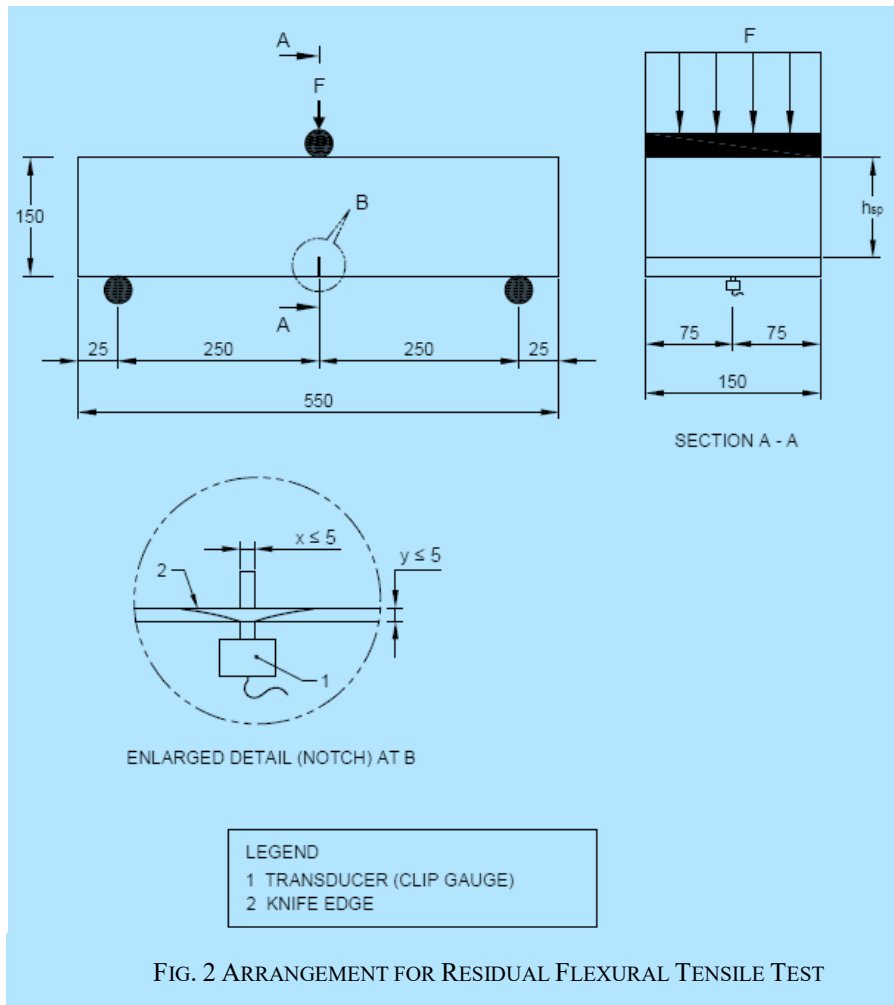
The reference factor $k_{R,4}$ shall be determined as:

$$k_{R,4} = f_{1.5m}/f_{R,4m}$$

where

- $$f_{1.5m} = \text{mean residual tensile strengths corresponding to a COD of 1.5 mm, determined in accordance with [D-3.3.7](#).}$$
- $$f_{R,4m} = \text{mean residual flexural tensile strengths corresponding to a CMOD of 3.5 mm, determined in accordance with [D-3.3.8](#).}$$

NOTE — The residual tensile strength (as obtained from a direct tensile test) is a fundamental material property of SFRC. This clause allows the concrete producer and steel fibre manufacturer to correlate the direct tensile strength of their particular SFRC mix design with the flexural tensile strength.



D-3.3.7 Residual Tensile Strength Test

The residual tensile strength shall be obtained using the testing arrangement shown in [Fig. 3](#) and with the following criteria:

- The critical section shall be where the cross sectional area is a minimum;
- The SFRC mix shall be batched to ensure a uniform distribution of fibres, and shall be placed in the moulds in a manner that does not interfere with the distribution of the fibres. The SFRC shall be compacted using lightly applied external vibration;
- The specimen shall be connected to the testing machine in such a manner that the machine does not apply a load to the specimen during the process of tightening of the grips and prior to testing;
- One end of the specimen shall be connected to the testing machine through a universal joint such that no moment is applied to the end of the specimen;
- Displacement measurements shall be taken on each of the four sides with the COD taken as the average of these measurements;
- A minimum of twelve specimens shall be tested;
- Tests where the failure of the specimen is outside of the testing region, or where the results are influenced by the test specimen boundaries, shall be retested;
- The characteristic values of the tensile strength $f_{0.5}$ and $f_{1.5}$, corresponding to CODs of 0.5 mm and 1.5 mm, respectively, shall be determined statistically as the 95 percentile confidence value assuming the population is normally distributed; and

- j) The mean values of $f_{0.5m}$ and $f_{1.5m}$, corresponding to CODs of 0.5 mm and 1.5 mm, respectively, shall be determined statistically as the 50th percentile confidence value assuming the population is normally distributed.

The stress results obtained from the test shall be multiplied by the three-dimensional orientation factor k_1 , where,

$$k_1 = \frac{1}{0.94 + 0.6\ell_f/b} \leq 1$$

And ℓ_f is the length and b is taken as the average of the width and depth of the specimen taken at the critical section.

NOTE — The factor k_1 removes the influence of the boundaries on the fibre distribution and converts the results of the test to a state where the fibres can be considered to be randomly orientated in three-dimensional space.

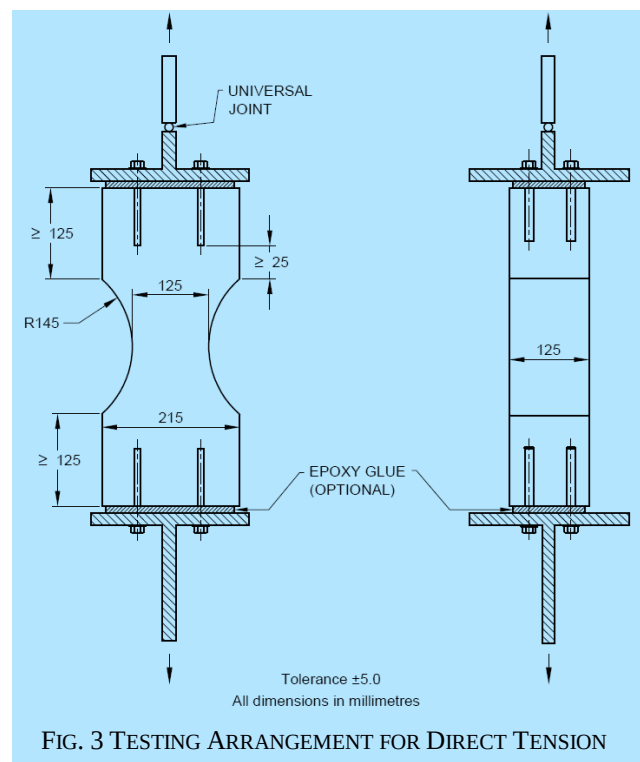


FIG. 3 TESTING ARRANGEMENT FOR DIRECT TENSION

NOTE — There is considerable discussion in the literature on whether the ends of the specimens should be fixed at each end, pinned at each end or fixed at one end and pinned at the other. The adoption of one pinned and one fixed end is a pragmatic decision between the ideal condition (fixed at each end) and limitations in manufacturing of specimens that are perfectly square at their ends for alignment within the testing machine. With both ends fixed any misalignment induces a moment in the specimen during gripping. This moment can induce a tensile stress on one side of the specimen, weakening the specimen. In some situations, the moment induced may be sufficient to crack the concrete. The inclusion of the universal joint at one end eliminates residual tensions that may result from the gripping procedure.

D-3.3.8 Residual Flexural Tensile Strength

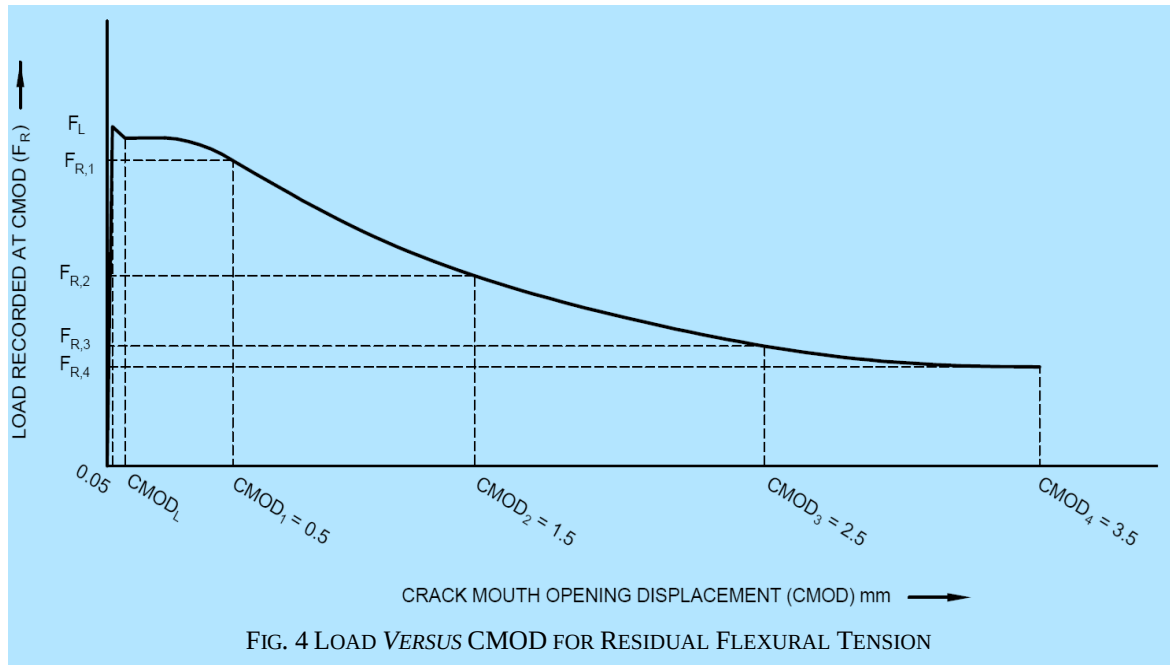
The residual flexural tensile strength (F_R) shall be determined from 3 point notched bending tests on 150 mm square section prisms. The notch depth shall be 25 mm and the test conducted in accordance with [D-3.3.6](#). The force F shall be plotted against the crack mouth opening displacement (CMOD), as shown in [Fig. 4](#) and the residual flexural stress calculated as:

$$F_R = \frac{3F_{Rj}L}{2bh_{sp}^2}$$

where

F_{Rj} = load recorded at CMOD_j (see [Fig. 4](#)).

- L = span, in mm;
 b = width, in mm of the specimen; and
 h_{sp} = distance, in mm between tip of the notch and top of cross section.



D-3.4 Modulus of Elasticity

The mean modulus of elasticity at the appropriate age (E_{ci}) shall be determined in accordance with accepted standard [C-5(38)].

NOTE — The addition of steel fibres to concrete in conventional dosages (100 kg/m^3 and less) does not change the elastic modulus of the concrete.

D-4 DESIGN OF REINFORCED SFRC MEMBERS

D-4.1 General

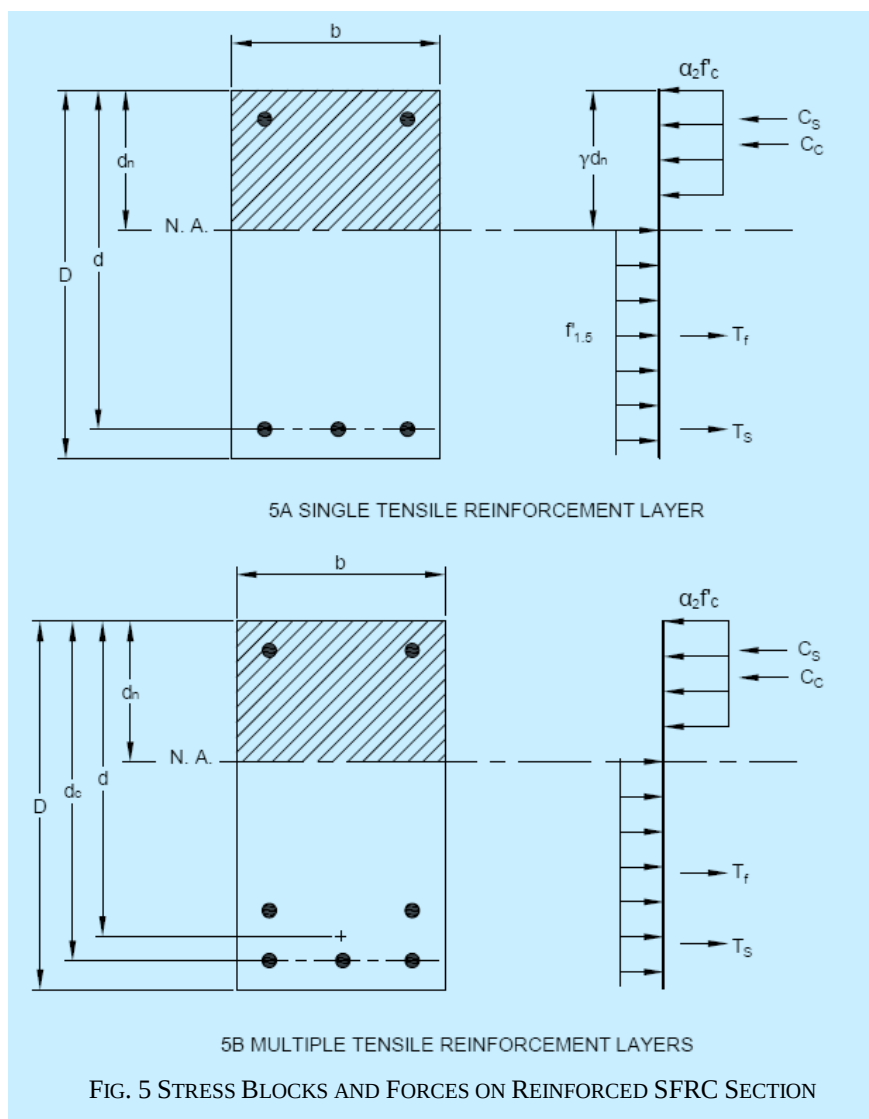
This clause applied to reinforced and prestressed beams subjected to any combination of shear force, bending moment and axial force. This clause does not apply when torsion acts in conjunction with shear or to non-flexural members.

D-4.2 Strength of Beams in Bending and Combined Bending and Axial Force

Calculations for strength of cross sections in bending shall incorporate equilibrium and strain-compatibility considerations and be consistent with the following assumptions:

- Plane sections normal to the axis shall remain plane after bending;
- Stress in the fibre reinforced concrete in that part of the cross section in tension shall be taken to be $f'_{1.5}$, where $f'_{1.5}$ is the characteristic residual tensile stress determined in accordance with [D-3.3.3](#); and
- Distribution of compressive stress shall be determined from a stress-strain relationship for the concrete.

The strength of a section in bending, or in combined bending and axial force, shall be determined using rectangular stress blocks for the concrete in compression and concrete in tension, as shown in [Fig. 5](#).



D-4.3 Minimum Reinforcement Requirements for Bending

The minimum tensile reinforcement shall be not less than that determined using the principles given in this annex, excluding fibres.

D-4.4 Strength of Beams in Shear

D-4.4.1 Design Shear Strength of a Beam

The design shear strength of a beam shall be taken as V_u

where

$$V_u = V_{uc} + V_{uf} + V_{us}$$

where V_{uc} and V_{us} are determined from 39 of accepted standard [C-5(38)] and V_{uf} is determined from [D-4.4.2](#).

Notwithstanding the above equations, the fibres component to the shear strength of a beam, V_{uf} , shall not exceed the greater of 0.3 V_u and that determined by [D-4.4.3](#) with V_{us} taken as zero.

D-4.4.2 Contribution to Shear Strength by Steel Fibres

The contribution of the fibres to the ultimate shear strength (V_{uf}) of an SFRC beam shall be calculated from the following equation:

$$V_{uf} = 0.7k_{\theta}b_vd_0f'_{1.5}$$

where

$$k_{\theta} = \cot \theta_v \leq 1.28$$

b_v = The effective width of the web (the vertical part of the beam).

d_0 = The effective depth of the beam.

$f'_{1.5}$ = characteristic residual tensile strength of concrete (at 28 days) at a Crack Opening Displacement (COD) of 0.5 mm.

θ_v = the angle between the axis of the concrete compression strut and the longitudinal axis of the member and shall be taken as not less than 38°.

D-4.4.3 Minimum Shear Reinforcement

The minimum contribution from the total of the transverse steel reinforcement and fibres shall satisfy the following:

$$(V_{us} + V_{uf})_{\min} \geq 0.1b_vd_0\sqrt{f'_c} \text{ and } \geq 0.6b_vd_0$$

D-4.5 Design for Serviceability Limit States

D-4.5.1 General

When an SFRC cross section is uncracked, the full cross section shall be assumed to be active and both concrete and steel are assumed to be elastic in tension as well as in compression.

When an SFRC cross section is cracked, the SFRC shall be assumed to be elastic in compression, and capable of sustaining a tensile stress equal to 1.1 $f'_{1.5}$.

NOTE — Prior to cracking, linear elastic behaviour is assumed for the SFRC, reinforcement and the bonded tendons in both tension and compression. After cracking, the SFRC in tension is assumed to carry a tensile stress of 1.1 $f'_{1.5}$, while the SFRC in compression, the reinforcement and the bonded tendons are assumed to be linear elastic.

D-4.5.2 Stress Limits

D-4.5.2.1 Concrete

The maximum compressive stress in the concrete at the serviceability limit states shall not exceed $0.6f_{cm}(t)$. Under permanent effect loading, the maximum compressive stress in the concrete shall not exceed $0.4f_{cm}(t)$.

NOTES

1 The satisfaction of limits on the concrete tensile stress at the serviceability limit state is not necessary if the member performance is satisfactory at the ultimate limit state.

2 Compressive stresses are limited to avoid large time-dependent deformations and to ensure creep behaviour remains linear with respect to stress. Restrictions on the tensile stress in SFRC are not necessary provided the member satisfies all other requirements at the strength and serviceability limit states.

D-4.5.2.2 Reinforcing steel

To avoid any inelastic deformation that could lead to large, permanently open cracks, tensile stresses in the reinforcement at the serviceability limit states shall not exceed $0.8f_{sy}$.

D-4.5.3 Minimum Reinforcement for Crack Control

The minimum amount of longitudinal reinforcement required to obtain controlled crack formation shall be:

$$A_{st,min} = k_1 k_c k_p f_{ct,ef} - 1.1 f'_{1.5} \frac{A_{ct}}{f_{s,max}} \geq 0.0 \text{ (mm}^2\text{)}$$

where

$A_{st,min}$ = minimum area of reinforcement required within the tensile zone (mm²). If $A_{st,min}$ is zero only steel fibres are necessary to control cracking;

k_1 = coefficient which allows for the effect of non-uniform self-equilibrating stresses due to non-linear shrinkage or temperature profiles through the member depth. In the absence of more detailed analysis, k_1 may be taken as 0.8;

k_c = coefficient that takes account of the nature of the stress distribution within the section immediately prior to cracking. The relevant stress distribution is that resulting from the combined effects of loading and restrained imposed deformations:

= 1.0 for pure tension ($e = M/N = 0$)

= 0.6 for pure bending

k_p = a coefficient that takes account of the level of prestress and is given by:

$$k_p = 1 - \frac{\sigma_{cp}}{k k_c f_{ct,ef}} \left(1 - k_c + 2.4 \frac{e}{D} - 6 k_c \frac{e}{D} \right)$$

$f_{ct,ef}$ = tensile strength of the concrete effective at the time when the cracks may first be expected to occur (MPa). Values of $f_{ct,ef}$ shall be obtained from $0.6 \sqrt{f_{cm}}$ but not less than 3.0 MPa and f_{cm} is the mean concrete compressive strength at the time cracking is expected to occur;

A_{ct} = area of concrete within the tensile zone (mm²). The tensile zone is that part of the cross section calculated to be in tension just before formation of the first crack;

$f_{s,max}$ = maximum stress permitted in the reinforcement immediately after formation of the crack;

e/D = ratio of the eccentricity of the prestressing force on the cross section, e , measured from the centroidal axis of the uncracked section to the overall depth of the cross section, D ; and

σ_{cp} = average intensity of the effective prestress (P_e/A_g).

D-4.5.4 Deflection Control

D-4.5.4.1 General

The deflection of an SFRC member shall be calculated using the procedures outlined in [D-4.5.4.2](#) and [D-4.5.4.3](#). Allowance shall be made for the expected load history, the expected construction procedure and any anticipated deflections resulting from deformation of forms or settlement of props.

D-4.5.4.2 Short-term deflection

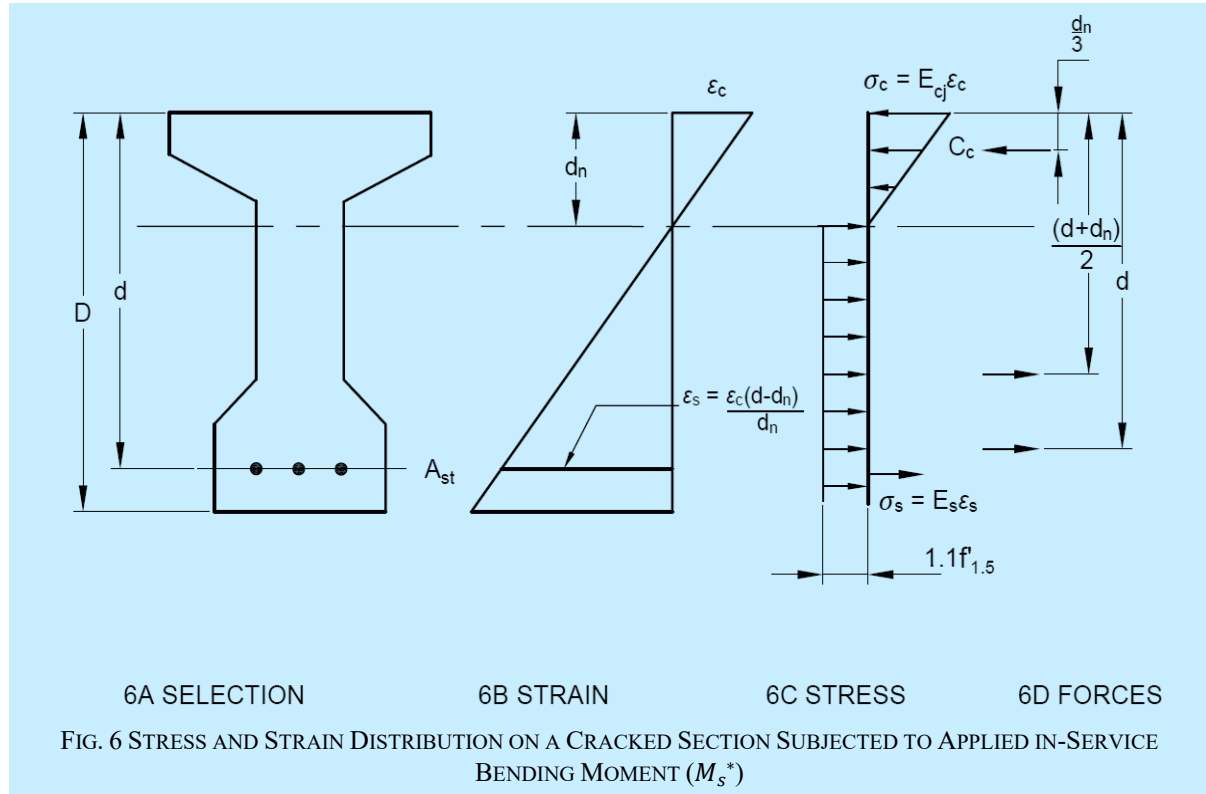
The short-term deflections due to external loads and prestressing, which occur immediately on their application, shall be calculated using the value of E_{cj} determined in accordance with this Annex and the value of the effective second moment of area of the member (I_{ef}). The value of I_{ef} may be determined from the values of I_{ef} at nominated cross sections as follows:

- a) For a simply supported span, the value at mid-span;
- b) In a continuous beam:
 - 1) For an interior span, half the mid-span value plus one quarter of each support value; or
 - 2) For an end span, half the mid-span value plus half the value at the continuous support.
- c) For a cantilever, the value at the support.

For the purpose of the above determinations, the value of I_{ef} at each of the cross sections nominated in items (a) to (c) is obtained from the instantaneous curvature $k_i = M_s^*/(E_{cj} I_{ef})$ calculated as the slope of the strain diagram in

[Fig. 6B](#) and obtained by satisfying the requirements for rotational and horizontal equilibrium of the stress distribution in [Fig. 6C](#).

NOTE — The procedure for calculating short-term deflection is consistent with the approach in this Annex. Short-term deflection of a cracked section is calculated based on the assumed stress and strain distribution shown in [Fig. 6](#). The assumption of a uniform tensile stress in the cracked region of the SFRC is considered to be reasonable for the determination of the neutral axis location and curvature and, hence, for the determination of the effective second moment of area (I_{ef}).



D-4.5.4.3 Long-term deflection

For reinforced and prestressed SFRC flexural members, that part of the deflection that occurs after the short term deflection shall be calculated as the sum of:

- The shrinkage component of the long-term deflection, determined from the design shrinkage strain of concrete (ϵ_{cs}) and the principles of mechanics; and
- The additional long-term creep deflections, determined from the design creep coefficient of concrete (ϕ_{cc}) and the principles of mechanics.

In the absence of more accurate calculations, the additional long-term deflection of a reinforced SFRC beam due to creep and shrinkage may be calculated by multiplying the short-term deflection caused by the sustained loads by a multiplier, k_{cs} , as follows:

$$k_{cs} = \left[2 - 1.2 \left(\frac{A_{sc}}{A_{st}} \right) \right] \geq 0.8$$

NOTES

1 For the calculation of long-term deflection, two alternative approaches are specified. For reinforced and prestressed beams, the shrinkage and creep components of the long-term deflection may be calculated separately using the material data specified in the Annex and the principles of mechanics. The shrinkage component of the long-term deflection may be readily determined from the shrinkage-induced curvatures calculated at the critical sections along the span of the beam using the design shrinkage strain, (ϵ_{cs}). The creep component of long-term deflection may be obtained from the creep-induced curvatures at the critical sections under the sustained service loads using the creep coefficient for concrete, (ϕ_{cc}).

2 Alternatively, for reinforced concrete beams and slabs, the additional long-term deflection caused by creep and shrinkage may be approximated by multiplying the short-term or immediate deflection caused by the sustained load by the deflection multiplier k_{cs} . The deflection multiplier depends on the ratio A_{sc}/A_{st} , where A_{sc} is the area of the longitudinal reinforcement in the compressive zone of

the cracked section (that is, between the neutral axis and the extreme compressive fibre of the cracked cross section) and A_{st} is the area of the tensile reinforcement.

D-5 DURABILITY

The minimum concrete grade and cover for SFRC in respective exposure classification shall be as for concrete without fibres and shall apply to the steel reinforcement only. SFRC shall not be used in exposure classification C1 or C2.

NOTES

- 1 Steel fibres do not require concrete cover as specified for steel reinforcement.
- 2 SFRC may not be suitable in some exposure classification U environments.
- 3 Fibres close to the surface are subject to corrosion and may result in surface staining. This will not impact durability.

D-6 FIRE

The structural performance of SFRC for fire shall be determined in accordance with this annex. The material properties for SFRC shall be as specified for concrete in this Annex except that to characteristic residual tensile stress of SFRC at elevated temperatures ($f'_{1.520}$) shall be either:

- a) Taken as $(f'_{1.520}) = k_{\theta 1} \times f'_{1.5}$ where $k_{\theta 1}$ is given in [Table 4](#); or
- b) Determined statistically from tests.

Table 4 Elevated Temperature Coefficient for Residual Tensile Stress of SFRC (Clause D-6)		
Sl No.	Temperature of SFRC °C	$k_{\theta 1}$
(1)	(2)	(3)
i)	0	1.0
ii)	100	1.0
iii)	500	0.6
iv)	700	0.1
v)	1200	0.0
NOTE — Linear interpolation between the values is permitted.		

D-7 PRODUCTION OF SFRC

D-7.1 Fibres

Steel fibres shall comply with accepted standard. The ‘Certificate of Conformity’ shall be supplied on request to the relevant authority.

NOTE — Till the availability of a separate Indian Standard specification for steel fibres, reference may be made to ASTM A820 : 2022 ‘Standard Specification for steel fibers for fiber-reinforced concrete’.

D-7.2 Mixing of Fibres

Fibres of the type and quantity specified shall be added in a controlled process ensuring that they are dispersed uniformly through the concrete mix. If added after the main mixing process, the concrete shall be remixed until the fibres have been completely dispersed throughout the batch.

NOTES

- 1 Balling of fibres should be avoided.
- 2 A record of fibre content should be recorded for each batch.

D-7.3 Pre-Construction Testing of Materials

Pre-construction testing in the form of initial type tests shall be undertaken in accordance with [Table 5](#). New initial type tests shall be undertaken when any of the following occur:

- a) The concrete composition changes;
- b) At least one source material of the concrete changes;
- c) The results of compressive tests undertaken as per this Annex do not meet the specification; and
- d) At least once in a period of 12 months.

The pre-construction tests shall be performed with the same personnel, materials and equipment that will be used during production.

Table 5 Pre-Construction Tests			
(Clause D-7.3)			
Sl No.	Material	Inspection/Test	Purpose
(1)	(2)	(3)	(4)
i)	Steel fibres	a) Check delivery note	Verify that the delivery is in accordance with the order, and is shipped from the correct source
		b) Check conformity label	Verify that the fibres have the correct conformity label which matches the corresponding certificate of conformity
ii)	Steel fibre content in the fresh concrete	Testing according to D-7.5 on the basis of 9 samples	Conformity with the target dosage. Verify homogeneous distribution of the steel fibres in the mix
iii)	Steel fibre concrete performance	Check limit of proportionality, and post-crack flexural strength in accordance with D-3.3.6 on a minimum of 12 beams	Verify, if the performance is in accordance to the specification. The performance level serves as the reference for continuous production control
NOTE — The strength of the population may be treated as normally distributed and the characteristic strength determined in accordance with this annex. A confidence level of 75 percent shall be used such that 95 percent of the population exceeds the characteristic value. For a sample of 12 specimens, the characteristic strength may be calculated from the mean strength using $\text{characteristic strength} = \text{mean strength} \times (1 - 1.84 \times \text{COV})$. The coefficient of variation (COV as a percentage) shall not exceed 25 percent.			

D-7.4 Factory Production Control

Factory production control in accordance with [Table 6](#) shall be undertaken to establish a production process for the steel fibre reinforced concrete, and shall include the following:

- a) Checking the correct concrete constituents are being used in production;
- b) Steel fibres are checked against specified requirements;
- c) Compressive tests undertaken in accordance with good practice [C-5 (40)]; and
- d) Fibre content and distribution.

Table 6 Routine Production Control

(Clause [D-7.4](#))

Sl No.	Subject	Inspection/Test	Purpose	Frequency
(1)	(2)	(3)	(4)	(5)
i)	<i>Equipment inspection</i>			
	a) Automatic dosing equipment for steel fibres	1) Visual inspection	Assure correct functioning of dosing device	Once per production day
		2) Control of accuracy	Avoid improper fibre dosage	On installation periodically in case of doubt
ii)	<i>Materials inspection</i>			
	a) Steel fibres	1) Check delivery note	Verify that the delivery is in accordance with the order, and is shipped from the correct source	Each delivery
		2) Check conformity requirements	Verify that the fibres matches the corresponding Certificate of Conformity	Each delivery
		3) Visual control, measure fibre dimensions	Compare the fibre geometry with the fibres used for ITT	Each delivery
iii)	<i>Production process inspection</i>			
	a) Fibres content-record	1) Record the quantity added	Check the content	Every batch
	b) Fibre content in the fresh concrete	2) Testing according to D-7.5	Conformity with the target dosage Verify homogenous distribution of the steel fibres in the mix	Beginning of each day and every/ 50 m ³ (manual dosing) and/150 m ³ (auto dosing)
	c) Concrete mix	3) Visual Check	Correct mixing with correct fibre type and even fibre distribution without balling	Daily
iv)	<i>Finished product inspection</i>			
	a) Steel fibre concrete performance	1) Check limit of proportionality, and post-crack flexural strength in accordance with D-3.3.6	Check performance level of the specification	2 beams every other day of production

D-7.5 Determining the Steel Fibre Content

Steel fibre content shall be measured from samples taken from the production concrete, by volume of concrete.

In addition:

- A sample shall be taken from the batch of concrete at unloading from the first third, middle third and final third of the batch;
- Each sample shall be a minimum of 10 litre;
- The sample container shall be filled in one continuous pour and where possible directly from the discharge chute; and
- Wash out or magnetic separation only shall be used.

The conformity of the correct steel fibre dosage is proven if the criteria in [Table 7](#) are met.

Table 7 Criteria of Acceptance for Steel Fibre Dosage (Clause D-7.5)			
Sl No.	Test Control	Test Control	Criteria
(1)	(2)	(3)	(4)
i)	Each sample	Each partial test	≥ 0.80 of the specified target dosage
ii)	Average of 3 samples from the batch	Each test	≥ 0.85 of the specified target dosage
iii)	Continuous control: average of > 3 tests	Continuous control average of > 3 tests	≥ 0.90 of the specified target dosage

D-7.6 Sampling, Testing and Assessment for Compliance of Hardened SFRC

When concrete is specified by parameters other than strength grade, the method of production control and, if required, project control shall be specified together with the relevant compliance criteria.

Methods of control and assessment shall provide a reliable operating characteristic curve so that:

- Concrete with a proportion defective of 0.05 has a probability of acceptance of at least 50 percent; and
- Concrete with a proportion defective of 0.30 has a probability of rejection of at least 98 percent.

ANNEX E

([Foreword](#) and [Clause 5](#))

CRITERIA FOR STRUCTURAL SAFETY OF TALL CONCRETE BUILDINGS

E-1 SCOPE

E-1.1 This Annex covers the following design aspects of reinforced concrete (RC) buildings of height greater than 50 m, but less than or equal to 250 m. Generally, these requirements are based on prescriptive approach and covers the following design aspects of tall buildings:

- Selection of appropriate structural system;
- Geometric proportioning of the building;
- Integrity of structural system;
- Resistance to wind and earthquake effects; and
- Other special considerations related to tall buildings.

E-1.2 This Annex is not applicable for tall buildings located in the near-field of seismogenic faults. For the purposes of this standard, near-field is taken as 10 km (shortest distance) from a seismogenic fault.

E-1.3 This Annex is applicable only for buildings that house 20 000 or fewer persons.

E-1.4 This Annex may also be used for design of buildings of height equal to or less than 50 m. The provisions of this Annex will add value to the design of such buildings. However, this Annex is not applicable for design of buildings of height more than 250 m.

E-1.5 This Annex addresses the following typical structural systems of tall concrete buildings:

- a) Structural wall systems;
- b) Moment frame systems;
- c) Moment frame — Structural wall systems;
- d) Structural wall — Flat slab floor systems with perimeter moment frame;
- e) Structural wall — Framed tube systems;
- f) Framed-tube system;
- g) Tube-in-tube system;
- h) Multiple tube system;
- j) Hybrid system; and
- k) Any of the above with additional framing systems, for example, outrigger trusses, belt trusses and braced frames.

E-1.6 This Annex shall be read along with all Indian Standards relevant to design and construction of buildings and structures. In case of conflict, clauses given in this Annex shall be applicable.

E-1.7 For buildings that do not conform to the prescriptive requirements of this Annex, a more rigorous procedure is necessary for design and review.

E-2 -

NOTE — (This clause is intentionally left blank.)

E-3 TERMINOLOGY

For the purpose of this Annex, the following definitions shall apply.

E-3.1 Building Height — The difference in levels between the base level of a building, at which inertia forces generated in the structure are transferred to the foundation, which then transfers these forces to the ground, and the highest level of a building, which is the roof of the topmost level which is structurally integral and continuous with the floor below (*see also* [E-4](#)).

E-3.2 Connecting Structure — A structure that links two or more towers, except at the podium levels.

NOTE — There are two distinctive types of connecting structures with each one requiring a different structural treatment, namely, (a) the connecting structure having the role of transferring gravity loads only and with no participation in the role of resisting the lateral loads on either of the towers; and (b) the connecting structure participating in the lateral load resisting structural system, where the two towers act together when one or both towers is subjected to lateral loads due to the interaction provided by the connecting tower.

E-3.3 Core — An assembly of structural walls coupled with link elements and offering lateral stiffness in two orthogonal directions.

E-3.4 Coupled Structural Wall Building — A building comprising of structural walls linked by in-plane beam elements, wherein the vertical and lateral loads are resisted by these structural walls, through axial load, in-plane bending moment and shear force, and the coupling action created by push-pull in the wall elements by framing action provided by link beams.

E-3.5 Frame Building — A building comprising beams and columns, wherein the vertical and lateral loads are resisted by these elements through moment-frame action.

E-3.6 Gravity Columns — Vertical members that are intended to carry primarily the vertical loads arising out of the dead and imposed loads on the building, and do not participate in the lateral load resisting system. Such columns shall be capable primarily of carrying the gravity loads and additional induced loads while complying with the relative lateral deformation effects imposed on them during lateral earthquake shaking.

E-3.7 Key Elements — The element such that its failure would cause the collapse of more than a limited area close to it. The limited area may be taken equal to 70 m² or 15 percent of the area of the storey, whichever is lesser. If key elements exist, it is preferable to modify the layout so that the key element is avoided.

E-3.8 Linking Beams (Also called coupling beams) — Horizontal members spanning two (vertical) structural walls.

E-3.9 Mixed Building or Hybrid Building — A building in which select primary elements resisting vertical and/or lateral loads comprise more than one material; composite elements comprising structural steel and reinforced concrete are common examples.

E-3.10 Multi-Tower Structures Linked by Podium — Two or more towers above a common podium structure that are not separated by a movement joint at the podium level.

E-3.11 Natural Period — The time taken (in seconds) by the structure to complete one cycle of oscillation in its natural mode of oscillation.

E-3.12 Structural Systems and Sub-Systems

E-3.12.1 Structural Wall System — A structural system comprising of inter-connected structural walls, wherein the vertical and lateral loads are resisted by the walls through axial load, in-plane bending moment and shear force. The wall elements are configured by a combination of structural wall elements connected integrally to a floor diaphragm. The wall elements form the primary lateral load resisting structural system for the building, and resist the loads imposed on them through axial, shear and flexural actions, and through coupling actions offered by the connecting link elements if provided.

E-3.12.2 Moment Frame System — A structural system comprising (beam-column) frames and resisting the vertical and lateral loads.

E-3.12.3 Moment Frame-Structural Wall System — A structural system comprising (beam-column) frames and structural walls resisting the vertical and lateral loads. The relative share of the lateral load resisted between these systems is dependent on their relative lateral stiffnesses.

E-3.12.4 Structural Wall System with Flat Slab Floor System — A structural system comprising structural walls, a beam-less floor system, and columns resisting the vertical and lateral loads. The relative share of the lateral load resisted between these systems is dependent on their relative lateral stiffnesses.

E-3.12.5 Core and Outrigger Structural System — A structural system comprising wall element(s) and perimeter columns, resisting the vertical and lateral loads. Essentially, the perimeter columns are for resisting gravity loads only. The core structure is connected to select perimeter column element(s) (often termed outrigger columns) by beam elements, known as outriggers, at discrete floor locations along the height of the building. This type of structure is an extension of the core structure, to enhance the lateral stiffness for taller structures, which mobilizes the perimeter columns, and offers increased leverage for push pull-action through the framing action offered by the deep beam(s) connecting the core to the outrigger columns. The global lateral stiffness is sensitive to: flexural stiffness of the core element, the flexural stiffness of the outrigger element(s) and the axial stiffness of the outrigger column(s).

E-3.12.6 Core, Outrigger and Belt Wall System — A structural system, which is an extension to the core and outrigger structure to enhance the lateral stiffness further, where the outrigger column(s) is linked to the adjacent columns by deep beam elements (often known as belt truss), typically at the same level as the outrigger elements. The sharing of loads between multiple columns has the dual function of enhancing the axial stiffness and mobilizing greater number of gravity columns to counteract the induced tension loads generated by the overall lateral loads.

E-3.12.7 Framed-Tube System — A structural system comprising closely spaced columns and deep beams in the perimeter frame for efficient tube action. The internal vertical elements, comprising core or columns are primarily utilized to resist gravity loads only. Effective utilization of the perimeter of the building maximizes the overall stiffness for a given building plan shape; this system is effective for very tall buildings.

E-3.12.8 Tube-in-Tube System — A structural system, which is an extension of the tube structure, where there is an internal tube, often a core element, supplementing the external perimeter described as the tube structure above, to enhance the overall lateral global stiffness. Just as the tube structure, typically, even this system is suited for very tall buildings.

E-3.12.9 Multiple-Tube System — A structural system, which is an extension of the tube structure and/or the tube-in-tube structure, where the architectural plan of the tower facilitates multiple tubes to be connected together, to enhance the lateral stiffness of the structure. Just as the tube structure and the tube-in-tube structure, typically, this suited is reserved for very tall buildings.

E-3.13 Super Tall Building — A building of height greater than 250 m.

E-3.14 Tall Building — A building of height greater than 50 m, but less than or equal to 250 m.

E-3.15 Transfer Structure — A structure, comprising of horizontal deep beams, trusses or thick slabs that transfers load actions and supports vertical elements above to vertical elements below that are not aligned with each other, through flexural and shear actions. Alternatively, it can be a trussed structure that fulfils the task through axial actions in the truss members.

E-4 SYMBOLS

For the purpose of this Annex, the following letter symbols shall have the meaning indicated against each; where other symbols are used, they are explained at appropriate places. All dimensions are in metres (m), loads in Newton (N), stresses in mega Pascals (MPa) and time in seconds (s), unless otherwise specified.

a	— Major axis of elliptic plan shape of the building
A_{cr}	— Cracked area of section
A_g	— Gross area of section
b	— Minor axis of elliptic plan shape of the building
h_i	— Inter-storey height of i^{th} floor in the building
l_s	— Clear span of coupling beam
B	— Smaller plan dimension of the building at its base
B_t	— Width of tower above podium level
D	— Dimension of the building in plan along the considered direction of earthquake shaking
d	— Overall depth of coupling beam
H	— Building height from its base to roof level: - excluding the height of basement storeys, if basement walls are connected with the ground floor slab or basement walls are fitted between the building columns, but; - including the height of basement storeys, if basement walls are not connected with the ground floor slab and basement walls are not fitted between the building columns
H_t	— Height of tower above podium level
h_w	— Unsupported height of structural wall
I_{eff}	— Effective or cracked moment of inertia of section
I_g	— Gross moment of inertia of section
L	— Larger plan dimension of the building at its base
L_t	— Length of the tower above podium level

- M_u — Factored bending moment at a cross-section in a vertical member of the building
 P_u — Factored axial load at a cross-section in the vertical member of the building
 P_{uz} — Factored pure axial load capacity (at $M_u = 0$) of a vertical member of the building
 T_a — Approximate fundamental translational lateral natural period
 Δ_{max} — Maximum relative lateral displacement within the storey

E-5 GENERAL REQUIREMENTS

E-5.1 Elevation

E-5.1.1 Height Limit for Structural Systems

The maximum building height (in m) shall not exceed values given in [Table 8](#) for buildings with different structural systems.

Table 8 Maximum values of Height H above Top of Base Level of Buildings with Different Structural Systems, in Metre

(Clause [E-5.1.1](#))

Sl No.	Seismic Zones	Structural System				
		Moment Frame	Structural Wall Well Distributed ¹⁾	Structural Wall + Moment Frame	Structural Wall + Perimeter Frame	Structural Wall + Framed Tube*
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	V	NA	120	150	150	180
ii)	IV	NA	150	200	200	225
iii)	III	60	200	225	225	250
iv)	II	80	250	250	250	250

¹⁾ Structural walls are considered to be well-distributed when structural walls that are outside of the core are capable of carrying at least 25 percent of the lateral loads.

* This includes systems covered under [E-1.5 \(g\)](#), [E-1.5 \(h\)](#) and [E-1.5 \(k\)](#).

E-5.1.2 Slenderness Ratio

The maximum values of the ratio of height to minimum base width B shall not exceed values given in [Table 9](#).

Table 9 Maximum Slenderness Ratio (H/B_t)

(Clause [E-5.1.2](#))

Sl No.	Seismic Zones	Structural System				
		Moment Frame	Structural Wall Well Distributed	Structural Wall + Moment Frame	Structural Wall + Perimeter Frame	Structural Wall + Framed Tube*
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	V	NA	8	8	9	9

Table 9 (Concluded)

(Clause [E-5.1.2](#))

Sl No.	Seismic Zones	Structural System				
		Moment Frame	Structural Wall Well Distributed	Structural Wall + Moment Frame	Structural Wall + Perimeter Frame	Structural Wall + Framed Tube*
(1)	(2)	(3)	(4)	(5)	(6)	(7)
ii)	IV	NA	8	8	9	9
iii)	III	4	8	8	9	10
iv)	II	5	9	9	10	10

* This includes systems covered under [E-1.5\(g\)](#), [E-1.5\(h\)](#) and [E-1.5\(k\)](#).

E-5.1.3 Aerodynamic Effects

Elevation profile, façade features of the building, and plan shape of the building shall be such as to attract minimum wind drag effects. Effects of features such as sharp corners, projected balconies, etc, shall be considered in design.

E-5.2 Plan

E-5.2.1 Plan Geometry

E-5.2.1.1 The plan shall preferably be rectangular (including square) or elliptical (including circular). In buildings with said plan geometries, structural members participate efficiently in resisting lateral loads without causing additional effects arising out of re-entrant corners and others.

E-5.2.2 Plan Aspect Ratio

The maximum plan aspect ratio (L_t/B_t) of the overall building shall not exceed 5.0. In case of an L-shaped building, L_t and B_t shall refer to the respective length and width of each leg of the building.

E-5.3 Storey Stiffness and Strength

Parameters influencing stiffness and strength of the building should be so proportioned, that the following are maintained:

- Lateral stiffness of any storey shall not be less than 70 percent of that of the storey above; and
- Lateral strength of any storey shall not be less than 90 percent that of the storey above.

E-5.4 Deformations

E-5.4.1 Lateral Drift

When design lateral forces are applied on the building, the maximum inter-storey lateral drift ratio (Δ_{max}/h_i) under serviceability loads (including wind load with return period of 20 years), which is estimated based on the sectional properties for serviceability loads mentioned in [E-7.2](#), shall be limited to 1/500. Total drift at the topmost usable floor shall be limited to $H/500$. For a single storey the drift limit may be relaxed to $h_i/400$.

E-5.5 Natural Modes of Vibration

E-5.5.1 The natural period of fundamental torsional mode of vibration shall not exceed 0.9 times the smaller of the natural periods of the fundamental translational modes of vibration in each of the orthogonal directions in plan.

E-5.5.2 The fundamental translational lateral natural period in any of the two horizontal plan directions, shall not exceed 8 s, considering sectional properties as per [E-7.2\(c\)](#) corresponding to serviceability loads.

E-5.6 Floor Systems

E-5.6.1 Material

All floor slabs shall be cast in-situ. For precast floor systems a minimum structural topping of 75 mm concrete with reinforcing mesh shall be used in Seismic Zones III and above, but can be reduced to 50 mm in Seismic Zone II.

E-5.6.2 Openings

E-5.6.2.1 Openings in floor diaphragm shall not be permitted along any floor diaphragm edge, unless perimeter members are shown to have stability and adequate strength.

E-5.6.2.2 The maximum area of openings in any floor diaphragm shall not exceed 30 percent of the plan area of diaphragm. Transfer of lateral forces from diaphragm to lateral load resisting vertical elements shall be ensured using collector elements, if required.

E-5.6.2.3 At any storey, the minimum width of floor slab along any section after deduction of openings shall not be less than 5 m, if there is no perimeter beam. Further, the cumulative width of the slab at any location shall preferably not be less than 50 percent of the floor width.

E-5.6.3 Natural Frequency of Floor System

The natural vertical vibration frequency of any floor system shall not be less than 3 Hz without demonstration of acceptability using rational procedures.

E-5.7 Materials

E-5.7.1 Concrete

E-5.7.1.1 The minimum grade of concrete shall be M 30.

E-5.7.1.2 The maximum grade of concrete shall be M 70. When higher grades are required, the designer shall ensure through experimentation that such concretes shall have at least a minimum crushing strain in compression of 0.002.

E-5.7.2 Reinforcing Steel

E-5.7.2.1 Reinforcing steel shall conform to provisions of good practice [C-5(5)].

E-5.7.2.3 Lapping of longitudinal bars shall be avoided in RC columns and walls that form a part of the lateral load resisting system; when diameter of bars is 20 mm or higher; mechanical couplers as per accepted standard [C-5(33)] shall be used to extend bars.

E-5.8 Progressive Collapse

Following are general guidelines to avoid progressive collapse of structure.

E-5.8.1 Possibilities of progressive collapse shall be precluded by:

- a) Selecting structural systems that are appropriate for ensuring structural integrity;
- b) Adopting rigorous structural investigations that verify acceptable structural behaviour, even when select critical members do not play their intended role; and
- c) Providing adequate redundancy and integrity to the structure.

E-5.8.2 Requirements of Key Elements

E-5.8.2.1 Key elements are members, joints or other components, whose failure would result in a disproportionate deterioration of the building and whose presence is vital to ensure ductile behaviour of the building. Vertical and lateral resistance of key elements shall be improved in many ways, including by the use of higher partial safety factors for loads and materials, to ensure that they do not yield before the designated ductile elements.

E-5.8.2.2 Elements adjoining key elements and capable of providing an alternative load transfer path, shall be suitably designed and detailed.

E-6 LOADS AND LOAD COMBINATIONS

E-6.1 The loads and load combinations specified in Part C ‘Structural Design/Section 1 Loads and Load Effects’ of this NBCS, the good practice [C-5(2)] and good practice [C-5(5)] shall be applicable for tall buildings also. In addition, requirements given in subsequent clauses shall be applicable.

E-6.2 Wind Effects

E-6.2.1 For buildings (a) with height greater than 150 m, or (b) with complexities in plan or elevation geometry, or (c) sited on complex topography with group effect or interference effect (existing and future potential), or (d) whose natural period is greater than 5 s, wind effects shall be determined by site-specific wind tunnel studies.

E-6.2.2 Site-Specific Wind Tunnel Studies

E-6.2.2.1 When wind tunnels studies result in higher storey shears and overturning moments than those calculated at based on 4 of Part C ‘Structural Design/Section 1 Loads and Load Effects’ of this NBCS, the results of wind tunnel studies may be used in design.

E-6.2.2.2 When wind tunnel studies result in lower story shears and moments than those calculated based on 4 of Part C/Section 1 of this NBCS,

- a) The minimum design wind base shear shall be at least 70 percent of that derived based on 4 of Part C/Section 1 of this NBCS; and
- b) The relative distribution of storey shears shall be as obtained from wind tunnel studies.

E-6.2.2.3 When wind tunnel studies indicate torsional motion, structural system of the building should be modified suitably to mitigate the torsional effects, so as to bring the torsional velocity below 0.003 rad/s for 10 year return period.

E-6.2.2.4 The damping ratio considered shall not be greater than 2 percent of critical for concrete buildings.

E-6.2.3 Lateral Acceleration

From serviceability considerations, under standard wind loads with return period of 10 years, the maximum structural peak combined horizontal acceleration $a_{\max}$ in the building for along and across wind actions at any floor level shall not exceed values given in Table 10, without or with the use of wind dampers in the building.

Table 10 Permissible Peak Combined Acceleration

(Clause E-6.2.3)

Sl No.	Building Use	Maximum Peak Combined Acceleration, $a_{\max}$ m/s ²
(1)	(2)	(3)
i)	Residential	0.18
ii)	Mercantile	0.25

E-6.3 Seismic Effects

E-6.3.1 Vertical shaking shall be considered simultaneously with horizontal shaking for tall buildings in Seismic Zone V.

E-6.3.2 For buildings in Seismic Zone V, deterministic site-specific design spectra may preferably be estimated and used in design. When site-specific investigations result in higher hazard estimation, site-specific investigation results shall be used.

E-6.3.3 Design base shear coefficient of a building under design lateral forces, shall not be taken less than that given in [Table 11](#).

Table 11 Minimum Design Base Shear Coefficient (Percent of Seismic Weight) (Clause E-6.3.3)					
SI No.	Building Height, <i>H</i>	Seismic Zone			
		II	III	IV	V
(1)	(2)	(3)	(4)	(5)	(6)
i)	$H \leq 120$ m	0.7	1.1	1.6	2.4
ii)	$H \geq 200$ m	0.5	0.75	1.25	1.75

NOTE — For buildings of intermediate height in the range 120 m to 200 m, linear interpolation shall be used.

E-6.3.4 Approximate Fundamental Natural Period

For buildings of height 50 m and more, the fundamental period *T* (in s) for a structure shall be determined by accounting for all structural properties and inherent stiffness of the building through rigorously validated structural analysis procedures. The fundamental period shall however not exceed the value obtained from the approximate fundamental translational natural period *T_a* (in s) of oscillation, estimated by following expression:

$$T_a = 0.0644H^{0.9} \text{ for concrete moment resisting frame systems}$$

and

$$T_a = 0.0672H^{0.75} \text{ for all other concrete structural systems}$$

E-7 STRUCTURAL ANALYSIS

E-7.1 Software

Structural analysis shall be carried out using standard 3-D computer model using well-established structural analysis software.

E-7.2 Considerations

Computer modelling shall consider the following:

- Rigid end offsets of linear members in the joint region, when centerline modelling is adopted;
- Floor diaphragm flexibility, as applicable;
- Cracked cross sectional area properties as per [Table 12](#); and
- The load combination for the initial analysis considering *P-Δ* effects shall be taken as: 1.2 DL + 0.5 IL ± 1.5 EL/WL.

E-7.3 Modelling

E-7.3.1 Modelling of buildings shall follow a simple approach, which reflects the distribution of mass and stiffness properties to properly account for all significant inertial forces under seismic actions and deformation shapes.

E-7.3.2 Analytical model of a building shall reflect the true behaviour of its members as well of the whole structure. One can adopt lumped modelling that is frame element modelling, distributed modelling that is finite element modelling or a combination of the two.

Table 12 Cracked Section Properties of RC Members					
(Clauses E-7.2 and E-7.3.6)					
Sl No.	Structural Element	Serviceability Design		Strength Design	
		Cross-Sectional Area	Moment of Inertia	Cross-Sectional Area	Moment of Inertia
(1)	(2)	(3)	(4)	(5)	(6)
i)	Slabs	1.0 A_g	0.35 I_g	1.00 A_g	0.25 I_g
ii)	Beams	1.0 A_g	0.7 I_g	1.00 A_g	0.35 I_g
iii)	Columns	1.0 A_g	0.9 I_g	1.00 A_g	0.70 I_g
iv)	Walls	1.0 A_g	0.9 I_g	1.00 A_g	0.70 I_g

E-7.3.3 In-plane stiffness of floor slabs shall be modelled, unless it is demonstrated that it is extremely stiff and sufficiently strong to remain elastic under seismic actions. See [5](#) of Part C 'Structural Design/Section 1 Loads and Load Effects' of this NBCS to identify when a floor slab may be considered to be extremely stiff in its own plane.

E-7.3.4 In moment frame buildings, when buildings with unreinforced masonry infill panels contribute to storey lateral stiffness, their effect shall be modelled as equivalent diagonal struts as per relevant provisions of [5](#) of Part C/Section 1 of this NBCS.

E-7.3.5 The analytical model for performing dynamic analysis of buildings with irregular configuration shall adequately represent irregularities in the configuration of the building.

E-7.3.6 Cracked sectional properties shall be used when representing concrete elements as per [Table 12](#).

E-7.3.7 In reinforced concrete buildings, lateral deflections resulting from unfactored lateral loads shall be estimated using section properties intended for use with unfactored lateral loads, and lateral deflections resulting from factored lateral loads using section properties intended for use with factored lateral loads.

E-7.3.8 Buildings may be considered to be fixed at their bases for determining lateral effects on buildings. When modelling flexibility of foundations, reference shall be made to [E-8](#) and [E-9](#).

When foundation flexibility is included in linear analysis, load-deformation characteristics of foundation-soil system shall be accounted for by equivalent linear stiffness, using soil properties that are consistent with soil strain levels associated with the design forces. A 50 percent increase and decrease in stiffness shall be incorporated in dynamic analysis, unless smaller variation can be justified; the largest value of response shall be used in the raft design.

E-7.3.9 Second order deformation effects ($P-\Delta$ effects) shall be considered.

E-7.3.10 The flexibility of the building shall be such that the inter-storey drift stability coefficient θ satisfies

$$\theta = \frac{P_i \Delta_i}{V_i h_{i-1} R} \leq 0.2$$

where

- P_i = total design vertical load at level i ;
- Δ_i = design storey drift at level i ;
- V_i = design shear force at level i ;
- h_{i-1} = storey height below level i ; and
- R = response reduction factor.

E-7.3.11 Stiffness of frames with flat slabs as horizontal members (that is, slab-column frames) shall be ignored in lateral load resistance, in all Seismic Zones. These shall be designed as gravity elements for deformation compatibility.

E-7.3.12 The model used in structural analysis of solid, coupled, perforated or punched structural walls shall represent stiffness, strength and deformation capacities of structural wall, structural wall segments and coupling beams or spandrel connections between structural walls. Stiffness of coupling beams and spandrel connections shall capture aspect ratio of these coupling beam and spandrel connections, extent of cracking anticipated, and reinforcement provided in them.

E-7.3.13 Effects shall be considered of construction sequence in buildings taller than 150 m.

E-7.3.14 Tower(s) connected by a podium shall be modelled separately and integrally.

E-7.4 Building Movements

For buildings taller than 150 m, and for buildings taller than 100 m with mass asymmetry, analysis shall be carried out for both vertical and horizontal long-term building movements.

E-7.4.1 Measures shall be taken in concrete and composite buildings to minimize adverse effects of shrinkage, creep, temperature variation and foundation settlement during the design life of the building (not less than 30 years).

E-7.4.2 Non-structural elements, such as curtain walls, cladding, partitions and finishes and service installations (for example, elevators, vertical pipes, ducts and cables), shall be required to withstand long-term movements of the building and associated differential effects.

E-7.4.3 Details of connections of non-structural elements with the structural elements of the building shall be planned, such that their relative movements are allowed without causing distress to both structural and non-structural elements.

E-7.4.4 Appropriate vertical compensation and sway compensation shall be accounted for during construction to minimize long term building movements for concrete and composite structures.

E-7.4.5 In gravity load analysis, internal forces shall be considered, which are developed due to differential vertical movement of vertical structural elements, due to shrinkage, creep, temperature, foundation settlement and construction compensation. In any case the total shrinkage strain of concrete shall not exceed 0.04 percent.

E-7.4.6 To calculate the effect of creep and shrinkage, selected strain prediction models of concrete shall be based on established principles of mechanics elaborated in specialist literature.

E-8 STRUCTURAL DESIGN

E-8.1 General Requirements

E-8.1.1 Method of Design

Limit state design method (as given in [C-5(2)]) shall be used in the design of RC members.

E-8.1.2 Staircase

Staircases built integrally with the structural system of the building and not confined by structural walls, shall be included in the three-dimensional structural model, and its elements designed as per forces induced in them under various load combinations. Alternately a sliding joint may be provided.

E-8.1.3 Multiple Tall Buildings connected with a Common Podium

E-8.1.3.1 This section deals with requirements for the following tall buildings with podium:

- a) Tall building with single tower and podium (see [Fig. 7A](#)); and

- b) Tall building with multiple towers and common podium (see [Fig. 7B](#)).

E-8.1.3.2 Modelling

E-8.1.3.2.1 Sensitivity analyses

- a) As part of collapse prevention evaluation, two sets of backstay sensitivity analyses shall be carried out using upper-bound and lower-bound cracked section properties of floor diaphragms and the stiffness parameters for those diaphragms and perimeter walls of podium and below the level of the backstay are given in [Table 13](#). These analyses shall be in addition to those required to be carried out using other cracked section properties described in [E-7.2\(c\)](#).

Table 13 Stiffness Parameters			
[Clauses E-8.1.3.2.1(a) and E-8.1.3.3.3(d)(2)]			
SI No.	Stiffness Parameter	Values to be Adopted	
(1)	(2)	Upper-Bound (3)	Lower-Bound (4)
i)	I_{eff} / I_g	0.5	0.15
ii)	A_{cr} / A_g	0.5	0.15

- b) Besides that of the floor diaphragms, flexibility of following structural elements in the structural analysis with appropriate modification to their stiffness:
- 1) Perimeter walls and their foundation supports; and
 - 2) Foundation supports under the tower lateral load resisting system.

E-8.1.3.3 Buildings with multiple towers

E-8.1.3.3.1 Backstay

Backstay transfers the forces from lateral load-resisting elements in the tower to additional structural elements provided within the podium and the basement, typically through one or more floor diaphragms. Lateral load resistance in the podium levels with assured force transfer path through floor diaphragms at these levels, helps the tall building to resist lateral overturning forces. This component of overturning resistance, referred as the backstay effect (also called shear reversal), is critical because shear force changes direction within the podium levels, and the same lateral load-resisting element helps resist the changing shear force.

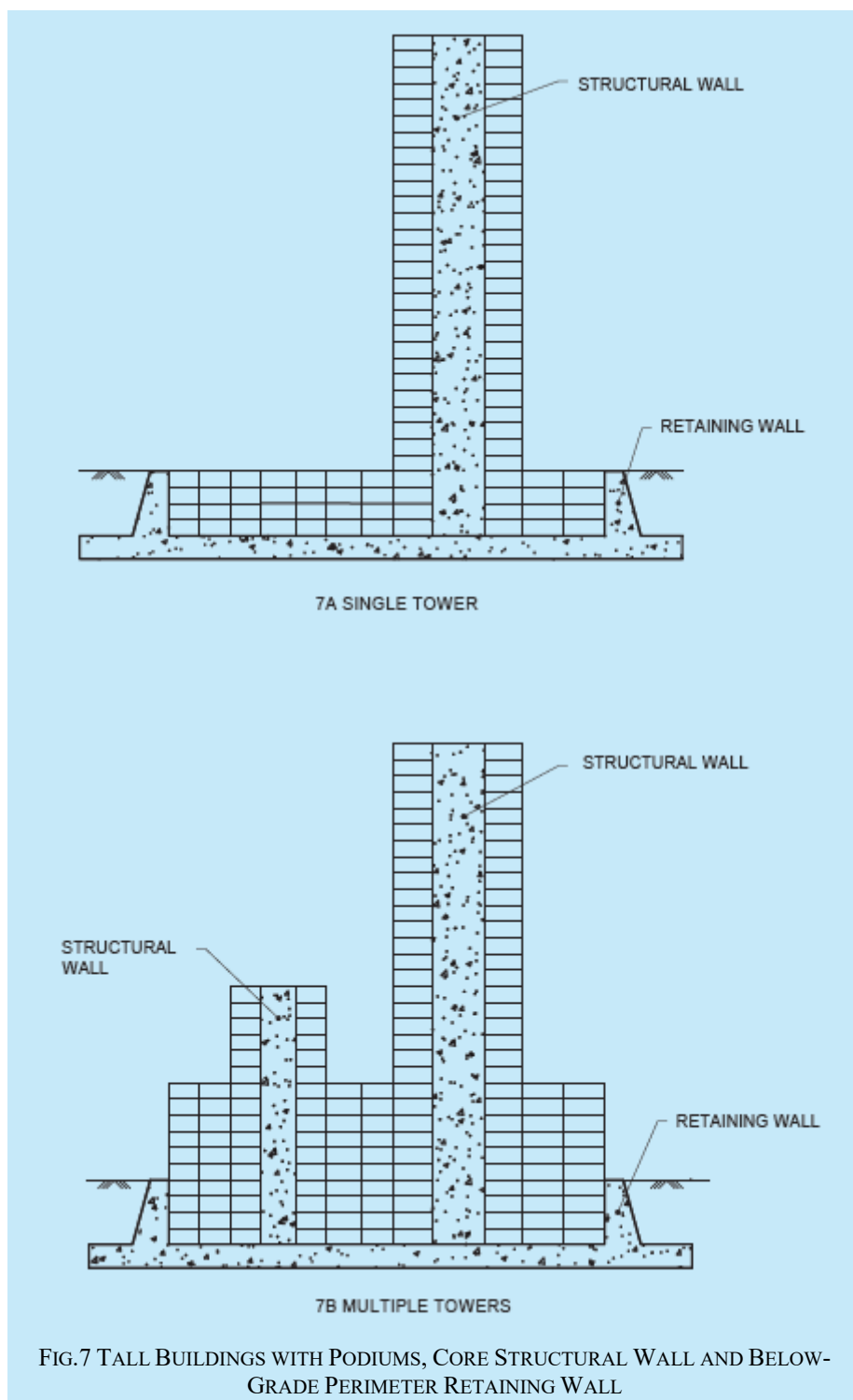
The following shall be considered:

- a) In estimating backstay effects, two lateral load paths shall be considered (see [Fig. 8](#)), namely:
- 1) *Direct load path* — where overturning resistance is provided by the tower core elements and foundation directly beneath the tower; and
 - 2) *Backstay load path* — where overturning resistance provided by in-plane forces in the backstay elements (lower floor diaphragm and perimeter walls).

In some tall buildings, backstay effects may not be considered. These include the following configurations:

- i) Buildings without below grade levels or buildings without significantly increased lateral load resisting systems at the base;
- ii) Buildings that extend below grade, but have structural separations between the superstructure and the podium portions of the building, and that accommodate lateral load deformations without transfer of lateral forces from superstructure portion to podium portion; and
- iii) Buildings with perimeter basement walls, where the walls are located directly below the lateral load resisting elements of the superstructure above. Here, there may be a marked change in

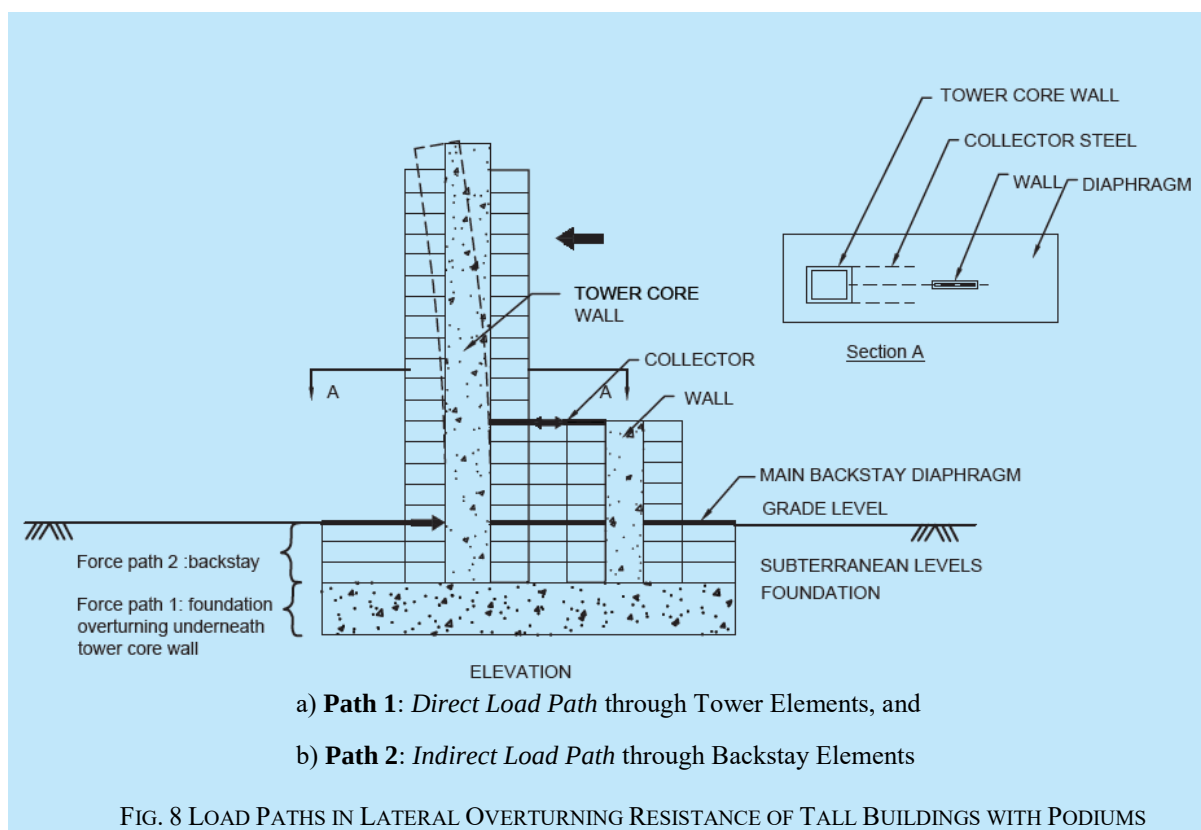
lateral strength and stiffness at this level, but lateral forces will not be transferred through the floor diaphragms.



- b) Backstay floor diaphragms shall be modelled considering their in-plane and out-of-plane floor flexibility. Any large discontinuity present in the slab shall be modelled;
- c) In direct load path case, vertical stiffness shall be considered of the piles, foundation and supporting soil;
- d) In backstay load path case, relative stiffness shall be considered of floor diaphragms and perimeter walls, along with vertical in-plane rocking stiffness of soil below the walls. Also, horizontal pressure imposed

by soil on retaining walls shall be considered. Axial stiffness of elements (representing backstay) along load path shall be reduced to account for cracking, bond slip, interface slip and other such effects;

- e) In tall buildings with backstay diaphragms, collector elements shall be provided (see Fig. 8), which are capable of transferring forces from the lateral load-resisting system of the tower to the additional elements providing the resistance and in turn to those forces of the podium. Such collector elements shall be provided for transferring lateral forces originating in the other portions of the structure;
- f) The backstay diaphragms shall be designed in accordance with the following:
 - 1) They shall be designed for the maximum of forces derived from sensitivity analysis;
 - 2) When the lateral force resisting system of a tall building has plan irregularity as per 5 of Part C 'Structural Design/Section 1 Loads and Load Effects' of this NBCS of Type I (torsion irregularity), Type II (re-entrant corners) and Type IV (out of plane offsets), and vertical irregularity as per 5 of Part C/Section 1 of this NBCS of Type IV (in-plane discontinuity of vertical elements resisting lateral forces), seismic forces shall be amplified by a factor of 1.5 in the design of:
 - i) Connections of diaphragms to vertical elements and to collectors; and
 - ii) Collectors and their connections, including connections to vertical elements, of the seismic forces resisting system.
 - 3) When the lateral force resisting system of a tall building has plan irregularity of Type III (diaphragm discontinuity) as per 5 of Part C/Section 1 of this NBCS at the backstay diaphragm levels, collector elements and their connections to vertical elements shall be designed to resist seismic forces amplified by an over-strength factor of 2.5;
 - 4) Backstay diaphragm floor shall be at least 150 mm thick, with double mesh reinforcement not less than 0.25 percent of cross section area in each direction (including both layers of reinforcement steel); and
 - 5) Adequate measures shall be taken to prevent shear sliding failure at connections of diaphragm to structural walls; here, the inclination of the strut shall be taken as 45° . Additional reinforcements shall be provided to resist the shear force at the interface between diaphragm and structural walls.



E-8.1.3.3.2 Structural walls

- a) Structural walls shown in [Fig. 8](#) can sustain plastic hinges at the level of the backstay diaphragm also. Such walls shall also be designed and detailed for plastic hinge development at that level; and
- b) All peripheral columns of the tower (irrespective of whether they are gravity columns or not) shall be provided with confinement reinforcements throughout the storeys adjoining (above and below) the backstay diaphragm level, as per the requirements of good practice [C-5(5)].

E-8.1.3.3.3 Towers connected by common podium

When buildings have two or more towers, they shall be designed considering the following:

- a) Such buildings shall be modelled as separate towers as well as integral towers. The podium shall be designed based on the worst of the two results;
- b) The estimation of natural period (for calculation of base shear) shall be based on individual building model;
- c) In the integral tower modelling:
 - 1) directional effects for all worst possibilities (that is, tower shaking in the same and in the opposite directions) should be considered in the design load combinations; and
 - 2) equivalent static seismic forces can be used, provided they are scaled to match base overturning moments obtained from response spectrum analysis.
- d) Where significant changes occur to mass or stiffness between the floors, the floor diaphragms of upper and lower levels shall be modelled to capture:
 - 1) diaphragm forces. Equivalent beam approach, finite element approach or strut and tie approach may be adopted to model the diaphragms; and
 - 2) potential cracking in the diaphragm by considering an upper bound and a lower bound axial stiffness. The lower and upper bound values of axial and flexural stiffnesses given in [Table 13](#) for sensitivity analysis, may be considered for the cracked section properties, to arrive at the design level earthquake demand on the RC diaphragms.
- e) Plan irregularities of Type I (torsion irregularity) and Type III (diaphragm discontinuity) as per 5 of Part C 'Structural Design/Section 1 Loads and Load Effects' of this NBCS, shall not be present in the first topmost connected floor and the first floor of the tower above the connected floor;
- f) Vertical irregularities of Type I (soft story) as per 5 of Part C/Sec 1 of this NBCS shall not be present in the first connected level and the level above it;
- g) Topmost slab between the towers of connected podium shall be at least 150 mm thick with double mesh reinforcements not less than 0.25 percent of cross-sectional area in each direction;
- h) Peripheral columns of the tower shall be provided with confinement reinforcements as per good practice [C-5(5)] at the first connected level and a level above;
- j) Structural walls of the tower shall be provided with boundary elements as per good practice [C-5(5)] at the uppermost connected level and a level above; and
- k) A transfer structure shall not be provided at the first connected floor.

E-8.2 Ductility

Notwithstanding any of the clauses of this standard, the designer shall take all measures to ensure that the building has:

- a) sufficient ductility capacity;
- b) acceptable energy dissipation mechanism; and
- c) desirable sequence of initiation of ductile behaviour in members.

E-8.3 Frame Buildings

Frame structure for seismic design shall have at least two planar frames with minimum 3 bays in the direction where the lateral load resistance is provided by the moment frames.

E-8.4 Moment Frame — Structural Wall Systems

E-8.4.1 Frame structural wall systems shall be designed as dual systems as per 5 of Part C/Section 1 of this NBCS.

E-8.4.2 In a moment frame–structural wall system, the moment frame shall comply with the requirements of [E-8.3](#), and the structural wall with the requirements of [E-8.5](#). In addition, the moment frames and structural walls shall comply with the requirements of good practice [C-5(5)].

E-8.4.3 *Special Requirements for Seismic Zone IV and above*

Special moment frame and shear walls shall not be discontinued in lower storeys and supported on less stiff and brittle elements.

E-8.5 Structural Wall Systems

E-8.5.1 The thickness of structural wall shall not be less than 160 mm or $h_t/20$, whichever is larger.

E-8.5.2 Opening in structural walls and the associated coupling beams shall meet the following requirements:

- a) For walls, when the opening size is less than 800 mm or one-third length of wall, whichever is lesser, in height or length, the influence of the opening may not be taken into account in the overall stiffness of the building;
- b) For beams, the opening size shall be less than one-third of the height or length of beam and location of the opening shall be such that the top and bottom one-third of height of the coupling beam is not disturbed; and
- c) In either case, all four sides of the opening shall be strengthened with additional reinforcements, and shall comply with requirements of good practice [C-5(5)]. Diameter of reinforcement bars used in this reinforcement shall not be less than 12 mm.

E-8.5.3 Gravity columns in structural wall buildings shall be designed as per requirements of deformation compatibility on non-seismic members given in 5 of Part C/Section 1 of this NBCS.

E-8.5.4 Beams carrying predominant vertical load shall not be supported on coupling beams. Also, columns or structural walls carrying predominant vertical load shall not be supported on coupling beam.

E-8.5.5 In a structural wall system, the structural wall shall comply with the requirements of good practice [C-5(5)].

E-8.5.6 Concentrated gravity loads applied on the wall above the design flexural section shall be assumed to be distributed over a width equal to the bearing width, plus a width on each side that increases at a slope of 2 vertical to 1 horizontal down to the design section, but:

- a) not greater than the spacing of the concentrated loads; and
- b) not extending beyond the edges of the wall panel.

E-8.5.7 Design of coupling beam shall comply with requirements of good practice [C-5(5)], unless it can be shown that loss of stiffness and strength of the coupling beams will not impair the vertical load carrying ability of the structure, the egress from the structure, or the integrity of non-structural components and their connections to the structure.

E-8.5.8 The nominal design shear stress shall be limited to $\tau_{c,max}$ in structural walls that do not sustain tensile axial stress under any load combination, while the nominal design shear stress shall be limited to $0.5\tau_{c,max}$ in

structural walls under tension in any load combination and coupling beams in structural walls under factored design loads, where $\tau_{c,max}$ is as per Table 20 in the good practice [C-5(2)].

E-8.5.9 The amount and distribution of the minimum reinforcement in structural walls shall be as per good practice [C-5(5)].

E-8.5.10 At locations where yielding of longitudinal reinforcements is likely to occur as a result of lateral displacement, development length of longitudinal reinforcement shall be 1.25 times the values calculated for the bar yielded in tension, that is at a stress level of f_y .

E-8.5.11 The maximum longitudinal reinforcement ratio in coupling beam shall be as given in [Table 14](#).

Table 14 Maximum Longitudinal Reinforcement in Coupling Beams (Each at Top and Bottom Surface) <i>(Clause E-8.5.11)</i>		
Sl No.	Span to Depth Ratio	Maximum Reinforcement, Percent
(1)	(2)	(3)
i)	$L_s/D \leq 1.0$	0.6
ii)	$1.0 \leq L_s/D < 2.0$	1.2
iii)	$2.0 \leq L_s/D$	1.5

E-8.5.12 *Requirements for Each Storey Resisting more than 35 percent of Design Base Shear*

Structural wall with a height-to-length ratio greater than 1.0 may be removed in any storey above, if it accounts for less than one-third of the storey shear strength. Further, the resulting structural system must not have any torsional irregularity as per 5 of Part C 'Structural Design/Section 1 Loads and Load Effects' of this NBCS .

E-8.5.13 *Special Requirements for Seismic Zones IV and Above*

- Structural walls shall be continuous to the base without being transferred in plane or out of plane at any level;
- The minimum longitudinal reinforcement in structural walls shall not be less than 0.4 percent of gross cross-sectional area;
- The minimum transverse reinforcement in structural walls shall not be less than 0.25 percent of gross cross-sectional area;
- The reinforcements shall be distributed in two curtains in each direction;
- Structural walls shall be fully embedded and anchored at their base in adequate basements or foundations, so that the wall does not rock. In this respect, walls supported by slabs or beams are not permitted; and
- All openings in structural walls shall preferably be aligned vertically. Random openings, arranged irregularly, shall not be permitted in coupled walls, unless their influence is insignificant.

E-8.6 Flat Slab — Structural Wall Systems

In structures with flat slab system as gravity load carrying system, structural walls shall carry all lateral loads on the building, and column strips of the flat slab system shall not be included in the lateral load resisting system. All requirements related to structures with flat slabs mentioned in 5 of Part C 'Structural Design/Section 1 Loads and Load Effects' of this NBCS shall be met with.

E-8.7 Framed Tube System, Tube-in-Tube System and Multiple Tube System

E-8.7.1 The plan shape of a tube-in-tube system shall be regular with a length to width ratio not more than 2 and, the inner tube shall be centered with the outer tube.

E-8.7.2 Reentrant corners and sharp changes to tubular form should be avoided.

E-8.7.3 Column spacing of framed tube shall preferably be not more than 5 m.

E-8.7.4 In a framed-tube system:

- a) Area of corner column shall be 1 to 2 times that of internal column; and
- b) Height to width ratio of the opening shall be similar to ratio of storey height to column spacing.

E-8.7.5 Due consideration shall be given to shear lag effects in the design of tube structures.

E-8.7.6 In Seismic Zones III and above:

- a) Single span frame shall not be adopted; and
- b) Axial compression ratio of columns shall be as per good practice [C-5(5)].

E-8.7.7 Beams carrying predominantly gravity load shall be directly supported on columns or on walls and not on frame beams.

E-8.7.8 The minimum requirements for reinforcement bar diameters in beams of moment frames of framed-tube structures are given in [Table 15](#).

Table 15 Reinforcement Requirements in Beams (Clause E-8.7.8) All dimensions in millimetres.			
SI No.	Reinforcement Type	Seismic Zone	
		II	III and above
(1)	(2)	(3)	(4)
i)	Stirrup diameter	≥ 8	≥ 10
ii)	Stirrup spacing	≤ 150	≤ 100
iii)	Main (longitudinal) reinforcements	≥ 16	≥ 16

E-9 FOUNDATIONS

E-9.1 Load paths and mechanisms shall be ensured explicitly for transferring vertical and lateral loads between structure and soil system underneath.

E-9.2 A factor of safety of 1.5 shall be provided against overturning and sliding under, (a) unfactored design wind load, gravity load and other permanent imposed load; and (b) 2.5 times design earthquake load, unfactored gravity load and other permanent imposed load .

E-9.3 Geotechnical Investigations

E-9.3.1 All geotechnical investigations needed to establish the safety of the building shall be conducted including for liquefaction potential analysis, and estimation of soil spring constants and modulus of subgrade reaction.

E-9.3.2 For geotechnical investigations, the good practice [C-5(39)] shall be referred.

E-9.4 Depth of Foundation

The embedded depth of the building shall be at least 1/15 of height of building for raft foundation and 1/20 of the height of building for pile and piled raft foundation (excluding pile length). But, this requirement may be relaxed:

- a) when the foundation rests on hard rock; or

- b) when there is no uplift under any portion of the raft in any service load combination, and provided the minimum competent founding strata requirement is fulfilled.

E-9.5 Podium and basement roof slab shall be capable of transferring in-plane shear from the tower to the foundation.

E-9.6 Expansion joints shall preferably be avoided in basements of tall buildings. If provided such joints shall be suitably detailed for water tightness.

E-9.7 Modelling of Soil

E-9.7.1 While modelling raft foundations through spring constant or modulus of sub-grade reaction, zoned spring constants or zoned modulus of sub-grade reaction shall be utilized for design, at least for the case of (dead load + live load) condition. For design of rafts for buildings taller than 150 m, a soil-structure interaction study shall be conducted, using actual column loads and column locations to obtain the zoned spring constants.

E-9.7.2 For piled raft foundations designed with settlement reducing piles, soil-structure interaction study shall be conducted with actual column loads and column locations. This analysis shall be conducted at a minimum for combinations of following loads:

- a) Dead;
- b) Imposed;
- c) Wind in X-direction;
- d) Wind in Y-direction;
- e) Seismic in X-direction; and
- f) Seismic in Y-direction.

where X and Y are orthogonal axes in plan.

E-9.8 Settlements of Foundations

Maximum vertical settlement of raft or piled raft foundations under gravity loads shall comply with requirements of good practice [C-5(40)] and good practice [C-5(41)]. The maximum vertical settlement is limited to 125 mm in raft or raft-pile foundation subject to maximum angular distortion of raft not exceeding 1/500 and 50 mm in rock.

E-10 RECOMMENDATIONS FOR NON-STRUCTURAL ELEMENTS

The non-structural elements (NSEs) of tall buildings shall comply with all relevant existing national standards and guidelines as laid down by the various statutory and non-statutory bodies as well as the client/owner of the building.

E-11 MONITORING DEFORMATIONS IN BUILDINGS

E-11.1 Earthquake Shaking

All tall buildings in Seismic Zone V and tall buildings exceeding 150 m in Seismic Zone III and IV shall be instrumented with tri-axial accelerometers to capture translational and twisting behavior of buildings during strong earthquake shaking.

E-11.2 Wind Oscillations

Buildings over 150 m in height may be instrumented with anemometers and accelerometers to measure wind speed, acceleration and direction on top of the buildings.

E-11.3 Foundation Settlement and Pressure Measurement

E-11.3.1 Permanent settlement markers (at corners and center) shall be provided at raft top level and referenced to a permanent benchmark. Records of settlement shall be maintained till completion of the building and preferably even after completion.

E-11.3.2 Raft or Piled-raft shall be instrumented for monitoring long-term pressure imposed by soil on the raft, at appropriate number (at least 5) of pressure pads below the raft. Alternatively, piles can be instrumented with strain gauges at their top to measure the load on them.

LIST OF STANDARDS

The following list records those standards which are acceptable as ‘good practice’ and ‘accepted standards’ in the fulfilment of the requirements of this NBCS. The latest version of a standard shall be adopted at the time of enforcement of this NBCS. The standards listed may be used as a guide in conformance with the requirements of the referred clauses in this NBCS.

In the following list, the number appearing in the first column within parentheses indicates the number of the reference in this Part C/Section 5.

<i>Sl No.</i>	<i>IS No.</i>	<i>Title</i>
(1)	IS 4845 : 1968	Definitions and terminology relating to hydraulic cement
	IS 6461	Cement concrete — Glossary of terms:
	(Part 1) : 2024	Concrete aggregates (<i>first revision</i>)
	(Part 2) : 2024	Materials (other than cement and aggregate) (<i>first revision</i>)
	(Part 3) : 2024	Concrete reinforcement (<i>first revision</i>)
	(Part 4) : 2024	Types of concrete (<i>first revision</i>)
	(Part 5) : 2024	Formwork for concrete (<i>first revision</i>)
	(Part 6) : 2024	Equipment, tool and plant (<i>first revision</i>)
	(Part 7) : 2024	Mixing, laying, compaction, curing and other construction aspect (<i>first revision</i>)
	(Part 8) : 2024	Properties of concrete (<i>first revision</i>)
	(Part 9) : 2024	Structural aspects (<i>first revision</i>)
	(Part 10) : 2024	Tests and testing apparatus (<i>first revision</i>)
	(Part 11) : 2024	Prestressed concrete (<i>first revision</i>)
	(Part 12) : 2024	Miscellaneous terms (<i>first revision</i>)
(2)	IS 456 : 2000	Plain and reinforced concrete — Code of practice (<i>fourth revision</i>)
(3)	IS 1343 : 2012	Prestressed concrete — Code of practice (<i>second revision</i>)
(4)	IS 16700 : 2023	Criteria for structural safety of tall concrete buildings (<i>first revision</i>)
(5)	IS 13920 : 2016	Ductile design and detailing of reinforced concrete structures subjected to seismic forces — Code of Practice (<i>first revision</i>)
(6)	IS 383 : 2016	Coarse and fine aggregate for concrete — Specification (<i>third revision</i>)
(7)	IS 9142 (Part 2) : 2018	Artificial lightweight aggregate for concrete — Specification: Part 2 Sintered fly ash coarse aggregate (<i>first revision</i>)
(8)	IS 269 : 2015	Ordinary Portland cement — Specification (<i>sixth revision</i>)
(9)	IS 455 : 2015	Portland slag cement — Specification (<i>fifth revision</i>)
(10)	IS 1489	Portland pozzolana cement — Specification
	Part 1 : 2015	Part 1 Fly ash based (<i>fourth revision</i>)

<i>Sl No.</i>	<i>IS No.</i>	<i>Title</i>
	Part 2 : 2015	Part 2 Calcined clay based (<i>fourth revision</i>)
(11)	IS 16415 : 2015	Composite cement — Specification
(12)	IS 18189 : 2023	Portland calcined clay limestone cement — Specification
(13)	IS 12330 : 1988	Specification for sulphate resisting Portland cement
(14)	IS 16993 : 2018	Microfine ordinary Portland cement — Specification
(15)	IS 8043 : 1991	Hydrophobic Portland cement — Specification (<i>second revision</i>)
(16)	IS 12600 : 1989	Portland cement, low heat — Specification
(17)	IS 8041 : 1990	Rapid hardening Portland cement — Specification (<i>second revision</i>)
(18)	IS 6452 : 1989	High alumina cement for structural use — Specification (<i>first revision</i>)
(19)	IS 6909 : 1990	Supersulphated cement — Specification (<i>first revision</i>)
(20)	IS 8042 : 2015	White Portland cement — Specification (<i>third revision</i>)
(21)	IS 10500 : 2012	Drinking water— Specification (<i>second revision</i>)
(22)	IS 3025 (Part 18) : 2022	Methods of sampling and test (physical and chemical) for water and wastewater: Part 18 Volatile and fixed solids (total, filterable and non-filterable at 550 °C (<i>second revision</i>))
(23)	IS 3025 (Part 24/Sec 1) : 2022	Methods of sampling and test (physical and chemical) for water and wastewater: Part 24 Sulphates, Section 1 Gravimetric and turbidity methods (<i>second revision</i>)
(24)	IS 3025 (Part 32) : 1988	Methods of sampling and test (physical and chemical) for water and wastewater: Part 32 Chloride (<i>first revision</i>)
(25)	IS 3025 (Part 17) : 2022	Methods of sampling and test (physical and chemical) for water and wastewater Part 17 Non- filterable residue total suspended solids at 103 °C to 105 °C (<i>second revision</i>)
(26)	IS 3812 (Part 1) : 2013	Pulverized fuel ash — Specification: Part 1 For use as pozzolana in cement, cement mortar and concrete (<i>third revision</i>)
(27)	IS 15388 : 2003	Silica fume — Specification
(28)	IS 16354 : 2015	Metakaolin for use in cement, cement mortar and concrete — Specification
(29)	IS 16714 : 2018	Ground granulated blast furnace slag for use in cement, mortar and concrete — Specification
(30)	IS 16715 : 2018	Ultrafine ground granulated blast furnace slag — Specification
(31)	IS 19058 : 2024	Ultrafine fly ash — Specification
(32)	IS 9103 : 1999	Concrete admixtures — Specification (<i>first revision</i>)
(33)	IS 16172 : 2023	Reinforcement couplers for mechanical splices of steel bars in concrete — Specification (<i>first revision</i>)
(34)	IS 18256 : 2023	Solid round glass fibre reinforced polymer (GFRP) bars for concrete reinforcement — Specification
(35)	IS 1566 : 1982	Specification for hard-drawn steel wire fabric for concrete reinforcement (<i>second revision</i>)
(36)	IS 1199 (Part 6) : 2018	Fresh concrete — methods of sampling, testing and analysis: Part 6 Tests on fresh self compacting concrete (<i>first revision</i>)
(37)	IS 10262 : 2019	Concrete mix proportioning — Guidelines (<i>second revision</i>)
(38)	IS 4031 (Part 5) : 1988	Methods of physical tests for hydraulic cement: Part 5 Determination of initial and final setting times (<i>first revision</i>)

<i>Sl No.</i>	<i>IS No.</i>	<i>Title</i>
(39)	IS 1892 : 2021	Subsurface investigation for foundations — Code of practice (<i>second revision</i>)
(40)	IS 1904 : 2021	General requirements for design and construction of foundations in soils — Code of practice (<i>fourth revision</i>)
(41)	IS 12070 : 1987	Code of practice for design and construction of shallow foundations on rocks

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART C STRUCTURAL DESIGN

SECTION 6 STEEL AND COMPOSITE CONSTRUCTION

SUBSECTION 6(A) STEEL

BUREAU OF INDIAN STANDARDS

CONTENTS

FOREWORD	...	3
1	SCOPE	...
2	-	...
3	TERMINOLOGY	...
4	SYMBOLS	...
5	UNITS	...
6	STANDARD DIMENSIONS, FORM AND WEIGHT	...
7	PLANS AND DRAWINGS	...
8	CONVENTION FOR MEMBER AXES	...
9	MATERIALS	...
10	GENERAL DESIGN REQUIREMENTS	...
11	METHODS OF STRUCTURAL ANALYSIS	...
12	LIMIT STATE DESIGN	...
13	DESIGN OF TENSION MEMBERS	...
14	DESIGN OF COMPRESSION MEMBERS	...
15	DESIGN OF MEMBERS SUBJECTED TO BENDING	...
16	MEMBER SUBJECTED TO COMBINED FORCES	...
17	CONNECTIONS	...
18	DESIGN AND DETAILING FOR EARTHQUAKE LOADS	...
19	FATIGUE	...
20	DESIGN ASSISTED BY TESTING	...
21	DURABILITY	...
22	FIRE RESISTANCE	...
23	SUSTAINABLE STEEL CONSTRUCTION	...
24	FABRICATION AND ERECTION	...
ANNEX A	SYMBOLS	...
ANNEX B	ANALYSIS AND DESIGN METHODS	...
ANNEX C	DESIGN AGAINST FLOOR VIBRATION	...
ANNEX D	DETERMINATION OF EFFECTIVE LENGTH OF COLUMNS	...
ANNEX E	ELASTIC LATERAL TORSIONAL BUCKLING	...
ANNEX F	CONNECTIONS	...
ANNEX G	PLATE GIRDERS WITH CORRUGATED WEB	...
ANNEX H	GENERAL RECOMMENDATIONS FOR STEELWORK TENDERS	...
LIST OF STANDARDS	...	212

National Building Code Sectional Committee, CED 46

FOREWORD

This NBCS (Part C/Subsection 6A) covers the structural design aspects of steel structures in buildings.

This Section covers the use of hot-rolled structural steel sections and steel tubes in buildings. It covers the design by limit state method and plastic theory.

This Section was first published in 1970 and revised in 1983, 2005 and 2016 as a chapter of the National Building Code of India. In the second revision of 2005, the detailed design provisions were not included as IS 800 : 1984 'Code of practice for general construction in steel (*second revision*)' was under revision at that time. This Section, therefore, then referred to the 1984 version of IS 800 with the remarks that the latest version would prevail, as and when published. It, however, covered the philosophical aspects of limit state design which was being introduced in the revision of IS 800. IS 800 was subsequently revised and published as IS 800 : 2007 and hence, became the prevailing version for use under this Section. In the 2016 version of the Code, this Section was brought in line with the revised version of IS 800 : 2007 read with its Amendment No. 1 and to take care of further improvements and incorporations required.

In the current revision, this publication has been renamed as National Building Construction Standards (NBCS).

This revision of the Section is intended to bring it in line with the revised version of IS 800 : 2026 (*under publication*) to take care of further improvements and incorporations required. In this revision, therefore, the following major modifications have been made:

- a) The working stress design has been discontinued due to the increasing familiarity of designers with the limit state design of steel structures;
- b) The table on tensile properties of structural steel products has been updated by including the latest Indian Standards relating to structural steels and aligning the corresponding provisions;
- c) Fire-resistant steel as per IS 15103 : 2002 'Fire resistant steel — Specification', structural weather-resistant steels as per IS 11587 : 1986 'Structural weather resistant steels — Specification', and structural steels for buildings and structures with improved seismic resistance as per IS 15962 : 2012 'Structural steels for buildings and structures with improved seismic resistance — Specification' have been included in the table of tensile properties of structural steel products;
- d) Provisions related to the requirement of temperature stress analysis have been updated;
- e) Provisions related to the requirement of fire load have been updated;
- f) [Table 2](#) Limiting width thickness ratio of elements has been updated;
- g) Provisions related to the consideration of second order effects have been updated;
- h) Provisions related to design for earthquake loads relating to load and load combinations, various types of frames and connections in the critical zones of frames have been updated;
- j) Provisions related to bracing connections have been updated;
- k) Provisions related to fabrication procedures have been updated;
- m) Certain terminologies, symbols and figures have been updated based on the changes incorporated;
- n) Buckling class of cross section table has been updated;
- p) Imperfection parameter for lateral torsional buckling of beam has been added;
- q) A new clause on sustainable steel construction has been added. This provision covers the following:
 - 1) Requirements from Engineers, contractors and supply chain members;
 - 2) Sustainable design process by minimizing the materials, fabrication; and avoidance of maintenance; and

3) Design for adaptability and deconstruction, etc.

- r) The detailed provision relating to design of corrugated web girder have been introduced in [Annex G](#).

The revised Section will enhance the confidence of designers, engineers, contractors, technical institutions, professional bodies and the industry and will open a new era in safe and economic construction in steel.

The provisions relating to using steel tubes in general building construction are based on IS 806 : 1968 'Code of practice for the use of steel tubes in general building construction', which is complementary to IS 800. The design using tubular steel in structural applications may be carried out using the limit state design provisions in this Section until IS 806 is revised as per the limit state design method.

The design of cold-formed steel members is based on the provisions of IS 801 : 1975 'Code of practice for use of cold formed light gauge steel structural members in general building construction'. The standard is currently under revision to be based on the limit state design practice. Until the revised IS 801 is published, the cold formed steel members may be designed using the current IS 801, based on the working stress method.

All standards, whether given herein above or cross-referred to in the main text of this section, are subject to revision. The parties to agreement based on this Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding of numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

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Important Explanatory Note for Users of the NBCS

In any Part/Section of this NBCS, where reference is made to ‘**good practice**’ in relation to **design, constructional procedures or other related information**, and where reference is made to ‘**accepted standard**’ in relation to **material specification, testing, or other related information**, the Indian Standards listed at the end of the Part/Section shall be used as a guide to the interpretation.

At the time of publication, the editions indicated in the standards were valid. All standards are subject to revision and parties to agreements based on any Part/ Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the list of standards given at the end of a Part/Section, the number appearing within parentheses in the first column indicates the number of the reference of the standard in the Part/Section. For example:

a) Good practices [C-6A(1)] refers to the Indian Standard(s) give at serial number (1) of the list of standards given at the end of this Part C/Subsection 6A, that is, IS 786 : 1967 ‘Conversion factors and conversion tables (*first revision*)’.

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART C STRUCTURAL DESIGN

SECTION 6 STEEL AND COMPOSITE CONSTRUCTION

SUBSECTION 6(A) STEEL

1 SCOPE

1.1 This NBCS (Part C/Subsection 6A) covers the structural design aspects of steel structures in buildings. This Section applies to general construction using hot rolled steel sections and steel tubes joined using riveting, bolting, and welding. Cold formed light gauge steel sections are covered in separate standards. This Section covers the design by limit state method and plastic theory.

1.2 This Section gives only general guidance as regards to the various loads to be considered in the design. For the actual loads, such as dead, imposed, snow, wind and earthquake loads and load combinations to be used, reference may be made to Part C 'Structural Design/Section 1 Loads and Load Effects' of this NBCS.

1.3 Fabrication and erection requirements covered in this Section are general and the minimum necessary quality of material and workmanship consistent with assumptions in the design rules. The actual requirements may be further developed as per other standards or the project specification, the type of structure and the method of construction.

1.4 For seismic design, recommendations pertaining to steel frames only are covered in this section. For more detailed information on seismic design of other structural and non-structural components, reference be made to 5 of Part C 'Structural Design/Section 1 Loads and Load Effects' of this NBCS and other special publications on the subject.

2 —

NOTE — This clause is intentionally left blank.

3 TERMINOLOGY

For the purpose of this standard, the following definitions shall apply.

3.1 Accidental Loads — Loads due to explosion, impact of vehicles, or other rare loads for which the structure is susceptible as per user.

3.2 Accompanying Load — Imposed load acting along with leading imposed load but Causing action/deflection lesser than the leading load.

3.3 Action Effect or Load Effect — Internal response of a structural element, including axial force, shear force, bending moment, and torsional moment, resulting from applied external loads, temperature variations, and other environmental influences such as seismic activity, settlement effects, and residual stresses.

3.4 Action — The primary cause for stress or deformations in a structure such as dead, imposed, wind, seismic or temperature loads.

3.5 Actual Length — The distance measured from centre to centre of intersection points with supporting members, applicable to beams, trusses, or other structural elements. In the case of a cantilevered member, it refers to the full projection from the fixed support.

3.6 Beam — A member subjected predominantly to bending.

3.7 Bearing Type Connection — A connection made using bolts in 'snug-tight' condition, or rivets where the load is transferred by bearing of bolts or rivets against plate inside the bolt hole.

3.8 Braced Member — A member in which the relative transverse displacement is effectively prevented by bracing.

3.9 Brittle Cladding — Claddings, such as cement fibre boards which get damaged before undergoing considerable deformation.

3.10 Buckling Load — The load at which an element, a member or a structure as a whole, either collapses in service or buckles in a load test and develops excessive lateral (out of plane) deformation or instability.

3.11 Buckling Strength or Resistance — Force or moment, which a member can withstand without buckling.

3.12 Built-up Section — A member fabricated by interconnecting more than one element to form a compound section acting as a single member.

3.13 Camber — Intentionally introduced pre-curving (usually upwards) in a system, member or any portion of a member with respect to the straight line joining the end points in place. Frequently, camber is introduced to compensate for deflections at a specific level of loads.

3.14 Characteristic Load (Action) — The value of specified load (action), above which not more than a specified percentage (usually 5 percent) of samples of corresponding load are expected to be encountered.

3.15 Characteristic Yield/Ultimate Stress — The minimum value of stress, below which not more than a specified percentage (usually 5 percent) of corresponding stresses of samples tested are expected to occur.

3.16 Column — A member in upright (vertical) position which supports a roof or floor system and predominantly subjected to compression.

3.17 Compact Section — A cross section, which can develop plastic moment, but has inadequate plastic rotation capacity needed for formation of a plastic collapse mechanism of the member or structure.

3.18 Constant Stress Range — The amplitude between which the stress ranges under cyclic loading is assumed to be constant during the life of the structure or a structural element.

3.19 Corrosion — An electrochemical process over the surface of steel, leading to oxidation of the metal.

3.20 Crane Load — The horizontal and vertical forces exerted by a crane on a structure, including dynamic effects due to crane movement, acceleration, and braking caused by its due to its self-weight and load lifted.

3.21 Cumulative Fatigue — Total damage due to fatigue loading of varying stress ranges.

3.22 Cut-off Limit — The stress range, corresponding to the particular detail, below which cyclic loading need not be considered in cumulative fatigue damage evaluation (corresponds to 10^8 number of cycles in most cases).

3.23 Dead Loads — The self-weights of all permanent constructions and installations including the self-weight of all walls, partitions, floors, roofs, and other permanent fixtures acting on a member.

3.24 Deflection — The deviation from the standard position of a member or structure.

3.25 Design Life — Time period for which a structure or a structural element is required to perform its function without damage.

3.26 Design Load/Factored Load — A load value obtained by multiplying the characteristic load with the partial safety factor for loads.

3.27 Design Spectrum — Frequency distribution of the stress ranges from all the nominal loading events during the design life (stress spectrum).

3.28 Detail Category — Designation given to a particular detail to indicate the S-N curve to be used in fatigue assessment.

3.29 Discontinuity — A sudden change in cross section of a loaded member, causing a stress concentration at the location.

3.30 Ductility — The property of the material or a structure indicating the extent to which it can deform beyond the limit of yield deformation before failure or fracture. The ratio of ultimate to yield deformation is usually termed as ductility.

3.31 Durability — The ability of a material to resist deterioration over long periods of time.

3.32 Earthquake effects — The inertia forces and relative deformations induced in a structure due to the ground shaking during an earthquake.

3.33 Edge Distance — Distance from the centre of a fastener hole to the nearest edge of an element measured perpendicular to the direction of load transfer.

3.34 Effective Lateral Restraint — Restraint that produces sufficient resistance to prevent deformation in the lateral direction.

3.35 Effective Length — Actual length of a member between points of effective restraint or effective restraint and free end, multiplied by a factor to take account of the end conditions in buckling strength calculations.

3.36 Elastic Cladding — Claddings, such as metal sheets that can undergo considerable deformation without damage.

3.37 Elastic Critical Moment — The elastic moment, which initiates lateral-torsional buckling of a laterally unsupported beam.

3.38 Elastic Design — Design, which assumes elastic behaviour of materials throughout the service load range.

3.39 Elastic Limit — The stress below which the material regains its original size and shape when the load is removed. In steel design, it is taken as the yield stress.

3.40 End Distance — Distance from the centre of a fastener hole to the edge of an element measured parallel to the direction of load transfer.

3.41 Erection Loads — The actions (loads and deformations) experienced by the structure exclusively during erection.

3.42 Erection Tolerance — Amount of deviation related to the plumbness, alignment, and level of the element as a whole in the erected position. The deviations are determined by considering the locations of the ends of the element.

3.43 Exposed Surface Area to Mass Ratio — The ratio of the surface area exposed to the fire (in mm²) to the mass of steel (in kg).

NOTE — In the case of members with fire protection material applied, the exposed surface area is to be taken as the internal surface area of the fire protection material.

3.44 Fabrication Tolerance — Amount of deviation allowed in the nominal dimensions and geometry in fabrication activities, such as cutting to length, finishing of ends, cutting of bevel angles, etc.

3.45 Factor of Safety — The factor by which the yield stress of the material of a member is divided to arrive at the permissible stress in the material.

3.46 Fatigue — Damage caused by repeated fluctuations of stress, leading to progressive cracking of a structural element.

3.47 Fatigue Loading — Set of nominal loading events, cyclic in nature, described by the distribution of the loads, their magnitudes and the numbers of applications in each nominal loading event.

3.48 Fatigue Strength — The stress range for a category of detail, depending upon the number of cycles it is required to withstand during design life.

3.49 Fire Exposure Condition

3.49.1 Three-Sided Fire Exposure Condition — Steel member incorporated in or in contact with a concrete or masonry floor or wall (at least against one surface); and

NOTES

1 Three-sided fire exposure condition is to be considered separately unless otherwise specified (see [22.9](#)).

2 Members with more than one face in contact with a concrete or masonry floor or wall may be treated as three-sided fire exposure.

3.49.2 Four-Sided Fire Exposure Condition — Steel member, which may be exposed to fire on all sides.

3.50 Fire Protection System — The fire protection material and its method of attachment to the steel member.

3.51 Fire Resistance — The ability of an element, component or structure, to fulfil for a stated period of time, the required stability, integrity, thermal insulation and/or other expected performance specified in a standard fire test.

3.52 Fire Resistance Level — The fire resistance grading period for a structural element or system, in minutes, which is required to be attained in the standard fire test.

3.53 Flexural Stiffness — Stiffness of a member against rotation as evaluated by the value of bending deformation moment required to cause a unit rotation while all other degrees of freedom of the joints of the member except the rotated one are assumed to be restrained.

3.54 Friction Type Connection — Connection effected by using pre-tensioned high strength bolts where shear force transfer is due to mobilization of friction between the connected plates due to clamping force developed at the interface of connected plates by the bolt pre-tension.

3.55 Gauge — The spacing between adjacent parallel lines of fasteners, transverse to the direction of load/stress.

3.56 Gravity Load — Loads arising due to gravitational effects.

3.57 Gusset Plate — The plate to which the members intersecting at a joint are connected.

3.58 High Shear — High shear condition is caused when the actual shear due to factored load is greater than a certain fraction of design shear resistance (see [16.2.2](#)).

3.59 Imposed (Live) Load — The load assumed to be produced by the intended use or occupancy including distributed, concentrated, impact, vibration and snow loads but excluding, wind, earthquake and temperature effects.

3.60 Instability — The phenomenon which disables an element, member or a structure to carry further load due to excessive deflection lateral to the direction of loading and vanishing stiffness.

3.61 Lateral Restraint for a Beam — See [3.34](#).

3.62 Leading Imposed Load — Imposed load causing higher action and/or deflection.

3.63 Limit State — Any limiting condition beyond which the structure ceases to fulfill its intended function (see also [3.86](#)).

3.64 Live Load — See [3.59](#).

3.65 Load — An externally applied force or action (see also [3.4](#)).

3.66 Main Member — A structural member, which is primarily responsible for carrying and distributing the applied load or action.

3.67 Mill Tolerance — Amount of variation allowed from the nominal dimensions and geometry, with respect to cross sectional area, non-parallelism of flanges, and out of straightness such as sweep or camber, in a product, as manufactured in a steel mill.

3.68 Normal Stress — Stress component acting normal to the face, plane or section.

3.69 Partial Safety Factor — The factor normally greater than unity by which either the loads (actions) are multiplied or the resistances are divided to obtain the design values.

3.70 Period of Structural Adequacy under Fire — The time (t), in minutes, for the member to reach the limit state of structural inadequacy in a standard fire test.

3.71 Permissible Stress — When a structure is being designed by the working stress method, the maximum stress that is permitted to be experienced in elements, members or structures under the nominal/service load (action).

3.72 Pitch — The centre to centre distance between individual fasteners in a line, in the direction of load/stress.

3.73 Plastic Collapse — The failure stage at which sufficient number of plastic hinges have formed due to the loads (actions) in a structure leading to a failure mechanism.

3.74 Plastic Design — Design against the limit state of plastic collapse.

3.75 Plastic Hinge — A yielding zone with significant inelastic rotation, which forms in a member, when the plastic moment is reached at a section.

3.76 Plastic Moment — Moment capacity of a cross section when the entire cross section has yielded due to bending moment.

3.77 Plastic Section — Cross section, which can develop a plastic hinge and sustain plastic moment over sufficient plastic rotation required for formation of plastic failure mechanism of the member or structure.

3.78 Poisson's Ratio — The absolute value of the ratio of lateral strain to longitudinal strain under uniaxial loading.

3.79 Proof Stress — The stress to which high strength friction grip (HSFG) bolts are pre-tensioned.

3.80 Proof Testing — The application of test loads to a structure, sub-structure, member or connection to ascertain the structural characteristics of only that specific unit.

3.81 Prototype Testing — Testing of structure, sub-structure, members or connections to ascertain the structural characteristics of that class of structures, sub-structures, members or connections that are nominally identical (full scale) to the units tested.

3.82 Prying Force — Additional tensile force developed in a bolt as a result of the flexing of a connection component such as a beam end plate or leg of an angle.

3.83 Rotation — The change in angle at a joint between the original orientation of two linear member and their final position under loading.

3.84 Secondary Member — Member which is provided for overall stability and or for restraining the main members from buckling or similar modes of failure.

3.85 Semi-compact Section — Cross section, which can attain the yield moment, but not the plastic moment before failure by plate buckling.

3.86 Serviceability Limit State — A limit state of acceptable service condition exceedance of which causes serviceability failure.

3.87 Shear Force — The in-plane force at any transverse cross section of a straight member of a column or beam.

3.88 Shear lag — The in plane shear deformation effect by which concentrated forces tangential to the surface of a plate gets distributed over the entire section perpendicular to the load over a finite length of the plate along the direction of the load.

3.89 Shear Stress — The stress component acting parallel to a face, plane or cross section.

3.90 Slender Section — Cross section in which the elements buckle locally before reaching yield moment.

3.91 Slenderness Ratio — The ratio of the effective length of a member to the radius of gyration of the cross section about the axis under consideration.

3.92 Slip Resistance — Limit shear that can be applied in a friction grip connection before slip occurs.

3.93 S-N Curve — The curve defining the relationship between the number of stress cycles to failure (N_{sc}) at a constant stress range (S_c), during fatigue loading of a structure.

3.94 Snow Load — Load on a structure due to the accumulation of snow and ice on surfaces such as roof.

3.95 Snug Tight — The tightness of a bolt achieved by a few impacts of an impact wrench or by the full effort of a person using a standard spanner.

3.96 Stability Limit State — A limit state corresponding to the loss of static equilibrium of a structure by excessive deflection transverse to the direction of predominant loads.

3.97 Stickability — The ability of the fire protection system to remain in place as the member deflects under load during a fire test.

3.98 Stiffener — An element used to retain or prevent the out-of-plane deformations of plates.

3.99 Strain — Deformation per unit length or unit angle.

3.100 Strain Hardening — The phenomenon of increase in stress with increase in strain beyond yielding.

3.101 Strength — Resistance to failure by yielding or buckling.

3.102 Strength Limit State — A limit state of collapse or loss of structural integrity.

3.103 Stress — The internal force per unit area of the original cross section.

3.104 Stress Analysis — The analysis of the internal force and stress condition in an element, member or structure.

3.105 Stress Cycle Counting — Sum of individual stress cycles from stress history arrived at using any rational method.

3.106 Stress Range — Algebraic difference between two extremes of stresses in a cycle of loading.

3.107 Stress Spectrum — Histogram of stress cycles produced by a nominal loading event design spectrum, during design life.

3.108 Structural Adequacy for Fire — The ability of the member to carry the test load exposed to the standard fire test.

3.109 Structural Analysis — The analysis of stress, strain, and deflection characteristics of a structure.

3.110 Strut — A compression member, which may be oriented in any direction.

3.111 Sway — The lateral deflection of a frame.

3.112 Sway Member — A member in which the transverse displacement of one end, relative to the other is not effectively prevented.

3.113 Tensile Stress — The characteristic stress corresponding to rupture in tension, specified for the grade of steel in the appropriate Indian Standard, as listed in [Table 1](#).

3.114 Test Load — The factored load, equivalent to a specified load combination appropriate for the type of test being performed.

3.115 Transverse — Direction along the stronger axes of the cross section of the member.

3.116 Ultimate Limit State — The state which, if exceeded can cause collapse of a part or the whole of the structure.

3.117 Ultimate Stress — See [3.113](#).

3.118 Wind Loads — Load experienced by member or structure due to wind pressure acting on the surfaces.

3.119 Yield Stress — The characteristic stress of the material in tension before the elastic limit of the material is exceeded, as specified in the appropriate Indian Standard, as listed in [Table 1](#).

3.120 Terms Relating to Sustainable Steel Design

3.120.1 Embodied Carbon — Life-cycle greenhouse gas emissions (expressed as carbon dioxide equivalents – CO₂e) that occur during the manufacture and transport of construction materials and components, as well as the construction process itself and end-of-life aspects of the building.

NOTE — In recent years, the term embodied carbon of construction materials and products has become synonymous with the term carbon footprint.

3.120.2 Deconstruction (or Disassembly, or Demounting) — Deconstruction is the process of taking a building apart into its components in such a way that they can be more readily reused; it minimizes the destructive aspects of the process of removing buildings, preserving components and materials not wasting them, and treating them as a resource.

3.120.3 Design for Deconstruction (DfD) — Designing for deconstruction is considering, at the design stage, how a building can be taken apart.

3.120.4 Environmental Product Declaration (EPD) — An EPD communicates verifiable, accurate, non-misleading environmental information for products and their applications; EPD information is expressed in information modules, which allow easy organization and expression of data packages throughout the life cycle of the product.

3.120.5 Material Efficiency — Optimizing the use of the steel by avoiding waste during fabrication and avoiding over specification of steel in the design.

3.120.6 Recycling — Steel involves re-melting of scrap to form new semi-finished products.

3.120.7 Responsible Sourcing — Management of sustainable development in the provision or procurement of a product.

3.120.8 Reuse — Use of previously used components with little or no reprocessing, largely in their original form; they may be reused for the original function (a conventional reuse scenario), or repurposed.

3.120.9 Sustainable Development — An enduring, balanced approach to economic activity, environmental responsibility and social progress.

3.120.10 Traceability — Ability to trace the history, application or location of an object.

4 SYMBOLS

For the purpose of this Subsection, the symbols (other than those used for load categorization) given in [Annex A](#) are applicable.

5 UNITS

For the purpose of design calculations the following units are recommended:

- a) forces and loads, in kN, kN/m, kN/m²;
- b) unit mass, in kg/m³;
- c) unit weight, in kN/m³;
- d) stresses and strengths, in N/mm² (= MN/m² or MPa); and
- e) moments (bending, etc), in kNm.

For conversion of one system of units to another system, [C-6A(1)] may be referred.

6 STANDARD DIMENSIONS, FORM AND WEIGHT

The dimensions, form, weight, tolerances of all rolled shapes, all rivets, bolts, nuts, studs, and welds and other members used in any steel structure shall conform to accepted standards [C-6A(2)], wherever applicable.

7 PLANS AND DRAWINGS

7.1 Plans, drawings and stress sheet shall be prepared according to accepted standards [C-6A(3)].

7.1.1 Plans

The plans (design drawings) shall show the sizes, sections, and the relative locations of the various members. Floor levels, column centres, and offsets shall be dimensioned. Plans shall be drawn to a scale large enough to convey the information adequately. Plans shall indicate the type of construction to be employed; and shall be supplemented by such data on the assumed loads, shears, moments and axial forces to be resisted by all members and their connections, as may be required for the proper preparation of shop drawings. Any special precaution to be taken in the erection of structure, from the design consideration shall also be indicated in the drawing.

7.1.2 Shop Drawings

Shop drawings, giving complete information necessary for the fabrication of the component parts of the structure including the location, type, size, length and detail of all welds and fasteners shall be prepared in advance of the actual fabrication. They shall clearly distinguish between shop and field rivets, bolts and welds. For additional information to be included on drawings for designs based on the use of welding, reference shall be made to appropriate Indian Standards. Shop drawings shall be made in conformity with accepted standard [C-6A(4)]. A marking diagram allotting distinct identification marks to each separate part of steel work shall be prepared. The diagram shall be sufficient to ensure convenient assembly and erection at site.

7.2 Symbols used for welding on plans and shop drawings shall be according to good practice [C-6A(5)].

8 CONVENTION FOR MEMBER AXES

Unless otherwise specified convention used for member axes is as follows ([Fig. 1](#)):

- a) x-x along the member;
- b) y-y an axis of the cross section:
 - 1) perpendicular to the flanges; and
 - 2) perpendicular to the smaller leg in an angle section.
- c) z-z an axis of the cross-section:

- 1) axis parallel to flanges; and
- 2) axis parallel to smaller leg in angle section.
- d) $u-u$ major axis (when it does not coincide with $z-z$ axis); and
- e) $v-v$ minor axis (when it does not coincide with $y-y$ axis).

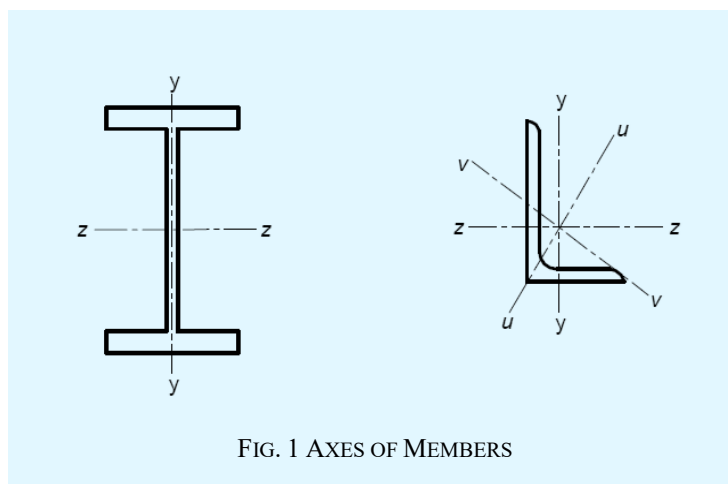


FIG. 1 AXES OF MEMBERS

9 MATERIALS

9.1 General

The material properties given in this clause are nominal values, to be accepted as characteristic values in design calculations.

9.2 Structural Steel

9.2.1 The provisions in this Section are applicable to the steels commonly used in steel construction namely, structural mild steel and high tensile structural steel.

9.2.2 All the structural steel used in general construction, coming under the purview of this standard shall before fabrication conform to accepted standard [C-6A(6)].

9.2.3 Structural steel other than those specified in [9.2.2](#) may also be used provided that the permissible stresses and other design provisions are suitably modified and the steel is also suitable for the type of fabrication adopted.

9.2.3.1 Steel that is not supported by mill test result may be used only in unimportant members and details, where their properties such as ductility and weldability would not affect the performance requirements of the members and the structure as a whole.

However, such steels may be used in structural system after confirming their quality by carrying out appropriate tests in accordance with the method specified in accepted standard [C-6A(7)].

9.2.4 Properties

The properties of structural steel for use in design may be taken as given in [9.2.4.1](#) and [9.2.4.2](#).

9.2.4.1 Physical properties of structural steel irrespective of its grade may be taken as:

- a) unit mass of steel, $\rho = 7\,850\text{ kg/m}^3$;
- b) modulus of elasticity, $E = 2.0 \times 10^5\text{ N/mm}^2\text{ (MPa)}$;
- c) Poisson's ratio, $\mu = 0.3$;
- d) modulus of rigidity, $G = 0.769 \times 10^5\text{ N/mm}^2\text{ (MPa)}$; and

e) coefficient of thermal expansion $\alpha_t = 12 \times 10^{-6} / ^\circ\text{C}$

9.2.4.2 Mechanical properties of structural steel

The principal mechanical properties of the structural steel important in design are the yield stress, f_y ; the tensile or ultimate stress, f_u ; the maximum percent elongation on a standard gauge length and notch toughness. Except for notch toughness, the other properties are determined by conducting tensile tests on samples cut from the plates, sections, etc., in accordance with accepted standard [C-6A(7)]. Commonly used properties for the common steel products of different specifications are summarized in [Table 1](#).

9.2.4.2.1 Steel tubes — Steel tubes used in building construction shall be hot finished tubes conforming to the requirements specified in accepted standard [C-6A(8)].

9.2.4.2.2 Tubes made by other than hot finishing processes, or which have been subjected to cold working, shall be regarded as hot finished if they have subsequently been heat-treated and are supplied in the normalized conditions.

9.2.4.3 Electrodes — The electrodes used for welding steel tubes shall conform to the requirements of accepted standard [C-6A(9)] or [C-6A(10)], as appropriate.

9.3 Rivets

9.3.1 Rivets shall be manufactured from steel conforming to accepted standard [C-6A(11)]. They may also be manufactured from steel conforming to accepted standard [C-6A(6)] provided that the steel meets the requirements given in accepted standard [C-6A(12)].

9.3.2 Rivets shall conform to accepted standards [C-6A(13)] as appropriate.

9.3.3 High Tensile Steel Rivets

High tensile steel rivets, shall be manufactured from steel conforming to accepted standard [C-6A(12)].

9.4 Bolts, Nuts and Washers

Bolts, nuts and washers shall conform as appropriate to accepted standards [C-6A(14)]. The recommendations in good practice [C-6A(15)] shall be followed.

9.5 Steel Casting

Steel casting shall conform to accepted standard [C-6A(16)] or accepted standard [C-6A(17)].

9.6 Welding Consumable

9.6.1 Covered electrodes shall conform to accepted standard [C-6A(9)] or [C-6A(10)], as appropriate.

9.6.2 Filler rods and wires for gas welding shall conform to accepted standard [C-6A(18)].

9.6.3 The supply of solid filler wires for submerged arc welding of structural steels shall conform to accepted standard [C-6A(19)].

9.6.4 The bare wire electrodes for submerged arc welding and the combination of wire and flux shall satisfy the requirements of accepted standard [C-6A(20)].

9.6.5 Filler rods and bare electrodes for gas shielded metal arc welding shall conform to accepted standards [C-6A(21)], as appropriate.

9.7 Other Materials

Other materials used in association with structural steel work shall conform to appropriate Indian standards.

Table 1 Tensile Properties of Structural Steel Products

(Clauses [3.113](#), [3.119](#) and [9.2.4.2](#))

Table 1 Tensile Properties of Structural Steel Products																			
(Clauses 3.113 , 3.119 and 9.2.4.2)																			
Sl No.	Accepted Standard	Grade/ Classification	Properties																
			Yield Stress, MPa <i>Min</i>				Ultimate Tensile Stress, MPa <i>Min</i>	Elongation, Percent <i>Min</i>											
(1)	(2)	(3)	(4)				(5)	(6)											
i)	[C-6A(22)]	CR1 CR2 CR3 CR4 CR5	280 240 220 210 190				410 370 350 350 350	Gauge Length											
								80 mm					50 mm						
								27 30 34 36 38					28 31 35 37 40						
								thickness, t, mm				Gauge Length-50mm and thickness, t, mm							
								< 0.40	≥ 0.40 and < 0.80	≥ 0.80 and < 1.00	≥ 1.00	< 0.40	≥ 0.40 and < 0.60	≥ 0.60 and <0.80	≥ 0.80 and < 1.00	≥ 1.00 and < 1.20	≥ 1.20 and < 1.60	≥ 1.60 and < 2.00 or ≥ 2.00 and < 2.50 or > 2.50	
								ISC590LA	—	430 to 580	420 to 570	410 to 560	590	—	—	17 to 32	17 to 32	18 to 33	18 to 33
		ISC490W	—	305 to 410	295 to 400	295 to 400	490	—	—	23 to 35	23 to 35	24 to 36	24 to 36	≥ 24					
		ISC540W	—	335 to 440	325 to 430	325 to 430	540	—	—	20 to 32	20 to 32	21 to 33	21 to 33	≥ 21					
		ISC590W	—	365 to 470	355 to 460	355 to 460	590	—	—	17 to 29	17 to 29	18 to 30	18 to 30	≥ 18					
		ISC450Y	—	270 to 360	260 to 370	250 to 340	450	—	—	26 to 42	27 to 43	28 to 44	29 to 45	≥ 29					
		ISC490Y	—	310 to 410	300 to 400	290 to 390	490	—	—	22 to 38	23 to 39	24 to 40	25 to 41	≥ 25					
		ISC540Y	—	340 to 450	330 to 440	320 to 430	540	—	—	18 to 35	19 to 36	20 to 37	21 to 38	≥ 21					

Table 1 (Continued)

Sl No.	Accepted Standard	Grade / Classification	Properties		
			Yield Stress, MPa <i>Min</i>	Ultimate Tensile Stress, MPa <i>Min</i>	Elongation, Percent <i>Min</i>
(1)	(2)	(3)	(4)	(5)	(6)
ii)	[C-6A(9)]	Ex40xx Ex41xx Ex42xx Ex43xx Ex44xx Ex50xx Ex51xx Ex52xx Ex53xx Ex54xx Ex55xx Ex56xx	330 330 330 330 330 360 360 360 360 360 360 360	410 to 540 410 to 540 410 to 540 410 to 540 410 to 540 510 to 610 510 to 610 510 to 610 510 to 610 510 to 610 510 to 610 510 to 610	16 20 22 24 24 16 18 18 20 20 20 20

Table 1 (Continued)

SI No.	Accepted Standard	Grade/ Classification	Properties								
			Yield Stress, MPa <i>Min</i>			Ultimate Tensile Stress, MPa <i>Min</i>	Elongation, Percent <i>Min</i>				
(1)	(2)	(3)	(4)			(5)	(6)				
iii)	[C-6A(23)]	HR1 HR2 HR3 HR4	— — — —			440 420 400 380	$t \leq 3$		$t > 3$		
							GL L ₀ = 80 mm	GL L ₀ = 50 mm	GL L ₀ = 5.65√S ₀	GL L ₀ = 50 mm	
							23	24	28	29	
							25	26	30	31	
							28	29	33	34	
							31	32	36	37	
		ISH270C ISH270D ISH270E	$t < 2$ $2 \leq t < 3.2$ $t \geq 3.2$			270 to 420 270 to 400 270 to 380	$t < 2$	$2 \leq t < 3.2$	$t \geq 3.2$	$t < 3$	$t > 3$
							GL Lo = 50 mm			GL Lo = 80 mm	GL L0 = 5.65√S ₀
			170, Min				26	26	31	25	30
			170, Min		165, Min		29	29	34	28	33
165, Min	155, Min		145, Min	32	32		37	31	36		

Table 1 (Continued)

Sl No.	Accepted Standard	Grade/Classification	Properties		
			Yield Stress, MPa <i>Min</i>	Ultimate Tensile Stress MPa <i>Min</i>	Elongation, Percent <i>Min</i>
(1)	(2)	(3)	(4)	(5)	(6)
iv)	[C-6A(24)]	3.6	180	330	25
		4.6	240	400	22
		4.8	320	420	—
		5.6	300	500	20
		5.8	400	520	—
		6.8	480	600	—
		8.8 ($d \leq 16$ mm)	640 ¹⁾	800	12
		8.8 ($d > 16$ mm)	660 ¹⁾	830	12
		9.8	720 ¹⁾	900	10
		10.9	940 ¹⁾	1 040	9
		12.9	1 100 ¹⁾	1 220	8
v)	[C-6A(25)]	1	200	370	26
		1A	220	410	25
		2	230	430	24
		2A	250	460	22
		3	270	490	21
		3A	280	540	20
		4	320	620	15
		5	350	710	13
		6	370	740	10

Table 1 (Continued)												
Sl No.	Accepted Standard	Grade/Classification	Properties									
			Yield Stress, MPa <i>Min</i>					Ultimate Tensile Stress, MPa <i>Min</i>			Elongation, Percent <i>Min</i>	
(1)	(2)	(3)	(4)					(5)			(6)	
vi)	[C-6A(26)]	S _t 37 S _t 42	<i>d</i> or <i>t</i>					360 to 440 410 to 500			26 23	
			≤ 20		> 20							
			{ 220 250		200 240							
vii)	[C-6A(27)]		<i>t</i>					<i>t</i>			<i>t</i>	
			≤ 16	> 16 and ≤ 40	> 40 and ≤ 60	> 60 and ≤ 100	>100 and ≤ 350	≤60	> 60 and ≤ 100	> 100 and ≤ 350	≤ 60	> 60 and ≤ 350
			235	225	215	200	185	360 to 480	360 to 480	350 to 480	24	23
			265	255	245	215	200	410 to 530	410 to 530	400 to 530	22	21
			290	285	280	255	230	460 to 580	450 to 570	440 to 570	21	20

Table 1 (Continued)

SI No.	Accepted Standard	Grade/Classification	Properties			
			Yield Stress, MPa <i>Min</i>		Ultimate Tensile Stress, MPa <i>Min</i>	Elongation Percent, <i>Min</i>
(1)	(2)	(3)	(4)		(5)	(6)
viii)	[C-6A(6)]	E250 (A, BR, B0, C) E275 (A, BR, B0, C) E300 (A, BR, B0, C) E350 (A, BR, B0, C) E410 (A, BR, B0, C) E450 (A, BR)	<i>d</i> or <i>t</i>			
			<20	20 to 40	> 40	
			250	240	230	
			275	265	255	
			300	290	280	
			350	330	320	
			410	390	380	
ix)	[C-6A(28)]	A B D E	235		400 to 520 ²⁾	$L_o = 5.65\sqrt{S_o}$
						22 ³⁾
						22 ³⁾
						22 ³⁾
						22 ³⁾
x)	[C-6A(29)]	—	240		350 to 450	25

Table 1 (Continued)						
Sl No.	Accepted Standard	Grade/Classification	Properties			
			Yield Stress, MPa <i>Min</i>		Ultimate Tensile Stress, MPa <i>Min</i>	Elongation Percent, <i>Min</i>
(1)	(2)	(3)	(4)		(5)	(6)
xi)	[C-6A(11)]	Annealed Condition	160		330 to 410	30
		As-Drawn Condition	190		410 to 490	20
xii)	[C-6A(30)]	HFC 210/CDS 210/ERW210	210		330	20
		HFC 240/CDS 240/ERW240	240		410	18
		HFC 310/CDS 310/ERW310	310		450	15
xiii)	[C-6A(31)]	1	170		290	30
		2	210		330	28
		3	240		410	25
		4	275		430	20
		5	310		490	15
xiv)	[C-6A(32)]	WR-Fe 480A	< 12	12 to 25	25 to 40	> 40
		WR-Fe 480A	345	325	325	—
		WR-Fe 480B	345	345	345	340
		WR-Fe 500	355	—	—	—
					355	20

Table 1 (Concluded)								
SI No.	Accepted Standard	Grade/Classification	Properties					
			Yield Stress, MPa <i>Min</i>		Ultimate Tensile Stress, MPa <i>Min</i>	Elongation Percent, <i>Min</i>		
(1)	(2)	(3)	(4)		(5)	(6)		
xv)	[C-6A(33)]		<12	12 to 25	25 to 40	> 40		
		Grade A (FR-Fe 410)	300	280	260	240	410	21
		Grade B (FR-Fe 490)	350	330	310	290	490	18
xvi)	[C-6A(34)]		< 20	20 to 40		> 40		
		E250S	250	240		230	410	23
		E300S	300	290		280	440	22
		E350S	350	330		320	490	22
		E450S	450	430		420	570	20
NOTE — Percentage of elongation shall be taken over the gauge length of $5.65\sqrt{S_0}$, where S_0 = original cross-sectional area of the test specimen.								

¹⁾ Stress at 0.2 percent non-proportional elongation, Min.

²⁾ For Grade A sections and bars, the upper limit of the tensile strength may be 550 MPa or as agreed between the purchaser and the manufacturer/the supplier.

³⁾ For full thickness flat test pieces with width 25 mm and gauge length 200 mm, the minimum elongation (percent) may be reduced as per specified in good practice [C-6A(29)].

10 GENERAL DESIGN REQUIREMENTS

10.1 Basis for Design

10.1.1 Design Objective

The objective of design is the achievement of an acceptable probability that structures will perform satisfactorily for the intended purpose during the design life. With an appropriate degree of safety, they should sustain all the loads and deformations, during construction and use and have adequate resistance to certain expected accidental loads and fire. Structure should be stable and have alternate load paths to prevent disproportionate overall collapse under accidental loading.

10.1.2 Methods of Design

Structure and its elements shall normally, be designed by the limit state method. Account should be taken of accepted theories, experimental information and experience and the need to design for durability. Calculations alone may not produce safe, serviceable and durable structures. Suitable materials, quality control, adequate detailing and good supervision are equally important.

10.1.3 Design Process

Structural design, including design for durability, construction and use should be considered as a whole. The realization of design objectives requires compliance with clearly defined standards for materials, fabrication, erection and in-service maintenance.

10.2 Loads and Forces

10.2.1 For the purpose of designing any element, member or a structure, the following loads (actions) and their effects shall be taken into account, where applicable, with partial safety factors for loads and their combinations given in Part C ‘Structural Design/Section 1 Loads and Load Effects’ of this NBCS (see [12.3.3](#)).

- a) Dead loads;
- b) Imposed loads (imposed load, crane load, snow load, dust load, wave load, earth pressures, etc);
- c) Wind loads;
- d) Earthquake effects;
- e) Fire Load, forces induced due to restraint against expansion during a fire;
- f) Erection loads;
- g) Accidental loads such as those due to blasts, impact of vehicles, etc; and
- h) Secondary effects due to contraction or expansion resulting from temperature changes, differential settlements of the structure as a whole or of its components, eccentric connections, rigidity of joints differing from design assumptions.

10.2.1.1 Dead loads should be assumed in design as specified in **2** of Part C ‘Structural Design/Section 1 Loads and Load Effects’ of this NBCS.

10.2.1.2 Imposed loads for different types of occupancy and function of structures shall be taken as recommended in **3** of Part C ‘Structural Design/Section 1 Loads and Load Effects’ of this NBCS. Imposed loads arising from equipment, such as cranes and machines should be assumed in design as per manufacturers/suppliers data (see [10.6.4](#)). Snow load shall be taken as per in **6** of Part C ‘Structural Design/Section 1 Loads and Load Effects’ of this NBCS.

10.2.1.3 Wind loads on structures shall be taken as per the recommendations of **4** of Part C ‘Structural Design/Section 1 Loads and Load Effects’ of this NBCS.

10.2.1.4 Earthquake effects shall be assumed as per the recommendations of **5** of Part C ‘Structural Design/Section 1 Loads and Load Effects’ of this NBCS.

10.2.1.5 The fire loads, erection loads and temperature effects shall be considered as specified in [10.3](#), [10.4](#) and [10.5](#), respectively.

10.3 Fire Loads

During a fire, the structural members around the fire experience higher temperatures. This causes expansion in members that is restrained by rest of the structure, inducing forces in the members. Simultaneously, due to high temperature both strength and stiffness of the members experiencing the higher temperature degrade. The forces in members during fire including that due to dead load and imposed load are checked against the reduced capacity of the members. As it is an extreme event, the design is done for loads without extra partial safety factors. The imposed load is taken as 100 percent of the characteristic Imposed loads in case of storage areas and otherwise taken as 50 percent of the characteristic Imposed load value, during fire.

Due to proper fire protection, the temperature felt by the members may be less than the ambient temperature, and this may be taken into consideration in evaluating temperature effects due to fire.

10.4 Erection Loads

All loads required to be carried by the structure or any part of it due to storage or positioning of construction material and erection equipment, including all loads due to operation of such equipment shall be considered as erection loads. Proper provision shall be made, including temporary bracings, to take care of all stresses developed during erection. Dead load, wind load and also such parts of the imposed load as would be imposed on the structure during the period of erection shall be taken as acting together with the erection loads. The structure as a whole and all parts of the structure in conjunction with the temporary bracings shall be capable of sustaining these loads during erection.

10.5 Temperature Effects

10.5.1 Expansion and contraction due to changes in temperature of the members and elements of a structure shall be considered and adequate provision made for such effect.

10.5.2 The temperature range varies for different localities and under different diurnal and seasonal conditions. The absolute maximum and minimum temperatures, which may be expected at the site, may be obtained from the Indian Metrological Department and used in assessing the maximum variations of temperature for which provision for expansion and contraction has to be made in the structure.

10.5.3 The range of variation in temperature of the building materials may be appreciably greater or lesser than the variation of air temperature and is influenced by the condition of exposure and the rate at which the materials composing the structure absorb or radiate heat. This difference in temperature variations of the material and air shall be given due consideration. The effect of differential temperature within an element or member, due to part exposure to direct sunlight shall also be considered.

10.5.4 The coefficient of thermal expansion for steel is as given in [9.2.4.1\(e\)](#).

10.6 Load Combinations

10.6.1 Load combinations for design purposes shall be those that produce maximum forces and effects and, consequently maximum stresses and deformations. The combination of loads with appropriate partial safety factors may be as per Part C/Section 1 of this NBCS.

NOTE — In the case of structures supporting cranes, imposed loads shall include the crane effects as given in [10.6.4](#).

10.6.2 Wind load and earthquake effects shall not be assumed to act simultaneously. The effect of each shall be considered separately.

10.6.3 The effect of cranes to be considered under imposed loads shall include the vertical loads, eccentricity effects induced by the vertical loads, impact factors, lateral (surge) and longitudinal (horizontal) thrusts, not acting simultaneously across and along the crane rail, respectively [see 3 of Part C ‘Structural Design/Section 1 Loads and Load Effects’ of this NBCS].

10.6.4 The crane loads and their combinations to be considered shall be as indicated by the customer. In the absence of any specific indications, the load combinations shall be in accordance with the provisions in [3](#) of Part C ‘Structural Design/Section 1 Loads and Load Effects’ of this NBCS or as given below:

- a) Vertical loads with full impact from one loaded crane or two cranes in case of tandem operation, together with vertical loads without impact from as many loaded cranes as may be positioned for maximum effect, along with maximum horizontal thrust from one crane only or two in case of tandem operation;
- b) Loads as specified in [10.6.4\(a\)](#), subject to cranes in maximum of any two bays of the building cross section shall be considered for multi-bay multi-crane gantries;
- c) The longitudinal thrust on a crane track rail shall be considered for a maximum of two loaded cranes on the track; and
- d) Lateral thrust (surge) and longitudinal thrust acting across and along the crane rail, respectively, shall be assumed not to act simultaneously. The effect of each force, shall however be investigated separately.

10.6.5 While investigating the effect of earthquake, the resulting effect from dead loads of all cranes parked in each bay, positioned to cause maximum effect shall be considered.

10.6.6 The crane runway girders supporting bumpers shall be checked for bumper impact loads also, as specified by the manufacturers.

10.6.7 Stresses developed due to secondary effects such as handling; erection, temperature and settlement of foundations, if any, shall be appropriately added to the stresses calculated from the combination of loads stated in [10.6.1](#), with appropriate partial safety factors.

10.7 Geometrical Properties

10.7.1 General

The geometrical properties of the gross and the effective cross sections of a member or part thereof, shall be calculated on the following basis:

- a) the properties of the gross cross section shall be calculated from the specified size of the member or part thereof or read from appropriate table; and
- b) the properties of the effective cross section shall be calculated by deducting from the area of the gross cross section, in the following:
 - 1) The sectional area in excess of effective plate width, in case of slender sections [see [10.8.2 \(d\)](#)]; and
 - 2) The sectional areas of all holes in the section except for parts in compression. In case of punched holes, hole size 2 mm in excess of the actual diameter may be deducted.

10.8 Classification of Cross Sections

10.8.1 Plate elements of a cross section may buckle locally due to compressive stresses. The local buckling can be avoided before the limit state is achieved by limiting the width to thickness ratio of each element of a cross section subjected to compression due to axial force, moment or shear.

10.8.1.1 When plastic analysis is used, the members shall be capable of forming plastic hinges with sufficient rotation capacity (ductility) without local buckling, to enable the redistribution of bending moment required before formation of the failure mechanism.

10.8.1.2 When elastic analysis is used, the member shall be capable of developing the yield stress under compression without local buckling.

10.8.2 On basis of the above, four classes of sections are defined as follows:

- a) *Class 1 (Plastic)* — Cross sections, which can develop plastic hinges and have the rotation capacity required for failure of the structure by formation of plastic mechanism. The width to thickness ratio of plate elements shall be less than that specified under class 1 (Plastic), in [Table 2](#);

- b) *Class 2 (Compact)* — Cross sections, which can develop plastic moment of resistance, but have inadequate plastic hinge rotation capacity for formation of plastic mechanism, due to local buckling. The width to thickness ratio of plate elements shall be less than that specified under class 2 (compact), but greater than that specified under class 1 (plastic), in [Table 2](#);
- c) *Class 3 (Semi-compact)* — Cross sections, in which the extreme fiber in compression can reach yield stress, but cannot develop the plastic moment of resistance, due to local buckling. The width to thickness ratio of plate elements shall be less than that specified under class 3 (semi-compact), but greater than that specified under class 2 (compact), in [Table 2](#); and
- d) *Class 4 (Slender)* — Cross sections in which the elements buckle locally even before reaching yield stress. The width to thickness ratio of plate elements shall be greater than that specified under class 3 (semi-compact), in [Table 2](#). In such cases, the effective sections for design shall be calculated either by following the provisions of good practice [C-6A(35)] to account for the post-local-buckling strength or by conservatively deducting width of the compression plate element in excess of the semi-compact section limit [see [Fig. 2\(C\)](#)].

The design of slender web elements in flexural members may be done as given in [15.2.1.1](#) for flexure and [15.4.2.2](#) for shear.

When different elements of a cross section fall under different classes, the section shall be classified as governed by the most critical element.

The maximum value of limiting width to thickness ratios of elements for different classifications of sections are given in [Fig. 2\(A\)](#). The effective section in case of class 4 (slender elements) may be conservatively taken as shown in [Fig. 2\(C\)](#).

10.8.3 Types of Elements

- a) *Internal elements* — These are elements attached along both longitudinal edges to other elements or to longitudinal stiffeners connected at suitable intervals to transverse stiffeners; for example, web of I-section and flanges and web of box section;
- b) *Outside elements or outstands* — These are elements attached along only one of the longitudinal edges to an adjacent element, the other edge being free to displace out of plane, for example flange overhang of an I-section, stem of T-section and legs of an angle section; and
- c) *Tapered elements* — These may be treated as flat elements having average thickness as defined in accepted standard [C-6A(36)].

The design of slender compression element (Class 4) considering the strength beyond elastic local buckling of element is outside the scope of this section. Reference may be made to in good practice [C-6A(35)] for such design provisions. The design of slender web elements may be made as given in [15.2.1.1](#) for flexure and [15.4.2.2](#) for shear.

10.8.4 Compound Elements in Built-up Section (see [Fig. 2](#))

In case of compound elements consisting of two or more elements bolted or welded together, the limiting width to thickness ratios as in [Table 2](#) should be considered on basis of the following:

- a) Outstanding width of compound element (b_e) to its own thickness;
- b) The internal width of each added plate between the lines of welds or fasteners connecting it to the original section to its own thickness; and
- c) Any outstand of the added plates beyond the line of welds or fasteners connecting it to original section to its own thickness.

10.9 Maximum Effective Slenderness Ratio

The maximum effective slenderness ratio, KL/r , values of a beam, strut or tension member shall not exceed those given in [Table 3](#). 'KL' is the effective length of the member and 'r' is appropriate radius of gyration based on the effective section as defined in [10.8](#).

Table 2 Limiting Width to Thickness Ratio

([Foreword](#) and [Clauses 10.8.2\(a\), 10.8.2\(b\), 10.8.2\(c\), 10.8.2\(d\), 10.8.4, 11.5.2.2, 14.3.1 and 15.6.3.1](#))

Sl No.	Compression Element			Ratio	Class of Section		
					Class 1 Plastic	Class 2 Compact	Class 3 Semi-compact
(1)	(2)			(3)	(4)	(5)	(6)
i)	Outstanding element	Under uniform compression		b/t_f	8.7ε	9.7ε	13.6ε (a in Fig. 2C)
		Non uniform stress	Tip in compression	b/t_f	$\frac{8.7 \varepsilon}{r_1}$	$\frac{9.7 \varepsilon}{r_1}$	$21\varepsilon\sqrt{k_\sigma}$
			Tip in tension	b/t_f	$\frac{8.7 \varepsilon}{r_1\sqrt{r_1}}$	$\frac{9.7 \varepsilon}{r_1\sqrt{r_1}}$	$21\varepsilon\sqrt{k_\sigma}$
ii)	Web of I, Channel, H, box or RHS	Axial compression		d/t_w	27ε	33ε	37ε (b in Fig. 2C)
		Non uniform stress	Neutral axis at mid-depth	d/t_w	70ε	80ε	117ε
			$r_1 > 0.5$ (or) $r_2 > -1$	d/t_w	$\frac{122 \varepsilon}{5.5 r_1 - 1}$	$\frac{182 \varepsilon}{6.53 r_1 - 1}$	$\frac{37 \varepsilon}{0.608 + 0.343 r_2 + 0.049 r_2^2}$
			$r_1 \leq 0.5$ (or) $r_2 \leq -1$	d/t_w	$\frac{35 \varepsilon}{r_1}$	$\frac{40 \varepsilon}{r_1}$	$59 \varepsilon (1 - r_2)$
iii)	Angle, compression due to bending (Both criteria should be satisfied)			b/t	8.7ε	9.7ε	14.5ε
				d/t	8.7ε	9.7ε	14.5ε
iv)	Single angle, or double angles with the components separated, axial compression (All three criteria should be satisfied)			b/t d/t $(b+d)/t$	Not applicable		14.5ε 14.5ε 22.3ε
v)	Outstanding leg of an angle in contact back-to-back in a double angle member or with another component			d/t	8.7ε	9.7ε	14.5ε
vi)	Circular, elliptical hollow tube (see Note 6), including welded tube	a) bending moment		D_o/t	$48 \varepsilon^2$	$68 \varepsilon^2$	$87 \varepsilon^2$
		b) axial compression		D_o/t	Not applicable		$87 \varepsilon^2$

NOTES

- Elements which exceed semi-compact limits are to be taken as of slender cross section
- $\varepsilon = \sqrt{(250 / f_y)}$
- Webs shall be checked for shear buckling in accordance with [15.4.2](#) when $d/t > 67\varepsilon$. Where, b is the width of the element may be taken as clear distance between lateral supports or between lateral support and free edge, as appropriate, t is the thickness of element, d is the depth of the web, D is the outer diameter of the element (see [Fig 2](#), [10.8.3](#) and [10.8.4](#)).
- Different elements of a cross section can be in different classes. In such cases the section is classified based on the least favourable classification.
- The stress ratio r_1 and r_2 are defined as:

$$r_1 = \frac{\text{depth under compression}}{\text{Overall depth}}$$

$$r_2 = \frac{\text{Minimum compressive stress (tensile if negative)}}{\text{Maximum design compressive stress of web alone}}$$

- for circular sections $D_e = D$,
 - Elliptical sections: (h = major axis, b = minor axis)
 - in compression: $D_e = h \left[1 + \left(1 - 2.3 \left(\frac{t}{h} \right)^{0.6} \right) \left(\frac{h}{b} - 1 \right) \right]$
 - conservatively $D_e = \frac{h^2}{b}$
 - In bending about the strong (minor) axis
 - For $h/b \leq 1.36$: $D_e = \frac{b^2}{h}$
 - For $h/b > 1.36$: $D_e = 0.4 \frac{h^2}{b}$
 - In compression and or bending about weak axis:

$$D_e = \frac{h^2}{b}$$

7 k_σ is the buckling factor given by:

Maximum compression at free edge	r_2	1	0	-1	-1 to -3
	K_σ	0.43	0.57	0.85	$0.57 - 0.21k_2 + 0.07r_2$
Maximum Compression at supported edge	r_2	1	$0 < r_2 < 1$	0	$-1 < r_2 < 0$
	K_σ	0.43	$0.578/(r_2 + 0.34)$	1.70	$1.7 - 5 r_2 + 17.1 r_2^2$

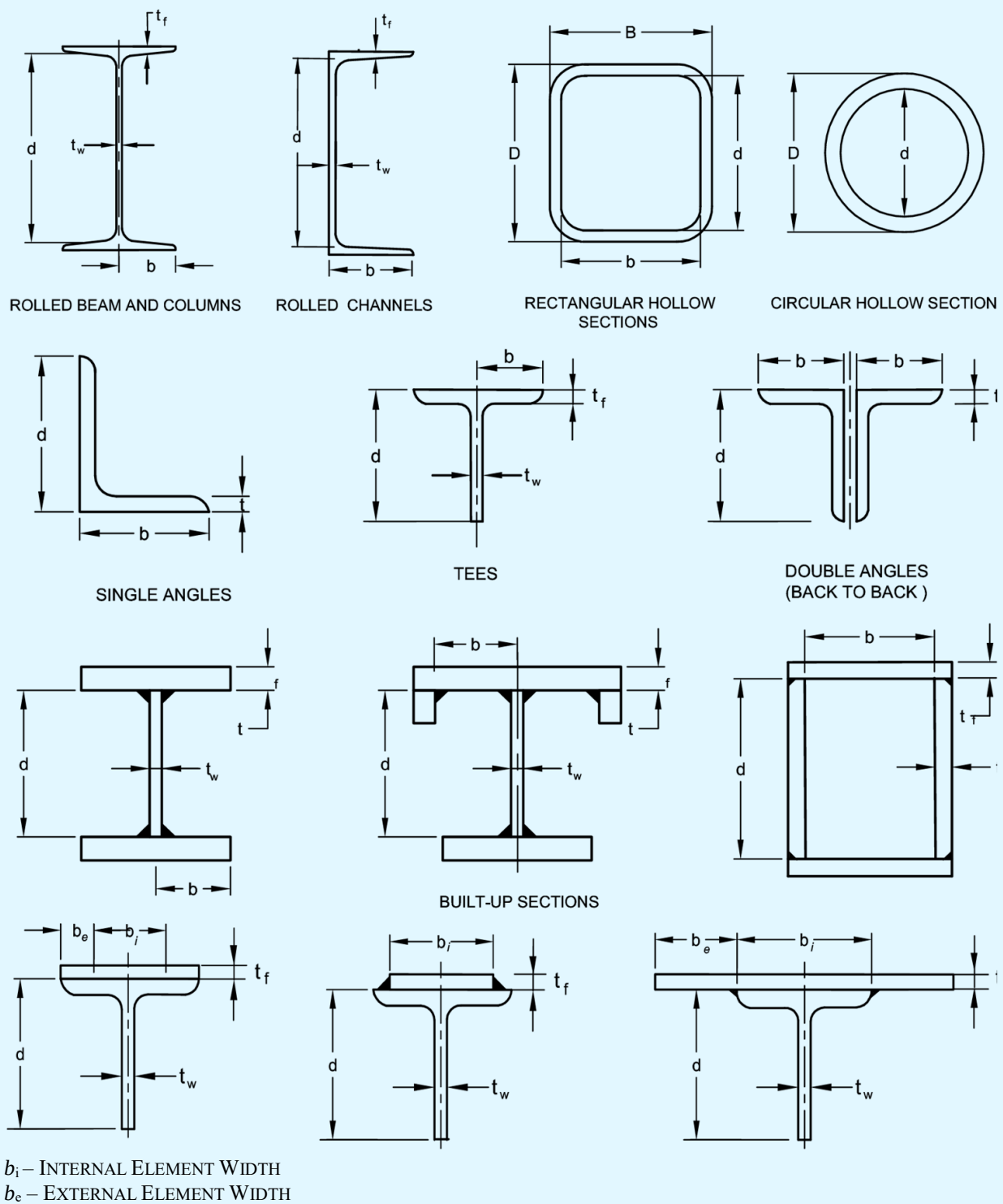


FIG. 2(A) DIMENSIONS OF SECTIONS

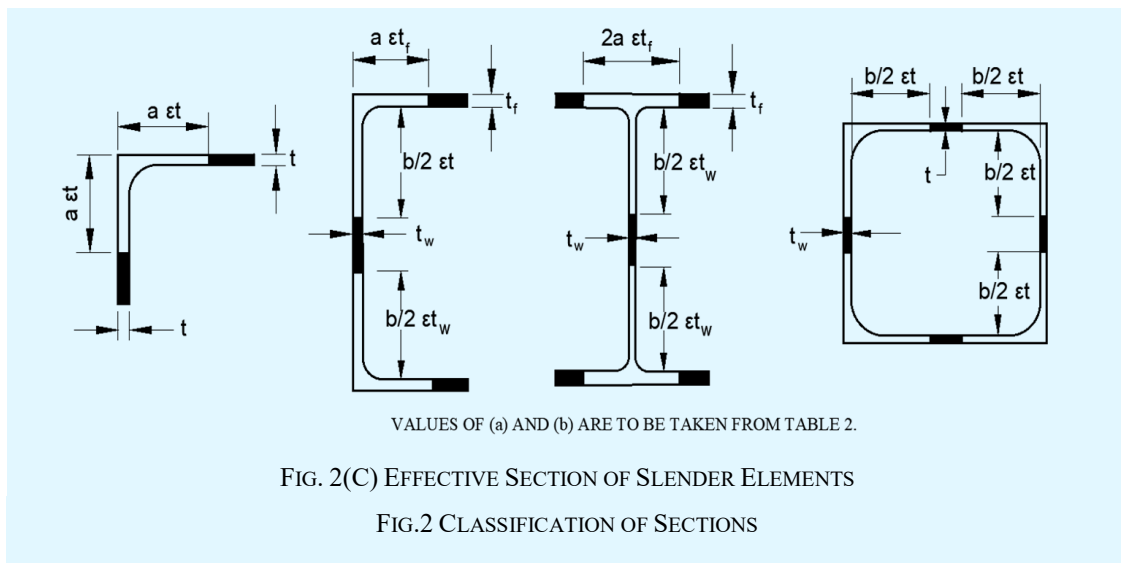
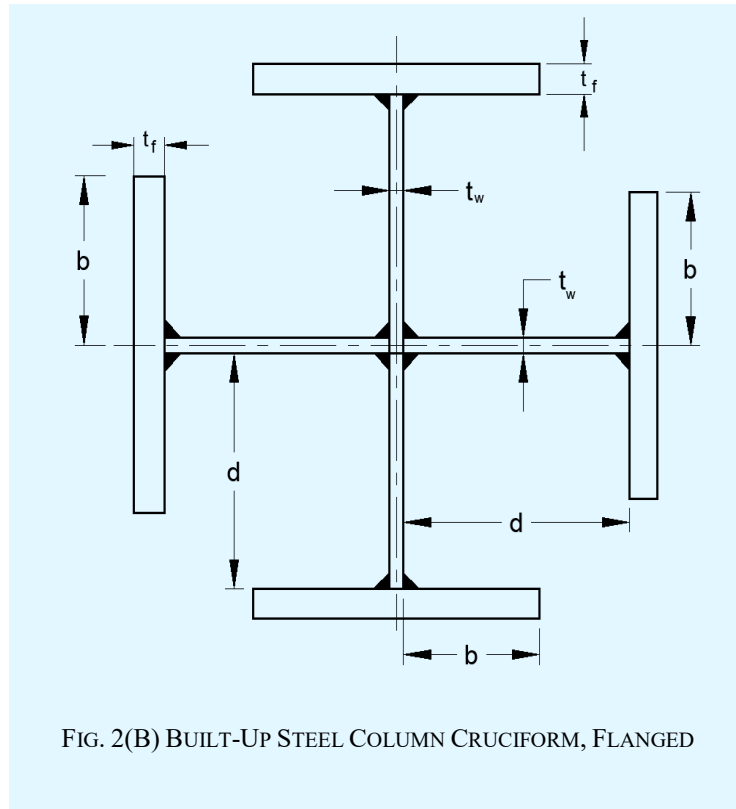


Table 3 Maximum Values of Effective Slenderness Ratios

(Clause 10.9)

Sl No.	Member	Maximum Effective Slenderness Ratio KL/r
(1)	(2)	(3)
i)	A member carrying compressive loads resulting from dead loads and imposed loads	180
ii)	A tension member in which a reversal of direct stress occurs due to loads other than wind or seismic forces	180
iii)	A member subjected to compression forces resulting only from combination with wind/earthquake effects, provided the deformation of such member does not adversely affect the stress in any part of the structure	250
iv)	Compression flange of a beam against lateral torsional buckling	300
v)	A member normally acting as a tie in a roof truss or a bracing system not considered effective when subject to possible reversal of stress into compression resulting from the action of wind or earthquake effects ¹⁾	350
vi)	Members always under tension ¹⁾ (other than pre-tensioned members)	400

¹⁾ Tension members, such as bracings, where sag is avoided, need not satisfy the maximum slenderness ratio limits.

10.10 Resistance to Horizontal Forces

10.10.1 In designing the steel frame work of a building, provision shall be made (by adequate moment connections or by a system of bracing) to effectively transmit to the foundations all the horizontal forces, giving due allowance for the stiffening effect of the walls and floors, where applicable.

10.10.2 When the walls, or walls and floors and/or roofs are capable of effectively transmitting all the horizontal forces directly to the foundations, the structural steel framework may be designed without considering the effects of wind or earthquake.

10.10.3 Wind and earthquake effects are reversible and therefore call for initial stiffness and strength under force reversal in both longitudinal and transverse directions. To resist torsional effects of wind and earthquake forces, bracings in plan should be provided and integrally connected with the longitudinal and transverse bracings, to impart adequate torsional resistance to the structure.

10.10.3.1 In shed type steel mill buildings, adequate bracings shall be provided to transfer the wind or earthquake loads from their points of action to the appropriate supporting members. Where the connections to the interior columns or frames are designed such that the wind or earthquake loads will not be transferred to the interior columns, the exterior columns or frames shall be designed to resist the total wind or earthquake loads. Where the connections to the interior columns and frames are designed such that the wind or earthquake effects are transferred to the interior columns also, and where adequate rigid diaphragm action can be mobilized as in the case of the cast-in place RC slab, both exterior and interior columns and frames may be designed on the assumption that the wind or earthquake load is divided among them in proportion to their relative stiffness. Columns also should be designed to withstand the net uplifting effect caused by excessive wind or earthquake shaking. Additional axial forces arising in adjacent columns due to the vertical component of bracings or due to frame action shall also be accounted for.

10.10.3.2 Earthquake forces are proportional to the seismic mass as defined in 5 of Part C 'Structural Design/Section 1 Loads and Load Effects' of this NBCS. Earthquake forces should be applied at the centre of

mass of all such components and their transfer to the foundation should be ensured. Other construction details, stipulated in good practice [C-6A(37)] should also be followed.

10.10.3.3 In buildings where high-speed traveling cranes are supported or where a building or structure is otherwise subjected to vibration or sway, triangulated bracing or rigid portal systems shall be provided to reduce the vibration or sway to an acceptable minimum.

10.10.4 Foundations

The foundations of a building or other structures shall be designed to provide the rigidity and strength that has been assumed in the analysis and design of the superstructure.

10.10.5 Eccentrically Placed Loads

Where a wall, or other gravity load, is placed eccentrically upon the flange of a supporting steel beam, the beam and its connections shall be designed for torsion, unless the beam is restrained laterally in such a way as to prevent the twisting of the beam.

10.11 Expansion Joints

Expansion joints in case of low-rise buildings (up to three storeys) shall comply with [10.11.1](#) to [10.11.4](#). For requirement of expansion joints in high-rise buildings, specialist literature shall be referred to.

10.11.1 In view of the large number of factors involved in deciding the location, spacing and nature of expansion joints, the decision regarding provision of expansion joints shall be left to the discretion of the designer.

10.11.2 Structures in which marked changes in plan dimensions take place abruptly, shall be provided with expansion joints at the section where such changes occur. Expansion joints shall be so provided that the necessary movement occurs with minimum resistance at the joint. The gap at the expansion joint should be such that:

- a) it accommodates the expected expansion/contraction due to seasonal and diurnal variation of temperature; and
- b) it avoids pounding of adjacent units under earthquake shaking. The structure adjacent to the joint should preferably be supported on separate columns but not necessarily on separate foundations.

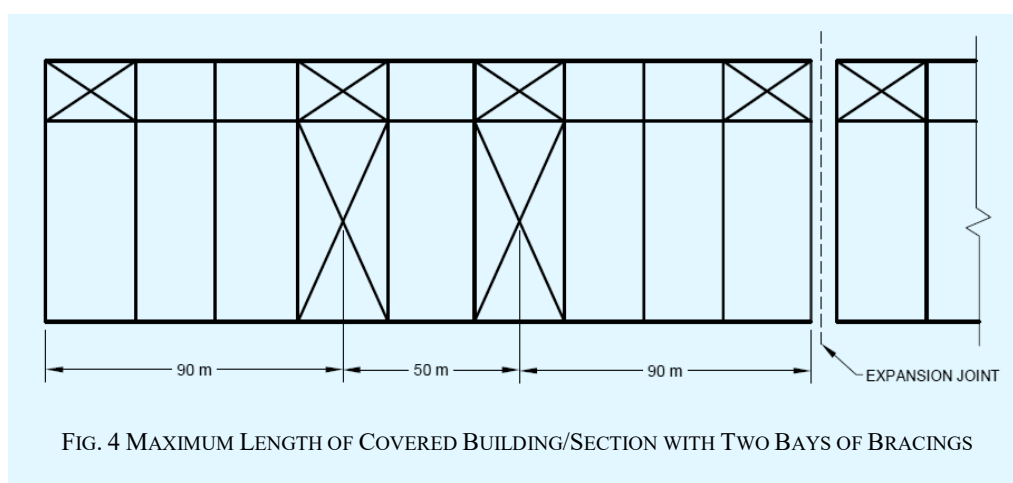
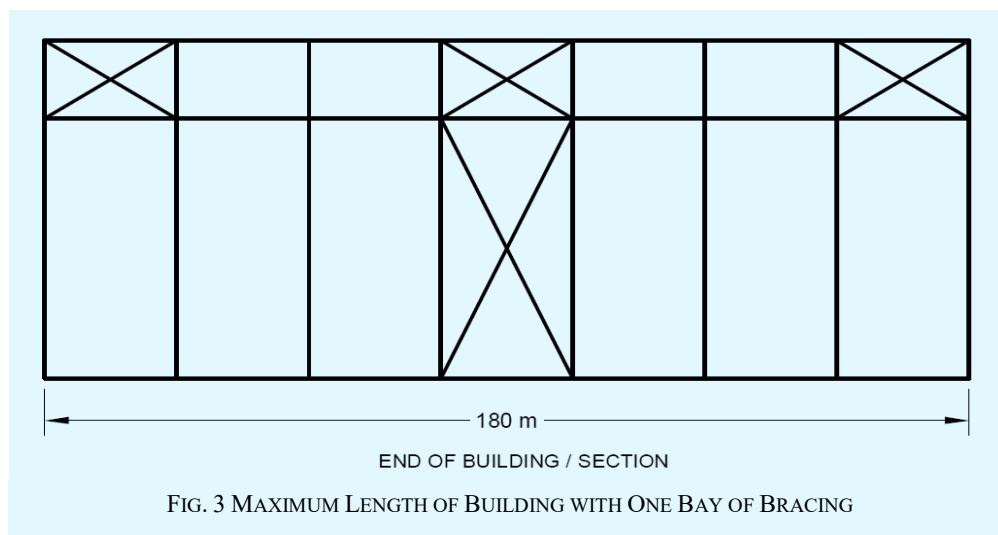
10.11.3 The details as to the length of a structure where expansion joints have to be provided may be determined after taking into consideration various factors such as temperature, exposure to weather and structural design. The provisions in [10.11.3.1](#) to [10.11.3.3](#) are given as general guidance. The provision for expansion may be made by providing double framing at the joint or providing for expansion at the connection to the single framing in detailing.

10.11.3.1 If one bay of longitudinal bracing is provided at the centre of the building or building section, the length of the building section may be restricted to 180 m in case of covered buildings and 120 m in case of open gantries (see [Fig. 3](#)).

10.11.3.2 If more than one bay of longitudinal bracing is provided near the centre of the building/section, the maximum centre line distance between the two lines of bracing may be restricted to 50 m for covered buildings (and 30 m for open gantries) and the maximum distance between the centre of the bracing to the nearest expansion joint/end of building or section may be restricted to 90 m (60 m in case of open gantries). The maximum length of the building section thus may be restricted to 230 m for covered buildings (150 m for open gantries). Beyond this, suitable expansion joints shall be provided (see [Fig.4](#)).

10.11.3.3 The maximum width of the covered building section should preferably be restricted to 150 m beyond which suitable provisions for the expansion joint may be made.

10.11.4 When the provisions of these sections are met for a building or open structure made only of steel frame with elastic steel claddings, the stress analysis due to temperature is not required. However, for buildings, specialist literature may be referred to decide on thermal stress design.



11 METHODS OF STRUCTURAL ANALYSIS

11.1 Methods of Determining Action Effects

11.1.1 General

For the purpose of complying with the requirements of the limit states of stability, strength and serviceability specified under Section 12 on the effects of design actions on a structure and its members and connections, shall be determined by structural analysis using the assumptions of 11.2 and 11.3 and one of the following methods of analysis:

- elastic analysis in accordance with 11.4;
- plastic analysis in accordance with 11.5;
- advanced analysis in accordance with Annex B; or
- dynamic analysis in accordance with 5 of Part C 'Structural Design/Section 1 Loads and Load Effects' of this NBCS.

The effects of design earthquake shaking specified in Section 18 shall be estimated by elastic analysis. Plastic analysis shall not be employed to obtain member forces corresponding to design earthquake shaking specified in Section 18. More information on analysis to resist earthquake effects is given in 5 of Part C 'Structural Design/Section 1 Loads and Load Effects' of this NBCS.

11.1.2 Non-sway and Sway Frames

For the purpose of analysis and design, the structural frames are classified as non-sway and sway frames as given below:

- a) *Non-sway frame* — One in which the transverse displacement of one end of the member relative to the other end is effectively prevented. This applies to triangulated frames and trusses or to frames where in-plane stiffness is provided by bracings, or by shear walls, or by floor slabs and roof decks secured horizontally to walls or to bracing systems parallel to the plane of loading and bending of the frame;
- b) *Sway frame* — One in which the transverse displacement of one end of the member relative to the other end is not effectively prevented. Such members and frames occur in structures which depend on flexural action of members to resist lateral loads and sway, as in moment resisting frames; and
- c) A rigid jointed multi-storey frame may be considered as a non-sway frame if in every individual storey, the deflection δ , over a storey height h_s , due to the notional horizontal loading given in [11.3.6](#) satisfies the following criteria:
 - 1) For clad frames, when the stiffening effect of the cladding is not taken into account in the deflection calculations:

$$\delta \leq \frac{h_s}{2\ 000}$$

- 2) For unclad frame or clad frames, when the stiffening effect of the cladding is taken into account in the deflection calculations:

$$\delta \leq \frac{h_s}{4\ 000}$$

- 3) A frame, which when analyzed considering all the lateral supporting system does not comply with the above criteria, should be classified as a sway frame, even if it is braced or otherwise laterally stiffened.

11.2 Forms of Construction Assumed for Structural Analysis

11.2.1 The effects of design action in the members and connections of a structure shall be determined by assuming singly or in combination of the following forms of construction (see [17.6.1](#)).

11.2.1.1 Rigid construction

In rigid construction, the connections between members (beam and column) at their junction shall be assumed to have sufficient rigidity to hold the original angles between the members connected at a joint unchanged under loading.

11.2.1.2 Semi-rigid construction

In semi-rigid construction, the connections between members (beam and column) at their junction may not have sufficient rigidity to hold the original angles between the members at a joint unchanged, but shall be assumed to have the capacity to furnish a dependable and known degree of flexural restraint. The relationship between the degree of flexural restraint and the level of the load effects shall be established by any rational method or based on test results (see [Annex F](#)).

11.2.1.3 Simple construction

In simple construction, the connections between members (beam and column) at their junction will not resist any appreciable moment and shall be assumed to be hinged.

11.2.2 Design of connections

The design of all connections shall be consistent with the form of construction, and the behaviour of the connections shall not adversely affect any other part of the structure beyond what is allowed for in design. Connections shall be designed in accordance with provisions under [17](#).

11.3 Assumptions in Analysis

11.3.1 The structure shall be analyzed in its entirety except as follows:

- a) Regular building structures, with orthogonal frames in plan, may be analyzed as a series of parallel two-dimensional sub-structures (part of a structure), the analysis being carried out in each of the two directions, at right angles to each other, except when there is significant load redistribution between the sub-structures (part of a structure);
- b) For vertical loading in a multi-storey building structure, provided with bracing or shear walls to resist all lateral forces, each level thereof, together with the columns immediately above and below, may be considered as a substructure, the columns being assumed fixed at the ends remote from the level under consideration; and
- c) Where beams at a floor level in a multi-bay building structure are considered as a substructure (part of a structure), the bending moment at the support of the beam due to gravity loads may be determined based on the assumption that the beam is fixed at the far end support, one span away from the span under consideration, provided that the floor beam is continuous beyond that support point.

11.3.2 Span Length

The span length of a flexural member in a continuous frame system shall be taken as the centre-to-centre distance between the supports.

11.3.3 Arrangements of Imposed Loads in Buildings

For building structures, the various arrangements of imposed loads considered for the analysis, shall include at least the following:

- a) Where the loading pattern is fixed, the arrangement concerned;
- b) Where the imposed load is variable and not greater than three-quarters of the dead load, the imposed load may be taken to be acting on all spans; and
- c) Where the imposed load is variable and exceeds three-quarters of the dead load, arrangements of imposed load acting on the floor under consideration shall include the following cases:
 - 1) imposed load on alternate spans;
 - 2) imposed load on two adjacent spans; and
 - 3) imposed load on all the spans.

11.3.4 Base Stiffness

In the analysis of all structures, the appropriate base stiffness about the axis under consideration shall be used. In the absence of the knowledge of the pedestal and foundation stiffness, the following may be assumed:

- a) When the column is rigidly connected to a suitable foundation, the stiffness of the pedestal shall be taken as the stiffness of the column above base plate. However, in case of very stiff pedestals and foundations the column may be assumed as fixed at base;
- b) When the column is nominally connected to the foundation, a pedestal stiffness of 10 percent of the column stiffness may be assumed;
- c) When an actual pin or rocker is provided in the connection between the steel column and pedestal, the column is assumed as hinged at base and the pedestal and foundation may be appropriately designed for the reactions from the column; and
- d) In case of (a) and (b), the bottom of the pedestal shall be assumed to have the following boundary condition in the absence of any detailed procedure based on theory or tests:
 - 1) When the foundation consists of a group of piles with a pile cap, raft foundation or an isolated footing resting on rock or very hard soil, the pedestal shall be assumed to be fixed at the level of the bottom of footing or at the top of the pile cap. Pedestal to be considered as fixed at top of pile cap, where

the Single Piles are tied at Pile cap level by means of ground beams (in both orthogonal directions) with required stiffness;

- 2) When the foundation consists of an isolated footing resting on other soils, the pedestal shall be assumed to be hinged at the level of the bottom of the footing;
- 3) When the pedestal is supported by a single pile, which is laterally surrounded by soil providing passive resistance, the pile shall be assumed to be fixed at a depth of 5 times the diameter of the pile below the ground level in case of compact ground or the top level of compact soil in case of poor soil overlying compact soil. Single pile cap, connected flexurally to orthogonal beams having adequate stiffness, shall be assumed to be fixed at the top of pile cap; and
- 4) When the column is founded into rock, it may be assumed to be fixed at the interface of the column and rock.

11.3.5 Simple Construction

Bending members may be assumed to have their ends connected for shear only and to be free to rotate. In triangulated structures, axial forces may be determined by assuming that all members are pin-connected. The eccentricity for stanchion and column shall be assumed in accordance with [14.3.3](#).

11.3.6 Notional Horizontal Loads

To analyze a frame subjected to gravity loads, considering the sway stability of the frame, notional horizontal forces should be applied. These notional horizontal forces account for practical imperfections and should be taken at each level as being equal to 0.5 percent of factored dead load plus vertical imposed loads applied at that level. The notional load should not be applied along with other lateral loads, such as wind and earthquake effects, in the analysis.

11.3.6.1 The notional forces should be applied on the whole structure, in both orthogonal directions, in one direction at a time, at roof and all floor levels or their equivalent. They should be taken as acting simultaneously with factored gravity loads.

11.3.6.2 The notional force should not be:

- a) applied when considering overturning or overall instability;
- b) combined with other horizontal (lateral) loads;
- c) combined with temperature effects; and
- d) taken to contribute to the net shear on the foundation.

11.3.6.3 The sway effect using notional load under gantry load case need not be considered if the ratio of height to lateral width of the building is less than unity.

11.4 Elastic Analysis

11.4.1 Assumptions

Individual members shall be assumed to remain elastic under the action of the factored design loads for all limit states.

The effect of haunching or any variation of the cross section along the axis of a member shall be considered, and where significant, shall be taken into account in the determination of the member stiffness.

11.4.2 First-order Elastic Analysis

In a first-order elastic analysis, the equilibrium of the frame in the undeformed geometry is considered, the changes in the geometry of the frame due to the loading are not accounted for, and changes in the effective stiffness of the members due to axial force are neglected. The conditions under which the first order elastic analysis results are adequate for design are given in Section [11.7](#).

11.5 Plastic Analysis

11.5.1 Application

The effects of design action throughout or on part of a structure may be determined by a plastic analysis, provided that the requirements of [11.5.2](#) are met. The distribution of design action effects shall satisfy equilibrium and the boundary conditions.

11.5.2 Requirements

When a plastic method of analysis is used, all of the following conditions shall be satisfied, unless adequate ductility of the structure and plastic rotation capacity of its members and connections are established for the design loading conditions by other means of evaluation:

- The yield stress of the grade of the steel used shall not exceed 450 MPa;
- The stress-strain characteristics of the steel shall not be significantly different from those obtained for steels complying with the accepted standard [C-6A(6)] or equivalent and shall be such as to ensure complete plastic moment redistribution. The stress-strain diagram shall have a plateau at the yield stress, extending for at least six times the yield strain. The ratio of the tensile strength to the yield stress specified for the grade of the steel shall not be less than 1.2. The elongation on a gauge length complying with the accepted standard [C-6A(6)] shall not be less than 15 percent; and the steel shall exhibit strain-hardening capability. Steels conforming to the accepted standard [C-6A(6)] shall be deemed to satisfy the above requirements;
- The members used shall be hot-rolled or fabricated using hot-rolled plates and sections;
- The cross section of members not containing plastic hinges should be at least that of compact section (see [10.8.2](#)), unless the members meet the strength requirements from elastic analysis;
- Where plastic hinges occur in a member, the proportions of its cross section should not exceed the limiting values for plastic section given in [10.8.2](#);
- The cross section should be symmetrical about its axis perpendicular to the axis of the plastic hinge rotation; and
- The members shall not be subject to impact loading, requiring fracture assessment or fluctuating loading, requiring a fatigue assessment [see [19](#)].

11.5.2.1 Restraints

If practicable, torsional restraint (against lateral buckling) should be provided at all plastic hinge locations. Where not feasible, the restraint should be provided within a distance of $D/2$ of the plastic hinge location, where D is the total depth of section.

The torsional restraint requirement at a section as above, need not be met at the last plastic hinge to form, provided it can be clearly identified.

Within a member containing a plastic hinge, the maximum distance L_m from the restraint at the plastic hinge to an adjacent restraint should be calculated by any rational method or the conservative method given below, so as to prevent lateral buckling.

Conservatively L_m (in mm) may be taken as:

$$L_m \leq \frac{38r_y}{\left[\frac{f_c}{130} + \left(\frac{f_y}{250} \right)^2 \left(\frac{x_t}{40} \right)^2 \right]^{1/2}}$$

where

- A = area of cross section;
- f_c = actual, in N/mm^2 , compressive stress on the cross section due to axial load;

- f_y = yield, in N/mm², stress;
 I_w, I_y, I_t = warping constant, second moment of the cross section about the minor axes and St. Venant's torsion constant, respectively;
 r_y = radius, in mm, of gyration about the minor axis; and
 χ_t = torsional index, $\chi_t = 1.132(AI_w / I_y I_t)^{0.5}$.

Where the member has unequal flanges, r_y should be taken as the lesser of the values of the compression flange only or the whole section.

Where the cross section of the member varies within the length L_m , the maximum value of r_y and the minimum value of χ_t should be used.

The spacing of restraints to member lengths not containing a plastic hinge should satisfy the recommendations of section on lateral buckling strength of beams (see [15.2.2](#)). Where the restraints are placed at the limiting distance L_m , no further checks are required.

11.5.2.2 Stiffeners at plastic hinge locations

Web stiffeners should be provided where a concentrated load, which exceeds 10 percent of the shear capacity of the member, is applied within $D/2$ of a plastic hinge location, (see [15.2.1.2](#)). The stiffener should be provided within a distance of half the depth of the member on either side of the hinge location and be designed to carry the applied load in accordance with [15.7.4](#). If the stiffeners are flat plates, the outstand width to the thickness ratio, b/t , should not exceed the values given in the plastic section (see [10.7](#) and [Table 2](#)). Where other sections are used, the ratio $\left(\frac{I_{so}}{I_t}\right)^{1/2}$, should not exceed the values given for plastic section (for simple outstand, as in [10.7](#));

where

- I_{so} = second moment of area of the stiffener about the face of the element perpendicular to the web;
 and
 I_t = St. Venant's torsion constant of the stiffener.

11.5.2.3 The frame shall be adequately supported against sway and out-of-plane buckling, by bracings, moment resisting frame or an independent system such as shear wall.

11.5.2.4 Fabrication restriction

Within a length equal to the member depth, on either side of a plastic hinge location, the following restrictions should be applied to the tension flange and noted in the design drawings. Holes, if required, should be drilled or else punched 2 mm undersize and reamed. All sheared or hand flame cut edges should be finished smooth by grinding, chipping or planning.

11.5.3 Assumptions in Analysis

The design action effects shall be determined using a rigid-plastic analysis.

It shall be permissible to assume full strength or partial strength connections, provided the capacities of these are used in the analysis, and provided that:

- In a full strength connection, the moment capacity of the connection shall be not less than that of the member being connected;
- In a partial strength connection, the moment capacity of the connection may be less than that of the member being connected; and

- c) In both cases the behavior of the connection shall be such as to allow all plastic hinges necessary for the collapse mechanism to develop, and shall be such that the required plastic hinge rotation does not exceed the rotation capacity at any of the plastic hinges in the collapse mechanism.

In the case of building structures, it is not normally necessary to consider the effect of alternating plasticity.

11.6 Frame Buckling Analysis

11.6.1 The elastic buckling load factor (λ_{cr}) shall be the ratio of the elastic buckling load set of the frame to the design load set for the frame, and shall be determined in accordance with [11.6.2](#).

NOTE — The value of λ_{cr} depends on the load set and has to be evaluated for each possible set of load combination.

11.6.2 In-plane frame buckling

The elastic buckling load factor (λ_{cr}) of a rigid-jointed frame shall be determined by using:

- One of the approximate methods of [11.6.2.1](#) and [11.6.2.2](#); or
- A rational elastic buckling analysis of the whole frame.

11.6.2.1 Regular non-sway frames (see [11.1.2](#))

In a rectangular non-sway frame with regular loading and negligible axial forces in the beams, the Euler buckling stress f_{cr} , for each column shall be determined in accordance with [14.1.2.1](#). The elastic buckling load factor (λ_{cr}) for the whole frame shall be taken as the lowest of the ratio of (f_{cr}/f_{cd}) for all the columns, where f_{cd} is the axial compressive stress in the column from the factored load analysis.

11.6.2.2 Regular sway frames

In a rectangular sway frame with regular loading and negligible axial forces in the beams, the buckling load, P_{cr} , for each column shall be determined as $P_{cr} = A f_{cr}$, where f_{cr} is the elastic buckling stress of the column in the plane of frame, obtained in accordance with [14.1.2.1](#). The elastic buckling load factor λ_{cr} , for the whole frame shall be taken as the lowest of all the ratios, λ_{scr} , calculated for each storey of the building, as given below:

$$\lambda_{scr} = \frac{\sum (P_{cr} / L)}{\sum (P / L)}$$

where

- L = column length and the summation includes all columns in the plane frame within a storey; and
 P = member axial force from the factored load analysis, with tension taken as negative.

11.7 Consideration of Second Order Effects

11.7.1 Any second-order effects of the loads acting on the structure in its deformed configuration may be neglected and first order elastic analysis or plastic analysis results may be used provided the following are satisfied:

- for clad frames, provided with the stiffening effects of masonry infill wall panels or diaphragms of profiled wall panel is not taken into account, and where elastic buckling load factor, λ_{cr} (see [11.6](#)) satisfies $\lambda_{cr}/\lambda_P \geq 10$;

If $10 > \lambda_{cr}/\lambda_P \geq 4.6$ the second-order effects may be considered by amplifying the design load effects obtained from elastic/plastic analysis by a factor as given in [11.7.3.1](#).

If $\lambda_{cr}/\lambda_P < 4.6$, second-order elastic analysis or second-order plastic analysis (see [11.7.3](#)) is to be carried out.

- b) for un-clad frames or for clad frames where the stiffening effects of masonry infill or diaphragms of profiled wall panel is taken into account, where elastic buckling load factor, λ_{cr} (see 11.6) satisfies $\lambda_{cr}/\lambda_p \geq 20$.

If $20 > \lambda_{cr}/\lambda_p \geq 5.75$ the second-order effects may be considered by amplifying the design load effects obtained from plastic analysis by a factor as given in [11.7.3.2](#).

If $\lambda_{cr}/\lambda_p < 5.75$, second-order elasto-plastic analysis or second-order elastic analysis (see [11.7.2](#)) shall be carried out.

where

λ_p = Ultimate load factor from elastic or plastic analysis.

11.7.2 Second-order Elastic Analysis

11.7.2.1 The analysis shall allow for the effects of the design loads acting on the structure and its members in their displaced and deformed configuration. These second-order effects shall be taken into account by using either:

- a) a first-order elastic analysis with moment amplification in accordance with [11.7.3.1](#), [11.7.3.2](#), provided the moment amplification factors, K_y and K_z are not greater than 1.4; or
- b) a second-order elastic analysis in accordance with [Annex B](#).

11.7.3 Moment Amplification

11.7.3.1 Moment amplification for members in non-sway frames

For a member with zero axial compression or a member subject to axial tension, the design bending moment is that obtained from the first-order analysis for factored loads, without any amplification.

For a braced member with a design axial compressive force P_d as determined by the first-order analysis, the design bending moment shall be calculated considering moment amplification as in [16.3.2.2](#).

11.7.3.2 Moment amplification for members in sway frames

The design bending moment shall be calculated as the product of moment amplification factor, [see [16.3.2.2](#) (K_y , K_z)] and the moment obtained from the first-order analysis of the sway frame, unless analysis considering second order effects is carried out (see [11.7.2](#)).

11.7.4 The calculated bending moments from the first order elastic analysis may be modified by redistribution up to 15 percent of the peak calculated moment of the member under factored load, provided that:

- a) in which the moments are reduced shall belong to plastic or compact section classification (see [10.8](#)), the internal forces and moments in the members of the frame are in equilibrium with applied loads; and
- b) all the members.

12 LIMIT STATE DESIGN

12.1 Basis for Design

12.1.1 In the limit state design method, the structure shall be designed to withstand safely all loads likely to act on it throughout its life. It shall not suffer total collapse under accidental loads such as from explosions or impact or due to consequences of human error to an extent beyond the local damages. The objective of the design is to achieve a structure that will remain fit for use during its life with acceptable target reliability. In other words, the probability of a limit state being reached during its lifetime should be very low. The acceptable limit for the safety and serviceability requirements before failure occurs is called a limit state. In general, the structure shall be designed on the basis of the most critical limit state and shall be checked for other limit states.

12.1.2 Steel structures are to be designed and constructed to satisfy the design requirements with regard to stability, strength, serviceability, brittle fracture, fatigue, fire, and durability such that they meet the following:

- a) Remain fit with adequate reliability and be able to sustain all actions (loads) and other influences experienced during construction and use;
- b) Have adequate durability under normal maintenance; and
- c) Do not suffer overall damage or collapse disproportionately under accidental events like explosions, vehicle impact or due to consequences of human error to an extent beyond local damage. The potential for catastrophic damage shall be limited or avoided by appropriate choice of one or more of the following:
 - 1) Avoiding, eliminating or reducing exposure to hazards, which the structure is likely to sustain;
 - 2) Choosing structural forms, layouts and details and designing such that;
 - i) the structure has low sensitivity to hazardous conditions; and
 - ii) the structure survives with only local damage even after serious damage to any one individual element by the hazard.
 - 3) Choosing suitable material, design and detailing procedure, construction specifications, and control procedures for shop fabrication and field construction as relevant to the particular structure.

12.1.3 The following conditions may be satisfied to avoid a disproportionate collapse:

- a) The building should be effectively tied together at each principal floor level and each column should be effectively held in position by means of continuous ties (beams) nearly orthogonal, except where the steel work supports only cladding weighing not more than 0.7 kN/m^2 along with imposed and wind loads. These ties must be steel members such as beams, which may be designed for other purposes, steel bar reinforcement anchoring the steel frame to concrete floor or steel mesh reinforcement in composite slab with steel profiled sheeting directly connected to beam with shear connectors. These steel ties and their end connections should be capable of resisting factored tensile force not less than the factored dead and imposed loads acting on the floor area tributary to the tie nor less than 75 kN . Such connection of ties to edge column should also be capable of resisting 1 percent of the maximum axial compression in the column at the level due to factored dead and imposed loads. All column splices should be capable of resisting a tensile force equal to the largest of a factored dead and imposed load reaction from a single floor level located between that column splice and the next column splice below that splice. Lateral load system to resist notional horizontal loads prescribed in [11.3.6](#) should be distributed throughout the building in nearly orthogonal directions so that no substantial portions is connected at only one point to such a system. Precast concrete or other heavy floor or roof units should be effectively anchored in the direction of their span either to each other over the support or directly to the support.
- b) Where the above conditions to tie the columns to the floor adequately are not satisfied each storey of the building should be checked to ensure that disproportionate collapse would not precipitate by the notional removal, one at a time, of each column.
- c) Where each floor is not laterally supported by more than one system, check should be made at each storey by removing one such lateral support system at a time to ensure that disproportionate collapse would not occur. The collapse is considered disproportionate, if more than 15 percent of the floor or roof area of 70 m^2 collapse at that level and at one adjoining level either above or below it, under a load equal to 1.05 or 0.9 times the dead load, 0.33 times temporary or full imposed load of permanent nature (as in storage buildings) and 0.33 times wind load acting together.

12.1.4 Structures designed for unusual or special functions shall comply with any other relevant additional limit state considered appropriate to that structure.

12.2 Limit State Design

12.2.1 For achieving the design objectives, the design shall be based on characteristic values for material strengths and applied loads (actions), which take into account the probability of variations in the material strengths and in the loads to be supported. The characteristic values shall be based on statistical data, if available. Where such data is not available, these shall be based on experience. The design values are derived from the characteristic values through the use of partial safety factors, both for material strengths and for loads. In the absence of special

considerations, these factors shall have the values given in this Section according to the material, the type of load and the limit state being considered. The reliability of design is ensured by satisfying the requirement:

$$\text{Design action} \leq \text{Design strength}$$

12.2.2 Limit states are the states beyond which the structure no longer satisfies the performance requirements specified. The limit states are classified as:

- a) limit state of strength; and
- b) limit state of serviceability.

12.2.2.1 The limit states of strength are those associated with failures (or imminent failure), under the action of probable and most unfavourable combination of loads on the structure using the appropriate partial safety factors, which may endanger the safety of life and property. The limit state of strength includes:

- a) loss of equilibrium of the structure as a whole or any of its parts or components;
- b) loss of stability of the structure (including the effect of sway where appropriate and overturning) or any of its parts including supports and foundations;
- c) failure by excessive deformation, rupture of the structure or any of its parts or components;
- d) fracture due to fatigue; and
- e) brittle fracture.

12.2.2.2 The limit state of serviceability include:

- a) deformation and deflections, which may adversely affect the appearance or effective use of the structure or may cause improper functioning of equipment or services or may cause damages to finishes and non-structural members;
- b) vibrations in the structure or any of its components causing discomfort to people, damages to the structure, its contents or which may limit its functional effectiveness. Special consideration shall be given to systems susceptible to vibration, such as large open floor areas free of partitions to ensure that such vibrations are acceptable for the intended use and occupancy (see [Annex C](#));
- c) repairable damage or crack due to fatigue;
- d) corrosion, durability; and
- e) fire.

12.3 Actions

The actions (loads) to be considered in design include direct actions (loads) experienced by the structure due to self-weight, external actions, etc, and imposed deformations such as that due to temperature and settlements.

12.3.1 Classification of Actions

Actions are classified by their variation with time as given below:

- a) *Permanent actions* (Q_p) — Actions due to self-weight of structural and non-structural components, fittings, ancillaries, and fixed equipment, etc;
- b) *Variable actions* (Q_v) — Actions due to construction and service stage loads such as imposed (live) loads (crane loads, snow loads, etc), wind loads, and earthquake effects; and
- c) *Accidental actions* (Q_a) — Actions expected due to explosions, and impact of vehicles, etc.

12.3.2 Characteristic Actions (Loads)

12.3.2.1 The characteristic actions, Q_c , are the values of the different actions that are not expected to be exceeded with more than 5 percent probability, during the life of the structure and they are taken as:

- a) the self-weight, in most cases calculated on the basis of nominal dimensions and unit weights [see 2 of Part C 'Structural Design/Section 1 Loads and Load Effects' of this NBCS];
- b) the variable loads, values of which are specified in relevant standard [(see Part C 'Structural design/Section 1 Loads and Load Effects' of this NBCS)];
- c) the upper limit with a specified probability (usually 5 percent) not exceeding during some reference period (design life); and
- d) specified by client, or by designer in consultation with client, provided they satisfy the minimum provisions of the relevant loading standard.

12.3.2.2 The characteristic values of accidental loads generally correspond to the value specified by relevant code, standard or client. The design for accidental load is generally not required in building unless it is required by client or approving authority in which case, generally recommendation in [12.1.2 \(c\)](#) or specialist literature shall be followed.

12.3.3 Design Actions

The design action, Q_d , is expressed as:

$$Q_d = \sum_k \gamma_{Lk} Q_{ck}$$

where

γ_{Lk} = partial safety factor for different loads k , given in [Table 4](#) to account for.

- 1) possibility of unfavourable deviation of the load from the characteristic value;
- 2) possibility of inaccurate assessment of the load;
- 3) uncertainty in the assessment of effects of the load; and
- 4) uncertainty in the assessment of the limit states being considered.

The loads, Load combinations and the relevant partial safety factors for loads in a combination shall be as given in Part C/Section 1 of this NBCS. The loads or load effects shall be multiplied by the relevant γ_L factors, given in [Table 4](#), to get the design loads or design load effects, when such a value is not available in Part C/Section 1 of this NBCS.

12.4 Strength

The ultimate strength calculation may require consideration of the following:

- a) Loss of equilibrium of the structure or any part of it, considered as a rigid body; and
- b) Failure by excessive deformation, rupture or loss of stability of the structure or any part of it including support and foundation.

12.4.1 Design Strength

The design strength, S_d , is obtained as given below from ultimate strength, S_u and partial safety factors for materials, γ_m given in [Table 5](#).

$$S_d = S_u / \gamma_m$$

where partial safety factor for materials, γ_m account for:

- a) possibility of unfavourable deviation of material strength from the characteristic value;
- b) possibility of unfavourable variation of member sizes;
- c) possibility of unfavourable reduction in member strength due to fabrication and tolerances; and
- d) uncertainty in the calculation of strength of the members.

Table 4 Partial Safety Factors for Loads, γ_L , for Limit States

(Clauses [10.6.1](#), [12.3.3](#), [12.6](#), [12.6.1](#), [18.2](#) and [22.5](#))

Sl No.	Combination	Limit State of Strength						Limit State of Serviceability			
		DL	IL ¹⁾		WL/EL	FL	AL	DL	IL ¹⁾		WL/EL/FL
			Leading	Accompanying					Leading	Accompanying	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
i)	DL + IL + CL	1.5	1.5	1.05	—	—	—	1.0	1.0	1.0	—
ii)	DL + IL + CL + WL	1.2 1.2	1.2 1.2	1.05 0.53	0.6 1.2	—	—	1.0	0.8	0.8	0.8
iii)	DL + WL	1.5 (0.9) ²⁾	—	—	1.5	—	—	1.0	—	—	1.0
iv)	DL + IL + CL + EL	1.2 1.2	1.2 1.2	1.05 0.53	1.0 1.0	—	—	1.0	0.8	0.8	1.0
v)	DL + EL	1.5 (0.9) ²⁾	—	—	1.0	—	—	1.0	—	—	1.0
vi)	DL + ER	1.2 (0.9) ²⁾	1.2	—	—	—	—	—	—	—	—
vii)	DL + IL ³⁾ + FL	1.0	1.0/0.5	—	—	1.0	—	1.0	1.0/0.5	—	1.0
viii)	DL + IL + AL	1.0	0.35 ⁴⁾	0.35 ⁴⁾	—	—	1.0	—	—	—	—

Abbreviations:
DL= Dead load, IL= Imposed load, WL= Wind load, CL= Crane load (Vertical/Horizontal), AL=Accidental load, ER= Erection load, EL= Earthquake effects, FL = Forces due to restraints during fire.

NOTE — The effects of actions (loads) in terms of stresses or stress resultants may be obtained from an appropriate method of analysis as in [11](#).

¹⁾When action of different imposed loads is simultaneously considered, the leading imposed load shall be considered to be the one causing the higher load effects in the member/section.

²⁾This value is to be considered when the dead load contributes to stability against overturning is critical or the dead load causes reduction in stress due to other loads.

³⁾ Imposed load shall be taken as the full characteristic specified values in case of storage loads and 50 percent of the specified characteristic values in other cases.

⁴⁾ 0.35 shall not be used when the imposed load is not permeant load as in earth pressure.

Table 5 Partial Safety Factor for Materials, γ_m
(Clauses [12.4.1](#), [13.2](#), [13.3.1](#), [17.1.6](#) and [17.5.7.1.1](#))

Sl No.	Definition	Partial Safety Factor	
(1)	(2)	(3)	
i)	Resistance, governed by yielding, γ_{m0}	1.10	
ii)	Resistance of member to buckling, γ_{m0}	1.10	
iii)	Resistance, governed by ultimate stress, γ_{m1}	1.25	
iv)	Resistance of connection:	Shop Fabrications	Field Fabrications
	a) bolts-Friction Type, γ_{mf} ;	1.25	1.25
	b) bolts-Bearing Type, γ_{mb} ;	1.25	1.25
	c) rivets, γ_{mr} ; and	1.25	1.25
	d) welds, γ_{mw}	1.25	1.50

12.5 Factors Governing the Ultimate Strength

12.5.1 Stability

Stability shall be ensured for the structure as a whole and for each of its elements. This should include, overall frame stability against overturning and sway, as given in [12.5.1.1](#) and [12.5.1.2](#).

12.5.1.1 Stability against overturning

The structure as a whole or any part of it shall be designed to prevent instability due to overturning, uplift or sliding under factored load as given below:

- The actions shall be divided into components aiding instability and components resisting instability;
- The permanent and variable actions and their effects causing instability shall be combined using appropriate partial safety factors for loads as per the limit state requirements, to obtain maximum destabilizing effect;
- The permanent actions (loads) and effects contributing to resistance shall be multiplied with a partial safety factor 0.9 and added together with design resistance (after multiplying with appropriate partial safety factor). Variable actions and their effects contributing to resistance shall be disregarded; and
- The resistance effect shall be greater than or equal to the destabilizing effect. Combination of imposed and dead loads should be such as to cause most severe effect on overall stability.

12.5.1.2 Sway stability

The whole structure, including portions between expansion joints, shall be adequately stiff against sway. To ensure this, in addition to designing for applied horizontal loads, a separate check should be carried out for notional horizontal loads such as given in [11.3.6](#) to evaluate the sway under gravity loads.

12.5.2 Fatigue

Generally, fatigue need not be considered unless a structure or element is subjected to numerous significant fluctuations of stress. Stress changes due to fluctuations in wind loading normally need not be considered. Fatigue design shall be in accordance with the provisions under [19](#). When designing for fatigue, the partial safety factor for load, γ_L , equal to unity shall be used for the load causing stress fluctuation and stress range.

12.5.3 Plastic Collapse

Plastic analysis and design may be used, if the requirement specified under the plastic method of analysis (see [11.5](#)) are satisfied.

12.6 Limit State of Serviceability

Serviceability limit state is related to the criteria governing normal use. Serviceability limit state is limit state beyond which the service criteria specified below, are no longer met:

- a) deflection limit;
- b) vibration limit;
- c) durability consideration; and
- d) fire resistance.

Unless specified otherwise (see [Table 4](#)), partial safety factor for loads, γ_L , of value equal to unity shall be used for all loads leading to serviceability limit states to check the adequacy of the structure under serviceability limit states.

12.6.1 Deflection

The deflection under serviceability loads of a building or a building component should not impair the strength of the structure or components or cause damage to finishing. Deflections are to be checked for the most adverse but realistic combination of service loads and their arrangement, by elastic analysis, using partial safety factors in [Table 4](#) for loads, as required, in [Table 6](#). [Table 6](#) gives recommended limits of deflections for certain structural members and systems. In [Table 6](#), imposed load should include all post construction loads including superimposed dead loads. Circumstances may arise where greater or lesser values would be more appropriate, depending upon the nature of material in element to be supported (vulnerable to cracking or not) and intended use of the structure, as required by client.

Where the deflection due to the combination of dead load and imposed load is likely to be excessive, consideration should be given to pre-camber the beams, trusses and girders. The value of desired camber shall be specified in design drawing. Generally, for spans greater than 25 m, a camber approximately equal to the deflection due to dead loads plus half the imposed load may be used. The deflection of a member shall be calculated without considering the impact factor or dynamic effect of the loads on deflection. Roofs, which are very flexible, shall be designed to withstand any additional load that is likely to occur as a result of ponding of water or accumulation of snow or ice.

12.6.2 Vibration

Suitable provisions in the design shall be made for the dynamic effects of imposed loads, impact loads and vibration due to machinery operating loads. In severe cases possibility of resonance, fatigue or unacceptable vibrations shall be investigated. Unusually flexible structures (generally the height to effective width of lateral load resistance system exceeding 5 : 1) shall be investigated for lateral vibration under dynamic wind loads. Structures subjected to large number of cycles of loading shall be designed against fatigue failure, as specified under [19](#). Floor vibration effect shall be considered using specialist literature (see [Annex C](#)).

12.6.3 Durability

Factors that affect the durability of the buildings, under conditions relevant to their intended life, are listed below:

- a) Environment;
- b) Degree of exposure;
- c) Shape of the member and the structural detail;
- d) Protective measure; and
- e) Ease of maintenance.

The durability of steel structures shall be ensured by following recommendations given under [21](#). Specialist literature may be referred to for more detailed and additional information in design for durability.

Table 6 Deflection Limits

(Clause 12.6.1)

Sl No.	Type of Building	Deflection	Design Load	Member	Supporting	Maximum Deflection		
(1)	(2)	(3)	(4)	(5)	(6)	(7)		
Industrial buildings	Vertical	{	Imposed load and wind load	Purlins and Girts	{ Elastic cladding Brittle cladding	Span/150 Span/180		
			Imposed load	Simple span	{ Elastic cladding Brittle cladding	Span/240 Span/300		
			Imposed load	Cantilever span	{ Elastic cladding Brittle cladding	Span/120 Span/150		
			Imposed load, wind load and crane load	Rafter supporting	{ Profiled metal sheeting Plastered sheeting	Span/180 Span/240		
			Crane load (Manual operation)	Gantry	Wheels of crane	Span/500		
			Crane load (Electric operation up to 50 t)	Gantry	Wheels of crane	Span/750		
			Crane load (Electric operation over 50 t)	Gantry	Wheels of crane	Span/1000		
			Lateral	{	No cranes (due to wind)	Column	{ Elastic cladding Masonry/Brittle cladding	Height/150 Height/240
					Crane load and wind load	Gantry (lateral)	{ (absolute) Relative displacement between rails supporting wheels of crane	Span/400 25 mm ¹⁾
					Crane load and wind load	Column/Frame	Gantry (Elastic cladding)	Height/200
Gantry (Brittle cladding; cab operated)	Height/400							
Other Buildings	Vertical	{	Imposed load	Floor and roof	Elements not susceptible to cracking	Span/300		
			Imposed load	Cantilever	Elements susceptible to cracking	Span/360		
					{ Elements not susceptible to cracking Elements susceptible to cracking	Span/150 Span/180		
	Lateral	{	Wind load	Building	Elastic cladding Brittle cladding	Height/300 Height/500		
			Wind load	Inter-storey drift	—	Storey height/300		

¹⁾ Crane supplier may be consulted for higher values

12.6.4 Fire Resistance

Fire resistance of a steel member is a function of its mass, its geometry, and the actions to which it is subjected, its structural support condition, fire protection measures adopted and the fire to which it is exposed. Design provisions to resist fire are briefly discussed under [22](#). Specialist literature may be referred to for more detailed information in design of fire resistance of steel structures.

13 DESIGN OF TENSION MEMBERS

13.1 Tension Members

Tension members are linear members in which axial forces act to cause elongation (stretch). Such members can sustain loads up to the ultimate load, at which stage they may fail by rupture at a critical section. However, if the gross area of the member yields over a major portion of its length before the rupture load is reached, the member may become non-functional due to excessive elongation. Plates and other rolled sections in tension may also fail by block shear of end bolted regions (see [13.4.1](#)).

The factored design tension T , in the members shall satisfy the following requirement:

$$T < T_d$$

where

T_d = Design strength of the member.

The design strength of a member under axial tension, T_d , is the lowest of the design strength due to yielding of gross section, T_{dg} , rupture strength of critical section, T_{dn} , and block shear T_{db} , given in [13.2](#), [13.3](#) and [13.4](#), respectively.

13.2 Design Strength Due to Yielding of Gross Section

The design strength of members under axial tension, T_{dg} , as governed by yielding of gross section, is given by:

$$T_{dg} = A_g f_y / \gamma_{m0}$$

where

- A_g = gross area of cross section;
- f_y = yield stress of the material; and
- γ_{m0} = partial safety factor for failure in tension by yielding (see [Table 5](#)).

13.3 Design Strength Due to Rupture of Critical Section

13.3.1 Plates

The design strength in tension of a plate, T_{dn} , as governed by rupture of net cross sectional area, A_n , at the holes is given by:

$$T_{dn} = 0.9 A_n f_u / \gamma_{m1}$$

where

- A_n = net effective area of the member given by;
- f_u = ultimate stress of the material; and
- γ_{m1} = partial safety factor for failure at ultimate stress (see [Table 5](#)).

$$A_n = \left[b - nd_h + \sum \frac{p_{si}^2}{4g_i} \right] t$$

where

- b, t = width and thickness of the plate, respectively;
- d_h = diameter of the bolt hole (2 mm in addition to the diameter of the hole, in case of directly punched holes);
- g = gauge length between the bolt holes, as shown in [Fig. 5](#);
- i = subscript for summation of all the inclined legs;
- n = number of bolt holes in the critical section; and
- p_s = staggered pitch length between line of bolt holes, as shown in [Fig. 5](#).

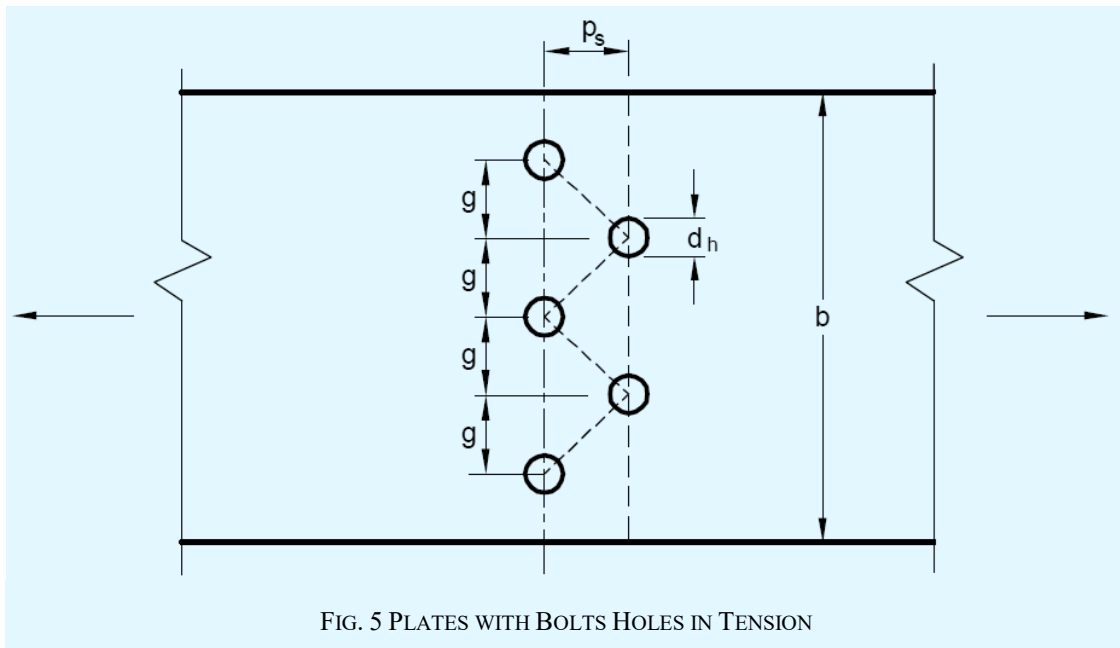


FIG. 5 PLATES WITH BOLTS HOLES IN TENSION

13.3.2 Threaded Rods

The design strength of threaded rods in tension, T_{dn} , as governed by rupture is given by:

$$T_{dn} = 0.9 A_c f_u / \gamma_{m1}$$

where

- A_c = net root area at the threaded section.

13.3.3 Single Angles

The rupture strength of an angle connected through one leg is affected by shear lag. The design strength, T_{dn} , as governed by rupture at net section is given by:

$$T_{dn} = 0.9 A_{nc} f_u / \gamma_{m1} + \beta A_{go} f_y / \gamma_{m0}$$

where

$$\beta = 1.4 - 0.076 (w/t) (f_y/f_u) (b_s/L_c) \leq (0.9f_u\gamma_{m0}/f_y\gamma_{m1}) \geq 0.7$$

where

A_{go} = gross area of the outstanding leg;

A_n = net area of the total cross section

A_{nc} = net area of the connected leg;

b_s = shear lag width, as shown in Fig. 6;

L_c = length of the end connection, that is the distance between the outermost bolts in the end joint measured along the load direction or length of the weld along the load direction;

t = thickness of the leg; and

w = outstand leg width.

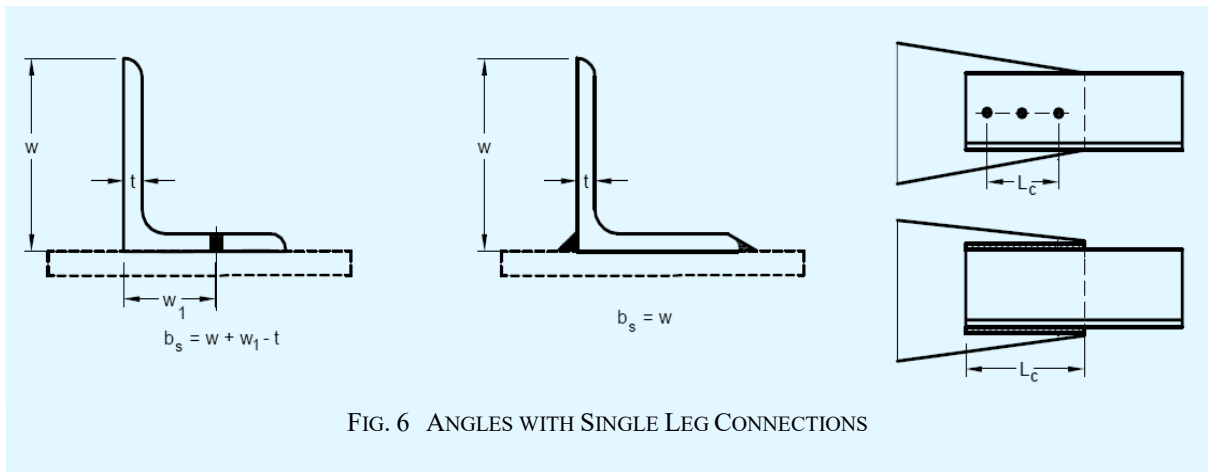


FIG. 6 ANGLES WITH SINGLE LEG CONNECTIONS

13.3.4 Other Section

The rupture strength, T_{dn} , of the double angles, channels, I-sections and other rolled steel sections, connected by one or more elements to an end gusset is also governed by shear lag effects. The design tensile strength of such sections as governed by tearing of net section may also be calculated using equation in 13.3.3, where β is calculated based on the shear lag distance, b_s , taken from the farthest edge of the outstanding leg to the nearest bolt/weld line in the connected leg of the cross section.

13.4 Design Strength Due to Block Shear

The strength as governed by block shear at an end connection of plates and angles is calculated as given in 13.4.1.

13.4.1 Bolted Connections

The block shear strength, T_{db} of connection shall be taken as the smaller of,

$$T_{db} = [A_{vg} f_y / (\sqrt{3} \gamma_{m0}) + 0.9A_{tn} f_u / \gamma_{m1}]$$

or

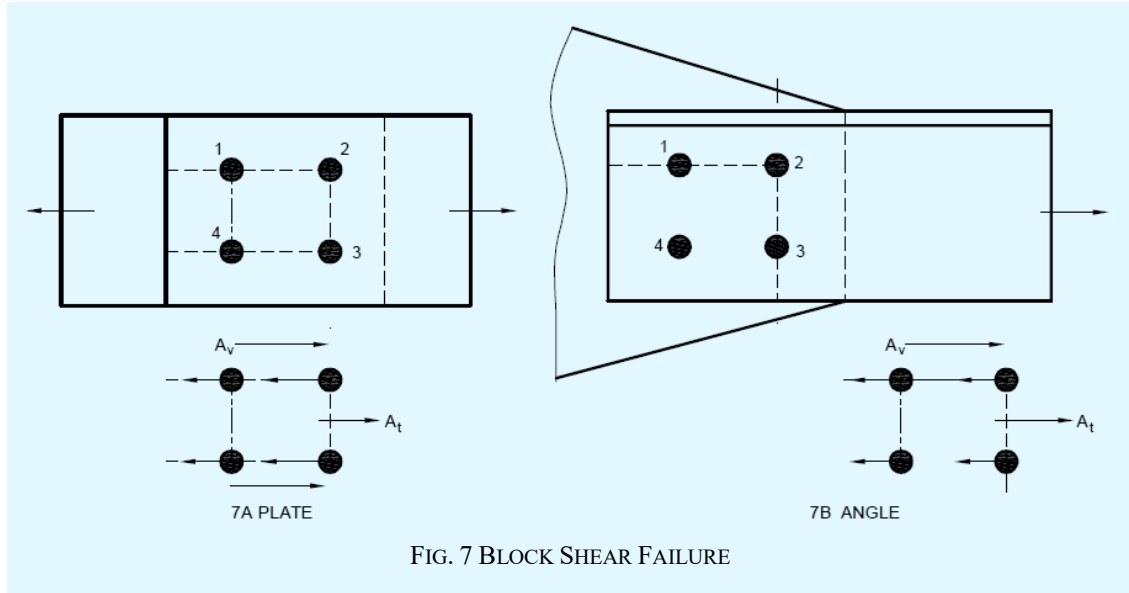
$$T_{db} = [0.9A_{vn} f_u / (\sqrt{3} \gamma_{m1}) + A_{tg} f_y / \gamma_{m0}]$$

where

A_{tg}, A_{tn} = minimum gross and net area in tension from the bolt hole to the toe of the angle, end bolt line, perpendicular to the line of force, respectively (2-3 as shown in [Fig. 7\(B\)](#));

A_{vg}, A_{vn} = minimum gross and net area in shear along bolt line parallel to external force, respectively (1-2 and 3-4 as shown in [Fig. 7\(A\)](#) and 1-2 as shown in [Fig. 7\(B\)](#); and

f_u, f_y = ultimate and yield stress of the material, respectively.



13.4.2 Welded Connection

The block shear strength, T_{db} shall be checked for welded end connections by taking an appropriate section in the member around the end weld, which can shear off as a block.

14 DESIGN OF COMPRESSION MEMBERS

14.1 Design Strength

14.1.1 Common hot rolled and built-up steel members used for carrying axial compression, usually fail by flexural buckling. The buckling strength of these members is affected by residual stresses, initial bow and accidental eccentricities of load. To account for all these factors, the strength of members subjected to axial compression is defined by buckling class a_0 , a , b , c , or d as given [Table 7](#).

14.1.2 The factored design compression, P in members shall satisfy the following requirement:

$$P < P_d$$

$$P_d = A_e f_{cd}$$

where

P_d = design strength of the member

A_e = effective sectional area as defined in [14.3.2](#); and

f_{cd} = design compressive stress, given by.

$$f_{cd} = \frac{\chi f_y}{\gamma_{m0}} \leq f_y / \gamma_{m0}$$

χ = stress reduction factor for different buckling class, slenderness ratio and yield stress and mode of buckling as given in [14.1.2.1](#) or [14.1.2.2](#), as applicable (see [Table 8](#)); and

γ_{m0} = partial safety factor for material strength.

NOTE — Calculated values of design compressive stress, f_{cd} for different buckling classes are given in [Table 9](#).

14.1.2.1 Reduction factor for flexural buckling (See also [Table 8](#))

$$\chi = \frac{1}{\phi + \sqrt{\phi^2 + \lambda^2}}$$

$$\phi = 0.5[1 + \alpha(\lambda - 0.2) + \lambda^2]$$

λ = non-dimensional effective slenderness ratio:

$$= \sqrt{f_y / f_{cr}} = \sqrt{f_y \left(\frac{KL}{r} \right)^2 / \pi^2 E}$$

$$f_{cr} = \text{Euler buckling stress about the relevant axis} = \frac{\pi^2 E}{(KL/r)^2}$$

where

KL/r = effective slenderness ratio or ratio of effective length, KL to appropriate radius of gyration, r ;
and

α = imperfection factor given in [Table 7](#).

14.1.2.2 Reduction factor for torsional and torsional flexural buckling

$$\chi = \frac{1}{\phi + \sqrt{\phi^2 + \lambda_{TF}^2}}$$

$$\phi = 0.5 \left[1 + \left(\frac{\lambda_{TF}}{\lambda_y} \right)^2 \cdot \alpha_{TF} (\lambda_y - 0.2) + \lambda_{TF}^2 \right]$$

$$\alpha_{TF} = \alpha \sqrt{\frac{A f_y (i_p^2 + d_y^2)}{6.25 G I_T}}$$

$$\lambda_{TF} = \sqrt{f_y / f_{crTF}}$$

$$\lambda_y = \sqrt{f_y / f_{cr,y}}$$

where

α = is the imperfection factor given in [Table 7](#);

d_y = is the distance between intermediate lateral support and shear centre at the section;

i_p = is the polar radius of gyration given by;

i_p^2 = $(I_y + I_z)/A$;

I_T = is the torsional constant of the cross section;

- I_y, I_z = is the moment of inertia about y and z axis, respectively and shear centre;
 λ_{TF} = non-dimensional effective slenderness ratio; and
 λ_y = non-dimensional effective slenderness ratio.
 f_{cry} = flexural buckling stress about the y axis ; and
 f_{crTF} = torsional flexural buckling stress at the extreme compression fibre ([Annex E](#)).

14.1.2.3 The classification of different sections under different buckling class a_0, a, b, c or d , is given in [Table 10](#). The stress reduction factor χ , and the design compressive stress f_{cd} , for different buckling class, yield stress, and effective slenderness ratio is given in [Table 8](#) and [Table 9](#) for convenience. The curves corresponding to different buckling class are presented in non-dimensional form, in [Fig. 8](#).

Table 7 Imperfection Factor, α						
(Clauses 14.1.1 , 14.1.2.1 and 14.1.2.2)						
SI No.	Buckling Class	a_0	a	b	c	d
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	α	0.13	0.21	0.34	0.49	0.76

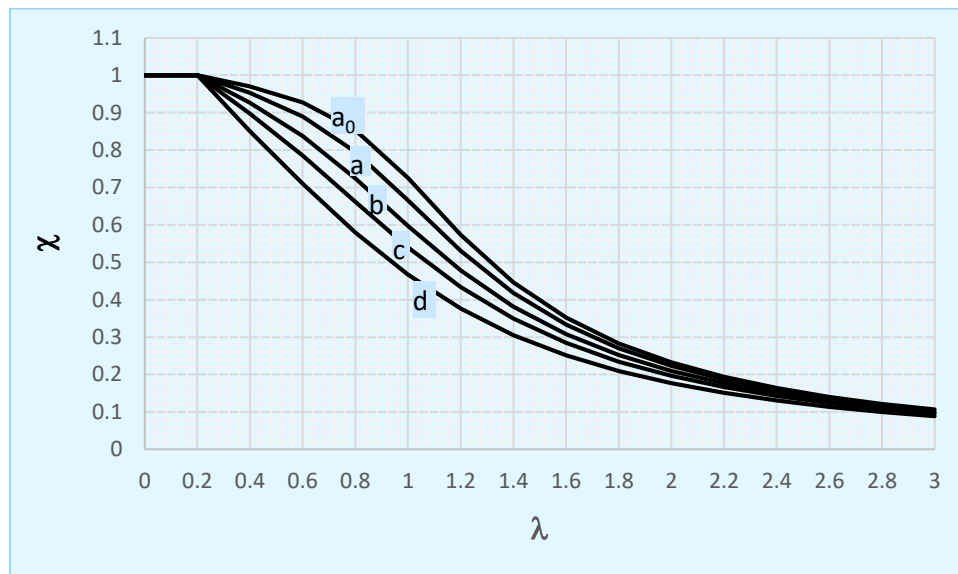


FIG. 8 COLUMN BUCKLING CURVES

14.2 Effective Length of Compression Members

14.2.1 The effective length KL , is calculated from the actual length L , of the member, considering the rotational and relative translational boundary conditions at the ends. The actual length shall be taken as the length from centre to centre of its intersections with the supporting members in the plane of the buckling deformation. In the case of a member with a free end, the free standing length from the centre of the intersecting member at the supported end, shall be taken as the actual length.

14.2.2 Effective Length

Where the boundary conditions in the plane of buckling can be assessed, the effective length, KL can be calculated on the basis of Table 11. Where frame analysis does not consider the equilibrium of a framed structure in the

deformed shape (second-order analysis or advanced analysis), the effective length of compression members in such cases can be calculated using the procedure given in [D-1](#). The effective length of stepped column in single storey buildings can be calculated using the procedure given in [D-2](#).

14.2.3 Eccentric Beam Connection

In cases where the beam connections are eccentric in plan with respect to the axes of the column, the same conditions of restraint as in concentric connection shall be deemed to apply, provided the connections are carried across the flange or web of the columns as the case may be, and the web of the beam lies within, or in direct contact with the column section. Where practical difficulties prevent this, the effective length shall be taken as equal to the distance between points of restraint in non-sway frames.

14.2.4 Compression Members in Trusses

In the case of bolted, riveted or welded trusses and braced frames, the effective length, KL , of the compression members shall be taken as 0.7 to 1.0 times the distance between centres of connections, depending on the degree of end restraint provided. In the case of members of trusses buckling in the plane perpendicular to the plane of the truss, the effective length, KL shall be taken as the distance between the centres of intersection. The design of angle struts shall be as specified in [14.5](#).

14.3 Design Details

14.3.1 Thickness of Plate Elements

Classification of members on the basis of the thickness of constituent plate elements shall satisfy the width-thickness ratio requirements specified in [Table 2](#).

14.3.2 Effective Sectional Area, A_e

Except as modified in [10.8.2](#) (Class 4), the gross sectional area shall be taken as the effective sectional area for all compression members fabricated by welding, bolting and riveting so long as the section is semi-compact or better. Holes not fitted with rivets, bolts or pins shall be deducted from gross area to calculate effective sectional area.

14.3.3 Eccentricity for Stanchions and Columns

14.3.3.1 For the purpose of determining the stress in a stanchion or column section, the beam reactions or similar loads shall be assumed to be applied at an eccentricity of 100 mm from the face of the section or at the centre of bearing whichever dimension gives the greater eccentricity, and with the exception of the following two cases:

- a) In the case of cap connection, the load shall be assumed to be applied at the face of the column or stanchion section or at the edge of packing, if used towards the span of the beam; and
- b) In the case of a roof truss bearing on a cap, no eccentricity be taken for simple bearings without connections capable of developing any appreciable moment. In case of web member connection with face, actual eccentricity is to be considered.

Table 8(A) Stress Reduction Factor, χ for Flexural Column Buckling Class, a_0

(Clauses [14.1.2.1](#) and [14.1.2.3](#))

KL/r ↓	Yield Stress, f_y (MPa)																		
	200	210	220	230	240	250	260	280	300	320	340	360	380	400	420	450	480	510	540
10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
20	1.000	0.999	0.998	0.998	0.997	0.997	0.996	0.995	0.994	0.992	0.991	0.990	0.989	0.988	0.987	0.986	0.984	0.983	0.981
30	0.986	0.985	0.983	0.982	0.981	0.980	0.979	0.977	0.975	0.973	0.971	0.969	0.967	0.965	0.964	0.961	0.958	0.955	0.952
40	0.970	0.968	0.966	0.965	0.963	0.961	0.960	0.956	0.953	0.949	0.946	0.943	0.939	0.936	0.932	0.927	0.921	0.915	0.909
50	0.951	0.948	0.945	0.943	0.940	0.937	0.934	0.929	0.923	0.917	0.911	0.905	0.899	0.892	0.885	0.874	0.863	0.852	0.840
60	0.927	0.922	0.918	0.914	0.909	0.905	0.900	0.891	0.881	0.870	0.860	0.848	0.837	0.824	0.812	0.792	0.772	0.752	0.731
70	0.895	0.888	0.881	0.874	0.867	0.859	0.852	0.836	0.819	0.802	0.784	0.766	0.747	0.728	0.710	0.682	0.655	0.629	0.605
80	0.851	0.841	0.830	0.819	0.808	0.796	0.784	0.760	0.736	0.712	0.687	0.664	0.641	0.619	0.598	0.568	0.540	0.515	0.491
90	0.792	0.777	0.762	0.747	0.731	0.716	0.701	0.670	0.641	0.614	0.587	0.563	0.539	0.518	0.498	0.470	0.445	0.422	0.401
100	0.721	0.702	0.683	0.665	0.647	0.629	0.612	0.580	0.550	0.523	0.498	0.475	0.453	0.434	0.416	0.391	0.369	0.349	0.332
110	0.643	0.622	0.602	0.583	0.564	0.547	0.530	0.499	0.471	0.446	0.423	0.403	0.384	0.366	0.350	0.329	0.310	0.293	0.278
120	0.568	0.547	0.527	0.509	0.491	0.475	0.459	0.431	0.405	0.383	0.363	0.344	0.328	0.312	0.299	0.280	0.264	0.249	0.236
130	0.500	0.480	0.462	0.445	0.429	0.414	0.399	0.374	0.351	0.331	0.313	0.297	0.282	0.269	0.257	0.241	0.227	0.214	0.203
140	0.441	0.423	0.406	0.390	0.376	0.362	0.350	0.327	0.307	0.289	0.273	0.259	0.246	0.234	0.223	0.209	0.197	0.186	0.176
150	0.391	0.374	0.359	0.345	0.332	0.319	0.308	0.288	0.270	0.254	0.240	0.227	0.216	0.205	0.196	0.183	0.172	0.163	0.154
160	0.348	0.333	0.319	0.306	0.294	0.283	0.273	0.255	0.239	0.225	0.212	0.201	0.191	0.181	0.173	0.162	0.152	0.144	0.136
170	0.311	0.298	0.285	0.273	0.263	0.253	0.244	0.227	0.213	0.200	0.189	0.179	0.170	0.161	0.154	0.144	0.135	0.128	0.121
180	0.280	0.268	0.256	0.246	0.236	0.227	0.219	0.204	0.191	0.179	0.169	0.160	0.152	0.145	0.138	0.129	0.121	0.114	0.108
190	0.253	0.242	0.231	0.222	0.213	0.205	0.197	0.184	0.172	0.162	0.152	0.144	0.137	0.130	0.124	0.116	0.109	0.103	0.097
200	0.230	0.219	0.210	0.201	0.193	0.186	0.179	0.166	0.156	0.146	0.138	0.131	0.124	0.118	0.112	0.105	0.099	0.093	0.088
210	0.209	0.200	0.191	0.183	0.176	0.169	0.163	0.151	0.142	0.133	0.125	0.119	0.113	0.107	0.102	0.095	0.090	0.084	0.080
220	0.191	0.183	0.175	0.167	0.161	0.154	0.149	0.138	0.129	0.122	0.115	0.108	0.103	0.098	0.093	0.087	0.082	0.077	0.073
230	0.176	0.168	0.160	0.154	0.147	0.142	0.136	0.127	0.119	0.112	0.105	0.099	0.094	0.090	0.086	0.080	0.075	0.071	0.067
240	0.162	0.155	0.148	0.141	0.136	0.131	0.126	0.117	0.109	0.103	0.097	0.091	0.087	0.083	0.079	0.074	0.069	0.065	0.061
250	0.150	0.143	0.136	0.131	0.125	0.121	0.116	0.108	0.101	0.095	0.089	0.084	0.080	0.076	0.073	0.068	0.064	0.060	0.057

Table 8(B) Stress Reduction Factor, χ for Flexural Column Buckling Class a

(Clauses [14.1.2.1](#) and [14.1.2.3](#))

KL/r ↓	Yield Stress, f_y (MPa)																		
	200	210	220	230	240	250	260	280	300	320	340	360	380	400	420	450	480	510	540
10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
20	1.000	0.999	0.998	0.997	0.995	0.994	0.993	0.993	0.990	0.988	0.986	0.984	0.983	0.981	0.979	0.977	0.975	0.972	0.970
30	0.977	0.975	0.974	0.972	0.970	0.969	0.967	0.965	0.961	0.957	0.954	0.951	0.948	0.946	0.943	0.938	0.934	0.930	0.925
40	0.952	0.949	0.947	0.944	0.942	0.939	0.937	0.934	0.926	0.921	0.916	0.911	0.906	0.901	0.896	0.888	0.881	0.873	0.865
50	0.923	0.919	0.915	0.911	0.908	0.904	0.900	0.896	0.884	0.876	0.867	0.859	0.851	0.842	0.834	0.820	0.807	0.794	0.780
60	0.888	0.883	0.877	0.871	0.865	0.859	0.853	0.847	0.828	0.816	0.803	0.790	0.777	0.763	0.750	0.730	0.710	0.690	0.671
70	0.846	0.837	0.829	0.820	0.811	0.803	0.794	0.785	0.758	0.740	0.722	0.703	0.686	0.668	0.651	0.626	0.602	0.579	0.557
80	0.793	0.781	0.769	0.757	0.746	0.734	0.722	0.710	0.675	0.653	0.631	0.610	0.589	0.570	0.551	0.525	0.501	0.478	0.458
90	0.730	0.715	0.700	0.685	0.671	0.657	0.643	0.628	0.590	0.565	0.542	0.520	0.500	0.481	0.463	0.439	0.416	0.396	0.377
100	0.661	0.644	0.627	0.610	0.594	0.579	0.564	0.549	0.510	0.486	0.463	0.443	0.424	0.407	0.390	0.368	0.348	0.331	0.314
110	0.591	0.573	0.555	0.538	0.522	0.507	0.492	0.478	0.440	0.418	0.397	0.379	0.362	0.346	0.332	0.312	0.295	0.279	0.265
120	0.525	0.507	0.489	0.473	0.458	0.443	0.429	0.416	0.381	0.361	0.343	0.326	0.311	0.297	0.284	0.267	0.252	0.238	0.226
130	0.466	0.448	0.432	0.416	0.402	0.388	0.376	0.364	0.332	0.314	0.298	0.283	0.269	0.257	0.246	0.231	0.217	0.206	0.195
140	0.413	0.397	0.382	0.368	0.355	0.342	0.331	0.320	0.291	0.275	0.260	0.247	0.235	0.224	0.214	0.201	0.189	0.179	0.170
150	0.368	0.353	0.339	0.326	0.314	0.303	0.293	0.283	0.257	0.243	0.229	0.218	0.207	0.197	0.189	0.177	0.166	0.157	0.149
160	0.329	0.316	0.303	0.291	0.280	0.270	0.261	0.252	0.229	0.215	0.204	0.193	0.184	0.175	0.167	0.157	0.147	0.139	0.132
170	0.296	0.283	0.272	0.261	0.251	0.242	0.233	0.225	0.204	0.192	0.182	0.172	0.164	0.156	0.149	0.140	0.131	0.124	0.117
180	0.267	0.255	0.245	0.235	0.226	0.218	0.210	0.203	0.184	0.173	0.163	0.155	0.147	0.140	0.134	0.125	0.118	0.111	0.105
190	0.242	0.231	0.222	0.213	0.205	0.197	0.190	0.183	0.166	0.156	0.147	0.140	0.133	0.126	0.121	0.113	0.106	0.100	0.095
200	0.220	0.210	0.202	0.193	0.186	0.179	0.172	0.166	0.151	0.142	0.134	0.127	0.120	0.115	0.109	0.102	0.096	0.091	0.086
210	0.201	0.192	0.184	0.177	0.170	0.163	0.157	0.152	0.137	0.129	0.122	0.115	0.110	0.104	0.099	0.093	0.087	0.083	0.078
220	0.184	0.176	0.169	0.162	0.155	0.149	0.144	0.139	0.126	0.118	0.111	0.106	0.100	0.095	0.091	0.085	0.080	0.075	0.071
230	0.170	0.162	0.155	0.149	0.143	0.137	0.132	0.128	0.115	0.108	0.102	0.097	0.092	0.088	0.083	0.078	0.073	0.069	0.065
240	0.157	0.149	0.143	0.137	0.132	0.127	0.122	0.118	0.106	0.100	0.094	0.089	0.085	0.081	0.077	0.072	0.068	0.064	0.060
250	0.145	0.138	0.132	0.127	0.122	0.117	0.113	0.109	0.098	0.092	0.087	0.082	0.078	0.074	0.071	0.066	0.062	0.059	0.056

Table 8(C) Stress Reduction Factor, χ for Flexural Column Buckling Class b																			
(Clauses 14.1.2.1 and 14.1.2.3)																			
KL/r ↓	Yield Stress, f_y (MPa)																		
	200	210	220	230	240	250	260	280	300	320	340	360	380	400	420	450	480	510	540
10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
20	1.000	0.998	0.996	0.994	0.993	0.991	0.990	0.986	0.983	0.981	0.978	0.975	0.972	0.970	0.967	0.963	0.960	0.956	0.953
30	0.963	0.961	0.958	0.955	0.953	0.950	0.948	0.943	0.938	0.933	0.929	0.924	0.920	0.915	0.911	0.904	0.898	0.892	0.886
40	0.925	0.921	0.917	0.913	0.909	0.906	0.902	0.895	0.887	0.880	0.873	0.866	0.859	0.852	0.845	0.835	0.825	0.815	0.805
50	0.883	0.877	0.872	0.866	0.861	0.855	0.850	0.839	0.829	0.818	0.808	0.798	0.787	0.777	0.767	0.752	0.737	0.722	0.708
60	0.835	0.827	0.820	0.812	0.805	0.798	0.790	0.775	0.761	0.746	0.732	0.718	0.704	0.691	0.677	0.657	0.638	0.620	0.602
70	0.781	0.771	0.761	0.751	0.742	0.732	0.722	0.703	0.685	0.667	0.649	0.632	0.615	0.599	0.584	0.561	0.540	0.520	0.502
80	0.721	0.709	0.697	0.685	0.673	0.661	0.650	0.627	0.606	0.585	0.566	0.547	0.529	0.512	0.496	0.474	0.453	0.434	0.416
90	0.657	0.643	0.629	0.615	0.602	0.589	0.576	0.552	0.530	0.508	0.488	0.470	0.452	0.436	0.421	0.400	0.380	0.363	0.346
100	0.593	0.577	0.562	0.548	0.534	0.520	0.507	0.483	0.461	0.440	0.421	0.403	0.387	0.372	0.358	0.339	0.321	0.306	0.291
110	0.531	0.515	0.500	0.485	0.471	0.458	0.445	0.422	0.401	0.381	0.364	0.348	0.333	0.319	0.306	0.289	0.274	0.260	0.247
120	0.474	0.458	0.443	0.429	0.416	0.403	0.391	0.370	0.350	0.332	0.316	0.301	0.288	0.276	0.265	0.249	0.236	0.223	0.212
130	0.423	0.408	0.394	0.380	0.368	0.356	0.345	0.325	0.307	0.291	0.276	0.263	0.251	0.240	0.230	0.217	0.204	0.194	0.184
140	0.378	0.364	0.350	0.338	0.327	0.316	0.306	0.287	0.271	0.256	0.243	0.231	0.221	0.211	0.202	0.190	0.179	0.169	0.161
150	0.339	0.325	0.313	0.302	0.291	0.281	0.272	0.255	0.241	0.227	0.215	0.205	0.195	0.186	0.178	0.167	0.158	0.149	0.142
160	0.305	0.292	0.281	0.271	0.261	0.252	0.243	0.228	0.215	0.203	0.192	0.182	0.174	0.166	0.158	0.149	0.140	0.133	0.126
170	0.275	0.264	0.253	0.244	0.235	0.227	0.219	0.205	0.193	0.182	0.172	0.163	0.155	0.148	0.142	0.133	0.125	0.118	0.112
180	0.249	0.239	0.229	0.220	0.212	0.205	0.198	0.185	0.174	0.164	0.155	0.147	0.140	0.133	0.128	0.120	0.113	0.106	0.101
190	0.227	0.217	0.208	0.200	0.193	0.186	0.179	0.168	0.157	0.148	0.140	0.133	0.127	0.121	0.115	0.108	0.102	0.096	0.091
200	0.207	0.198	0.190	0.183	0.176	0.169	0.163	0.153	0.143	0.135	0.128	0.121	0.115	0.110	0.105	0.098	0.092	0.087	0.083
210	0.190	0.182	0.174	0.167	0.161	0.155	0.149	0.140	0.131	0.123	0.117	0.110	0.105	0.100	0.096	0.090	0.084	0.080	0.075
220	0.174	0.167	0.160	0.154	0.148	0.142	0.137	0.128	0.120	0.113	0.107	0.101	0.096	0.092	0.088	0.082	0.077	0.073	0.069
230	0.161	0.154	0.147	0.141	0.136	0.131	0.126	0.118	0.111	0.104	0.098	0.093	0.088	0.084	0.080	0.075	0.071	0.067	0.063
240	0.149	0.142	0.136	0.131	0.126	0.121	0.117	0.109	0.102	0.096	0.091	0.086	0.082	0.078	0.074	0.070	0.065	0.062	0.058
250	0.138	0.132	0.126	0.121	0.117	0.112	0.108	0.101	0.095	0.089	0.084	0.080	0.076	0.072	0.069	0.064	0.060	0.057	0.054

Table 8(D) Stress Reduction Factor, χ for Column Flexural Buckling Class c

(Clauses 14.1.2.1 and 14.1.2.3)

KL/r ↓	Yield Stress, f_y (MPa)																		
	200	210	220	230	240	250	260	280	300	320	340	360	380	400	420	450	480	510	540
10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
20	0.999	0.997	0.994	0.992	0.990	0.987	0.985	0.981	0.976	0.972	0.968	0.964	0.961	0.957	0.953	0.948	0.943	0.938	0.933
30	0.948	0.944	0.941	0.937	0.933	0.930	0.926	0.920	0.913	0.907	0.901	0.895	0.889	0.883	0.877	0.869	0.861	0.853	0.845
40	0.896	0.891	0.885	0.880	0.875	0.870	0.866	0.856	0.847	0.838	0.829	0.820	0.812	0.803	0.795	0.783	0.771	0.760	0.748
50	0.841	0.834	0.827	0.821	0.814	0.807	0.801	0.788	0.776	0.763	0.752	0.740	0.729	0.717	0.706	0.690	0.675	0.660	0.645
60	0.783	0.774	0.765	0.757	0.748	0.740	0.732	0.716	0.700	0.685	0.670	0.656	0.642	0.628	0.615	0.596	0.578	0.561	0.544
70	0.722	0.711	0.700	0.690	0.680	0.670	0.660	0.641	0.623	0.605	0.588	0.572	0.557	0.542	0.528	0.508	0.489	0.471	0.454
80	0.659	0.646	0.634	0.622	0.611	0.600	0.589	0.568	0.548	0.529	0.512	0.495	0.479	0.464	0.450	0.430	0.412	0.395	0.379
90	0.596	0.583	0.569	0.557	0.544	0.533	0.521	0.499	0.479	0.460	0.443	0.426	0.411	0.397	0.383	0.365	0.348	0.332	0.318
100	0.536	0.522	0.508	0.495	0.483	0.471	0.459	0.438	0.418	0.400	0.384	0.368	0.354	0.341	0.328	0.311	0.296	0.282	0.269
110	0.480	0.466	0.453	0.440	0.428	0.416	0.405	0.385	0.366	0.349	0.333	0.319	0.306	0.294	0.283	0.268	0.254	0.242	0.230
120	0.430	0.416	0.403	0.391	0.379	0.368	0.358	0.339	0.321	0.306	0.291	0.278	0.267	0.256	0.246	0.232	0.220	0.209	0.199
130	0.385	0.372	0.360	0.348	0.337	0.327	0.317	0.299	0.283	0.269	0.256	0.244	0.234	0.224	0.215	0.203	0.192	0.182	0.173
140	0.346	0.333	0.322	0.311	0.301	0.291	0.282	0.266	0.251	0.238	0.227	0.216	0.206	0.197	0.189	0.178	0.168	0.160	0.152
150	0.311	0.300	0.289	0.279	0.269	0.261	0.252	0.237	0.224	0.212	0.202	0.192	0.183	0.175	0.168	0.158	0.149	0.141	0.134
160	0.281	0.270	0.260	0.251	0.242	0.234	0.227	0.213	0.201	0.190	0.180	0.172	0.164	0.156	0.150	0.141	0.133	0.126	0.120
170	0.255	0.245	0.236	0.227	0.219	0.212	0.205	0.192	0.181	0.171	0.162	0.154	0.147	0.140	0.134	0.126	0.119	0.113	0.107
180	0.232	0.223	0.214	0.206	0.199	0.192	0.186	0.174	0.164	0.155	0.147	0.139	0.133	0.127	0.121	0.114	0.107	0.102	0.096
190	0.212	0.203	0.195	0.188	0.181	0.175	0.169	0.158	0.149	0.140	0.133	0.126	0.120	0.115	0.110	0.103	0.097	0.092	0.087
200	0.194	0.186	0.179	0.172	0.166	0.160	0.154	0.144	0.136	0.128	0.121	0.115	0.110	0.105	0.100	0.094	0.089	0.084	0.079
210	0.178	0.171	0.164	0.158	0.152	0.146	0.141	0.132	0.124	0.117	0.111	0.105	0.100	0.096	0.092	0.086	0.081	0.076	0.072
220	0.164	0.157	0.151	0.145	0.140	0.135	0.130	0.122	0.114	0.108	0.102	0.097	0.092	0.088	0.084	0.079	0.074	0.070	0.066
230	0.152	0.145	0.140	0.134	0.129	0.124	0.120	0.112	0.105	0.099	0.094	0.089	0.085	0.081	0.077	0.073	0.068	0.065	0.061
240	0.141	0.135	0.129	0.124	0.120	0.115	0.111	0.104	0.098	0.092	0.087	0.082	0.078	0.075	0.071	0.067	0.063	0.060	0.056
250	0.131	0.125	0.120	0.115	0.111	0.107	0.103	0.096	0.090	0.085	0.081	0.076	0.073	0.069	0.066	0.062	0.058	0.055	0.052

Table 8(E) Stress Reduction Factor, χ for Flexural Column Buckling Class d
(Clauses 14.1.2.1 and 14.1.2.3)

KL/r ↓	Yield Stress, f_y (MPa)																		
	200	210	220	230	240	250	260	280	300	320	340	360	380	400	420	450	480	510	540
10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
20	0.999	0.995	0.991	0.988	0.984	0.980	0.977	0.970	0.964	0.958	0.952	0.946	0.940	0.935	0.930	0.922	0.915	0.908	0.901
30	0.922	0.916	0.911	0.906	0.901	0.896	0.891	0.881	0.872	0.863	0.855	0.847	0.839	0.831	0.823	0.813	0.802	0.792	0.782
40	0.848	0.841	0.834	0.828	0.821	0.815	0.808	0.796	0.784	0.773	0.762	0.751	0.741	0.731	0.721	0.707	0.694	0.681	0.668
50	0.777	0.768	0.760	0.752	0.744	0.736	0.728	0.713	0.699	0.685	0.672	0.659	0.647	0.635	0.624	0.608	0.592	0.577	0.563
60	0.707	0.697	0.687	0.678	0.668	0.659	0.651	0.634	0.617	0.602	0.587	0.573	0.560	0.547	0.535	0.517	0.501	0.486	0.471
70	0.640	0.629	0.618	0.607	0.597	0.587	0.578	0.559	0.542	0.526	0.510	0.496	0.482	0.469	0.456	0.439	0.423	0.408	0.394
80	0.576	0.564	0.553	0.542	0.531	0.521	0.511	0.492	0.474	0.458	0.442	0.428	0.414	0.402	0.390	0.373	0.358	0.344	0.330
90	0.517	0.505	0.493	0.482	0.471	0.461	0.451	0.432	0.415	0.399	0.384	0.370	0.357	0.345	0.334	0.319	0.304	0.292	0.280
100	0.464	0.451	0.440	0.428	0.418	0.408	0.398	0.380	0.363	0.348	0.334	0.321	0.309	0.298	0.288	0.274	0.261	0.249	0.239
110	0.416	0.404	0.392	0.381	0.371	0.361	0.352	0.335	0.319	0.305	0.292	0.281	0.270	0.259	0.250	0.237	0.226	0.215	0.206
120	0.373	0.361	0.350	0.340	0.330	0.321	0.313	0.297	0.282	0.269	0.257	0.246	0.236	0.227	0.219	0.207	0.197	0.187	0.179
130	0.336	0.325	0.314	0.305	0.295	0.287	0.279	0.264	0.251	0.239	0.228	0.218	0.209	0.200	0.193	0.182	0.173	0.164	0.157
140	0.303	0.292	0.283	0.274	0.265	0.257	0.250	0.236	0.224	0.213	0.203	0.194	0.185	0.178	0.171	0.161	0.153	0.145	0.138
150	0.274	0.264	0.255	0.247	0.239	0.231	0.224	0.212	0.201	0.190	0.181	0.173	0.165	0.159	0.152	0.144	0.136	0.129	0.123
160	0.249	0.240	0.231	0.223	0.216	0.209	0.203	0.191	0.181	0.171	0.163	0.155	0.149	0.142	0.137	0.129	0.122	0.116	0.110
170	0.227	0.218	0.210	0.203	0.196	0.190	0.184	0.173	0.164	0.155	0.147	0.140	0.134	0.128	0.123	0.116	0.110	0.104	0.099
180	0.207	0.199	0.192	0.185	0.179	0.173	0.167	0.157	0.149	0.141	0.134	0.127	0.122	0.116	0.111	0.105	0.099	0.094	0.089
190	0.190	0.183	0.176	0.169	0.164	0.158	0.153	0.144	0.136	0.128	0.122	0.116	0.111	0.106	0.101	0.095	0.090	0.085	0.081
200	0.175	0.168	0.162	0.156	0.150	0.145	0.140	0.132	0.124	0.118	0.112	0.106	0.101	0.097	0.093	0.087	0.082	0.078	0.074
210	0.161	0.155	0.149	0.143	0.138	0.134	0.129	0.121	0.114	0.108	0.102	0.097	0.093	0.089	0.085	0.080	0.075	0.071	0.068
220	0.149	0.143	0.138	0.133	0.128	0.123	0.119	0.112	0.105	0.100	0.094	0.090	0.086	0.082	0.078	0.074	0.069	0.066	0.062
230	0.138	0.133	0.128	0.123	0.118	0.114	0.110	0.104	0.097	0.092	0.087	0.083	0.079	0.075	0.072	0.068	0.064	0.061	0.058
240	0.129	0.123	0.119	0.114	0.110	0.106	0.103	0.096	0.090	0.085	0.081	0.077	0.073	0.070	0.067	0.063	0.059	0.056	0.053
250	0.120	0.115	0.110	0.106	0.102	0.099	0.095	0.089	0.084	0.079	0.075	0.071	0.068	0.065	0.062	0.058	0.055	0.052	0.049

Table 9(A) Design Compressive Stress, f_{cd} (MPa) For Column Flexural Buckling Class a_0
(Clauses [14.1.2](#) and [14.1.2.3](#))

$KL/r \downarrow$	Yield Stress, f_y (MPa)																		
	200	210	220	230	240	250	260	280	300	320	340	360	380	400	420	450	480	510	540
10	182	191	200	209	218	227	236	255	273	291	309	327	345	364	382	409	436	464	491
20	182	191	200	209	218	226	235	253	271	289	306	324	342	359	377	403	429	456	482
30	179	188	197	205	214	223	231	249	266	283	300	317	334	351	368	393	418	443	467
40	176	185	193	202	210	218	227	243	260	276	292	308	324	340	356	379	402	424	446
50	173	181	189	197	205	213	221	236	252	267	282	296	310	324	338	358	377	395	412
60	168	176	184	191	198	206	213	227	240	253	266	278	289	300	310	324	337	349	359
70	163	170	176	183	189	195	201	213	223	233	242	251	258	265	271	279	286	292	297
80	155	160	166	171	176	181	185	194	201	207	212	217	221	225	228	232	236	239	241
90	144	148	152	156	160	163	166	171	175	178	182	184	186	188	190	192	194	196	197
100	131	134	137	139	141	143	145	148	150	152	154	155	157	158	159	160	161	162	163
110	117	119	120	122	123	124	125	127	129	130	131	132	133	133	134	135	135	136	136
120	103	104	105	106	107	108	109	110	111	111	112	113	113	114	114	115	115	115	116
130	91.0	91.7	92.4	93.0	93.5	94.0	94.4	95.2	95.8	96.3	96.8	97.2	97.6	97.9	98.2	98.6	98.9	99.2	99.4
140	80.3	80.8	81.2	81.6	82.0	82.3	82.6	83.2	83.6	84.0	84.3	84.6	84.9	85.1	85.3	85.6	85.9	86.1	86.3
150	71.1	71.5	71.8	72.1	72.3	72.6	72.8	73.2	73.5	73.8	74.0	74.3	74.5	74.6	74.8	75.0	75.2	75.4	75.5
160	63.3	63.6	63.8	64.0	64.2	64.4	64.6	64.8	65.1	65.3	65.5	65.7	65.8	66.0	66.1	66.3	66.4	66.5	66.7
170	56.6	56.8	57.0	57.2	57.3	57.5	57.6	57.8	58.0	58.2	58.3	58.5	58.6	58.7	58.8	58.9	59.1	59.2	59.3
180	50.9	51.1	51.2	51.3	51.5	51.6	51.7	51.9	52.0	52.1	52.3	52.4	52.5	52.6	52.6	52.7	52.8	52.9	53.0
190	46.0	46.1	46.2	46.3	46.4	46.5	46.6	46.8	46.9	47.0	47.1	47.2	47.3	47.3	47.4	47.5	47.6	47.6	47.7
200	41.7	41.9	41.9	42.0	42.1	42.2	42.2	42.4	42.5	42.6	42.6	42.7	42.8	42.8	42.9	43.0	43.0	43.1	43.2
210	38.0	38.1	38.2	38.3	38.3	38.4	38.5	38.6	38.6	38.7	38.8	38.8	38.9	39.0	39.0	39.1	39.1	39.2	39.2
220	34.8	34.9	34.9	35.0	35.1	35.1	35.2	35.2	35.3	35.4	35.4	35.5	35.5	35.6	35.6	35.7	35.7	35.8	35.8
230	32.0	32.0	32.1	32.1	32.2	32.2	32.3	32.3	32.4	32.4	32.5	32.5	32.6	32.6	32.7	32.7	32.7	32.8	32.8
240	29.4	29.5	29.5	29.6	29.6	29.7	29.7	29.8	29.8	29.9	29.9	29.9	30.0	30.0	30.0	30.1	30.1	30.2	30.2
250	27.2	27.3	27.3	27.3	27.4	27.4	27.4	27.5	27.5	27.6	27.6	27.6	27.7	27.7	27.7	27.8	27.8	27.8	27.9

Table 9(B) Design Compressive Stress, f_{cd} (MPa) For Column Flexural Buckling Class a
(Clauses 14.1.2 and 14.1.2.3)

KL/r ↓	Yield Stress, f_y (MPa)																		
	200	210	220	230	240	250	260	280	300	320	340	360	380	400	420	450	480	510	540
10	182	191	200	213	218	227	236	255	273	291	309	327	345	364	382	409	436	464	491
20	182	191	200	208	217	226	235	252	270	287	305	322	339	357	374	400	425	451	476
30	178	186	195	203	212	220	229	245	262	279	295	311	328	344	360	384	408	431	454
40	173	181	189	197	205	213	221	237	253	268	283	298	313	328	342	363	384	405	425
50	168	176	183	191	198	205	213	227	241	255	268	281	294	306	318	336	352	368	383
60	162	169	175	182	189	195	202	214	226	237	248	258	268	278	286	299	310	320	329
70	154	160	166	171	177	182	188	197	207	215	223	230	237	243	249	256	263	268	274
80	144	149	154	158	163	167	171	178	184	190	195	199	204	207	210	215	219	222	225
90	133	137	140	143	146	149	152	157	161	164	168	170	173	175	177	179	182	184	185
100	120	123	125	128	130	132	133	136	139	141	143	145	146	148	149	151	152	153	154
110	107	109	111	112	114	115	116	118	120	121	123	124	125	126	127	128	129	129	130
120	95.5	96.7	97.9	98.9	100	101	101	103	104	105	106	107	107	108	109	109	110	110	111
130	84.6	85.5	86.3	87	87.7	88.3	88.8	89.8	90.6	91.3	92.0	92.5	93.0	93.5	93.9	94.4	94.9	95.3	95.7
140	75.2	75.8	76.4	76.9	77.4	77.8	78.2	78.9	79.5	80.0	80.5	80.9	81.3	81.6	81.9	82.3	82.6	83.0	83.2
150	67.0	67.4	67.9	68.2	68.6	68.9	69.2	69.7	70.2	70.6	70.9	71.2	71.5	71.8	72.0	72.3	72.6	72.9	73.1
160	59.9	60.3	60.6	60.9	61.1	61.4	61.6	62.0	62.4	62.7	62.9	63.2	63.4	63.6	63.8	64.0	64.3	64.5	64.6
170	53.8	54.1	54.3	54.6	54.8	55.0	55.1	55.5	55.7	56.0	56.2	56.4	56.6	56.7	56.9	57.1	57.3	57.4	57.6
180	48.6	48.8	49.0	49.2	49.3	49.5	49.6	49.9	50.1	50.3	50.5	50.6	50.8	50.9	51.0	51.2	51.3	51.5	51.6
190	44.0	44.2	44.3	44.5	44.6	44.7	44.9	45.1	45.3	45.4	45.6	45.7	45.8	45.9	46.0	46.2	46.3	46.4	46.5
200	40.0	40.2	40.3	40.4	40.5	40.7	40.7	40.9	41.1	41.2	41.3	41.4	41.5	41.6	41.7	41.8	41.9	42.0	42.1
210	36.6	36.7	36.8	36.9	37.0	37.1	37.2	37.3	37.4	37.6	37.7	37.8	37.8	37.9	38.0	38.1	38.2	38.3	38.3
220	33.5	33.6	33.7	33.8	33.9	34.0	34.0	34.2	34.3	34.4	34.5	34.5	34.6	34.7	34.7	34.8	34.9	35.0	35.0
230	30.8	30.9	31.0	31.1	31.2	31.2	31.3	31.4	31.5	31.6	31.6	31.7	31.8	31.8	31.9	31.9	32.0	32.1	32.1
240	28.5	28.5	28.6	28.7	28.7	28.8	28.8	28.9	29.0	29.1	29.1	29.2	29.3	29.3	29.4	29.4	29.5	29.5	29.6
250	26.3	26.4	26.5	26.5	26.6	26.6	26.7	26.7	26.8	26.9	26.9	27.0	27.0	27.1	27.1	27.2	27.2	27.3	27.3

Table 9(C) Design Compressive Stress, f_{cd} (MPa) for Column Flexural Buckling Class b
(Clauses 14.1.2 and 14.1.2.3)

KL/r ↓	Yield Stress, f_y (MPa)																		
	200	210	220	230	240	250	260	280	300	320	340	360	380	400	420	450	480	510	540
10	182	191	200	209	218	227	236	255	273	291	309	327	345	364	382	409	436	464	491
20	182	190	199	208	217	225	234	251	268	285	302	319	336	353	369	394	419	443	468
30	175	183	192	200	208	216	224	240	256	271	287	302	318	333	348	370	392	414	435
40	168	176	183	191	198	206	213	228	242	256	270	283	297	310	323	342	360	378	395
50	161	167	174	181	188	194	201	214	226	238	250	261	272	283	293	308	322	335	347
60	152	158	164	170	176	181	187	197	207	217	226	235	243	251	259	269	279	287	295
70	142	147	152	157	162	166	171	179	187	194	201	207	213	218	223	230	236	241	246
80	131	135	139	143	147	150	154	160	165	170	175	179	183	186	190	194	198	201	204
90	120	123	126	129	131	134	136	141	144	148	151	154	156	159	161	163	166	168	170
100	108	110	112	114	116	118	120	123	126	128	130	132	134	135	137	139	140	142	143
110	96.5	98.3	100	101	103	104	105	107	109	111	112	114	115	116	117	118	119	121	121
120	86.2	87.5	88.6	89.7	90.7	91.7	92.5	94.1	95.4	96.6	97.7	98.6	100	100	101	102	103	104	104
130	76.9	77.8	78.7	79.5	80.3	81.0	81.6	82.7	83.7	84.6	85.4	86.1	86.8	87.3	87.9	88.6	89.2	89.8	90.3
140	68.7	69.4	70.1	70.7	71.3	71.8	72.3	73.1	73.9	74.6	75.2	75.7	76.2	76.6	77.1	77.6	78.1	78.5	78.9
150	61.6	62.1	62.6	63.1	63.6	64.0	64.3	65.0	65.6	66.1	66.6	67.0	67.4	67.7	68.1	68.5	68.9	69.2	69.5
160	55.4	55.8	56.2	56.6	56.9	57.3	57.5	58.1	58.5	59.0	59.3	59.7	60.0	60.3	60.5	60.9	61.2	61.5	61.7
170	50.0	50.3	50.7	51.0	51.2	51.5	51.7	52.2	52.5	52.9	53.2	53.5	53.7	53.9	54.1	54.4	54.7	54.9	55.1
180	45.3	45.6	45.9	46.1	46.3	46.5	46.7	47.1	47.4	47.7	47.9	48.1	48.3	48.5	48.7	48.9	49.2	49.3	49.5
190	41.2	41.5	41.7	41.9	42.1	42.2	42.4	42.7	42.9	43.2	43.4	43.6	43.7	43.9	44.0	44.2	44.4	44.6	44.7
200	37.6	37.8	38.0	38.2	38.3	38.5	38.6	38.9	39.1	39.3	39.5	39.6	39.8	39.9	40.0	40.2	40.3	40.5	40.6
210	34.5	34.7	34.8	35.0	35.1	35.2	35.3	35.5	35.7	35.9	36.0	36.2	36.3	36.4	36.5	36.6	36.8	36.9	37.0
220	31.7	31.9	32.0	32.1	32.2	32.3	32.4	32.6	32.8	32.9	33.0	33.1	33.2	33.3	33.4	33.6	33.7	33.8	33.9
230	29.2	29.4	29.5	29.6	29.7	29.8	29.9	30.0	30.1	30.3	30.4	30.5	30.6	30.7	30.7	30.8	30.9	31.0	31.1
240	27.1	27.2	27.3	27.3	27.4	27.5	27.6	27.7	27.8	27.9	28.0	28.1	28.2	28.3	28.3	28.4	28.5	28.6	28.7
250	25.1	25.2	25.3	25.3	25.4	25.5	25.6	25.7	25.8	25.9	26.0	26.0	26.1	26.2	26.2	26.3	26.4	26.5	26.5

Table 9(D) Design Compressive Stress, f_{cd} (MPa) for Column Flexural Buckling Class c

(Clauses 14.1.2 and 14.1.2.3)

KL/r	Yield Stress, f_y , (MPa)																		
↓	200	210	220	230	240	250	260	280	300	320	340	360	380	400	420	450	480	510	540
10	182	191	200	209	218	227	236	255	273	291	309	327	345	364	382	409	436	464	491
20	182	190	199	207	216	224	233	250	266	283	299	316	332	348	364	388	412	435	458
30	172	180	188	196	204	211	219	234	249	264	278	293	307	321	335	355	376	395	415
40	163	170	177	184	191	198	205	218	231	244	256	268	280	292	304	320	337	352	367
50	153	159	165	172	178	183	189	201	212	222	232	242	252	261	270	282	295	306	317
60	142	148	153	158	163	168	173	182	191	199	207	215	222	228	235	244	252	260	267
70	131	136	140	144	148	152	156	163	170	176	182	187	192	197	202	208	213	218	223
80	120	123	127	130	133	136	139	145	149	154	158	162	165	169	172	176	180	183	186
90	108	111	114	116	119	121	123	127	131	134	137	140	142	144	146	149	152	154	156
100	97.5	100	102	104	105	107	109	112	114	116	119	120	122	124	125	127	129	131	132
110	87.3	89.0	90.5	92.0	93.3	94.6	95.7	97.9	100	102	103	104	106	107	108	110	111	112	113
120	78.2	79.4	80.6	81.7	82.7	83.7	84.6	86.2	87.6	88.9	90.1	91.1	92.1	93.0	93.8	94.9	95.9	96.8	97.6
130	70.0	71.0	71.9	72.8	73.5	74.3	75.0	76.2	77.3	78.3	79.2	80.0	80.7	81.4	82.0	82.9	83.6	84.3	84.9
140	62.9	63.6	64.4	65.0	65.6	66.2	66.7	67.7	68.6	69.3	70.0	70.7	71.2	71.8	72.3	72.9	73.5	74.1	74.6
150	56.6	57.2	57.8	58.3	58.8	59.2	59.7	60.4	61.1	61.7	62.3	62.8	63.3	63.7	64.1	64.6	65.1	65.5	65.9
160	51.1	51.6	52.1	52.5	52.9	53.3	53.6	54.2	54.8	55.3	55.7	56.1	56.5	56.9	57.2	57.6	58.0	58.4	58.7
170	46.4	46.8	47.1	47.5	47.8	48.1	48.4	48.9	49.3	49.8	50.1	50.5	50.8	51.1	51.3	51.7	52.0	52.3	52.6
180	42.2	42.5	42.8	43.1	43.4	43.6	43.9	44.3	44.7	45.0	45.3	45.6	45.8	46.1	46.3	46.6	46.9	47.1	47.3
190	38.5	38.8	39.0	39.3	39.5	39.7	39.9	40.3	40.6	40.9	41.1	41.4	41.6	41.8	42.0	42.2	42.5	42.7	42.9
200	35.3	35.5	35.7	35.9	36.1	36.3	36.5	36.8	37.0	37.3	37.5	37.7	37.9	38.1	38.2	38.4	38.6	38.8	39.0
210	32.4	32.6	32.8	33.0	33.1	33.3	33.4	33.7	33.9	34.1	34.3	34.5	34.7	34.8	34.9	35.1	35.3	35.4	35.6
220	29.9	30.1	30.2	30.4	30.5	30.6	30.8	31.0	31.2	31.4	31.5	31.7	31.8	31.9	32.1	32.2	32.4	32.5	32.6
230	27.6	27.8	27.9	28.0	28.2	28.3	28.4	28.6	28.8	28.9	29.1	29.2	29.3	29.4	29.5	29.7	29.8	29.9	30.0
240	25.6	25.7	25.9	26.0	26.1	26.2	26.3	26.4	26.6	26.7	26.9	27.0	27.1	27.2	27.3	27.4	27.5	27.6	27.7
250	23.8	23.9	24.0	24.1	24.2	24.3	24.4	24.5	24.7	24.8	24.9	25.0	25.1	25.2	25.3	25.4	25.5	25.6	25.7

Table 9(E) Design Compressive Stress, f_{cd} (MPa) for Column Buckling Class d

(Clauses [14.1.2](#) and [14.1.2.3](#))

KL/r ↓	Yield Stress, f_y (MPa)																		
	200	210	220	230	240	250	260	280	300	320	340	360	380	400	420	450	480	510	540
10	182	191	200	209	218	227	236	255	273	291	309	327	345	364	382	409	436	464	491
20	182	190	198	206	215	223	231	247	263	279	294	310	325	340	355	377	399	421	442
30	168	175	182	189	197	204	211	224	238	251	264	277	290	302	314	332	350	367	384
40	154	161	167	173	179	185	191	203	214	225	235	246	256	266	275	289	303	316	328
50	141	147	152	157	162	167	172	182	191	199	208	216	224	231	238	249	258	268	277
60	129	133	137	142	146	150	154	161	168	175	182	188	193	199	204	212	219	225	231
70	116	120	124	127	130	133	137	142	148	153	158	162	167	171	174	180	184	189	193
80	105	108	111	113	116	118	121	125	129	133	137	140	143	146	149	153	156	159	162
90	94.1	96.4	98.6	101	103	105	107	110	113	116	119	121	123	126	128	130	133	135	137
100	84.3	86.2	87.9	89.6	91.1	92.6	94.0	96.7	99.1	101	103	105	107	108	110	112	114	116	117
110	75.6	77.0	78.4	79.7	81.0	82.1	83.2	85.3	87.1	88.8	90.4	91.8	93.1	94.4	95.5	97.1	98.5	100	101
120	67.8	69.0	70.1	71.1	72.1	73.0	73.9	75.5	77.0	78.3	79.5	80.6	81.7	82.6	83.5	84.7	85.8	86.9	87.8
130	61.0	62.0	62.8	63.7	64.5	65.2	65.9	67.2	68.3	69.4	70.4	71.2	72.1	72.8	73.5	74.5	75.4	76.2	76.9
140	55.0	55.8	56.5	57.2	57.8	58.4	59.0	60.0	61.0	61.8	62.6	63.3	64.0	64.6	65.2	66.0	66.7	67.3	67.9
150	49.8	50.4	51.0	51.6	52.1	52.6	53.1	53.9	54.7	55.4	56.0	56.6	57.2	57.7	58.1	58.8	59.3	59.9	60.4
160	45.2	45.7	46.2	46.7	47.1	47.5	47.9	48.6	49.3	49.9	50.4	50.9	51.3	51.7	52.1	52.7	53.1	53.6	54.0
170	41.2	41.6	42.1	42.4	42.8	43.1	43.5	44.1	44.6	45.1	45.5	45.9	46.3	46.7	47.0	47.4	47.8	48.2	48.6
180	37.7	38.0	38.4	38.7	39.0	39.3	39.6	40.1	40.5	41.0	41.3	41.7	42.0	42.3	42.6	43.0	43.3	43.6	43.9
190	34.5	34.9	35.2	35.4	35.7	35.9	36.2	36.6	37.0	37.4	37.7	38.0	38.2	38.5	38.7	39.1	39.4	39.6	39.9
200	31.8	32.0	32.3	32.5	32.8	33.0	33.2	33.6	33.9	34.2	34.5	34.7	35.0	35.2	35.4	35.7	35.9	36.2	36.4
210	29.3	29.6	29.8	30.0	30.2	30.4	30.5	30.9	31.2	31.4	31.7	31.9	32.1	32.3	32.5	32.7	32.9	33.1	33.3
220	27.1	27.3	27.5	27.7	27.9	28.0	28.2	28.5	28.7	29.0	29.2	29.4	29.6	29.7	29.9	30.1	30.3	30.5	30.6
230	25.2	25.3	25.5	25.7	25.8	26.0	26.1	26.4	26.6	26.8	27.0	27.1	27.3	27.5	27.6	27.8	27.9	28.1	28.2
240	23.4	23.6	23.7	23.9	24.0	24.1	24.2	24.5	24.7	24.8	25.0	25.2	25.3	25.4	25.5	25.7	25.9	26.0	26.1
250	21.8	22.0	22.1	22.2	22.3	22.5	22.6	22.8	22.9	23.1	23.2	23.4	23.5	23.6	23.7	23.9	24.0	24.1	24.2

Table 10 Buckling Class of Cross Sections

(Clause [14.1.2.3](#))

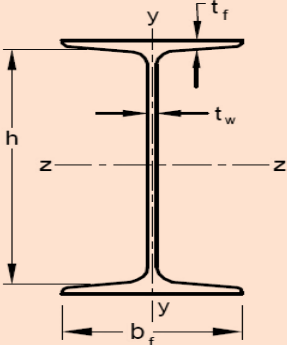
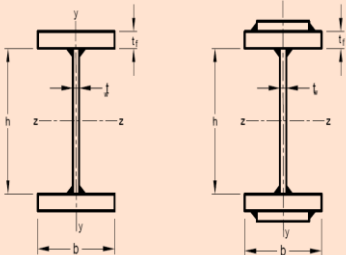
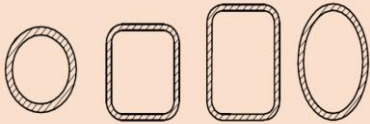
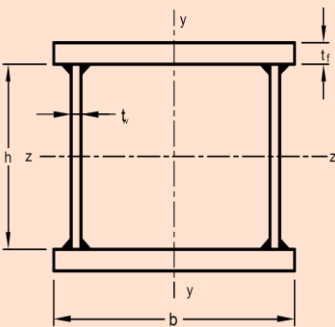
Cross Section	Limits	Buckling About Axis	Buckling Class	
			$f_y \leq 420 \text{ MPa}$	$f_y > 420 \text{ MPa}$
(1)	(2)	(3)	(4)	(5)
Rolled I-Sections 	$h/b_f > 1.2$	$t_f \leq 40 \text{ mm}$	z-z a	a ₀ a
		$t_f > 40 \text{ mm}$	z-z b y-y c	a b
	$h/b_f \leq 1.2$	$t_f \leq 100 \text{ mm}$	z-z b y-y c	a b
		$t_f > 100 \text{ mm}$	z-z d y-y d	c c
Welded I-Section 	$t_f \leq 40 \text{ mm}$		z-z b y-y c	b c
	$t_f > 40 \text{ mm}$		z-z c y-y d	c d
Hollow Section 	Hot rolled		a	a ₀
	Cold formed		c	c
Welded Box Section, Cruciform column 	Generally (except as below)		b	b
	Thick welds $s_w > 0.5t_f$ and $b/t_f < 30$ and $h/t_w < 30$		c	c

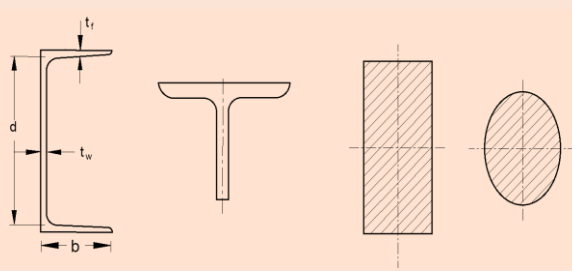
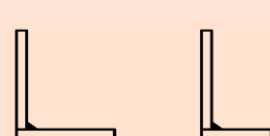
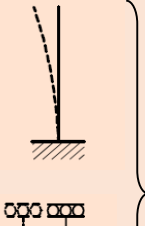
Table 10 (Concluded)				
Cross Section	Limits	Buckling About Axis	Buckling Class	
			$f_y \leq 420 \text{ MPa}$	$f_y > 420 \text{ MPa}$
(1)	(2)	(3)	(4)	(5)
Channel, T, and Solid Sections 		any	c	c
Rolled angle or L sections 		any	b	a
Welded angle or L sections $t \leq 40 \text{ mm}$ 		any	c	c

Table 11 Effective Length of Prismatic Compression Members

(Clause [14.2.2](#))

Boundary Conditions				Schematic Representation	Effective Length
At One End		At the Other End			
Translation	Rotation	Translation	Rotation		
(1)	(2)	(3)	(4)		
Restrained	Restrained	Free	Free		2.0L
Restrained	Free	Free	Restrained		
Restrained	Free	Restrained	Free		1.0L
Restrained	Restrained	Free	Restrained		1.2L
Restrained	Restrained	Restrained	Free		0.8L
Restrained	Restrained	Restrained	Restrained		0.65L

NOTE — L is the unsupported length of the compression member (see [13.2.1](#)).

14.3.3.2 In continuous columns, the bending moments due to eccentricities of loading on the columns at any floor may be divided equally between the columns above and below that floor level, provided that the moment of inertia of one column section, divided by its effective length does not exceed 1.5 times the corresponding value of the other column. Where this ratio is exceeded, the bending moment shall be divided in proportion to the moment of inertia of the column sections divided by their respective effective lengths.

14.3.4 Splices

14.3.4.1 Where the ends of compression members are prepared for bearing over the whole area, they shall be spliced to hold the connected members accurately in position, and to resist bending or tension, if present. Such splices should maintain the intended member stiffness about each axis. Splices should be located as close to the point of inflection as possible. Otherwise their capacity should be adequate to carry magnified moment (see [16.3.2.2](#)). The ends of compression members faced for bearing shall invariably be machined to ensure perfect contact of surfaces in bearing.

14.3.4.2 Where such members are not faced for complete bearing, the splices shall be designed to transmit all the forces to which the members are subjected.

14.3.4.3 Wherever possible, splices shall be proportioned and arranged so that the centroidal axis of the splice coincides as nearly as possible with the centroidal axes of the members being jointed, in order to avoid eccentricity; but where eccentricity is present in the joint, the resulting stress shall be accounted for.

14.4 Column Bases

14.4.1 General

Column bases should have sufficient stiffness and strength to transmit axial force, bending moments and shear forces at the base of the columns to their foundation without exceeding the load carrying capacity of the supports. Anchor bolts and shear keys should be provided wherever necessary. Shear resistance at the proper contact surface between steel base and concrete/grout may be calculated using a friction coefficient of 0.45. Design recommendations of anchor bolts in concrete foundations may be referred as per good practice [C-6A(38)].

The nominal bearing pressure between the base plate and the support below may be determined on the basis of linearly varying distribution of pressure. The maximum bearing pressure on concrete under the limit states design load should not exceed the bearing strength equal to $0.3f_{ck} [\sqrt{A_1/A_2}]$ and $0.9f_{ck}$, whichever is smaller, where f_{ck} is the smaller of characteristic cube strength of concrete or bedding material, A_1 is the area of concrete pedestal on which the base plate is sitting, and A_2 is the surface area of the steel base plate transferring the load to the concrete pedestal.

If the size of the base plate is larger than that required to limit the bearing pressure on the base support, an equal projection, c of the base plate beyond the face of the column and gusset may be taken as effective in transferring the column load as given in [Fig. 9](#), such that bearing pressure on the effective area does not exceed bearing capacity of concrete base.

14.4.2 Gusseted Bases

For stanchion with gusseted bases, the gusset plates, angle cleats, stiffeners, fastenings, etc, in combination with the bearing area of the shaft, shall be sufficient to take the loads, bending moments and reactions to the base plate without exceeding specified strength. All the bearing surfaces shall be machined to ensure perfect contact.

14.4.2.1 Where the ends of the column shaft and the gusset plates are not faced for complete bearing, the welds, fastenings connecting them to the base plate shall be sufficient to transmit all the forces to which the base is subjected.

14.4.2.2 Column and base plate connections

Where the end of the column is connected directly to the base plate by means of full penetration butt welds, the connection shall be deemed to transmit to the base all the forces and moments to which the column is subjected.

14.4.3 Slab Bases

Columns with slab bases need not be provided with gussets, but sufficient fastenings shall be provided to retain the parts securely in place and to resist all moments and forces, other than direct compression, including those arising during transit, unloading and erection. Minimum of four number of anchor bolts shall be used to anchor column bases to foundation and minimum of six numbers in the case of slab bases transferring earthquake forces.

14.4.3.1 The minimum thickness, t_s , of rectangular slab bases, supporting columns under axial compression shall be

$$t_s = \sqrt{2.5 w (a^2 - 0.3b^2) \gamma_{m0} / f_y} > t_f$$

where

- a, b = larger and smaller projection, respectively of the slab base beyond the rectangle circumscribing the column;
- t_f = flange thickness of compression member; and
- w = uniform pressure from below on the slab base under the factored load axial compression.

When only the effective area of the base plate is used as in [14.4.1](#), c^2 may be used in the above equation (see [Fig. 9](#)) instead of $(a^2 - 0.3b^2)$.

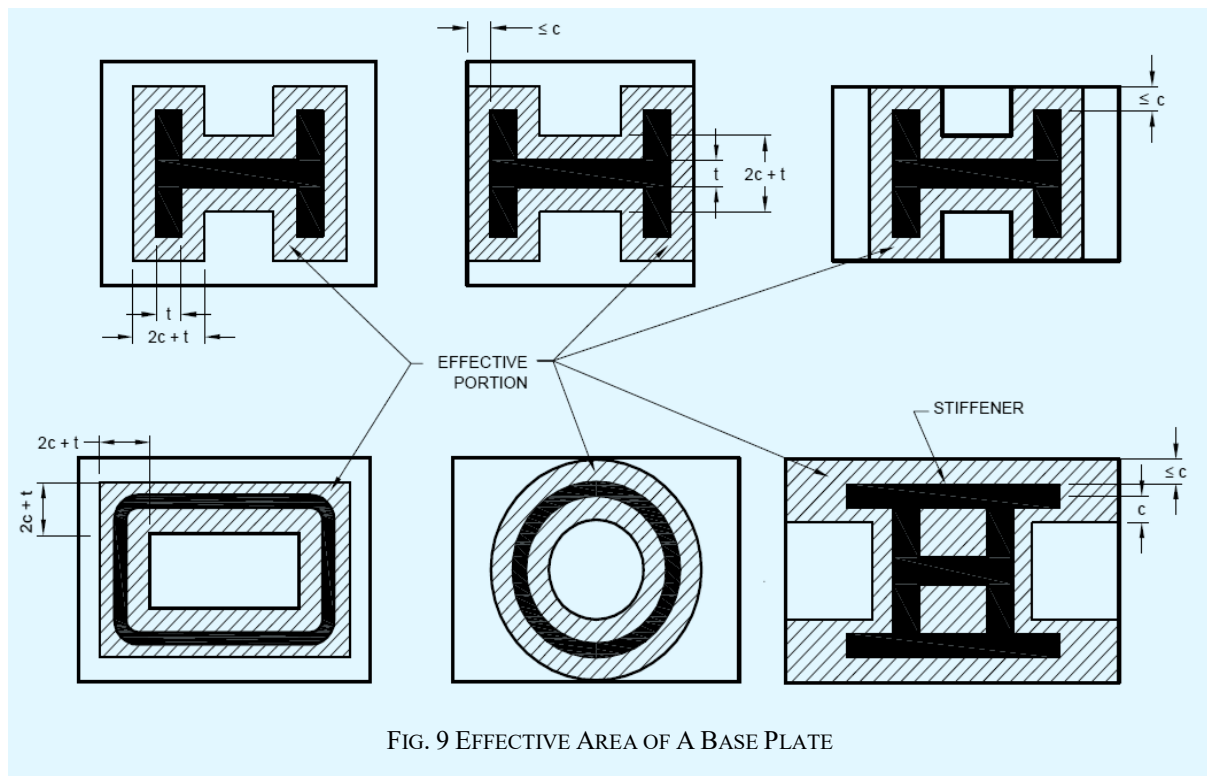


FIG. 9 EFFECTIVE AREA OF A BASE PLATE

14.4.3.2 When the slab does not distribute the column load uniformly, due to eccentricity of the load etc, special calculation shall be made to show that the base is adequate to resist the moment due to the non-uniform pressure from below.

14.4.3.3 Bases for bearing upon concrete or masonry need not be machined on the underside.

14.4.3.4 In cases where the cap or base is fillet welded directly to the end of the column without boring and shouldering, the contact surfaces shall be machined to give a perfect bearing and the welding shall be sufficient

to transmit the forces as required in [14.4.3](#). Where full strength butt welds are provided, machining of contact surfaces is not required.

14.5 Angle Struts

14.5.1 Single Angle Struts

The compression in single angles may be transferred either concentrically to its centroid through the end connections or eccentrically by connecting one of its legs to a gusset or adjacent member.

14.5.1.1 Concentric loading

When a single angle is concentrically loaded in compression, the design strength may be evaluated using [14.1.2](#), by calculating the non-dimensional slenderness ratio, λ_{vv} , for buckling about the minor axis, v-v, as given by:

$$\lambda_{vv} = \frac{\left(\frac{l_{vv}}{r_{vv}} \right)}{\varepsilon \sqrt{\frac{\pi^2 E}{250}}}$$

r_{vv} = radius of gyration of the angle section about the minor axis; and

l_{vv} = laterally unsupported length preventing translation normal to the minor axis

14.5.1.2 Loaded through one leg

The design strength of single angle members loaded in compression through one of its legs, f_{cde} , is affected by flexural torsional buckling and bending. The design compressive strength of such eccentrically loaded angle members, *in lieu of* a more exact second order design under combined bending axial compression, may be evaluated approximately as given below.

The design compressive strength, of single angles loaded through gusset connection to one leg parallel to the *a-a* axis (*a-a* axis being either the z-z or the y-y axis, depending on which leg is connected to the gusset) of the member may be calculated as:

$$f_{cde} = K_f \chi_{aa} f_y / \gamma_{m0} \leq f_{cd} \text{ (given in [14.1.2](#))}$$

where

f_{cde} = Design compressive stress of single angle.

K_f = $k_1 + k_2 \lambda_{aa} + k_3 \lambda_{\phi}$;

χ_{aa} = Stress reduction factor (see Table 8) for buckling class 'b', calculated as per [14.1.2.1](#), using the non-dimensional slenderness ratio, λ_{aa} .

where

k_1, k_2, k_3 = constants depending upon the end condition, as given in [Table 12](#); and

K_f = modification factor to account for eccentric end connection.

$$\lambda_{aa} = \frac{\left(\frac{l_{aa}}{r_{aa}} \right)}{\varepsilon \sqrt{\frac{\pi^2 E}{250}}}$$

$$\lambda_{\phi} = \frac{(b_1 + b_2)/2t}{\varepsilon \sqrt{\frac{\pi^2 E}{250}}}$$

- b_1, b_2 = width of the connected and outstanding legs of the angle, respectively;
- l_{aa} = centre to centre length of the lateral support preventing translation of the member perpendicular to a-a axis (a-a axis being either the z-z or the y-y axis, depending on which leg is connected to the gusset);
- r_{aa} = radius of gyration of the angle member about the a-a axis (a-a axis being either the z-z or the y-y axis, depending on which leg is connected to the gusset), parallel to the plane of the end gusset plates;
- t = thickness of the leg; and
- ε = is the yield stress ratio, $\sqrt{\frac{250}{f_y}}$, f_y being in MPa

Table 12 Constants k_1 , k_2 , and k_3

(Clause [14.5.1.2](#))

SI No.	End Connection	Gusset/Connecting Member Fixity ¹⁾	k_1	k_2	k_3
(1)	(2)	(3)	(4)	(5)	(6)
i)	Fully welded or connected with two or more bolts	Fixed	0.798	0.563	-2.072
		Hinged	0.401	0.420	-1.040
ii)	Single bolt	Fixed	0.418	0.547	-1.400
		Hinged	0.374	0.415	-2.072

¹⁾ In-plane rotational restraint provided to the gusset by the supporting member. For partial restraint, k_f can be interpolated between the results for fixed and hinged cases.

14.5.2 Double Angle Struts

14.5.2.1 For double angle discontinuous struts, connected back to back, on opposite sides of the gusset or a section, by not less than two bolts or rivets in line along the angles at each end, or by the equivalent in welding, the load may be regarded as applied axially. The effective length, KL , in the plane of end gusset shall be taken as between 0.7 times and 0.85 times the distance between intersections, depending on the degree of the restraint provided. The effective length, KL , in the plane perpendicular to that of the end gusset, shall be taken as equal to the distance between centres of intersections. The calculated average compressive stress shall not exceed the values based on [14.1.2](#). The angles shall be connected together over their lengths so as to satisfy the requirements of [14.8](#) and [17.2.5](#).

14.5.2.2 Double angle discontinuous struts connected back-to-back, to one side of a gusset or section by one or more bolts or rivets in each angle, or by the equivalent in welding, shall be designed in accordance with [14.5.1](#) and the angles shall be connected together over their lengths so as to satisfy the requirements of [14.8](#) and [17.2.5](#).

14.5.3 Continuous Members

Double angle continuous struts such as those forming the flanges, chords or ties of trusses or trussed girders, or the legs of towers shall be designed as axially loaded compression members, and the effective length shall be taken in accordance with [14.2.4](#).

14.5.4 Combined Stresses

In addition to axial loads, if the struts carry loads which cause transverse bending, the combined bending and axial stresses shall be checked in accordance with [16.3](#). For determining the permissible axial and bending stresses, the effective length shall be taken in accordance with the [14.2](#) and [15.3](#).

14.6 Laced Columns

14.6.1 General

14.6.1.1 Members comprising two main components laced and tied, should where practicable, have a radius of gyration about the axis perpendicular to the plane of lacing not less than the radius of gyration about the axis parallel to the plane of lacing [see [Fig.10\(A\)](#) and [Fig. 10\(B\)](#)].

14.6.1.2 As far as practicable, the lacing system shall be uniform throughout the length of the column.

14.6.1.3 Except for tie plates as specified in [14.7](#), double laced systems [see [Fig. 10\(B\)](#)] and single laced systems [see [Fig. 10\(A\)](#)] on opposite sides of the main components shall not be combined with cross members (ties) perpendicular to the longitudinal axis of the strut [see [Fig. 10\(C\)](#)], unless all forces resulting from deformation of the strut members are calculated and provided for in the design of lacing and its fastenings.

14.6.1.4 Single laced systems, on opposite faces of the components being laced together shall preferably be in the same direction so that one is the shadow of the other, instead of being mutually opposed in direction.

14.6.1.5 The effective slenderness ratio, $(KL/r)_e$, of laced columns shall be taken as 1.05 times the $(KL/r)_0$, the actual maximum slenderness ratio, in order to account for shear deformation effects.

14.6.2 Width of Lacing Bars

In bolted/riveted construction, the minimum width of lacing bars shall be three times the nominal diameter of the end bolt/rivet.

14.6.3 Thickness of Lacing Bars

The thickness of flat lacing bars shall not be less than one-fortieth of its effective length for single lacings and one-sixtieth of the effective length for double lacings.

Rolled sections or tubes of equivalent strength may be permitted instead of flats, for lacings.

14.6.4 Angle of Inclination

Lacing bars, whether in double or single systems, shall be inclined at an angle not less than 40° nor more than 70° to the axis of the built-up member.

14.6.5 Spacing

14.6.5.1 The maximum spacing of lacing bars, whether connected by bolting, riveting or welding, shall also be such that the maximum slenderness ratio of the components of the main member (a_1/r_1) , between consecutive lacing connections is not greater than 50 or 0.7 times the most unfavourable slenderness ratio of the member as a whole, whichever is less, where a_1 is the unsupported length of the individual member between lacing points, and r_1 is the minimum radius of gyration of the individual member being laced together.

14.6.5.2 Where lacing bars are not lapped to form the connection to the components of the members, they shall be so connected that there is no appreciable interruption in the triangulation of the system.

14.6.6 Design of Lacings

14.6.6.1 The lacing shall be proportioned to resist a total transverse shear, V_t , at any point in the member, equal to at least 2.5 percent of the axial force in the member and shall be divided equally among all transverse-lacing systems in parallel planes.

14.6.6.2 For members carrying calculated bending stress due to eccentricity of loading, applied end moments and/or lateral loading, the lacing shall be proportioned to resist the actual shear due to bending, in addition to that specified in [14.6.6.1](#).

14.6.6.3 The slenderness ratio, KL/r , of the lacing bars shall not exceed 145. In bolted/riveted construction, the effective length of lacing bars for the determination of the design strength shall be taken as the length between the inner end fastener of the bars for single lacing, and as 0.7 of this length for double lacings effectively connected at intersections. In welded construction, the effective lengths shall be taken as 0.7 times the distance between the inner ends of welds connecting the single lacing bars to the members.

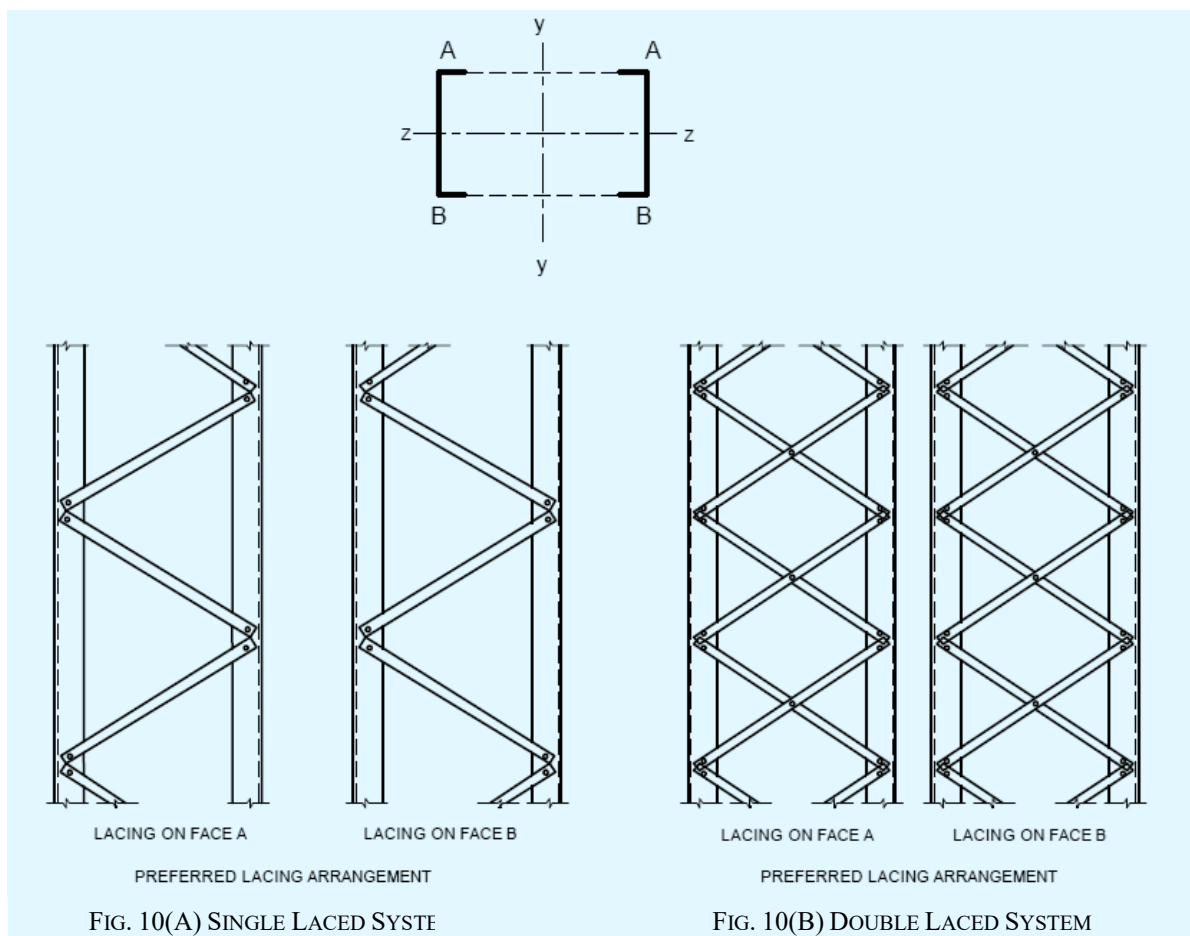
NOTE — The required section for lacing bars as compression/tension members shall be determined by using the appropriate design stresses, f_{cd} subject to the requirements in [14.6.3](#) to [14.6.6](#) and T_d in [13.1](#).

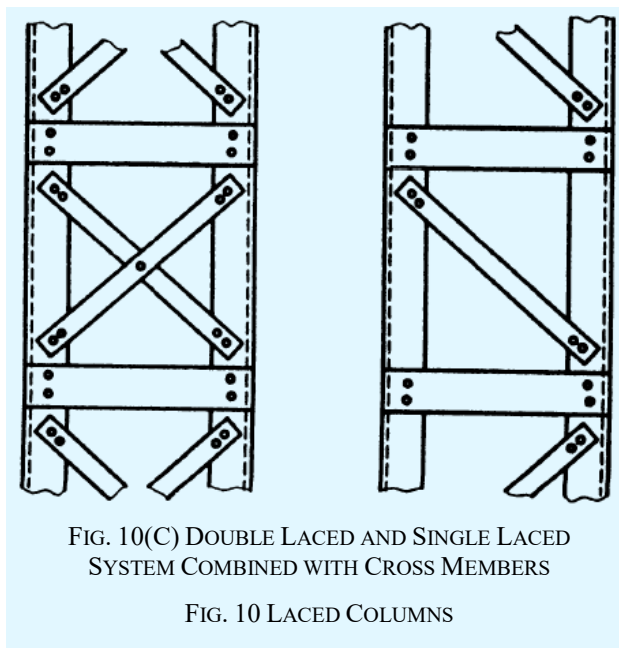
14.6.7 Attachment to Main Members

The bolting, riveting or welding of lacing bars to the main members shall be sufficient to transmit the force calculated in the bars. Where welded lacing bars overlap the main members, the amount of lap measured along either edge of the lacing bar shall be not less than four times the thickness of the bar or the thickness of the element of the members to which it is connected, whichever is less. The welding should be sufficient to transmit the load in the bar and shall, in any case, be provided along each side of the bar for the full length of lap.

14.6.8 End Tie Plates

Laced compression members shall be provided with tie plates as per [14.7](#) at the ends of lacing systems and at intersection with other members/stays and at points where the lacing systems are interrupted.

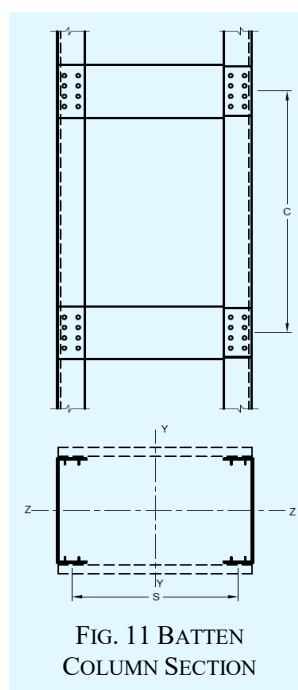




14.7 Battened Columns

14.7.1 General

14.7.1.1 Compression members composed of two main components battened should preferably have the individual members of the same cross section and symmetrically disposed about their major axis. Where practicable, the compression members should have a radius of gyration about the axis perpendicular to the plane of the batten not less than the radius of gyration about the axis parallel to the plane of the batten (see [Fig. 11](#)).



14.7.1.2 Battened compression members, not complying with the requirements specified in this Section or those subjected to eccentricity of loading, applied moments or lateral forces in the plane of the battens (see Fig. 11), shall be designed according to the exact theory of elastic stability or empirically, based on verification by tests.

NOTE — If the column section is subjected to eccentricity or other moments about an axis perpendicular to battens, the battens and the column section should be specially designed for such moments and shears.

14.7.1.3 The battens shall be placed opposite to each other at each end of the member and at points where the member is stayed in its length and as far as practicable, be spaced and proportioned uniformly throughout. The number of battens shall be such that the member is divided into not less than three bays within its actual length from centre to centre of end connections.

14.7.1.4 The effective slenderness ratio $(KL/r)_e$ of battened columns, shall be taken as 1.1 times the $(KL/r)_0$, the maximum actual slenderness ratio of the column, to account for shear deformation effects.

14.7.2 Design of Battens

14.7.2.1 Battens

Battens shall be designed to carry the bending moments and shear forces arising from transverse shear force V_t equal to 2.5 percent of the total axial force on the whole compression member, at any point in the length of the member, divided equally between parallel planes of battens. Battened member carrying calculated bending moment due to eccentricity of axial loading, calculated end moments or lateral loads parallel to the plane of battens, shall be designed to carry actual shear in addition to the above shear. The main members shall also be checked for the same shear force and bending moments as for the battens.

Battens shall be of plates, angles, channels, or I-sections and at their ends shall be riveted, bolted or welded to the main components so as to resist simultaneously a shear $V_b = V_t C/NS$ along the column axis and a moment $M = V_t C/2N$ at each connection,

where

- C = distance between centre-to-centre of battens, longitudinally;
- N = number of parallel planes of battens;
- S = minimum transverse distance between the centroid of the rivet/bolt group/welding connecting the batten to the main member; and
- V_t = transverse shear force as defined above.

14.7.2.2 Tie plates

Tie plates are members provided at the ends of battened and laced members, and shall be designed by the same method as battens. In no case shall a tie plate and its fastenings be incapable of carrying the forces for which the lacing or batten has been designed.

14.7.2.3 Size

When plates are used for battens, the end battens and those at points where the member is stayed in its length shall have an effective depth, longitudinally, not less than the perpendicular distance between the centroids of the main members. The intermediate battens shall have an effective depth of not less than three quarters of this distance, but in no case shall the effective depth of any batten be less than twice the width of one member, in the plane of the battens. The effective depth of a batten shall be taken as the longitudinal distance between outermost bolts, rivets or welds at the ends. The thickness of batten or the tie plates shall be not less than one fiftieth of the distance between the innermost connecting lines of rivets, bolts or welds, perpendicular to the main member.

14.7.2.4 The requirement of bolt size and thickness of batten specified above does not apply when angles, channels or I-sections are used for battens with their legs or flanges perpendicular to the main member. However, it should be ensured that the ends of the compression members are tied to achieve adequate rigidity.

14.7.3 Spacing of Battens

In battened compression members where the individual members are not specifically checked for shear stress and bending moments, the spacing of battens, centre-to-centre of its end fastenings, shall be such that the slenderness ratio (KL/r) of any component over that distance shall be not greater than 50 nor greater than 0.7 times the slenderness ratio of the member as a whole about its z-z axis (parallel to the battens).

14.7.4 Attachment to Main Members

14.7.4.1 Welded connections

Where tie or batten plates overlap the main members, the amount of lap shall be not less than four times the thickness of the plate. The length of weld connecting each edge of the batten plate to the member shall, in aggregate, be not less than half the depth of the batten plate. At least one-third of the weld shall be placed at each end of this edge. The length of weld and depth of batten plate shall be measured along the longitudinal axis of the main member.

In addition, the welding shall be returned along the other two edges of the plates transversely to the axis of the main member for a length not less than the minimum lap specified above.

14.8 Compression Members Composed of Two Components Back-to-Back

14.8.1 Compression members composed of two angles, channels, or tees back-to-back in contact or separated by a small distance, shall be connected together by riveting, bolting or welding so that the ratio of most unfavorable slenderness of each member between the intermediate connections is not greater than 40 or 0.6 times the most unfavorable ratio of slenderness of the strut as a whole, whichever is less [see [17](#)].

14.8.2 In no case shall the ends of the strut be connected together with less than two rivets or bolts or their equivalent in welding, and there shall be not less than two additional connections in between, spaced equidistant along the length of the strut. Where the members are separated back-to-back, the rivets or bolts through these connections shall pass through solid washers or packing in between. Where the legs of the connected angles or the connected tees are 125 mm wide or more, or where webs of channels are 150 mm wide or over, not less than two rivets or bolts shall be used in each connection, one on line of each gauge mark.

14.8.3 Where these connections are made by welding, solid packing shall be used to effect the jointing unless the members are sufficiently close together to permit direct welding, and the members shall be connected by welding along both pairs of edges of the main components.

14.8.4 The rivets, bolts or welds in these connections shall be sufficient to carry the shear force and moments, if any, specified for battened struts (see [14.7.3](#)), and in no case shall the rivets or bolts be less than 16 mm diameter for members up to and including 10 mm thick; 20 mm diameter for members up to and including 16 mm thick; and 22 mm diameter for members over 16 mm thick.

Compression members connected by such riveting, bolting or welding shall not be subjected to transverse loading in a plane perpendicular to the riveted, bolted or welded surfaces.

14.8.5 Where the components are in contact back-to-back, the spacing of the rivets, bolts or intermittent welds shall not exceed the maximum spacing for compression members as given under [17](#).

15 DESIGN OF MEMBERS SUBJECTED TO BENDING

15.1 General

Members subjected to predominant bending shall have adequate design strength to resist bending moment, shear force, and concentrated forces imposed upon and their combinations. Further, the members shall satisfy the deflection limitation presented under [12](#), as serviceability criteria. Member subjected to other forces in addition to bending or biaxial bending shall be designed in accordance with the provisions under [16](#).

15.1.1 Effective Span of Beams

The effective span of a beam shall be taken as the distance between the centres of the supports, except where the point of application of the reaction is taken as eccentric at the support, when it shall be permissible to take the effective span as the length between the assumed lines of the reactions.

15.2 Design Strength in Bending (Flexure)

The design bending strength of beam, adequately supported against lateral torsional buckling (laterally supported beam) is governed by the yield stress (see 15.2.1). When a beam is not adequately supported against lateral buckling (laterally un-supported beams) the design bending strength may be governed by lateral torsional buckling strength (see 15.2.2).

The factored design moment, M at any section, in a beam due to external actions, shall satisfy:

$$M \leq M_d$$

where

M_d = design bending strength of the section, calculated as given in 15.2.1.2.

15.2.1 Laterally Supported Beam

A beam may be assumed to be adequately supported at the supports, provided the compression flange has full lateral restraint and nominal torsional restraint at supports supplied by web cleats, partial depth end plates, fin plates or continuity with the adjacent span. Full lateral restraint to compression flange may be assumed to exist if the frictional or other positive restraint of a floor connection to the compression flange of the member is capable of resisting a lateral force not less than 2.5 percent of the maximum force in the compression flange of the member. This may be considered to be uniformly distributed along the flange, provided gravity loads constitute the dominant loading on the member and the floor construction is capable of resisting this lateral force.

The design bending strength of a section which are not susceptible to web buckling under shear before yielding (where $d/t_w \leq 67\epsilon$) shall be determined according to 15.2.1.2.

15.2.1.1 Section with webs susceptible to shear buckling before yielding

When the flanges are plastic, compact or semi-compact but the web is susceptible to shear buckling before yielding (see 15.4.2.1), the design bending strength shall be calculated using one of the following methods:

- the bending moment and axial force acting on the section may be assumed to be resisted by flanges only and the web is designed only to resist shear (see 15.4); and
- the bending moment and axial force acting on the section may be assumed to be resisted by the whole section. In such a case, the web shall be designed for combined shear and normal stresses using simple elastic theory in case of semi-compact webs and simple plastic theory in the case of compact and plastic webs.

15.2.1.2 When the factored design shear force does not exceed $0.6 V_d$, where V_d is the design shear strength of the cross section (see 15.4), the design bending strength, M_d shall be taken as:

$$M_d = \beta_b Z_p f_y / \gamma_{m0}$$

To avoid irreversible deformation under serviceability loads, M_d shall be less than $1.2 Z_e f_y / \gamma_{m0}$ in case of simply supported and cantilever beams and $1.5 Z_e f_y / \gamma_{m0}$ in all other cases. of;

where

- β_b = 1.0 for plastic and compact sections;
- β_b = Z_e / Z_p for semi-compact sections;

- f_y = yield stress of the material;
 γ_{m0} = partial safety factor (see [12.4.1](#)); and
 Z_p, Z_e = plastic and elastic section moduli of the cross section, respectively

15.2.1.3 When the design shear force (factored), V exceeds $0.6V_d$, where V_d is the design shear strength of the cross section (see [15.4](#)) the design bending strength, M_d shall be taken as:

$$M_d = M_{dv}$$

where

M_{dv} = design bending strength under high shear as defined in [16.2](#).

15.2.1.4 Holes in the tension zone

- a) The effect of holes in the tension flange, on the design bending strength need not be considered if:

$$(A_{nf}/A_{gf}) \geq (f_y/f_u) (\gamma_{m1}/\gamma_{m0}) / 0.9$$

where

- A_{nf}/A_{gf} = ratio of net to gross area of the flange in tension;
 f_y/f_u = ratio of yield and ultimate stress of the material; and
 γ_{m1}/γ_{m0} = ratio of partial safety factors against ultimate to yield stress (see [12.4.1](#)).

When the A_{nf}/A_{gf} does not satisfy the above requirement, the reduced effective flange area, A_{ef} satisfying the above equation may be taken as the effective flange area in tension, instead of A_{gf} .

- b) The effect of holes in the tension region of the web on the design flexural strength need not be considered, if the limit given in [15.2.1.4\(a\)](#) above is satisfied for the complete tension zone of the cross section, comprising the tension flange and tension region of the web; and
c) Fastener holes in the compression zone of the cross section need not be considered in design bending strength calculation, except for oversize and slotted holes or holes without any fastener.

15.2.1.5 Shear lag effects

The shear lag effects in flanges may be disregarded provided:

- a) For outstand elements (supported along one edge), $b_o \leq L_o / 20$; and
b) For internal elements (supported along two edges), $b_i \leq L_o / 10$.

where

- b_o = width of the flange with outstand;
 b_i = width of the flange as an internal element; and
 L_o = length between points of zero moment (inflection) in the span.

Where these limits are exceeded, the effective width of flange for design strength may be calculated using specialist literature, or conservatively taken as the value satisfying the limit given above.

15.2.2 Laterally Unsupported Beams

Resistance to lateral torsional buckling need not be checked separately (member may be treated as laterally supported, see [15.2.1](#)) in the following cases:

- a) bending is about the minor axis of the section;

- b) section is hollow (rectangular/ tubular) or solid bars; and
- c) in case of major axis bending, λ_{LT} (as defined herein) is less than 0.4.

The design bending strength of laterally unsupported beam as governed by lateral torsional buckling is given by:

$$M_d = \beta_b Z_p f_{bd}$$

where

$$\begin{aligned} \beta_b &= 1.0 \text{ for plastic and compact sections;} \\ &= Z_e/Z_p \text{ for semi-compact sections (Class 3);} \\ &= Z_{eff}/Z_p \text{ for slender sections (Class 4);} \\ f_{bd} &= \chi_{LT} f_y / \gamma_{m0}; \\ &= \text{design bending compressive stress;} \\ f_m &= \text{the moment gradient effect parameter given in the } \text{Table 14}; \\ \chi_{LT} &= \text{bending stress reduction factor to account for lateral torsional buckling, in doubly} \\ &\quad \text{symmetrical sections with lateral support at the ends given by;} \\ &= \frac{f_m}{\{\varphi_{LT} + [\varphi_{LT}^2 - f_m \lambda_{LT}^2]^{0.5}\}} \leq 1.0 ; \\ Z_p, Z_e, Z_{eff} &= \text{plastic section modulus and elastic section modulus (Class 3) and effective section} \\ &\quad \text{modulus (class 4) with respect to extreme compression fibre;} \\ \alpha_{LT} &= \text{the imperfection parameter is given in } \text{Table 13}; \text{ and} \\ \varphi_{LT} &= 0.5 \left[1 + f_m \left(\left(\frac{\lambda_{LT}}{\lambda_y} \right)^2 \alpha_{LT} (\lambda_y - 0.2) + \lambda_{LT}^2 \right) \right]; \end{aligned}$$

The non-dimensional slenderness ratio, λ_{LT} , is given by:

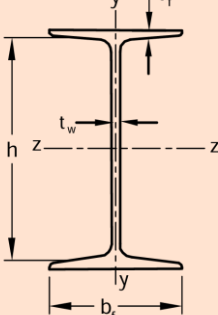
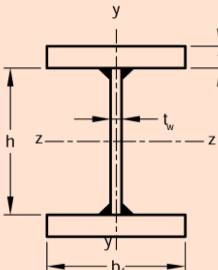
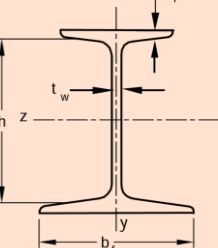
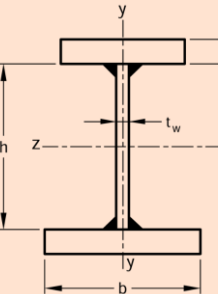
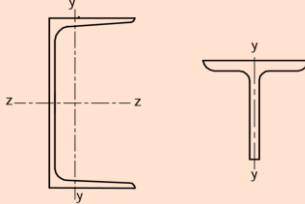
$$\lambda_{LT} = \sqrt{\beta_b Z_p f_y / M_{crLT}} \leq \sqrt{1.2 Z_e f_y / M_{crLT}} = \sqrt{f_y / f_{crLT}} \quad \lambda_y = \sqrt{f_y / f_{cry}}$$

where

$$\begin{aligned} f_{cry} &= \text{the non-dimensional weak axis slenderness ratio for laterally supported length equal to distance} \\ &\quad \text{between lateral supports to the compression flange;} \\ f_{crLT} &= \text{extreme fibre bending compressive stress corresponding to elastic lateral torsional buckling} \\ &\quad \text{moment (see } \text{15.2.2.1}); \text{ and} \\ M_{crLT} &= \text{elastic lateral torsional buckling moment calculated in accordance with } \text{15.2.2.1}. \end{aligned}$$

Table 13 Imperfection Parameter, α_{LT} for Lateral Torsional Buckling of Beams

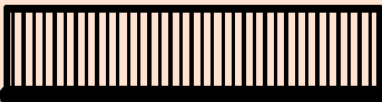
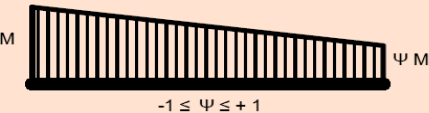
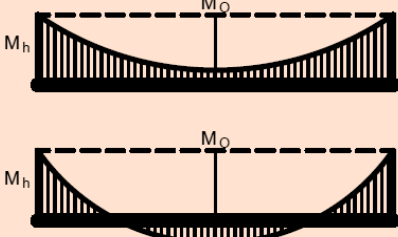
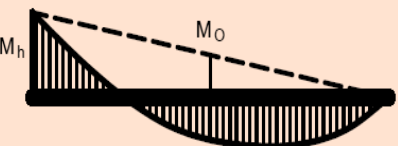
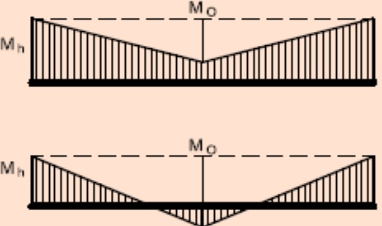
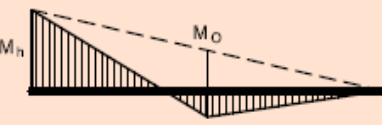
(Clause 15.2.2)

Cross Section	Limits	α_{LT}
Rolled I or H Sections 	$h/b > 1.2$	$t_f \leq 40 \text{ mm}$ $0.12 \sqrt{\frac{Z_{ez}}{Z_{ey}}} \leq 0.34$
		$t_f > 40 \text{ mm}$ $0.16 \sqrt{\frac{Z_{ez}}{Z_{ey}}} \leq 0.49$
	$h/b \leq 1.2$	— $0.16 \sqrt{\frac{Z_{ez}}{Z_{ey}}} \leq 0.49$
Welded I Sections 	$t_f \leq 40 \text{ mm}$	$0.21 \sqrt{\frac{Z_{ez}}{Z_{ey}}} \leq 0.64$
	$t_f > 40 \text{ mm}$	$0.25 \sqrt{\frac{Z_{ez}}{Z_{ey}}} \leq 0.76$
Rolled unequal I or H Sections 	$h/b_{\min} \leq 2.0$	0.21
	$h/b_{\min} > 2.0$	0.34
Welded unequal I section 	$h/b_{\min} \leq 2.0$	0.64
	$h/b_{\min} > 2.0$	0.76
Other rolled Sections 	-----	0.76

NOTE — The selection of lateral torsional buckling imperfection factor is based on $b_{\min}$ irrespective of it being tension or compression flange.

Table 14 Moment Gradient Effect Parameter

(Clauses 15.2.2 and 15.2.2.2)

Load Case	Factor f_M (Clause 15.2.2)	Factor k_c (Clause 15.2)
(1)	(2)	(3)
 M = uniform	1.0	1.0
 $-1 \leq \psi \leq +1$	$1.25 - 0.1\psi - 0.15\psi^2$	$\frac{1}{1.33 - 0.33\psi}$
	1.05	0.94
	For $0 \leq \frac{M_0}{M_h} < 2.0$: $1.0 + 1.35 \frac{M_0}{M_h} - 0.33 \left[\frac{M_0}{M_h} \right]^3$ For $\frac{M_0}{M_h} \geq 2.0$: 1.05	$\frac{M_0}{M_h} < 1.0$: 1.0 $\frac{M_0}{M_h} \geq 1.0$: 0.9
	For $0 \leq \frac{M_0}{M_h} < 1.47$: $1.25 + 0.5 \left[\frac{M_0}{M_h} \right]^2 - 0.275 \left[\frac{M_0}{M_h} \right]^4$ For $\frac{M_0}{M_h} \geq 1.47$: 1.05	$\frac{M_0}{M_h} < 0.5$: 0.75 $\frac{M_0}{M_h} \geq 0.5$: 0.91
	1.10	0.86
	For $0 \leq \frac{M_0}{M_h} < 2.0$: $1.0 + 1.25 \frac{M_0}{M_h} - 0.30 \left[\frac{M_0}{M_h} \right]^3$ For $\frac{M_0}{M_h} \geq 2.0$: 1.10	$\frac{M_0}{M_h} < 1.0$: 1.0 $\frac{M_0}{M_h} \geq 1.0$: 0.77
	For $0 \leq \frac{M_0}{M_h} < 1.5$: $1.25 + 0.325 \left[\frac{M_0}{M_h} \right]^2 - 0.175 \left[\frac{M_0}{M_h} \right]^4$ For $\frac{M_0}{M_h} \geq 1.50$: 1.10	$\frac{M_0}{M_h} < 0.5$: 0.75 $\frac{M_0}{M_h} \geq 0.5$: 0.82

Where

M_0 is the bending moment at midspan, and may be determined by subtracting the average of the two hogging end moments, from the midspan sagging moment for simply supported boundary condition.

For $\frac{M_0}{M_h} < 0.0$, f_m and k_c may be conservatively taken as 1.0.

15.2.2.1 Elastic lateral torsional buckling moment

In case of simply supported, prismatic members with doubly symmetric cross section subjected to uniform moment, the elastic lateral torsional buckling moment, M_{crLT} , can be determined from:

$$M_{crLT} = \sqrt{\left\{ \left(\frac{\pi^2 E I_y}{(L_{LT})^2} \right) \left(G I_t + \frac{\pi^2 E I_w}{(L_{LT})^2} \right) \right\}}$$

where

L_{LT} = effective lateral unbraced length for lateral torsional buckling (see [15.3](#));

I_t = torsional constant = $\sum b_i t_i^3 / 3$ for open section;

I_w = warping constant; and

I_y = moment of inertia about the weaker, (y), axis.

M_{cr} for different beam sections, considering loading, support condition, and asymmetric section, shall be more accurately calculated using the method given in [Annex E](#). The above equations for calculating M_{cr} may be conservatively used for channel sections.

15.2.2.2 Simplified method for lateral torsional buckling

A simplified method based on effective compression flange lateral buckling as given below may be used for monosymmetric and doubly symmetric I and H sections with and without lateral restraints, to determine the equivalent compression flange lateral buckling stress reduction factor, χ_{LTc} , that is substituted in equation given in [15.2.2](#) for calculating the design bending strength, M_d ,

The stress reduction factor for lateral buckling of equivalent compression flange, χ_{LTc} , is evaluated using equations given in [14.1.2.1](#) for compression members, by treating the equivalent compression flange as a strut and substituting equivalent non-dimensional slenderness, λ_{LTc} , in the place of λ .

where

$$\lambda_{LTc} = k_c \beta_c \lambda_{c,z}$$

k_c is a slenderness correction factor for moment gradient between restraints, as given in [Table 14](#).

$$\beta_c = \sqrt{\frac{0.06 \frac{h}{t_{f,max}}}{\lambda_{c,z} + \frac{t_{f,max}}{t_{f,min}}}}$$

$t_{f,max}$ is the maximum thickness of the top and bottom flange of the section

$t_{f,min}$ is the minimum thickness of the top and bottom flange of the section

$$\lambda_{c,z} = \sqrt{\frac{f_y}{f_{cr,cz}}}$$

where

A_c = is the area of equivalent compression flange, given by;

A_c = $A_{cf} + A_w/2$ for load applied at compression flange;

A_c = $A_{cf} + A_w/6$ for load applied at shear centre of the beam section;

A_c = A_{cf} for load applied at tension flange;

- A_{cf} = is the area of compression flange ($= b_c t_{fc}$);
- A_w = is the area of the web ($= h t_w$); and
- $f_{cr,cz}$ = is the elastic critical axial stress of the equivalent compression flange of area of the beam, A_c , as given below, buckling about the centre line axis of the web as a strut.

15.3 Effective Length for Lateral Torsional Buckling

15.3.1 For simply supported beams and girders of span length, L , where no lateral restraint to the compression flanges is provided, but where each end of the beam is restrained against torsion, the effective length L_{LT} of the lateral buckling to be used in [15.2.2.1](#) shall be taken as in [Table 15](#).

In simply supported beams with intermediate lateral restraints against lateral torsional buckling, the effective length for lateral torsional buckling, L_{LT} to be used in [15.2.2.1](#), may be conservatively taken as the length of the relevant segment in between the lateral restraints, or can be calculated using other known methods or rigorous analysis. In the case of intermediate partial lateral restraints, the effective length, L_{LT} shall be taken as equal to 1.2 times the length of the relevant segment in between the partial lateral restraints.

Restraint against torsional rotation at supports in these beams may be provided by:

- web or flange cleats; or
- bearing stiffeners acting in conjunction with the bearing of the beam; or
- lateral end frames or external supports providing lateral restraint to the compression flanges at the ends; or
- their being built into walls.

Table 15 Effective Length for Simply Supported Beams, L_{LT}

(Clause [15.3.1](#))

SI No.	Conditions of Restraint at Supports		Loading Condition	
	Torsional Restraint	Warping Restraint	Normal	Destabilizing
(1)	(2)	(3)	(4)	(5)
i)	Fully restrained	Both flanges fully restrained	$0.70 L$	$0.85 L$
ii)	Fully restrained	Only compression flange fully restrained	$0.75 L$	$0.90 L$
iii)	Fully restrained	Both flanges partially restrained	$0.80 L$	$0.95 L$
iv)	Fully restrained	Only compression flange partially restrained	$0.85 L$	$1.00 L$
v)	Fully restrained	No restraint in both flanges	$1.00 L$	$1.20 L$
vi)	Partially restrained by bottom flange support connection	No restraint in both flanges	$1.0 L + 2D$	$1.2 L + 2D$
vii)	Partially restrained by bottom flange bearing support	No restraint in both flanges	$1.2 L + 2D$	$1.4 L + 2D$
NOTES				
1 Torsional restraint prevents rotation about the longitudinal axis.				
2 Warping restraint prevents rotation of the flange in its plane.				
3 D is the overall depth of the beam.				
4 In case of continuous beams, L shall be taken as the distance between points of inflection, and the restraint conditions at the point of inflections should be considered.				

15.3.2 For beams, which are provided with members giving effective lateral restraint to the compression flange at intervals along the span, in addition to the end torsional restraint required in [15.3.1](#), the effective length for lateral torsional buckling shall be taken as the distance, centre-to-centre of the restraint members in the critical segment under normal loading condition and 1.2 times this distance, where the load is acting on the beam above the shear centre and towards the shear centre, so as to have greater destabilizing effect during lateral torsional buckling deformation.

15.3.3 For cantilever beams of projecting length L , the effective length L_{LT} to be used in [15.2.2.1](#) shall be taken as in [Table 16](#) for different support conditions.

15.3.4 Where a member is provided intermediate lateral supports to improve the lateral buckling strength, these restraints should have sufficient strength and stiffness to prevent lateral movement of the compression flange at that point, relative to the end supports. The intermediate lateral restraints should be either connected to an appropriate bracing system capable of transferring the restraint force to the effective lateral support at the ends of the member, or should be connected to an independent robust part of the structure capable of transferring the restraint force. Two or more parallel member requiring such lateral restraint shall not be simply connected together assuming mutual dependence for the lateral restraint.

The intermediate lateral restraints should be connected to the member as close to the compression flange as practicable. Such restraints should be closer to the shear centre of the compression flange than to the shear centre of the section. However, if torsional restraint preventing relative rotation between the two flanges is provided, the intermediate lateral restraint may be connected at any appropriate level. The point of contraflexure in beams with double curvature should be additionally effectively restrained against torsion in order to be considered it as a brace location.

For beams which are provided with members giving effective lateral restraint at intervals along the span, the effective lateral restraint shall be capable of resisting a force of 2.5 percent of the maximum force in the compression flange taken as divided equally between the points at which the restraint members are provided. Further, each restraint point should be capable of resisting 1 percent of the maximum force in the compression flange.

15.3.4.1 In a series of such beams, with solid webs, which are connected together by the same system of restraint members, the sum of the restraining forces required shall be taken as 2.5 percent of the maximum flange force in one beam only.

15.3.4.2 In the case of a series of latticed beams, girders or roof trusses which are connected together by the same system of restraint members, the sum of the restraining forces required shall be taken as 2.5 percent of the maximum force in the compression flange plus 1.25 percent of this force for every member of the series other than the first, up to a maximum total of 7.5 percent.

15.3.5 Purlins adequately restrained by sheeting need not be normally checked for the restraining forces required by rafters, roof trusses or portal frames that carry predominately roof loads provided there is bracing of adequate stiffness in the plane of rafters or roof sheeting which is capable of acting as a stressed skin diaphragm.

15.3.6 In case of beams with double curvature bending, adequate direct lateral support to the compression flange in the hogging moment region may be provided as given above for simply supported beam. The effect of support to the tension (top) flange in the hogging moment region on lateral restraint to the compression flange may be considered as per specialist literature.

15.3.7 In case of cantilever beams with tip loading, at the free end the tension flange also shall be effectively braced against lateral movement.

15.4 Shear

The design of plated webs with and without stiffeners are treated in this section. The design recommendations of Corrugated plate girders are presented in [Annex G](#).

The factored design shear force, V , in a beam due to external actions shall satisfy:

$$V \leq V_d$$

where

$$\begin{aligned} V_d &= \text{design strength} \\ &= V_n / \gamma_{m0} \end{aligned}$$

where

$$\gamma_{m0} = \text{partial safety factor against shear failure (see [12.4.1](#))}$$

The nominal shear strength of a cross section, V_n , may be governed by plastic shear resistance (see [15.4.1](#)) or strength of the web as governed by shear buckling (see [15.4.2](#)).

15.4.1 The nominal plastic shear resistance under pure shear is given by:

$$V_n = V_p$$

where

$$V_p = \frac{A_v f_{yw}}{\sqrt{3}}$$

A_v = shear area; and

f_{yw} = yield strength of the web.

15.4.1.1 The shear area may be calculated as given below:

a) I-and channel sections:

1) *Major Axis Bending:*

Hot-rolled : $h t_w$

Welded : $d t_w$

2) *Minor Axis Bending:*

Hot-rolled or welded : $2b t_f$

b) Rectangular hollow sections of uniform thickness:

Loaded parallel to depth (h) : $A h / (b+h)$

Loaded parallel to width (b) : $A b / (b+h)$

c) Circular hollow tubes of uniform thickness : $2A / \pi$

d) Plates and solid bars : A

where

A = cross section area;

b = overall breadth of tubular section, breadth of I-section flanges;

d = clear depth of the web between flanges;

h = overall depth of the section;

t_f = thickness of the flange; and

t_w = thickness of the web.

NOTE — Fastener holes need not be accounted for in plastic design shear strength calculation provided that:

$$A_{vn} \geq (f_y / f_u) (\gamma_{m1} / \gamma_{m0}) A_v / 0.9$$

If A_{vn} does not satisfy the above condition, the effective shear area may be taken as that satisfying the above limit. Block shear failure criteria may be verified at the end connections. Provisions given under [16](#) may be referred to for design strength under combined high shear and bending.

15.4.2 Resistance to Shear Buckling

15.4.2.1 Resistance to shear buckling shall be verified as specified, when

$$d/t_w > 67 \varepsilon_w \text{ for a web without stiffeners, and}$$

$$> 67 \varepsilon_w \sqrt{\frac{K_v}{5.35}} \text{ for a web with stiffeners}$$

where

$$K_v = \text{shear buckling coefficient (see [15.4.2.2](#)); and}$$

$$\varepsilon_w = \sqrt{250 / f_{yw}}$$

15.4.2.2 Shear buckling design methods

The nominal shear strength, V_n , of webs with or without intermediate stiffeners as governed by buckling may be evaluated using one of the following methods:

- a) *Simple post-critical method* — The simple post critical method, based on the shear buckling strength can be used for webs of I-section girders, with or without intermediate transverse stiffener, provided that the web has transverse stiffeners at the supports. The nominal shear strength is given by:

$$V_n = V_{cr}$$

where

$$V_{cr} = \text{shear force corresponding to web buckling; and}$$

$$= A_v \tau_b$$

where

τ_b = shear stress corresponding to web buckling, determined as follows:

- 1) when $\lambda_w \leq 0.8$, $\tau_b = f_{yw} / \sqrt{3}$
- 2) when $0.8 < \lambda_w < 1.2$, $\tau_b = [1 - 0.8(\lambda_w - 0.8)](f_{yw} / \sqrt{3})$
- 3) when $\lambda_w \geq 1.2$, $\tau_b = f_{yw} / (\sqrt{3} \lambda_w^2)$

where

λ_w = Non-dimensional web slenderness ratio for shear buckling stress, given by:

$$\lambda_w = \sqrt{f_{yw} / (\sqrt{3} \tau_{cr,e})}$$

$$\tau_{cr,e} = \text{the elastic critical shear stress of the web} = \frac{K_v \pi^2 E}{12 (1 - \mu^2) [d / t_w]^2}$$

where

$$\mu = \text{Poisson's ratio; and}$$

$$\begin{aligned}
 K_v &= 5.35 \text{ when transverse stiffeners are provided only at supports} \\
 &= 4.0 + 5.35 / (c/d)^2 \text{ for } c/d < 1.0 \\
 &= 5.35 + 4.0 / (c/d)^2 \text{ for } c/d \geq 1.0
 \end{aligned}$$

where c, d are the spacing of transverse stiffeners and depth of the web, respectively.

- b) *Tension field method* — The tension field method, based on the post-shear buckling strength, may be used for webs with intermediate transverse stiffeners, in addition to the transverse stiffeners at supports, provided the panels adjacent to the panel under tension field action, or the end posts provide anchorage for the tension fields and if $c/d \geq 1.0$, where c, d are the spacing of transverse stiffeners and depth of the web, respectively.

In the tension field method, the nominal shear resistance, V_n , is given by:

$$V_n = V_{tf}$$

where

$$V_{tf} = [A_v \tau_b + 0.9 w_{tf} t_w f_v \sin \phi] \leq V_p$$

where

τ_b = shear stress corresponding to buckling of web [15.4.2.2\(a\)](#);

f_v = yield strength of the tension field obtained from $\left[f_{yw}^2 - 3 \tau_b^2 + \psi^2 \right]^{0.5} - \psi$;

ψ = $1.5 \tau_b \sin 2\phi$;

ϕ = inclination of the tension field nearly $= \tan^{-1} \left(\frac{d}{c} \right) / 1.5$;

w_{tf} = width of the tension field, given by:

$$= d \cos \phi - (c - s_c - s_t) \sin \phi;$$

f_{yw} = yield stress of the web;

d = depth of the web;

c = spacing of transverse stiffeners in the web; and

s_c, s_t = anchorage lengths of tension field along the compression and tension flange respectively, obtained from:

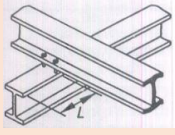
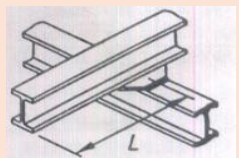
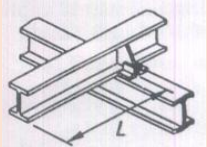
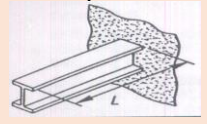
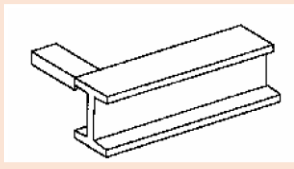
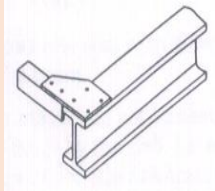
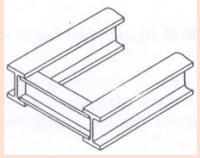
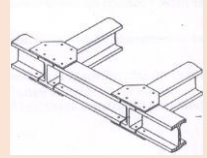
$$s = \frac{2}{\sin \phi} \left[\frac{M_{fr}}{f_{yw} t_w} \right]^{0.5} \leq c$$

M_{fr} = reduced plastic moment capacity of the respective flange plate (disregarding any edge stiffener) after accounting for the axial force, N_f in the flange, due to overall bending and any external axial force in the cross section, and is calculated as :

$$M_{fr} = 0.25 b_f t_f^2 f_{yf} \left[1 - \left\{ N_f / (b_f t_f f_{yf} / \gamma_{m0}) \right\}^2 \right]$$

b_f, t_f = width and thickness of the relevant flange respectively; and

f_{yf} = yield stress of the flange.

Table 16 Effective Length, L_{LT} for Cantilever of Length, L (Clause 15.3.3)			
Restraint Condition		Loading Conditions	
At Support	At tip	Normal	Destabilizing
(2)	(3)	(4)	(5)
a) Continuous, with lateral restraint to top flange 	1) Free 2) Lateral restraint to top flange 3) Torsional restraint 4) Lateral and torsional restraint	3.0L 2.7L 2.4L 2.1L	7.5L 7.5L 4.5L 3.6L
b) Continuous, with partial torsional restraint 	1) Free 2) Lateral restraint to top flange 3) Torsional restraint 4) Lateral and torsional restraint	2.0L 1.8L 1.6L 1.4L	5.0L 5.0L 3.0L 2.4L
c) Continuous, with lateral and torsional restraint 	1) Free 2) Lateral restraint to top flange 3) Torsional restraint 4) Lateral and torsional restraint	1.0L 0.9L 0.8L 0.7L	2.5L 2.5L 1.5L 1.2L
d) Restrained laterally, torsionally and against rotation on plan 	1) Free 2) Lateral restraint to top flange 3) Torsional restraint 4) Lateral and torsional restraint	0.8L 0.7L 0.6L 0.5L	1.4L 1.4L 0.6L 0.5L
Tip Restraint Conditions			
a) Free 	b) Lateral restraint to top flange 	c) Torsional restraint 	d) Lateral and torsional restraint 

15.5 Stiffened Web Panels

15.5.1 End Panels Design (see Fig.12)

The girders in which the interior panel (panel A) is designed using tension field action, the end panels shall be designed using only simple post critical method, according to [15.4.2.2\(a\)](#).

Additionally, the end panel along with the stiffeners should be checked as a beam spanning between the flanges to resist a shear force, R_{tf} , and a moment, M_{tf} , due to tension field forces as given in [15.5.3](#). Further, end stiffener should be capable of resisting the reaction plus a compressive force due to the moment, equal to M_{tf} [see [Fig. \(12\)](#)].

15.5.2 End Panels Designed Using Tension Field Action (see Fig. 13 and Fig. 14)

The end panel (Panel A) shall be designed according to [15.4.2.2\(b\)](#).

Additionally, it should be provided with an end post consisting of a single or double stiffener, (see [Fig. 13](#) and [Fig. 14](#)), satisfying the following.

- a) *Single stiffener* (see [Fig. 13](#)) — The top of the end post should be rigidly connected to the flange using full strength welds.

The end post should be capable of resisting the reaction plus a moment from the anchor forces equal to $2/3 M_{tf}$ due to tension field forces, where M_{tf} is obtained from [15.5.3](#). The width and thickness of the end post are not to exceed the width and thickness of the flange.

- b) *Double stiffener* (see [Fig. 14](#)) — The end post should be checked as a beam spanning between the flanges of the girder and capable of resisting a shear force R_{tf} and a moment, M_{tf} due to the tension field forces as given in [15.5.3](#).

15.5.3 Anchor Forces

The resultant longitudinal shear, R_{tf} , and a moment M_{tf} from the anchor of tension field forces are evaluated as given below:

$$R_{tf} = \frac{H_q}{2} \text{ and } M_{tf} = \frac{H_q d}{10}$$

where

$$H_q = 1.25V_p \left(1 - \frac{V_{cr}}{V_p} \right)^{1/2}$$

$$V_p = \frac{dtf_y}{\sqrt{3}}$$

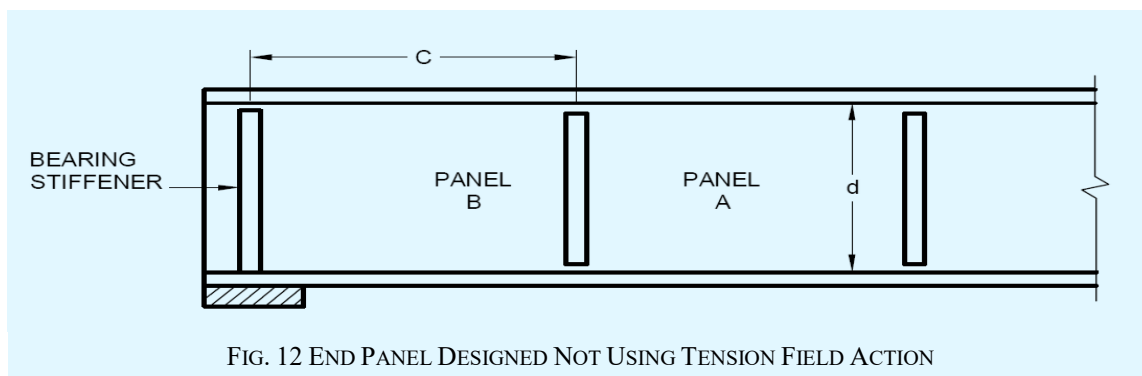
d = web depth,

If the actual factored shear force, V in the panel designed using tension field approach is less than the shear strength, V_{tf} as given in [15.4.2.2\(b\)](#), then the values of H_q may be reduced by the ratio $\frac{V - V_{cr}}{V_{tf} - V_{cr}}$

where

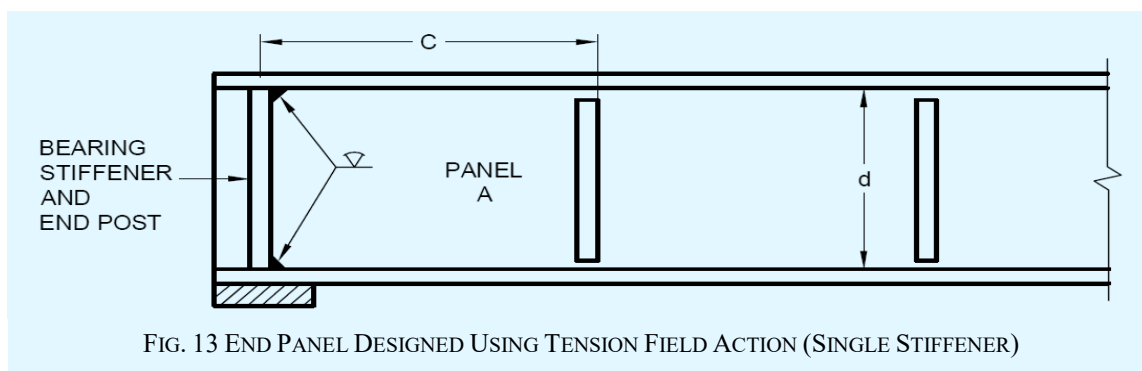
V_{cr} = critical shear strength for the panel designed utilizing tension field action as given in [15.4.2.2\(a\)](#); and

V_{tf} = basic shear strength for the panel utilizing tension field action as given in [15.4.2.2\(b\)](#).



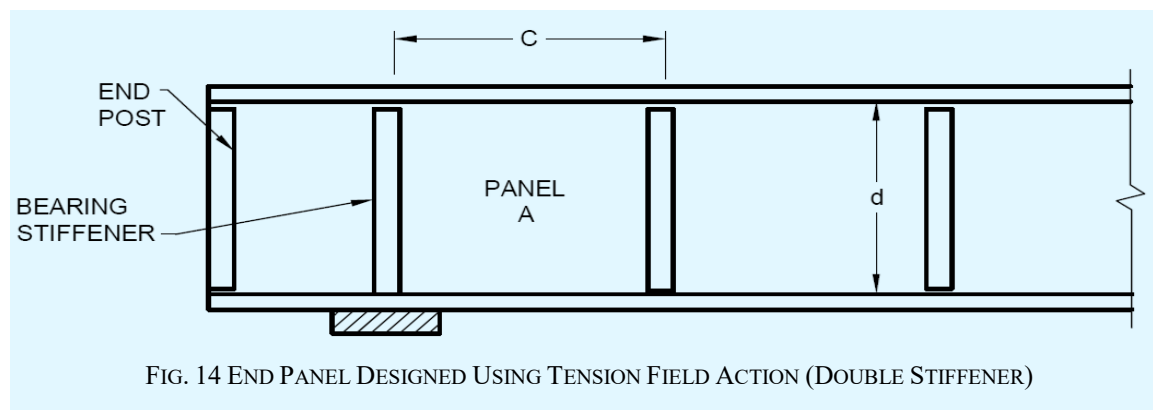
NOTES

- 1 Panel A is designed utilizing tension field action as given in [15.4.2.2\(b\)](#).
- 2 Panel B is designed using simple post critical method as given in [15.4.2.2\(a\)](#).
- 3 Bearing stiffener is designed for the compressive force due to bearing plus compressive force due to the moment M_{tf} as given in [15.5.3](#).



NOTES

- 1 Panel A is designed utilizing tension field action as given in [15.4.2.2\(b\)](#).
- 2 Bearing stiffener and end post is designed for combination of compressive loads due to bearing and a moment equal to $2/3 M_{tf}$ as given in [15.5.3](#).



NOTES

- 1 Panel A is designed utilizing tension field action as given in [15.4.2.2\(b\)](#).
- 2 Bearing stiffener is designed for compressive force due to bearing as given in [15.4.2.2\(a\)](#).
- 3 End post is designed for horizontal shear R_{tf} and moment M_{tf} as given in [15.5.3](#).

15.5.4 Panels with Openings — Panels with opening of dimension greater than 10 percent of the minimum panel dimension should be designed without using tension field action as given in [15.4.2.2\(b\)](#). The adjacent panels should be designed as an end panel as given in [15.5.1](#) or [15.5.2](#), as appropriate.

15.6 Design of Beams and Plate Girders with Solid Webs

15.6.1 Minimum Web Thickness

The thickness of the web in a section shall satisfy the requirements given in [15.6.1.1](#) and [15.6.1.2](#).

15.6.1.1 Serviceability requirement

a) when transverse stiffeners are not provided:

- 1) $\frac{d}{t_w} \leq 200\epsilon_w$ (web connected to flanges along both longitudinal edges); and
- 2) $\frac{d}{t_w} \leq 90\epsilon_w$ (web connected to flanges along one longitudinal edge only).

b) when only transverse stiffeners are provided (in webs connected to flanges along both longitudinal edges):

1) when $3d \geq c \geq d$

$$\frac{d}{t_w} \leq 200\epsilon_w;$$

2) when $0.74 d \leq c < d$

$$\frac{c}{t_w} \leq 200\epsilon_w;$$

3) when $c < 0.74d$

$$\frac{d}{t_w} \leq 270\epsilon_w; \text{ and}$$

4) when $c > 3d$, the web shall be considered as unstiffened.

c) when transverse stiffeners and longitudinal stiffeners at one level only are provided (distance from the compression flange equal to 2/5 of the distance from the compression flange angle, plate or tongue plate to the neutral axis) according to [15.7.13\(a\)](#):

1) when $2.4d \geq c \geq d$

$$\frac{d}{t_w} \leq 250\epsilon_w$$

2) when $0.74 d \leq c \leq d$

$$\frac{c}{t_w} \leq 250\epsilon_w$$

3) when $c < 0.74 d$

$$\frac{d}{t_w} \leq 340\epsilon_w$$

- d) when a second longitudinal stiffener (located at neutral axis is provided) $\frac{d}{t_w} \leq 400\epsilon_w$

where

- c = spacing of transverse stiffener (see [Fig. 12](#) and [Fig. 13](#));
 d = depth of the web;
 f_{yw} = yield stress of the web;
 t_w = thickness of the web; and
 ϵ_w = yield stress ratio of web = $\sqrt{250/f_{yw}}$.

15.6.1.2 Compression flange buckling requirement

In order to avoid buckling of the compression flange into the web, the web thickness shall satisfy the following:

- a) When transverse stiffeners are not provided;

$$\frac{d}{t_w} \leq 345\epsilon_f^2$$

- b) When transverse stiffeners are provided, and

- 1) when $c \geq 1.5d$

$$\frac{d}{t_w} \leq 345\epsilon_f^2$$

- 2) when $c < 1.5d$

$$\frac{d}{t_w} \leq 345\epsilon_f$$

where

- c = spacing of transverse stiffener (see [Fig. 12](#) and [Fig. 13](#));
 d = depth of the web;
 f_{yf} = yield stress of compression flange;
 t_w = thickness of the web; and
 ϵ_f = yield stress ratio of flange = $\sqrt{250/f_{yf}}$.

15.6.2 Sectional Properties

The effective sectional area of compression flanges shall be the gross area with deductions for excessive width of plates as specified for compression members [see [14](#)] and for open holes occurring in a plane perpendicular to the direction of stress at the section being considered (see [15.2.1.4](#)).

The effective sectional area of tension flanges shall be the gross sectional area with deductions for holes as specified in [15.2.1.4](#).

The effective sectional area for parts in shear shall be taken as specified in [15.4.1.1](#).

15.6.3 Flanges

15.6.3.1 In riveted or bolted construction, flange angles shall form as large a part of the area of the flange as practicable (preferably not less than one-third) and the number of flange plates shall be kept to a minimum.

In exposed situations, where flange angles are used, at least one plate of the top flange shall extend over the full length of the girder, unless the top edge of the web is machined flush with the flange angles. Where two or more flange plates are used, tacking rivets shall be provided, if necessary to comply with the requirements of [17](#).

Each flange plate shall extend beyond its theoretical cut-off point, and the extension shall contain sufficient rivets, bolts or welds to develop in the plate, the load calculated for the bending moment on the girder section (taken to include the curtailed plate) at the theoretical cut-off point.

The outstand of flange plates, that is the projection beyond the outer line of connections to flange angles, channel or joist flanges or in the case of welded constructions their projection beyond the face of the web or tongue plate, shall not exceed the values given in [10.8.2](#) (see [Table 2](#)).

In the case of box girders, the thickness of any plate, or the aggregate thickness of two or more plates, when these plates are tacked together to form the flange, shall satisfy the requirements given in [10.8.2](#) (see [Table 2](#)).

15.6.3.2 Flange splices

Flange splices should preferably, not be located at points of maximum stress. Where splice plates are used, their area shall be not less than 5 percent in excess of the area of the flange element spliced; their centre of gravity shall coincide, as nearly as possible, with that of the element spliced. There shall be enough bolts, rivets or welds on each side of the splice to develop the load in the element spliced plus 5 percent but in no case should the strength developed be less than 50 percent of the effective strength of the material spliced. In welded construction, flange plates shall be joined by complete penetration butt welds, wherever possible. These butt welds shall develop the full strength of the plates.

15.6.3.3 Connection of flanges to web

The flanges of plate girders shall be connected to the web by sufficient rivets, bolts or welds to transmit the maximum horizontal shear force resulting from the bending moment gradient in the girder, combined with any vertical loads which are directly applied to the flange. If the web is designed using tension field method as given in [15.4.2.2 \(b\)](#), the weld should be able to transfer the tension field stress, f_{yw} acting on the web.

15.6.3.4 Bolted/Riveted construction

For girders in exposed situations and which do not have flange plates for their entire length, the top edge of the web plate shall be flush with or above the angles, and the bottom edge of the web plate shall be flush with or set back from the angles.

15.6.3.5 Welded construction

The gap between the web plates and flange plates shall be kept to a minimum and for fillet welds shall not exceed 1 mm at any point before welding.

15.6.4 Webs

15.6.4.1 Effective sectional area of web of plate girder

The effective cross sectional area shall be taken as the full depth of the web plate multiplied by the thickness.

NOTE — Where webs are varied in thickness in the depth of the section by the use of tongue plates or the like, or where the proportion of the web included in the flange area is 25 percent or more of the overall depth, the above approximation is not permissible and the maximum shear stress shall be computed on theory.

15.6.4.2 Splices in webs

Splices and cutouts for service ducts in the webs should preferably not be located at points of maximum shear force and heavy concentrated loads.

Splices in the webs of the plate girders and rolled sections shall be designed to resist the shears and moments at the spliced section (see [Annex F](#)).

In riveted or bolted construction, splice plates shall be provided on each side of the web. In welded construction, web splices shall preferably be made with complete penetration butt welds.

15.6.4.3 Where additional plates are required to augment the strength of the web, they shall be placed on each side of the web and shall be equal in thickness. The proportion of shear force assumed to be resisted by these plates shall be limited by the amount of horizontal shear which they can transmit to the flanges through their fastenings, and such reinforcing plates and their fastenings shall be carried up to the points at which the flange without the additional plates is adequate.

15.7 Stiffener Design

15.7.1 General

15.7.1.1 When the web of a member acting alone (that is, without stiffeners) proves inadequate, stiffeners for meeting the following requirements should be provided:

- Intermediate transverse web stiffener* — To improve the buckling strength of a slender web due to shear (see [15.7.2](#)).
- Load carrying stiffener* — To prevent local buckling of the web due to concentrated loading (see [15.7.3](#) and [15.7.5](#)).
- Bearing stiffener* — To prevent local crushing of the web due to concentrated loading (see [15.7.4](#) and [15.7.6](#)).
- Torsion stiffener* — To provide torsional restraint to beams and girders at supports (see [15.7.9](#)).
- Diagonal stiffener* — To provide local reinforcement to a web under shear and bearing (see [15.7.7](#)).
- Tension stiffener* — To transmit tensile forces applied to a web through a flange (see [15.7.8](#)).

The same stiffeners may perform more than one function and their design should comply with the requirements of all the functions for which designed.

15.7.1.2 Outstand of web stiffeners

Unless the outer edge is continuously stiffened, the outstand from the face of the web should not exceed $20t_q\epsilon$.

When the outstand of web stiffener is between $14t_q\epsilon$ and $20t_q\epsilon$, then the stiffener design should be on the basis of a core section with an outstand of only $14t_q\epsilon$, where t_q is the thickness of the stiffener.

15.7.1.3 Stiff bearing length

The stiff bearing length of any element, b_1 , is that length which cannot deform appreciably in bending. To determine b_1 , the dispersion of load through a steel bearing element should be taken as 45° through solid material, such as bearing plates, flange plates, etc (see [Fig.15](#)).

15.7.1.4 Eccentricity

Where a load or reaction is applied eccentric to the centreline of the web or where the centroid of the stiffener does not lie on the centreline of the web, the resulting eccentricity of loading should be accounted for in the design of the stiffener.

15.7.1.5 Buckling resistance of stiffeners

The buckling resistance F_{qd} should be based on the design compressive stress f_{cd} (see [14.1.2](#)) of a strut (curve c), the radius of gyration being taken about the axis parallel to the web. The effective section is the full area or core area of the stiffener (see [15.7.1.2](#)) together with an effective length of web on each side of the centreline of the stiffeners, limited to 20 times the web thickness. The design strength used should be the minimum value obtained for buckling about the web or the stiffener.

The effective length for intermediate transverse stiffeners used in calculating the buckling resistance, F_{qd} , should be taken as 0.7 times the length, L of the stiffener.

The effective length for load carrying web stiffeners used in calculating the buckling resistance, F_{xd} , assumes that the flange through which the load or reaction is applied is effectively restrained against lateral movement relative to the other flange, the effective length should be taken as:

- $KL = 0.7L$, when flange is restrained against rotation in the plane of the stiffener (by other structural elements); and
- $KL = L$, when flange is not so restrained.

where

L = length of the stiffener.

If the load or reaction is applied to the flange by a compression member, then unless effective lateral restraint is provided at that point, the stiffener should be designed as part of the compression member applying the load, and the connection between the column and beam flange shall be checked for the effects of the strut action.

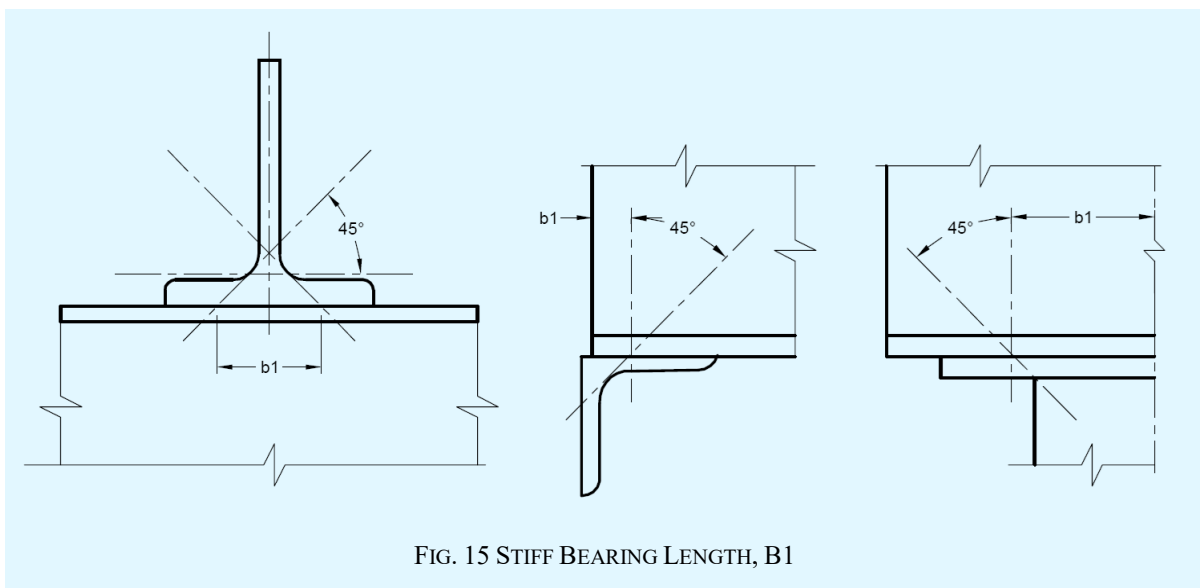


FIG. 15 STIFF BEARING LENGTH, B1

15.7.2 Design of Intermediate Transverse Web Stiffeners

15.7.2.1 General

Intermediate transverse stiffeners may be provided on one or both sides of the web.

15.7.2.2 Spacing

Spacing of intermediate stiffeners, where provided, shall comply with [15.6.1](#) depending on the thickness of the web.

15.7.2.3 Outstand of stiffeners

The outstand of the stiffeners should comply with [15.7.1.2](#).

15.7.2.4 Minimum stiffeners

Transverse web stiffeners not subject to external loads or moments should have a second moment of area, I_s about the centreline of the web, if stiffeners are on both sides of the web and about the face of the web, if single stiffener

on only one side of the web is used such that:

$$\begin{aligned} \text{if } c/d &\geq \sqrt{2}, & I_s &\geq 0.75 d t_w^3, \text{ and} \\ \text{if } c/d &< \sqrt{2}, & I_s &\geq \frac{1.5 d^3 t_w^3}{c^2} \end{aligned}$$

where

- c = actual stiffener spacing;
- d = depth of the web; and
- t_w = minimum required web thickness for spacing using tension field action, as given in [15.4.2.1](#).

15.7.2.5 Buckling check on intermediate transverse web stiffeners

Stiffeners not subjected to external loads or moments should be checked for a stiffener force:

$$F_q = V - V_{cr} / \gamma_{m0} \leq F_{qd}$$

where

- F_{qd} = design resistance of the intermediate stiffeners;
- V_{cr} = shear buckling resistance of the web panel designed without using tension field action as given in [15.4.2.2 \(a\)](#); and
- V = factored shear force adjacent to the stiffener.

Stiffeners subject to external loads and moments should meet the conditions for load carrying web stiffeners in [15.7.3](#). In addition they should satisfy the following interaction expression:

$$\frac{F_q - F_x}{F_{qd}} + \frac{F_x}{F_{xd}} + \frac{M_q}{M_{yq}} \leq 1$$

where

- F_q = stiffener force given above;
- F_{qd} = design resistance of an intermediate web stiffener corresponding to buckling about an axis parallel to the web (see [15.7.1.5](#));
- F_x = external load or reaction at the stiffener;
- F_{xd} = design resistance of a load carrying stiffener corresponding to buckling about axis parallel to the web (see [15.7.1.5](#));
- M_q = moment on the stiffener due to eccentrically applied load and transverse load, if any; and
- M_{yq} = yield moment capacity of the stiffener based on its elastic modulus about its centroidal axis parallel to the web.

If $F_q < F_x$, then $(F_q - F_x)$ should be taken as zero;

15.7.2.6 Connection of intermediate stiffeners to web

Intermediate transverse stiffeners not subject to external loading should be connected to the web so as to withstand a shear between each component of the stiffener and the web (in kN/mm) of not less than:

$$t_w^2 / (5b_s)$$

where

t_w = web thickness, in mm; and

b_s = outstand width of the stiffener, in mm.

For stiffeners subject to external loading, the shear between the web and the stiffener due to such loading has to be added to the above value.

Stiffeners not subject to external loads or moments may terminate clear of the tension flange and in such a situation the distance cut short from the line of the weld should not be more than $4t_w$.

15.7.3 Load Carrying Stiffeners

15.7.3.1 Web check

Load carrying web stiffeners should be provided where compressive forces applied through a flange by loads or reactions exceed the buckling strength, F_{cdw} , of the unstiffened web, calculated using the following:

The effective length of the web for evaluating the slenderness ratio is calculated as in [15.7.1.5](#). The area of cross section is taken as $(b_1 + n_1) t_w$.

where

b_1 = width of stiff bearing on the flange (see [15.7.1.3](#)); and

n_1 = dispersion of the load through the web at 45° , to the level of half the depth of the cross-section.

The buckling strength of this web about axis parallel to the web is calculated as given in [14.1.2.1](#), using curve 'c'.

15.7.4 Bearing Stiffeners

Bearing stiffeners should be provided for webs where forces applied through a flange by loads or reactions exceeding the local capacity of the web at its connection to the flange, F_w , given by:

$$F_w = (b_1 + n_2) t_w f_{yw} / \gamma_{m0}$$

where

b_1 = Stiff bearing length (see [15.7.1.3](#));

n_2 = Length obtained by dispersion through the flange to the web junction at a slope of 1 : 2.5 to the plane of the flange;

t_w = Thickness of the web; and

f_{yw} = Yield stress of the web.

15.7.5 Design of Load Carrying Stiffeners

15.7.5.1 Buckling check

The external load or reaction, F_x on a stiffener should not exceed the buckling resistance, F_{xd} of the stiffener as given in [15.7.1.5](#).

Where the stiffener also acts as an intermediate stiffener it should be checked for the effect of combined loads in accordance with [15.7.2.5](#).

15.7.5.2 Bearing check

Load carrying web stiffeners should also be of sufficient size that the bearing strength of the stiffener, F_{psd} , given below is not less than the load transferred, F_x

$$F_{psd} = A_q f_{yq} / (0.8 \gamma_{m0}) \geq F_x$$

where

F_x = external load or reaction,

A_q = area of the stiffener in contact with the flange, and

f_{yq} = yield stress of the stiffener.

15.7.6 Design of Bearing Stiffeners

Bearing stiffeners should be designed for the applied load or reaction less the local capacity of the web as given in [15.7.4](#). Where the web and the stiffener material are of different strengths the lesser value should be assumed to calculate the capacity of the web and the stiffener. Bearing stiffeners should project nearly as much as the overhang of the flange through which load is transferred.

15.7.7 Design of Diagonal Stiffeners

Diagonal stiffeners should be designed to carry the portion of the applied shear and bearing that exceeds the capacity of the web.

Where the web and the stiffener are of different strengths, the value for design should be taken as given in [15.7.6](#).

15.7.8 Design of Tension Stiffeners

Tension stiffeners should be designed to carry the portion of the applied load or reaction less the capacity of the web as given in [15.7.4](#) for bearing stiffeners.

Where the web and the stiffener are of different strengths, the value for design should be taken as given in [15.7.6](#).

15.7.9 Torsional Stiffeners

Where bearing stiffeners are required to provide torsional restraint at the supports of the beam, they should meet the following criteria:

a) Conditions of [15.7.4](#); and

b) Second moment of area of the stiffener section about the centreline of the web, I_s should be such that:

$$I_s \geq 0.34 \alpha_s D^3 T_{cf}$$

where

$$\alpha_s = 0.006 \text{ for } L_{LT} / r_y \leq 50;$$

$$= 0.3 / (L_{LT} / r_y) \text{ for } 50 < L_{LT} / r_y \leq 100;$$

$$= 30 / (L_{LT} / r_y)^2 \text{ for } L_{LT} / r_y > 100;$$

D = Overall depth of beam at support;

T_{cf} = Maximum thickness of compression flange in the span under consideration;

L_{LT} = Laterally unsupported effective length of the compression flange of the beam; and

r_y = Radius of gyration of the beam about the minor axis.

15.7.10 Connection to Web of Load Carrying and Bearing Stiffeners — Stiffeners, which resist loads or reactions applied through a flange, should be connected to the web by sufficient welds or fasteners to transmit a design force equal to the lesser of:

a) tension capacity of the stiffener; and

b) sum of the forces applied at the two ends of the stiffener when they act in the same direction or the larger of the forces when they act in opposite directions.

Stiffeners, which do not extend right across the web, should be of such length that the shear stress in the web due to the design force transmitted by the stiffener does not exceed the shear strength of the web. In addition, the capacity of the web beyond the end of the stiffener should be sufficient to resist the applied force.

15.7.11 Connection to Flanges

15.7.11.1 In tension

Stiffeners required to resist tension should be connected to the flange transmitting the load by continuous welds or non-slip fasteners.

15.7.11.2 In compression

Stiffeners required to resist compression should either be fitted against the loaded flange or connected by continuous welds or non-slip fasteners.

The stiffener should be fitted against or connected to both flanges when:

- a) a load is applied directly over a support; or
- b) it forms the end stiffener of a stiffened web; or
- c) it acts as a torsion stiffener.

15.7.12 Hollow Sections

Where concentrated loads are applied to hollow sections consideration should be given to local stresses and deformations and the section reinforced as necessary.

15.7.13 Horizontal Stiffeners

Where horizontal stiffeners are used in addition to vertical stiffeners, they shall be as follows:

- a) One horizontal stiffener shall be placed on the web at a distance from the compression flange equal to $2/5$ of web depth, d when the thickness of the web is less than the limits specified in [15.6.1](#). The stiffener shall be designed so that I_s is not less than $4ct_w^3$ where I_s and t_w are as defined in [15.7.2.4](#) and c is the actual distance between the vertical stiffeners;
- b) A second horizontal stiffener (single or double) shall be placed at the neutral axis of the girder when the thickness of the web is less than the limit specified in [15.6.1](#). This stiffener shall be designed so that I_s is not less than $d_2 t_w^3$ where I_s and t_w are as defined in [15.7.2.4](#) and d_2 is twice the clear distance from the compression flange angles, plates or tongue plates to the neutral axis;
- c) Horizontal web stiffeners shall extend between vertical stiffeners, but need not be continuous over them; and
- d) Horizontal stiffeners may be in pairs arranged on each side of the web, or single located on one side of the web.

15.8 Box Girders

The design and detailing of box girders shall be such as to give full advantage of its higher load carrying capacity. Box girder shall be designed in accordance with specialist literature. The diaphragms and horizontal stiffeners should conform to [15.7.12](#) and [15.7.13](#).

15.8.1 All diaphragms shall be connected so as to transfer the resultant shears to the web and flanges.

15.8.2 Where the concentrated or moving load does not act directly on top of the web, the local effect shall be considered in the design of flanges and the diaphragms.

15.9 Purlins and Sheeting Rails (Girts)

All purlins shall be designed in accordance with the requirements for beams as specified in [15.2.1](#) and [15.2.2](#). The

limitations of bending stress based on lateral instability of the compression flange shall be considered if the purlin does not receive adequate support from bracings and sheathings in the plane of compression flange. In any case the maximum bending moment shall not exceed the values specified in [15.2.1](#). The calculated deflections should not exceed those permitted for the type of roof cladding used, as specified in [12.6.1](#). In calculating the bending moment, advantage may be taken of the continuity of the purlin over supports. The bending about the two axes shall be determined separately and checked according to the biaxial bending requirements specified under [16](#).

15.10 Bending in a Non-Principal Plane

15.10.1 When the flexural deflection of a member is constrained to a non-principal plane by lateral restraints preventing lateral deflection, then the force exerted by the restraints shall be determined, and the principal axes bending moments acting on the member shall be calculated from these forces and applied forces, by a rational analysis. The combined effect of bending about the principal axes shall satisfy the requirements given under [16](#).

15.10.2 When the deflections of a member loaded in a non-principal plane are unconstrained; the principal axes bending moments shall be calculated by a rational analysis. The combined effect of bending about the principal axes shall satisfy the requirements given under [16](#).

16 MEMBER SUBJECTED TO COMBINED FORCES

16.1 General

This Section governs the design of members subjected to combined forces, such as shear force and bending, axial force and bending, or shear force, axial force and bending.

16.2 Combined Shear and Bending

16.2.1 No reduction in moment capacity of the section is necessary as long as the cross section is not subjected to high shear force (factored value of applied shear force is less than or equal to 60 percent of the shear strength of the section as given in [15.4](#)). The moment capacity may be taken as, M_d (see [15.2](#)) without any reduction.

16.2.2 When the factored value of the applied shear force is high (exceeds the limit specified in [16.2.1](#)), the factored moment of the section should be less than the moment capacity of the section under higher shear force, M_{dv} , calculated as given below:

- a) Plastic or compact section:

$$M_{dv} = M_d - \beta(M_d - M_{fd}) \leq 1.2 Z_e f_y / \gamma_{m0}$$

where

$$\beta = (2V/V_d - 1)^2;$$

M_d = plastic design moment of the whole section disregarding high shear force effect (see [15.2.1.2](#)); considering web buckling effects (see [15.2.1.1](#));

V = factored applied shear force as governed by web yielding or web buckling;

V_d = design shear strength as governed by web yielding or web buckling (see [15.4.1](#) or [15.4.2](#));

M_{fd} = plastic design strength of the area of the cross section excluding the shear area, considering partial safety factor γ_{m0} ; and

Z_e = elastic section modulus of the whole section.

- b) Semi-compact section:

$$M_{dv} = Z_e f_y / \gamma_{m0}$$

16.3 Combined Axial Force and Bending Moment

Under combined axial force and bending moment, section strength as governed by material failure and member strength as governed by buckling failure shall be checked in accordance with [16.3.1](#) and [16.3.2](#) respectively.

16.3.1 Section Strength

16.3.1.1 Plastic and compact sections

In the design of members subjected to combined axial force (tension or compression) and bending moment, the following should be satisfied:

$$\left(\frac{M_y}{M_{ndy}} \right)^{\alpha_1} + \left(\frac{M_z}{M_{ndz}} \right)^{\alpha_2} \leq 1.0$$

Conservatively, the following equation may also be used under combined axial force and bending moment:

$$\frac{N}{N_d} + \frac{M_y}{M_{dy}} + \frac{M_z}{M_{dz}} \leq 1.0$$

where

- α_1, α_2 = constants as given in [Table 17](#);
- A_g = gross area of the cross section;
- M_{dy}, M_{dz} = design strength under corresponding moment acting alone (see [15.2](#));
- M_{ndy}, M_{ndz} = design reduced flexural strength under combined axial force and the respective uniaxial moment acting alone (see [16.3.1.2](#));
- M_y, M_z = factored applied moments about the minor and major axis of the cross section, respectively;
- N = factored applied axial force (Tension, T or Compression, P);
- N_d = design strength in tension, T_d as obtained from [12](#) or in compression due to yielding given by $N_d = A_g f_y / \gamma_{m0}$; and
- γ_{m0} = partial factor of safety in yielding.

16.3.1.2 For plastic and compact sections without bolts holes, the following approximations may be used for evaluating M_{ndy} and M_{ndz} :

a) Plates:

$$M_{nd} = M_d (1 - n^2)$$

b) Welded I or H-sections:

$$M_{ndy} = M_{dy} \left[1 - \left(\frac{n-a}{1-a} \right)^2 \right] \leq M_{dy} \quad (n \geq a)$$

$$M_{ndz} = M_{dz} (1 - n) / (1 - 0.5a) \leq M_{dz}$$

where

$$n = N / N_d \text{ and } a = (A - 2 b t_f) / A \leq 0.5$$

c) For standard I or H sections:

$$M_{ndz} = 1.11 M_{dz} (1 - n) \leq M_{dz}$$

$$\text{for } n \leq 0.2, M_{ndy} = M_{dy}$$

$$\text{for } n > 0.2, M_{ndy} = 1.56 M_{dy} (1 - n) (n + 0.6)$$

d) For rectangular hollow sections and welded box sections:

When the section is symmetric about both axes and without bolt holes:

$$M_{ndy} = M_{dy} (1 - n) / (1 - 0.5a_f) \leq M_{dy}$$

$$M_{ndz} = M_{dz} (1 - n) / (1 - 0.5a_w) \leq M_{dz}$$

where

$$a_f = (A - 2 h t_w) / A \leq 0.5$$

$$a_w = (A - 2 b t_f) / A \leq 0.5$$

e) Circular hollow tubes without bolt holes:

$$M_{nd} = 1.04 M_d (1 - n^{1.7}) \leq M_d$$

Table 17 Constants α_1 and α_2 (Clause 16.3.1.1)			
Sl No. (1)	Section (2)	α_1 (3)	α_2 (4)
i)	I and Channel	$5n \geq 1$	2
ii)	Circular tubes	2	2
iii)	Rectangular tubes	$1.66/(1 - 1.13n^2) \leq 6$	$1.66/(1 - 1.13n^2) \leq 6$
iv)	Solid rectangles	$1.73 + 1.8n^3$	$1.73 + 1.8n^3$
NOTE — $n = N/N_d$			

16.3.1.3 Semi-compact section

In the absence of high shear force (see 16.2.1), semi-compact section design is satisfactory under combined axial force and bending, if the maximum longitudinal stress under combined axial force and bending, f_x satisfies the following criteria:

$$f_x \leq f_y / \gamma_{m0}$$

For cross section without holes, the above criterion reduces to,

$$\frac{N}{N_d} + \frac{M_y}{M_{dy}} + \frac{M_z}{M_{dz}} \leq 1.0$$

where

N_d, M_{dy}, M_{dz} are as defined in 16.3.1.1.

16.3.2 Overall Member Strength

Members subjected to combined axial force and bending moment shall be checked for overall buckling failure as given in this section.

16.3.2.1 Bending and axial tension

The reduced effective moment, M_{eff} , under tension and bending calculated as given below, should not exceed the bending strength due to lateral torsional buckling, M_d (see 15.2.2),

$$M_{eff} = [M - \psi T Z_{ec} / A] \leq M_d$$

where

- A = area of cross section;
 M, T = factored applied moment and tension, respectively;
 Z_{ec} = elastic section modulus of the section with respect to extreme compression fibre;
 ψ = 0.8, if T and M can vary independently, or otherwise; and
= 1.0.

16.3.2.2 Bending and axial compression

Members subjected to combined axial compression and biaxial bending shall satisfy the following interaction relationships:

$$\frac{P}{P_{dy}} + K_{yy} \frac{M_y}{M_{dy}} + K_{yz} \frac{M_z}{M_{dz}} \leq 1.0$$

$$\frac{P}{P_{dz}} + K_{zy} \frac{M_y}{M_{dy}} + K_{zz} \frac{M_z}{M_{dz}} \leq 1.0$$

where

- $K_{yy}, K_{yz}, K_{zy}, K_{zz}$ = interaction factors as given in [Table 18](#);
 M_{dy}, M_{dz} = design bending strength about y (minor) or z (major) axis, considering laterally unsupported length of the cross section [see [15](#)];
 M_y, M_z = maximum factored applied bending moments about y and z-axis of the member, respectively;
 P = applied axial compression under factored load; and
 P_{dy}, P_{dz} = design strength under axial compression as governed by buckling about minor (y) and major (z) axis respectively.

Table 18 Interaction Factors (Section 10.3.2.2)				
(Clause 16.3.2.2)				
Sl No.	Type of Section	Slenderness Ratio	Plastic Sections (Class 1, Class 2)	Elastic Sections (Class 3, Class 4)
(1)	(2)	(3)	(4)	(5)
i)	All	$\lambda_z < 1.0$	$K_{zz} = C_{mz} [1 + (\lambda_z - 0.2) n_z]$	$K_{zz} = C_{mz} [1 + 0.6 \lambda_z n_z]$
		$\lambda_z \geq 1.0$	$K_{zz} = C_{mz} [1 + 0.8 n_z]$	$K_{zz} = C_{mz} [1 + 0.6 n_z]$
ii)	I sections	$\lambda_y < 1.0$	$K_{yy} = C_{my} [1 + (2\lambda_y - 0.6) n_y]$	$K_{yy} = C_{my} [1 + 0.6 \lambda_y n_y]$
		$\lambda_y \geq 1.0$	$K_{yy} = C_{my} [1 + 1.4 n_y]$	$K_{yy} = C_{my} [1 + 0.6 n_y]$
iii)	Hollow sections	$\lambda_y < 1.0$	$K_{yy} = C_{my} [1 + (\lambda_y - 0.2) n_y]$	$K_{yy} = C_{my} [1 + 0.6 \lambda_y n_y]$
		$\lambda_y \geq 1.0$	$K_{yy} = C_{my} [1 + 0.8 n_y]$	$K_{yy} = C_{my} [1 + 0.6 n_y]$
iv)	All	All	$K_{zy} = 0.66 K_{yy}$	$K_{zy} = K_{yy}$

Table 18 (Concluded)				
Sl No.	Type of Section	Slenderness Ratio	Plastic Sections (Class 1, Class 2)	Elastic Sections (Class 3, Class 4)
(1)	(2)	(3)	(4)	(5)
v)	Not susceptible to lateral torsional buckling	All	$K_{yz} = 0.6 K_{zz}$	$K_{yz} = 0.8 K_{zz}$
vi)	Susceptible to lateral torsional buckling	$\lambda_y < 0.4$	$K_{yz} = 1 - \frac{0.1 \lambda_y n_y}{C_{mLT} - 0.25} \leq 0.6 \lambda_y$	$K_{yz} = 1 - \frac{0.05 \lambda_y n_y}{C_{mLT} - 0.25}$
		$0.4 \leq \lambda_y < 1.0$	$K_{yz} = 1 - \frac{0.1 \lambda_y n_y}{C_{mLT} - 0.25}$	$K_{yz} = 1 - \frac{0.05 \lambda_y n_y}{C_{mLT} - 0.25}$
		$\lambda_y \geq 1.0$	$K_{yz} = 1 - \frac{0.1 n_y}{C_{mLT} - 0.25}$	$K_{yz} = 1 - \frac{0.05 n_y}{C_{mLT} - 0.25}$
Where n_y, n_z = ratio of actual applied axial force to the design axial strength for buckling about the y and z-axis, respectively C_{my}, C_{mz}, C_{mLT} = Equivalent uniform moment factor for lateral torsional buckling as per Table 19 .				

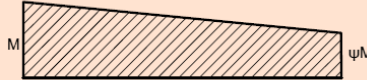
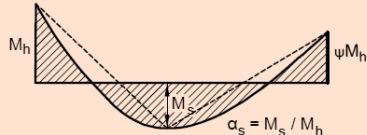
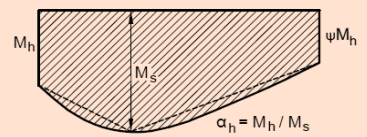
Table 19 Equivalent Uniform Moment Factor				
(Table 18)				
Bending Moment Diagram	Range		C_{my}, C_{mz}, C_{mLT}	
			Uniform Loading	Concentrated Load
(1)	(2)		(3)	(4)
	$-1 \leq \psi \leq 1$		$0.6 + 0.4 \psi \geq 0.4$	
	$0 \leq \alpha_s \leq 1$	$-1 \leq \psi \leq 1$	$0.2 + 0.8 \alpha_s \geq 0.4$	$0.2 + 0.8 \alpha_s \geq 0.4$
	$-1 \leq \alpha_s \leq 0$	$0 \leq \psi \leq 1$	$0.1 - 0.8 \alpha_s \geq 0.4$	$-0.8 \alpha_s \geq 0.4$
		$-1 \leq \psi \leq 0$	$0.1(1-\psi) - 0.8 \alpha_s \geq 0.4$	$-0.2\psi - 0.8 \alpha_s \geq 0.4$
	$0 \leq \alpha_h \leq 1$	$-1 \leq \psi \leq 1$	$0.95 - 0.05 \alpha_h$	$0.90 + 0.10 \alpha_h$
	$-1 \leq \alpha_h \leq 0$	$0 \leq \psi \leq 1$	$0.95 + 0.05 \alpha_h$	$0.90 + 0.10 \alpha_h$
		$-1 \leq \psi \leq 0$	$0.95 + 0.05 \alpha_h (1+2 \psi)$	$0.90 + 0.1 \alpha_h (1+2 \psi)$
For members with sway buckling mode, the equivalent uniform moment factor $C_{my} = C_{mz} = 0.9$.				

Table 19 (Concluded)

Bending Moment Diagram	Range	C_{my}, C_{mz}, C_{mLT}	
		Uniform Loading	Concentrated Load
(1)	(2)	(3)	(4)
C_{my}, C_{mz}, C_{mLT} shall be obtained according to the bending moment diagram between the relevant braced points			
Moment factor direction C_{my} C_{mz} C_{mLT}	Bending axis Z-Z Y-Y Z-Z	Points braced in Y-Y Z-Z Z-Z	

17 CONNECTIONS

17.1 General

17.1.1 This Section deals with the design and detailing requirements for joints between members. Connection elements consist of components such as cleats, gusset plates, brackets, connecting plates and connectors such as rivets, bolts, pins, and welds. The connections in a structure shall be designed so as to be consistent with the assumptions made in the analysis of the structure and comply with the requirements specified in this section. Connections shall be capable of transmitting the calculated design actions.

17.1.2 Where members are connected to the surface of a web or the flange of a section, the ability of the web or the flange to transfer the applied forces locally should be checked and where necessary, local stiffening provided.

17.1.3 Ease of fabrication and erection should be considered in the design of connections. Attention should be paid to clearances necessary for field erection, tolerances, tightening of fasteners, welding procedures, subsequent inspection, surface treatment and maintenance.

17.1.4 The ductility of steel assists the distribution of forces generated within a joint. Effects of residual stresses and stresses due to tightening of fasteners and normal tolerances of fit-up need not therefore be considered in connection design, provided ductile behaviour is ensured.

17.1.5 In general, use of different forms of fasteners to transfer the same force shall be avoided. However, when different forms of fasteners are used to carry a shear load or when welding and fasteners are combined, then one form of fastener shall be normally designed to carry the total load. Nevertheless, fully tensioned friction grip bolts may be designed to share the load with welding, provided the bolts are fully tightened to develop necessary pretension after welding.

17.1.6 The partial safety factor in the evaluation of design strength of connections shall be taken as given in [Table 5](#).

17.2 Location Details of Fasteners

17.2.1 Clearances for Holes for Fasteners

Bolts may be located in standard size, over size, short slotted or long slotted hole, as follows:

- Standard clearance hole** — Except where fitted bolts, bolts in low-clearance or oversize holes are specified, the diameter of standard clearance holes for fasteners shall be as given in [Table 20](#);
- Oversize hole** — Holes of size larger than the standard clearance holes, as given in [Table 20](#) may be used in slip resistant connections and hold down bolted connections, only where specified, provided the oversize;

- c) Holes in the outer ply is covered by a cover plate of sufficiently large size and thickness and having a hole not larger than the standard clearance hole (and hardened washer in slip resistant connections); and
- d) *Short and long slots* — Slotted holes of size larger than the standard clearance hole, as given in [Table 20](#) may be used in slip resistant connections and hold down bolted connections, only where specified, provided the oversize holes in the outer ply is covered by a cover plate of sufficiently large size and thickness and having a hole of size not larger than the standard clearance hole (and hardened washer in slip resistant connection).

Table 20 Clearances for Fastener Holes

(Clauses [17.2.1](#) and [24.2.2.1](#))

Sl No.	Nominal Size of Fastener, d mm	Size of the Hole = Nominal Diameter of the Fastener + Clearances mm			
		Standard clearance in diameter and width of slot	Over size clearance in diameter	Clearance in the length of the slot	
				Short Slot	Long Slot
(1)	(2)	(3)	(4)	(5)	(6)
i)	12 to 14	1.0	3.0	4.0	2.5 d
ii)	16 to 22	2.0	4.0	6.0	2.5 d
iii)	24	2.0	6.0	8.0	2.5 d
iv)	Larger than 24	3.0	8.0	10.0	2.5 d

17.2.2 Minimum Spacing

The distance between centres of fasteners shall not be less than 2.5 times the nominal diameter of the fastener.

17.2.3 Maximum Spacing

17.2.3.1 The distance between the centres of any two adjacent fasteners shall not exceed $32t$ or 300 mm, whichever is less, where t is the thickness of the thinner plate.

17.2.3.2 The distance between the centres of two adjacent fasteners (pitch) in a line lying in the direction of stress, shall not exceed $16t$ or 200 mm, whichever is less, in tension members and $12t$ or 200 mm, whichever is less, in compression members; where t is the thickness of the thinner plate. In the case of compression members wherein forces are transferred through butting faces, this distance shall not exceed 4.5 times the diameter of the fasteners for a distance equal to 1.5 times the width of the member from the butting faces.

17.2.3.3 The distance between the centres of any two consecutive fasteners in a line adjacent and parallel to an edge of an outside plate shall not exceed 100 mm plus $4t$ or 200 mm, whichever is less, in compression and tension members; where t is the thickness of the thinner outside plate.

17.2.3.4 When fasteners are staggered at equal intervals and the gauge does not exceed 75 mm, the spacing specified in [17.2.3.2](#) and [17.2.3.3](#) between centres of fasteners may be increased by 50 percent, subject to the maximum spacing specified in [17.2.3.1](#).

17.2.4 Edge and End Distances

17.2.4.1 The edge distance is the distance at right angles to the direction of stress from the centre of a hole to the adjacent edge. The end distance is the distance in the direction of stress from the centre of a hole to the end of the element.

In slotted holes, the edge and end distances should be measured from the edge or end of the material to the centre of its end radius or the centre line of the slot, whichever is smaller. In oversize holes, the edge and end distances should be taken as the distance from the relevant edge/end plus half the diameter of the standard clearance hole corresponding to the fastener, less the nominal diameter of the oversize hole.

17.2.4.2 The minimum edge and end distances from the centre of any hole to the nearest edge of a plate shall not be less than 1.7 times the hole diameter in case of sheared or hand-flame cut edges; and 1.5 times the hole diameter in case of rolled, machine-flame cut, sawn and planed edges.

17.2.4.3 The maximum edge distance to the nearest line of fasteners from an edge of any un-stiffened part should not exceed $12 t\epsilon$, where $\epsilon = (250/f_y)^{1/2}$ and t is the thickness of the thinner outer plate. This would not apply to fasteners interconnecting the components of back to back tension members. Where the members are exposed to corrosive influences, the maximum edge distance shall not exceed 40 mm plus $4t$, where t is the thickness of thinner connected plate.

17.2.5 Tacking Fasteners

17.2.5.1 In case of members covered under [17.2.4.3](#), when the maximum distance between centres of two adjacent fasteners as specified in [17.2.4.3](#) is exceeded, tacking fasteners not subjected to calculated stress shall be used.

17.2.5.2 Tacking fasteners shall have spacing in a line not exceeding 32 times the thickness of the thinner outside plate or 300 mm, whichever is less. Where the plates are exposed to the weather, the spacing in line shall not exceed 16 times the thickness of the thinner outside plate or 200 mm, whichever is less. In both cases, the distance between the lines of fasteners shall not be greater than the respective pitches.

17.2.5.3 All the requirements specified in [17.2.5.2](#) shall generally apply to compression members, subject to the stipulations given under [14](#) affecting the design and construction of compression members.

17.2.5.4 In tension members [see [13](#)] composed of two flats, angles, channels or tees in contact back to back or separated back to back by a distance not exceeding the aggregate thickness of the connected parts, tacking fasteners with solid distance pieces shall be provided at a spacing in line not exceeding 1 000 mm.

17.2.5.5 For compression members covered under [14](#), tacking fasteners in a line shall be spaced at a distance not exceeding 600 mm.

17.2.6 Countersunk Heads

For countersunk heads, one-half of the depth of the countersinking shall be neglected in calculating the length of the fastener in bearing in accordance with [17.3.3](#). For fasteners in tension having countersunk heads, the tensile strength shall be reduced by 33.3 percent. No reduction is required to be made in shear strength calculations.

17.3 Bearing Type Bolts

17.3.1 Effective Areas of Bolts

17.3.1.1 Since threads can occur in the shear plane, the area A_e for resisting shear should normally be taken as the net tensile stress area, A_n of the bolts. For bolts where the net tensile stress area is not defined, A_n shall be taken as the area at the root of the threads.

17.3.1.2 Where it can be shown that the threads do not occur in the shear plane, A_e may be taken as the cross section area, A_s at the shank.

17.3.1.3 In the calculation of thread length, allowance should be made for tolerance and thread run off.

17.3.2 A bolt subjected to a factored shear force (V_{sb}) shall satisfy the condition

$$V_{sb} \leq V_{db}$$

where V_{db} is the design strength of the bolt, taken as the smaller of the value as governed by shear, V_{dsb} (see [17.3.3](#)) and bearing, V_{dpb} (see [17.3.4](#)).

17.3.3 Shear Capacity of Bolt

The design strength of the bolt, V_{dsb} as governed by shear strength is given by:

$$V_{dsb} = V_{nsb} / \gamma_{mb}$$

where

V_{nsb} = nominal shear capacity of a bolt, calculated as follows.

$$V_{nsb} = \frac{f_u}{\sqrt{3}} (n_n A_{nb} + n_s A_{sb})$$

where

A_{nb} = net shear area of the bolt at threads, may be taken as the area corresponding to root diameter at the thread;

A_{sb} = nominal plain shank area of the bolt;

f_u = ultimate tensile strength of a bolt;

n_n = number of shear planes with threads intercepting the shear plane; and

n_s = number of shear planes without threads intercepting the shear plane.

17.3.3.1 Long joints

When the length of the joint, l_j of a splice or end connection in a compression or tension element containing more than two bolts (that is, the distance between the first and last rows of bolts in the joint, measured in the direction of the load transfer) exceeds $15d$ in the direction of load, the nominal shear capacity (see [17.3.2](#)), V_{db} shall be reduced by the factor β_{lj} , given by:

$$\beta_{lj} = 1.075 - l_j / (200 d) \text{ but } 0.75 \leq \beta_{lj} \leq 1.0 = 1.075 - 0.005(l_j / d)$$

where

d = Nominal diameter of the fastener.

NOTE — This provision does not apply when the distribution of shear over the length of joint is uniform, as in the connection of web of a section to the flanges.

17.3.3.2 Large grip lengths

When the grip length, l_g (equal to the total thickness of the connected plates) exceeds 5 times the diameter, d of the bolts, the design shear capacity shall be reduced by a factor β_g , given by:

$$\beta_g = 8d / (3d + l_g) = 8 / (3 + l_g/d)$$

β_g shall not be more than β_{lj} given in [17.3.3.1](#). The grip length, l_g shall in no case be greater than $8d$.

17.3.3.3 Packing plates

The design shear capacity of bolts carrying shear through a packing plate in excess of 6 mm shall be decreased by a factor, β_{pk} given by:

$$\beta_{pk} = (1 - 0.0125 t_{pk})$$

where

t_{pk} = thickness of the thicker packing, in mm.

17.3.4 Bearing Capacity of the Bolt

The design bearing strength of a bolt on any plate, V_{dpb} as governed by bearing is given by:

$$V_{dpb} = V_{npb} / \gamma_{mb}$$

where

$$\begin{aligned} V_{npb} &= \text{Nominal bearing strength of a bolt; and} \\ &= 2.5 k_b d t f_u. \end{aligned}$$

where

$$\begin{aligned} d_0 &= \text{diameter of the hole;} \\ d &= \text{nominal diameter of the bolt;} \\ f_{ub}, f_u &= \text{Ultimate tensile stress of the bolt and the ultimate tensile stress of the plate, respectively;} \\ e, p &= \text{end and pitch distances of the fastener along bearing direction;} \\ k_b &= \text{is smaller of } \frac{e}{3d_0}, \frac{p}{3d_0} - 0.25, \frac{f_{ub}}{f_u}, 1.0; \text{ and} \\ t &= \text{summation of the thicknesses of the connected plates experiencing bearing stress in the same} \\ &\quad \text{direction, or if the bolts are countersunk, the thickness of the plate minus one half of the depth} \\ &\quad \text{of countersinking.} \end{aligned}$$

The bearing resistance (in the direction normal to the slots in slotted holes) of bolts in holes other than standard clearance holes may be reduced by multiplying the bearing resistance obtained as above, V_{npb} , by the factors given below:

- Over size and short slotted holes – 0.7; and
- Long slotted holes – 0.5.

NOTE — The block shear of the edge distance due to bearing force may be checked as given in [12.4](#).

17.3.5 Tension Capacity

A bolt subjected to a factored tensile force, T_b shall satisfy,

$$T_b \leq T_{db}$$

where

$$\begin{aligned} T_{db} &= T_{nb} / \gamma_{mb}; \text{ and} \\ T_{nb} &= \text{nominal tensile capacity of the bolt, calculated as.} \end{aligned}$$

$$T_{nb} = 0.90 f_{ub} A_n < f_{yb} A_{sb} (\gamma_{mb} / \gamma_{m0})$$

where

$$\begin{aligned} A_{sb} &= \text{shank area of the bolt;} \\ A_n &= \text{net tensile stress area as specified in the appropriate Indian Standard (for bolts where the} \\ &\quad \text{tensile stress area is not defined, } A_n \text{ shall be taken as the area at the bottom of the threads);} \\ f_{ub} &= \text{ultimate tensile stress of the bolt; and} \\ f_{yb} &= \text{yield stress of the bolt.} \end{aligned}$$

17.3.6 Bolt Subjected to Combined Shear and Tension

A bolt required to resist both design shear force (V_{sd}) and design tensile force (T_b) at the same time shall satisfy,

$$\left(\frac{V_{sb}}{V_{db}}\right)^2 + \left(\frac{T_b}{T_{db}}\right)^2 \leq 1.0$$

where

- V_{db} = design shear capacity (see [17.3.2](#));
- V_{sb} = factored shear force acting on the bolt;
- T_b = factored tensile force acting on the bolt, and
- T_{db} = design tension capacity (see [17.3.5](#)).

17.4 Friction Grip Type Bolting

17.4.1 In friction grip type bolting, initial pretension in bolt (usually high strength) develops clamping force at the interfaces of elements being joined. The frictional resistance to slip between the plate surfaces subjected to clamping force opposes slip due to externally applied shear. Friction grip type bolts and nuts shall conform to the accepted standard [C-6A(39)]. Their installation procedures shall conform to good practice [C-6A(15)].

17.4.2 Where slip between bolted plates cannot be tolerated at working loads/ strength limit states loads (slip critical connections), the requirements of [17.4.3](#) shall be satisfied at service and strength limit states loads, respectively. However, the requirements of [17.4.4](#) shall be satisfied by all connections at the strength limit states.

17.4.3 Slip Resistance

Design for friction type bolting, where slip resistance is required at factored design force V_{sf} , shall satisfy the following:

$$V_{sf} \leq V_{dsf}$$

where

- V_{dsf} = V_{nsf} / γ_{mf} ; and
- V_{nsf} = nominal shear capacity of a bolt as governed by slip for friction type connection, calculated as follows.

$$V_{nsf} = \mu_f n_e K_h F_o$$

where

- A_{nb} = net area of the bolt at threads;
- f_o = proof stress ($= 0.70 f_{ub}$);
- F_o = minimum bolt tension (proof load) at installation and may be taken as $A_{nb} f_o$;
- K_h = 1.0 for fasteners in clearance holes;
= 0.85 for fasteners in oversized and short slotted holes and for fasteners in long slotted holes loaded perpendicular to the slot;
- n_e = number of effective interfaces offering frictional resistance to slip;
- μ_f = coefficient of friction (slip factor) as specified in [Table 21](#) ($\mu_f \leq 0.55$);
= 0.7 for fasteners in long slotted holes loaded parallel to the slot;
- γ_{mf} = 1.10 (if slip resistance is designed at service load); and
= 1.25 (if slip resistance is designed at ultimate load).

NOTE — V_{nsf} may be evaluated at a service load or ultimate load using appropriate partial safety factors, depending upon whether slip resistance is required at service load or ultimate load.

Table 21 Typical Average Values for Coefficient of Friction (μ)

(Clause [17.4.3](#))

Sl No.	Treatment of Surface	Coefficient of Friction, μ
(1)	(2)	(3)
i)	Surfaces not treated	0.20
ii)	Surfaces blasted with short or grit with any loose rust removed, no pitting	0.50
iii)	Surfaces blasted with shot or grit and hot-dip galvanized	0.10
iv)	Surfaces blasted with shot or grit and spray-metallized with zinc (thickness 50 μ m to 70 μ m)	0.25
v)	Surfaces blasted with shot or grit and painted with ethyl zinc silicate coat (thickness 30 μ m to 60 μ m)	0.30
vi)	Sand blasted surface, after light rusting	0.52
vii)	Surfaces blasted with shot or grit and painted with ethyl zinc silicate coat (thickness 60 μ m to 80 μ m)	0.30
viii)	Surfaces blasted with shot or grit and painted with alkali zinc silicate coat (thickness 60 μ m to 80 μ m)	0.30
ix)	Surface blasted with shot or grit and spray metallized with aluminium (thickness > 50 μ m)	0.50
x)	Clean mill scale	0.33
xi)	Sand blasted surface	0.48
xii)	Red lead painted surface	0.1

17.4.3.1 Long Joints

The provision for the long joints in [17.3.3.1](#) shall apply to friction grip connections also.

17.4.4 Capacity after Slipping

When friction type bolts are designed not to slip only under service loads, the design capacity at ultimate load may be calculated as per bearing type connection (see [17.3.2](#) and [17.3.3](#)).

NOTE — The block shear resistance of the edge distance due to bearing force may be checked as given in [13.4](#).

17.4.5 Tension Resistance

A friction bolt subjected to a factored tension force (T_f) shall satisfy:

$$T_f \leq T_{df}$$

where

$$T_{df} = T_{nf} / \gamma_{mf}$$

T_{nf} = nominal tensile strength of the friction bolt, calculated as:

$$T_{nf} = 0.9 f_{ub} A_n \leq f_{yb} A_{sb} (\gamma_{m1} / \gamma_{m0})$$

where

A_n = net tensile stress area as specified in various parts of the accepted standard [C-6A(40)] (for bolts where the tensile stress area is not defined, A_n shall be taken as the area at the root of the threads);

A_{sb} = shank area of the bolt;

- f_{ub} = ultimate tensile stress of the bolt; and
 γ_{mf} = partial factor of safety.

17.4.6 Combined Shear and Tension

Bolts in a connection for which slip in the serviceability limit state shall be limited, which are subjected to a tension force, T , and shear force, V , shall satisfy:

$$\left(\frac{V_{sf}}{V_{df}} \right)^2 + \left(\frac{T_f}{T_{df}} \right)^2 \leq 1.0$$

where

- V_{df} = design shear strength;
 V_{sf} = applied factored shear at design load;
 T_{df} = design tension strength; and

17.4.7 Where prying force, Q as illustrated in [Fig. 16](#) shall be calculated as given below and added to the tension in the bolt:

$$Q = \frac{l_v}{2 l_e} \left[T_e - \frac{\beta \eta f_o b_e t^4}{27 l_e l_v^2} \right]$$

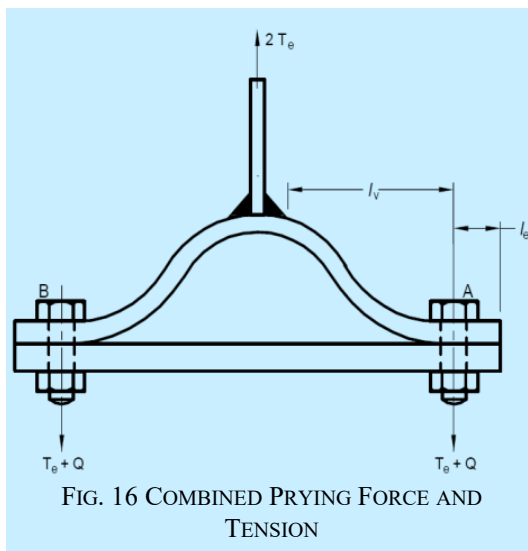
where

- l_e = distance between prying force and bolt centreline and is the minimum of either the end distance or the value given by; and
 l_v = distance from the bolt centreline to the toe of the fillet weld or to half the root radius for a rolled section.

$$l_e = 1.1 t \sqrt{\frac{\beta f_o}{f_y}}$$

where

- β = 2 for non pre-tensioned bolt and 1 for pre-tensioned bolt;
 b_e = effective width of flange per pair of bolts;
 f_o = proof stress in consistent units;
 η = 1.5; and
 t = thickness of the end plate.



17.5 Welds and Welding

17.5.1 General

Requirements of welds and welding shall conform to the accepted standards [C-6A(41)], as appropriate.

17.5.1.1 End returns

Fillet welds terminating at the ends or sides of parts should be returned continuously around the corners for a distance of not less than twice the size of the weld, unless it is impractical to do so. This is particularly important on the tension end of parts carrying bending loads.

17.5.1.2 Lap joint

In the case of lap joints, the minimum lap should not be less than four times the thickness of the thinner part joined or 40 mm, whichever is more. Single end fillet should be used only when lapped parts are restrained from openings. When end of an element is connected only by parallel longitudinal fillet welds, the length of the weld along either edge should not be less than the transverse spacing between longitudinal welds.

17.5.1.3 A single fillet weld should not be subjected to moment about the longitudinal axis of the weld.

17.5.2 Size of Weld

17.5.2.1 The size of normal fillets shall be taken as the minimum weld leg size. For deep penetration welds, where the depth of penetration beyond the root run is a minimum of 2.4 mm, the size of the fillet should be taken as the minimum leg size plus 2.4 mm.

17.5.2.2 For fillet welds made by semi-automatic or automatic processes, where the depth of penetration is considerably in excess of 2.4 mm, the size shall be taken considering actual depth of penetration subject to agreement between the purchaser and the contractor.

17.5.2.3 The size of fillet welds shall not be less than 3 mm. The minimum size of the first run or of a single run fillet weld shall be as given in [Table 22](#), to avoid the risk of cracking in the absence of preheating.

17.5.2.4 The size of butt weld shall be specified by the effective throat thickness.

17.5.3 Effective Throat Thickness

17.5.3.1 The effective size of a fillet weld shall not be less than 3 mm and shall generally not exceed $0.7t$, or $1.0t$ under special circumstances, where t is the thickness of the thinner plate of elements being welded.

17.5.3.2 For the purpose of stress calculation in fillet welds joining faces inclined to each other, the effective throat thickness shall be taken as K times the fillet size, where K is a constant, depending upon the angle between fusion faces, as given in [Table 23](#).

Table 22 Minimum Size of First Run or of a Single Run Fillet Weld

(Clause [17.5.2.3](#))

Sl No.	Thickness of Thicker Part mm		Minimum Size of First run mm
	Over	Up to and Including	
(1)	(2)	(3)	(4)
i)	—	10	3
ii)	10	20	5
iii)	20	32	6

Table 22 (Concluded)

Sl No.	Thickness of Thicker Part mm		Minimum Size of First run mm
	Over	Up to and Including	
(1)	(2)	(3)	(4)
iv)	32	50	8 of first run 10 for minimum size of weld
NOTES			
1 When the minimum size of the fillet weld given in the table is greater than the thickness of the thinner part, the minimum size of the weld should be equal to the thickness of the thinner part. The thicker part shall be adequately preheated to prevent cracking of the weld.			
2 Where the thicker part is more than 50 mm thick, special precautions like preheating should be taken.			

Table 23 Values of K for Different Angles Between Fusion Faces

(Clause [17.5.3.2](#))

Sl No.	Angle Between Fusion Faces	Constant, K
(1)	(2)	(3)
i)	60° to 90°	0.70
ii)	91° to 100°	0.65
iii)	101° to 106°	0.60
iv)	107° to 113°	0.55
v)	114° to 120°	0.50

17.5.3.3 The effective throat thickness of a complete penetration butt weld shall be taken as the thickness of the thinner part joined, and that of an incomplete penetration butt weld shall be taken as the minimum thickness of the weld metal common to the parts joined, excluding reinforcements.

17.5.4 Effective Length or Area of Weld

17.5.4.1 The effective length of fillet weld shall be taken as only that length which is of the specified size and required throat thickness. In practice the actual length of weld is made of the effective length shown in drawing plus two times the weld size, but not less than four times the size of the weld.

17.5.4.2 The effective length of butt weld shall be taken as the length of the continuous full size weld, but not less than four times the size of the weld.

17.5.4.3 The effective area of a plug weld shall be considered as the nominal area of the hole in the plane of the faying surface. These welds shall not be designed to carry stresses.

17.5.4.4 If the maximum length l_j of the side welds transferring shear along its length exceeds 150 times the throat size of the weld, t_t , the reduction in weld strength as per the long joint (see [17.5.7.3](#)) should be considered. For flange to web connection, where the welds are loaded for the full length, the above limitation would not apply.

17.5.5 Intermittent Welds

17.5.5.1 Unless otherwise specified, the intermittent fillet welding shall have an effective length of not less than four times the weld size, with a minimum of 40 mm.

17.5.5.2 The clear spacing between the effective lengths of intermittent fillet weld shall not exceed 12 and 16 times the thickness of thinner plate joined, for compression and tension joint, respectively, and in no case be more than 200 mm.

17.5.5.3 Unless otherwise specified, the intermittent butt weld shall have an effective length of not less than four times the weld size and the longitudinal space between the effective length of welds shall not be more than 16 times the thickness of the thinner part joined. The intermittent welds shall not be used in positions subject to dynamic, repetitive and alternating stresses.

17.5.6 Weld Types and Quality

For the purpose of this NBCS, weld shall be fillet, butt, slot or plug or compound welds. Welding electrodes shall conform to IS 814.

17.5.7 Design Stresses in Welds

17.5.7.1 Shop welds

17.5.7.1.1 Fillet welds

Design strength of a fillet weld, f_{wd} shall be based on its throat area and shall be given by:

$$f_{wd} = f_{wn} / \gamma_{mw}$$

where

f_u = smaller of the ultimate stress of the weld or of the parent metal;

$f_{wn} = f_u / \sqrt{3}$; and

γ_{mw} = partial safety factor (see [Table 5](#)).

17.5.7.1.2 Butt welds

Butt welds shall be treated as parent metal with a thickness equal to the throat thickness, and the stresses shall not exceed those permitted in the parent metal.

17.5.7.1.3 Slot or plug welds

The design shear stress on slot or plug welds shall be as per [17.5.7.1.1](#).

17.5.7.2 Site welds

The design strength in shear and tension for site welds made during erection of structural members shall be calculated according to [17.5.7.1](#) but using a partial safety factor γ_{mw} of 1.5.

17.5.7.3 Long joints

When the length of the welded joint, l_j of a splice or end connection in a compression or tension element is greater than $150 t_t$, the design capacity of weld (see [17.5.7.1.1](#)), f_{wd} shall be reduced by the factor:

$$\beta_{lw} = 1.2 - \frac{0.2l_j}{150t_t} \leq 1.0$$

where

l_j = length of the joint in the direction of the force transfer; and

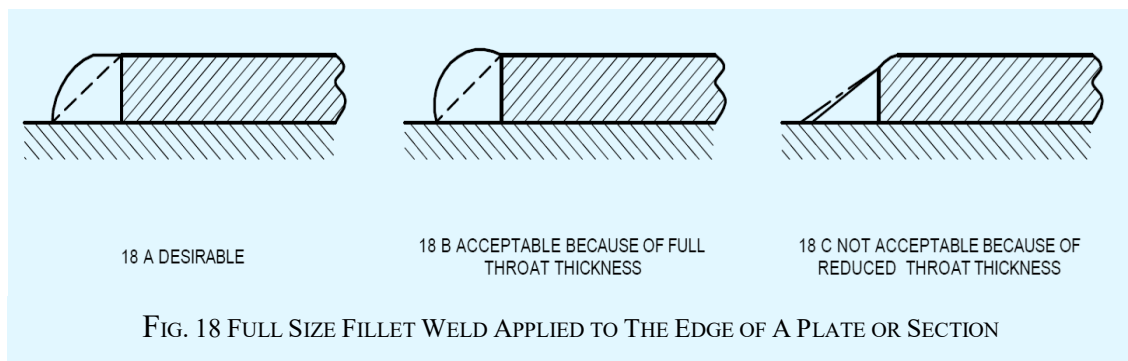
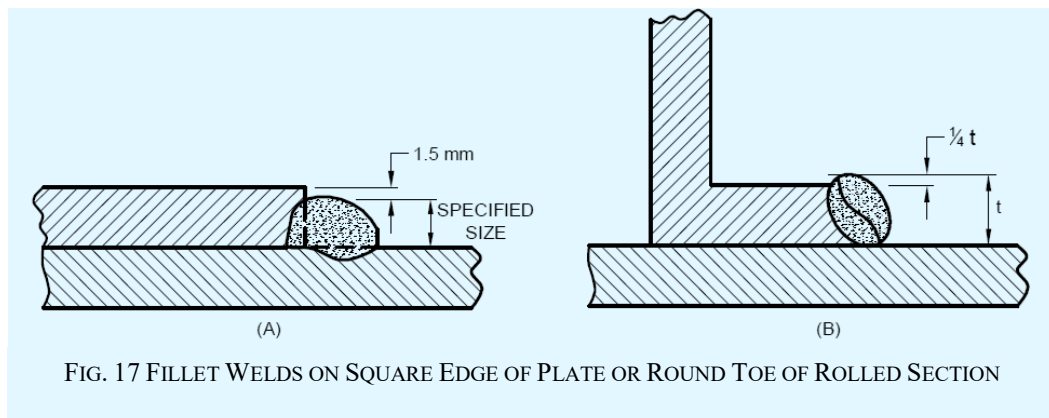
t_t = throat size of the weld.

17.5.8 Fillet Weld Applied to the Edge of a Plate or Section

17.5.8.1 Where a fillet weld is applied to the square edge of a part, the specified size of the weld should generally be at least 1.5 mm less than the edge thickness in order to avoid washing down of the exposed corner [see Fig. 17(A)].

17.5.8.2 Where the fillet weld is applied to the rounded toe of a rolled section, the specified size of the weld should generally not exceed 3/4 of the thickness of the section at the toe [see Fig. 17(B)].

17.5.8.3 Where the size specified for a fillet weld is such that the parent metal will not project beyond the weld, no melting of the outer cover or covers shall be allowed to occur to such an extent as to reduce the throat thickness (see Fig. 18).



17.5.8.4 When fillet welds are applied to the edges of a plate or section in members subject to dynamic loading, the fillet weld shall be of full size with its leg length equal to the thickness of the plate or section, with the limitations specified in 17.5.8.3.

17.5.8.5 End fillet weld, normal to the direction of force shall be of unequal size with a throat thickness not less than $0.5t$, where t is the thickness of the part, as shown in Fig. 19. The difference in thickness of the welds shall be negotiated at a uniform slope.

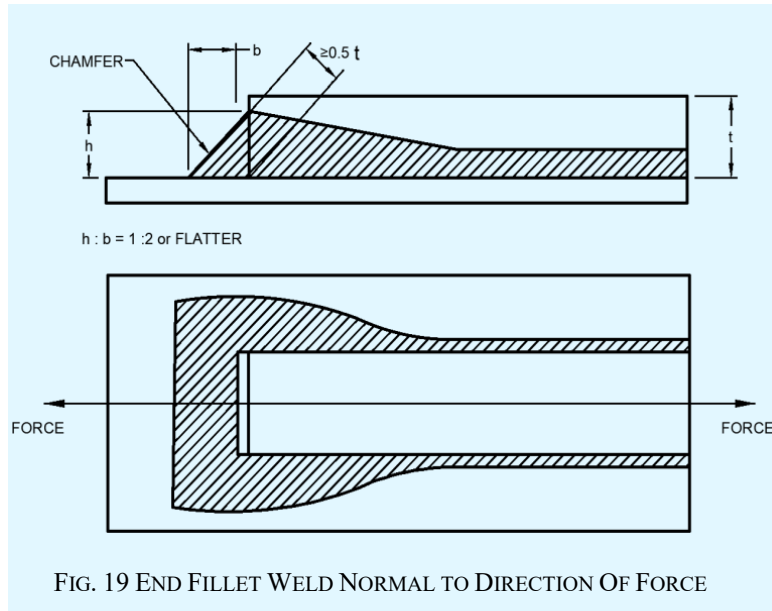
17.5.9 Stresses Due to Individual Forces

When subjected to either compressive or tensile or shear force alone, the stress in the weld is given by and shall be less than or equal to design strength of the weld, f_{ws} , given in 17.5.7.1.1:

$$f_a \text{ or } q = \frac{P}{t_l l_w}$$

where

- f_a = Calculated, in N/mm^2 , normal stress due to axial force;
 l_w = Effective, in mm, length of weld;
 P = force transmitted (axial force N or the shear force Q);
 q = shear, in N/mm^2 , stress; and
 t_t = Effective, in mm; throat thickness of weld.



17.5.10 Combination of Stresses

17.5.10.1 Fillet welds

17.5.10.1.1 When subjected to a combination of normal and shear stress, the equivalent stress f_e shall satisfy the following:

$$f_e = \sqrt{f_a^2 + 3q^2} \leq \frac{f_u}{\gamma_{mw}}$$

where

- f_a = normal stresses, compression or tension, due to axial force or bending moment (see [17.5.9](#)); and
 q = shear stress due to shear force or tension (see [17.5.9](#)).

17.5.10.1.2 Check for the combination of stresses need not be done for:

- side fillet welds joining cover plates and flange plates; and
- fillet welds where algebraic sum of normal and shear stresses does not exceed f_{wd} (see [17.5.7.1.1](#)).

17.5.10.2 Butt welds

17.5.10.2.1 Check for the combination of stresses in butt welds need not be carried out provided that:

- butt welds are axially loaded; and
- in single and double bevel welds the algebraic sum of normal and shear stresses does not exceed the design normal stress, and the shear stress does not exceed 50 percent of the design shear stress.

17.5.10.2.2 Combined bearing, bending and shear

Where bearing stress, f_{br} is combined with bending (tensile or compressive), f_b and shear stresses, q under the most unfavorable conditions of loading in butt welds, the equivalent stress, f_e as obtained from the following formula, shall not exceed the values allowed for the parent metal:

$$f_e = \sqrt{f_b^2 + f_{br}^2 - f_b f_{br} + 3q^2}$$

where

- f_b = calculated, in N/mm²; stress due to bending;
- f_{br} = calculated, in N/mm², stress due to bearing;
- f_e = equivalent stress; and
- q = shear, in N/mm², stress.

17.5.11 Where a packing is welded between two members and is less than 6 mm thick, or is too thin to allow provision of adequate welds or to prevent buckling, the packing shall be trimmed flush with the edges of the element subject to the design action and the size of the welds along the edges shall be increased over the required size by an amount equal to the thickness of the packing. Otherwise, the packing shall extend beyond the edges and shall be fillet welded to the pieces between which it is fitted.

17.6 Design of Connections

Each element in a connection shall be designed so that the structure is capable of resisting the design actions. Connections and adjacent regions of the members shall be designed by distributing the design action effects such that the following requirements are satisfied:

- a) Design action effects distributed to various elements shall be in equilibrium with the design action effects on the connection;
- b) Required deformations in the elements of the connections are within their deformations capacities;
- c) All elements in the connections and the adjacent areas of members shall be capable of resisting the design action effects acting on them; and
- d) Connection elements shall remain stable under the design action effects and deformations.

17.6.1 Connections can be classified as rigid, semi-rigid and flexible for the purpose of analysis and design as per the recommendation in [Annex E](#). Connections with sufficient rotational stiffness may be considered as rigid. Examples of rigid connections include flush end-plate connection and extended end-plate connections. Connections with negligible rotational stiffness may be considered as flexible (pinned). Examples of flexible connections include single and double web angle connections and header plate connections. Where a connection cannot be classified as either rigid or flexible, it shall be assumed to be semi-rigid. For Example, of semi-rigid connections include top and seat angle connection and top and seat angle with single/double web angles.

17.6.2 Design shall be on the basis of any rational method supported by experimental evidence. Residual stresses due to installation of bolts or welding normally need not be considered in statically loaded structures. Connections in cyclically loaded structures shall be designed considering fatigue as given under [19](#). For earthquake load combinations, the connections shall be designed to withstand the calculated design action effects as specified in [18](#).

17.6.3 Beam and column splice shall be designed in accordance with the recommendation given in [F-2](#) and [F-3](#).

17.7 Minimum Design Action on Connection

Connections carrying design action effects, except for lacing connections, connections of sag rods, purlins and girts, shall be designed to transmit the greater of:

- a) the design action in the member; and

- b) The minimum design action effects expressed either as the value or the factor times the member design capacity for the minimum size of member required by the strength limit state, specified as follows:
- 1) *Connections in rigid construction* — A bending moment of at least 0.5 times the member design moment capacity;
 - 2) *Connections to beam in simple construction* — A shear force of at least 0.15 times the member design shear capacity or 40 kN, whichever is lesser;
 - 3) *Connections at the ends of tensile or compression member* — A force of at least 0.3 times the member design capacity;
 - 4) *Splices in members subjected to axial tension* — A force of at least 0.3 times the member design capacity in tension;
 - 5) *Splices in members subjected to axial compression* — For ends prepared for full contact in accordance with [24.7.1](#), it shall be permissible to carry compressive actions by bearing on contact surfaces. When members are prepared for full contact to bear at splices, there shall be sufficient fasteners to hold all parts securely in place. The fasteners shall be sufficient to transmit a force of at least 0.15 times the member design capacity in axial compression;

When members are not prepared for full contact, the splice material and its fasteners shall be arranged to hold all parts in line and shall be designed to transmit a force of at least 0.3 times the member design capacity in axial compression.

In addition, splices located between points of effective lateral support shall be designed for the design axial force, P_d plus a design bending moment, not less than the design bending moment $M_d = P_d l_s / 1\,000$ where, l_s is the distance between points of effective lateral support.

- 6) *Splices in flexural members* — A bending moment of 0.3 times the member design capacity in bending. This provision shall not apply to splices designed to transmit shear force only; and

A splice subjected to a shear force only shall be designed to transmit the design shear force together with any bending moment resulting from the eccentricity of the force with respect to the centroid of the group.

- 7) *Splices in members subject to combined actions* — A splice in a member subject to a combination of design axial tension or design axial compression and design bending moment shall satisfy requirements in (4), (5) and (6) above, simultaneously.

17.8 Intersections

Members or components meeting at a joint shall be arranged to transfer the design actions between the parts, wherever practicable, with their centroidal axes meeting at a point. Where there is eccentricity at joints, the members and components shall be designed for the design bending moments which result due to eccentricity.

The disposition of fillet welds to balance the design actions about the centroidal axis or axes for end connections of single angle, double angle and similar type members is not required for statically loaded members but is required for members, connection components subject to fatigue loading.

Eccentricity between the centroidal axes of angle members and the gauge lines for their bolted end connections may be neglected in statically loaded members, but shall be considered in members and connection components subject to fatigue loading.

17.9 Choice of Fasteners

Where slip in the serviceability limit state is to be avoided in a connection, high-strength bolts in a friction-type joint, fitted bolts or welds shall be used.

Where a joint is subjected to impact or vibration, either high strength bolts in a friction type joint or ordinary bolts with locking devices or welds shall be used.

17.10 Connection Components

Connection components (cleats, gusset plates, brackets and the like) other than connectors, shall have their capacities assessed using the provisions of [12](#), [13](#), [14](#), [15](#) and [16](#), as applicable.

17.11 Analysis of a Bolt/Weld Group

17.11.1 Bolt/Weld Group Subject to In-plane Loading

17.11.1.1 General method of analysis

The design force in a bolt/weld or design force per unit length in a bolt/weld group subject to in-plane loading shall be determined in accordance with the following:

- a) The connection plates shall be considered to be rigid and to rotate relative to each other about a point known as the instantaneous centre of rotation of the group;
- b) In the case of a group subject to a pure couple only, the instantaneous centre of rotation coincides with the group centroid. In the case of in-plane shear force applied at the group centroid, the instantaneous centre of the rotation is at infinity and the design force is uniformly distributed throughout the group. In all other cases, either the results of independent analyses for a pure couple alone and for an in-plane shear force applied at the group centroid shall be superposed, or a recognized method of analysis shall be used; and
- c) The design force in a bolt or design force per unit length at any point in the group shall be assumed to act at right angles to the radius from that point to the instantaneous centre, and shall be taken as proportional to that radius.

17.11.2 Bolt/Weld Group Subject to Out-of-Plane Loading

17.11.2.1 General method of analysis

The design force of a bolt in bolt group or design force per unit length in the fillet weld group subject to out-of-plane loading shall be determined in accordance with the following:

- a) Design force in the bolts or per unit length in the fillet weld group resulting from any shear force or axial force shall be considered to be equally shared by all bolts in the group or uniformly distributed over the length of the fillet weld group; and
- b) Design force resulting from a design bending moment shall be considered to vary linearly with distance from the relevant centroidal axes. The following shall also be considered:
 - 1) in bearing type of bolt group plates in the compression side of the neutral axis and only bolts in the tension side of the neutral axis may be considered for calculating the neutral axis and second moment of area;
 - 2) in the friction grip bolt group, only the bolts shall be considered in the calculation of neutral axis and second moment of area; and
 - 3) the fillet weld group shall be considered in isolation from the connected element; for the calculation of centroid and second moment of the weld length.

17.11.2.2 Alternative analysis

The design force per unit length in a fillet weld/bolt group may alternatively be determined by considering the fillet weld group as an extension of the connected member and distributing the design forces among the welds of the fillet weld group so as to satisfy equilibrium between the fillet weld group and the elements of the connected member.

17.11.3 Bolt/Weld Group Subject to In-plane and Out-of-Plane Loading

17.11.3.1 General method of analysis

The design force in a bolt or per unit length of the weld shall be determined by the superposition of analysis for in-plane and out-of-plane cases discussed in [17.11.1](#) and [17.11.2](#).

17.11.3.2 Alternative analysis

The design force in a bolt or per unit length in the fillet weld group may alternatively be determined by considering the fillet weld group as an extension of the connected member and proportioning the design force per bolt or unit length in the weld group to satisfy equilibrium between the bolt/weld group and the elements of the connected member.

Force calculated in the most stressed bolt or highest force per unit length of the weld shall satisfy the strength requirements of [17.3](#), [17.4](#) or [17.5](#), as appropriate.

17.12 Lug Angles

17.12.1 Lug angles connecting outstanding leg of a channel-shaped member shall, as far as possible, be disposed symmetrically with respect to the section of the member.

17.12.2 In the case of angle members, the lug angles and their connections to the gusset or other supporting member shall be capable of developing a strength not less than 20 percent in excess of the force in the outstanding leg of the member, and the attachment of the lug angle to the main angle shall be capable of developing a strength not less than 40 percent in excess of the force in the outstanding leg of the angle.

17.12.3 In the case of channel members and the like, the lug angles and their connection to the gusset or other supporting member shall be capable of developing a strength of not less than 10 percent in excess of the force not accounted for by the direct connection of the member, and the attachment of the lug angles to the member shall be capable of developing 20 percent in excess of that force.

17.12.4 In no case shall fewer than two bolts, rivets or equivalent welds be used for attaching the lug angle to the gusset or other supporting member.

17.12.5 The effective connection of the lug angle shall, as far as possible terminate at the end of the member connected, and the fastening of the lug angle to the main member shall preferably start in advance of the direct connection of the member to the gusset or other supporting member.

17.12.6 Where lug angles are used to connect an angle member, the whole area of the member shall be taken as effective notwithstanding the requirements given under [13](#).

18 DESIGN AND DETAILING FOR EARTHQUAKE EFFECTS

18.1 General

18.1.1 Earthquake is a ‘displacement loading’ on a structure. This implies that ‘displacements’ are ‘imposed’ under a structure during earthquake shaking and ‘forces’ are induced in it. Thus, a structure is allowed to be designed to have lesser strength capacity than that required if it were to remain elastic. And, only chosen types of damage are permitted to be accrued in select components of a structure during strong earthquake shaking.

Thus, reliance is placed on the nonlinear response of a structure, especially on its deformation and ductility capacities. These must be achieved by adopting special design and detailing practices. Therefore, steel frames shall be so designed and detailed as to give them stability, and adequate stiffness, strength, and ductility capacities to resist severe earthquake shaking without collapse in all earthquake zones mentioned in 5 of Part C ‘Structural Design/Section 1 Loads and Load Effects’ of this NBCS.

18.1.2 Steel frames required to resist earthquake effects in all earthquake zones shall satisfy the requirements specified in good practice [C-6A(42)]. Frames are classified into two types, one which have minimal inelastic deformation and ductility capacities, called ordinary frames, and one with higher deformation and ductility capacities, called special frames. For earthquake zones III and above, only special frames are permitted to be used.

18.1.3 For earthquake zone II, ordinary frames, namely ordinary moment frames (OMF) and ordinary concentrically braced frames (OCBF), are permitted to be used. In design and detailing of OMFs and OCBFs, specific requirements of this Section shall be satisfied along with other provisions of this NBCS, in addition to the earthquake demand and design criteria requirements of 5 of Part C ‘Structural design/Section 1 Loads and Load Effects’ of this NBCS.

18.1.4 Frames, which are a part of the gravity load resisting system and are not intended to resist the lateral earthquake forces, need not satisfy the requirements of this section, provided they can accommodate the resulting earthquake shaking induced deformation without premature failure.

18.2 Loads and Load Combinations

18.2.1 Earthquake loads shall be calculated as per 5 of Part C 'Structural design/Section 1 Loads and Load Effects' of this NBCS, except that the reduction factors recommended in [18.3](#) may be used.

18.2.2 In the limit state design of frames resisting earthquake loads, the load combinations shall conform to [Table 4](#).

18.2.3 In addition the following load combination shall be considered as required in [18.5.1.1](#), [18.7.3.1](#), [18.11.2.2](#) and [18.11.3.4](#):

- a) $1.2 \text{ Dead Load (DL)} + 0.5 \text{ Imposed Load (IL)} \pm 2.5 \text{ Earthquake Force (E)}$; and
- b) $0.9 \text{ Dead Load (DL)} \pm 2.5 \text{ Earthquake Force (E)}$.

18.3 Response Reduction Factor

18.3.1 For structures designed and detailed in accordance with the provisions of this Section, the response reduction factors specified herein may be used for calculating the design earthquake forces, in conjunction with the provisions of 5 of Part C 'Structural Design/Section 1 Loads and Load Effects' of this NBCS.

18.3.2 For braced frame systems, the response reduction factor (R) for ordinary concentrically braced frames (OCBF), for special concentrically braced frames (SCBF) and for special eccentrically braced frames (SEBF) shall be taken from 5 of Part C 'Structural Design/Section 1 Loads and Load Effects' of this NBCS. The response reduction factor (R) for ordinary eccentrically braced frames (OEBF) shall be taken as 4.5.

18.3.3 For moment frame systems, the response reduction factor (R) for special moment frames (SMF) shall be taken from 5 of Part C 'Structural Design/Section 1 Loads and Load Effects' of this NBCS. For the Ordinary Moment Frames (OMF) designed as per [18.10](#), the response reduction factor (R) shall be taken as 4.0.

18.3.4 For non-ductile frames (NDF), the response reduction factor (R) shall be taken as 1.5.

18.3.5 The braced and moment resisting frames in single storey industrial buildings can be designed as non-ductile frames (NDF), if their overall height to overall width ratio does not exceed 1.0. Such frames need not meet the ductility requirements of other types of frames as specified in [18.4](#) to [18.12](#).

18.4 Connections, Joints and Fasteners

18.4.1 All bolts used in critical zones of frames designed to resist earthquake effects shall be fully tensioned high strength friction grip (HSFG) bolts or turned and fitted bolts. The critical zones of frames are zones of plastic hinges of moment resisting frames and beam to columns as well as bracing and their connections of braced frames. The plastic hinge zones shall be taken as a distance equal to 2 times the depth of the member from the section of maximum bending moment.

18.4.2 All welds used in frames designed to resist earthquake effects shall be complete penetration butt welds, except in column splices, which shall conform to [18.5.2](#).

18.4.3 Bolted joints shall be designed not to share load in combination with welds on the same faying surface.

18.5 Columns

18.5.1 Column Strength

When P_u/P_d is greater than 0.4, the requirements in [18.5.1.1](#) and [18.5.1.2](#) shall be met.

Where

P_u = required ultimate compressive strength of the member; and

P_d = design resistance in axial compression as obtained from [14.1.2](#).

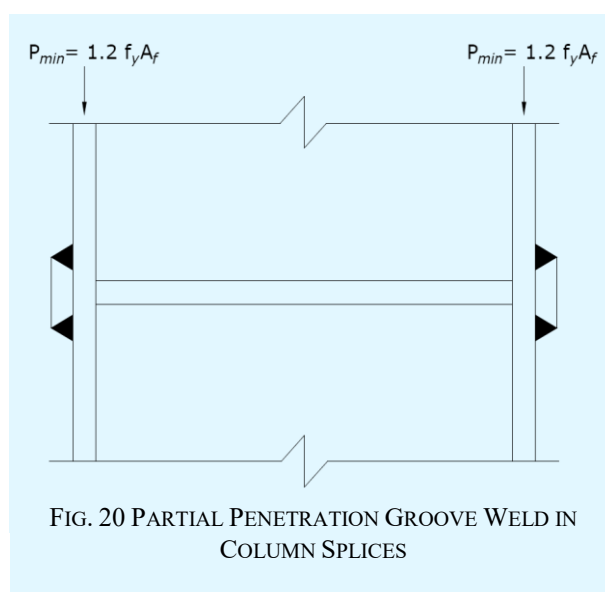
18.5.1.1 The required axial compressive and axial tensile strength in the absence of applied moment, shall be determined from the load combination in [18.2.3](#).

18.5.1.2 The required strength determined in [18.5.1.1](#) need not exceed either of the maximum load transferred to the column considering 1.2 times the nominal strength of the connecting beam or brace element, or the resistance of the foundation to uplift.

18.5.2 Column Splice

18.5.2.1 A partial-joint penetration groove weld may be provided in column splice, such that the design strength of the joints shall be at least equal to 200 percent of the required strength.

18.5.2.2 The minimum required strength for each flange splice shall be 1.2 times $A_f f_y$ as shown [Fig. 20](#), where A_f is the area of each flange in the smaller connected column.



18.6 Storey Drift

The storey drift limits shall conform to 5 of Part C ‘Structural design/Section 1 Loads and Load Effects’ of this NBCS. The deformation compatibility of members not designed to resist seismic lateral load shall also conform to 5 of Part C ‘Structural design/Section 1 Loads and Load Effects’ of this NBCS.

18.7 Ordinary Concentrically Braced Frames (OCBF)

18.7.1 Ordinary concentrically braced frames (OCBF) should be shown to withstand inelastic deformation a moment of 1.2 times the full plastic moment of the corresponding to a joint rotation of at least 0.02 radians without

degradation in strength and stiffness below the full yield value. Ordinary concentrically braced frames meeting the requirements of this section shall be deemed to satisfy the required inelastic deformation.

18.7.1.1 Ordinary concentrically braced frames shall not be used in seismic zones IV and above and for buildings with importance factor greater than unity ($I > 1.0$) in seismic zone III.

18.7.1.2 The provision in this section apply for diagonal and X-bracing only. Specialist literature may be consulted for V and inverted V-type bracing. K-bracing shall not be permitted in systems to resist earthquake.

18.7.2 Bracing Members

18.7.2.1 The slenderness of bracing members shall not exceed 120.

18.7.2.2 The required compressive strength of bracing member shall not exceed 0.8 times P_d , where P_d is the design strength in axial compression (see [14.1.2](#)).

18.7.2.3 Along any line of bracing, braces shall be provided such that for lateral loading in either direction, the tension braces will have to resist between 30 to 70 percent of the total lateral load.

18.7.2.4 Bracing cross-section can be plastic, compact or semi-compact, but not slender, as defined in [10.8.2](#).

18.7.2.5 For all built-up braces, the spacing of tack fasteners shall be such that the unfavorable slenderness ratio of individual element, between such fasteners, shall not exceed 0.4 times the governing slenderness ratio of the brace itself. Bolted connections shall be avoided within the middle one-fourth of the clear brace length (0.25 times the length in the middle).

18.7.2.6 The bracing members shall be designed so that gross area yielding (see [13.2](#)) and not the net area rupture (see [13.3](#)) would govern the design tensile strength.

18.7.3 Bracing Connections

18.7.3.1 End connections in bracings shall be designed to withstand the minimum of the following:

- a) A tensile force in the bracing equal to $1.2 f_y A_g$;
- b) Force in the brace due to load combinations in [18.2.3](#); and
- c) Maximum force that can be transferred to the brace by the system.

18.7.3.2 The connection should be checked for tension rupture and block shear under the load determined in [18.7.3.1](#).

18.7.3.3 The connection shall be designed to withstand should be shown to withstand inelastic deformation a moment of 1.2 times the full plastic moment of the corresponding to a joint rotation of at least 0.02 radians braced section about the buckling axis.

18.7.3.4 Gusset plates shall be checked for buckling out of their plane.

18.8 Special Concentrically Braced Frames (SCBF)

18.8.1 Special concentrically braced frames (SCBF) should be shown to withstand inelastic deformation corresponding to a joint rotation of at least 0.04 radians without degradation in strength and stiffness below the full yield value. Special concentrically braced frames meeting the requirements of this section shall be deemed to satisfy the required inelastic deformation.

18.8.1.1 Special concentrically braced frames (SCBF) may be used in any seismic zone as defined in 5 of Part C 'Structural design/Section 1 Loads and Load Effects' of this NBCS and for any building.

18.8.1.2 The provision in this section apply for diagonal and X-bracing only. Specialist literature may be consulted for V and inverted V-type bracing. K-bracing shall not be permitted in system to resist earthquake.

18.8.2 Bracing Members

18.8.2.1 Bracing members shall be made of E 250B steel of accepted standard [C-6A(6)] only.

18.8.2.2 The slenderness of bracing members shall not exceed 160 (only hangers).

18.8.2.3 The required compressive strength of bracing member shall not exceed the design strength in axial compression P_d (see [14.1.2](#)).

18.8.2.4 Along any line of bracing, braces shall be provided such that for lateral loading in either direction, the tension braces will resist between 30 percent to 70 percent of the load.

18.8.2.5 Braced cross-section shall be plastic as defined in [10.8.2](#).

18.8.2.6 In built-up braces, the spacing of tack connections shall be such that the slenderness ratio individual element between such connections shall not exceed 0.4 times the governing slenderness ratio of the brace itself. Bolted connection shall be avoided.

18.8.3.4 Gusset plates shall be checked for buckling out of their plane.

18.8.4 Column

18.8.4.1 The column sections used in special concentrically braced frames (SCBF) shall be plastic as defined in [10.8.2](#).

18.8.4.2 Splices shall be located within the middle one-third of the column clear height. Splices shall be designed for the forces that can be transferred to it. In addition, splices in columns shall be designed to develop at least the nominal shear strength of the smaller connected section.

18.9 Eccentrically Braced Frames (EBF)

18.9.1 Ordinary eccentrically braced frames (OEBF) and Special eccentrically braced frames (SEBF) shall be designed in accordance with specialist literature.

18.10 Ordinary Moment Frames (OMFs)

Ordinary moment frames (OMFs) of structural steel shall be designed to satisfy the requirements of this section.

18.10.1 Basis of Design

Ordinary moment frames designed in accordance with these provisions are expected to possess limited inelastic deformation capacity through flexural yielding of the beams, panel zones, and columns.

18.10.2 Analysis

When two moment frames oriented in orthogonal directions intersect, the possibility of beam yielding in both orthogonal directions simultaneously shall be considered in the estimation of design demand on the common column.

18.10.3 Member Requirements

The requirements specified hereunder shall be satisfied by the component or member.

18.10.3.1 Beams

Beam cross-section shall be plastic or compact, as defined in [10.8.2](#).

18.10.3.2 Columns

Column cross-section shall be plastic or compact, as defined in [10.8.2](#).

18.10.4 System Requirements

It is preferable to plan buildings to have independent lateral load resisting moment frames in the vertical plane along each principal plan direction. Also, the requirements given hereunder shall be satisfied by the building system.

18.10.4.1 Beam-to-column connections

Beam-to-column connections may be rigid or semi-rigid moment connections, but shall satisfy the following criteria:

- a) Rigid moment connections shall be designed to withstand a moment of at least 1.2 times of either the full plastic moment of the connected beam or the maximum moment that can be delivered by the beam to the joint due to the induced weakness at the ends of the beam, whichever is less.
- b) Semi-rigid connections shall be designed and to withstand either a moment of at least 0.5 times the full plastic moment of the connected beam or the maximum moment that can be delivered by the system, whichever is less. The design moment shall be achieved within a rotation of 0.01 radians. The information given in [Annex F](#) may be used for checking.
- c) The stiffness and strength of semi-rigid connections shall be accounted for in the design and the overall stability of the frame shall be ensured.
- d) The rigid and semi-rigid connections shall be designed to withstand a shear resulting from the load combination $1.2 \text{ DL} + 0.5 \text{ LL}$ plus the shear corresponding to the design moment defined in **(a)** and **(b)**, respectively.
- e) In rigid fully welded connections, continuity plates of thickness equal to or greater than the thickness of the beam flange shall be provided and welded to the column flanges and web.

18.10.4.2 Column to beam strength ratio

At a beam-column joint, strong column - weak beam design shall be adopted, suitably accounting for the reduction in moment capacity of columns due to axial load.

18.11 Special Moment Frames (SMF)

18.11.1 Special moment frames (SMF) shall be made of E250B steel of accepted standard [C-6A(6)] and should be shown to withstand inelastic deformation corresponding to a joint rotation of 0.04 radians without degradation in strength and stiffness below the full yield value (M_p). Special moment frames meeting the requirements of this section shall be deemed to satisfy the required inelastic deformation.

18.11.1.1 Special moment frames (SMF) maybe used in any seismic zone and for any buildings.

18.11.2 Beam-to-Column Joints and Connections

18.11.2.1 All beam-to-column connections shall be rigid (see [Annex F](#)) and designed to withstand a moment of at least 1.2 times the full plastic moment of the connected beam. When a reduced beam section is used, its minimum flexural strength shall be at least equal to 0.8 times the full plastic moment of the unreduced section.

18.11.2.2 The connection shall be designed to withstand a shear resulting from the load combination $1.2 \text{ DL} + 0.5 \text{ LL}$ plus the shear resulting from the application of $1.2M_p$ in the same direction, at each end of the beam (causing double curvature bending). The shear strength need not exceed the required value corresponding to the load combination in [18.2.3](#).

18.11.2.3 In column strong axis connections (beam and column web in the same plane), the panel zone shall be checked for shear buckling in accordance with [15.4.2](#) at the design shear defined in [18.11.2.2](#). Column web doubler plates or diagonal stiffeners may be used to strengthen the web against shear buckling.

18.11.2.4 The individual thickness of the column webs and doubler plates (see [Fig. 21](#)), shall satisfy the following:

$$t \geq (d_p + b_p)/90$$

where

t = Thickness of the column web or doubler plate

d_p = Panel zone depth between continuity plates

b_p = Panel zone width between column flanges

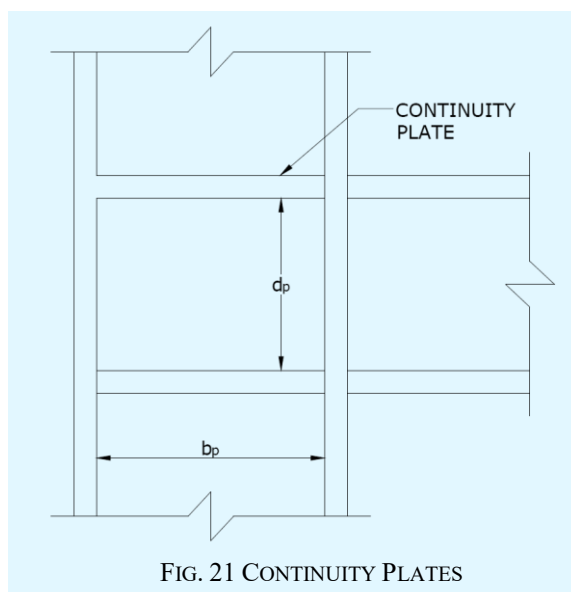


FIG. 21 CONTINUITY PLATES

18.11.2.5 Continuity plates (tension stiffener) (see [15.7](#)) shall be provided in all strong axis welded connections except in end plate connection.

18.11.3 Beam and Column Limitation

18.11.3.1 Beam and column sections shall be either plastic or compact as defined in [10.8.2](#). At potential plastic hinge locations, they shall necessarily be plastic.

18.11.3.2 The section selected for beams and columns shall satisfy the following relation:

$$\frac{\sum M_{Pc}}{\sum M_{Pb}} \geq 1.2$$

where

M_{Pc} = sum of the moment capacity in the column above and below the beam centreline; and

M_{Pb} = sum of the moment capacity in the beams at the intersection of the beam and column centrelines.

In tall buildings, higher mode effects shall be accounted for in accordance with specialist literature.

18.11.3.3 Lateral support to the column at both top and bottom beam flange levels shall be provided so as to resist at least 2 percent of the beam flange strength, except for the case described in [18.11.3.4](#).

18.11.3.4 A plane frame designed as non-sway in the direction perpendicular to its plane, shall be checked for buckling, under the load combination specified in [18.2.3](#).

18.12 Column Bases

18.12.1 Fixed column bases and their anchor bolts should be designed to withstand a moment of 1.2 times the full plastic moment capacity of the column section. The anchor bolts shall be designed to withstand the combined action of shear and tension as well as prying action, if any.

18.12.2 Both fixed and hinged column bases shall be designed to withstand the full shear under any load case or 1.2 times the shear capacity of the column section, whichever is higher.

18.13 Industrial Structures

18.13.1 Single storey industrial buildings with frame height to span ratio less than 1.0 and with regular configuration need not meet the ductility requirements of other types of frames specified in [18.3](#), [18.4](#) or in good practice[C-6A(42)], provided they satisfy the requirements in [18.13.2](#) and [18.13.3](#).

18.13.2 Braced frames or moment frames are permitted as lateral force resisting system and shall be designed using $R = 1.5$ as prescribed for NDF.

18.13.3 Connections in lateral force resisting systems to resist earthquake shall be designed for full member capacity except those in earthquake zone V, where the connections shall be designed for 50 percent extra force.

19 FATIGUE

19.1 General

Structure and structural elements subject to loading that could lead to fatigue failure shall be designed against fatigue as given in this section. This shall however not cover the following:

- a) Corrosion fatigue;
- b) Low cycle (high stress) fatigue;
- c) Thermal fatigue;
- d) Stress corrosion cracking;
- e) Effects of high temperature ($> 150\text{ }^{\circ}\text{C}$); and
- f) Effects of low temperature ($<$ brittle transition temperature).

For the purpose of design against fatigue, different details (of members and connections) are classified under different fatigue class. The design stress range corresponding to various number of cycles, are given for each fatigue class. The requirements of this Section shall be satisfied with, at each critical location of the structure subjected to cyclic loading, considering relevant number of cycles and magnitudes of stress range expected to be experienced during the life of the structure.

19.2 Design

19.2.1 Reference Design Condition

The standard S-N curves for each detail category are given for the following conditions:

- a) Detail is located in a redundant load path, wherein local failure at that detail alone will not lead to overall collapse of the structure;
- b) Nominal stress history at the local point in the detail is estimated/evaluated by a conventional method without taking into account the local stress concentration effects due to the detail;
- c) Load cycles are not highly irregular;
- d) Details are accessible for and subject to regular inspection;
- e) Structure is exposed to only mildly corrosive environment as in normal atmospheric condition and suitably protected against corrosion (pit depth $< 1\text{ mm}$);

- f) Structure is not subjected to temperature exceeding 150 °C; and
- g) Transverse fillet or butt weld connects plates of thickness not greater than 25 mm.

19.2.1.1 Thickness correction factor

The values obtained from the standard S-N curve shall be modified by a capacity reduction factor μ_t , when plates greater than 25 mm in thickness are joined together by transverse fillet or butt welding, given by:

$$\mu_t = (25/t_p)^{0.25} \leq 1.0$$

where

t_p = actual thickness of the thicker plate being joined, in mm.

No thickness correction is necessary when full penetration butt weld reinforcements are machined flush and proved free of defect through nondestructive testing.

19.2.1.2 Necessity for fatigue assessment

- a) Fatigue assessment is not normally required for building structures except as follows; and
 - 1) Members supporting lifting or rolling loads;
 - 2) Member subjected to repeated stress cycles from vibrating machinery;
 - 3) Members subjected to wind induced oscillations of a large number of cycles in life; and
 - 4) Members subjected to crowd induced oscillations of a large number of cycles in life.
- b) No fatigue assessment is necessary if the following conditions are satisfied.

- 1) The highest normal stress range $f_{t\max}$ satisfies:

$$f_{t\max} \leq 27 \mu_r / \gamma_{mf}$$

- 2) The highest shear stress range $\tau_{t\max}$ satisfies:

$$\tau_{t\max} \leq 67 \mu_r / \gamma_{mf}$$

- 3) The total number of actual stress cycles N_{SC} , satisfies:

$$N_{SC} \leq 5 \times 10^6 \left(\frac{27 \mu_r}{\gamma_{mf} f_{feq}} \right)^3$$

where

f_{feq} = equivalent constant amplitude stress range, in MPa, given by.

$$f_{feq} = \left[\frac{\sum_{i=1}^{r5} n_i f_i^3 + \sum_{j=r5}^r n_j f_j^5}{n} \right]^{1/3}$$

where

f_i, f_j = stress ranges falling above and below the f_{fn} , the stress range corresponding to the detail at 5×10^6 number of life cycles.

$$n = \sum_{i=1}^r n_i$$

19.2.2 Design Spectrum

19.2.2.1 Stress evaluation

Design stress shall be determined by elastic analysis of the structure to obtain stress resultants and the local stresses may be obtained by a conventional stress analysis method. The normal and shear stresses shall be determined considering all design actions on the members, but excluding stress concentration due to the geometry of the detail. The stress concentration effect is accounted for in detail category classification (see [Table 26](#)). The stress concentration, however, not characteristic of the detail shall be accounted for separately in the stress calculation.

In the fatigue design of trusses made of members with open sections, in which the end connections are not pinned, the stresses due to secondary bending moments shall be taken into account, unless the slenderness ratio (KL/r), of the member is greater than 40.

In the determination of stress range at the end connections between hollow sections, the effect of connection stiffness and eccentricities may be disregarded, provided:

- the calculated stress range is multiplied by appropriate factor given in [Table 24\(A\)](#) in the case of circular hollow section connections and [Table 24\(B\)](#) in the case of rectangular hollow section connections; and
- the design throat thickness of fillet welds in the joints is greater than the wall thickness of the connected member.

19.2.2.2 Design stress spectrum

In the case of loading events producing non-uniform stress range cycle, the stress spectrum may be obtained by a rational method, such as 'rain flow counting' or an equivalent method.

**Table 24(A) Multiplying Factors for Calculated Stress Range
(Circular Hollow Sections)**

(Clause [19.2.2.1](#))

Sl No.	Type of Connection		Chords	Verticals	Diagonals
(1)	(2)		(3)	(4)	(5)
i)	Gap connections	{ K type	1.5	1.0	1.3
		{ N type	1.5	1.8	1.4
ii)	Overlap connections	{ K type	1.5	1.0	1.2
		{ N type	1.5	1.65	1.25

**Table 24(B) Multiplying Factors for Calculated Stress Range
(Rectangular Hollow Sections)**

(Clause [19.2.2.1](#))

Sl No.	Types of Joint		Chords	Verticals	Diagonals
(1)	(2)		(3)	(4)	(5)
i)	Gap connections	{ K type	1.5	1.0	1.5
		{ N type	1.5	2.2	1.6
ii)	Overlap connections	{ K type	1.5	1.0	1.3
		{ N type	1.5	2.0	1.4

19.2.3 Partial Safety Factors

19.2.3.1 Partial safety factor for actions and their effects (γ_{fi})

Unless and otherwise the uncertainty in the estimation of the applied actions and their effects demand a higher value, the partial safety factor for loads in the evaluation of stress range in fatigue design shall be taken as 1.0.

19.2.3.2 Partial safety factor for fatigue strength (γ_{mft})

Partial safety factor for strength is influenced by consequences of fatigue damage and level of inspection capabilities.

19.2.3.3 Based on consequences of fatigue failure, component details have been classified as given in the [Table 25](#) and the corresponding partial safety factor for fatigue strength shall be used:

- failsafe structural component/detail is the one where local failure of one component due to fatigue crack does not result in the failure of the structure due to availability of alternate load path (redundant system); and
- non-failsafe structural component/detail is the one where local failure of one component leads rapidly to failure of the structure due to its non-redundant nature.

Table 25 Partial Safety Factors for Fatigue Strength (γ_{mft}) <i>(Clauses 19.2.3.3, 19.5.1 and 19.5.2.3)</i>					
SI No.	Inspection and Access			Consequence of Failure	
				Failsafe	Non-failsafe
(1)	(2)			(3)	(4)
i)	Periodic inspection, maintenance and accessibility to detail is good			1.00	1.25
ii)	Periodic inspection, maintenance and accessibility to detail is poor			1.15	1.35

19.3 Detail Category

[Table 26\(A\)](#) to [Table 26 \(D\)](#) indicate the classification of different details into various categories for the purpose of assessing fatigue strength. Details not classified in the table may be treated as the lowest detail category of a similar detail, unless superior fatigue strength is proved by testing and/or analysis.

Holes in members and connections subjected to fatigue loading shall not be made:

- using punching in plates having thickness greater than 12 mm unless the holes are sub-punched and subsequently reamed to remove the affected material around the punched hole; and
- using gas cutting unless the holes are reamed to remove the material in the heat affected zone.

19.4 Fatigue Strength

The fatigue strength of the standard detail for the normal or shear fatigue stress range, not corrected for thickness effects discussed in [19.2.1](#), is given below (see also [Fig. 22](#) and [Fig. 23](#)):

- Normal stress range:

- when $N_{sc} \leq 5 \times 10^6$

$$f_t = f_{fn} \sqrt[3]{5 \times 10^6 / N_{sc}}$$

- when $5 \times 10^6 \leq N_{sc} \leq 10^8$

$$f_t = f_{fn} \sqrt[5]{5 \times 10^6 / N_{sc}}$$

b) Shear stress:

$$\tau_t = \tau_{fn} \sqrt[5]{5 \times 10^6 / N_{sc}}$$

where

f_t, τ_t = design normal and shear fatigue stress range of the detail, respectively, for life cycle of N_{sc} ; and

f_{fn}, τ_{fn} = normal and shear fatigue strength of the detail for 5×10^6 cycles, for the detail category (see [Table 26](#)).

Table 26(A) Detail Category Classification, Group 1 Non-Welded Details (f_m)

(Clauses [19.2.2.1](#), [19.3](#) and [19.4](#))

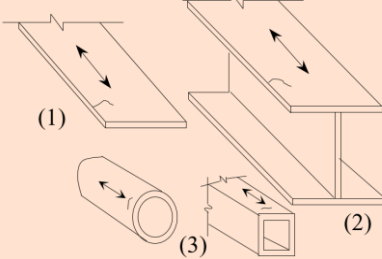
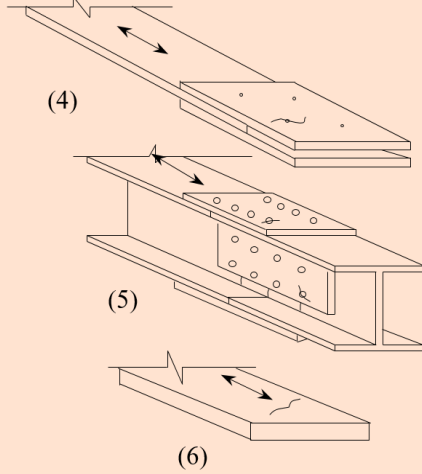
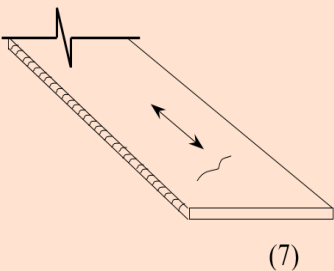
Sl No.	Detail Category	Constructional Details	
		Illustration (see Note)	Description
(1)	(2)	(3)	(4)
i)	118		Rolled and extruded products: a) Plates and flats (1); b) Rolled sections (2); and c) Seamless tubes (3) Sharp edges, surface and rolling flaws to be removed by grinding in the direction of applied stress
ii)	103		a) Bolted connections: (4) and (5) — Stress range calculated on the gross section and on the net section Unsupported one-sided cover plate connections shall be avoided or the effect of the eccentricity taken into account in calculating stresses b) Material with gas-cut or sheared edges with no draglines: (6) — All hardened material and visible signs of edge discontinuities to be removed by machining or grinding in the direction of applied stress
iii)	92		c) Material with machine gas-cut edges with draglines or manual gas-cut material: (7) — Corners and visible signs of edge discontinuities to be removed by grinding in the direction of the applied stress
NOTE — The arrow indicates the location and direction of the stresses acting in the basic material for which the stress range is to be calculated on a plane normal to the arrow.			

Table 26(B) Detail Category Classification, Group 2 Welded Details - Not in Hollow Sections (f_m)

(Clauses [19.2.2.1](#), [19.3](#) and [19.4](#))

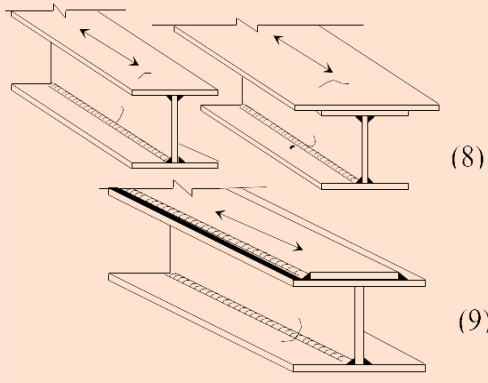
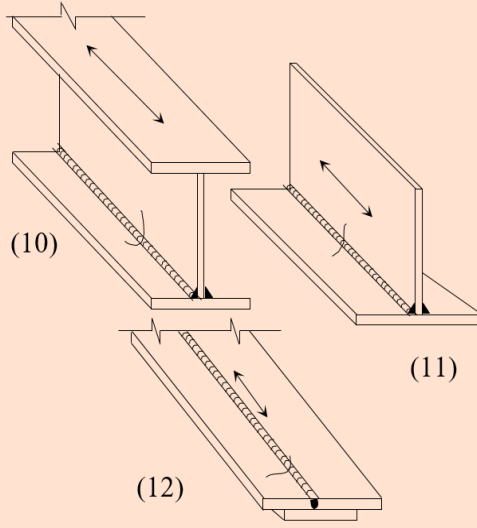
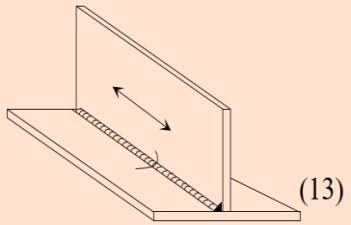
Sl No.	Detail Category	Constructional Details	
		Illustration (see Note)	Description
(1)	(2)	(3)	(4)
i)	92	 (8) and (9) — Zones of continuous automatic longitudinal fillet or butt welds carried out from both sides and all welds not having un-repaired stop-start positions	Welded plate I-section and box girders with continuous longitudinal welds: (8) and (9) — Zones of continuous automatic longitudinal fillet or butt welds carried out from both sides and all welds not having un-repaired stop-start positions
ii)	83	 (10) and (11) — Zones of continuous automatic butt welds made from one side only with a continuous backing bar and all welds not having un-repaired stop-start positions (12) — Zones of continuous longitudinal fillet or butt welds carried out from both sides but containing stop-start positions. For continuous manual longitudinal fillet or butt welds carried out from both sides, use Detail Category 92	Welded plate I-section and box girders with continuous longitudinal welds: (10) and (11) — Zones of continuous automatic butt welds made from one side only with a continuous backing bar and all welds not having un-repaired stop-start positions (12) — Zones of continuous longitudinal fillet or butt welds carried out from both sides but containing stop-start positions. For continuous manual longitudinal fillet or butt welds carried out from both sides, use Detail Category 92
iii)	66	 (13) — Zones of continuous longitudinal welds carried out from one side only, with or without stop-start positions	Welded plate I-section and box girders with continuous longitudinal welds: (13) — Zones of continuous longitudinal welds carried out from one side only, with or without stop-start positions

Table 26(B) (Continued)

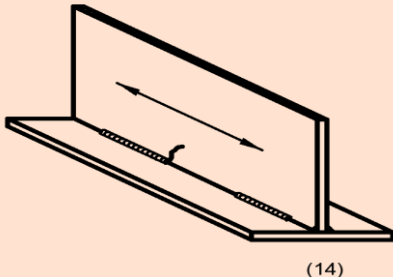
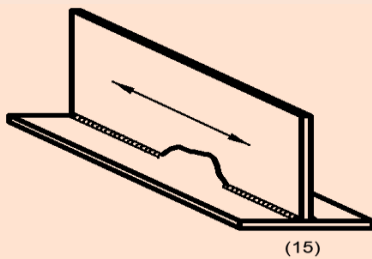
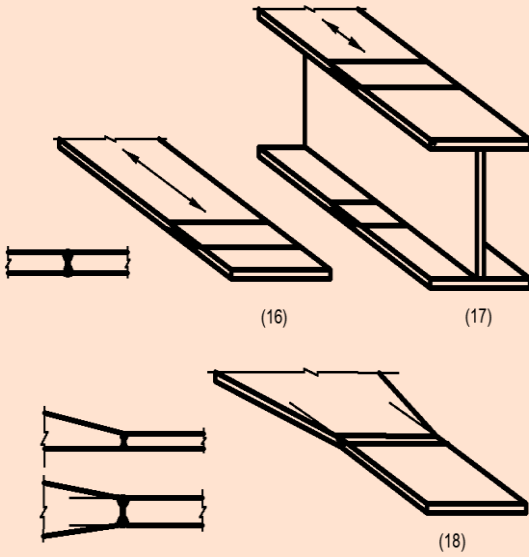
Sl No.	Detail Category	Constructional Details	
		Illustration (see Note)	Description
(1)	(2)	(3)	(4)
iv)	59	 (14)	Intermittent longitudinal welds: (14) — Zones of intermittent longitudinal welds
v)	52	 (15)	Intermittent longitudinal welds: (15) — Zones containing cope holes in longitudinally welded T-joints. Cope hole not to be filled with weld
vi)	83	 (16) (17) (18)	Transverse butt welds (complete penetration): Weld run-off tabs to be used, subsequently removed and ends of welds ground flush in the direction of stress. Welds to be made from two sides (16) — Transverse splices in plates, flats and rolled sections having the weld reinforcement ground flush to plate surface. 100 percent NDT inspection, and weld surface to be free of exposed porosity in the weld metal (17) — Plate girders welded as in (16) before assembly (18) — Transverse splices as in (16) with reduced or tapered transition with taper $\leq 1 : 4$

Table 26(B) (Continued)

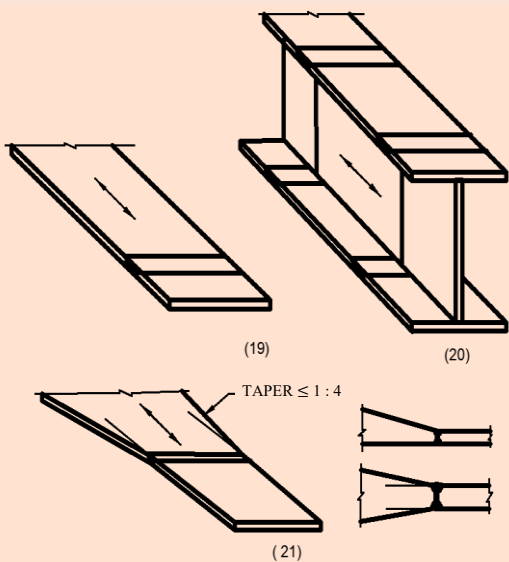
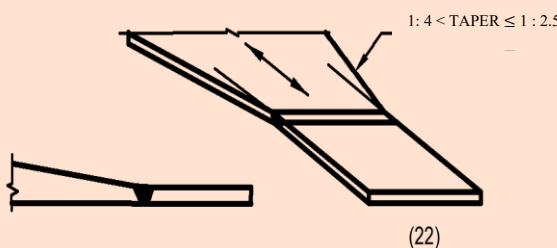
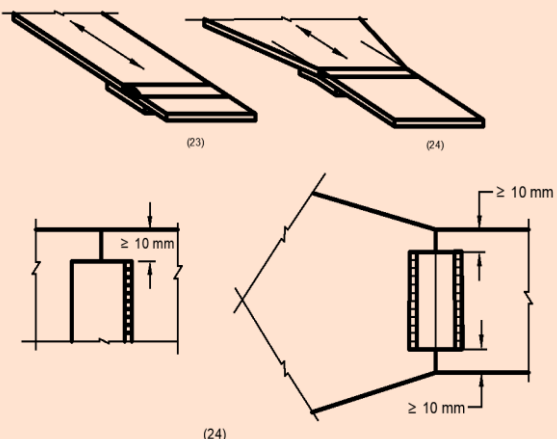
Sl No.	Detail Category	Constructional Details	
		Illustration (see Note)	Description
(1)	(2)	(3)	(4)
vii)	66	 (19) — Transverse splices of plates, rolled sections or plate girders (20) — Transverse splice of rolled sections or welded plate girders, without cope hole. With cope hole use Detail Category 52, as in (15) (21) — Transverse splices in plates or flats being tapered in width or in thickness where the taper is $\leq 1 : 4$	Transverse butt welds (complete penetration): Welds run-off tabs to be used, subsequently removed and ends of welds ground flush in the direction of stress. Welds to be made from two sides
viii)	59	 (22)	Transverse butt welds (complete penetration): Weld run-off tabs to be used, subsequently removed and ends of welds ground flush in the direction of stress. Welds to be made from two sides (22) — Transverse splices as in (21) with taper in width or thickness $> 1 : 4$ but $\leq 1 : 2.5$
ix)	52	 (23) (24)	Transverse butt welds (complete penetration): (23) — Transverse butt-welded splices made on a backing bar. The end of the fillet weld of the backing strip shall stop short by more than 10 mm from the edges of the stressed plate (24) — Transverse butt welds as per (23) with taper on width or thickness $< 1 : 2.5$

Table 26(B) (Continued)

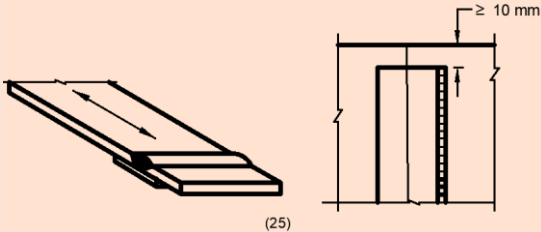
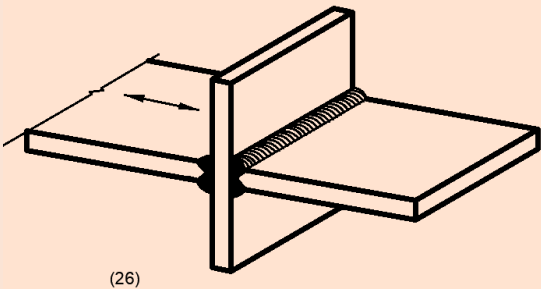
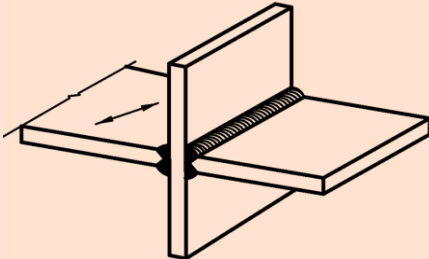
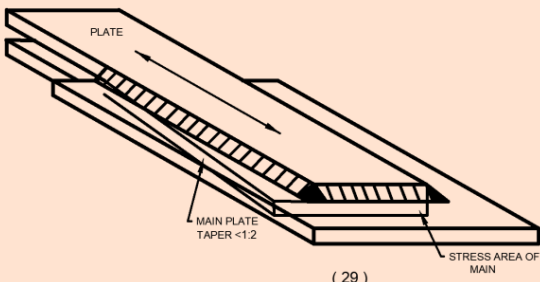
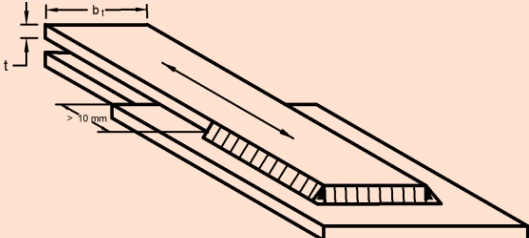
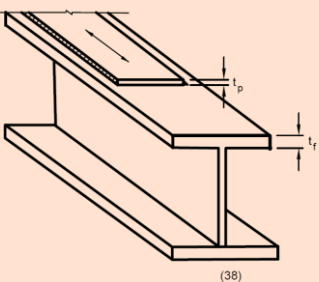
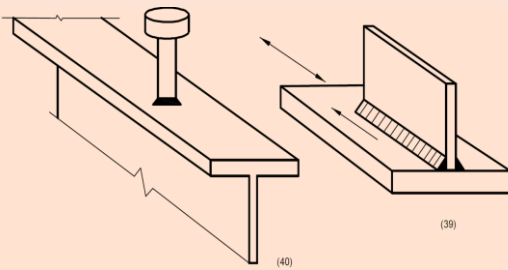
Sl No.	Detail Category		Constructional Details	
			Illustration (see Note)	Description
(1)	(2)		(3)	(4)
x)	37		 (25)	Transverse butt welds (complete penetration): (25) — Transverse butt welds as in (23) where fillet welds end closer than 10 mm to plate edge
xi)	52		 (26)	Cruciform joints with load-carrying welds: (26) — Full penetration welds with intermediate plate NDT inspected and free of defects. Maximum misalignment of plates either side of joint to be < 0.15 times the thickness of intermediate plate
xii)	41	(27)	 (27) and (28)	(27) — Partial penetration or fillet welds with stress range calculated on plate area
	27	(28)		(28) — Partial penetration or fillet welds with stress range calculated on throat area of weld
xiii)	46		 (29)	Overlapped welded joints: (29) — Fillet welded lap joint, with welds and overlapping elements having a design capacity greater than the main plate. Stress in the main plate to be calculated on the basis of area shown in the illustration
xiv)	41	(30)	 (30) and (31)	Overlapped weld joints: (30) — Fillet welded lap joint, with welds and main plate both having a design capacity greater than the overlapping elements
xv)	33	31		31 — Fillet welded lap joint, with main plate and overlapping elements both having a design capacity greater than the weld

Table 26(B) (Continued)

Sl No.	Detail Category	Constructional Details			
		Illustration (see Note)		Description	
(1)	(2)	(3)		(4)	
xvi)	66	(32)	(33)		Welded attachments (non-load carrying welds) — Longitudinal welds: (32) — Longitudinal fillet welds. Class of detail varies according to the length of the attachment weld as noted (33) — Gusset welded to the edge of a plate or beam flange. Smooth transition radius (r), formed by machining or flame cutting plus grinding. Class of detail varies according to r/b ratio as noted
		—	$1/3 \leq r/b$		
	59	$l \leq 50\text{mm}$	—		
	52	$50 < l \leq 100\text{ mm}$	$1/6 \leq r/b < 1/3$		
	37	$100\text{ mm} < l$	—		
	33	—	$r/b < 1/6$		
xvii)	59			Welded attachments: (34) — Shear connectors on base material (failure in base material)	
		(34)			
xviii)	59	$t \leq 12\text{ mm}$		Transverse welds: (35) — Transverse fillet welds with the end of the weld $\geq 10\text{ mm}$ from the edge of the plate (36) — Vertical stiffeners welded to a beam or plate girder flange or web by continuous or intermittent welds. In the case of webs carrying combined bending and shear design actions, the fatigue strength shall be determined using the stress range of the principal stresses (37) — Diaphragms of box girders welded to the flange or web by continuous or intermittent welds	
	52	$t > 12\text{ mm}$			

Table 26(B) (Continued)

Sl No.	Detail Category	Constructional Details		
		Illustration (see Note)	Description	
(1)	(2)	(3)		(4)
xix)	37	$t_f \text{ or } t_p \leq 25\text{mm}$		Cover plates in beams and plate girders:
	27	$t_f \text{ or } t_p > 25\text{ mm}$		(38) — End zones of single or multiple welded cover plates, with or without a weld across the end. For a reinforcing plate wider than the flange, an end weld is essential
xx)	67			Welds loaded in shear:
				(39) — Fillet welds transmitting shear. Stress range to be calculated on weld throat area
				(40) — Stud welded shear connectors (failure in the weld) loaded in shear (the shear stress range to be calculated on the nominal section of the stud)

NOTE — The arrow indicates the location and direction of the stresses acting in the basic material for which the stress range is to be calculated on a plane normal to the arrow.

Table 26(C) Detail Category Classification, Group 3 Bolts

(Clauses [19.2.2.1](#), [19.3](#) and [19.4](#))

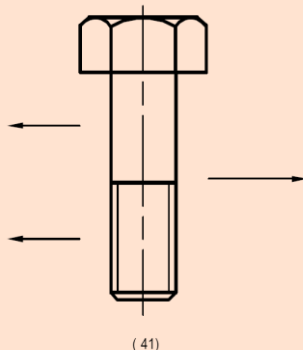
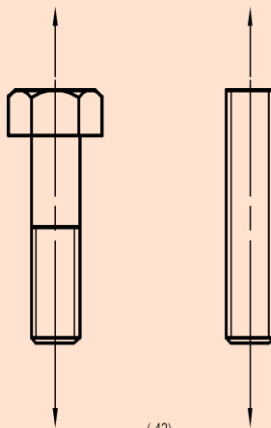
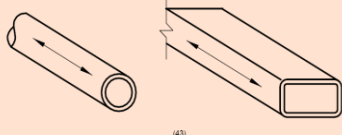
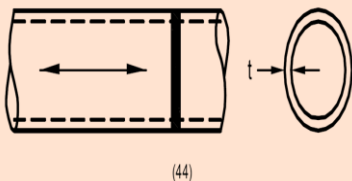
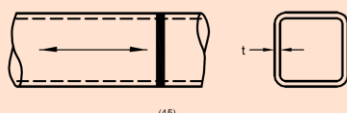
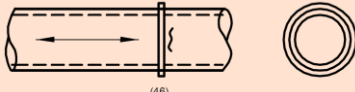
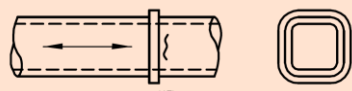
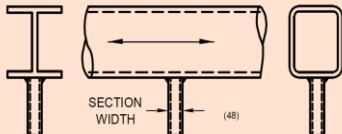
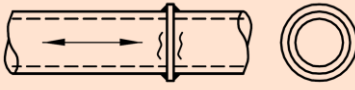
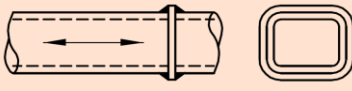
Sl No.	Detail Category	Constructional Details	
		Illustration (see Note)	Description
(1)	(2)	(3)	(4)
i)	83	 (41)	Bolts in shear (8.8/TB Bolting category only): (41) — Shear stress range calculated on the minor diameter area of the bolt (A_c) NOTE — If the shear on the joint is insufficient to cause slip of the joint the shear in the bolt need not be considered in fatigue
ii)	27	 (42)	Bolts and threaded rods in tension (tensile stress to be calculated on the tensile stress area, A_t): (42) — Additional forces due to prying effects shall be taken into account. For tensional bolts, the stress range depends on the connection geometry NOTE — In connections with tensioned bolts, the change in the force in the bolts is often less than the applied force, but this effect is dependent on the geometry of the connection. It is not normally required that any allowance for fatigue be made in calculating the required number of bolts in such connections
NOTE — The arrow indicates the location and direction of the stresses acting in the basic material for which the stress range is to be calculated on a plane normal to the arrow.			

Table 26(D) Detail Category Classification, Group 4 Welded details — in Hollow Sections (f_{in})

(Clauses [19.2.2.1](#), [19.3](#) and [19.4](#))

Sl No.	Detail Category	Constructional Details	
		Illustration (see Note)	Description
(1)	(2)	(3)	(4)
i)	103		Continuous automatic longitudinal welds: (43) — No stop-starts, or as manufactured, proven free to detachable discontinuities.
ii)	66 ($t \geq 8$ mm)		Transverse butt welds: (44) — Butt-welded end-to-end connection of circular hollow sections NOTE — Height of the weld reinforcement less than 10 percent of weld with smooth transition to the plate surface. Welds made in flat position and proven free to detachable discontinuities)
	52 ($t < 8$ mm)		
iii)	52 ($t \geq 8$ mm)		(45) — Butt-welded end-to-end connection of rectangular hollow sections
	41 ($t < 8$ mm)		
iv)	41 ($t \geq 8$ mm)		Butt welds to intermediate plate: (46) — Circular hollow sections, end-to-end butt-welded with an intermediate plate
	37 ($t < 8$ mm)		
v)	37 ($t \geq 8$ mm)		(47) — Rectangular hollow sections, end-to-end butt welded with an intermediate plate
	30 ($t < 8$ mm)		
vi)	52		Welded attachments (non-load-carrying): (48) — Circular or rectangular hollow section, fillet welded to another section. Section width parallel to stress direction ≤ 100 mm
vii)	33 ($t < 8$ mm)		Fillet welds to intermediate plate: (49) — Circular hollow sections, end-to-end fillet welded with an intermediate plate
	29 ($t < 8$ mm)		
viii)	29 ($t \geq 8$ mm)		(50) — Rectangular hollow sections, end-to-end fillet welded with an intermediate plate
	27 ($t < 8$ mm)		
NOTE — The arrow indicates the location and direction of the stresses acting in the basic material for which the stress range is to be calculated on a plane normal to the arrow.			

19.5 Fatigue Assessment

19.5.1 The design fatigue strength for N_{SC} life cycles (f_{fd} , τ_{fd}) may be obtained from the standard fatigue strength for N_{SC} cycles by multiplying with correction factor, μ_r , for thickness, as mentioned in [19.2.1](#) and dividing by partial safety factor given in [Table 25](#).

19.5.2 Stress Limitations

19.5.2.1 The maximum (absolute) value of the normal and shear stresses shall never exceed the elastic limit (f_y , τ_y) for the material under cyclic loading.

19.5.2.2 The maximum stress range shall not exceed $1.5 f_y$ for normal stresses and $1.5 f_y / \sqrt{3}$ for the shear stresses under any circumstance.

19.5.2.3 Constant stress range

The actual normal and shear stress range f and τ at a point of the structure subjected to N_{SC} cycles in life shall satisfy:

$$f \leq f_{fd} = \mu_r f_f / \gamma_{mft}$$

$$\tau \leq \tau_{fd} = \mu_r \tau_f / \gamma_{mft}$$

where

f_f , τ_f = normal and shear fatigue strength ranges for the actual life cycle, N_{SC} , obtained from [19.4](#);

μ_r = correction factor (see [19.2.1](#)); and

γ_{mf} = partial safety factor against fatigue failure, given in [Table 25](#).

19.5.2.4 Variable stress range

Fatigue assessment at any point in a structure, wherein variable stress ranges f_i or τ_i for n_i number of cycles ($i=1$ to r) are encountered, shall satisfy the following:

a) For normal stress (f):

$$\frac{\sum_{i=1}^{r_5} n_i f_i^3}{5 \times 10^6 (\mu_r f_{fn} / \gamma_{mft})^3} + \frac{\sum_{j=r_5}^r n_j f_j^5}{5 \times 10^6 (\mu_r f_{fn} / \gamma_{mft})^5} \leq 1.0$$

b) For shear stresses (τ):

$$\sum_{i=0}^r n_i \tau_i^5 \leq 5 \times 10^6 (\mu_r \tau_{fn} / \gamma_{mft})$$

Where r_5 is the summation upper limit of all the normal stress ranges (f_i) having magnitude lesser than $(\mu_r f_{fn} / \gamma_{mft})$ for that detail and the lower limit of all the normal stress ranges (f_j) having magnitude greater than $(\mu_r f_{fn} / \gamma_{mft})$ for the detail. In the above summation all normal stress ranges, f_i , and shear stress ranges τ_i having magnitude less than $0.55 \mu_r f_{fn}$, and $0.55 \mu_r \tau_{fn}$, respectively may be disregarded.

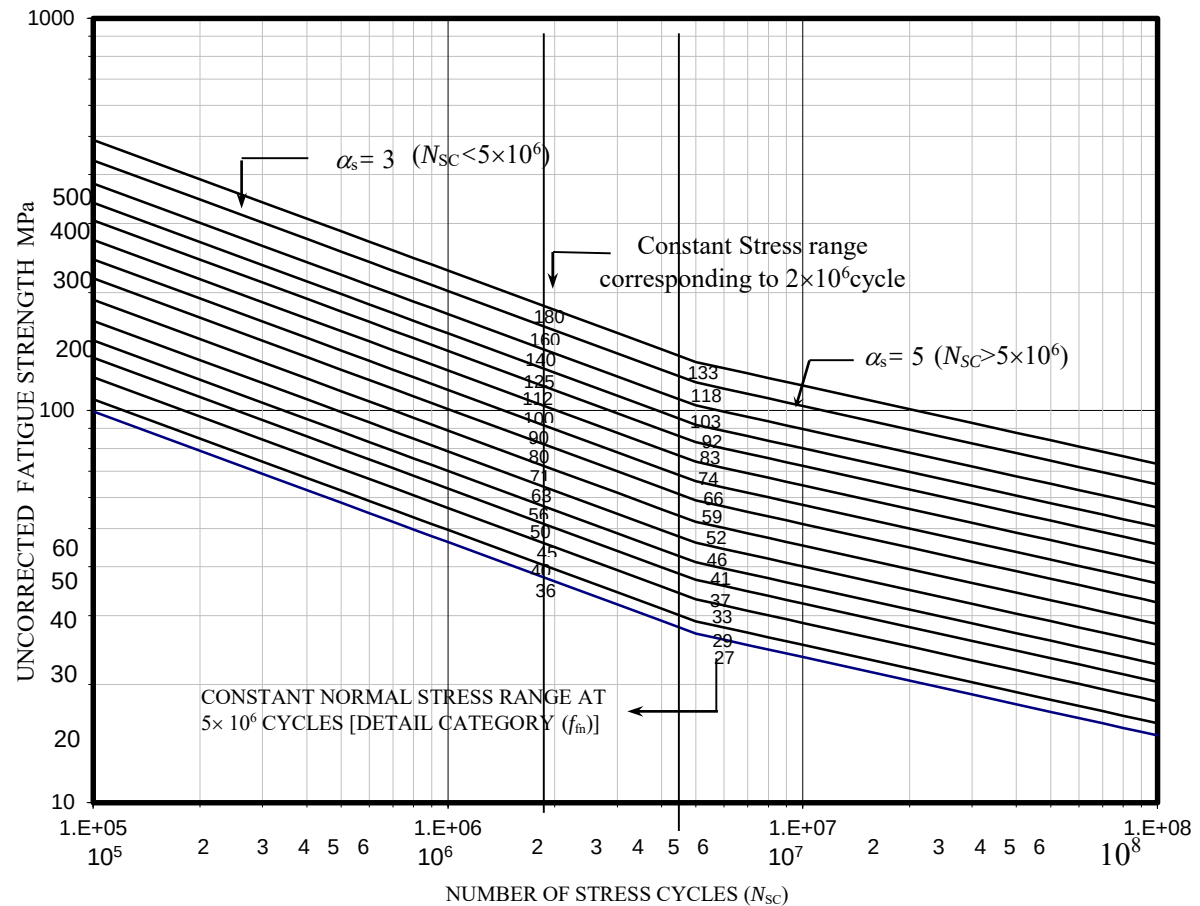


FIG. 22 S-N CURVE FOR NORMAL STRESS

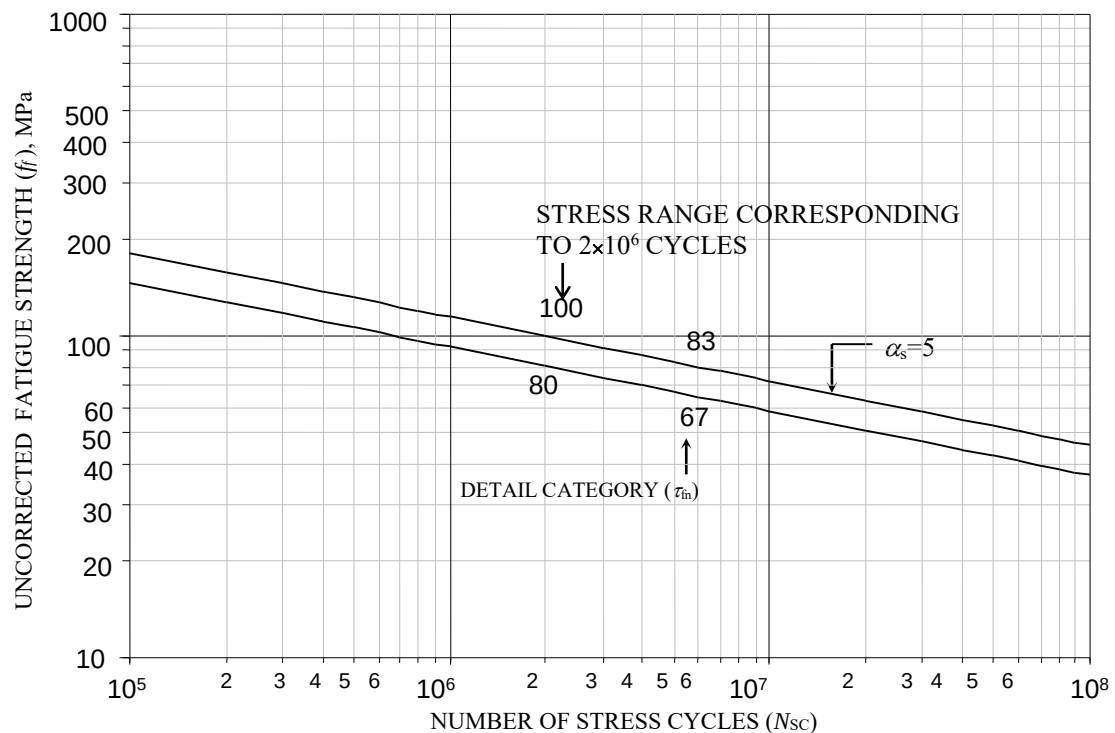


FIG. 23 S-N CURVE FOR SHEAR STRESS

20 DESIGN ASSISTED BY TESTING

20.1 Need for Testing

Testing of structures, members or components of structures is not required when designed in accordance with this standard. Testing may be accepted as an alternative to calculations or may become necessary in special circumstances.

Testing of a structural system, member or component may be required to assist the design in the following cases:

- when the calculation methods available are not adequate for the design of a particular structure, member or component, testing shall be undertaken in place of design by calculation or to supplement the design by calculation;
- where rules or methods for design by calculation would lead to uneconomical design, experimental verification may be undertaken to avoid conservative design;
- when the design or construction is not entirely in accordance with sections of this standard, experimental verification is recommended;
- when confirmation is required on the consistency of production of material, components, members or structures originally designed by calculations or testing; and
- when the actual performance of an existing structure capacity is in question, testing shall be used to confirm it.

20.1.1 Testing of structural system, member or component shall be of the following categories:

- Proof testing* — The application of test loads to a structure, sub-structure, member or connection to ascertain the structural characteristics of only that specific unit; and
- Prototype testing* — Testing of structures, sub-structures, members or connections is done to ascertain the structural characteristics of a class of such structures, sub-structures members or connection, which are nominally identical to the units tested.

20.2 Types of Test

20.2.1 Acceptance Test

This is intended as a non-destructive test for confirming structural performance. It should be recognized that the loading applied to certain structures might cause permanent distortions. Such effects do not necessarily indicate structural failure in acceptance test. However, the possibility of their occurrence should be agreed to before testing.

The load for acceptance test, $F_{\text{test,a}}$ shall be determined from:

$$F_{\text{test,a}} = (\text{Self weight}) + (1.15 \times \text{Remainder of the permanent load}) + (1.25 \times \text{Variable load})$$

The assembly shall satisfy the following criteria:

- it shall demonstrate substantially linear behaviour under test loading; and
- on removal of the test load, the residual deflection shall not exceed 20 percent of the maximum-recorded deflection.

If the above criteria are not satisfied the test may be repeated one more time only, when the assembly shall satisfy the following criteria:

- it shall demonstrate substantially linear behaviour on the second application of test loading; and
- corresponding recorded residual deflection in the second test shall not exceed 10 percent of the maximum deflection during the test.

20.2.2 Strength Test

Strength test is used to confirm the calculated resistance of a structure or component. Where a number of items are to be constructed to a common design, and one or more prototypes are tested to confirm their strength, the others may be accepted without any additional test, provided they are similar in all relevant respects to the prototype.

Before carrying out the strength test, the specimen should first be subjected to and satisfy the acceptance test. Since the resistance of the assembly under test depends on the material properties, the actual yield strength of all the steel materials in the assembly shall be determined from coupon {test piece as defined in the accepted standard [C-6A(7)]} tests. The mean value of the yield strength, f_{ym} , taken from such tests shall be determined with due regard to the importance of each element in the assembly. The strength test load $F_{test, s}$ (including self weight) shall be determined from:

$$F_{test, s} = \gamma_{mi} F_d (f_{ym} / f_y)$$

where

- F_d = factored design load for the ultimate limit state;
- f_y = characteristic yield stress of the material as assumed in the design; and
- γ_{mi} = partial safety factor for the type of failure, as prescribed in this standard.

At this load there shall be no failure by buckling or rupture of any part of the structure or component tested. On removal of the test load, the deflection should decrease by at least 20 percent of the maximum deflection at $F_{test, s}$.

20.2.3 Test to Failure (Ultimate Strength Test)

The objective of a test to failure is to determine the design resistance from the ultimate resistance. In this situation it is still desirable to carry out the acceptance and strength tests, before test to failure.

Not less than three tests shall be carried out on nominally identical specimens. An estimate should be made of the anticipated ultimate resistance as a basis for such tests. During a test to failure, the loading shall first be applied in increments up to the strength test load. Subsequent load increments shall then be determined from consideration of the principal load deflection plot. The test load resistance, $F_{test, R}$ shall be determined as that load at which the specimen is unable to sustain any further increase in load. At this load, gross permanent distortion is likely to have occurred and in some cases such large gross deformation may define the test limit. If the deviation of any individual test result exceeds 10 percent of the mean value obtained for all the three tests, at least three more tests shall be carried out. When the deviation from the mean does not exceed 10 percent of the mean, the design resistance may be evaluated as given below:

- a) When the failure is ductile, the design resistance, F_d may be determined from:

$$F_d = 0.9 F_{test, Min} (f_y / f_{ym}) / \gamma_{m0}$$

where

- $F_{test, Min}$ = minimum test result from the tests to failure;
- f_{ym} = average yield strength as obtained from the material tests; and
- f_y = characteristic yield stress of the grade of steel.

In the case of a sudden (brittle) rupture type failure, the design resistance may be determined from:

$$F_d = 0.9 F_{test, Min} (f_u / f_{um}) / \gamma_{m1}$$

where

- f_u = characteristic ultimate stress of the grade of steel used; and
- f_{um} = average ultimate tensile strength of the material obtained from tests.

b) In the case of a sudden (brittle) buckling type failure, the design resistance shall be determined from:

$$F_d = 0.75 F_{\text{test,Min}} (f_y / f_{ym}) / \gamma_{m0}$$

c) In ductile buckling type failure in which the relevant slenderness λ can be reliably assessed, the design resistance may be determined from:

$$F_d = 0.9 F_{\text{test,Min}} [(\chi_f / \chi_m f_{ym}) / \gamma_{m0}]$$

where

- χ = reduction factor for the relevant buckling curve; and
- χ_m = value of χ when the yield strength is f_{ym} .

20.2.4 Check Tests

Where a component or assembly is designed on the basis of strength tests or tests to failure and a production run is carried out of such items, an appropriate number of samples (not less than two) shall be selected from each production batch at random for check tests.

20.2.4.1 The samples shall be carefully examined to ensure that they are similar in all respects to the prototype tested, particular attention being given to the following:

- a) Dimensions of components and connections;
- b) Tolerance and workmanship; and
- c) Quality of steel used, checked with reference to mill test certificates.

20.2.4.2 Where it is not possible to determine either the variations or the effect of variations from the prototype, an acceptance test shall be carried out as a check test.

20.2.4.3 In this check test, the deflections shall be measured at the same positions as in the acceptance test of the prototype. The maximum measured deflection shall not exceed 120 percent of the deflection recorded during the acceptance test on the prototype and the residual deflection should not be more than 105 percent of that recorded for the prototype.

20.3 Test Conditions

Followings are the conditions for the test:

- a) loading and measuring devices shall be calibrated in advance;
- b) the design of the test rig shall be such that:
 - 1) loading system adequately simulates the magnitude and distribution of the loading;
 - 2) it allows the specimen to perform in a manner representative of service conditions;
 - 3) lateral and torsional restraint, if any, should be representative of those in service;
 - 4) specimen should be free to deflect under load according to service condition;
 - 5) loading system shall be able to follow the movements of the specimen without interruption or abnormal restraints; and
 - 6) inadvertent eccentricities at the point of application of the test loads and at the supports are avoided.

- c) test load shall be applied to the unit at a rate as uniform as practicable;
- d) deflections should be measured at sufficient points of high movements to ensure that the maximum value is determined;
- e) if the magnitude of stresses in a specimen is to be determined, the strain at the desired location may be measured and the corresponding stress calculated; and
- f) prior to any test, preliminary loading (not exceeding the characteristic values of the relevant loads) may be applied and then removed, in order to set the test specimen on to the test rig.

20.4 Test Loading

20.4.1 Where the self-weight of the specimen is not representative of the actual permanent load in service, allowance for the difference shall be made in the calculation of test loads to be applied.

20.4.2 On the attainment of maximum load for either acceptance or strength tests, this load shall be maintained as given below. Reading of load and deflection shall be taken at intervals of 15 min and the loading shall be maintained constant until there is no significant increase in deflection during a 15 min period or until at least 1 h has elapsed after application of maximum load.

20.4.3 The test load shall be equal to the design load for the relevant limit state in proof testing.

20.4.4 The test load in prototype testing shall be equal to the design load for the relevant limit state as multiplied by the appropriate factor given in [Table 27](#).

20.5 Criteria for Acceptance

20.5.1 Acceptance for Strength

The test structure, sub-structure, member or connection shall be deemed to comply with the requirements for strength if it is able to sustain the strength test load as required in [20.4.2](#).

It shall then be inspected to determine the nature and extent of any damage incurred during the test. The effects of the damage shall be considered and if necessary appropriate repairs to the damaged parts carried out.

20.5.2 Acceptance for Serviceability

The maximum deformation of the structure or member under the serviceability limit state test load shall be within the serviceability limit values appropriate to the structure.

Table 27 Factors to Allow for Variability of Structural Units

(Clause [20.4.4](#))

Sl No.	No. of Similar Units to be Tested	For Strength Limit State	For Serviceability Limit State
(1)	(2)	(3)	(4)
i)	1	1.5	1.2
ii)	2	1.4	1.2
iii)	3	1.3	1.2
iv)	4	1.3	1.1
v)	5	1.3	1.1
vi)	10	1.2	1.1

21 DURABILITY

21.1 General

A durable steel structure is one that performs satisfactorily the desired function in the working environment under the anticipated exposure condition during its service life, without deterioration of the cross sectional area and loss of strength due to corrosion.

The material used, the detailing, fabrication, erection and surface protection measures should all address the corrosion protection and durability requirements.

21.2 Requirements for Durability

21.2.1 Shape, Size, Orientation of Members, Connections and Details

The design, fabrication and erection details of exposed structures should be such that good drainage of water is ensured. Standing pool of water, moisture accumulation and rundown of water for extended duration shall be avoided.

The details of connections should ensure that:

- a) all exposed surfaces are easily accessible for inspection and maintenance; and
- b) all surfaces, not so easily accessible are completely sealed against ingress of moisture.

21.2.2 Exposure Condition

21.2.2.1 General environment

The general environment, to which a steel structure is exposed during its working life is classified into five levels of severity, as given in [Table 28](#).

21.2.2.2 Abrasion

Specialist literature may be referred for durability of surfaces exposed to abrasive action as in machinery, conveyor belt support system, storage bins for grains or aggregates.

21.2.2.3 Exposure to sulphate attack

Appropriate coatings may be used when surfaces of structural steel are exposed to concentration of sulphates (SO_3) in soil, ground water, etc.

When exposed to very high sulphate concentrations of more than 2 percent in soil and 5 percent in water, some form of lining such as polyethylene, polychloroprene sheet or surface coating based on asphalt, chlorinated rubber, epoxy or polymethane material should be used to completely avoid access of the solution to the steel surface.

21.2.3 Corrosion Protection Methods

The methods of corrosion protection are governed by actual environmental conditions as specified in good practices [C-6A(43)]. The main corrosion protection methods are given below:

- a) Controlling the electrode potential;
- b) Inhibitors; and
- c) Inorganic/metal coatings or organic/paint systems.

Table 28 Environmental Exposure Conditions

(Clause [21.2.2.1](#))

Sl No.	Environmental Classifications	Exposure Conditions
(1)	(2)	(3)
i)	Mild	Surfaces normally protected against exposure to weather or aggressive condition as in interior of buildings, except when located in coastal areas
ii)	Moderate	Structural steel surfaces: a) exposed to condensation and rain; b) continuously under water; c) exposed to non-aggressive soil/groundwater; and d) sheltered from saturated salt air in coastal areas
iii)	Severe	Structural steel surfaces: a) exposed to severe frequent rain; b) exposed to alternate wetting and drying; c) severe condensation; d) completely immersed in sea water; and e) exposed to saturated salt air in coastal area
iv)	Very severe	Structural steel surface exposed to: a) sea water spray; b) corrosive fumes; and c) aggressive sub-soil or ground water.
v)	Extreme	Structural steel surfaces exposed to: a) tidal zones and splash zones in the sea; and b) aggressive liquid or solid chemicals

21.2.4 Surface Protection

21.2.4.1 In the case of mild exposure, a coat of primer after removal of any loose mill scale may be adequate. As the exposure condition becomes more critical, more elaborate surface preparations and coatings become necessary. In case of extreme environmental classification, protection shall be as per specialist literature. [Table 29](#) gives guidance to protection of steelwork for different desired lives.

21.2.4.2 Steel surfaces shall be provided with at least one coat of primer immediately after its surface preparation, such as by sand blasting to remove all mill scale and rust and to expose the steel.

21.2.4.3 Steel without protective coating shall not be stored for long duration in outdoor environment.

21.2.4.4 Surfaces to transfer forces by friction as in HSFG connections shall not be painted. However, it shall be ensured that moisture is not trapped on such surfaces after pre-tensioning of bolts by proper protective measures.

21.2.4.5 Members to be assembled by welding shall not be pre-painted at regions adjacent to the location of such welds. However, after welding, appropriate protective coatings shall be applied in the region, as required by the exposure conditions. If the contact surfaces cannot be properly protected against ingress of moisture by surface coating, they may be completely sealed by appropriate welds.

21.2.4.6 Pre-painted members shall be protected against abrasion of the coating during transportation, handling and erection.

21.2.5 Special Steels

Steels with special alloying elements and production process to obtain better corrosion resistance may be used as per specialist literature.

Table 29(A) Protection Guide for Steel Work Application — Desired Life of Coating System in Different Environment

(Clause [21.2.4.1](#))

Sl No.	Atmospheric Condition/Environmental Classification	Coating System					
		1	2	3	4	5	6
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	Normal inland (rural and urban areas), mild	12 years	18 years	20 years	About 20 years	About 20 years	Above 20 years
ii)	Polluted inland (high airborne sulphur dioxide), moderate	10 years	15 years	12 years	About 18 years	15 to 20 years	Above 20 years
iii)	Normal coastal (as normal inland plus high airborne salt levels), severe	10 years	12 years	20 years	About 20 years	About 20 years	Above 20 years
iv)	Polluted coastal (as polluted inland plus high airborne salt levels), very severe or extreme	8 years	10 years	10 years	About 15 years	15 to 20 years	Above 20 years

Table 29(B) (i) Protection Guide for Steel Work Application — Specification for Different Coating System (Shop Applied Treatments)

(Clause [21.2.4.1](#))

Sl No.	Protection	Coating System					
		1	2	3	4	5	6
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	Surface preparation	Blast clean	Blast clean	Blast clean	Blast clean	Girt blast	Blast clean
ii)	Pre-fabrication primer	Zinc phosphate epoxy, 20 µm	2 pack zinc-rich epoxy, 20 µm	—	2 pack zinc-rich epoxy, 20 µm	—	Ethyl zinc silicate, 20 µm
iii)	Post fabrication primer	High-build zinc phosphate modified alkyd, 60 µm	2 pack zinc-rich epoxy, 20 µm	Hot dip galvanized, 85 µm	2 pack zinc-rich epoxy, 25 µm	Sprayed zinc or sprayed aluminium	Ethyl zinc silicate, 60 µm

Table 29(B) (i) (Concluded)							
Sl No.	Protection	Coating System					
		1	2	3	4	5	6
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
iv)	Intermediate coat	—	High-build zinc phosphate, 25 µm	—	2 pack epoxy micaceous iron oxide	Sealer	Chlorinated rubber alkyd, 35 µm
v)	Top coat	—	—	—	2 pack epoxy micaceous iron oxide, 85 µm	Sealer	—

Table 29(B) (ii) Protection Guide for Steel Work Application — Specification for Different Coating System (Site Applied Treatments)

(Clause [21.2.4.1](#))

Sl No.	Protection	Coating System					
		1	2	3	4	5	6
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	Surface preparation	As necessary	As necessary	No site treatment	As necessary	No site treatment	As necessary
ii)	Primer	Touch in	Touch in	—	—	—	Touch in
iii)	Intermediate coat	—	Modified Alkyd micaceous iron oxide, 50 µm	—	Touch In	—	High-build micaceous iron oxide chlorinated rubber micaceous, 75 µm
iv)	Top coat	High-build Alkyd finish, 60µm	Modified alkyd micaceous iron oxide, 50 µm	—	High-build chlorinated rubber	—	High-build iron oxide chlorinated rubber, 75 µm

22 FIRE RESISTANCE

22.1 Requirements

The requirements shall apply to steel building elements designed to exhibit a required fire-resistance level (FRL) as per the relevant specifications. The provisions of this section are quite rudimentary and simplistic. Specialist literature for design of steel structures for fire resistance and design based on testing may be necessary for a more comprehensive design of steel buildings to resist fire.

22.1.1 For protected steel members and connections, the thickness of protection material (h_i) shall be greater than or equal to that required to give a period of structural adequacy (PSA) greater than or equal to the required FRL.

22.1.2 For unprotected steel members and connections, the exposed surface area to mass ratio (k_{sm}) shall be less than or equal to that required to give a PSA equal to the required FRL.

22.2 Fire Resistance Level

The required FRL shall be as prescribed in Part F 'Fire and Life Safety' of this NBCS and relative fire norms/standards, as appropriate or in building specifications or as required by the user or the city ordinance. The FRL specified in terms of the duration (in min) of standard fire load without collapse depends upon;

- a) the purpose for which structure is used; and
- b) the time taken to evacuate in case of fire.

22.3 Period of Structural Adequacy (PSA)

22.3.1 The calculation of PSA involves:

- a) calculation of the strength of the element as a function of temperature of the element and the determination of limiting temperature;
- b) calculation of the thermal response of the element, that is calculation of the variation of the temperature of the element or the parts of the element with time, when exposed to fire; and
- c) determination of PSA at which the temperature of the element or parts of the element reaches the limiting temperature.

22.3.2 Determination of Period of Structural Adequacy

The period of structural adequacy (PSA) shall be determined using one of the following methods:

- a) *By calculation:*
 - 1) determining the limiting temperature of the steel (T_l) in accordance with [22.5](#); and
 - 2) determining the PSA as the time (t) (in min) from the start of the test to the time at which the limiting steel temperature is attained, in accordance with [22.6](#) for protected members and [22.7](#) for unprotected members.
- b) By direct application of a single test in accordance with [22.8](#); or
- c) By calculation of the temperature of the steel member by using a rational method of analysis confirmed by test data or by methods available in specialist literature.

22.4 Variation of Mechanical Properties of Steel with Temperature

22.4.1 Variation of Yield Stress with Temperature

The influence of temperature on the yield stress of steel shall be taken as follows for structures of mild steels and high strength low alloy steels:

where

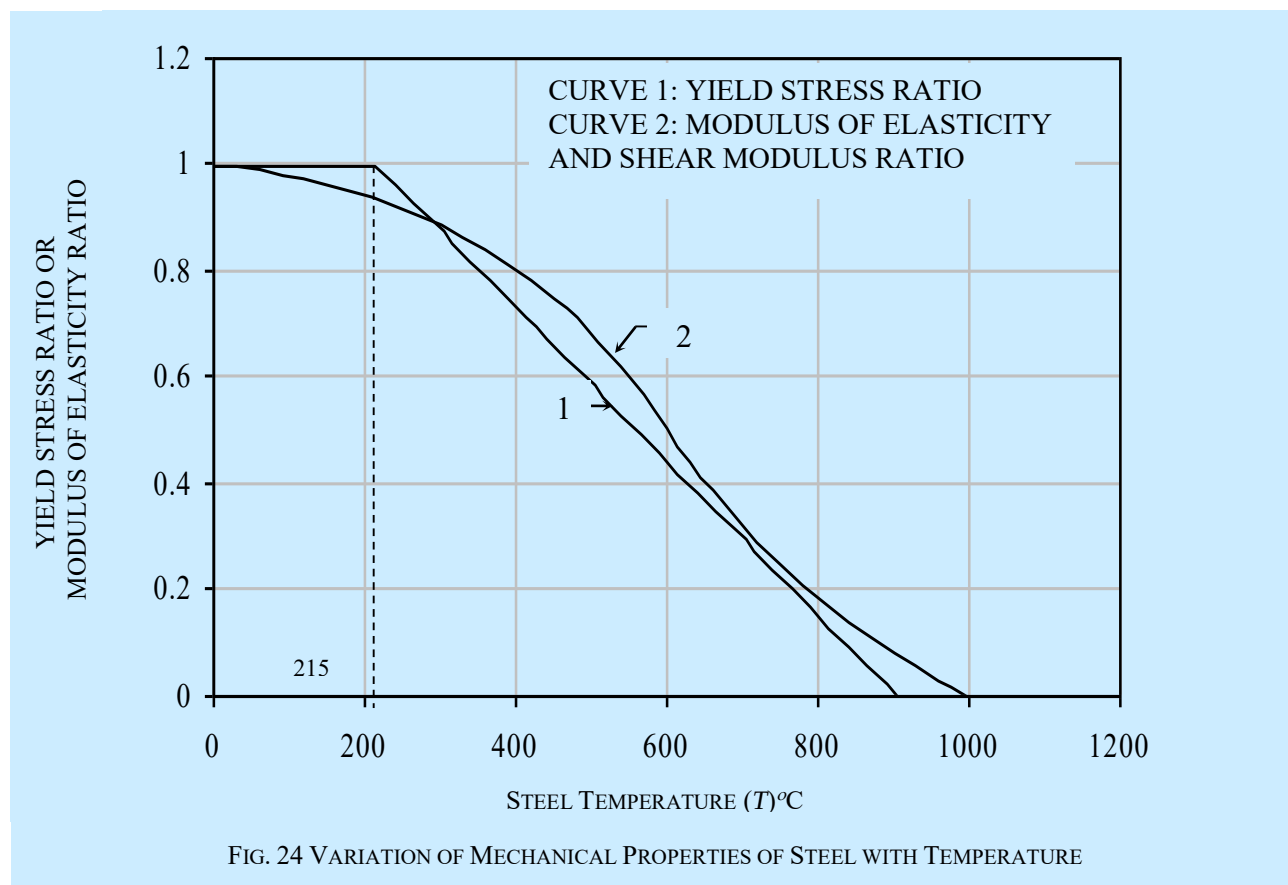
$$\frac{f_y(T)}{f_y(20)} = \frac{905 - T}{690} \leq 1.0$$

$f_y(T)$ = yield stress of steel at T °C;

$f_y(20)$ = yield stress of steel at 20 °C (room temperature); and

T = temperature of the steel, in °C.

This relationship is shown by curve 1 in [Fig. 24](#).



For temperature less than 215 °C no reduction in the yield stress need to be considered.

22.4.2 Variation of Modulus of Elasticity and Shear Modulus with Temperature

The influence of temperature on the modulus of elasticity shall be taken as follows for structures of mild steels and high strength low alloy steels:

$$\frac{E(T)}{E(20)} \text{ and } \frac{G(T)}{G(20)} = 1.0 + \left[\frac{T}{2000 \left[\ln \left(\frac{T}{100} \right) \right]} \right] \text{ when } 0^\circ\text{C} < T \leq 600^\circ\text{C}$$

$$= \frac{690 \left(1 - \frac{T}{1000} \right)}{T - 53.5} \text{ when } 600^\circ\text{C} < T \leq 1000^\circ\text{C}$$

where

$E(T)$ = Modulus of elasticity of steel at T °C, and

$E(20)$ = Modulus of elasticity of steel at 20 °C (room temperature).

$G(T)$ = Shear Modulus of steel at T °C, and

$G(20)$ = Shear Modulus of steel at 20 °C (room temperature).

This relationship is shown by curve 2 in [Fig. 24](#).

22.4.3 For special steel with higher temperature resistance, such as TMCP steels, the manufacturer's recommendation shall be used to obtain the variation of yield strength and modulus of elasticity of steel with temperature.

22.5 Limiting Steel Temperature

The limiting steel temperature (T_l), in °C, in the case of structural steels, shall be calculated by substituting f_y , f_u , E , and G values with $f_y(T)$, $f_u(T)$, $E(T)$, and $G(T)$ calculated in accordance with [22.4](#), respectively in [13](#), [14](#), [15](#), and [16](#) to calculate the design capacity of the member as a function of steel temperature. The limiting temperature (T_l) will be the temperature at which the design capacity is equal to the design action on the member.

The design action under fire shall consider the following:

- Loads as per the load combinations for fire conditions given in [Table 4](#);
- Effects of restraint to expansion of the elements during fire; and
- Reduced bond likely under fire.

Limiting steel temperature for special steels may be appropriately calculated using the thermal characteristics of the material obtained from the supplier of the steel.

22.6 Temperature Increase with Time in Protected Members

22.6.1 The time (t) at which the limiting temperature (T_l) is attained shall be determined by calculation on the basis of a suitable series of fire tests in accordance with [22.6.2](#) or from the results of a single test in accordance with [22.6.3](#).

22.6.1.1 For beams and for all members with a four-sided fire exposure condition, the limiting temperature (T_l) shall be taken as the average of all of the temperatures measured at the thermocouple locations on all sides.

22.6.1.2 For columns with a three-sided fire exposure condition, the limiting temperature (T_l) shall be taken as the average of the temperatures measured at the thermocouple locations on the face farthest from the wall. Alternatively, the temperatures from members with a four-sided fire exposure condition and having the same surface area to mass ratio may be used.

22.6.2 Temperature Based on Test Series

Calculation of the variation of steel temperature with time shall be by interpolation of the results of a series of fire tests using the regression analysis equation specified in [22.6.2.1](#), subject to the limitations and conditions of [22.6.2.3](#).

22.6.2.1 Regression analysis

The relationship between temperature (T) and time (t) for a series of tests on a group shall be calculated by least-square regression as follows:

$$t = k_0 + k_1 h_i + k_2 \left(\frac{h_i}{k_{sm}} \right) + k_3 T + k_4 h_i T + k_5 \left(\frac{h_i T}{k_{sm}} \right) + k_6 \left(\frac{T}{k_{sm}} \right)$$

where

h_i = thickness of fire protection material, in mm;

- k_0 to k_6 = regression coefficients from test data (see [22.6.2.2](#));
 k_{sm} = exposed surface area to mass ratio, in $10^3 \text{ mm}^2/\text{kg}$;
 t = time from the start of the test, in min; and
 T = steel temperature, in $^\circ\text{C}$ obtained from test as given in [22.6.1](#), $T > 250^\circ\text{C}$.

22.6.2.2 *In lieu of test results*, the values for coefficients in [Table 30](#) may be used in the equation [22.6.2.1](#) when the test satisfies the conditions specified in [22.6.2.3](#).

Table 30 Regression Coefficients, k (Clause 22.6.2.2)							
Sl No.	k_0	k_1	k_2	k_3	k_4	k_5	k_6
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	- 25.90	1.698	- 13.71	0.030 0	0.000 5	0.514 4	6.633

22.6.2.3 *Limitations and conditions on use of regression analysis*

Test data to be utilized in accordance with [22.6.2.1](#), shall satisfy the following:

- Steel members shall be protected with board, sprayed blanket or similar insulation materials having a dry density less than 1000 kg/m^3 ;
- All tests shall incorporate the same fire protection system;
- All members shall have the same fire exposure condition;
- Test series shall include at least nine tests;
- Test series may include prototypes which have not been loaded provided that stickability has been demonstrated; and
- All members subject to a three-sided fire exposure condition shall be within a group in accordance with [22.9](#).

The regression equation obtained for one fire protection system may be applied to another system using the same fire protection material and the same fire exposure condition provided that stickability has been demonstrated for the second system.

A regression equation obtained using prototypes with a four-sided fire exposure condition may be applied to a member with a three-sided fire exposure condition provided that stickability has been demonstrated for the three-sided case.

22.6.3 *Temperature Based on Single Test*

The variation of steel temperature with time measured in a standard fire test may be used without modification provided:

- fire protection system is the same as the prototype;
- fire exposure condition is the same as the prototype;
- fire protection material thickness is equal to or greater than that of the prototype;
- surface area to mass ratio is equal to or less than that of the prototype; and
- where the prototype has been submitted to a standard fire test in an unloaded condition, stickability has been separately demonstrated.

22.6.4 Parameters of Importance in the Standard Fire Test

- a) Specimen type, loading, configuration;
- b) Exposed surface area to mass ratio;
- c) Insulation type, thermal properties and thickness; and
- d) Moisture content of the insulation material.

22.7 Temperature Increase with Time in Unprotected Members

The time (t) at which the limiting temperature (T_l) is attained shall be calculated using the following equations:

- a) Three-sided fire exposure condition:

$$t = -5.2 + 0.022 \, T + \left(\frac{0.433 T_l}{k_{sm}} \right)$$

- b) Four-sided fire exposure condition:

$$t = -4.7 + 0.0263 \, T + \left(\frac{0.213 T_l}{k_{sm}} \right)$$

where

- k_{sm} = exposed surface area to mass ratio, $2 \times 10^3 \, \text{mm}^2/\text{kg} \leq k_{sm} \leq 35 \times 10^3 \, \text{mm}^2/\text{kg}$;
 t = time, in min, from the start of the test; and
 T = steel temperature, in °C, $500 \, ^\circ\text{C} \leq T \leq 750 \, ^\circ\text{C}$.

For temperatures below 500 °C, linear interpolation shall be used, based on the time at 500 °C and an initial temperature of 20 °C at t equals 0.

22.8 Determination of PSA from a Single Test

The period of structural adequacy (PSA) determined from a single test may be applied without modification provided:

- a) conditions, specified in [22.6.3](#) are satisfied;
- b) conditions of support are the same as the prototype and the restraints are not less favourable than those of the prototype; and
- c) ratio of the design load for fire to the design capacity of the member is less than or equal to that of the prototype.

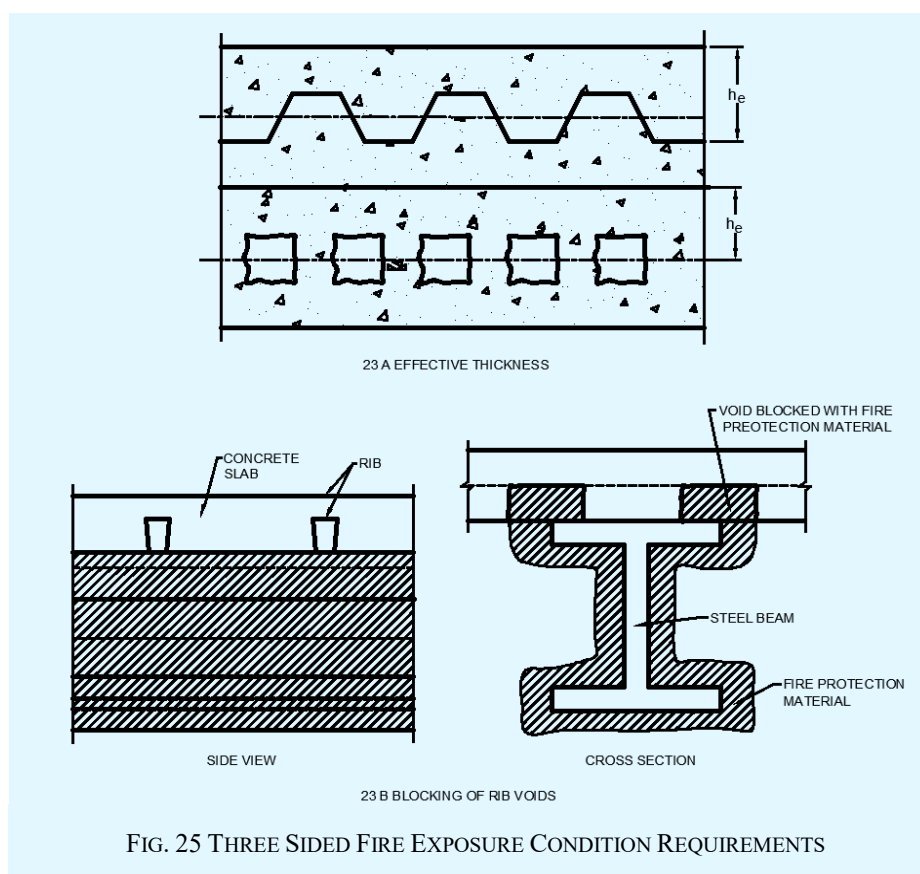


FIG. 25 THREE SIDED FIRE EXPOSURE CONDITION REQUIREMENTS

22.9 Three-sided Fire Exposure Condition

Members subject to a three-sided fire exposure condition shall be considered in separate groups unless the following conditions are satisfied:

- a) the characteristics of the members of a group as given below, shall not vary from one another by more than:
 - 1) Concrete density: $\left(\frac{\text{highest in group}}{\text{lowest in group}} \right) \leq 1.25$; and
 - 2) Effective thickness (h_e): $\left(\frac{\text{largest in group}}{\text{smallest in group}} \right) \leq 1.25$

where the effective thickness (h_e) is equal to the cross sectional area excluding voids per unit width, as shown in [Fig. 25 \(a\)](#).

- b) Rib voids shall either be:
 - 1) all open; or
 - 2) all blocked as shown in [Fig. 25 \(b\)](#).
- c) Concrete slabs may incorporate permanent steel deck formwork.

22.10 Special Considerations

22.10.1 Connections

Connections shall be protected with the maximum thickness of fire protection material required for any of the members framing into the connection to achieve their respective fire-resistance levels. This thickness shall be maintained over all connection components, including bolt heads, welds and splice plates.

22.10.2 Web Penetrations

The thickness of fire protection material at and adjacent to web penetrations shall be the greatest of that required, when:

- area above the penetration is considered as a three-sided fire exposure condition (k_{sm1}) (see Fig. 26);
- area below the penetration is considered as a four-sided fire exposure condition (k_{sm2}) (see Fig. 26); and
- section as a whole is considered as a three-sided fire exposure condition (k_{sm}) (see Fig. 26).

This thickness shall be applied over the full beam depth and shall extend on each side of the penetration for a distance at least equal to the beam depth and not less than 300 mm.

22.11 Fire Resistance Rating

The fire resistance rating of various building components such as walls, columns, beams, and floors are given in Table 31 and Table 32. Fire damage assessment of various structural elements of the building and adequacy of the structural repairs can be done by the fire resistance rating for encased steel column and beam (see Table 31 and Table 32).

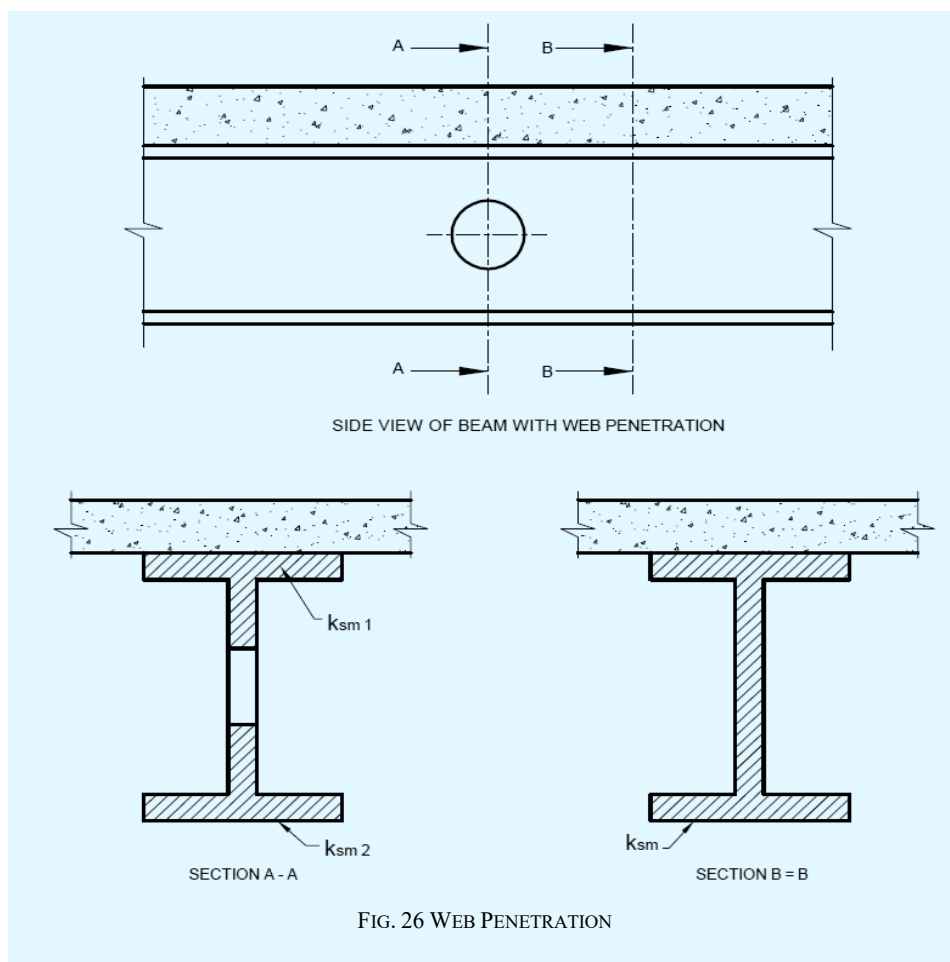


Table 31 Encased Steel Columns, 203 mm × 203 mm (Protection Applied on Four Sides)

(Clause 22.11)

Sl No.	Nature of Construction and Materials	Minimum Dimensions (in mm), Excluding Any Finish, for a Fire Resistance of				
		1 h (3)	1 ½ h (4)	2 h (5)	3 h (6)	4 h (7)
(1)	(2)					
i)	Hollow protection (without an air cavity over the flanges):					
	a) Metal lathing with trowelled lightweight aggregate gypsum plaster ¹⁾	13	15	20	32	—
	b) Plasterboard with 1.6 mm wire binding at 100 mm pitch, finished with lightweight aggregate gypsum plaster less than the thickness specified:					
	1) 9.5 mm plaster board; and	10	15	—	—	—
	2) 19 mm plaster board.	10	13	20	—	—
	c) Asbestos insulating boards, thickness of board:					
	1) single thickness of board, with 6 mm cover fillets at transverse joints; and	—	19	25	—	—
	2) two layers, of total thickness.	—	—	—	38	50
	d) Solid bricks of clay, composition or sand lime, reinforced in every horizontal joint, unplastered	50	50	50	75	100
	e) Aerated concrete blocks	60	60	60	—	—
	f) Solid blocks of lightweight concrete hollow protection (with an air cavity over the flanges)	50	50	50	60	75
ii)	Asbestos insulating board screwed to 25 mm asbestos battens	12	19	—	—	—
iii)	Solid protections:					
	a) Concrete, not leaner than 1:2:4 mix (unplastered):					
	1) Concrete not assumed to be load bearing, reinforced ²⁾ ; and	25	25	25	50	75
	2) Concrete assumed to be load bearing	50	50	50	75	75
	b) Lightweight concrete, not leaner than 1 : 2 : 4 mix (unplastered) concrete not assumed to be load bearing, reinforced ²⁾	25	25	25	25	25

¹⁾ So fixed or designed, as to allow full penetration for mechanical bond.

²⁾ Reinforcement shall consist of steel binding wire not less than 2.3 mm diameter, or a steel mesh weighing not less than 0.5 kg/m². In concrete protection, the spacing of that reinforcement shall not exceed 200 mm in any direction.

Table 32 Encased Steel Beams, 406 mm × 176 mm (Protection Applied on Three Sides)

(Clause [22.11](#))

Sl No.	Nature of Construction and Materials	Minimum Thickness (in mm) of Protection for a Fire Resistance of					
		½ h	1 h	1½ h	2 h	3 h	4 h
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	Hollow protection (without an air cavity beneath the lower flanges):						
	a) Metal lathing with trowelled lightweight aggregate gypsum plaster ¹⁾	13	13	15	20	25	—
	b) Plasterboard with 1.6 mm wire binding at 100 mm pitch, finished with lightweight aggregate gypsum plaster less than the thickness specified ²⁾ :						
	1) 9.5 mm plaster board; and	10	10	15	—	—	—
	2) 19 mm plaster board	10	10	13	20	—	—
	c) Asbestos insulating boards, thickness of board:						
	1) single thickness of board, with 6 mm cover fillets at transverse joints	—	—	19	25	—	—
	2) two layers, of total thickness	—	—	—	—	38	50
ii)	Hollow protection (with an air cavity below the lower flange):						
	a) Asbestos insulating board screwed to 25 mm asbestos battens	9	12	—	—	—	—
iii)	Solid protections:						
	a) Concrete, not leaner than 1 : 2 : 4 mix (unplastered):						
	1) concrete not assumed to be load bearing, reinforced ³⁾ ; and	25	25	25	25	50	75
	2) concrete assumed to be load bearing	50	50	50	50	75	75
	b) Lightweight concrete, not leaner than 1 : 2 : 4 mix (unplastered) ⁴⁾	25	25	25	25	40	60

¹ So fixed or designed, as to allow full penetration for mechanical bond.

² Where wire binding cannot be used, expert advice should be sought regarding alternative methods of support to enable the lower edges of the plasterboard to be fixed together and to the lower flange, and for the top edge of the plasterboard to be held in position.

³ Reinforcement shall consist of steel binding wire not less than 2.3 mm in diameter, or a steel mesh weighing not less than 0.5 kg/m². In concrete protection, the spacing of that reinforcement shall not exceed 200 mm in any direction.

⁴ Concrete not assumed to be load bearing, reinforced.

23 SUSTAINABLE STEEL CONSTRUCTION

23.1 General

This Chapter specifies general requirements and practices for achieving environmentally sustainable steelwork building construction. This supplements the requirements of the other chapters of this specification.

23.2 Principles of Sustainable Steel Construction

- a) The structural steel design should provide a structure capable of fulfilling its intended function while sustaining the specified loads for its intended life with due regard to sustainability, robustness, economy, health and safety. Any expected change in the intended function during its life time should also be considered;
- b) In environmentally sustainable construction all stake holders should consider the broader and holistic implications of the decision, which include the whole life cycle sustainability, impacts on other parts of the building and the project, externalities, maintenance, societal cost and consequences, without impinging upon safety and efficiency of the structure;
- c) All parties in the supply chain for structural steelwork should take practical steps to reduce the greenhouse gas emissions of the processes over which they have control, thus contributing to sustainable construction. Further, construction operations should avoid/minimize polluting the environment with dust, wastewater, chemicals and leaching debris;
- d) Steelmaking should make full use of available secondary material (scrap) globally;
- e) Structural steelwork should be designed, detailed, and constructed to minimize material use and wastage of material and facilitate its recovery for reuse at its end-of-life, as a part of the circular economy, to the extent that this is reasonably practicable; and
- f) The design should consider practical aspects of construction to avoid repair and rework.

23.3 Specification Requirements from Supply Chain Members

23.3.1 Requirements from Employer

- a) Appointment of key contractors (Architects, contractors for Steel work, cladding, glazing, Mechanical, Electrical, coating systems, etc) as early as practical to ensure decisions are fully coordinated;
- b) Maximization of structural zones that facilitate lean and efficient designs; and
- c) Flexibility in design, if use of reclaim materials is considered.

23.3.2 Requirements from Main Contractor

- a) Early engagement of supply chain and subcontractors with emphasis on sustainability; and
- b) Judicious site planning to receive material, use of labour and location of machinery for efficient handling and construction.

23.3.3 Requirements from the Engineering Group

- a) Design for material efficiency, equipment efficiency, and minimal embodied energy and longevity;
- b) Consider holistically the environmental impact implication of temporary works and supports;
- c) Design with inbuilt flexibility and adaptability;
- d) Consider the economic and environmental implications of the type of connections chosen; and
- e) Assess the requirement of protective treatment against corrosion and fire to minimize the life time maintenance and avoid over-specification.

23.3.4 Requirements from Steelwork Contractor

Establish a carbon reduction plan in steelwork. This may include use of low carbon emission transport, maximize the steelwork loads, optimizing haulage routes, minimizing empty runs, etc.

23.4 Sustainable Design

23.4.1 General

The environmental impacts associated with steelwork building construction be greatly reduced during the design process by:

- a) reducing the material in the structure;
- b) minimizing fabrication, and avoidance of maintenance during service life; and
- c) Identifying environmental improvements at the conceptual design stage requires the involvement of all the stakeholders such as architects, steelwork contractor during all design stages. Collaborative design processes result in:
 - 1) materials being used more efficiently;
 - 2) fewer on-site changes; and
 - 3) reduced environmental impacts.

Greenhouse gas or carbon emissions currently being the priority, action is required by all parts of the supply chain for the sector to decarbonize in line with global targets. This includes supply-side measures, in particularly decarbonizing steel-making and also demand-side measures including more efficient design and design for longevity, deconstruction and reuse.

Greenhouse gas emission should be based on life cycle assessment, which should include emissions in material manufacturing, transport, fabrication, erection, finishing, commissioning, maintenance during life and finally at the end of useful life deconstruction and disposal of the materials as waste or through recycling and reuse.

Further, environmentally friendly designs should be implemented in the construction projects and pollution of the environment during the process of construction should be avoided, if not minimized.

23.4.2 Minimizing Material Consumption through Design

Minimizing material consumption through design may be achieved by:

- a) Avoiding over-specification of design loads;
- b) Optimizing spans and grids: and adapting the building layout to take full advantage of the structural steelwork's attributes;
- c) Minimizing complex load paths and transfers in order to maximize direct load paths to ground;
- d) Ensuring that structural deflection and vibration criteria, coordination zones and member depths are appropriate and allow for efficient structural solutions that avoid excessive material use;
- e) Aiming to design for utilizations as close to 1.0 as practically possible, with due regard to permanent and temporary conditions;
- f) Specifying higher strength steel grades in members or part of members not governed by serviceability criteria;
- g) Considering key load paths, anticipated connection types and accurate loading criteria, in order to facilitate lean design principles;
- h) Avoiding the over rationalization of steel member sizes to ensure efficient use of material, with due consideration of minimum economic quantities for procurement;
- j) Reviewing section sizes and minimizing weight, whilst complying with safe methods of erection and fire protection requirements; and
- k) Semi-continuous design, which assumes that the joints transfer moments from the beam to the column and allow rotation between the connected members as the load is resisted.

NOTE — Historically, steelwork design and fabrication has mainly been driven by cost however, sustainability and particularly the climate emergency, are likely to change this with more clients willing or mandated to pay more for sustainable and low carbon buildings. Reducing the carbon impact of steel structures through lean and less conservative design is technically achievable but can require more design effort and more complex fabrication.

23.4.3 Design for Adaptability and Deconstruction

Design engineer should review design options with the steelwork contractor that may contribute to future flexibility, adaptability, deconstruction and reuse of the building structure, such as:

- a) how the building design and information records about the materials and members can facilitate future, remodelling, deconstruction and recovery for reuse and recycling;
- b) ensuring accessibility to members and connections to allow for deconstruction and for ease of strengthening in the future; and
- c) greater standardization of sections sizes, lengths and connections, if appropriate from adaptability and deconstruction.

23.5 Sustainable Fabrication

23.5.1 Information Records

A non-exhaustive checklist of information to be communicated in sustainable fabrication are given below:

- a) Sustainability information to be provided to the steelwork contractor:
 - 1) any part of the steelwork that is to incorporate reclaimed steel products;
 - 2) any part of the steelwork that is to be left untreated;
 - 3) any details of the fire design, namely the design utilization per section size in the fire condition;
 - 4) any workshop waste targets;
 - 5) a life cycle assessment of the whole building; and
 - 6) any specific material sourcing requirements.
- b) Sustainability information to be provided by the steelwork contractor:
 - 1) All relevant EPDs for the steel and other products;
 - 2) The embodied carbon of the fabricated steelwork;
 - 3) Digital records of the 'As erected' structure, which should include all condition of structures, survey, metadata, etc;
 - 4) Material quality and quantity data, including batch traceability for main structural members to facilitate future reuse (to be included in the operation and maintenance manual; and
 - 5) Documentation of evidence of an active environmental management system.

23.5.2 Fabrication Waste Management

Strategies for fabrication waste management include:

- a) ordering directly from mills to minimize offcuts;
- b) using stock lengths as ordered efficiently to minimize waste. Where possible, splice locations in steel members should be coordinated to fall within standard stock length sizes;
- c) recycling and reusing steel fabrication waste, for example, off-cuts, swarf, etc;
- d) considering use off-cuts of material, particularly from plate, elsewhere, for example, temporary works, connection fittings, shims, packs, rather than simply scrapping for recycling. Always prioritized reuse over recycling;
- e) considering use of machines that facilitate common line cutting of fittings;
- f) minimizing coating waste and coating packaging waste, by using intermediate bulk; and
- g) Minimizing/ avoiding overspray of coatings should ensure operators are properly trained and the appropriate and properly maintained equipment is used.

23.5.3 Sustainable Welding

Strategies for sustainable welding include:

- a) Avoiding low productivity processes and using automatic and semi-automatic processes with continuous wire feed, thus using material efficiently and avoiding energy waste;
- b) Adopting energy-saving processes such as, inverter-based power sources instead of older regulator/rectifier-based sources, electrically powered pre-heating with temperature control instead of gas flame;
- c) Minimizing pre-heat in the welding procedure specification and welding continuously to avoid the need to re-heat; and
- d) Periodically reviewing welding techniques and weld procedures to make the best use of the latest advancements in welding equipment and consumables, including the selection of joint set-ups and consumables that allow for deeper penetration fillet welds to be achieved, thus potentially reducing the number of weld runs required.

23.5.4 Sustainable Fire and Corrosion Protection Systems

23.5.4.1 Untreated steelwork

Any part of the steelwork such as steelwork in a less corrosivity category, that is to be left untreated shall be identified in the Project Specification.

23.5.4.2 Paint systems

The protective coating system shall satisfy the requirements given in [21](#).

23.5.4.3 Intumescent systems

For the same fire resistance water-based acrylic intumescent likely to have a lower environmental impact, when compared to solvent-based acrylic intumescent.

23.5.4.4 Hot-dip galvanizing

Requirements for thicker coatings, for example, by prior grit blasting, should be avoided by prior assessment of the predicted galvanized coating life and the minimum thickness required to avoid over-specification. For certain steel members (for example, with higher reactivity during galvanizing and/or larger section thickness), thicker coatings may also be achieved without grit blasting. Design and fabrication considerations for hot dip galvanizing should ensure adequate drainage and optimize material efficiency. Plan should optimize transport distances and provide direct delivery to the construction works, where feasible.

24 FABRICATION AND ERECTION

24.1 General

Tolerances for fabrication of steel structures shall conform to the accepted standard [C-6A(44)]. Tolerances for erection of steel structures shall conform to the accepted standard [C-6A(45)]. For general guidance on fabrication by welding, reference may be made to good practice [C-6A(46)].

In addition to the above, the following requirements shall also be considered:

- a) The designer shall, in conjunction with the floor framing and vertical framing/stacking arrangements of the project, provide suitable tolerance mitigation measures. Such measures may include the use of slotted or oversized holes in beams and columns, use of HSFG bolts, or other appropriate detailing. These provisions need not be adopted for all members; however, they shall be judiciously incorporated in selected connections and splices to achieve the desired tolerance accommodation.

- b) The structural designer shall coordinate with other disciplines, such as façade, building services, and transportation systems, and ensure that the specified tolerances for steel structures are compatible with their requirements. Where the design necessitates tolerances more stringent than those specified in the relevant codes, such requirements shall be clearly indicated in the drawings and specifications. However, in no case shall the specified tolerances be more stringent than the applicable mill rolling tolerances.
- c) The designer shall specify the permissible overall horizontal and vertical tolerances for the completed building structure, in accordance with the design intent. While [C-6A(45)] specifies tolerances for erected elements, it does not address the overall geometrical accuracy of the completed structure. Therefore, the structural designer shall, in coordination with the architect and other relevant disciplines, define the acceptable limits of overall as-built accuracy. These limits shall be clearly specified so as to enable proper planning and execution by the contractor at all stages of construction.

24.2 Fabrication Procedures

24.2.1 Straightening

Material shall be straightened or formed to the specified configuration by methods that will not reduce the properties of the material below the values used in design. Local application of pressure at room or at elevated temperature or other thermal means may be used for straightening, provided the above is satisfied.

24.2.2 Clearances

The erection clearance for cleated ends of members, connecting steel to steel should preferably be not greater than 2.0 mm at each end. The erection clearance at ends of beams without web cleats should be not more than 3 mm at each end. Where for practical reasons, greater clearance is necessary, suitably designed seating should be provided.

24.2.2.1 The holes for bolts may be made as given in [Table 20](#) of [17.2.1](#), unless otherwise specified by the engineer. The hole diameter in base plates shall not exceed the anchor bolt diameter by more than 6 mm.

24.2.2.2 In friction type of connection clearance may be maintained, unless specified otherwise in the design document.

24.2.3 Cutting

Cutting shall be effected by sawing, shearing, cropping, machining or thermal cutting process. Shearing, cropping and gas cutting shall be clean, reasonably square, and free from any distortion. Should the inspector find it necessary, the edges shall be ground after cutting. Planning or finishing of sheared or gas-cut edges of plates or shapes shall not be required, unless specially noted on drawing or included in stipulated edge preparation for welding or when specifically required in the following section.

Re-entrant corners shall be free from notches and shall have largest practical radii with a minimum radius of 15 mm.

24.2.3.1 Shearing

Shearing of items over 16 mm thick to be galvanized and subject to tensile force or bending moment shall not be carried out, unless the item is stress relieved subsequently.

The use of sheared edges in the tension area shall be avoided in location subject to plastic hinge rotation at factored loading.

24.2.3.2 Thermal cutting

Gas cutting of high tensile steel by mechanically controlled torch may be permitted, provided special care is taken to leave sufficient metal to be removed by machining, so that all metal that has been hardened by flame is removed. Hand flame cutting may be permitted only subject to the approval of the inspecting authority.

Except where the material is subsequently joined by welding, no load shall be transmitted through a gas cut surface.

Thermally cut free edges, which shall be subject to calculated static tensile stress shall be free from round bottom gouges greater than 5 mm deep. Gouges greater than 5 mm deep and notches shall be removed by grinding.

24.2.4 Holing

24.2.4.1 Holes through more than one thickness of material for members, such as compound stanchion and girder flanges, shall be where possible, drilled after the members are assembled and tightly clamped or bolted together. A round hole for a bolt shall either be machine flame cut, or drilled full size, or sub-punched 3 mm undersize and reamed to size or punched full size.

Hand flame cutting of a bolt hole shall not be permitted except as a site rectification measure for holes in column base plates.

24.2.4.2 Punching

A punched hole shall be permitted only in material whose yield stress (f_y) does not exceed 360 MPa and where thickness does not exceed $(5600/f_y)$ mm. In cyclically loaded details, punching shall be avoided in plates with thickness greater than 12 mm.

For greater thickness and cyclically loaded details, holes shall be either drilled from the solid or sub-punched or sub drilled and reamed.

The die for all sub-punched holes or the drill for all sub-drilled holes shall be at least 3 mm smaller than the required diameter of finished hole.

24.2.4.3 Oversize holes

A special plate washer of minimum thickness 4 mm shall be used under the nut, if the hole diameter is larger than the bolt diameter by 3 mm or more.

Oversize hole shall not exceed $1.25d$ or $(d + 8)$ mm in diameter, where d is the nominal bolt diameter, in mm.

A short slotted hole shall not exceed the appropriate hole size in width and $1.33d$ in length. A long slotted hole shall not exceed the appropriate hole size in width and $2.5d$, in length. If the slot length is larger than those specified, shear transfer in the direction of slot is not admissible even in friction type of connection.

Slotted holes shall be punched either in one operation or else formed by punching or drilling two round holes apart and completed by high quality mechanically controlled flame cutting and dressing to ensure that bolt can freely travel the full length of the slot.

24.2.4.4 Fitted bolt holes

Holes for turned and fitted bolts shall be drilled to a diameter equal to the nominal diameter of the shank or barrel subject to tolerance specified in the accepted standards [C-6A(47)]. Preferably, parts to be connected with close tolerance or barrel bolts shall be firmly held together by tacking bolts or clamps and the holes drilled through all the thicknesses at one operation and subsequently reamed to size. All holes not drilled through all thicknesses at one operation shall be drilled to a smaller size and reamed out after assembly. Where this is not practicable, the parts shall be drilled and reamed separately through hard bushed steel jigs.

24.2.4.5 Holes for rivets or bolts shall not be formed generally by gas cutting process. However, advanced gas cutting processes such as plasma cutting may be used to make holes in statically loaded members only. In cyclically loaded members subjected to tensile stresses which are vulnerable under fatigue, gas cutting shall not be used unless subsequent reaming is done to remove the material in the heat affected zone around the hole.

24.3 Assembly

All parts of bolted members shall be pinned or bolted and rigidly held together during assembly.

The component parts shall be assembled and aligned in such a manner that they are neither twisted nor otherwise damaged, and shall be so prepared that the specified camber, if any, is provided.

24.3.1 Holes in Assembly

When holes are drilled in one operation through two or more separable parts, these parts, when so specified by the engineer, shall be separated after drilling and the burrs removed.

Matching holes for rivets and black bolts shall register with each other so that a gauge of 1.5 mm or 2.0 mm (as the case may be, depending on whether the diameter of the rivet or bolt is less than or more than 25 mm) less in diameter than the diameter of the hole will pass freely through the assembled members in the direction at right angle to such members.

Drilling done during assembly to align holes shall not distort the metal or enlarge the holes.

Holes in adjacent part shall match sufficiently well to permit easy entry of bolts. If necessary, holes except oversize or slotted holes may be enlarged to admit bolts, by moderate amount of reaming.

24.3.2 Thread Length

When design is based on bolts with unthreaded shanks in the shear plane, appropriate measures shall be specified to ensure that, after allowing for tolerance, neither the threads nor the thread run-out be in the shear plane.

The length of bolt shall be such that at least one clear thread shows above the nut and at least one thread plus the thread run out is clear beneath the nut after tightening. One washer shall be provided under the rotated part.

24.3.3 Assembly Subjected to Vibration

When non-preloaded bolts are used in a structure subject to vibration, the nuts shall be secured by locking devices or other mechanical means. The nuts of preloaded bolts may be assumed to be sufficiently secured by the normal tightening procedure.

24.3.4 Washers

Washers are not normally required on non-preloaded bolts, unless specified otherwise. Tapered washers shall be used where the surface is inclined at more than 3° to a plane perpendicular to the bolt axis.

Hardened washer shall be used for preloaded bolts or the nut, whichever is to be rotated.

All material within the grip of the bolt shall be steel and no compressible material shall be permitted in the grip.

24.4 Riveting

24.4.1 Rivets shall be heated uniformly throughout their length, without burning or excessive scaling, and shall be of sufficient length to provide a head of standard dimensions. These shall, when driven, completely fill the holes and, if countersunk, the countersinking shall be fully filled by the rivet. If required, any protrusion of the countersunk head shall be dressed off flush.

24.4.2 Riveted member shall have all parts firmly drawn and held together before and during riveting, and special care shall be taken in this respect for all single-riveted connections. For multiple riveted connections, a service bolt shall be provided in every third or fourth hole.

24.4.3 Wherever practicable, machine riveting shall be carried out by using machines of the steady pressure type.

24.4.4 All loose, burned or otherwise defective rivets shall be cut out and replaced before the structure is loaded, and special care shall be taken to inspect all single riveted connections.

24.4.5 Special care shall be taken in heating and driving long rivets.

24.5 Bolting

24.5.1 In all cases where the full bearing area of the bolt is to be developed, the bolt shall be provided with a washer of sufficient thickness under the nut to avoid any threaded portion of the bolt being within the thickness of the parts bolted together, unless accounted for in design.

24.5.2 Pre-tensioned bolts shall be subjected to initial tension (the proof stress) by an appropriate pre-calibrated method.

24.6 Welding

24.6.1 Welding shall be in accordance with good practices [C-6A(48)], as appropriate.

24.6.2 For welding of any particular type of joint, welders shall give evidence acceptable to the purchaser of having satisfactorily completed appropriate tests as prescribed in good practices [C-6A(49)], as relevant.

24.6.3 Assembly and welding shall be carried out in such a way to minimize distortion and residual stress and that the final dimensions are within appropriate tolerances.

24.7 Machining of Butts, Caps and Bases

24.7.1 Column splices and butt joints of struts and compression members, depending on contact for stress transmission, shall be accurately machined and close-butted over the whole section with a clearance not exceeding 0.2 mm locally, at any place. Sum of all such clearance shall not be more than 30 percent of the contact area for stress transmission. In column caps and bases, the ends of shafts together with the attached gussets, angles, channels, etc; after connecting together should be accurately machined so that clearance between the contact surfaces shall not exceed 2 mm locally, subject further to the condition that sum total of all such clearance shall not exceed 30 percent of the total contact area for stress transmission. Care should be taken that these gussets, connecting angles or channels are fixed with such accuracy that they are not reduced in thickness by machining by more than 2.0 mm.

24.7.2 Where sufficient gussets and rivets or welds are provided to transmit the entire loading (*see* provisions under [11](#)), the column ends need not be machined.

24.7.3 Slab Bases and Caps

Slab bases and slab caps, except when cut from material with true surfaces, shall be accurately machined over the bearing surfaces and shall be in effective contact with the end of the stanchion, the bearing face which is to be grouted to fit tightly at both top and bottom, unless welds are provided to transmit the entire column face.

24.7.4 To facilitate grouting, sufficient gap shall be left between the base plates and top of pedestal and holes shall be provided where necessary in stanchion bases for the escape of air.

24.8 Painting

24.8.1 Painting shall be done in accordance with the good practice good practices [C-6A(50)].

24.8.2 All surfaces, which are to be painted, oiled or otherwise treated, shall be dry and thoroughly cleaned to remove all loose scale and loose rust.

24.8.3 Shop contact surfaces need not be painted unless specified. If so specified, they shall be brought together while the paint is still wet.

24.8.4 Surfaces not in contact, but inaccessible after shop assembly, shall receive the full specified protective treatment before assembly. This does not apply to the interior of sealed hollow sections.

24.8.5 Chequered plates shall be painted but the details of painting shall be specified by the purchaser.

24.8.6 In case of surfaces to be welded, the steel shall not be painted or metal coated within a suitable distance of any edge to be welded, if the paint specified or the metal coating is likely to be harmful to welders or impair the quality of the welds.

24.8.7 Welds and adjacent parent metal shall not be painted prior to de-slagging, inspection and approval.

24.8.8 Parts to be encased in concrete shall not be painted or oiled.

24.8.9 Contact surface in friction type connection shall not be painted in advance.

24.9 Marking

Each piece of steel work shall be distinctly marked before dispatch, in accordance with a marking diagram and shall bear such other marks as will facilitate erection.

24.10 Shop Erection

24.10.1 The steel work shall be temporarily shop erected complete or as arranged with the inspection agency so that accuracy of fit may be checked before dispatch. The parts shall be shop assembled with sufficient numbers of parallel drifts to bring and keep the parts in place.

24.10.2 In the case of parts drilled or punched, through steel jigs with bushes resulting in all similar parts being interchangeable, the steelwork may be shop erected in such position as arranged with the inspection agency.

24.10.3 In case of shop fabrication using numerically controlled machine data generated by computer software (like CAD), the shop erection may be dispensed with at the discretion of the inspector.

24.11 Packing

All projecting plates or bars and all ends of members at joints shall be stiffened, all straight bars and plates shall be bundled, all screwed ends and machined surfaces shall be suitably packed and all rivets, bolts, nuts, washers and small and loose parts shall be packed separately in cases, so as to prevent damage or distortion during transit.

24.12 Inspection and Testing

24.12.1 The inspecting authority shall have free access at all reasonable times to those parts of the manufacturer's works which are concerned with the fabrication of the steelwork and shall be afforded all reasonable facilities for satisfying himself that the fabrication is being undertaken in accordance with the provisions of this section.

24.12.2 Unless specified otherwise, inspection shall be made at the place of manufacture prior to dispatch and shall be conducted so as not to interfere unnecessarily with the operation of the work.

24.12.3 The manufacturer shall guarantee compliance with the provisions of this standard, if required to do so by the purchaser.

24.12.4 Should any structure or part of a structure be found not to comply with any of the provisions of this section, it shall be liable to rejection. No structure or part of the structure, once rejected shall be resubmitted for test, except in cases where the purchaser or his authorized representative considers the defect as rectifiable.

24.12.5 Defects, which may appear during fabrication, shall be made good with the consent of and according to the procedure laid down by the inspecting authority.

24.12.6 All gauges and templates necessary to satisfy the inspection authority shall be supplied by the manufacturer. The inspecting authority may, at his discretion, check the test results obtained at the manufacturer's works by independent testing at outside laboratory, and should the material so tested be found to be unsatisfactory, the cost of such tests shall be borne by the manufacturer, and if found satisfactory, the cost shall be borne by the purchaser.

24.13 Site Erection

24.13.1 Plant and Equipment

The suitability and capacity of all plant and equipment used for erection shall be to the satisfaction of the engineer.

24.13.2 Storing and Handling

All structural steel should be so stored and handled at the site that the members are not subjected to excessive stresses and damage by corrosion due to exposure to environment.

24.13.3 Setting Out

The positioning and leveling of all steelwork, the plumbing of stanchions and the placing of every part of the structure with accuracy shall be in accordance with the approved drawings and to the satisfaction of the engineer in accordance with the deviation permitted below.

24.13.3.1 Erection tolerances

Unloaded steel structure, as erected, shall satisfy the tolerance criteria specified in the accepted standards [C-6A(47)].

24.13.4 Safety during Fabrication and Erection

24.13.4.1 All steel materials including fabricated structures, either at fabrication shop or at erection site, shall be handled only by a worker skilled in such jobs; where necessary with load tested lifting devices, having tested wire rope slings of correct size. The devices should be well maintained and operated by experienced operators.

24.13.4.2 Oxygen and acetylene cylinders and their hoses shall have distinctive colours. Cylinders should be stored in upright position in well-ventilated rooms or in open air, not exposed to flames, naked lights or extreme heat and should also be in upright position when they are being used. All gas cutting works shall be done only by experienced skilled gas cutters, equipped with gloves, boots, aprons, goggles and good cutting sets of approved make.

24.13.4.3 While doing any welding work, it should be ensured that the welding machine is earthed and the welding cables are free from damage. The welder and his assistant shall use a face shield or head shield with a welding lens and clear cover glass and their hands, legs and bodies shall be well protected by leather gloves, shoes and aprons. Combustible materials should be kept away from the sparks and globules of molten metals generated in any arc welding. In case of welding in a confined place, it should be provided with an exhaust system to take care of the harmful gases, fumes and dusts generated.

24.13.4.4 In addition to precautions against all the hazards mentioned above, erection workers shall also be protected in the following manner:

- a) all workers shall wear helmets and shall also be provided with gloves and shoes. In addition those working at heights shall use safety belts;
- b) all structures shall be so braced/guyed during erection that there is no possibility of collapse before erection work is completed; and
- c) warning signs such as 'DANGER', 'CAUTION', '440 V', 'DO NOT SMOKE', 'LOOK AHEAD', etc; should be displayed at appropriate places.

24.13.4.5 For detailed safety precautions during erection, reference shall be made to the good practice [C-6A(51)].

24.13.5 Field Connections

24.13.5.1 Field riveting

Rivets driven at the site shall be heated and driven with the same care as those driven in the shop.

24.13.5.2 Field bolting

Field bolting shall be carried out with the same care as required for shop bolting.

24.13.5.3 Fillet welding

Field assembly and welding shall be executed in accordance with the requirements for shop fabrications excepting such as manifestly apply to shop conditions only. Where the steel has been delivered painted, the paint shall be removed for a distance of at least 50 mm on either side of the joint.

24.14 Painting after Erection

24.14.1 Before painting of such steel which is delivered unpainted is commenced, all surfaces to be painted shall be dry and thoroughly cleaned from all loose scale and rust, as required by the surface protection specification.

24.14.2 The specified protective treatment shall be completed after erection. All rivet and bolt heads and the site welds after de-slugging shall be cleaned. Damaged or deteriorated paint surfaces shall first be made good with the same type of paint as the shop coat. Where specified, surfaces which will be in contact after site assembly, shall receive a coat of paint (in addition to any shop priming) and shall be brought together while the paint is still wet. No painting shall be used on contact surfaces in the friction connection, unless specified otherwise by the design document.

24.14.3 Where the steel has received a metal coating in the shop, this coating shall be completed on site so as to be continuous over any welds and site rivets or bolts, subject to the approval of the engineer. Painting on site may complete protection. Bolts, which have been galvanized or similarly treated, are exempted from this requirement.

24.14.4 Surface, which will be inaccessible after site assembly, shall receive the full-specified protective treatment before assembly.

24.14.5 Site painting should not be done in frosty or foggy weather, or when humidity is such as to cause condensation on the surfaces to be painted.

24.15 Bedding Requirement

24.15.1 Bedding shall be carried out with Portland cement grout or mortar, as described under [24.15.4](#) or fine cement concrete in accordance with good practice [C-6A(52)].

24.15.2 For multistoried buildings, this operation shall not be carried out until a sufficient number of bottom lengths of stanchions have been properly lined, levelled and plumbed and sufficient floor beams are in position.

24.15.3 Whatever method is employed, the operation shall not be carried out until the steelwork has been finally levelled and plumbed, stanchion bases being supported meanwhile by steel wedges or nuts; and immediately before grouting, the space under the steel shall be thoroughly cleaned.

24.15.4 Bedding of structure shall be carried out with grout or mortar, which shall be of adequate strength and shall completely fill the space to be grouted and shall either be placed under pressure or by ramming against fixed supports. The grouts or mortar used shall be non-shrinking variety.

24.16 Steelwork Tenders and Contracts

A few recommendations are given in [Annex H](#) for general information.

ANNEX A

(Clause 4)

SYMBOLS

Symbols used in this section shall have the following meanings with respect to the structure or member or condition, unless otherwise defined elsewhere in this Section:

A	=	area of cross section;
A_c	=	area at root of threads;
A_e	=	effective cross sectional area;
A_{ef}	=	reduced effective flange area;
A_f	=	total flange area;
A_g	=	gross cross sectional area;
A_{gf}	=	gross cross sectional area of flange;
A_{go}	=	gross cross sectional area of outstanding (not connected) leg of a member;
A_n	=	net area of the total cross section;
A_{nb}	=	net tensile cross sectional area of bolt;
A_{nc}	=	net cross sectional area of the connected leg of a member;
A_{nf}	=	net cross sectional area of each flange;
A_{no}	=	net cross sectional area of outstanding (not connected) leg of a member;
A_{pb}	=	nominal bearing area of bolt on any plate;
A_q	=	cross sectional area of a bearing (load carrying) stiffener in contact with the flange;
A_s	=	tensile stress area;
A_{sb}	=	gross cross sectional area of a bolt at the shank;
A_{tg}	=	gross sectional area in tension from the centre of the hole to the toe of the angle section/channel section, etc (block shear failure, see 13.4) perpendicular to the line of force;
A_{tn}	=	net sectional area in tension from the centre of the hole to the toe of the angle perpendicular to the line of force block shear failure (see 13.4);
A_v	=	shear area;
A_{vg}	=	gross cross sectional area in shear along the line of transmitted force (see 13.4);
A_{vn}	=	net cross sectional area in shear along the line of transmitted force (see 13.4);
a, b	=	larger and smaller projection of the slab base beyond the rectangle circumscribing the column, respectively (see 14.4);
a_o	=	peak acceleration;
a_1	=	unsupported length of individual elements being laced between lacing points;
B	=	length of side of cap or base plate of a column;
b	=	outstand/width of the element;
b_1	=	stiff bearing length;
b_e	=	effective width of flange between pair of bolts;
b_f	=	width of the flange;
b_i	=	width of flange as an internal element;
b_o	=	width of flange outstand;

b_p	=	panel zone width between column flanges at beam-column junction;
b_s	=	shear lag distance;
b_t	=	width of tension field;
b_w	=	width of outstanding leg;
C	=	centre to centre longitudinal distance of battens;
C_m	=	coefficient of thermal expansion;
c	=	spacing of transverse stiffener;
c_b	=	moment amplification factor for braced member;
c_m	=	moment reduction factor for lateral torsional buckling strength calculation;
c_s	=	moment amplification factor for sway frame;
D	=	overall depth/diameter of the cross section;
d	=	depth of web, Nominal diameter;
d_2	=	twice the clear distance from the compression flange angles, plates or tongue plates to the neutral axis;
d_h	=	diameter of a bolt/rivet hole;
d_o	=	nominal diameter of the pipe column or the dimensions of the column in the depth direction of the base plate;
d_p	=	panel zone depth in the beam-column junction;
E	=	modulus of elasticity for steel;
$E(T)$	=	modulus of elasticity of steel at T °C;
$E(20)$	=	modulus of elasticity of steel at 20 °C;
E_p	=	modulus of elasticity of the panel material;
F_{cdw}	=	buckling strength of un-stiffened beam web under concentrated load;
F_d	=	factored design load;
F_n	=	normal force;
F_o	=	minimum proof pretension in high strength friction grip bolts;
F_{psd}	=	bearing capacity of load carrying stiffener;
F_q	=	stiffener force;
F_{qd}	=	stiffener buckling resistance;
F_{test}	=	test load;
$F_{test,a}$	=	load for acceptance test;
$F_{test, Min}$	=	minimum test load from the test to failure;
$F_{test, R}$	=	test load resistance;
$F_{test, S}$	=	strength test load;
F_w	=	design capacity of the web in bearing;
F_x	=	external load, force or reaction;
F_{xd}	=	buckling resistance of load carrying web stiffener;
f	=	actual normal stress range for the detail category;
f_1	=	frequency for a simply supported one way system;
f_2	=	frequency of floor supported on steel girder perpendicular to the joist;

f_a	=	calculated stress due to axial force at service load;
f_{abc}	=	permissible bending stress in compression at service load;
f_{ac}	=	permissible compressive stress at service load;
f_{abt}	=	permissible bending stress in tension at service load;
f_{apb}	=	permissible bearing stress of the bolt at service load;
f_{asb}	=	permissible stress of the bolt in shear at service load;
f_{at}	=	permissible tensile stress at service load;
f_{atb}	=	permissible tensile stress of the bolt at service load;
f_{aw}	=	permissible stress of the weld at service load;
f_b	=	actual bending stress at service load;
f_{bc}	=	actual bending stress in compression at service load;
f_{bd}	=	design bending compressive stress corresponding to lateral buckling;
f_{br}	=	actual bearing stress due to bending at service load;
f_{bt}	=	actual bending stress in tension at service load;
f_{bs}	=	permissible bending stress in column base at service load;
f_c	=	actual axial compressive stress at service load;
f_{cr}	=	elastic buckling stress of a column, Euler buckling stress;
f_{cd}	=	design compressive stress;
$f_{cr,b}$	=	extreme fibre compressive stress corresponding to elastic lateral buckling moment;
f_e	=	equivalent stress at service load;
f_f	=	fatigue stress range corresponding to 5×10^6 cycles of loading;
f_{feq}	=	equivalent constant amplitude stress;
f_{fd}	=	design normal fatigue strength;
$f_{f, \text{Max}}$	=	highest normal stress range;
f_{fn}	=	normal fatigue stress range;
f_{nw}	=	normal stress in weld at service load;
f_o	=	proof stress;
f_p	=	actual bearing stress at service load;
f_{pb}	=	actual bearing stress in bending at service load;
f_{psd}	=	bearing strength of the stiffeners;
f_r	=	frequency;
f_{sb}	=	actual shear stress in bolt at service load;
f_t	=	actual tensile stress at service load;
f_{tb}	=	actual tensile stress of the bolt at service load;
f_u	=	characteristic ultimate tensile stress;

f_{ub}	=	characteristic ultimate tensile stress of the bolt;
f_{um}	=	average ultimate stress of the material as obtained from test;
f_{up}	=	characteristic ultimate tensile stress of the connected plate;
f_v	=	applied shear stress in the panel designed utilizing tension field action;
f_w	=	actual stress of weld at service load;
f_{wd}	=	design stress of weld at service load;
f_{wn}	=	nominal strength of fillet weld;
f_x	=	maximum longitudinal stress under combined axial force and bending;
f_y	=	characteristic yield stress;
$f_y(T)$	=	yield stress of steel at T °C;
$f_y(20)$	=	yield stress of steel at 20 °C ;
f_{yb}	=	characteristic yield stress of bolt;
f_{yf}	=	characteristic yield stress of flange;
f_{ym}	=	average yield stress as obtained from test;
f_{yp}	=	characteristic yield stress of connected plate;
f_{yq}	=	characteristic yield stress of stiffener material;
f_{yw}	=	characteristic yield stress of the web material;
G	=	modulus of rigidity for steel;
g	=	gauge length between centres of the holes perpendicular to the load direction, acceleration due to gravity;
h	=	depth of the section;
h_b	=	total height from the base to the floor level concerned;
h_c	=	height of the column;
h_e	=	effective thickness;
h_f	=	centre to centre distance of flanges;
h_i	=	thickness of fire protection material;
h_L	=	height of the lip;
h_s	=	storey height;
h_y	=	distance between shear centre of the two flanges of a cross section;
I	=	moment of inertia of the member about an axis perpendicular to the plane of the frame;
I_{fc}	=	moment of inertia of the compression flange of the beam about the axis parallel to the web;
I_f	=	moment of inertia of the tension flange of the beam about minor axis;
I_q	=	moment of inertia of a pair of stiffener about the centre of the web, or a single stiffener about the face of the web;
I_s	=	second moment of inertia;
I_{so}	=	second moment of inertia of the stiffener about the face of the element perpendicular to the web;
I_T	=	transformed moment of inertia of the one way system (in terms of equivalent steel, assuming the concrete flange of width equal to the spacing of the beam to be effective);
I_t	=	St. Venant's torsion constant;
I_w	=	warping constant;

I_y	=	moment of inertia about the minor axis of the cross section;
I_z	=	moment of inertia about the major axis of the cross section;
K_b	=	effective stiffness of the beam and column;
K_h	=	reduction factor to account for the high strength friction grip connection bolts in oversized and slotted holes Effective length of the member;
KL	=	appropriate effective slenderness ratio of the section;
KL/r	=	effective slenderness ratio of the section about the minor axis of the section;
KL/r	=	effective slenderness ratio of the section about the major axis of the section;
$\left(\frac{KL}{r}\right)_o$	=	actual maximum effective slenderness ratio of the laced column;;
$\left(\frac{KL}{r}\right)_e$	=	effective slenderness ratio of the laced column accounting for shear deformation;
K_v	=	shear buckling co-efficient;
K_w	=	warping restraint factor;
k	=	regression coefficient;
k_{sm}	=	exposed surface area to mass ratio;
K_y, K_z, K_{LT}	=	moment amplification factors;
L	=	actual length, unsupported length, length centre to centre distance of the intersecting members, cantilever length;
L_c	=	length of end connection in bolted and welded members, taken as the distance between outermost fasteners in the end connection, or the length of the end weld, measured along the length of the member;
L_{LT}	=	effective length for lateral torsional buckling;
L_m	=	maximum distance from the restraint to the compression flange at the plastic hinge to an adjacent restraint (limiting distance);
L_o	=	length between points of zero moment (inflection) in the span;
l	=	centre to centre length of the supporting member;
l_e	=	distance between prying force and bolt centre line;
l_g	=	grip length of bolts in a connection;
l_j	=	length of the joint;
l_s	=	length between points of lateral support to the compression flange in a beam;
l_v	=	distance from bolt centre line to the toe of fillet weld or to half the root radius for a rolled section;
l_w	=	length of weld;
M	=	bending moment;
M_a	=	applied bending moment;
M_{cr}	=	elastic critical moment corresponding to lateral torsional buckling of the beam;
M_d	=	design flexural strength;
M_{dv}	=	moment capacity of the section under high shear;
M_{dy}	=	design bending strength about the minor axis of the cross section;
M_{dz}	=	design bending strength about the major axis of the cross section;
M_{eff}	=	reduced effective moment;

M_{fr}	=	reduced plastic moment capacity of the flange plate;
M_{fd}	=	design plastic resistance of the flange alone;
M_{nd}	=	design bending strength under combined axial force and uniaxial moment;
$M_{ndy},$ M_{ndz}	=	design bending strength under combined axial force and the respective uniaxial moment acting alone;
M_p	=	plastic moment capacity of the section;
M_{pb}	=	moment in the beam at the intersection of the beam and column centre lines;
M_{pc}	=	moments in the column above and below the beam surfaces;
M_{pd}	=	plastic design strength;
M_{pdf}	=	plastic design strength of flanges only;
M_q	=	applied moment on the stiffener;
M_s	=	Moment at service (working) load
M_{tf}	=	moment resistance of tension flange;
M_y	=	factored applied moment about the minor axis of the cross section;
M_{yq}	=	moment capacity of the stiffener based on its elastic modulus;
M_z	=	factored applied moment about the major axis of the cross section;
N	=	number of parallel planes of battens;
N_d	=	design strength in tension or in compression;
N_f	=	axial force in the flange;
N_{SC}	=	number of stress cycles;
n	=	number of bolts in the bolt group/critical section;
n_e	=	number of effective interfaces offering frictional resistance to slip;
n_n	=	number of shear planes with the threads intercepting the shear plane in the bolted connection;
n_s	=	number of shear planes without threads intercepting the shear plane in the bolted connection;
P	=	factored applied axial force;
P_{cr}	=	elastic buckling load
P_d	=	design axial compressive strength;
P_{dy}, P_{dz}	=	design compression strength as governed by flexural buckling about the respective axis;
P_e	=	elastic Euler buckling load;
P_{min}	=	minimum required strength for each flange splice;
P_r	=	required compressive strength;
P_s	=	actual compression at service load;
P_y	=	yield strength of the cross section under axial compression;
p	=	pitch length between centres of holes parallel to the direction of the load;
p_s	=	staggered pitch length along the direction of the load between lines of the bolt holes (see Fig. 5);
Q	=	prying force;
Q_a	=	accidental load (Action);
Q_c	=	characteristic loads (Action);

Q_d	=	design load (Action);
Q_p	=	permanent loads (Action);
Q_v	=	variable loads (Action);
q	=	shear stress at service load;
R	=	ratio of the mean compressive stress in the web (equal to stress at mid depth) to yield stress of the web; reaction of the beam at support ;
R_d	=	design strength of the member at room temperature ;
R_i	=	net shear in bolt group at bolt 'i';
R_r	=	response reduction factor;
R_{tf}	=	flange shear resistance;
R	=	ultimate strength of the member at room temperature;
u	=	appropriate radius of gyration;
r	=	minimum radius of gyration of the individual element being laced together;
r_l	=	ratio of the design action on the member under fire to the design capacity;
r_f	=	radius of gyration about the minor axis (v-v) of angle section;
r_{vv}	=	radius of gyration about the minor axis;
r_y	=	radius of gyration about the major axis;
r_z	=	minimum transverse distance between the centroid of the rivet or bolt group or weld group;
S	=	constant stress range;
S_c	=	design strength;
S_d	=	original cross sectional area of the test specimen;
S_o	=	spring stiffness;
S_p	=	ultimate strength;
S_u	=	anchorage length of tension field along the compression flange;
s_c	=	anchorage length of tension field along the tension flange;
s_t	=	actual stiffener spacing;
s_a	=	ratio of the mean compressive stress in the web (equal to stress at mid depth) to yield stress of the web; reaction of the beam at support;
T	=	temperature in degree Celsius; Factored tension;
T_b	=	applied tension in bolt;
T_{cf}	=	thickness of compression flange;
T_d	=	design strength under axial tension;
T_{dg}	=	yielding strength of gross section under axial tension;
T_{dn}	=	rupture strength of net section under axial tension;
T_{db}	=	design strength of bolt under axial tension, Block shear strength at end connection;
T_e	=	externally applied tension;
T_f	=	factored tension force of friction type bolt;
T_l	=	limiting temperature of the steel;
T_{nb}	=	nominal strength of bolt under axial tension;
T_{nd}	=	design tension capacity;

T_{ndf}	=	design tension capacity of friction type bolt;
T_{nf}	=	nominal tensile strength of friction type bolt;
T_s	=	actual tension under service load;
t	=	thickness of element/angle, time in minutes;
t_f	=	thickness of flange;
t_p	=	thickness of plate;
t_{pk}	=	thickness of packing;
t_q	=	thickness of stiffener;
t_s	=	thickness of base slab;
t_t	=	effective throat thickness of welds;
t_w	=	thickness of web;
V	=	factored applied shear force;
V_b	=	shear in batten plate;
V_{bf}	=	factored frictional shear force in friction type connection;
V_{cr}	=	critical shear strength corresponding to web buckling;
V_d	=	design shear strength;
V_{db}	=	block shear strength;
V_n	=	nominal shear strength of bolt;
V_{nbf}	=	bearing capacity of bolt for friction type connection;
V_p	=	plastic shear resistance under pure shear;
V_n	=	nominal shear strength;
V_{npb}	=	nominal bearing strength of bolt;
V_{nsb}	=	nominal shear capacity of a bolt;
V_{nsf}	=	nominal shear capacity of bolt as governed by slip in friction type connection;
V_s	=	transverse shear at service load;
V_{sb}	=	factored shear force in the bolt;
V_{sd}	=	design shear capacity;
V_{sdf}	=	design shear strength in friction type bolt;
V_{sf}	=	factored design shear force of friction bolts;
V_t	=	applied transverse shear;
V_{tf}	=	shear resistance in tension field;
W	=	total load;
w	=	uniform pressure from below on the slab base due to axial compression under the factored load;
w_{tf}	=	width of tension field;
x_t	=	torsional index;
Z_e	=	elastic section modulus;
Z_{ec}	=	elastic section modulus of the member with respect to extreme compression fibre;
Z_{et}	=	elastic section modulus of the member with respect to extreme tension fibre;
Z_p	=	plastic section modulus;

Z_v	=	contribution to the plastic section modulus of the total shear area of the cross section;
y_g	=	distance between point of application of the load and shear centre of the cross section;
y_s	=	co-ordinate of the shear centre in respect to centroid;
α	=	imperfection factor for buckling strength in columns and beams;
α_t	=	coefficient of thermal expansion;
β_M	=	ratio of smaller to the larger bending moment at the ends of a beam column;
β_{My}, β_{Mz}	=	equivalent uniform moment factor for flexural buckling for y-y and axes, respectively;
β_{MLT}	=	equivalent uniform moment factor for lateral torsional buckling;
χ	=	strength reduction factor to account for buckling under compression;
χ_m	=	strength reduction factor, χ , at f_{ym} ;
χ_{LT}	=	strength reduction factor to account for lateral torsional buckling of beams;
δ	=	storey deflection;
δ_L	=	horizontal deflection of the bottom of storey due to combined gravity and notional load;
δ_p	=	load amplification factor;
δ_u	=	horizontal deflection of the top of storey due to combined gravity and notional load;
ϕ	=	inclination of the tension field stress in web;
γ	=	unit weight of steel;
γ_L	=	partial safety factor for load;
γ_m	=	partial safety factor for material;
γ_{m0}	=	partial safety factor against yield stress and buckling;
γ_{m1}	=	partial safety factor against ultimate stress;
γ_{mb}	=	Partial safety factor for bolted connection with bearing type bolts;
γ_{mf}	=	partial safety factor for bolted connection with high strength friction grip bolts;
γ_{ft}	=	partial safety factor for fatigue load;
γ_{mft}	=	partial safety factor for fatigue strength;
γ_{mv}	=	partial safety factor against shear failure;
γ_{mw}	=	partial safety factor for strength of weld;
ε	=	yield stress ratio, $(250 / f_y)^{1/2}$;
λ	=	Non dimensional slenderness ratio $= \sqrt{f_y (KL/r)^2 / \pi^2 E} = \sqrt{f_y / f_{cr}} = \sqrt{P_y / P_{cr}}$
λ_{cr}	=	elastic buckling load factor;
λ_e	=	equivalent slenderness ratio;
λ_{LT}	=	non-dimensional slenderness ratio in lateral bending;
λ_{scr}	=	elastic buckling load factor of each storey;
μ	=	poisson's ratio;
μ_c	=	correction factor;
μ_f	=	coefficient of friction (slip factor);
μ_r	=	capacity reduction factor;
θ	=	ratio of the rotation at the hinge point to the relative elastic rotation of the far end of the beam segment containing plastic hinge;

ρ	=	unit mass of steel;
τ	=	actual shear stress range for the detail category;
τ_b	=	buckling shear stress;
τ_{ab}	=	permissible shear stress at the service load;
$\tau_{cr,e}$	=	elastic critical shear stress;;
τ_f	=	fatigue shear stress range;
$\tau_{f, Max}$	=	highest shear stress range;
τ_{fd}	=	design shear fatigue strength;
τ_{fn}	=	fatigue shear stress range at N_{sc} cycle for the detail category;
τ_v	=	actual shear stress at service load;
ψ	=	ratio of the moments at the ends of the laterally unsupported length of a beam; and
Γ	=	frame buckling load factor.

NOTE — The subscripts y, z denote the y-y and z-z axes of the section, respectively. For symmetrical sections, y-y denotes the minor principal axis whilst z-z denotes the major principal axis (see 8).

ANNEX B

[Clauses [11.1.1\(c\)](#) and [11.7.2.1\(b\)](#)]

ANALYSIS AND DESIGN METHODS

B-1 ADVANCED STRUCTURAL ANALYSIS AND DESIGN

B-1.1 Analysis

For a frame, comprising members of compact section with full lateral restraint, an advanced structural analysis may be carried out, provided the analysis can be shown to accurately model the actual behaviour of that class of frames. The analysis shall take into account the relevant material properties, residual stresses, geometrical imperfections, reduction in stiffness due to axial compression, second-order effects, section strength and ductility, erection procedures and interaction with the foundations.

Advanced structural analysis for earthquake loads shall take into account provisions of 5 of Part C ‘Structural design/Section 1 Loads and Load Effects’ of this NBCS.

B-1.2 Design

For the strength limit state, it shall be sufficient to satisfy the section capacity requirements given under [15](#) for the members subjected to bending, under [14](#) for axial members, under [16](#) for combined forces and under [17](#) for connections. Effect of moment magnification given in [16](#), instability given in [14](#) and lateral buckling given in [15](#) need not be considered while designing the member, since advanced analysis methods directly consider these.

B-2 SECOND ORDER ELASTIC ANALYSIS AND DESIGN

B-2.1 Analysis

In a second-order elastic analysis, the members shall be assumed to remain elastic, and changes in frame geometry under the design load and changes in the effective stiffness of the members due to axial forces shall be accounted for. In a frame where the elastic buckling load factor (λ_{cr}) of the frame as determined in accordance with [11.6](#) is greater than 5, the changes in the effective stiffness of the members due to axial forces may be neglected.

B-2.2 Design Bending Moment

The design bending moment under factored load shall be taken as the maximum bending moment in the length of the member. It shall be determined either:

- directly from the second-order analysis; or
- approximately, if the member is divided into a sufficient number of elements, as the greatest of the element end bending moments, or;
- by amplifying the calculated design bending moment, taken as the maximum bending moment along the length of a member as obtained by superposition of the simple beam bending moments determined by the analysis.

For a member with zero axial force or a member subject to axial tension, the factored design bending moment shall be calculated as the moment obtained from second-order analysis without any amplification.

For a member with a design axial compressive force as determined from the analysis, the factored design bending moment shall be calculated as follows:

$$M = \delta_b M_m$$

where

δ_b = moment amplification factor for a braced member determined in accordance with [16](#).

B-3 FRAME INSTABILITY ANALYSIS

B-3.1 Analysis

Frame instability, as treated here, is related to the design of multi-storey rigid-jointed frames subject to side sway. The elastic critical load factor, λ_{cr} may be determined using the deflection method as given in [B-3.2](#) or any other recognized method. This is used to calculate the amplified sway moments for elastic designs and to check frame stability in plastic designs. The elastic critical load factor, λ_{cr} of a frame is the ratio by which each of the factored loads would have to be increased to cause elastic instability.

B-3.2 Deflection Method

An accurate method of analysis (ordinary linear elastic analysis) should be used to determine the horizontal deflections of the frame due to horizontal forces applied at each floor level, which is equal to the notional horizontal load in [11.3.6](#). Allowance should be made for the degree of rigidity of the base as given in [B-3.2](#) in this deflection calculation.

The base stiffness should be determined by reference to [11.3.4](#).

The elastic critical load factor, λ_{cr} is calculated as:

$$\lambda_{cr} = \frac{1}{200 \phi_{s, \text{Max}}}$$

where

$\phi_{s, \text{Max}}$ = largest value of the sway index where, ϕ_s given by:

$$\phi_{si} = \frac{\delta_{ui} - \delta_{Li}}{h_i}$$

where

h_i = storey height;

δ_{ui} = horizontal deflection of the top of the storey due to the combined gravity and notional loads; and

δ_{Li} = horizontal deflection of the bottom of the storey due to gravity and notional load.

B-3.3 Partial Sway Bracing

In any storey the stiffening effect of infill wall panels may be allowed for by introducing a diagonal strut in that storey of area A , given by:

$$A = \frac{k_3 \sum K_c}{h(h/b)} \left[1 + (h/b)^2 \right]^{3/2}$$

where

h = storey height;

b = width of the braced bay;

$\sum K_c$ = sum of the stiffness I/L , of the columns in that storey;

$$k_3 = \frac{h^2 \sum S_p}{80E \sum K_c} \leq 2; \text{ and}$$

$\sum S_p$ = Sum of spring stiffness (horizontal force per unit horizontal deflection of all the panels in that storey determined from:

$$S_p = \frac{0.6h/b}{\left[1 + (h/b)^2 \right]^2} t_p E_p$$

where

t_p = thickness of the wall panel; and

E_p = modulus of elasticity of the panel material.

ANNEX C

[Clauses [12.2.2.2](#) and [12.6.2](#)]

DESIGN AGAINST FLOOR VIBRATION

C-1 GENERAL

Floor with longer spans of lighter construction and less inherent damping are vulnerable to vibrations under normal human activity. Natural frequency of the floor system corresponding to the lowest mode of vibration, damping characteristics, are important characteristics in floor vibration. Open web steel joists (trusses) or steel beams on the concrete deck may experience walking vibration problem. Fatigue, overloading of floor systems and vibrations due to rhythmic activities such as aerobic or dance classes are not within the scope of this annex.

C-2 ANNOYANCE CRITERIA

In the frequency range of 2 to 8 Hz in which people are most sensitive to vibration, the threshold level corresponds approximately to 0.5 percent g , where g is the acceleration due to gravity. Continuous vibration is generally more annoying than decaying vibration due to damping. Floor systems with the natural frequency less than 8 Hz in the case of floors supporting rhythmic activity and 5 Hz in the case of floors supporting normal human activity should be avoided. However, if it is unavoidable, recommendations of specialist literature may be followed.

C-3 FLOOR FREQUENCY

The fundamental natural frequency can be estimated by assuming full composite action, even in non-composite construction. This frequency, f_1 for a simply supported one way system is given by:

$$f_1 = 156 \sqrt{EI_T / WL^4}$$

where

E = modulus of elasticity of steel, in MPa;

I_T = transformed moment of inertia of the one way system (in terms of equivalent steel) assuming the concrete flange of width equal to the spacing of the beam to be effective, in mm⁴;

L = span length, in mm; and

W = dead load of the one way joist, in N/mm.

If the one way joist system is supported by a flexible beam running perpendicular with the natural frequency f_2 , the floor frequency may be reduced to f_r , given by:

$$\frac{1}{f_r^2} = \frac{1}{f_1^2} + \frac{1}{f_2^2}$$

C-4 DAMPING

The percentage of critical damping may be assumed approximately as given below:

<i>Sl No.</i>	<i>System</i>	<i>Critical Damping Percent</i>
(1)	(2)	(3)
i)	Fully composite construction	2
ii)	Bare steel beam and concrete deck	3 to 4
iii)	Floor with finishes, false ceiling, fire proofing, ducts, furniture	6
iv)	Partitions not located along a support or not spaced farther apart than 6 m and partitions oriented in orthogonal directions	Up to 12

C-5 ACCELERATION

The peak acceleration a_o , from heel impact for floors of spans greater than 7m and natural frequency f_1 , less than 10 Hz may be calculated as:

$$a_o/g = 600 f_r / W$$

where

W = total weight of floors plus contents over the span length and equivalent floor width (b), in N;

$b = 40t_s$ (20 t_s when overhang is only on one side of the beam);

t_s = equivalent thickness of the slab, averaging concrete in slab and ribs; and

g = acceleration due to gravity.

ANNEX D

(Clause 14.2.2)

DETERMINATION OF EFFECTIVE LENGTH OF COLUMNS

D-1 METHOD FOR DETERMINING EFFECTIVE LENGTH OF COLUMNS IN FRAMES

In the absence of a more exact analysis, the effective length of columns in framed structures may be obtained by multiplying the actual length of the column between the centres of laterally supporting members (beams) given in Fig. 27 and Fig. 28 with the effective length factor K , calculated by using the equations given below, provided the connection between beam and column is rigid type:

a) *Non-sway frames (braced frame)* [(see 11.1.2 (a))]

A frame is designated as non-sway frame if the relative displacement between the two adjacent floors is restrained by bracings or shear walls (see 11.1.2). The effective length factor, K , of column in non-sway frames is given by (see Fig. 27):

$$K = \frac{[1 + 0.145(\beta_1 + \beta_2) - 0.265\beta_1\beta_2]}{[2 - 0.364(\beta_1 + \beta_2) - 0.247\beta_1\beta_2]}$$

b) *Sway frames (moment resisting frames)* [see 11.1.2 (b)]

The effective length factor K , of column in sway frames is given by (see Fig. 28):

$$K = \left[\frac{1 - 0.2(\beta_1 + \beta_2) - 0.12\beta_1\beta_2}{1 - 0.8(\beta_1 + \beta_2) + 0.6\beta_1\beta_2} \right]^{0.5}$$

where

$$\beta_1, \beta_2 \text{ are given, } \beta = \frac{\sum K_c}{\sum K_c + \sum K_b}$$

K_c, K_b = Effective flexural stiffness of the columns and beams meeting at the joint at the two ends of the columns and rigidly connected at the joints, and these are calculated by:

$$K_c \text{ and } K_b = C (I/L)$$

I = moment of inertia of the member about an axis perpendicular to the plan of the frame,

L = length of the member equal to centre to centre distance of the intersecting member, and

C = correction factor as shown in Table 33.

Table 33 Correction Factors for Effective Flexural Stiffness

(Clause D-1)

Sl No.	Far End Condition	Correction Factor, C	
		Braced Frame	Unbraced Frame
(1)	(2)	(3)	(4)
i)	Pinned	$0.67(1 - \bar{n})$	$0.75(1 - \bar{n})$
ii)	Rigidly connected to column	$0.5(1 - \bar{n})$	$1.5(1 - 0.2\bar{n})$
iii)	Fixed	$1.0(1 - 0.4\bar{n})$	$1.0(1 - 0.4\bar{n})$

NOTE — $\bar{n} = \frac{P}{P_e}$

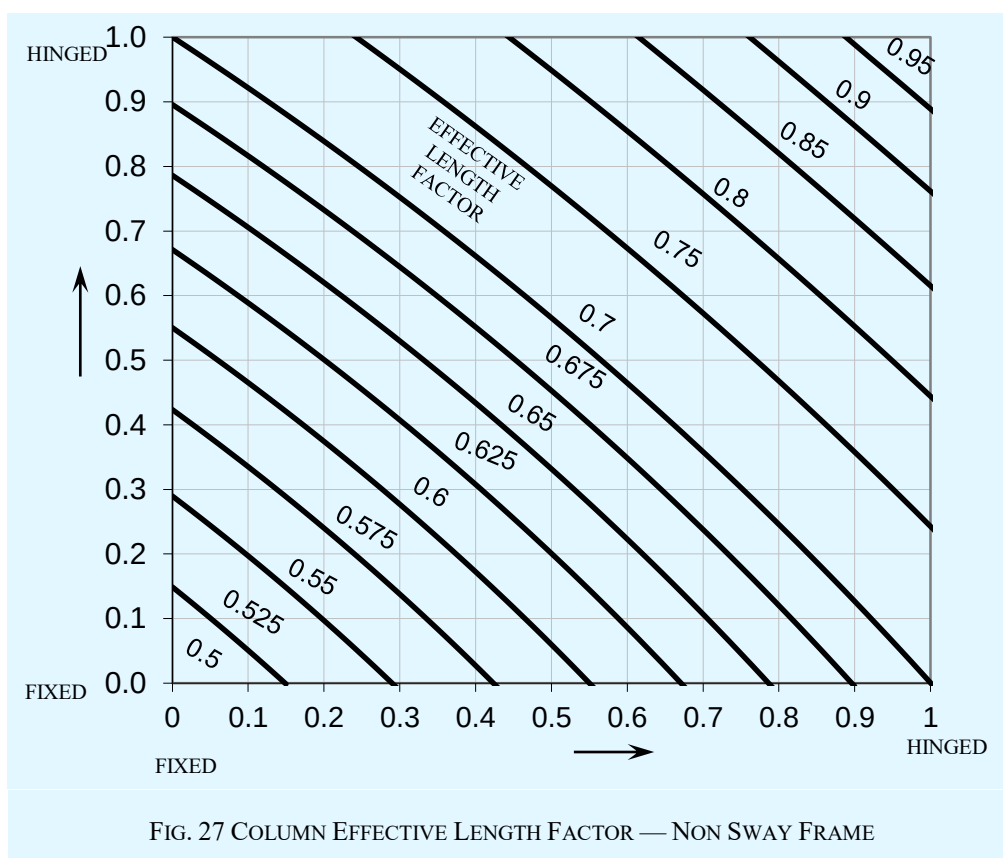
where
 P_e = elastic buckling load, and
 P = applied load.

D-2 METHOD FOR DETERMINING EFFECTIVE LENGTH FOR STEPPED COLUMNS (see 14.2.2)

D-2.1 Single Stepped Columns

Effective length in the plane of stepping (bending about axis z-z) for bottom and top parts for single stepped column shall be taken as given in Table 34.

NOTE — The provisions of D-2.1 are applicable to intermediate columns as well with stepping on either side, provided appropriate values of I_1 and I_2 are taken.



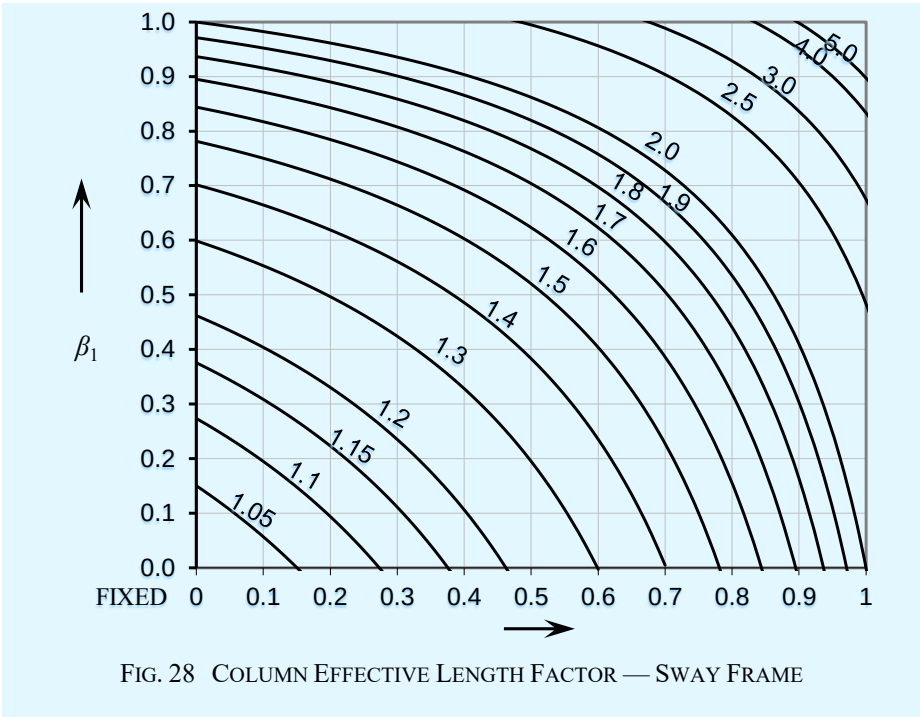


FIG. 28 COLUMN EFFECTIVE LENGTH FACTOR — SWAY FRAME

Table 34 Effective Length of Singled Stepped Columns (Clause D-2.1)				
Sl No.	Degree of End Restraint	Sketch	Effective Length Coefficients	Column Parameters for all Cases
(1)	(2)	(3)	(4)	(5)
i)	Effectively held in position and restrained against rotation at both ends		$K_1 = \sqrt{\frac{K_{12}^2 + K_{11}^2(\alpha - 1)}{\alpha}}$ $K_2 = \frac{K_1}{C_1} \leq 3$ where K_{12} and K_{11} are to be taken as per Table 35	
ii)	Effectively held in position at both ends and restrained against rotation at bottom end only		$K_1 = \sqrt{\frac{K_{12}^2 + K_{11}^2(\alpha - 1)}{\alpha}}$ $K_2 = \frac{K_1}{C_1} \leq 3$ where K_{12} and K_{11} are to be taken as per Table 36	

Table 34 (Concluded)

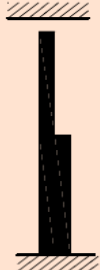
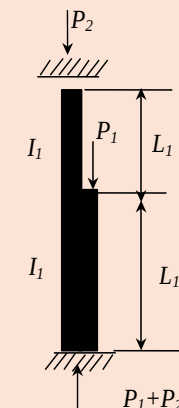
Sl No.	Degree of End Restraint	Sketch	Effective Length Coefficients	Column Parameters for all Cases
(1)	(2)	(3)	(4)	(5)
iii)	Effectively held in position and restrained against rotation at bottom end, and top end held against rotation but not held in position		K_1 to be taken as per Table 37 $K_2 = \frac{K_1}{C_1} \leq 3$	$\alpha = \frac{P_1 + P_2}{P_2}$ $C_1 = \frac{L_2}{L_1} \sqrt{\frac{I_1}{I_2 \alpha}}$ $\frac{i_2}{i_1} = \frac{I_2}{L_2} \times \frac{L_1}{I_1}$ Effective length of bottom part of column in plane of stepping = $K_1 L_1$
iv)	Effectively held in position and restrained against rotation at bottom end, and top end neither held against rotation nor held in position		K_1 to be taken as per Table 38 $K_2 = \frac{K_1}{C_1} \leq 3$	Effective length of top part of column in plane of stepping = $K_2 L_2$

Table 35 Coefficients of Effective Lengths K_{12} And K_{11} for Columns with Both Ends Effectively Held in Position and Restrained Against Rotation

(Table 34)

	Coefficients K_{12} and K_{11} for L_2/L_1 Equal to														
I_2/I_1	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.2	1.4	1.6	1.8	2.0
	Coefficient K_{12} ($P_1 = 0$)														
0.05	0.74	0.94	1.38	1.60	1.87	2.07	2.23	2.39	2.52	2.67	3.03	3.44	3.85	4.34	4.77
0.1	0.67	0.76	1.00	1.20	1.42	1.61	1.78	1.92	2.04	2.20	2.40	2.60	2.86	3.18	3.41
0.2	0.64	0.70	0.79	0.93	1.07	1.23	1.41	1.50	1.60	1.72	1.92	2.11	2.28	2.45	2.64
0.3	0.62	0.68	0.74	0.85	0.95	1.06	1.18	1.28	1.39	1.48	1.67	1.82	1.96	2.12	2.20
0.4	0.60	0.66	0.71	0.77	0.82	0.93	0.99	1.08	1.17	1.23	1.39	1.53	1.66	1.79	1.92
0.5	0.59	0.65	0.70	0.77	0.82	0.93	0.99	1.08	1.17	1.23	1.39	1.53	1.66	1.79	1.92
1.0	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.90	0.95	1.00	1.10	1.20	1.30	1.40	1.50
	Coefficient K_{11} ($P_2 = 0$)														
0.05	0.65	0.67	0.71	0.85	1.01	1.17	1.31	1.41	1.50	1.57	1.67	1.74	1.78	1.82	1.86
0.1	0.64	0.65	0.65	0.65	0.78	0.92	1.05	1.15	1.25	1.33	1.45	1.55	1.62	1.68	1.71
0.2	0.62	0.64	0.65	0.65	0.66	0.73	0.83	0.92	1.01	1.09	1.23	1.33	1.41	1.48	1.54
0.3	0.60	0.63	0.64	0.65	0.66	0.67	0.73	0.81	0.89	0.94	1.09	1.20	1.28	1.35	1.41
0.4	0.58	0.63	0.63	0.64	0.64	0.66	0.68	0.75	0.82	0.88	1.01	1.10	1.19	1.26	1.32
0.5	0.57	0.61	0.63	0.64	0.64	0.65	0.68	0.72	0.77	0.83	0.94	1.04	1.12	1.19	1.25
1.0	0.55	0.58	0.60	0.61	0.62	0.63	0.65	0.67	0.70	0.73	0.80	0.88	0.93	1.01	1.05
NOTE — Intermediate value may be obtained by interpolation															



(Table 34)

Table 36 Coefficients of Effective Lengths K_{12} And K_{11} for Columns with Both Ends Effectively Held in Position and Restrained Against Rotation at Bottom End Only															
Coefficients K_{12} and K_{11} for L_2/L_1 Equal to															
I_2/I_1	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.2	1.4	1.6	1.8	2.0
Coefficient K_{12} ($P_1 = 0$)															
0.05	0.97	1.66	2.10	2.43	2.72	2.92	3.08	3.24	3.48	3.73	4.22	4.85	5.36	6.00	6.58
0.1	0.83	1.21	1.57	1.95	2.14	2.33	2.46	2.60	2.76	2.91	3.28	3.61	4.03	4.43	4.85
0.3	0.78	0.90	1.09	1.27	1.44	1.60	1.74	1.86	1.98	2.11	2.35	2.51	2.76	2.99	3.25
0.5	0.78	0.86	0.99	1.10	1.22	1.35	1.47	1.57	1.67	1.76	1.96	2.15	2.34	2.50	2.76
1.0	0.78	0.85	0.92	0.99	1.06	1.13	1.20	1.27	1.34	1.41	1.54	1.68	1.82	1.97	2.10
Coefficient K_{11} ($P_2 = 0$)															
0.05	0.67	0.67	0.82	1.16	1.35	1.48	1.58	1.65	1.69	1.74	1.81	1.84	1.86	1.88	1.90
0.1	0.67	0.67	0.73	0.93	1.11	1.25	1.36	1.45	1.52	1.57	1.66	1.72	1.77	1.80	1.82
0.3	0.67	0.67	0.67	0.71	0.80	0.90	0.99	1.08	1.15	1.22	1.33	1.41	1.48	1.54	1.59
0.5	0.67	0.67	0.67	0.69	0.73	0.81	0.87	0.94	1.01	1.07	1.17	1.26	1.33	1.39	1.44
1.0	0.67	0.67	0.67	0.68	0.71	0.74	0.78	0.82	0.87	0.91	0.99	1.07	1.13	1.19	1.24
NOTE — Intermediate value may be obtained by interpolation.															

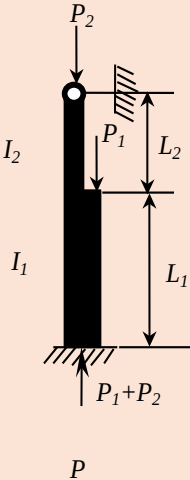


Table 37 Coefficients of Effective Lengths K_1 for Columns Effectively Held in Position and Restrained Against Rotation at Bottom End and Top End Held Against Rotation but not Held in Position

(Table 34)

C_1	Coefficients K_1 for I_2/I_1 Equal to														
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.2	1.4	1.6	1.8	2.0
0	2.0	1.86	1.7	1.67	1.6	1.55	1.50	1.4	1.43	1.4	1.37	1.3	1.1	1.10	1.0
0.5	2.0	1.90	1.8	1.74	1.6	1.65	1.61	1.5	1.55	1.5	---	---	---	---	---
1.0	2.0	2.00	2.0	2.00	2.0	2.00	---	---	---	---	---	---	---	---	---
1.5	2.0	2.25	2.3	2.48	---	---	---	---	---	---	---	---	---	---	---
2.0	2.0	2.66	2.91	---	---	---	---	---	---	---	---	---	---	---	---
2.5	2.5	3.17	3.50	---	---	---	---	---	---	---	---	---	---	---	---
3.0	3.0	3.70	4.12	---	---	---	---	---	---	---	---	---	---	---	---

NOTE — Intermediate values may be obtained by interpolation.

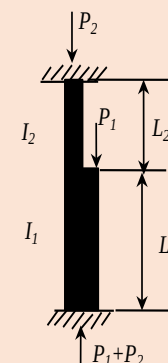
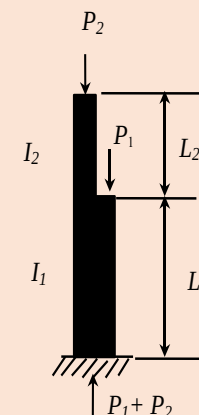


Table 38 Coefficients of Effective Lengths K_1 for Columns with Top Ends Free and Bottom End Effectively Held in Position and Restrained Against Rotation

(Table 34)

C_1	Coefficients K_1 for I_2/I_1 Equal to														
	0.1	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0	2.5	5.0	10	20
0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
0.5	2.0	2.14	2.24	2.36	2.47	2.57	2.67	2.76	2.85	2.94	3.02	---	---	---	---
1.0	2.0	2.73	3.13	3.44	3.74	4.00	---	---	---	---	---	---	---	---	---
1.5	3.0	3.77	4.35	4.86	---	---	---	---	---	---	---	---	---	---	---
2.0	4.0	4.90	5.67	---	---	---	---	---	---	---	---	---	---	---	---
2.5	5.0	6.08	7.00	---	---	---	---	---	---	---	---	---	---	---	---
3.0	6.0	7.25	---	---	---	---	---	---	---	---	---	---	---	---	---

NOTE — Intermediate values may be obtained by interpolation.



D-3 Effective Length for Double Stepped Columns

Effective lengths in the plane of steppings (bending about axis z-z) for bottom, middle and top parts for a double stepped column shall be taken as follows (see also [Fig. 29](#)):

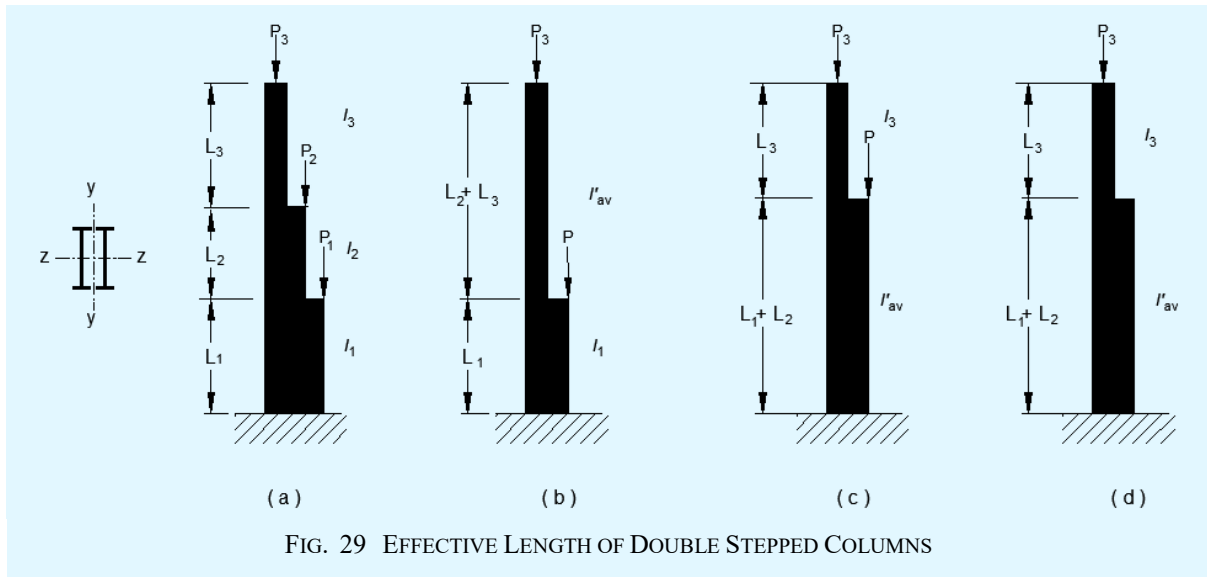


FIG. 29 EFFECTIVE LENGTH OF DOUBLE STEPPED COLUMNS

Coefficient K_1 for effective length of bottom part of double stepped column shall be taken from the formula:

$$K_1 = \sqrt{\frac{t_1 K_1^2 + (t_2 K_2^2 + K_3^2) \times (1 + n_2)^2 \times \frac{I_1}{I'_{av}}}{1 + t_1 + t_2}}$$

where

K_1 , K_2 , and K_3 are taken from [Table 39](#).

$$t_1 = \frac{P_1}{P_3}, \quad t_2 = \frac{P_2}{P_3},$$

I'_{av} = average value of moment of inertia for the lower and middle parts

$$= \frac{I_1 L_1 + I_2 L_2}{L_1 + L_2}$$

I'_{av} = average value of moment of inertia for the middle and top parts

$$= \frac{I_2 L_2 + I_3 L_3}{L_2 + L_3}$$

Value of coefficient K_2 for middle part of column is given by formula:

$$K_2 = \frac{K_1}{C_2}, \text{ and}$$

Value of coefficient K_3 for the top part of the column is given by:

$$K_3 = \frac{K_1}{C_3} \leq 3$$

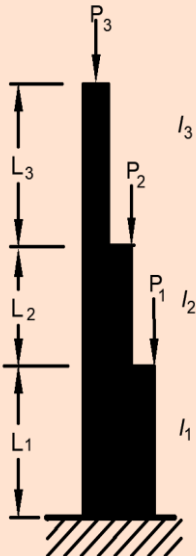
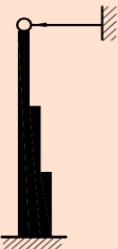
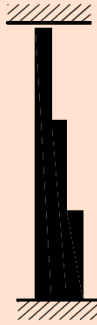
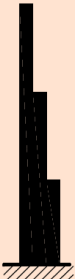
where

$$C_2 = \frac{L_2}{L_1} \sqrt{\frac{I_1(P_2 + P_3)}{I_2(P_1 + P_2 + P_3)}}, C_3 = \frac{L_3}{L_1} \sqrt{\frac{I_1 P_3}{I_3(P_1 + P_2 + P_3)}}$$

NOTE — The provisions of [D-3](#) are applicable to intermediate columns as well with steppings on either side, provided appropriate values of I_1 , I_2 and I_3 are taken.

Table 39 Values of K_1 , K_2 and K_3

(Clause [D-3](#))

Sl No.	Degree of End Restraint	Sketch	K_1	K_2	K_3	Column Parameters for all Cases
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Effectively held in position and restrained against rotation at both ends		$K_1 = K_{11}$ where K_{11} is taken from Table 35	$K_2 = K_{11}$ where K_{11} is taken from Table 35	$K_3 = K_{12}$ where K_{12} is taken from Table 35	
ii)	Effectively held in position at both ends and restrained against rotation at bottom end only		$K_1 = K_{11}$ where K_{11} is taken from Table 36	$K_2 = K_{11}$ where K_{11} is taken from Table 36	$K_3 = K_{12}$ where K_{12} is taken from Table 36	
iii)	Effectively held in position and restrained against rotation at bottom end, and top end held against rotation but not held in position		$K_1 = K_1$ where K_1 is taken from Table 37 with $C_1 = 0$	$K_2 = K_1$ where K_1 is taken from Table 37 with $C_1 = 0$	$K_3 = K_1$ where K_1 is taken from Table 37 with $C_1 = \frac{L_3}{L_1 + L_2} \sqrt{\frac{I_{av}}{I_3}}$	
iv)	Effectively held in position and restrained against rotation at bottom end, and top end neither held against rotation nor against translation		$K_1 = 2$	$K_2 = 2$	$K_3 = K_1$ where K_1 is taken from Table 38 with $C_1 = \frac{L_3}{L_1 + L_2} \sqrt{\frac{I_{av}}{I_3}}$	

ANNEX E

(Clause [15.2.2.1](#))

ELASTIC LATERAL TORSIONAL BUCKLING

E-1 ELASTIC CRITICAL MOMENT

E-1.1 General

The elastic critical moment is affected by the following:

- Moment gradient in the unsupported length;
- Boundary conditions at the lateral support points;
- Non-symmetric and non-prismatic nature of the member; and
- Location of transverse load with respect to shear centre.

The boundary conditions at the lateral supports have two components:

- Torsional restraint* — where the cross section is prevented from rotation about the shear centre; and
- Warping restraint* — where the flanges are prevented from rotating in their own plane about an axis perpendicular to the flange.

The elastic critical moment corresponding to lateral torsional buckling of a doubly symmetric prismatic beam subjected to uniform moment in the unsupported length and torsionally restraining lateral supports is given by:

$$M_{cr} = \frac{\pi^2 E I_y}{(L_{LT})^2} \left[\frac{I_w}{I_y} + \frac{G I_t (L_{LT})^2}{\pi^2 E I_y} \right]^{0.5}$$

where

I_y, I_w, I_t = moment of inertia about the minor axis, warping constant and St.Venant's torsion constant of the cross section, respectively (see accepted standards [C-6A(36)] for properties of sections);

G = modulus of rigidity; and

L_{LT} = effective length against lateral torsional buckling (see [15.3](#)).

This equation in simplified form for I-section has been presented in [15.2.2.1](#).

While the simplified equation is generally on the safe side, there are many situations where this may be very conservative. More accurate calculation of the elastic critical moment for general case of unsymmetrical sections, tapered members, loading away from shear centre and beams with moment gradient can be obtained from specialist literature, by using an appropriate computer program or equations given below.

E-1.2 Elastic Critical Moment of a Section Symmetrical About Minor Axis

In case of a beam which is symmetrical only about the minor axis, and bending about major axis, the elastic critical moment for lateral torsional buckling is given by the general equation below,

$$M_{cr} = c_1 \frac{\pi^2 E I_y}{(L_{LT})^2} \left\{ \left[\left(\frac{K}{K_w} \right)^2 \frac{I_w}{I_y} + \frac{G I_t (L_{LT})^2}{\pi^2 E I_y} + (c_2 y_g - c_3 y_j)^2 \right]^{0.5} - (c_2 y_g - c_3 y_j) \right\}$$

where

c_1, c_2, c_3 = factors depending upon the loading and end restraint conditions (see [Table 40](#));

K = effective length factors of the unsupported length accounting for boundary conditions at the end lateral supports. The effective length factor K varies from 0.5 for complete

restraint against rotation about weak axis to 1.0 for free rotate about weak axis, with 0.7 for the case of one end fixed and other end free. It is analogous to the effective length factors for compression members with end rotational restraint;

- K_w = warping restraint factor. Unless special provisions to restrain warping of the section at the end lateral supports are made, K_w should be taken as 1.0;
- y_g = y distance between the point of application of the load and the shear centre of the cross section and is positive when the load is acting towards the shear centre from the point of application;
- y_j = $y_s - 0.5 \int_A (z^2 + y^2)^2 y dA / I_z$;
- y_s = coordinate of the shear centre with respect to centroid, positive when the shear centre is on the compression side of the centroid; and
- y, z = coordinates of the elemental area with respect to centroid of the section.

y_j can be calculated by using the following approximation:

a) *Plain flanges:*

- 1) $y_j = 0.8 (2\beta_f - 1) h_y / 2.0$ (when $\beta_f > 0.5$)
- 2) $y_j = 1.0 (2\beta_f - 1) h_y / 2.0$ (when $\beta_f \leq 0.5$)

b) *Lipped flanges:*

- 1) $y_j = 0.8 (2\beta_f - 1) (1 + h_L/h) h_y / 2$ (when $\beta_f > 0.5$)
- 2) $y_j = (2\beta_f - 1) (1 + h_L/h) h_y / 2$ (when $\beta_f \leq 0.5$)

where

h_L = height of the lip,

h = overall height of the section,

h_y = distance between shear centre of the two flanges of the cross section, and

$\beta_f = I_{fc} / (I_{fc} + I_{ft})$ where I_{fc} , I_{ft} are the moment of inertia of the compression and tension flanges, respectively, about the minor axis of the entire section.

I_t = St. Venant's Torsion constant, given by:

$$I_t = \sum b_i t_i^3 / 3 \quad (\text{for open section})$$

$$= 4A_e^2 / \sum (b/t) \quad (\text{for hollow section})$$

where

A_e = area enclosed by the section, and

b, t = breadth and thickness of the elements of the section, respectively.

I_w = The warping constant, given by:

$(1 - \beta_f) \beta_f I_y h_y^2$ (for I-sections mono-symmetric about weak axis) = 0 (for angle, Tee, narrow rectangle section and approximately for hollow sections)

Table 40 Constants c_1 , c_2 , and c_3

(Clause E-1.2)

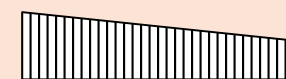
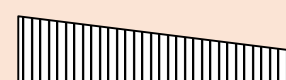
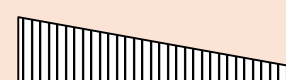
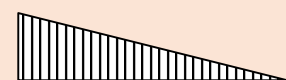
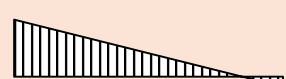
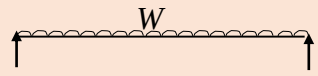
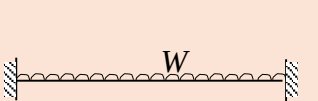
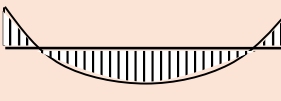
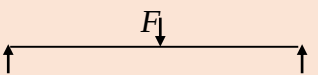
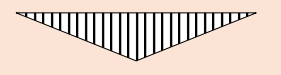
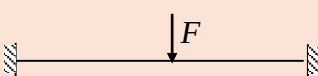
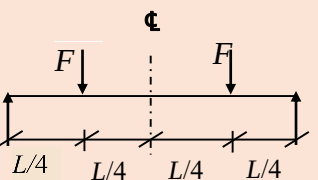
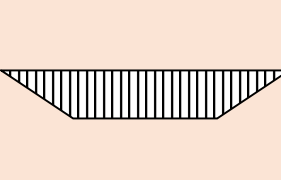
Loading and Support Conditions	Bending Moment Diagram	Value of K	Constants		
			c_1	c_2	c_3
(1)	(2)	(3)	(4)	(5)	(6)
	$\psi = +1$	1.0	1.000		1.000
		0.7	1.000	---	1.113
		0.5	1.000		1.144
	$\psi = +\frac{3}{4}$	1.0	1.141		0.998
		0.7	1.270	---	1.565
		0.5	1.305		2.283
	$\psi = +\frac{1}{2}$	1.0	1.323		0.992
		0.7	1.473	---	1.556
		0.5	1.514		2.271
	$\psi = +\frac{1}{4}$	1.0	1.563		0.977
		0.7	1.739	---	1.531
		0.5	1.788		2.235
	$\psi = 0$	1.0	1.879		0.939
		0.7	2.092	---	1.473
		0.5	2.150		2.150
	$\psi = -\frac{1}{4}$	1.0	2.281		0.855
		0.7	2.538	---	1.340
		0.5	2.609		1.957
	$\psi = -\frac{1}{2}$	1.0	2.704		0.676
		0.7	3.009	---	1.059
		0.5	3.093		1.546
	$\psi = -\frac{3}{4}$	1.0	2.927		0.366
		0.7	3.009	---	0.575
		0.5	3.093		0.837

Table 40 (Concluded)					
Loading and Support Conditions	Bending Moment Diagram	Value of K	Constants		
			c_1	c_2	c_3
(1)	(2)	(3)	(4)	(5)	(6)
	$\psi = -1$ 	1.0 0.7 0.5	2.752 3.063 3.149	---	0.000 0.000 0.000
		1.0 0.5	1.132 0.972	0.459 0.304	0.525 0.980
		1.0 0.5	1.285 0.712	1.562 0.652	0.753 1.070
		1.0 0.5	1.365 1.070	0.553 0.432	1.730 3.050
		1.0 0.5	1.565 0.938	1.267 0.715	2.640 4.800
		1.0 0.5	1.046 1.010	0.430 0.410	1.120 1.890

ANNEX F

(Clauses [11.2.1.2](#), [15.6.4.2](#), [17.6.1](#), [18.10.4.1](#) and [18.11.2.1](#))

CONNECTIONS

F-1 GENERAL

The requirement for the design of splice and beam to column connection as well as recommendation for their design shall be as given below.

F-2 BEAM SPLICES

F-2.1 For rolled section beam splices located away from the point of maximum moment, it may be assumed that the flange splice carries all the moment and the web splice carries the shear (see [Fig. 30](#)). However, in the case of a deep girder, the total moment may be divided between the flange and the web in accordance with the stress distribution. The web connection should then be designed to resist its share of moment and shear. Even web splice is designed to carry only shear force, the moment about the centroid of the bolt group on either side of the splice should be designed for moment due to eccentricity.

F-2.2 Flange joints should preferably not be located at points of maximum stress. Where splice plates are used, their area shall not be less than 5 percent in excess of the area of the flange element spliced; and their centre of gravity shall coincide, as nearly as possible with that of the element spliced. There shall be enough fasteners on each side of the splice to develop the load in the element spliced plus 5 percent but in no case should the strength developed be less than 50 percent of the effective strength of the material spliced. Wherever possible in welded construction, flange plates shall be joined by complete penetration butt welds. These butt welds shall develop the full strength of the plates. Whenever the flange width or thickness changes at the splice location, gradual transition shall be made in the width/thickness of the larger flange.

F-2.3 When beam splice is located at the point of inflection of a continuous beam, the flange splicing requirement given above may be relaxed appropriately.

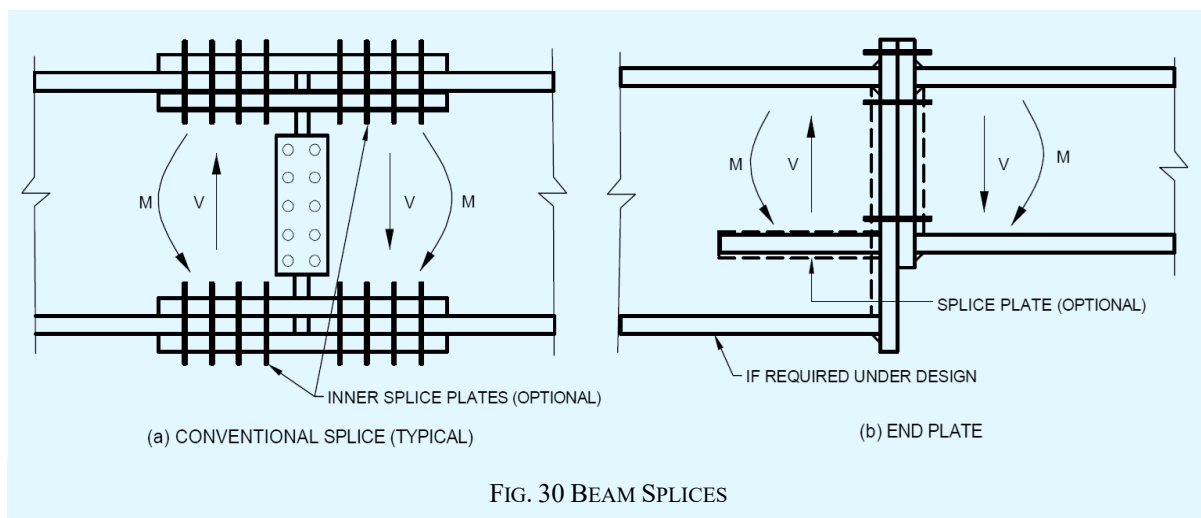


FIG. 30 BEAM SPLICES

F-3 COLUMN SPLICE

F-3.1 Where the ends of compression members are faced for bearing over the whole area, they shall be spliced to hold the connected parts aligned. The ends of compression members faced for bearing shall invariably be machined to ensure perfect contact of surfaces in bearing (see [Fig. 31](#)).

F-3.2 Where such members are not faced for complete bearing the splices shall be designed to transmit all the forces to which the members is subjected at the splice location.

F-3.3 Wherever possible, splices shall be proportioned and arranged so that centroidal axis of the splice coincides as nearly as possible with the centroidal axes of the members joined, in order to avoid eccentricity; but where eccentricity is present in the joint, the resulting stress considering eccentricity shall be provided for.

F-3.4 If a column flange is subjected to significant tension or if the faces are not prepared for bearing, or if full continuity is required without slip, only HSFG bolts shall be used.

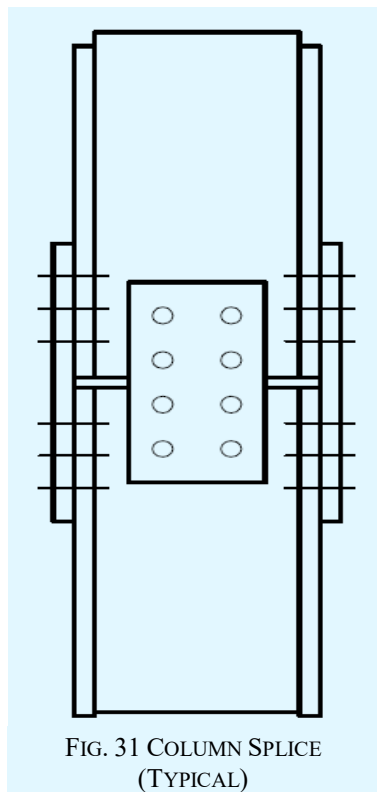


FIG. 31 COLUMN SPLICE
(TYPICAL)

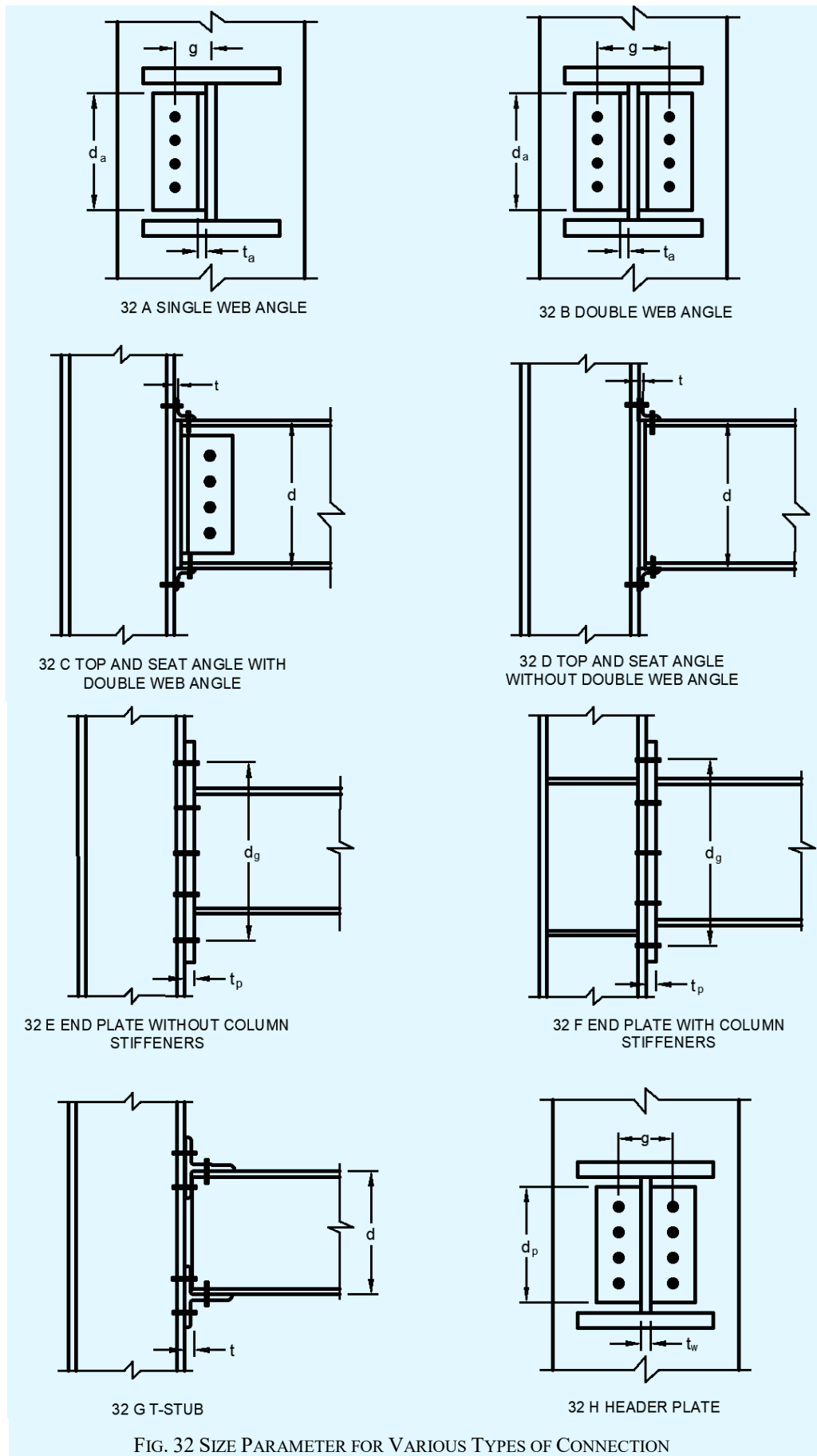
F-4 BEAM-TO-COLUMN CONNECTIONS

F-4.1 Simple Connections

Simple connections are assumed to transfer only shear at some nominal eccentricity and typically used in frames up to about five stories in height, where strength rather than stiffness govern the design. In such frames separate lateral load resisting system is to be provided in the form of bracings or shear walls. The connections shown in [Fig. 32 \(A\)](#), [Fig. 32 \(B\)](#), [Fig. 32 \(C\)](#) and [Fig. 32 \(D\)](#) can be assumed as simple connections in framed analysis and need to be checked only for the transfer of shear from beam to column.

F-4.2 Rigid Connections

In high-rise and slender structures, stiffness requirements may warrant the use of rigid connections. Rigid connections transfer significant moments to the columns and are assumed to undergo negligible deformations at the joint. These are necessary in sway frames for stability and also contribute in resisting lateral loads. The connections shown in [Fig. 32 \(E\)](#), [Fig. 32 \(F\)](#), [Fig. 32 \(G\)](#) can be assumed as rigid connection in frame analysis and need to be checked for both shear and moment transfer from beam to the column. Fully welded connections can also be considered as rigid beam to column connections.



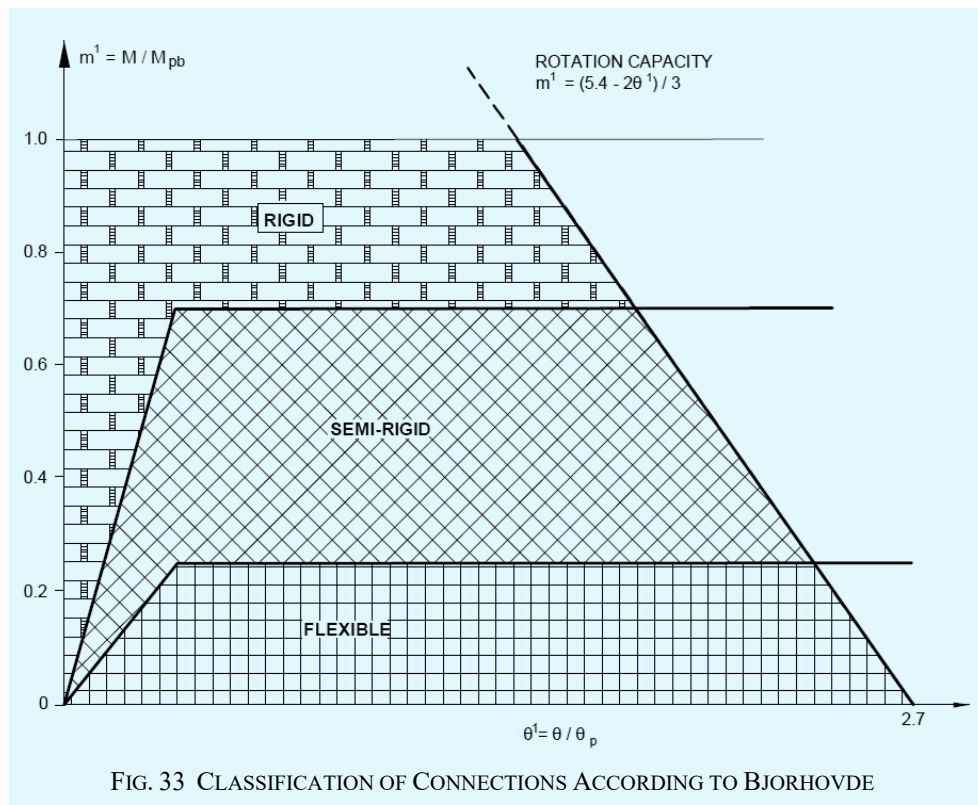
F-4.3 Semi-Rigid Connections

Semi-rigid connections fall between the two types mentioned above. The fact is that simple connections do have some degree of rotational rigidity as in the semi-rigid connections. Similarly, rigid connections do experience some degree of joint deformation and this can be utilized to reduce the joint design moments.

The moment-rotation relationship of the connections have to be determined based on experiments conducted for the specific design or based on the relationship derived from tests, presented in specialist literature. The simplest method of analysis will be to idealize the connection as an equivalent rotational spring with either a bilinear or non-linear moment-rotation characteristics. The classification proposed by Bjorhovde combined with the Frye-Morris model can be used with convenience to model semi-rigid connections, as given in the next section.

F-4.3.1 Connection Classification

Connections are classified according to their ultimate strength or in terms of their initial elastic stiffness and Bjorhovde's classification. It is based on the non-dimensional moment parameter ($m^1 = M_u / M_{pb}$) and the non-dimensional rotation parameter ($\theta^1 = \theta_r / \theta_p$), where θ_p is the plastic rotation. The Bjorhovde's classification is based on a reference length of the beam equal to 5 times the depth of the beam. The limits used for connection classification are shown in [Table 41](#) and are graphically represented in [Fig. 33](#).



F-4.3.2 Connection Models

Frye-Morris has derived the following polynomial model for the moment curvature relationship of semi-rigid connections:

$$\theta_r = C_1(KM)^1 + C_2(KM)^3 + C_3(KM)^5$$

where

M = moment at the joint, in kN.m;

K = standardization parameter which depend on the connection type and geometry; and

C_1, C_2, C_3 = curve fitting constants Table 42 shows the curve fitting constants and standardization constants for Frye-Morris Model [All size parameters in the table are in mm, (see Fig. 32)].

Table 41 Connection Classification Limits

(Clause F-4.3.1)

Sl No. (1)	Nature of the Connection (2)	In Terms of Strength (3)	In Terms of Stiffness (4)
i)	Rigid connection	$m^1 \geq 0.7$	$m^1 \geq 2.5\theta^1$
ii)	Semi-rigid connection	$0.7 > m^1 > 0.2$	$2.5\theta^1 > m^1 > 0.5\theta^1$
iii)	Flexible connection	$m^1 \leq 0.2$	$m^1 \leq 0.5\theta^1$

Table 42 Connection Constants in Frye-Morris Model

(Clause F-4.3.2)

Sl No. (1)	Type (2)	Connection Type (3)	Curve-Fitting Constants (4)	Standardization Constants (5)
i)	A	Single web angle connection	$C_1 = 1.91 \times 10^4$ $C_2 = 1.30 \times 10^{11}$ $C_3 = 2.70 \times 10^{17}$	$K = d_a^{-2.4} t_c^{-1.81} g^{0.15}$
ii)	B	Double web angle connection	$C_1 = 1.64 \times 10^3$ $C_2 = 1.03 \times 10^{14}$ $C_3 = 8.18 \times 10^{25}$	$K = d_a^{-2.4} t_c^{-1.81} g^{0.15}$
iii)	C	Top and seat angle connection with double web angle	$C_1 = 2.24 \times 10^{-1}$ $C_2 = 1.86 \times 10^4$ $C_3 = 3.23 \times 10^8$	$K = d^{-1.287} t_a^{-1.128} t_c^{-0.415} l_a^{-0.694} (g - 0.5d_b)^{1.35}$
iv)	D	Top and seat angle connection without double web angle	$C_1 = 1.63 \times 10^3$ $C_2 = 7.25 \times 10^{14}$ $C_3 = 3.31 \times 10^{23}$	$K = d^{-1.5} t_a^{-0.5} l_a^{-0.7} d_b^{-1.1}$
v)	E	End plate connection without column stiffeners	$C_1 = 1.78 \times 10^4$ $C_2 = -9.55 \times 10^{16}$ $C_3 = 5.54 \times 10^{29}$	$K = d_g^{-2.4} t_p^{-0.4} t_f^{-1.5}$
vi)	F	End plate connection with column stiffeners	$C_1 = 2.60 \times 10^2$ $C_2 = 5.37 \times 10^{11}$ $C_3 = 1.31 \times 10^{22}$	$K = d_g^{-2.4} t_p^{-0.6}$
vii)	G	T-stub connection	$C_1 = 4.05 \times 10^2$ $C_2 = 4.45 \times 10^{13}$ $C_3 = -2.03 \times 10^{23}$	$K = d^{-1.5} t_f^{-0.5} l_t^{-0.7} d_b^{-1.1}$

Table 42 (Concluded)

Sl No.	Type	Connection Type	Curve-Fitting Constants	Standardization Constants
(1)	(2)	(3)	(4)	(5)
viii)	H	Header-plate connection	$C_1 = 3.87$ $C_2 = 2.71 \times 10^5$ $C_3 = 6.06 \times 10^{11}$	$K = t_p^{-1.6} g^{1.6} d_b^{-2.3} t_w^{-0.5}$

where

d = depth, in mm, of beam;

d_a = depth, in mm, of the angle;

d_b = diameter, in mm, of the bolt;

d_g = centre, in mm, to centre of the outermost bolt of the end plate connection;

g = gauge, in mm, distance of bolt line;

t_a = thickness, in mm, of the top angle;

t_c = thickness, in mm, of the web angle;

t_f = thickness, in mm, of flange T-stub connector;

t_w = thickness, in mm, of web of the beam in the connection;

t_p = thickness, in mm, of end plate, header plate;

l_a = length, in mm, of the angle; and

l_t = length, in mm of the T-stub connector.

NOTE — For preliminary analysis using a bilinear moment curvature relationship, the stiffness given in [Table 43](#) may be assumed depending on the type of connection. The values are based on the secant stiffness at a rotation of 0.01 radian and typical dimension of connecting angle and other components as given in the table.

Table 43 Secant Stiffness

([Table 42](#))

SI No.	Type of Connection	Dimension mm	Secant Stiffness kNm/Radian
(1)	(2)	(3)	(4)
i)	Single web connection angle	$d_a = 250, t_a = 10, g = 35$	1 150
ii)	Double web-angle connection	$d_a = 250, t_a = 10, g = 77.5$	4 450
iii)	Top and seat angle connection without double web angle connection	$d_a = 300, t_a = 10, l_a = 140, d_b = 20$	2 730
iv)	Header plate	$d_p = 175, t_p = 10, g = 75, t_w = 7.5$	2 300

F-5 COLUMN BASES

F-5.1 Base Plates

Columns shall be provided with base plates capable of distributing the compressive forces in the compressed parts of the column over a bearing area such that the bearing pressure on the foundation does not exceed the design strength of the point. The design strength of the joint between the base plate and the foundation shall be determined taking account of the material properties and dimensions of both the grout and the concrete foundation.

F-5.2 Holding Down Bolts (Anchor Bolts)

F-5.2.1 Holding down bolts shall be provided if necessary to resist the effects of the design loads. They shall be designed to resist tension due to uplift forces and tension due to bending moments as appropriate.

F-5.2.2 When calculating the tension forces due to bending moments, the lever arm shall not be taken as more than the distance between the centroid of the bearing area on the compression side and the centroid of the bolt group on the tension side, taking the tolerances on the positions of the holding down bolts into account.

F-5.2.3 Holding down bolts shall either be anchored into the foundation by a hook or by a washer plate or by some other appropriate load distributing member embedded in the concrete.

F-5.2.4 If no special elements for resisting the shear force are provided, such as block or bar shear connectors, it shall be demonstrated that sufficient resistance to transfer the shear force between the column and the foundation is provided by one of the following:

- Frictional resistance of the joint between the base plate and the foundation;
- Shear resistance of the holding down bolts;
- Shear resistance of the surrounding part of the foundation; and
- Shear and bearing resistance of the shear key plates welded to the base plate and embedded in the pedestal/foundation.

ANNEX G

(Foreword)

PLATE GIRDERS WITH CORRUGATED WEB

G-1 FLEXURAL RESISTANCE

The overhang of the flange beyond the farthest face of the web at any section may be limited to that required by class 1, 2 and 3 sections, as given in [Table 2](#). Contribution of web in flexural resistance is disregarded as given below.

The flexural design resistance is given by

$$M_d = \min \left\{ \frac{b_c t_c f_{fc}}{\gamma_{m0}} \left(h_w + \frac{t_t + t_c}{2} \right); \frac{b_t t_t f_{ft}}{\gamma_{m0}} \left(h_w + \frac{t_t + t_c}{2} \right) \right\}$$

where

The dimensional symbols are shown in [Fig. 34](#).

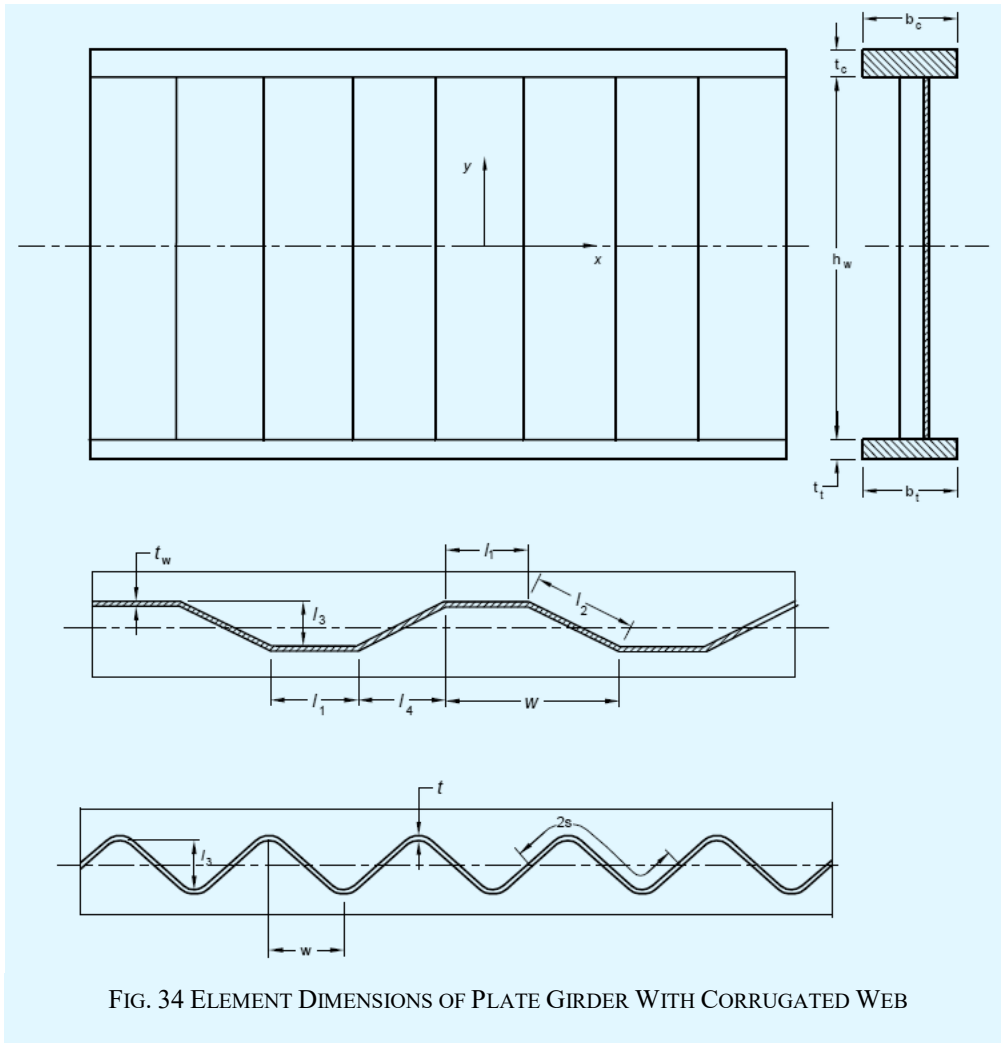


FIG. 34 ELEMENT DIMENSIONS OF PLATE GIRDER WITH CORRUGATED WEB

The design compressive stress, f_{fc} , is the minimum of yielding compressive stress after accounting for transverse (in-plane) bending of compression flange due to eccentric shear flow in the flange due to corrugated web as given below, or by lateral buckling of compression flange as given in [15.2.2](#).

The yield strength after accounting for eccentric shear flow in the flanges is given by

$$f_{fc} = f_{yc} \sqrt{1 - \frac{4 M_{vc} \gamma_{m0}}{b_c^2 t_c f_{yc}}}; \quad f_{ft} = f_{yt} \sqrt{1 - \frac{M_{vt} \gamma_{m0}}{M_{pf}}}$$

where

f_{yc}, f_{yt} = Yield strength of the material in the tension and compression flanges respectively

M_{pf} = Ultimate design resistance of the flange under in-plane bending

= $t b^3 f_{yf} / 4$ for plastic and compact flanges, $t b^3 f_{yf} / 6$ for semi compact flanges

M_{vc}, M_{vt} = in-plane transverse bending moment in the tension and compression flange, respectively due to eccentric shear force from the web as shown in [Fig. 35](#), and is evaluated from

$$M_{vc} = \frac{V y_c b_c t_c l_1 l_3}{2 I_z}; \quad M_{vt} = \frac{V y_t b_t t_t l_1 l_3}{2 I_z}$$

where

V = Shear force at the section of the girder;

I_z = Moment of inertia about the major axis of the section disregarding the corrugated web; and
 y_c, y_t = Distance of the centroid of the compression and tension flange, respectively, from the neutral axis of the girder.

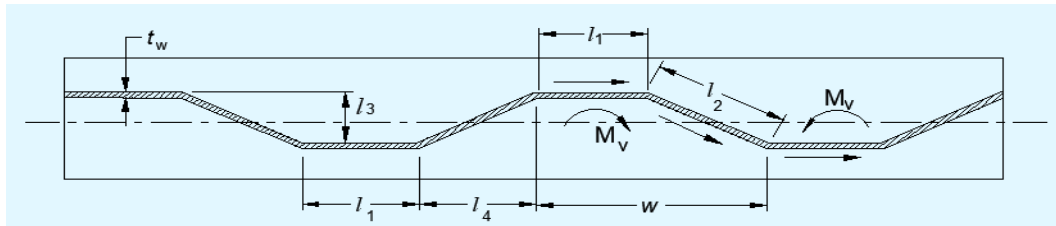


FIG. 35 IN-PLANE TRANSVERSE BENDING MOMENT IN FLANGES DUE TO ECCENTRIC SHEAR FLOW FROM WEB

G-2 SHEAR RESISTANCE

The design shear resistance of the web, V_d , is given by

$$V_d = V_n / \gamma_{m0}$$

where

$$V_n = h_w t_w \chi_v f_{yw} / \sqrt{3}$$

where

h_w, t_w, f_{yw} = depth, thickness and yield strength of the web;

χ_v = shear stress reduction factor to account for web buckling, and is the smallest of local buckling χ_{vl} , global buckling, χ_{vg} , or interaction buckling, χ_{vlg} , of the web under shear, evaluated as given below; and

$$\chi_v = \min (\chi_{vl}, \chi_{vg}, \chi_{vlg}).$$

The shear stress reduction factor for local buckling of the web is evaluated as,

$$\begin{aligned} \chi_{vl} &= 1.0 & \text{when } \lambda_{wl} \leq 0.8, \\ \chi_{vl} &= [1 - 0.8(\lambda_{wl} - 0.8)] & \text{when } 0.8 < \lambda_{wl} < 1.2, \\ \chi_{vl} &= \frac{1}{\sqrt{3}\lambda_{wl}^2} & \text{when } \lambda_{wl} \geq 1.2, \end{aligned}$$

where

$$\lambda_{wl} = \sqrt{f_{yw} / (\sqrt{3}\tau_{cr,l})}$$

$$\tau_{cr,l} = \frac{k_v \pi^2 E}{12(1 - \mu^2) (h_w / t_w)^2}$$

$$k_v = 4.0 + 5.35 \left(\frac{c}{h_w} \right) \quad \text{when } (c/h_w) < 1.0$$

$$k_v = 5.35 + 4.0 \left(\frac{c}{h_w} \right) \quad \text{when } (c/h_w) \geq 1.0$$

where

c is larger of l_1 and l_2 of corrugated web (see Fig. 35).

In case of sinusoidally corrugated webs $\tau_{cr,e}$ may be evaluated using:

$$\tau_{cr,l} = \left(5.35 + \frac{l_3 s}{h_w t_w} \right) \frac{\pi^2 E}{12(1 - \mu^2)(s/t_w)^2}$$

The shear stress reduction factor for global buckling of the web is evaluated as:

$$\chi_{vg} = \frac{1.5}{(0.5 + \lambda_{wg}^2)} \leq 1.0$$

$$\lambda_{wg} = \sqrt{f_{yw} / (\sqrt{3} \tau_{cr,g})}$$

$$\tau_{cr,g} = \frac{32.4}{t_w h_w^2} \sqrt{D_x D_z^3}$$

$$D_x = \frac{Et_w^3}{12(1-\mu^2)} \frac{w}{s} \text{ for sinusoidal corrugation}$$

$$D_x = \frac{Et_w^3}{12(1-\mu^2)} \frac{w}{(l_1 + l_2)} \text{ for profiled corrugation}$$

$$D_z = \frac{EI_z}{w}$$

I_z = second moment of area of one corrugation of length w

The shear stress reduction factor for global buckling of the web is evaluated as:

$$\chi_{vlg} = \frac{\chi_{vl} \times \chi_{vg}}{\sqrt{\chi_{vl}^2 + \chi_{vg}^2}} \leq 1.0$$

The reduction in shear resistance of the web due to the moment at the section need not be considered, since its contribution to flexure is not considered.

G-3 RESISTANCE TO CONCENTRATED LOAD

The design for bearing resistance under concentrated load and reactions may be carried out as required in [15.7](#).

ANNEX H

(Clause [24.16](#))

GENERAL RECOMMENDATIONS FOR STEELWORK TENDERS AND CONTRACTS

H-1 GENERAL

H-1.1 The recommendations given in this annex are in line with those generally adopted for steelwork construction and are meant for general information.

H-1.2 These recommendations do not form part of the requirements of the Section and compliance with these is not necessary for the purpose of complying with this Section.

H-1.3 The recommendations are unsuitable for inclusion as a block requirement in a contract, but in drawing up a contract the points mentioned should be given consideration.

H-2 EXCHANGE OF INFORMATION

Before the steelwork design is commenced, the building designer should be satisfied that the planning of the building, its dimensions and other principal factors meet the requirements of the building owner and comply with regulations of all authorities concerned. Collaboration of building designer and steelwork designer should begin at the outset of the project by joint consideration of the planning and of such questions as the stanchion spacing, materials to be used for the construction, and depth of basement.

H-3 INFORMATION REQUIRED BY THE STEELWORK DESIGNER

H-3.1 General

Site plans showing in plan and elevation of the proposed location and main dimensions of the building or structure;

- a) Ground levels, existing and proposed;
- b) Particulars of buildings or other constructions which may have to remain on the actual site of the new building or structure during the erection of the steelwork;
- c) Particulars of adjacent buildings affecting, or affected by, the new work;
- d) Stipulation regarding the erection sequence or time schedule;
- e) Conditions affecting the position or continuity of members;
- f) Limits of length and weight of steel members in transit and erection;
- g) Drawings of the substructure, proposed or existing, showing:
 - 1) levels of stanchion foundations, if already determined;
 - 2) any details affecting the stanchion bases or anchor bolts;
 - 3) permissible bearing pressure on the foundation; and
 - 4) provisions for grouting.

NOTE — In the case of new work, the substructure should be designed in accordance with the relevant standards dealing with foundations and substructure.

- h) The maximum wind velocity appropriate to the site (*see* Part C ‘Structural Design/Section 1 Loads and Load Effects’ of this NBCS); and
- j) Environmental factors, such as proximity to sea coast, and corrosive atmosphere. Reference to bye-laws and regulations affecting the steelwork design and construction.

H-3.2 Further Information Relating to Buildings

- a) Plans of the floors and roof with principal dimensions, elevations and cross sections showing heights between floor levels;
- b) The occupancy of the floors and the positions of any special loads should be given;
- c) The building drawings, which should be fully dimensioned, should preferably be to the scale of 1 to 100 and should show all stairs, fire escapes, lifts, etc, suspended ceilings, flues and ducts for heating and ventilating. Doors and windows should be shown, as the openings may be taken into account in the computations of dead load;
 - 1) Requirements should be given in respect of any maximum depth of beams or minimum head room; and
 - 2) Large-scale details should be given of any special features affecting the steelwork.
- d) The inclusive weight per m² of walls, floors, roofs, suspended ceilings, stairs and partitions, or particulars of their construction and finish for the computation of dead load;
 - 1) The plans should indicate the floors, which are to be designed to carry partitions. Where the layout of partitions is not known, or a given layout is liable to alteration, these facts should be specially noted so that allowance may be made for partitions in any position (*see* Part C ‘Structural Design/Section 1 Loads and Load Effects’ of this NBCS).
- e) The superimposed loads on the floors appropriate to the occupancy, as given in Part C ‘Structural Design/Section 1 Loads and Load Effects’ of this NBCS or as otherwise required;
- f) Details of special loads from cranes, runways, tips, lifts, bunkers, tanks, plant and equipment; and
- g) The grade of fire resistance appropriate to the occupancy as may be required.

H-4 INFORMATION REQUIRED BY TENDERER (IF NOT ALSO DESIGNER)

H-4.1 General

All information listed under [H-3.1](#);

- a) Climatic conditions at site-seasonal variations of temperature, humidity, wind velocity and direction;
- b) Nature of soil. Results of the investigation of sub-soil at site of building or structure;
- c) Accessibility of site and details of power supply;
- d) Whether the steelwork contractor will be required to survey the site and set out or check the building or structure lines, foundations and levels;
- e) Setting-out plan of foundations, stanchions and levels of bases;
- f) Cross sections and elevations of the steel structure, as necessary, with large-scale details of special features;
- g) Whether the connections are to be bolted, riveted or welded. Particular attention should be drawn to connections of a special nature, such as turned bolts, high strength friction grip bolts, long rivets and overhead welds;
- h) Quality of steel, and provisions for identification;
- j) Requirements in respect of protective paintings at works and on site, galvanizing or cement wash;
- k) Approximate dates for commencement and completion of erection;
- m) Details of any tests which have to be made during the course of erection or upon completion; and
- n) Schedule of quantities. Where the tenderer is required to take off quantities, a list should be given of the principal items to be included in the schedule.

H-4.2 Additional Information Relating to Buildings

- a) Schedule of stanchions giving sizes, lengths and typical details of brackets, joints, etc;
- b) Plan of grillages showing sizes, lengths and levels of grillage beams and particulars of any stiffeners required;
- c) Plans of floor beams showing sizes, lengths and levels eccentricities and end moments. The beam reactions and details of the type of connection required should be shown on the plans;
- d) Plan of roof steelwork. For a flat roof, the plan should give particulars similar to those of a floor plan. Where the roof is pitched, details should be given of trusses, portals, purlins, bracing, etc;
- e) The steelwork drawings should preferably be to a scale of 1 to 100 and should give identification marks against all members; and
- f) Particulars of holes required for services, pipes, machinery fixings, etc Such holes should preferably be drilled at works.

H-4.3 Information Relating to Execution of Building Work

- a) Supply of materials;
- b) Weight of steelwork for payment;
- c) Wastage of steel;
- d) Insurance, freight and transport from shop to site;
- e) Site facilities for erection;
- f) Tools and plants;
- g) Mode and terms of payment;
- h) Schedules;
- j) Force majeure (sections and provisions for liquidation and damages for delay in completion); and
- k) Escalation sections.

H-5 DETAILING

In addition to the number of copies of the approved drawings or details required under the contract, dimensioned shop drawings or details should be submitted in duplicate to the engineer who should retain one copy and return the other to the steel supplier or fabricators with his comments, if any.

H-6 TIME SCHEDULE

As the dates on which subsequent trades can commence, depend on the progress of erection of the steel framing, the time schedule for the latter should be carefully drawn up and agreed to by the parties concerned at a joint meeting.

H-7 PROCEDURE ON SITE

The steelwork contractor should be responsible for the positioning and levelling of all steelwork. Any checking or approval of the setting out by the general contractor or the engineer should not relieve the steelwork contractor of his responsibilities in this respect.

H-8 INSPECTION

References may be made to the good practice [C-6A(44)] for general guidance.

H-8.1 Access to Contractor's Works

The contractor should offer facilities for the inspection of the work at all stages.

H-8.2 Inspection of Fabrication

Unless otherwise, agreed, the inspection should be carried out at the place of fabrication. The contractor should be responsible for the accuracy of the work and for any error, which may be subsequently discovered.

H-8.3 Inspection on Site

To facilitate inspection, the contractor should during all working hours, have a foreman or properly accredited charge hand available on the site, together with a complete set of contract drawings and any further drawings and instructions which may have been issued from time to time.

H-9 MAINTENANCE

H-9.1 General

Where steelwork is to be encased in solid concrete, brickwork or masonry, the question of maintenance should not arise, but where steelwork is to be housed in hollow fire protection or is to be unprotected, particularly where the steelwork is exposed to a corroding agent, the question of painting or protective treatment of the steelwork should be given careful consideration at the construction stage, having regard to the special circumstances of the case.

H-9.2 Connections

Where connections are exposed to a corroding agent, they should be periodically inspected, and any corroded part should be thoroughly cleaned and painted.

Where bolted connections are not solidly encased and are subject to vibratory effects of machinery or plant, they should be periodically inspected and all bolts tightened.

LIST OF STANDARDS

The following list records those standards which are acceptable as 'good practice' and 'accepted standards' in the fulfilment of the requirements of this NBCS. The latest version of a standard shall be adopted at the time of enforcement of this NBCS. The standards listed may be used by the Authority as a guide in conformance with the requirements of the referred clauses in this NBCS.

In the following list, the number appearing in the first column within parentheses indicates the number of the reference in this Subsection of this NBCS.

<i>Sl No.</i>	<i>IS No.</i>	<i>Title</i>
(1)	IS 786 : 1967	Conversion factors and conversion tables (<i>first revision</i>)
(2)	IS 808 : 2021	Hot rolled steel beam, column, channel and angle sections — Dimensions and properties (<i>fourth revision</i>)
	IS 1852 : 1985	Specification for rolling and cutting tolerance for hot rolled steel products (<i>fourth revision</i>)
	IS 12779 : 1989	Rolling and cutting tolerances for hot rolled parallel flange beam and column sections — Specification
(3)	IS 8000	Geometrical product specifications (GPS) — Geometrical tolerancing
	(Part 1) : 2019	Tolerances of form, orientation, location and run-out (<i>second revision</i>)
	(Part 2) : 2024	Maximum material requirement (MMR), least material requirement (LMR) and reciprocity requirement (RPR) (<i>third revision</i>)
	(Part 3) : 2023	Profile tolerancing (<i>third revision</i>)
	(Part 4) : 1976	Practical examples of indications on drawings
	IS 8976 : 1978	Guide for preparation and arrangement of sets of drawings and parts list
	IS 962 : 1989	Code of practice for architectural and building drawings (<i>second revision</i>)
(4)	IS 962 : 1989	Code of practice for architectural and building drawings (<i>second revision</i>)
(5)	IS 813 (Part 1) : 2018/ ISO 2553 : 2013	Welding and allied processes: Part 1 Symbolic representation on drawings — Welded Joints (<i>second revision</i>)
(6)	IS 2062 (Part 1) : 2025	Structural steel: Part 1 Hot rolled medium and high tensile steel (<i>eighth revision</i>)
(7)	IS 1608	Metallic materials — Tensile testing:
	(Part 1) : 2022/ ISO 6892-1 : 2019	Method of test at room temperature (<i>fifth revision</i>)
	(Part 2) : 2020/ ISO 6892-2 : 2018	Method of test at elevated temperature (<i>fourth revision</i>)
(8)	IS 1161 : 2014	Steel tubes for structural purposes — Specification (<i>fifth revision</i>)
(9)	IS 814 : 2004	Covered electrodes for manual metal arc welding of carbon and carbon manganese steel — Specification (<i>sixth revision</i>)
(10)	IS 1395 : 1982	Specification for low and medium alloy steel covered electrodes for manual metal arc welding (<i>third revision</i>)
(11)	IS 7557 : 1982	Specification for steel wire (up to 20 mm) for the manufacture of cold-forged rivets (<i>first revision</i>)
(12)	IS 1148 : 2009	Steel rivet bars (medium and high tensile) for structural purposes (<i>fourth revision</i>)
(13)	IS 1929 : 1982	Specification for hot forged steel rivets for hot closing (12 mm to 36 mm diameter) (<i>first revision</i>)

<i>Sl No.</i>	<i>IS No.</i>	<i>Title</i>
	IS 2155 : 1982	Specification for cold forged solid steel rivets for hot closing (6 mm to 16mm diameter) (<i>first revision</i>)
(14)	IS 1363	Hexagon head bolts, screws and nuts of product grade C:
	(Part 1) : 2023/ ISO 4016 : 2022	Hexagon head bolts (size range M5 to M64) (<i>sixth revision</i>)
	(Part 2) : 2023/ ISO 4018 : 2022	Hexagon head screws (size range M5 to M64) (<i>sixth revision</i>)
	(Part 3) : 2018/ ISO 4034 : 2012	Hexagon nuts, Style 1 (size range M5 to M64) (<i>fifth revision</i>)
	IS 1364	Hexagon head bolts, screws and nuts of product grades A and B:
	(Part 1) : 2023/ ISO 4014 : 2022	Hexagon head bolts (size range M1.6 to M64) (<i>sixth revision</i>)
	(Part 2) : 2023/ ISO 4017 : 2022	Hexagon head screws (size range M1.6 to M64) (<i>sixth revision</i>)
	(Part 3) : 2018/ ISO 4032 : 2012	Hexagon nuts, Style 1 (size range M1.6 to M64) (<i>fifth revision</i>)
	(Part 4) : 2025/ ISO 4035 : 2023	Hexagon nuts , Style 0 (size range M1.6 to M64) (<i>fifth revision</i>)
	(Part 5) : 2002/ ISO 4036 : 1999	Hexagon thin nuts — Product grade B (unchamfered) (size range M1.6 to M10) (<i>fourth revision</i>)
	IS 1367	Technical supply conditions for threaded steel fasteners:
	(Part 1) : 2014/ ISO 8992 : 2005	General requirements for bolts, screws, studs and nuts (<i>fourth revision</i>)
	(Part 2) : 2002/ ISO 4759-1 : 2000	Tolerance for fasteners — Bolts, screws, studs and nuts — Product grades A, B and C (<i>third revision</i>)
	(Part 3) : 2017/ ISO 898-1 : 2013	Mechanical properties of fasteners made of carbon steel and bolts, screws and studs (<i>fifth revision</i>)
	(Part 5) : 2018/ ISO 898-5 : 2012	Mechanical properties of fasteners made of carbon steel and alloy steel — Set screws and similar threaded fasteners with specified hardness classes — Coarse thread and fine patch thread (<i>fourth revision</i>)
	(Part 6) : 2025/ ISO 898-2 : 2022	Mechanical properties of fasteners made of carbon steel and alloy steel — Nuts with specified property classes (<i>fifth revision</i>)
	(Part 7) : 1980	Mechanical properties and test methods for nuts without specified proof loads (<i>second revision</i>)
	(Part 8) : 20020/ ISO 2320 : 2015	Prevailing torque type steel nuts — functional properties (<i>fifth revision</i>)
	(Part 9)	Surface discontinuities:
	Sec 1 : 1993/ ISO 6157-1 : 1988	Bolts, screws and studs for general application (<i>third revision</i>)
	Sec 2 : 1993/ ISO 6167-3 : 1988	Bolts, screws and studs for special applications (<i>third revision</i>)

<i>Sl No.</i>	<i>IS No.</i>	<i>Title</i>
	(Part 10) : 2002/ ISO 6157-2 : 1995	Surface discontinuities — Nuts (<i>third revision</i>)
	(Part 11) : 2024/ ISO 4042 : 2022	Electroplated coating systems (<i>fifth revision</i>)
	(Part 12) : 1981	Phosphate coatings on threaded fasteners (<i>second revision</i>)
	(Part 13) : 2020/ ISO 10684 : 2004	Hot dip galvanized coatings on threaded fasteners (<i>third revision</i>)
	(Part 14)	Mechanical properties of corrosion-resistant stainless-steel fasteners:
	Sec 1 : 2023/ ISO 3506-1 : 2020	Bolts, screws and studs with specified grades and property classes (<i>fifth revision</i>)
	Sec 2 : 2023/ ISO 3506-2 : 2020	Nuts with specified grades and property classes (<i>fifth revision</i>)
	Sec 3 : 2018/ ISO 3506-3 : 2009	Set screws and similar fasteners not under tensile stress (<i>fourth revision</i>)
	(Part 16) : 2002/ ISO 8991 : 1986	Designation system for fasteners (<i>third revision</i>)
	(Part 17) : 2023/ ISO 3269 : 2019	Inspection, sampling and acceptance procedure (<i>fifth revision</i>)
	(Part 18) : 1996	Packaging (<i>third revision</i>)
	(Part 19) : 1997/ ISO 3800 : 1993	Axial load fatigue testing of bolts, screws and studs
	(Part 20) : 1996/ ISO 898-7 : 1992	Torsional test and minimum torques for bolts and screws with nominal diameters 1 mm to 10 mm
	IS 3640 : 1982	Specification for hexagon fit bolts (<i>first revision</i>)
	IS 3757 : 1985	Specification for high strength structural bolts (<i>second revision</i>)
	IS 5370 : 1969	Specification for plain washers with outside diameter $\approx 3 \times$ inside diameter
	IS 5372 : 1975	Taper washer for channels (ISMC) (<i>first revision</i>)
	IS 5374 : 1975	Taper washers for I-beams (ISMB) (<i>first revision</i>)
	IS 5624 : 2021	Foundation bolts - Specification (<i>second revision</i>)
	IS 6610 : 1972	Specification for heavy washers for steel structures
	IS 6623 : 2004	High strength structural nuts — Specification (<i>second revision</i>)
	IS 6639 : 1972	Specification for hexagon bolts for steel structures
	IS 6649 : 1985	Specification for hardened and tempered washers for high strength structural bolts and nuts (<i>first revision</i>)
(15)	IS 4000 : 1992	High strength bolts in steel structures — Code of practice (<i>first revision</i>)
(16)	IS 1030 : 1998	Carbon steel castings for general engineering purposes (<i>fifth revision</i>)
(17)	IS 2708 : 1993	1.5 percent manganese steel castings for general engineering purpose (<i>third revision</i>)
(18)	IS 1278 : 1972	Specification for filler rods for gas welding (<i>second revision</i>)

<i>Sl No.</i>	<i>IS No.</i>	<i>Title</i>
(19)	IS 1387 : 1993	General requirements for the supply of metallurgical materials (<i>second revision</i>)
(20)	IS 15977 : 2013	Classification and acceptance test for bare solid wire electrodes and wire flux combination for submerged arc welding of structural steel — Specification
(21)	IS 6419 : 1996	Welding rods and bare electrodes for gas shielded arc welding of structural steels — Specification (<i>first revision</i>)
	IS 6560 : 2017/ ISO 21952 : 2012	Welding consumables — wire electrodes, wires, rods and deposits for gas shielded arc welding of creep resisting steels — Classification (<i>second revision</i>)
(22)	IS 513	Cold reduced carbon steel sheet and strip:
	(Part 1) : 2016	Cold forming and drawing purpose (<i>sixth revision</i>)
	(Part 2) : 2016	High tensile and multi-phase steel (<i>sixth revision</i>)
(23)	IS 1079 : 2017	Hot rolled carbon steel sheet, plate and strip - Specification (<i>seventh revision</i>)
(24)	IS 1367 (Part 3) : 2017/ISO 898-1 : 2013	Technical supply conditions for threaded steel fasteners: Part 3 Mechanical properties of fasteners made of carbon steel and bolts, screws and studs (<i>fifth revision</i>)
(25)	IS 1875 : 1992	Carbon steel billets, blooms, slabs and bars for forgings — Specification (<i>fifth revision</i>)
(26)	IS 1990 : 2009	Steel rivet and stay bars for boilers (<i>second revision</i>)
(27)	IS 2002 : 2024	Steel plates for pressure vessels for intermediate and high temperature service including boilers — Specification (<i>fourth revision</i>)
(28)	IS 3039 : 2024	Structural steel for ship construction — Specification (<i>third revision</i>)
(29)	IS 6240 : 2008	Hot rolled steel plate (up to 6 mm) sheet and strip for the manufacture of low pressure liquefiable gas cylinders (<i>fourth revision</i>)
(30)	IS 9295 : 2024	Steel tubes for idlers for belt conveyors — Specification (<i>second revision</i>)
(31)	IS 10748 : 2004	Hot rolled steel strip for welded tubes and pipes — Specification (<i>second revision</i>)
(32)	IS 11587 : 1986	Specification for structural weather resistant steels
(33)	IS 15103 : 2002	Fire resistant steel — Specification
(34)	IS 15962 : 2012	Structural steels for building and structures with improved seismic resistance
(35)	IS 801: 1975	Code of practice for use of cold-formed light gauge steel structural members in general building construction (<i>first revision</i>)
(36)	IS 808 : 2021	Hot-rolled steel beam, column, channel and angle sections — Dimensions and Properties (<i>fourth revision</i>)
(37)	IS 4326 : 2013	Earthquake resistance design and construction of buildings — Code of practice (<i>third revision</i>)
(38)	IS 1946 : Part 2 : 2025	Fixing Devices in Walls, Ceilings and Floors of Solid Construction: Part 2 Design of post-installed anchorage to concrete — Code of practice
(39)	IS 3757 : 1985	Specification for high strength structural bolts (<i>second revision</i>)
(40)	IS 1367	Technical supply conditions for threaded steel fasteners:
	(Part 1) : 2014/ ISO 8992 : 2005	General requirements for bolts, screws, studs and nuts (<i>fourth revision</i>)

<i>Sl No.</i>	<i>IS No.</i>	<i>Title</i>
	(Part 2) : 2002/ ISO 4759-1 : 2000	Tolerance for fasteners — Bolts, screws, studs and nuts – Product grades A, B and C (<i>third revision</i>)
	(Part 3) : 2017/ ISO 898-1 : 2013	Mechanical properties of fasteners made of carbon steel and bolts, screws and studs (<i>fifth revision</i>)
	(Part 5) : 2018/ ISO 898-5 : 2012	Mechanical properties of fasteners made of carbon steel and alloy steel — Set screws and similar threaded fasteners with specified hardness classes — Coarse thread and fine patch thread (<i>fourth revision</i>)
	(Part 6) : 2025/ ISO 898-2 : 2022	Mechanical properties of fasteners made of carbon steel and alloy steel — Nuts with specified property classes (<i>fifth revision</i>)
	(Part 7) : 1980	Mechanical properties and test methods for nuts without specified proof loads (<i>second revision</i>)
	(Part 8) : 20020/ ISO 2320 : 2015	Prevailing torque type steel nuts — Functional properties (<i>fifth revision</i>)
	(Part 9)	Surface discontinuities
	Sec 1 : 1993/ ISO 6157-1: 1988	Bolts, screws and studs for general application (<i>third revision</i>)
	Sec 2 : 1993/ ISO 6167-3 : 1988	Bolts, screws and studs for special applications (<i>third revision</i>)
	(Part 10) : 2002/ ISO 6157-2 : 1985	Surface discontinuities — Nuts (<i>third revision</i>)
	(Part 11) : 2024/ ISO 4042 : 2022	Electroplated coating systems (<i>fifth revision</i>)
	(Part 12) : 1981	Phosphate coatings on threaded fasteners (<i>second revision</i>)
	(Part 13) : 2020/ ISO 10684 : 2004	Hot dip galvanized coatings on threaded fasteners (<i>third revision</i>)
	(Part 14)	Mechanical properties of corrosion-resistant stainless-steel fasteners
	Sec 1 : 2023/ ISO 3506-1 : 2020	Bolts, screws and studs with specified grades and property classes (<i>fifth revision</i>)
	Sec 2 : 2023/ ISO 3506-2 : 2020	Nuts with specified grades and property classes (<i>fifth revision</i>)
	Sec 3 : 2018/ ISO 3506-3 : 2009	Set screws and similar fasteners not under tensile stress (<i>fourth revision</i>)
	(Part 16) : 2002/ ISO 8991 : 1986	Designation system for fasteners (<i>third revision</i>)
	(Part 17) : 2023/ ISO 3269 : 2019	Inspection, sampling and acceptance procedure (<i>fifth revision</i>)
	(Part 18) : 1996	Packaging (<i>third revision</i>)
	(Part 19) : 1997/ ISO 3800 : 1993	Axial load fatigue testing of bolts, screws and studs

<i>Sl No.</i>	<i>IS No.</i>	<i>Title</i>
	(Part 20) : 1996/ ISO 898-7 : 1992	Torsional test and minimum torques for bolts and screws with nominal diameters 1 mm to 10 mm
(41)	IS 816 : 1969	Code of practice for use of metal arc welding for general construction in mild steel (<i>first revision</i>)
	IS 9595:1996	Metal arc welding of carbon and carbon manganese steels — Recommendations (<i>first revision</i>)
(42)	IS 18168 : 2023	Earthquake resistant design and detailing of steel buildings — Code of practice
(43)	IS 9077 : 1979	Code of practice for corrosion protection of steel reinforcement in RB and RCC construction
	IS 9172 : 1979	Recommended design practice for corrosion prevention of steel structures
(44)	IS 7215 : 1974	Tolerances for fabrication of steel structures
(45)	IS 12843 : 1989	Tolerances for erection of steel structures
(46)	IS 9595 : 1996	Metal arc welding of carbon and carbon manganese steels- Recommendations (<i>first revision</i>)
(47)	IS 919	Geometrical product specifications (GPS) — ISO code system for tolerances on linear sizes:
	(Part 1) : 2014/ ISO 286-1 : 2010	Bases of tolerance, deviations and fits (<i>third revision</i>)
	(Part 2) : 2014/ ISO 286-2 : 2010	Tables of standard tolerance grades and limit deviations for holes and shafts (<i>second revision</i>)
(48)	IS 816 : 1969	Code of practice for use of metal arc welding for general construction in mild steel (<i>first revision</i>)
	IS 819 : 1957	Code of practice for resistance spot welding for light assemblies in mild steel
	IS 1024 : 2025	Use of welding in bridges and structures subject to dynamic loading — Code of practice (<i>third revision</i>)
	IS 1261 : 1959	Code of practice for seam welding in mild steel
	IS 1323 : 1982	Code of practice for oxy-acetylene welding for structural work in mild steels (<i>second revision</i>)
	IS 9595 : 1996	Metal arc welding of carbon and carbon manganese steels- Recommendations (<i>first revision</i>)
(49)	IS 817	Training of welders — Code of Practice
	(Part 1) : 1992	Manual metal arc welding (<i>second revision</i>)
	(Part 2) : 1996	Oxyfuel welding (<i>second revision</i>)
	IS 1393 : 1961	Code of practice for training and testing of oxy-acetylene welders
	IS 7307 (Part 1) : 1974	Approval tests for welding procedures: Part 1 Fusion welding of steel
	IS 7310 (Part 1) : 2019/ISO 9606-1 : 2012	Qualification testing of welders — Fusion welding: Part 1 Steels (<i>first revision</i>)
	IS 7318 (Part 1) : 1974	Approval tests for welders when welding procedure approval is not required: Part 1 Fusion welding of steel

<i>Sl No.</i>	<i>IS No.</i>	<i>Title</i>
(50)	IS 1477	Code of practice for painting of ferrous metals in buildings:
	(Part 1) : 1971	Pre-treatment (<i>first revision</i>)
	(Part 2) : 1971	Painting (<i>first revision</i>)
(51)	IS 7205 : 1974	Safety code for erection of structural steelwork
(52)	IS 456 : 2000	Plain and reinforced concrete — Code of practice (<i>fourth revision</i>)

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART C STRUCTURAL DESIGN

SECTION 6 STEEL AND COMPOSITE CONSTRUCTION

6B COMPOSITE CONSTRUCTION IN STRUCTURAL STEEL AND CONCRETE

BUREAU OF INDIAN STANDARDS

CONTENTS

FOREWORD	...	3
1 SCOPE	...	7
2 -	...	7
3 TERMINOLOGY	...	7
4 SYMBOLS	...	9
5 MATERIALS AND WORKMANSHIP	...	9
6 STRUCTURAL ANALYSIS	...	13
7 LIMIT STATE DESIGN	...	19
8 DESIGN FOR ULTIMATE LIMIT STATES	...	23
9 DESIGN FOR SERVICEABILITY LIMIT STATES	...	32
10 DESIGN FOR FATIGUE LIMIT	...	35
11 SHEAR CONNECTORS	...	35
12 COMPOSITE COLUMNS	...	46
13 CONSTRUCTION AND ERECTION	...	51
ANNEX A SYMBOLS	...	64
ANNEX B BENDING RESISTANCE OF COMPOSITE BEAMS	...	68
ANNEX C COMPOSITE SLABS	...	82
ANNEX D PLASTIC DESIGN RESISTANCE OF SYMMETRIC COMPOSITE COLUMNS	...	90
ANNEX E TESTING	...	101
ANNEX F STRENGTH DETAILS AND ELASTIC MODULUS OF NORMAL CONCRETE	...	103
LIST OF STANDARDS	...	104

National Building Code Sectional Committee, CED 46

FOREWORD

This NBCS (Part C/Subsection 6B) covers the composite construction consists of the use of prefabricated steel structural units like steel beams and steel open web joists (trusses) in combination with concrete elements and often with steel reinforcements.

The design and construction should ensure monolithic action between the structural steel and concrete components so that they act as a single structural unit.

The Section 6 ‘Steel’ was first published in 1970 and was subsequently revised in 1983, 2005 and 2016 as a chapter of the National Building Code of India. In the second revision in 2005, the detailed design provisions were not included as IS 800 : 1984 ‘Code of practice for general construction in steel (*second revision*)’ was under revision at that time. This Section, therefore, then referred to the 1984 version of IS 800 with the remarks that the latest version would prevail, as and when published. It, however, covered the philosophical aspects of limit state design which was being introduced in the revision of IS 800. IS 800 was subsequently revised and published as IS 800 : 2007 and hence, became the prevailing version for use under this Section. In the third revision of 2016, the Section 6 was brought in line with the revised version of IS 800 : 2007 and its amendment No. 1 and to take care of further improvements and incorporations required.

In the current revision, this publication has been renamed as National Building Construction Standards (NBCS).

In this revision of Section 6, the Section has been subdivided into the following subsections with the chapter on ‘composite construction in structural steel and concrete’ being introduced for the first time:

- a) 6A Steel; and
- b) 6B Composite construction in structural steel and concrete.

The provisions of this Subsection (Part C/Subsection 6B) are largely based on IS 11384 : 2022 ‘Composite construction in structural steel and concrete — Code of practice (*first revision*)’. The growing use of steel-concrete composites in building construction necessitated this new chapter in this NBCS.

The provisions of this Subsection conform to the limit state design philosophy, ensuring consistency with IS 456 : 2000 ‘Plain and reinforced concrete — Code of practice’ for reinforced concrete and Part C/Sec 6A for structural steel. The Subsection includes comprehensive provisions for the design of key structural components such as beams, slabs, and columns in composite construction. It specifically addresses two types of composite columns — Concrete-encased steel columns, which may be either fully or partially encased, and concrete-filled steel columns — Both of which enhance load-bearing capacity and fire resistance. For composite slabs, the Subsection includes two types: profiled sheeting, which serves as formwork for reinforced concrete, and embossed profile sheeting, which functions as both formwork and tension reinforcement, with or without shear connectors. It also provides guidelines for the design and testing of shear connectors to ensure efficient load transfer. Serviceability limit states regulate deflection, vibration, and fire performance. Additionally, the subsection includes specifications for light gauge steel and lightweight concrete, broadening the scope of composite construction applications. This subsection is restricted to the design of steel-concrete composite components and systems used in buildings.

Though the common methods of designs have been covered in this NBCS, special systems of design and construction, not covered by this NBCS, may be permitted on production of satisfactory evidence regarding their design adequacy and safety based on specialist literature or by analysis, test, or both.

All requirements of Part C/Sec 6A in so far as they apply, shall be deemed to form a part of this subsection, except where otherwise laid down in this subsection.

All standards, whether given herein above or cross-referred to in the main text of this Section, are subject to revision. The parties to agreement based on this Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

As the regulation of land development and buildings is a State subject and is dealt by the States and local bodies such as urban local bodies within their jurisdiction through building regulatory documents like building regulations, building byelaws, and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

For the purpose of deciding whether a particular requirement of this Subsection is complied with the final value, observed or calculated, expressing the result of a test or analysis shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded-off value should be the same as that of the specified value in this Subsection.

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Important Explanatory Note for Users of the NBCS

In any Part/Section of this NBCS, where reference is made to ‘**good practice**’ in relation to **design, constructional procedures or other related information**, and where reference is made to ‘**accepted standard**’ in relation to **material specification, testing, or other related information**, the Indian Standards listed at the end of the Part/Section shall be used as a guide to the interpretation.

At the time of publication, the editions indicated in the standards were valid. All standards are subject to revision and parties to agreements based on any Part/Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the list of standards given at the end of a Part/Section, the number appearing within parentheses in the first column indicates the number of the reference of the standard in the Part/Section. For example:

- a) Good practices [C-6B(1)] refers to the Indian Standard(s) given at serial number (1) of the list of standards given at the end of this Part C/Subsection 6B, that is, IS 456 : 2000 ‘Plain and reinforced concrete — Code of practice (*fourth revision*)’ and IS 800 : 2007 ‘General construction in steel — Code of practice (*third revision*)’; and
- b) Accepted standards [C-6B(4)] refers to the Indian Standards given at serial number (4) of the list of standards given at the end of this Part C/Subsection 6B, that is, IS 2062 (Part 1) : 2025 ‘Structural steel: Part 1 Hot rolled medium and high tensile steel (*eighth revision*)’.

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART C STRUCTURAL DESIGN

SECTION 6 STEEL AND COMPOSITE CONSTRUCTION

6B COMPOSITE CONSTRUCTION IN STRUCTURAL STEEL AND CONCRETE

1 SCOPE

1.1 This NBCS (Part C/Subsection 6B) deals with the design and construction of composite structures made up of structural steel and cast-in-situ/precast concrete, joined together to act integrally.

1.2 This Subsection is applicable to simply supported as well as continuous beams slabs and supporting column systems. The design provisions in this code are based on the limit states method of design.

2 -

NOTE — This clause is intentionally left blank.

3 TERMINOLOGY

For the purpose of this Subsection, the definitions given in as per good practices [C-6B(1)] and the following shall apply.

3.1 Accidental Load — The load is not normally expected in design life but has a major impact if it ever occurs, such as ramming of vehicles against columns or any other member of the frame like bracings, blast loading, etc.

3.2 Accompanying Load — Imposed load acting along with leading imposed load but causing lower action and/or deflections.

3.3 Action — The primary cause for stress or deformations in a structure, such as dead, imposed, wind, seismic, or temperature loads.

3.4 Built-up Section — A member fabricated by inter-connecting more than one element to form a compound section acting as a single member.

3.5 Camber — intentionally introduced pre-curving (generally upwards) in a system, member or any portion of a member with respect to its chord. These are generally introduced to compensate for deflections at a specific level of loads.

3.6 Composite Action — Integral action of primary supporting steel member and supported concrete deck with or without limited slip at their interface to ensure greater strength and rigidity. In composite columns, it is the integral action between steel and its encasement or infill concrete. The shear transfer is to be ensured through the use of mechanical devices known as shear connectors in composite beams and columns, as required (*see* [Fig. 8](#)).

3.7 Design Load/Factored Load — A load value obtained by multiplying the characteristic load with the partial safety factor for loads.

3.8 Design Service Life — The time period during which the structure or its components should satisfy the design objectives and functions.

3.9 Detail Category — Designation given to a particular detail to indicate the S-N curve to be used in fatigue assessment.

3.10 Differential Shrinkages — It is entirely due to shrinkage of concrete from the time composite action comes into effect. When the coefficient of thermal expansion varies significantly between steel and concrete (concrete with limestone or granite aggregate), it also includes the difference in thermal strain between the steel and concrete. Differential shrinkage may lead to an increase in stresses and is more pronounced in continuous beams.

3.11 Fatigue — Damage caused by repeated fluctuations of stress, leading to progressive cracking of a structural element.

3.12 Fatigue Limit State — The state of failure through fatigue damage due to repeated application of loads.

3.13 Fatigue Strength — Stress range that can be endured by a category of detail, depending upon the number of cycles.

3.14 Flexible Shear Connector — It resists shear forces by bending, tension or shearing in the root at the connection point of the steel member. It is subjected to plastic deformations as it reaches the ultimate strength, and the failure mode is ductile. It consists of studs, channels, etc ([Fig. 8](#)) welded to steel members to develop composite action with the concrete.

3.15 Initial Dead Load — The combination of weight of steel structure and the portion of the concrete deck that is supported by the steel structure alone before the development of full composite action with concrete reaching 75 percent of its 28 days strength.

3.16 Limit State — The load state beyond which the structure is incapable of performing its desired function.

3.17 Loads — Applied forces as per loads indicated in relevant standards (*see* Part C ‘Structural Design/ Section 1 Loads and Load effects’ of this NBCS) for dead, imposed, wind, snow loads, etc, and also for seismic loads that the structure is subjected to during its lifetime.

3.18 Partial Safety Factor for Loads — The factors multiplied by the loads or their combinations to obtain design loads while checking performance under various limit states.

3.19 m-k Factors — Physically, in a composite slab, ‘*m*’ is a broad measure of the mechanical interlock between the embossed profiled decking sheet and the reinforced cement concrete on top of the deck, and ‘*k*’ represents the frictional resistance between the two elements that is, steel and concrete (*see* [Fig. 38](#)).

3.20 Rigid Shear Connectors — These are shear hoops welded to bars, angles, horseshoes or tees welded to the flange of the fabricated steel units, as shown in [Fig. 8\(b\)](#) and [Fig. 8\(c\)](#). They resist shear forces by concrete bearing on the vertical face and shearing. They exhibit negligible deformation under shear transfer (that is, brittle failure). These are not usually recommended for ultimate limit state design.

3.21 Service Limit — The loading state beyond which the structure or its components becomes incapable of performing its intended function due to excessive deformation, or deflection, or vibration.

3.22 Serviceability Loads — The loads on the structure against which the serviceability of the structure needs to be checked.

3.23 Shear Connectors — These are the mechanical attachments to steel members to transfer interface shear between steel and concrete to develop composite action and are composed of flexible shear connectors [[Fig. 8\(a\)](#), [Fig. 8\(c\)](#)], rigid shear connectors [[Fig. 8\(b\)](#)], etc.

3.24 S-N Curve — Curve, defining the relationship between the numbers of stress cycles to failure N_{sc} at a constant stress range S_c , during fatigue loading on parts of a structure.

3.25 Strength Factors — The factors by which the specified strength is divided to obtain design strength; while assessing the safety under limit states of strength.

3.26 Stress Range — Algebraic difference between two extremes of stresses in a cycle of loading at a location in a member.

3.27 Superimposed Dead Load — The dead loads added subsequent to concrete hardening of concrete that is resisted by composite action.

3.28 Transient Load — The loads that are assumed to be varying over a short time interval like imposed load, loads with dynamic effect, temperature effects, wind loads on the structure, earthquake loads, accident loads, etc.

3.29 Ultimate Limit State — The state at which the structure fails and loses its integrity leading to its collapse.

4 SYMBOLS

For the purpose of this Subsection, the symbols (other than those used for load categorization) given in [Annex A](#) are applicable.

5 MATERIALS AND WORKMANSHIP

5.1 The main materials for composite construction are structural steel, reinforcing steel, and concrete. The materials and workmanship of structural steel shall generally comply with specifications laid down in as per good practice [C-6B(3)], and that of reinforced concrete shall comply with specifications laid down in as per good practice [C-6B(2)]. However, the general properties and specifications of materials for composite construction are detailed as given in [5.2](#) to [5.4](#).

5.2 Structural Steel

All the structural steel used in general construction, coming under the purview of this NBCS, shall before fabrication conform to the accepted standard [C-6B(4)].

Structural steel other than those complying with the accepted standard [C-6B(4)] may also be used provided that the limiting stresses and other design provisions are suitably modified and the steel is also suitable for the type of fabrication adopted.

Steel that is not supported by mill test results may be used only in unimportant members and details, where their properties such as ductility and weldability would not affect the performance requirements of the members and the structure as a whole. However, such steels may be used in the structural system after confirming their quality is as per the accepted standard [C-6B(4)], by carrying out appropriate tests in accordance with the method specified in the accepted standard [C-6B(5)].

5.2.1 Properties of Structural Steel

The following physical properties shall be assumed for all grades of steel for design purposes:

Unit mass of steel	= 7 850 kg/m ³
Young's modulus (modulus of elasticity)	= 2.0×10^5 MPa
Shear modulus	= 0.769×10^5 MPa
Poisson's ratio	= 0.30
Coefficient of thermal expansion	= 0.000 012/°C

The mechanical properties of steel like yield stress, ultimate stress and, elongation shall be as per values indicated in as per good practice [C-6B(3)].

5.2.2 Specifications of Structural Steel

Unless otherwise permitted herein, structural steel used shall, before fabrication, comply with the requirements of the accepted standards [C-6B(6)].

The use of structural steel not covered by the above standards may be permitted with the specific approval of the competent authority.

5.2.3 Other Steels

Except where permitted with the specific approval of the authority, steels for machined parts and for uses in features other than structural members or elements shall comply with the relevant accepted standards [C-6B(7)], as appropriate.

5.2.4 Castings and Forgings

Steel casting and forgings shall comply with the requirements of the Indian Standards the accepted standard [C-6B(8)] as appropriate.

5.2.5 Fasteners

Bolts, nuts, washers and rivets shall comply with the following or relevant accepted standard [C-6B(9)], as appropriate.

5.2.6 Welding Consumables and Practices

Welding consumables and practices shall comply with the following accepted standard [C-6B(10)], as appropriate.

5.2.7 Wire Ropes and Cables

These shall conform to the relevant accepted standard [C-6B(11)] except where the use of other types is specifically permitted by the authority.

5.3 Concrete

5.3.1 All structural reinforced concrete shall be of minimum grade M20 and shall be in accordance with material specification and workmanship as stipulated in as per good practice [C-6B(2)]. The strengths shall be specified in terms of the characteristic compressive strengths of cubes, f_{ck} , measured at 28 days. The design provisions in this Subsection are applicable for concrete strength between M20 to M75. Specialist literature shall be adopted in composite design, while using concrete strength outside this range.

5.3.2 Concrete grade shall be designated based on its characteristic strength. The three main categories of concrete strength grade are given below, and the recommended design properties of concrete are correlated to 28 day characteristic compressive strength unless specified otherwise. The mechanical properties of concrete, namely, tensile strength and modulus of elasticity, shall be determined as per good practice [C-6B(2)]. Additional data is given in [Annex F](#).

- a) *Ordinary Concrete* — Concrete grades up to M20 are included in this type. It could be prepared by a nominal mix proportioned by the weight of its main ingredients;
- b) *Standard Concrete* — This type comprises of concrete grades from M25 to M55. It is made based on a design mix proportioned by the weight of its main ingredients, along with chemical admixtures to achieve certain target values; and
- c) *High Strength Concrete* — Concrete grades from M60 to M80 are included in this type. Even though usage of high strength concrete is allowed in composite construction, capacity equations specified in this provision shall not be directly used in the design. Specialist literature and experimental results are required for using this concrete.

5.3.3 Lightweight Concrete

Lightweight concrete may be used in composite construction, and the design provisions in this NBCS shall be used within a strength range of M20 to M60. The mean tensile strength $f_{ctm,l}$ and modulus of elasticity $E_{cm,l}$ of lightweight concrete is to be calculated as modifications over the provisions in [C-6B(2)], as follows:

$$f_{ctm,l} = \eta_l f_{ctm}$$

$$E_{cm,l} = \eta_E E_{cm}$$

The factors η_l and η_E are determined using the following equation:

$$\eta_l = 0.4 + \frac{0.6 \rho}{2200}$$

$$\eta_E = \left(\frac{\rho}{2200} \right)^2$$

Here, ρ is the upper limit of the oven-dry density of the relevant class of lightweight concrete, as given in [Table 1](#).

Table 1 Density Classes and Corresponding Design Densities for Lightweight Concrete

(Clause [5.3.3](#))

Sl No.	Density Class, ρ	1.0	1.2	1.4	1.6	1.8	2.0
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	Density range (kg/m ³)	801 to 1 000	1 001 to 1 200	1 201 to 1 400	1 401 to 1 600	1 601 to 1 800	1 801 to 2 000
ii)	Density (kg/m ³)	Plain concrete	1 050	1 250	1 450	1 650	2 050
		Reinforced concrete	1150	1 350	1 550	1 750	2 150

5.3.4 Design of concrete components of a composite structure against creep, shrinkage and temperature stress may be done as indicated in as per good practice [C-6B(2)] or as per specialist literature.

5.4 Reinforcement Steel

5.4.1 Reinforcement steel shall consist of hot rolled, thermo-mechanically treated, or heat-treated rods, de-coiled rods, or cold-worked steel of various grades as given in [Table 2](#). The grade designations and strength properties are given in [Table 3](#).

Table 2 Grades of Reinforcing Steel

(Clause [5.4.1](#))

Sl No.	Types of Steel	Grade/Designation	Relevant Standard
(1)	(2)	(3)	(4)
i)	Mild steel (MS)	Grade I	[C-6B(12)]
ii)	High yield strength deformed (HYSD)	Fe 415, Fe 415S, Fe 415D Fe 500, Fe 500S, Fe 500D, Fe 550, Fe 550D, Fe 600	[C-6B(13)]

5.4.2 The minimum strength of reinforcing steel as specified in as per good practice [C-6B(2)] is either the yield stress in case of mild steel or 0.2 percent proof strength in case of high yield strength steel and it is notionally taken as the characteristic strength of reinforcement f_{yk} .

5.4.3 The steel may be coated or galvanized to improve its corrosion resistance. The following corrosion resistive steel may be used as reinforcements:

- Galvanized Reinforcements* — The strength, elongation and bond properties are not adversely affected by galvanizing;
- Epoxy-coated Reinforcements* — These are reinforcements conforming to the accepted standard [C-6B(13)] coated by fusion bonding epoxy conforming to the accepted standard [C-6B(14)]. The bond of coated reinforcements is lowered by up to 20 percent compared to un-coated reinforcements. The lap length and anchorage length shall be increased by 25 percent while using these steel bars; and

Stainless Steel Reinforcements — Properties of stainless steel reinforcement shall not be inferior to carbon steel reinforcement of corresponding strength and class. These are reinforcements conforming to the accepted standard [C-6B(15)].

Table 3 Strength of Reinforcing Steel

(Clause [5.4.1](#))

SI No.	Type of Steel	Grade/ Designation	Minimum Yield Stress/0.2 Percent Proof Stress,	Minimum Tensile Strength, as Percent of the Actual 0.2 Percent Proof Stress/Yield Stress But Not Less Than	Minimum Percent Elongation
(1)	(2)	(3)	(4)	(5)	(6)
i)	Mild steel	Grade - I	Bars up to and including 20 mm dia = 250 MPa	410 MPa	23.0
			Bar dia > 20 mm ≤ 50 mm = 240 MPa	410 MPa	23.0
ii)	High yield strength deformed steel (HYSD)	Fe 415	415 MPa	110 Percent (≥ 485 MPa)	14.5
		Fe 415D/S		112 Percent (≥ 500 MPa)	18.0
		Fe 500	500 MPa	108 Percent (≥ 545 MPa)	12.0
		Fe 500D/S		110 Percent (≥ 565 MPa)	16.0
		Fe 550	550 MPa	110 Percent (≥ 585 MPa)	10.0
		Fe 550 D		108 Percent (≥ 600 MPa)	14.5
		Fe 600	600 MPa	106 Percent (≥ 600 MPa)	10.0
NOTES					
1 Elongation on a gauge length of 5.65√A, where A is the cross-sectional area of the test piece, when tested in accordance with [C-6B(5)].					
2 For Seismic Zone III, IV and V; HYSD steel bars having a minimum elongation of 14.5 percent and conforming to other requirements of [C-6B(13)] shall be used.					
3 For Seismic Zone III, IV and V; Structural steel maximum yield strength shall not exceed the specified minimum value by more than 20 percent and conforming to other requirements of [C-6B(13)] shall be used.					

6 STRUCTURAL ANALYSIS

This section is applicable to composite structures in which most of the members are composite. Where the structural behaviour is essentially that of reinforced or prestressed concrete, with only a few composite members, analysis provisions of as per good practice [C-6B(2)] or as per good practice [C-6B(16)] respectively, shall be used. Similarly, where the structural behaviour is essentially that of structural steel, with only a few composite members, analysis provisions of as per good practice [C-6B(3)] shall be used.

6.1 Modelling and Basic Assumptions

6.1.1 Local Buckling of Plate Elements of Steel Section and Section Classification

The types of elements that are encountered in steel sections are listed below:

- a) Internal elements are elements attached along both longitudinal edges to other elements or to longitudinal stiffeners connected at suitable intervals to transverse stiffeners, for example, web of I-section and flanges and web of box section;
- b) Outside elements or outstands are elements attached along only one of the longitudinal edges to an adjacent element, the other edge being free to displace out of plane for example, flange overhang of an I-section, the stem of T-section and legs of an angle section; and
- c) Tapered elements may be treated as a flat elements having an average thickness as defined in the accepted standard [C-6B(17)].

6.1.2 The section strength at the ultimate limit state should be considered based on their ability to resist local buckling before full plastic strength is developed. In this respect, the structural steel sections may be classified as given in as per good practice [C-6B(3)]. Guidelines for sectional classification of composite sections with and without concrete encasements are given below:

- a) *Composite Sections*
 - 1) Steel compression elements (plates) in a composite section should be classified according to the least favourable class of steel elements in compression;
 - 2) The class of a composite section depends on the direction of the bending moment at that section;
 - 3) A steel compression element restrained by a reinforced concrete element through shear connectors may be placed in a more favourable class, after ensuring its improved local buckling resistance due to the above connection;
 - 4) Plastic stress distribution over the cross section should be used for section classifications 1 and 2. In classification 3, the elastic stress distribution should be used taking into account the sequence of construction and the effects of creep and shrinkage;
 - 5) For classification, design values of the strength of materials should be taken. Concrete in tension should be neglected. The stress distribution should be established for the gross cross-section of the steel web and the effective concrete flanges;
 - 6) The tension capacity of steel reinforcements satisfying the ductility requirements specified in [5.4](#), only can be used in evaluating the strength in the hogging moment regions of the composite members. When the design resisting moment of the full or partial composite members is evaluated based on either simple rectangular stress block or parabolic stress block, the minimum area of steel reinforcement, A_{st} , used within the effective width of the concrete flange should satisfy the following:

$$A_{st} = A_c \propto \frac{1}{\varepsilon^2} \frac{f_{ctm}}{f_{yk}} \sqrt{k_c}$$

where

$$\varepsilon = \sqrt{250/f_y};$$

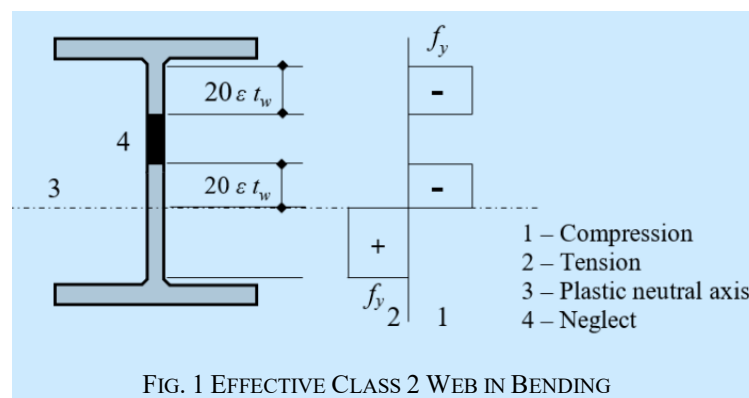
$$A_c = \text{Effective area of the concrete flange};$$

- f_y = Nominal value of yield strength of structural steel, in MPa;
 f_{yk} = Characteristic yield strength of tension reinforcement, in MPa;
 f_{ctm} = Mean tensile strength of concrete, in MPa (Annex F);
 k_c = $\frac{1}{(1+d_s/(2 z_0))} + 0.3 \leq 1.0$;
 α = 1.06 for compact section, 1.17 for plastic section;
 d_s = thickness of concrete slab excluding haunch if any; and
 z_0 = vertical distance between the centroid of the uncracked concrete slab and the centroid of the uncracked effective composite section using the modular ratio m for short-term loading.

- 7) The welded mesh should not be included in the effective section unless it has sufficient ductility before fracture when embedded in concrete; and
- 8) Account should be taken of the class of steel section at every stage of construction in the global elastic analysis.

b) *Composite Sections without Concrete Encasements*

- 1) A steel compression flange that is restrained from buckling by effective attachment to a concrete flange by shear connectors may be assumed to be in class 1 if the maximum spacing of the connectors is in accordance with [11.6.3](#);
- 2) Other steel flanges and webs in compression in the composite beams should be classified on the basis of width to thickness ratios and susceptibility to local buckling of steel only unless they are also restrained by concrete as in columns. Accordingly, sections are categorized into three groups as indicated in as per good practice [C-6B(3)]; and
- 3) In beams, cross-sections with webs in Class 3 and flanges in Class 1 or 2 may be treated as an effective cross-section in Class 2 with an effective web in accordance with [Fig. 1](#). The proportion of the web in compression should be replaced by a part of $20\epsilon t_w$ adjacent to the compression flange, with another part of $20\epsilon t_w$ adjacent to the plastic neutral axis of the effective cross-section where $\epsilon = \sqrt{250/f_y}$ and t_w is the thickness of the web.



c) *Composite section with concrete encasements*

- 1) A steel outstand flange shall be classified as per [Table 4](#); and
- 2) In partially concrete-encased I-section ([Fig. 2](#)), wherein the concrete is effectively attached to the web with stud or any reinforcement and covers on each side of the at least 80 percent of the flange overhang, the limiting overhang width of the flange to thickness ratio, b/t_f , for section classification may be taken as given in [Table 4](#), where $\epsilon = \sqrt{250/f_y}$.

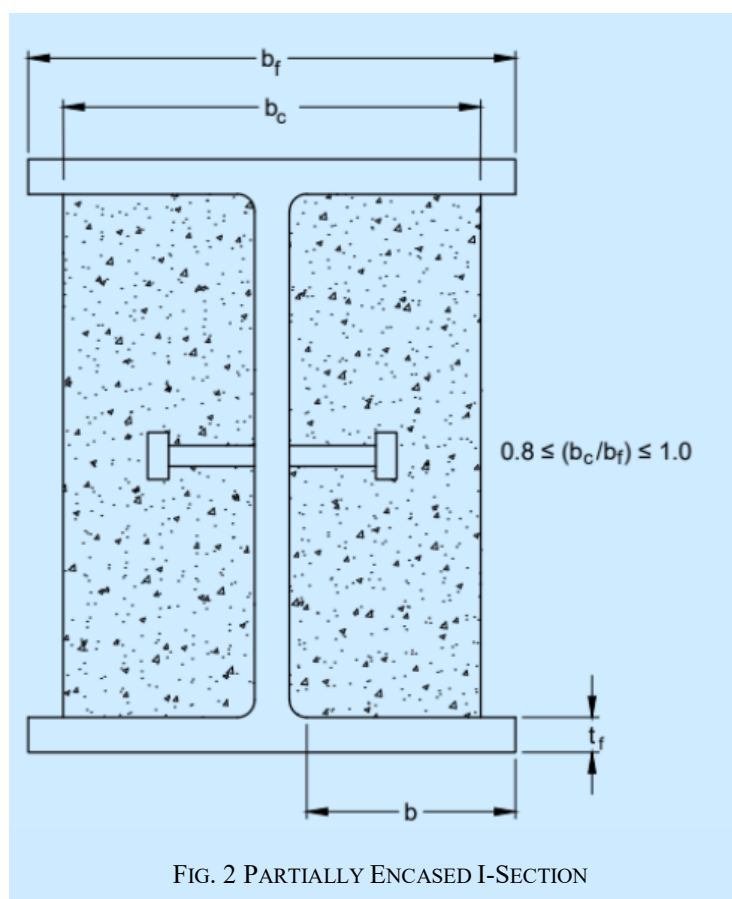


FIG. 2 PARTIALLY ENCASED I-SECTION

Table 4 Encased I Section Classification

[Clause [6.1.2\(c\)](#)]

Sl No.	Section	Flange Overhang to Thickness Ratio (b/t_f)		
		Class 1 Plastic	Class 2 Compact	Class 3 Semi-compact
(1)	(2)	(3)	(4)	(5)
i)	Rolled section	9.4ϵ	10.5ϵ	15.7ϵ
ii)	Welded section	8.4ϵ	9.4ϵ	13.6ϵ

6.1.3 Joints

Joints in a composite structure may be modelled as:

- Hinged, wherein the connection transfers only axial force and shear force, and negligible moment may be assumed to be transferred;
- Rigid, wherein relative deformation between the members being joined is negligible due to the rigid nature of the joint, and the compatibility of deformation of the adjacent members at the joint may be assumed in the analysis; and
- Semi-rigid, wherein the joint deformation under the force transferred across the joint should be modelled.

The effects of the deformation behaviour of the joints on the internal force resultants and deformations in the members and structure may be generally neglected. However, where such effects may be significant (such as in semi-rigid connections), they shall be considered as discussed in as per good practice [C-6B(3)].

6.1.4 Foundation Support

Normally the support from the foundation may be assumed as hinged or rigid, depending on the soil condition and foundation type. Effects of support deformation may be considered, where significant.

6.1.5 Deformation of the Structure

6.1.5.1 If under the external actions, the deformations are small, then the analysis can be done by studying the equilibrium of the undeformed structure. Such an analysis is referred to as first-order analysis or linear elastic analysis.

6.1.5.2 If under the external actions, the deformations are large enough to significantly affect or modify the structural behaviour, then the analysis should be performed considering the equilibrium of the structure under deformed configuration. Such an analysis is referred to as second-order analysis.

6.1.5.3 First-order analysis is adequate if the increase in the internal forces or moments due to actions on the deformation from the first-order analysis is less than 10 percent. This may be assumed to be satisfied if the ratio of the P_{cr}/P_d is greater than or equal to 10, where P_{cr} is the load corresponding to elastic instability of the member or structure as a whole and P_d is the factored design compressive load. In evaluating the elastic critical load, P_{cr} , effect of cracking and creep of concrete, deformation of the joint shall be considered.

6.1.6 Imperfections

Effects of imperfections in the member (local imperfections) and imperfections in the overall geometry of the structure should be considered if they are significant. The first order analysis does not account for these, and hence the design method used should consider these effects. The second-order analysis should include the imperfections in the modelling, if the ratio of elastic critical load to the factored design loads, $P_{cr}/P_d < 4.0$. If the second-order analysis does not include the imperfection effects (when $P_{cr}/P_d \geq 4.0$), the design method used should consider the imperfection effects. If the second-order analysis includes imperfection modelling, then the design method shall not include the effects of imperfections.

6.1.7 Shear Lag and Effective Width of Flanges

The effectiveness of steel and concrete flanges is reduced due to shear deformation in their plane either through rigorous analysis or by using effective width of the concrete as given in section [8.3.3](#) and that of steel flanges as given in as per good practice [C-6B(3)].

6.1.8 Creep, Shrinkage and Temperature

The effect of creep of concrete on the internal forces can be accounted by using effective modulus of elasticity of concrete, as recommended in as per good practice [C-6B(2)], by considering the creep and the age of concrete at the time of loading. The shrinkage strain may be taken as recommended in as per good practice [C-6B(2)] or as per specialized literature.

When permanent loads are imposed in several stages, one mean value of time may be used for calculating creep coefficient. Such a mean value may also be used for evaluating prestressing deformations if the age of the relevant portion of the concrete at the time of prestressing is more than 14 days.

If a prefabricated and prestressed concrete slab is made composite with structural steel subsequently, the creep and shrinkage values only after the composite action becomes effective are to be taken.

The effects of creep, shrinkage and differential temperature, in terms of internal forces, moments and deformation in determinate structure and indeterminate structures, where compatibility is not enforced, are referred to as 'primary effects'. In indeterminate structures, wherein the compatibility of deformation is ensured in the analysis, the additional effects of enforcing compatibility are referred to as 'secondary effects'.

The effects of creep, shrinkage, and temperature are normally neglected in analysis for the ultimate limit state other than fatigue for composite members, in which steel sections used should meet either plastic or compact classification.

The characteristic values of indirect actions due to the controlled imposed deformations (for example jacking of supports) may be calculated using the characteristic or nominal values of material properties in prestressed indeterminate composite beams and slabs unless a more accurate method is followed.

6.1.9 Cracking of Concrete

The effect of cracking of concrete in tension zones, as obtained from elastic analysis of the uncracked concrete model, may be considered by using effective stiffness of cracked concrete as given in as per good practice [C-6B(2)].

6.1.10 Sequence of Construction

Depending upon the type of construction (propped/unpropped, prestressed, precast, etc) the model for different stages of loading may differ. For the resultant internal forces, stresses and deflection at service load considering the effects of the staged construction may be obtained by superposing the analysis results for different loads obtained from different appropriate models for the stage of construction.

The effect of the sequence of construction may be neglected in structures made of plastic and compact steel sections and the model corresponding to the final stage of construction may be utilized for the ultimate limit state, other than fatigue.

6.2 Methods of Global Analysis

Different methods of global analysis, as discussed in the following sections, are permitted. The design and member strength check methods are chosen to be consistent with assumptions in analysis and actual global/member behaviour. Linear elastic first order method is most commonly used, but the second order effects and member imperfection effects, etc. Have to be taken care of in the design and member check stages.

6.2.1 Linear Elastic Analysis

6.2.1.1 Effect of cracking of concrete in tension, creep, shrinkage, temperature, prestressing and sequence of construction should be considered in the linear elastic analysis, as discussed in [6.1](#) and [7](#). Forces and moments from the linear elastic analysis may be used even when the design is based on the ultimate limit state. In the case of continuous structures with hogging moments over supports, evaluation of the effective section at supports may be necessary, as mentioned in [8.3.3](#). Appropriate load combinations with corresponding load factors are to be used to find out the maximum design values of moments and shears.

The stability of a structure as a whole against overturning shall be ensured under the limit state as per provisions of as per good practice [C-6B(2)]. The foundation components of the structure shall also be safe against sliding under adverse conditions of the applied characteristic loads. The following factor of safety shall be ensured:

- a) *Overturning* — The stability of a structure as a whole against overturning shall be ensured so that the restoring moment shall not be less than the sum of 1.2 times the maximum overturning moment due to the characteristic dead load and 1.4 times the maximum overturning moment due to the characteristic imposed loads. In cases where dead load provides the restoring moment, only 0.9 times the characteristic dead load as per good practice [C-6B(2)] shall be considered. Restoring moments due to imposed loads shall be ignored;
- b) *Sliding* — The structure shall have a factor against sliding of not less than 1.4 under the most adverse combination of the applied characteristic forces. In this case also, only 0.9 times the characteristic dead load shall be taken into account if it assists in resisting sliding; and
- c) *Instability* — Linear elastic instability analysis of member and the global system is possible, wherein first order assumptions and the resulting eigenvalue problem leads to eigenvalues as the instability load,

6.2.1.2 Redistribution of moments from first order analysis

Bending moments obtained from the linear elastic analysis of indeterminate slabs, beams and frames in composite

structures in buildings may be redistributed to a limited extent as given below while ensuring the equilibrium is still satisfied:

- a) concrete members are subjected primarily to flexure in accordance with as per good practice [C-6B(2)];
- b) steel members satisfy the plastic section requirements of as per good practice [C-6B(3)];
- c) composite members have partial or full shear connection as given in [11](#);
- d) redistributed internal forces and moments should satisfy equilibrium;
- e) inelastic behaviour of the material, and local and lateral-torsional buckling should allow such a redistribution;
- f) at least one end of the beam should be connected by a rigid or full-strength joint to the adjacent member; and
- g) each span is of uniform depth.

[Table 5](#) gives the maximum percentage reduction in the hogging bending moments obtained from linear elastic global analysis, in composite beams, unless it is verified that the plastic rotation capacity permits a higher value.

Table 5 Limits to Percentage Redistribution of Hogging Bending Moments Obtained from Linear Elastic Analysis

(Clause [6.2.1.2](#))

SI No.	Classification of the Cross-Section in the Hogging Moment Region	Plastic	Compact	Semi-compact	Slender
(1)	(2)	(3)	(4)	(5)	(6)
i)	For un-cracked section analysis moment	40	30	20	10
ii)	For cracked section analysis moment	25	15	10	0

NOTES

1For grades of steel having yield strength greater than 360 MPa, redistribution is applicable to beams with plastic or compact cross-sections. In the redistribution, the reduction in hogging moment should not exceed the values for compact sections given in [Table 5](#), unless it is demonstrated that rotation capacity permits a higher value.

2The limits in [Table 5](#) for semi-compact and slender sections relate to bending moments to be resisted by composite members and moments to be resisted by bare steel section should not be redistributed.

6.2.2 Non-linear Global Analysis

The non-linear analysis may be performed as specified in as per good practice [C-6B(3)]. The non-linear behaviour of the shear connection shall be taken into account. Effects of deformed geometry of the structure, imperfections, joint and foundations, creep, shrinkage, temperature, stages of loading shall be taken into account as discussed in [6.1](#).

6.2.3 Rigid Plastic Analysis

Rigid plastic global analysis may be used in composite structures to obtain ultimate limit state design values other than fatigue if the conditions given below are satisfied. Second-order effects, imperfections, creep, shrinkage, temperature effects, etc. need not be considered in the rigid plastic analysis. The effect of alternating plasticity due to variable loading need not be normally considered in buildings:

- a) All the members and joints of the frame are either composite or structural steel;
- b) Material and cross-section should satisfy requirements of plastic analysis as given in [C-6B(3)];

- c) Joints are able to sustain the plastic resisting moments over a large plastic hinge rotation;
- d) Critical plastic hinge location shall satisfy the following requirements:
 - 1) Cross-section of the structural steel section shall be symmetric about the plane of bending;
 - 2) restraints are provided to prevent lateral torsional buckling;
 - 3) lateral restraint is provided to the compression flange at all critical plastic hinge sections under any loading;
 - 4) The rotation capacity of the section, considering any axial compression at the section, should be adequate to permit the plastic hinge rotation to form the plastic collapse mechanism; and
 - 5) When rotation requirements are not calculated and ensured, all effective sections at potential plastic hinge locations should be of plastic classification;
- e) The plastic hinge rotation capacity in composite sections may be assumed to be adequate where:
 - 1) the yield strength of structural steel does not exceed 450 MPa;
 - 2) the concrete slab is of normal density concrete, having a characteristic strength within the range of 25 MPa to 50 MPa;
 - 3) contribution of reinforced concrete encasement in compression is neglected in calculating design moment resistance;
 - 4) all effective steel sections at potential plastic hinge rotation should be plastic, and all the other sections should be plastic or compact;
 - 5) all beam connections to adjacent beams/columns should have adequate plastic rotation capacity or should have a design resisting moment of at least 1.2 times the design plastic resistance of the adjacent beam;
 - 6) structural steel compression flange at plastic hinge location should be laterally restrained;
 - 7) adjacent span lengths do not differ by more than 50 percent of the shorter span length;
 - 8) the end span length is not more than 115 percent of the adjacent span length;
 - 9) at plastic sections with the concrete slab is under compression, not more than 15 percent of the total depth of the section shall be under compression at the ultimate moment unless the plastic hinge at the section is the last to form (not requiring much plastic hinge rotation);
 - 10) plastic hinges shall not be located in composite columns unless plastic rotation capacity can be verified;
 - 11) the lateral support to the compression flange shall be located within half the depth of the steel section from the plastic hinge section; and
 - 12) sections away from plastic hinge regions should satisfy at least compact section requirements.

7 LIMIT STATE DESIGN

7.1 General

In the limit state design method, the structure shall be designed to withstand safely all loads likely to act on it throughout its life. The objective of the design is to arrive at a structure that will remain fit for use during its life with acceptable target reliability. In other words, the probability of a limit state being reached during its lifetime should be very low. The acceptable limit for the safety and serviceability requirements before failure occurs is called a limit state. In general, the structure shall be designed on the basis of the most critical limit state and shall be checked for other limit states.

For satisfactory functioning of a structure during its design life conditions, stipulations and directives laid down in [C-6B(1)] shall be adequately satisfied for all steel-concrete composite structures.

7.1.1 Effect of Construction Sequence

The linear elastic method is valid for the analysis of the structure after considering load history, sequence of

concrete casting and development of composite strength. In the case of propped construction, most of the initial dead load is resisted through the beam-prop system and the steel beam remains basically unstressed at this stage. In the case of un-propped construction, the steel beams alone have to carry the initial dead load and construction loads. Consequently, stresses and deflections at this stage shall not exceed specified design limits. The necessary distinction has to be made in the analysis about the stage of loading and the effectiveness of the system resisting the load. In the ultimate limit state, however, this distinction is not necessary while checking for flexural resistance. For the design of steel components and concrete deck, stipulations of as per good practice [C-6B(1)] and this Subsection shall be applied.

7.1.2 The sequence of construction should be considered as an integral part of the design process, for example, when calculating the stresses under serviceability limit criteria. The sequence and method of construction need to be mentioned in the drawings. Additional information such as the position of construction joints, if any, may also be included in the drawings.

Where the composite section is loaded before the concrete has attained its 28 days characteristic cube strength, the elastic properties and limiting compressive strength of the concrete, and the nominal strengths of shear connectors should be based upon the cube strength of the concrete at the time of loading. Where a partially cast slab is assumed to act compositely, the shear connection should be designed for this condition as well as for the final condition. In calculating deflections, consideration should be given to the sequence of construction and, where appropriate, proper account should be taken of the pre-camber of steel section, the deflections of the steel section due to loads applied to it prior to the development of composite action and of partial composite action where deck slabs are cast in stages. In execution, the rate and sequence of concreting should be required to be such that partly matured concrete is not damaged as a result of limited composite action occurring from the deformation of the steel beams under subsequent concreting operations.

7.2 Limit States

A composite structure or part of it is considered unfit for use when it exceeds a particular state called the limit state, beyond which it infringes on one of the criteria governing its performance or use. The limit states can be classified into the following categories:

- a) Ultimate limit state;
- b) Serviceability limit state; and
- c) Fatigue limit state.

7.2.1 Ultimate Limit State

It is the state when under the worst combination of factored loads the structure or its components either reach design strength or becomes unstable. Both stability and strength need to be checked under the ultimate limit state.

In steel-concrete composite structures used in buildings or general constructions, the significant ultimate limit states to be considered are as follows:

- a) Collapse due to flexural, shear or bearing failure of one or more critical sections or components;
- b) Collapse due to horizontal shear failure at the interface between the steel beam and the concrete slab or composite slab system involving concrete slab and embossed profiled sheets;
- c) Collapse due to the vertical separation of the concrete/composite slab from the steel beams; and
- d) Collapse due to shear failure between steel and concrete components of the composite column or due to buckling of both fully/partially concrete encased steel columns or concrete filled hollow sections used as columns.

7.2.2 Serviceability Limit States

It is the state at which any of the following conditions occur during the loads encountered under construction and service:

- a) Stress in structural steel has reached the prescribed limit;

- b) Stress in concrete has reached the prescribed limit;
- c) Deflection of a structure or its component reaches the prescribed limit;
- d) Concrete crack width reaches the prescribed limit;
- e) Slip at the interface between steel and concrete exceeds the prescribed limit;
- f) Vibration becomes excessive, especially at overhangs;
- g) Excessive corrosion affecting the durability of the structure; and
- h) Unacceptable effects due to fire.

7.2.3 Fatigue Limit State

It is the state at which stress range due to the application of imposed loads reaches the prescribed limit, corresponding to the number of load cycles and detail configuration.

7.3 Design Philosophy

For ensuring the design objectives, the design should be based on the characteristic values for material strengths (resistance) and applied loads (actions), which take into account the probability of variations in the material strengths and in the loads to be supported. The characteristic values should be based on statistical data, if available. Where such data is not available, they should be based on experiments. The design values are derived from the characteristic values through the use of partial safety factors, both for material strengths and for loads. In the absence of special considerations, these factors should have the values given in this section according to the material, the type of load and the limit state being considered. The reliability of the design is achieved by ensuring that:

$$\text{Design Action (Load)} \leq \text{Design Strength (Resistance)}$$

Design action refers to the external actions or load which act on the structure and the design resistance refers to the maximum resistance the structure and its components provides to resist the actions without causing failure of the structure and its components or causing hindrances to the smooth operation of the structure for which it is intended.

7.4 Design Actions (Loads)

7.4.1 Actions (Loads) are Classified into Three Main Categories:

- a) *Permanent or dead loads* — Loads due to self-weight of structural and non-structural components, fittings, ancillaries, and fixed equipment etc. Dead loads shall be calculated on the basis of unit weights which shall be established taking into consideration the materials specified for construction. Alternatively, the dead loads may be calculated on the basis of nominal dimensions and unit weights of materials given in (see Part C ‘Structural Design/Section 1 Loads and Load effects’ of this NBCS);
- b) *Variable or imposed loads* — Construction and service stage loads such as imposed loads (for example, crane loads, impact, etc.), wind loads, snow loads and other loads shall be assumed in accordance with (see Part C ‘Structural Design/Section 1 Loads and Load effects’ of this NBCS) and the earthquake forces shall be calculated in accordance with the same; and
- c) *Accidental loads* — Accidental loads are actions or loads expected due to explosions and the impact of vehicles, etc. The characteristic values of accidental loads generally correspond to the value specified by relevant code, standard or client. Design for the accidental load is generally not required in every building unless it is required by the client or approving authority, in which case, general recommendations given in as per good practice [C-6B(3)] or specialist literature shall be followed.

7.4.2 Other than the actions due to the externally applied loads as given in [7.4.1](#), if the effects of shrinkage, creep and temperature are liable to affect material safety and serviceability of the structure, these shall be accounted for in the design calculations (see Part C ‘Structural Design/Section 1 Loads and Load effects’ of this NBCS).

7.4.3 Load Combination

The different combinations of loads considered shall be as given in (see Part C ‘Structural Design/Section 1 Loads and Load effects’ of this NBCS). For each combination, different partial safety factors for loads are assigned to different loads, γ_f to account for:

- Possibility of unfavourable deviation of the load from the characteristic value;
- Possibility of inaccurate assessment of the load;
- Uncertainty in the assessment of effects of the load; and
- Uncertainty in the assessment of the limit states being considered.

The loads or load effects shall be multiplied by the relevant γ_f factors, given in [Table 6](#), to calculate the design loads or design load effects.

Table 6 Partial Safety Factor for Loads, γ_f for Limit States

(Clause [7.4.3](#) and [9.3.2.1](#))

Sl No.	Combination	Limit State of Strength					Limit State of Serviceability			
		DL	LL ²⁾		WL/EL	AL	DL	LL ²⁾		WL/EL
			Leading	Accompanying				Leading	Accompanying	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
i)	DL+LL+CL	1.5	1.5	1.05	—	—	1.0	1.0	1.0	—
ii)	DL+LL+CL+ WL/EL	1.2	1.2	1.05	0.6	—	—	—	—	—
		1.2	1.2	0.53	1.2	—	1.0	0.8	0.8	0.8
iii)	DL+WL/EL	1.5	—	—	1.5	—	1.0	—	—	1.0
		(0.9) ¹⁾								
iv)	DL+ER	1.2	1.2	—	—	—	—	—	—	—
		(0.9) ¹⁾								
v)	DL+LL+AL	1.0	0.35	0.35	—	1.0	—	—	—	—

¹⁾ This value is to be considered when the dead load contribution to stability against overturning is critical or the dead load causes a reduction in stress due to other loads.

²⁾ When the action of different imposed loads is simultaneously considered, the leading imposed load shall be considered to be the one causing the higher load effects in the member/section.

NOTE — The effects of actions (loads) in terms of stresses or stress resultants may be obtained from an appropriate method of analysis.

Abbreviations: DL= Dead Load, LL= Imposed Load, WL= Wind Load, CL= Crane Load (Vertical / horizontal), AL=Accidental Load, ER= Erection Load, EL= Earthquake Load.

7.5 Material Strength and Partial Safety Factor

The design strength, S_d of a structural component, is obtained as given below from nominal ultimate strength, S_u , and partial safety factors for material strength, γ_m .

$$S_d = S_u / \gamma_m$$

where the partial safety factor for materials, γ_m , (see [Table 7](#)), account for:

- Possibility of unfavorable deviation of material strength from the characteristic value;

¹⁾ This value is to be considered when the dead load contribution to stability against overturning is critical or the dead load causes a reduction in stress due to other loads.

²⁾ When the action of different imposed loads is simultaneously considered, the leading imposed load shall be considered to be the one causing the higher load effects in the member/section.

- b) Possibility of unfavorable variation of member sizes;
- c) Possibility of unfavorable reduction in member strength due to fabrication and tolerances; and
- d) Uncertainty in the calculation of the strength of the members.

Table 7 Partial Safety Factor for Material γ_m

(Clauses [7.5](#), [8.3.5.1](#), [8.3.6.1](#) and [9.1.2](#))

SI No.	Material	Item	Partial Safety Factor γ_m
(1)	(2)	(3)	(4)
i)	Structural steel, and steel sheeting	Resistance against yield stress, γ_{m0}	1.10
		Resistance of member to buckling, γ_{m0}	1.10
		Resistance against ultimate stress γ_{m1}	1.25
ii)	Reinforcement	Reinforcement resistance against yield stress, γ_k	1.15
iii)	Shear connectors	Yield stress, γ_{mv}	1.25
iv)	Connections	Bolts-friction type, γ_{mf}	1.25
		Bolts-bearing type, γ_{mb}	1.25
		Rivets, γ_{mr}	1.25
		Welds for shop fabrication, γ_{mw}	1.25
		Welds for site fabrication, γ_{mw}	1.50
v)	Concrete	For basic and seismic load combinations, γ_c	1.50

NOTE — The partial safety factor for resistance for all materials in case of accidental loads = 1.0

8 DESIGN FOR ULTIMATE LIMIT STATES

8.1 Composite Beams

Composite beams shall be checked for the following:

- a) resistance to flexure;
- b) resistance to lateral-torsional buckling;
- c) resistance to web shear buckling;
- d) resistance to longitudinal shear (between steel and concrete components).

8.1.1 Assumptions

Design for the limit state of collapse in flexure shall be based on the assumptions given below:

- a) Plane sections normal to the neutral axis remain plane and normal after bending;
- b) The maximum strain in concrete at the outermost compression fiber at collapse is taken as 0.003 5 in bending as per good practice [C-6B(2)];
- c) The tensile strength of the concrete is ignored;
- d) The stress-strain curve for the reinforcing steel shall be assumed to be the same as given in as per good practice [C-6B(2)];

- e) The properties of structural steel shall be taken as given in as per good practice [C-6B(3)]; and
- f) The stress-strain curve for concrete may be taken to be the same as given in as per good practice [C-6B(2)]. The corresponding parabolic stress block may be used to evaluate the design bending resistance of composite beams and slabs. The bending resistance of a composite beam for the value of concrete stress using an equivalent rectangular stress block, which simplifies the design resistance equation, may be used as an alternative method. Both methods and the design resistance equations are given in [Annex B](#).

NOTE — Critical cross-sections include the following:

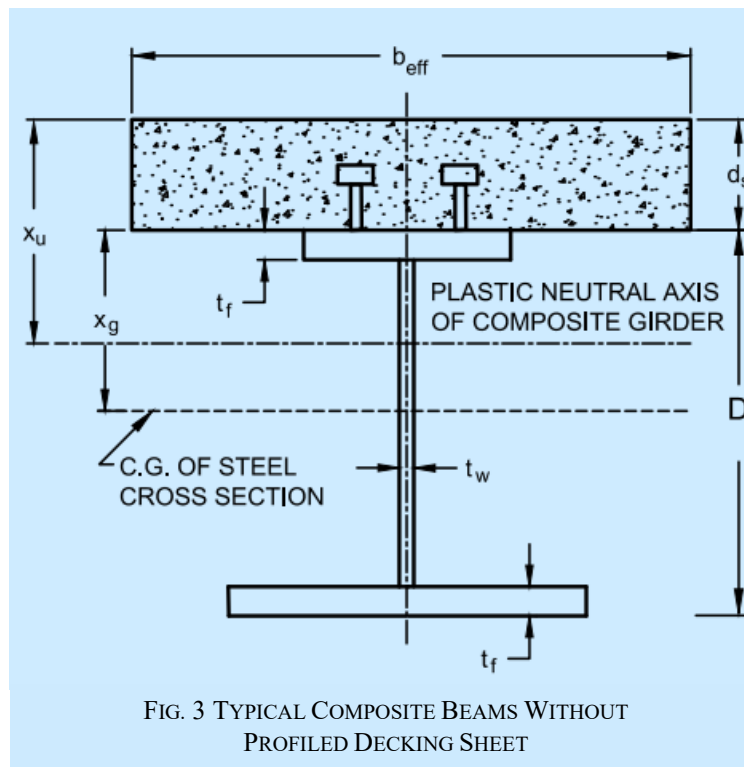
- a) the sections where the bending moment is maximum;
- b) the section at the faces of the support;
- c) sections adjacent to concentrated force or reactions; and
- d) locations where a sudden change of cross-section occurs; (ratio of greater to the lesser resisting moment at adjacent sections exceeds 1.2) (changes due to cracking of concrete not to be included).

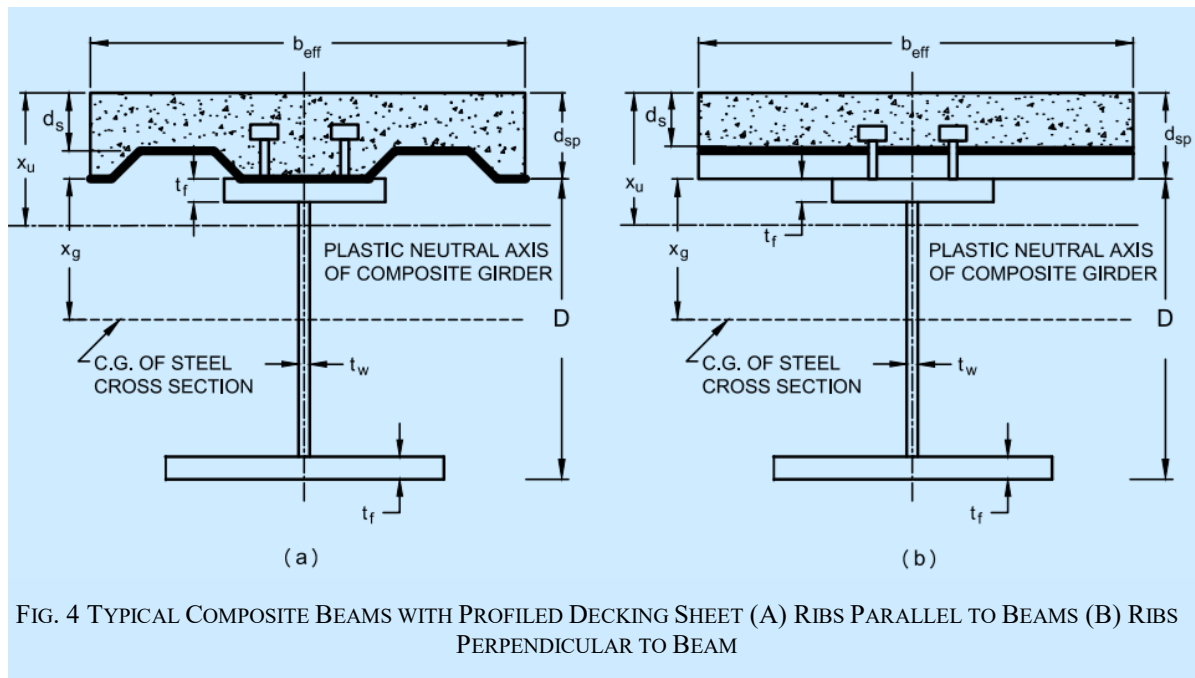
8.1.2 Ultimate Bending Resistance

For determining the position of the plastic neutral axis and the ultimate moment of resistance of composite beams, the guidelines given in [Annex B](#) may be used.

8.1.2.1 General

Composite constructions of types: (a) Top flange of steel beam supporting the concrete slab ([Fig. 3](#)) and (b) composite slab resting on structural steel beam ([Fig. 4](#)) are included in this design philosophy. A typical composite beam arrangement is shown in [Fig. 3](#), where cast in place or precast RC slab is directly resting on the steel beam. [Fig. 4](#) depict a composite slab consisting of RC and profiled deck sheet with or without embossments) and supported on a structural steel beam, parallel and perpendicular to the steel beam, respectively. The neutral axis may be in the concrete slab, in the top flange of the steel section, or in the web of the steel section.





Bending moments and shears due to the application of factored loads may be analyzed in indeterminate structures by elastic theory assuming the concrete in the slab as un-cracked and unreinforced.

Hogging moments over internal supports as calculated above should be checked against section resistance assuming steel beam acting integrally with concrete (considering un-cracked and unreinforced). If the flexural tensile stress in concrete thus calculated exceeds the tensile strength of concrete, $f_{ctk, 0.05}$ as in [Annex F](#) then a new analysis, neglecting concrete but including reinforcements over the effective width of the slab (see [8.3.3](#)) and over 15 percent of the span on each side of the support should be done to calculate the required design resistance, provided adjacent spans do not differ appreciably.

The redistribution of elastic analysis moment of indeterminate structure, as recommended in [6.2.1.2](#), is allowed to arrive at the design moments in the case of plastic beams. Design for plastic analysis moments (as given in [6.2.3](#)) is permitted in the case of beams classified as plastic.

The bottom flange of the beam in the hogging moment zone should be adequately braced against lateral buckling. Otherwise, lateral buckling resistance has to be evaluated as recommended in [B-4](#).

8.2 Precast Slab on Steel Beam

The use of precast slab, both full depth and partial depth, is allowed for composite construction as one of the components of composite beams. Precast slabs shall be erected and connected to the steel section so as to ensure composite action along with the steel beam.

8.2.1 Full Depth Precast Slab

Full-depth precast concrete deck panels may be used for new construction as well as for the replacement of deteriorated concrete decks on existing steel beams so as to obtain composite action. This shall be ensured by proper shear connection during the erection of these precast panels. The typical requirements for these types of beams are as given below:

- Panels shall either span the full width of the concreting deck or shall be in lengths that span between two or more parallel beams. The minimum thickness of the slab shall be 150 mm;
- The panels shall be connected to the beams using shear connectors in pockets, which consist of mechanical connectors, such as shear studs encapsulated in non-shrinking grouted pockets. These connections cause the panels to develop composite action with the beams; and

- c) The contact between the precast panels at their longitudinal edge should ensure transfer of compression between the panels, necessary for composite action.

8.2.2 Partial Depth Precast Slab

Partial depth precast concrete deck panels are generally thin RCC/prestressed concrete panels that span between beams and also serve as forms/shuttering for the cast-in-place concrete deck. The typical geometrical parameters that govern the use of these panels as a composite unit for the floor system are as given below:

- a) The minimum thickness of the precast panels shall be 75 mm;
- b) As a composite floor system, the cast in-situ concrete and the partial-depth panels together create the total thickness of the slab, with the panel reinforcing steel serving as the positive moment reinforcement in the bending direction of the combined slab;
- c) Dimensions of the precast panels shall be chosen from consideration of easy handling, ease of lifting by cranes and for catering to the construction loads, including load of wet cast in-situ concrete;
- d) Partial-depth panels must be capable of developing sufficient composite action with the cast *in-situ* concrete to be an effective floor system;
- e) To ensure full bond between the cast *in-situ* concrete and precast panels, it is recommended that the top surface of the precast panel is intentionally roughened while casting it and is cleaned by removing the laitance or other contaminants on the surface and other measures may be taken, before the placement of the cast *in-situ* concrete so as to ensure a good bond between precast and cast in place concrete; and
- f) After the precast panels are in place, the top layer of the reinforcing steel shall be placed over it, and the cast-in-situ concrete shall be placed on top of the panels.

8.2.3 General Design Principles

The design of precast slabs is based on the following principles:

- a) The precast slab together with any *in-situ* concrete (for partial depth slab) should be designed as continuous in both the longitudinal and the transverse direction;
- b) The joints between slabs should be designed to transmit membrane forces as well as bending moment and shear forces;
- c) The effective width of precast slab in the composite beam action shall be calculated as per [8.3.3](#);
- d) The design principles of composite beams involving either full depth or partial depth precast slabs are similar to standard composite decks using cast *in-situ* reinforced concrete;
- e) Vertical shear check of the composite beam shall be done as per [8.3.6](#); and
- f) For serviceability limit states, guidelines given in [7.2.2](#) and [9](#) shall be followed.

8.2.4 Joints between Steel Beam and Precast Concrete Slab

- a) Where precast slabs are supported on steel beams without bedding, the influence of the vertical tolerances of the bearing surfaces shall be considered; and
- b) The shear transfer between steel flange and precast concrete through mechanical shear connector shall be designed as per [11](#) with the following precautions:
 - 1) If shear connectors are welded to the steel beam projecting into the recesses within the slabs or joints between slabs, which are filled with concrete after erection, the detailing and the properties of the concrete should be such that it can be cast properly.
 - 2) The minimum infill around the shear connectors should be at least 25 mm.
 - 3) If shear connectors are arranged in groups, sufficient reinforcement should be provided near each group to prevent premature local failure in either the precast or the *in-situ* concrete.
 - 4) Special provision for protection against corrosion shall be adopted, wherein the steel flange under

precast slabs without bedding should have the same corrosion protection as the rest of the steelwork in addition to the top coating provided after erection.

- 5) Bedding with the purpose of protecting against corrosion may be designed to be non-load bearing.

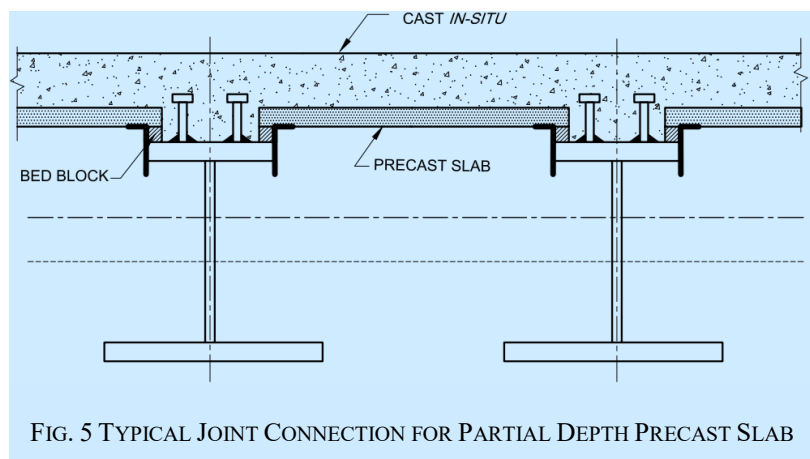
8.2.5 Joints between Precast members

The critical sections of members close to joints should be designed to resist the most adverse combination of shear, axial force and bending caused by the ultimate vertical and horizontal forces. When the design of the precast members is based on the assumption that the joint between them is not capable of transmitting bending moment (see 8.3.4), suitable precautions should be taken to ensure that if any crack developed is not excessive, and reduce the shear or axial force resistance of the member and should not be aesthetically or functionally objectionable.

Where space is left between two or more precast units, to be filled later with *in-situ* concrete or non-shrink grout, the gap should be large enough for easy placement and adequate compaction of the filling material to fill the gap completely.

8.2.6 Structural Connection at Joints

When designing and detailing the connections across joints between precast members, the overall stability of the structure, including its stability during construction, shall be considered. A typical joint connection is shown in Fig. 5.



8.3 Design Method

Connections should, where possible, be designed in accordance with the generally accepted methods applicable to reinforced concrete or structural steel.

8.3.1 General Considerations in Design Details

In addition to ultimate resistance requirements, the following should be considered:

- a) *Protection* — Connections should be designed so that the standard of protection against weather and corrosion required for the rest of the structure is sufficiently maintained;
- b) *Appearance* — Where connections are to be exposed, they should be designed to achieve and maintain the quality of appearance required for the rest of the structure; and
- c) *Manufacture, assembly, and erection* — Methods of manufacture and erection should be considered during design. Care should be taken during erection and the following precautions are mandatory:
 - 1) Where projecting bars or sections are required, they should be kept to a minimum and made as simple as possible. The lengths of such projections should be not more than necessary;
 - 2) Fixing devices should be located in concrete sections of adequate strength;
 - 3) The practicability of both casting and assembly should be considered; and

- 4) Most connections require the introduction of suitable jointing material. Sufficient space should be allowed in the design for such material to ensure that the proper filling of the joint is possible.

8.3.2 Reinforcement Continuity at Joint

Where continuity of reinforcement is required through the connection, the joining method used should be such that the assumptions made in analyzing the structure and critical sections are realized. The standard methods applicable for achieving continuity of reinforcements are lapping, butt welding of bars and grout couplers.

8.3.3 Effective Width of Concrete Slab

- a) For resistance calculation, the width of the slab, which is effective as the compression flange of the composite beam on each side of the steel beam, b_{eff} , shall be as per the equation given below:

$$b_{eff} = b_o + (b_{e1} + b_{e2}) = b_o + \frac{L_e}{4} \leq b_o + (b_1 + b_2)$$

where

- b_o = centre to centre between the outer row of shear connectors;
- b_{e1}, b_{e2} = are the value of the effective width of the concrete flange on each side of the web and taken as $L_e/8$ but not greater than the geometric width b_1 and b_2 , respectively as shown in [Fig. 6](#);
 $L_e/8 \leq b_1$ and b_2 (respectively);
- L_e = the effective equivalent span length of the corresponding composite beam;
centre-to-centre distance between the supports for simply supported beam; and
as shown in [Fig. 7](#) in a continuously supported beam.

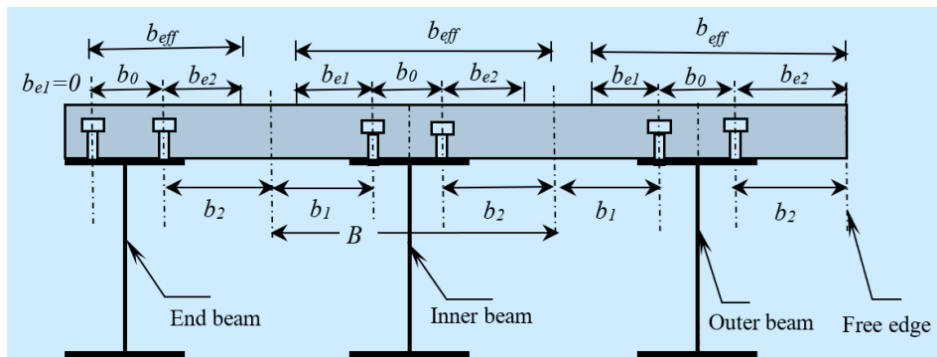


FIG. 6 EFFECTIVE WIDTHS FOR CONCRETE FLANGE OF COMPOSITE BEAMS

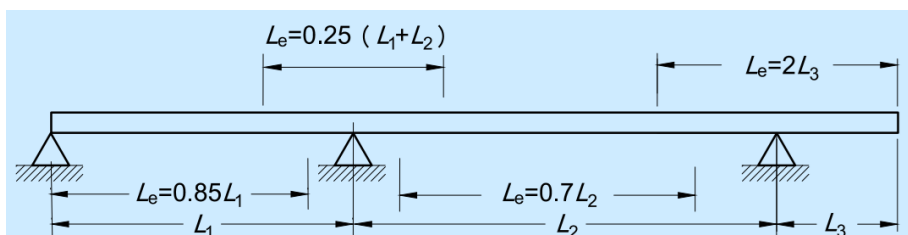


FIG. 7 EQUIVALENT SPANS FOR BEAM CONTINUITY

Reinforcements placed parallel to the steel beam within the effective width of the concrete slab should be considered for hogging moment resistance calculation of the composite beam at the continuous support.

- b) The effective width at an end beam may be determined as:

$$b_{\text{eff}} = b_0 + \beta_{e1} b_{e1} + \beta_{e2} b_{e2}$$

where

$$\beta_{e1} = 0.55 + 0.025 (L_e/b_{e1}) \leq 1.0$$

$$\beta_{e2} = 0.55 + 0.025 (L_e/b_{e2}) \leq 1.0$$

$$L_e = \text{Equivalent effective end span; and}$$

$$b_{e1} = 0 \text{ as shown in Fig. 6.}$$

8.3.4 Effective Cross Section for Strength Calculation

In calculating the strength of the cross section of the composite beams, the following should be considered:

- For Sagging Moment — Concrete in the effective width and structural steel beam to be included but not the steel reinforcements in the effective width of concrete; and
- For Hogging Moment — Concrete to be neglected but longitudinal steel reinforcement along the beam length within the effective width of the concrete in tension and structural steel beam to be included.

8.3.5 Design of Structure for Bending Moment

8.3.5.1 The factored design flexural strength, M_d , in a beam and the external action, M , shall satisfy

$$M \leq M_d$$

where

$$M_d = M_n / \gamma_{m0} = \text{design flexural resistance calculated as given below;}$$

$$\gamma_{m0} = \text{partial safety factor against flexural failure (Table 7); and}$$

$$M_n = \text{nominal ultimate flexural resistance.}$$

8.3.5.2 Considering local buckling, sections are to be analyzed as plastic, compact or semi-compact with the following additional consideration:

- Load history and development in composite action are to be taken into consideration with appropriate values of the modular ratio, m at each stage, and stresses and deflections are to be the summation of values over successive stages;
- Determination of bending resistance for a beam before composite action has set in or during the construction stage shall be done as per good practice [C-6B(3)];
- The structural steel reinforcement and concrete fully interact with full shear transfer between the concrete and structural steel under composite action;
- The effective width of concrete in composite action may be as mentioned in 8.3.3;
- For calculating the bending resistance of plastic, compact and semi-compact composite beams, the procedure presented in Annex B may be followed; and
- The partial shear connection may be used in plastic and compact sections either for attaining economy without losing much in the moment capacity of the composite section or under conditions where the number of shear connectors required for full shear interactions cannot be provided. Requirements on the method of evaluating the bending resistance of partial shear connection are presented in B-1.2.

The bending moment and shear force distribution in continuous beams for secondary, as well as primary moment resistant frames/beams have to be determined by structural analysis.

8.3.5.3 Design of structure for effect of lateral buckling on moment resistance

Lateral buckling may govern the design under the following conditions:

- At the construction stage, in the top flange closer to mid-span in both simply supported and continuous beams; and
- At the construction and composite stage, in the bottom flange closer to support in continuous beams.

At the construction stage, the effect of lateral buckling on the bottom flange in a continuous beam shall be taken care of by considering cantilever action up to the point of inflection from the support.

If required, suitable horizontal bracings or members may be provided at the bottom flange to reduce the effective length of the compression flange near support. For beams that are provided with such bracings or members giving effective lateral restraint to the compression flange at intervals along the span, the effective lateral restraint shall be capable of resisting a force equal to 2.5 percent of the maximum force in the compression flange taken as divided equally between the numbers of points at which the restraint in bracing members occur. In beams supporting composite slabs, wherein the sheeting is welded to the top flange before concreting is done, the lateral restraint of the deck sheet may be considered.

The effective laterally unsupported length of compression flange of steel sections in sagging and hogging moment regions may be taken as given in as per good practice [C-6B(3)]. In sagging moment segments of composite beams, the concrete slab may be assumed to provide lateral support, provided an adequate shear connection between the slab and the beam is provided. In the hogging moment regions of composite beams, the effective length considering the rotational restraint provided by the slab to the top tension flange may be taken into account as given in [B-2](#).

8.3.5.4 Simplified design of continuous beams in buildings without lateral bracings

Continuous composite beams (or composite beams within a frame) with plastic, compact, or semi-compact sections may be designed without any additional lateral bracings to the compression flange when the following conditions are satisfied:

- The concrete slab on top should be adequately attached with shear connectors to the structural steel section and to at least one another parallel member top flange to facilitate inverted U-frame action that rotationally restrains bottom flange lateral buckling at hogging moment sections;
- Adjacent span lengths of continuous beams should not differ by more than 20 percent of the shorter span length. Cantilever span length should not exceed 15 percent of the adjacent span length;
- The loading on each span should be uniformly distributed, with the permanent design loads exceeding 40 percent of the total design load;
- At each support of steel member, the bottom flange should be laterally restrained with a web bearing stiffener at that location; and
- The maximum depth of steel beam sections should not exceed the value given in [Table 8](#).

Table 8 Maximum Depth of Steel Beams in Composite Construction (In mm)

[Clause [8.3.5.4 \(e\)](#)]

Sl No.	Type of Member	Nominal Yield Strength Limit of Structural Steel (MPa)							
		250		300		350		> 400	
		Un-cased	En-cased	Un-cased	En-cased	Un-cased	En-cased	Un-cased	En-cased
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
i)	I Section	800	1 000	700	900	650	850	500	650
ii)	Parallel flange	600	800	550	750	400	600	270	430

8.3.6 Design of Structural Steel Web against Vertical Shear Force

8.3.6.1 The factored design shear force, V_d , in a composite beam due to external action, V , shall satisfy:

$$V \leq V_d$$

where

V_d = design shear resistance calculated as given below.

$$V_d = V_n / \gamma_{m0}$$

γ_{m0} = partial safety factor as mentioned in [Table 7](#).

V_n = nominal ultimate shear resistance.

The vertical shear force is assumed to be resisted by the web of the steel section only unless the value for contribution from the reinforced concrete part of the beam has been established. The nominal shear resistance, V_n may be governed by plastic shear resistance or strength of the web or governed by shear buckling as given below:

a) Plastic Shear Resistance

The nominal plastic shear resistance of composite beams under pure shear shall be calculated as indicated in as per good practice [C-6B(3)], disregarding the contribution of the concrete slab. The shear resistance for I-sections, channels both for major axis bending and minor axis bending as well as for rectangular and circular hollow sections of uniform thickness shall be as per good practice [C-6B(3)].

b) Shear Buckling Resistance

Resistance to shear buckling shall be verified as specified in as per good practice [C-6B(3)]. The nominal shear resistance, V_n , of webs with or without intermediate stiffeners as governed by buckling may be evaluated as detailed in as per good practice [C-6B(3)] either using the simple post-critical method or tension field method. The contribution of encasing concrete to shear resistance may be considered and that of shear reinforcement can be considered provided they consist of closed-loop stirrups. The total shear may be shared between the structural steel section and encasing concrete in the ratio of their contribution to plastic moment capacity.

8.3.6.2 Reduction in bending resistance under high shear force

If V is less than $0.6V_d$, reduction in the plastic bending resistance of the section need not be considered. When $V > 0.6V_d$, the bending resistance is reduced as the contribution of the web to bending gets diminished. The reduced bending resistances of the following sections are given below:

a) Plastic or compact section

$$M_{dv} = M_d - \beta(M_d - M_{fd}) \leq (1.2 z_e f_y) / \gamma_{m0}$$

where

$$\beta = (2V/V_d - 1)^2$$

M_d = plastic design moment of the whole section disregarding high shear force effect considering web buckling effects

V = factored applied shear force.

V_d = design shear resistance as governed by web yielding or web buckling

M_{fd} = Plastic design resistance of the area of the cross section excluding the shear area, considering partial safety factor γ_{m0}

b) *Semi-compact Section*

$$M_{dv} = Z_e f_y / \gamma_{mo}$$

where

Z_e = elastic section modulus of the whole section.

8.3.7 Hybrid Sections

Use of hybrid steel sections consisting of different grade steel elements is permitted. The plastic and compact section design strength is obtained by using the appropriate value of yield stress in the different elements of the hybrid section. The design strength of the semi-compact section is evaluated with necessary adjustment (reduction) in stresses of the flange element in the cross-section with higher yield stress by the reduction factor, R_h . The R_h may be determined using the procedure as presented in [B-3](#) of [Annex B](#).

8.3.8 Partially Encased Sections

This section is applicable for partially encased composite plastic or compact beams, provided the web depth to thickness ratio, d/t_w , is less than or equal to 117ϵ .

The full shear connection between the structural steel section and concrete encasing the web shall be ensured. The steel reinforcement in the compression zone and concrete in the tension zone may be disregarded.

The partial shear connection may be used in the partially encased composite section, provided the requirements of [8.3.5.2](#) are satisfied.

9 DESIGN FOR SERVICEABILITY LIMIT STATES

9.1 General

9.1.1 Serviceability limit states are related to the criteria governing normal use. Serviceability limit state is the limit state beyond which the serviceability criteria specified below are no longer met:

- a) Stress and deflection limit;
- b) Vibration limit;
- c) Durability consideration; and
- d) Fire resistance

9.1.2 Linear elastic analysis is used for finding out design moments and stresses under various load combinations and load factors, as mentioned in [Table 7](#), for serviceability limit states. Concrete is assumed as unreinforced and un-cracked for the analysis.

9.1.3 Method of Construction

The stress and strain at serviceability limit state depend on whether the steel beam is propped or un-propped during construction, as given below:

- a) *Un-propped construction* — In un-propped construction, the steel beam has to carry the construction load, including shuttering, wet concrete and its own weight until the concrete hardens;
- b) *Propped construction* — In propped construction, both the dead and imposed load are resisted by the composite section. When props are used, they should be kept in place until the *in-situ* concrete has attained a strength equal to approximately twice the stress to which the concrete may be subjected to upon removal of props; and
- c) *Non-composite construction* — In non-composite construction, the concrete slab is designed according to [C-6B(2)] and the steel deck sheet act just as a permanent shuttering.

The difference in the method (a) and (b) of construction does not, however, affect the ultimate limit load, wherein the total load, including the transient loads, shall be resisted by the composite section.

9.2 Negative Moments

Negative moments over intermediate supports may be adjusted as mentioned in [8.1.2.1](#).

9.3 Stresses and Deflections

For calculating stresses at service load and deflection, the value of the modular ratio, m shall be taken as,

$$m = \frac{E_s}{E_{cm}} \geq 7.5 \text{ For short-term effect or loading.}$$

$$m = \frac{E_s}{K_c E_{cm}} \geq 15.0 \text{ For permanent or long-term loads.}$$

where

K_c = Creep factor, 0.5

E_s = Modulus of elasticity for steel = 2.0×10^5 N/mm²;

E_{cm} = Modulus of elasticity of cast *in-situ* concrete as per [Annex F](#);

f_{ck} = characteristic cube compressive strength of concrete in N/mm²; and

E_{ci} = Modulus of elasticity of cast *in-situ* concrete at i days ($i < 28$ days).

The equivalent area of concrete slab at any stage shall be determined by dividing the effective width of the concrete slab by the modular ratio as given below.

$$m = \frac{E_s}{E_{ci}}$$

Final stresses and deflection are to be worked out separately at each stage of load history with relevant modular ratios and section modulus as discussed above and then added together.

9.3.1 Limiting Stresses for Serviceability

Limiting stresses for different stages of construction are as indicated below:

- Concrete — The limiting compressive stress in concrete should not exceed one-third of the characteristic strength of concrete;
- Reinforcement Steel — The limiting tensile stress in steel reinforcement should not exceed f_{yk}/γ_k ; and
- Structural Steel — The limiting stress in steel beam considering the different stages of construction should not exceed f_y/γ_{m0} .

Where bearing stress is combined with tensile or compressive stress, bending and shear stresses under the most unfavourable conditions of loading, the equivalent stress obtained from the following equation shall not exceed $0.9f_y$:

$$f_{ec} = \sqrt{f_{bc}^2 + f_p^2 + f_{bc} \cdot f_p + 3\tau_b^2}$$

and

$$f_{et} = \sqrt{f_{bt}^2 + f_p^2 + f_{bt} \cdot f_p + 3\tau_b^2}$$

where

f_{ec} and f_{et} = equivalent compressive and tensile stress in steel section;

- f_{bc} and f_{bt} = actual compressive and tensile stress in steel section;
 f_p = actual bearing stress in steel section; and
 τ_b = actual shear stress in steel section.

The value of bending stresses f_{bc} about each axis, to be used in the above formula, shall be individually lesser than the values of the maximum allowable stresses in bending about the corresponding axis.

9.3.2 Limiting Deflection and Camber

9.3.2.1 Deflection limit

The deflection under serviceability loads of a building or a building component should not impair the strength of the structure or components or cause damage to finishing. Deflections are to be checked for the most adverse but realistic combination of service loads and their arrangements by elastic analysis using a load factor given Table 6. The deflection of a member shall be calculated without considering the impact factor or dynamic effect of the loads on deflection. As per good practice [C-6B(3)] gives recommended limits of deflections for certain structural members and systems. Circumstances may arise where greater or lesser values would be more appropriate depending upon the nature of the material in element to be supported (vulnerable to cracking or not) and intended use of the structure, as required by the client.

9.3.2.2 Provision of camber

Where the deflection due to the combination of dead load and imposed load is likely to be excessive, consideration should be given to pre-camber the steel beams. The values of desired camber shall be specified in the design drawing. Generally, for spans greater than 25 m, camber approximately equal to the deflection due to dead loads plus half the imposed load may be used.

9.4 Vibration

Suitable provisions in the design shall be made for the dynamic effects of imposed loads, impact loads and vibration due to machinery operating loads. In severe cases, the possibility of resonance, fatigue or unacceptable vibrations shall be investigated. Unusually flexible structures (generally the height to the effective width of lateral load resistance system exceeding 5 : 1) shall be investigated for lateral vibration under dynamic wind loads as per (see Part C 'Structural Design/Section 1 Loads and Load effects' of this NBCS). Structures subjected to a large number of cycles of loading shall be designed against fatigue failure in accordance with as per good practice [C-6B(3)]. The floor vibration effect shall be considered using specialist literature.

9.5 Durability

9.5.1 Several factors that affect the durability of the buildings under conditions relevant to their intended life are listed below:

- a) The environment;
- b) The degree of exposure;
- c) The shape of the member and the structural detail;
- d) The protective measure; and
- e) Ease of maintenance.

The durability of the structural steel component of a composite section shall be ensured by recommendations as given in as per good practice [C-6B(3)]. Specialist literature may be referred to for more detailed additional information in design for durability. For concrete, the durability shall be ensured by following the recommendations as given in as per good practice [C-6B(2)].

9.5.2 Profiled Steel Sheet for Composite Slabs

The durability criteria for profiled steel sheets of composite deck slab shall be met by the following procedures:

- a) The exposed surfaces of the steel sheeting shall be adequately protected to resist particular atmospheric conditions; and
- b) A zinc coating of a total mass 275 g/m² (including both sides) is sufficient for internal floors in a non-aggressive environment, but the specification may be varied depending on service conditions.

9.6 Fire Resistance

Fire resistance of a steel component of a composite member is a function of its mass, its geometry, and the actions to which it is subjected, its structural support conditions, fire protection measures adopted and the fire to which it is exposed. Design provisions to the resistance of fire for concrete shall be as per guidelines given in as per good practice [C-6B(2)]. For the design of structural steel components for fire resistance, as per good practice [C-6B(3)] shall be referred. Specialist literature may be referred to for more detailed information on the design for fire resistance of steel/composite structures. The aspect of fire resistance is given in [15](#).

9.7 Control of Cracking in Concrete and Crack Width Calculation

Minimum reinforcements in terms of diameter and spacing required for crack control at the top of concrete as per good practice [C-6B(2)] are to be provided in composite beams at the zone of the negative moment to prevent cracks adversely affecting the appearance and durability of the structure. Crack width calculation as well as limiting crack width as given in as per good practice [C-6B(2)] may be followed, subject to the discretion of engineers. The crack width in concrete shall be restricted to values as indicated in as per good practice [C-6B(2)].

10 DESIGN FOR FATIGUE LIMIT

This section applies to the design of structures and structural elements subject to loading which could lead to fatigue failure. The following effects are not considered in the section:

- a) Corrosion fatigue;
- b) Low cycle (high stress) fatigue;
- c) Thermal fatigue;
- d) Stress corrosion cracking;
- e) Effects of high temperature (> 150 °C); and
- f) Effects of low temperature (< below transition temperature).

The fatigue design of various components of composite structures like members, welded joints, bolts, shear lugs etc. shall be carried out as per the specifications laid down in as per good practice [C-6B(3)]. Fatigue provisions in the design of shear connectors are discussed in [11.3](#).

11 SHEAR CONNECTORS

The shear connectors shall fulfill the dual purpose of transferring shear force between concrete and structural steel as well as anchoring the two components relative to each other with minimum slip to ensure full or partial composite action as per the design requirement. The dimensional details of shear shall be as given in [Fig. 8](#).

11.1 Longitudinal Shear in Beams and Slabs

Longitudinal shear load on shear connectors in a composite section, irrespective of boundary conditions of the members, is to be calculated for service and fatigue limit states on the basis of elastic theory. Appropriate sectional properties based on effective widths and modular ratios as per the load history and development of composite action shall be considered for the design of the section for resistance against longitudinal shear between steel and concrete.

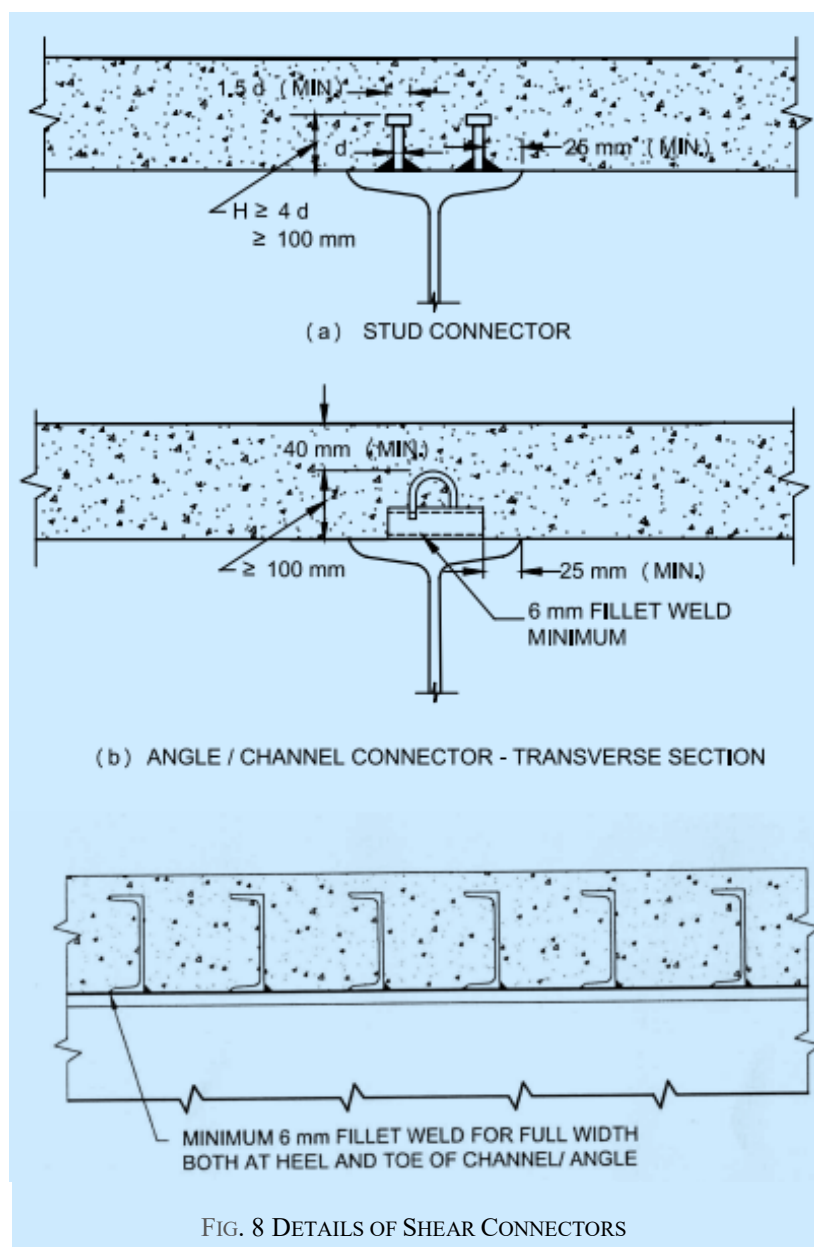


FIG. 8 DETAILS OF SHEAR CONNECTORS

NOTES

- 1 The diameter of the stud connector welded to the flange plate shall not exceed two and half times the flange plate thickness.
- 2 The height of the stud connectors shall not be less than four times their diameter nor 100 mm.
- 3 The diameter of the head of the stud shall not be less than one and a half times the diameter of the stud and the depth shall not be less than 0.4 d.
- 4 The size of the fillet weld joining other types of connectors to the flange plate shall not exceed half the thickness of the flange plate.
- 5 Channel and angle connectors shall have at least 6 mm fillet welds placed along the heel and toe of the channels/angles. The clear distance between the edge of the flange and the edge of the shear connectors shall not be less than 25 mm.
- 6 The overall height of a connector, including any hoop, which is an integral part of the connector, shall be at least 100 mm with a clear cover of 25 mm.

11.2 Design Strength of Shear Connectors

Shear connectors shall be checked for adequacy against failure in both ultimate limit states and fatigue limit states. The strength of shear connectors against failure under ultimate limit states and fatigue limit states shall be considered as per [11.2.1](#), [11.2.2](#) and [11.3](#), respectively.

11.2.1 Design Strength of Shear Connectors

Design static strengths of flexible shear connectors, mainly stud connectors and channel connectors can be determined by the following equations:

a) Stud connectors

The design resistance, Q_d of stud shears connectors shall be as given below:

$$Q_d = \frac{0.8 f_{us} \cdot \pi \cdot d^2 / 4}{\gamma_{mv}} \leq \frac{0.29 \alpha d^2 \sqrt{0.8 f_{ck} \cdot E_{cm}}}{\gamma_{mv}}$$

where

$$\alpha = 0.2 \left\{ \frac{h_s}{d} + 1 \right\} \text{ for } 3 \leq \frac{h_s}{d} \leq 4 \text{ and } \alpha = 1.0 \text{ for } \frac{h_s}{d} > 4;$$

Q_d = design strength of stud in, Newton (N);

γ_{mv} = partial safety factor for stud connector = 1.25;

d = diameter of the shank of the stud, in mm ($16 \text{ mm} \leq d \leq 25 \text{ mm}$);

f_{us} = ultimate tensile strength of the stud material $\leq 500 \text{ N/mm}^2$;

f_{ck} = Characteristic compressive strength of concrete of density not less than 1750 kg/m^3 ;

h_s = nominal height of stud in mm; and

E_{cm} = Secant modulus of elasticity of concrete (see [Annex F](#)).

b) Channel Connectors

Assuming that the web of the channel is placed vertical and the shear applied is nominally perpendicular to the web, the design resistance of a channel connector shall be determined as given below:

$$Q_d = \frac{[20b(h)^{3/4}] (0.8f_{ck})^{1/3}}{\gamma_{mv}}$$

where

Q_d = Design strength of channel in Newton (N);

h = height of the channel in mm;

b = length of the channel in mm;

f_{ck} = Characteristic compressive strength of concrete in MPa; and

γ_{mv} = factor for channel shear connector = 1.25.

While using the channel shear connectors, the following recommendations need to be followed:

- The height h of the channel should not exceed 20 times the channel web thickness or 150 mm, whichever is less;
- The width b of the channel should not exceed 300 mm;
- The underside of the top flange of the channel should not be less than 30 mm clear above the bottom reinforcement; and
- The size of the fillet weld connecting the channel to the flange plate should not exceed half the flange plate thickness.

11.2.2 The design strengths of some standard shear connectors are given in [Table 9](#). The following points have to be considered while using [Table 9](#):

- f_{ck} is the specified characteristic cube strength at 28 days;

- b) Connector strengths for concrete of intermediate grade may be obtained by linear interpolation;
- c) For channels of lengths different from those quoted above, the capacities are proportional to the lengths greater than 150 mm;
- d) For rolled angle and tee shear connectors, the values given for channel connectors are applicable, provided the height is at least equal to that of the channel;
- e) For stud connectors of overall height greater than 100 mm, the design static strength should be taken as the values given in [Table 9](#) for 100 mm high connectors;
- f) The above provisions of stud connectors are not applicable to composite slab using a profiled deck. The strength of the shear connector in such cases is given in [11.2.3](#);
- g) The number of shear connectors given by the above [Table 9](#) shall be distributed in the zone between the maximum and the zero moment sections. The number of connectors has to meet both ultimate strength consideration as well as fatigue consideration; and
- h) In order to avoid undesirable slip, the maximum interface shear per unit length due to superimposed dead load and imposed load under service conditions at any point in the beam should satisfy [11.3.1](#).

Table 9 Ultimate Static Design Strengths of Shear Connectors (Q_n for Different Concrete Strengths)
(Clause [11.2.2](#) and [11.3.1](#))

Sl No.	Type of Connector	Material	Size		Ultimate Static Strength in kN per connector			
			Nominal diameter mm	Nominal height mm	For concrete strengths, f_{ck}			
					MPa			
					25	30	40	50
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	Stud connectors	Material with a characteristic yield strength of 385 MPa, minimum elongation of 18 percent and a characteristic tensile strength of 495 MPa	25	100	112	125	149	156
			22	100	87	97	115	120
			20	100	72	80	90	100
			20	75	63	71	84	97
			19	100	65	72	86	90
			19	75	64	72	85	90
			16	75	46	51	61	64
			12	65	26	29	34	36
ii)	Channels 150 mm long (min)	As per [C-6B(4)] (E250 A/BR)	ISMC 125		244	259	285	307
			ISMC 100		206	219	241	260
			ISMC 75		166	176	194	209

11.2.3 Design Resistance of Studs Used with Profiled Steel Sheeting

11.2.3.1 Sheeting with ribs parallel to the supporting beams

The studs are located within a region of concrete that has the shape of a haunch ([Fig. 9](#)). When the sheeting is continuous across the beam, the width of the haunch b_0 is equal to the width of the trough, as given in [Fig. 10](#). Where the sheeting is not continuous, b_0 is as given in [Fig. 9](#). The depth of the haunch should be taken as d_p , the overall depth of the sheeting excluding embossments.

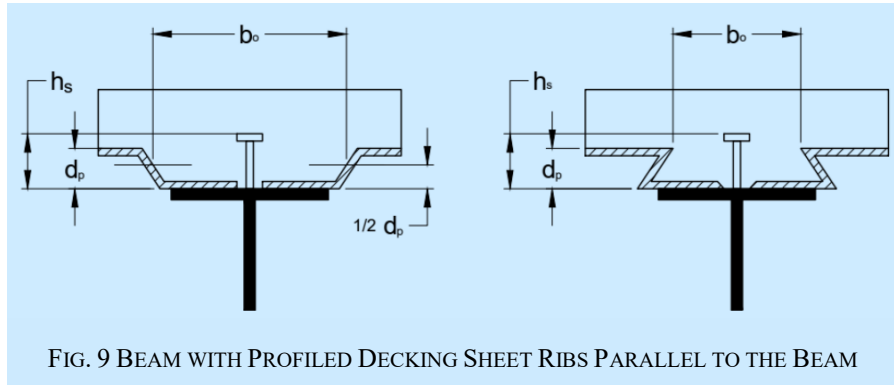


FIG. 9 BEAM WITH PROFILED DECKING SHEET RIBS PARALLEL TO THE BEAM

The design resistance of the shear connector should be taken as the resistance in a solid slab, multiplied by the reduction factor k_p given by the following expression:

$$k_p = 0.6 \frac{b_0}{d_p} \left(\frac{h_s}{d_p} - 1 \right) \leq 1.0$$

Where h_s is the overall height of the stud, but not greater than $d_p + 75$ mm.

Where the sheeting is not continuous across the beam, as shown in [Fig. 9](#), the side of the haunch and its reinforcement should be taken as the depth of the profile d_p excluding the embossments.

11.2.3.2 Sheeting with ribs transverse to the supporting beams

Provided that the conditions given in [11.2.2](#) are satisfied, the design shear resistance should be taken as the resistance in a solid slab, calculated as given by [11.2.1](#) (except that f_{us} should not be taken as greater than 450 N/mm²) multiplied by the reduction factor k_t given by:

$$k_t = \frac{0.7}{\sqrt{n_r}} \frac{b_0}{d_p} \left(\frac{h_s}{d_p} - 1 \right) \leq 1.0$$

where

n_r is the number of stud connectors in one rib at the beam intersection, not to exceed two in calculation of the reduction factor k_t and of the longitudinal shear resistance of the connection.

The factor k_t should not be taken greater than the appropriate value k_{tmax} given in [Table 10](#). The values for k_t are applicable provided that: the studs are placed in ribs with a height d_p not greater than 85 mm and a width b_0 not less than d_p and for through deck welding, the diameter of the studs is not greater than 20 mm, or for holes provided in the sheeting, the diameter of the studs is not greater than 22 mm.

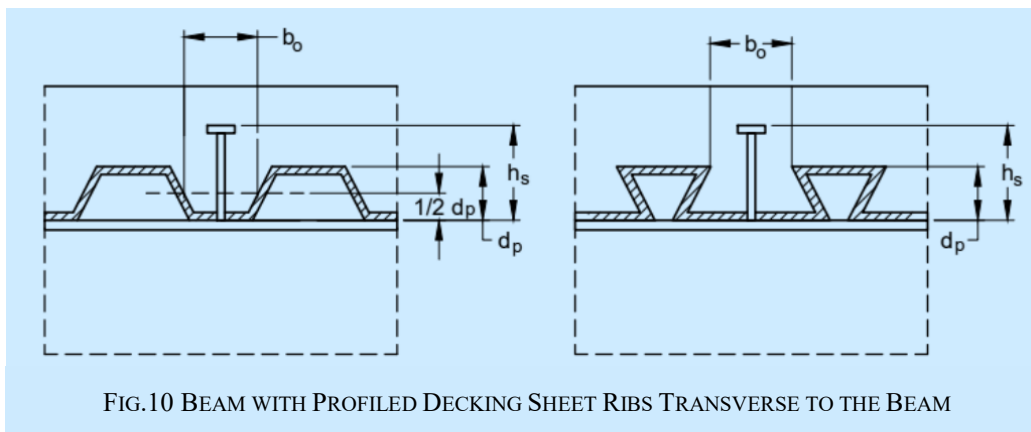


FIG.10 BEAM WITH PROFILED DECKING SHEET RIBS TRANSVERSE TO THE BEAM

Table 10 Upper Limits $K_{t, \max}$ for the Reduction Factor K_T

(Clause 11.2.3.2)

Sl No.	Number of Stud Connectors per Rib	Thickness t of Sheet (mm)	Studs 20 mm in Diameter and Welded Through Profiled Steel Sheeting	Profiled Sheeting with Holes and Studs 19 mm or 22 mm in Diameter
(1)	(2)	(3)	(4)	(5)
i)	$n_r=1$	≤ 1.0	0.85	0.75
		> 1.0	1.0	0.75
ii)	$n_r=2$	≤ 1.0	0.70	0.60
		> 1.0	0.8	0.60

11.3 Fatigue Strength of Shear Connectors

The fatigue shear stress range (fatigue Strength) of the shear connector shall be obtained from [C-6B(3)], corresponding to the design load life cycle, N_{SC} .

The strength shall be determined as given below:

$$\tau_f = \tau_{fn} \sqrt[5]{5 \times 10^6 / N_{SC}}$$

τ_{fn} = design normal and shear fatigue stress range respectively of the detail for 5×10^6 cycles as given in [C-6B(3)].

τ_{fn} = 67 N/mm² for stud connector as per good practice [C-6B(3)].

τ_{fn} = 59 N/mm² for channel connector as per good practice [C-6B(3)] [provided that the thickness of the top flange of steel beam is greater than or equal to 12 mm and the edge distance from the end of weld to the edge of the top flange is 10 mm].

The design fatigue strengths of some standard shear connectors are presented in Table 11.

Table 11 Fatigue Strengths Q_r (in kN)

(Clause 11.3)

Sl No.	Type of Connectors (Headed Stud – Diameter)	Connector Material	N = Nos. of Cycles in Millions				
			0.1	0.5	2	10	100
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	Headed studs ϕ 25	$f_{ys} = 385$ MPa $f_{us} = 495$ MPa Elongation = 18 percent	72	52	40	29	18
ii)	Headed studs ϕ 22		56	40	31	22	14
iii)	Headed studs ϕ 20		46	33	25	18	12
iv)	Headed studs ϕ 16		29	21	16	12	7
v)	Channel 150 long for a nominal weld of 8 mm	[C-6B(4)]	108	79	60	43	27

NOTE — For intermediate stress cycles the values may be interpolated from log scales (that is, the above equation). Other connectors, if used, should have their capacities established through tests.

11.3.1 Spacing of Shear Connectors

The design shear action per unit length of the steel concrete interface, V_L , is given by:

$$V_L = \sum \left[\frac{VA_{ec}y}{I} \right]_{DL,IL}$$

where

- V_L = Longitudinal shear per unit length;
- V = The factored vertical shear forces due to dead load and imposed load (including impact if any) separately at each state of load history;
- A_{ec} = The transformed compressive area of concrete above the plastic neutral axis of the composite section with the appropriate modular ratio depending on the nature of load (short term that is, imposed load, or long term, that is, dead load);
- y = C.G. distance of transformed concrete area from the plastic neutral axis;
- I = Moment of Inertia of the composite section using an appropriate modular ratio; and
- DL, IL = Different load history conditions, that is, sustained load or composite action dead load, transient load, or composite action imposed load. These loads are to be considered with appropriate load factors at this stage.

Maximum spacing of shear connectors, S_{L1} is given as

$$S_{L1} = \frac{\sum Q_n}{V_L}$$

Q_n is the ultimate static strength of one shear connector ([Table 9](#)) and the summation is over the number of shear studs in one section.

The maximum longitudinal force at the interface due to bending moment shall also be calculated over the shear span, L , equal to the distance from zero moment to maximum moment section and is given by the following equations:

$$H_1 = A_{sl}f_y \times 10^{-3} / \gamma_{m0}$$

$$H_2 = 0.36f_{ck}A_{ec} \times 10^{-3}$$

where

- H_1, H_2 = Longitudinal interface shear force due to bending (kN);
- A_{sl} = Area of tensile steel (mm²) in the longitudinal direction;
- A_{ec} = Effective area of concrete:
 $b_{eff}.x_u$ (for neutral axis within the slab); and
 $b_{eff}.d_s$ (for the neutral axis in the steel section)

Sufficient connectors should be provided to resist the longitudinal force H , the maximum compressive force action in the composite beam slab interface, which is the smaller of H_1 and H_2 .

The maximum spacing of shear connectors, S_{L2} , is given as:

$$S_{L2} = \frac{\sum Q_n}{H} \cdot L$$

11.3.2 Fatigue Strength of Shear Connectors

$$V_r = \Sigma \left[\frac{V_R \cdot A_{ec} \cdot y}{I} \right]_{LL}$$

where

V_R = The shear range in which is the difference between the maximum and minimum vertical shear envelop due to imposed load and impact; and

H = is imposed load with impact.

V , A_{ec} , y and I are as explained above

Spacing of shear connectors from fatigue consideration is given as:

$$S_R = \frac{\Sigma Q_r}{V_r}$$

Q_r is the design fatigue strength of one shear connector, which is to be taken from as per good practice [C-6B(3)] [Table 10](#).

For full shear connection, the lowest spacing of S_{L1} , S_{L2} , and S_R is to be provided as the actual spacing of the shear connectors.

11.4 Partial Shear Connection

11.4.1 Partial shear connection may be used either for attaining economy without losing much in the moment capacity of the composite section or where the number of shear connectors required for full shear cannot be provided without compromising minimum spacing provisions.

Partial shear connections may be used in plastic and compact sections. The number of connectors n_p shall then be determined by a partial shear connection theory, taking into account the deformation capacity of the shear connector.

$$S_c \text{ degree of shear connection} = \frac{n_p}{n_f}$$

n_p = Number of shear connectors provided for partial shear connection;

n_f = Number of shear connectors required for full shear connection;

M_R = Required reduced bending resistance of the section;

M_d = The fully plastic moment of resistance of the composite section; and

M_{ps} = Plastic moment of resistance of steel section alone.

$$n_p = \frac{M_R - M_{ps}}{M_d - M_{ps}} n_f$$

11.4.2 Limitation on the Use of Partial Shear Connection in Beams for Buildings

Headed studs with an overall length after welding not less than 4 times its diameter and with shank diameter not less than 16 mm and neither greater than 25 mm may be considered as ductile with the following limits for the degree of shear connection, S_c :

a) For steel sections with equal flanges:

$$L_e \leq 25: S_c \geq 1 - 1.42e^2 (0.75 - 0.03L_e); S_c \geq 0.4$$

$$L_e > 25: S_c = 1.0$$

- b) For steel sections having a bottom flange with an area of three times the area of top flange:

$$L_e \leq 20: S_c \geq 1 - 1.42\epsilon^2 (0.30 - 0.015L_e); S_c \geq 0.4$$

$$L_e > 20: S_c = 1.0$$

where

$\epsilon = \sqrt{250/f_y}$, L_e is the distance between points of zero bending moment in the sagging bending range in metres. For typical continuous beams, L_e may be assumed as shown in [Fig. 6](#) and [Fig. 7](#).

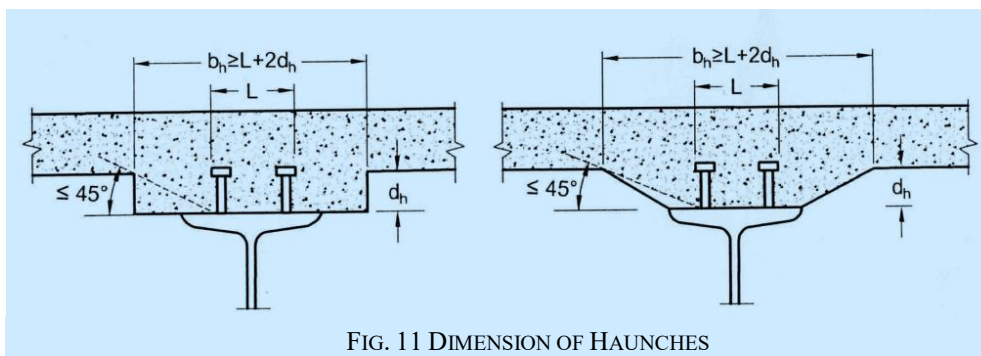
- c) For steel sections having a bottom flange with an area exceeding the area of the top flange but less than three times that area, the limit for S_c may be determined from expressions in (a) and (b) above by linear interpolation.

11.5 Precautions against Separation of Steel Beam from Concrete

- a) Where a concrete haunch is used, between the steel flange and the soffit of the slab, top of stud and top flange of channel shear connectors shall extend up to at least 40 mm above the transverse reinforcements in the haunches, provided the reinforcements are sufficient to transfer longitudinal shear; and
- b) Where shear connectors are placed adjacent to the longitudinal edge of the slab, transverse reinforcement provided in accordance with [11.7](#) shall be fully anchored in the concrete between the edge of the slab and the adjacent row of connectors.

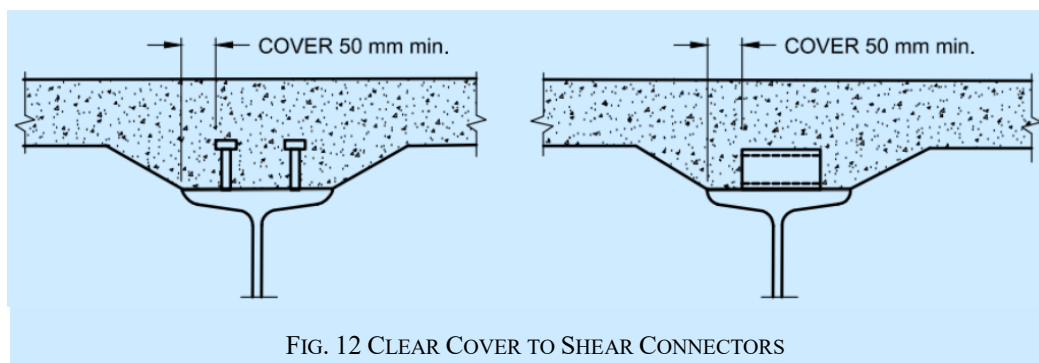
11.6 Detail of Haunches in Reinforced Concrete

11.6.1 [Figure 11](#) indicates the dimension of haunches as applicable for slabs with haunches resting on steel beam. The edge of haunches shall be located outside a line drawn at 45 degrees from the outside edge of the base of the connector, as shown in [Fig. 11](#).



11.6.2 Clear Cover to Shear Connectors

The clear depth of concrete cover over the top of the shear connectors shall not be less than 25 mm. The horizontal clear concrete cover to any shear connector shall not be less than 50 mm as shown in [Fig. 12](#).



11.6.3 Spacing of Shear Connectors

Where the compression flange of steel beam, in spite of being semi-compact is assumed to be compact or plastic based on the restraint provided by shear connectors, the centre-to-centre spacing of the shear connectors in the direction of the compression should satisfy the following:

- a) Where the slab is in contact over the full length (for example, solid slab)

$$S_L < 21 \cdot t_f \cdot \varepsilon$$

- b) Where the slab is not in contact over the full length (for example, slab with ribs transverse to the beam):

$$S_L < 14 \cdot t_f \cdot \varepsilon$$

where

$$\varepsilon = \sqrt{250/f_y}$$

t_f = thickness of the flange;

f_y = nominal yield strength of the flange in N/mm²; and

S_L = maximum spacing of the shear connector.

In addition, the clear distance from the edge of the compression flange to the nearest line of shear connectors should not be greater than $9t_f \varepsilon$ or 50 mm, whichever is less.

In all cases, a shear connector shall be provided throughout the length of the beam and may be uniformly spaced between critical cross sections. The maximum longitudinal spacing of the shear connectors shall be limited to the lesser of 600 mm or 3 times the total slab thickness or four times the height of the connector (including any hoop which is an integral part of the connector), whichever is least.

Minimum spacing should be such as to allow proper concrete flow and compaction around the connectors. In stud connectors, the minimum spacing should not be less than 75 mm.

11.7 Transverse Shear Check (Requirement of Bottom Steel in Concrete Slab)

Planes, which are critical for longitudinal shear failure in the process of transfer of longitudinal shear from the beam to the slab are of four main types as shown in [Fig. 13](#). The shear force Q in kN/m of beam is given by:

$$Q = \frac{N_r Q_u}{S_L}$$

where

N_r = Number of shear connection at a section;

Q_u = Design strength one shear connector; and

S_L = Longitudinal spacing of connector in meter.

The shear force transferred per metre length from steel beam to concrete slab above, V_L shall satisfy both the following conditions:

a) $V_L \leq 0.623 L \sqrt{f_{ck}}$; or

b) $V_L \leq 0.232 L \sqrt{f_{ck}} + 0.1 \cdot A_{st} \cdot f_{yk} \cdot n$

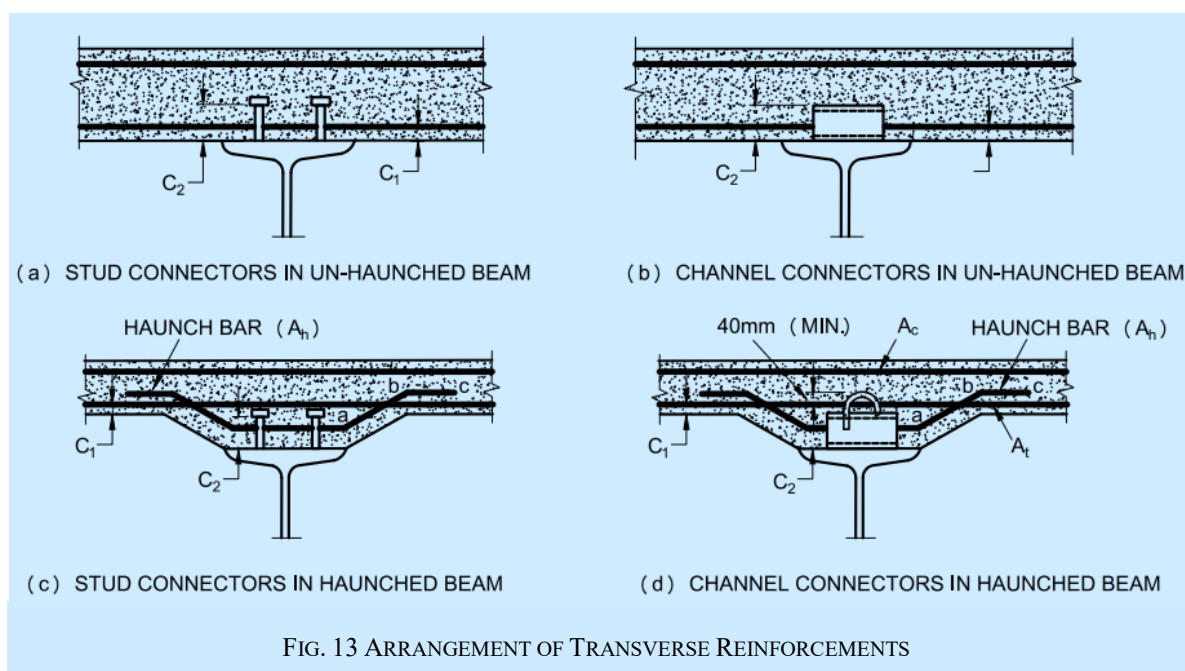
where

V_L = Longitudinal shear force per unit length calculated for the ultimate limit state;

- f_{ck} = Characteristic strength of concrete, in MPa;
- f_{yk} = Yield stress of transverse reinforcement, in MPa;
- L = Length (in mm) of possible shear planes; for two rows of shear studs as shown in Fig.15,
 $L = 2 h_s + b_0$;
- n = Number of times each lower transverse reinforcing bar is intersected by a shear surface (that is, the number of rows of shear the connector at one section of the beam). Generally, for T-beam $n = 2$ and for L-beam $n = 1$; and
- A_{st} = Sectional areas (in cm^2) of transverse reinforcements per meter run of the beam. The amount of transverse steel (cm^2/m) in the bottom of the slab shall not be less than $\frac{2.5Q}{f_{yk}}$.

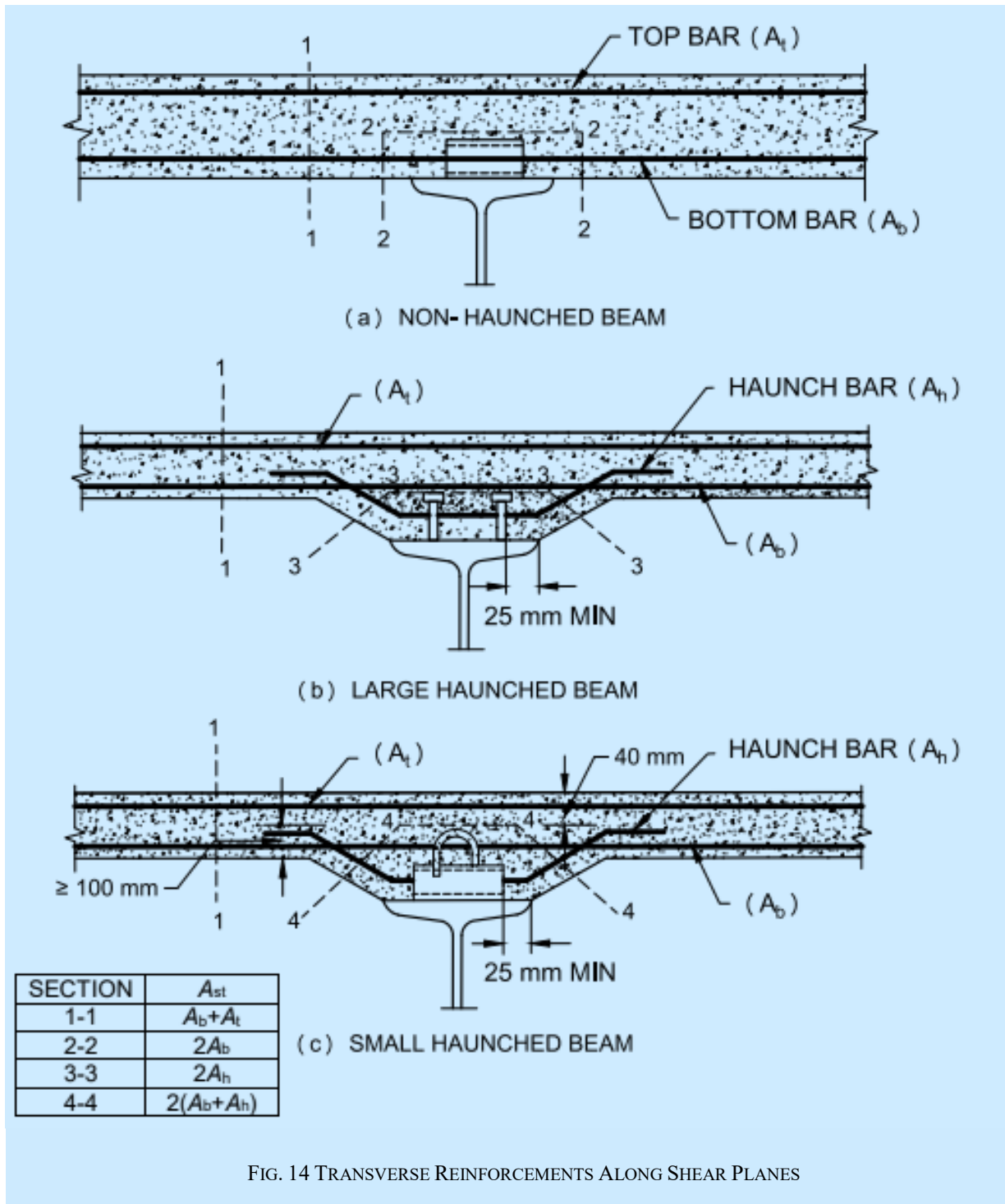
11.8 General Arrangements of Transverse Reinforcements

11.8.1 If the concrete by itself is insufficient to take the longitudinal shear, sufficient transverse reinforcements shall be provided to transfer longitudinal shear force from the beam to the effective width of the slab. The area of transverse reinforcement per unit length of the beam will be the sum total of all the reinforcement [A_t , A_b or A_h , as shown in Fig. 14(a), Fig. 14(b) and Fig. 14 (c)], which are intersected by the shear plane and are fully anchored on both the sides of the shear plane considered.



11.8.2 Total Transverse Reinforcements

The total transverse reinforcements A_{st} per unit length of the beam in case of shear plane 1 – 1, which crosses the whole thickness of the slab, will be the sum of ($A_t + A_b$) [Fig. 14(a)]. Area of reinforcements A_t and A_b include those provided for flexure. The total transverse reinforcements across plane 2-2 [Fig. 14(a)] is $A_{st} = 2 A_b$ and that across plane 3-3 [Fig. 14(b)] is $A_{st} = 2A_h$ as these planes do not cross the full thickness of the slab. In the case of shear plane 4 – 4 [Fig. 14(c)], the total transverse reinforcement is $A_{st} = 2(A_h + A_b)$. The transverse reinforcements shall be placed at locations as shown in Fig. 14. The haunch bars shall be extended beyond the junction of bottom bars by a length equal to the anchorage length.



12 COMPOSITE SLABS WITH PROFILED STEEL SHEETING

12.1 General

The provisions in this section deal with composite floor slabs spanning only in the direction of ribs of the decking. The provisions are applicable in buildings, wherein the slabs are subjected to predominantly static imposed loads. The application of the provision is limited to narrowly spaced web, which is defined by the ratio of the width of the sheet rib to the rib spacing $b_r/b_p \leq 0.6$ (see Fig. 15).

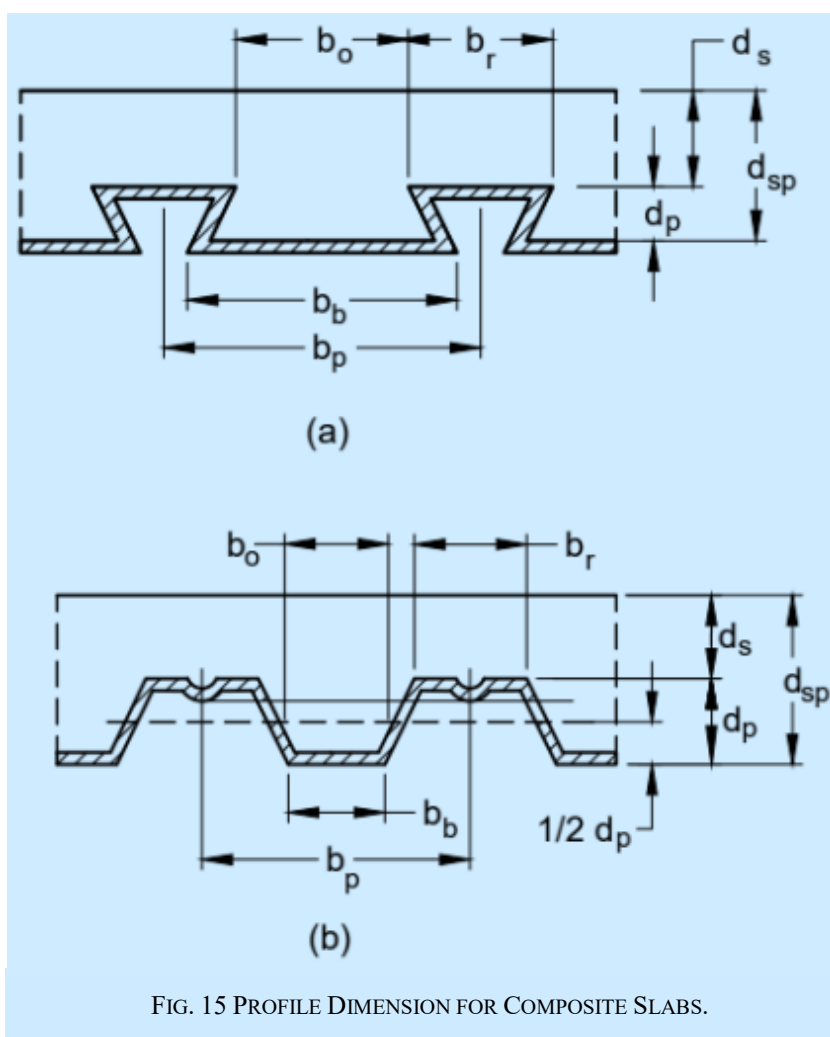


FIG. 15 PROFILE DIMENSION FOR COMPOSITE SLABS.

12.2 Prequalifications

12.2.1 Prequalification of Slab dimension

The prequalification of composite and non-composite slabs are presented below:

- a) Where the slab acts composite with a steel beam or used as a diaphragm:
 - 1) the overall depth of the slab should not be less than 90 mm; and
 - 2) the thickness of concrete above the main top flat surface of the sheeting ribs should not be less than 50 mm.
- b) Where the slab does not act composite with a steel beam or has no other stabilizing function:
 - 1) the overall depth of the slab should not be less than 80 mm; and
 - 2) the thickness of concrete above the main flat top surface of the sheeting ribs should not be less than 40 mm.

12.2.2 Bearing Length

The recommended minimum bearing lengths of steel decking on the support (L_{bs}) and the composite slab, including the cast in place concrete (L_{bc}) shall be as given in (see Fig. 16):

- a) For composite slab bearing on steel or concrete: $L_{bc}=75$ mm, $L_{bs}=50$ mm; and
- b) For composite slab bearing on other material: $L_{bc}=100$ mm, $L_{bs}=70$ mm.

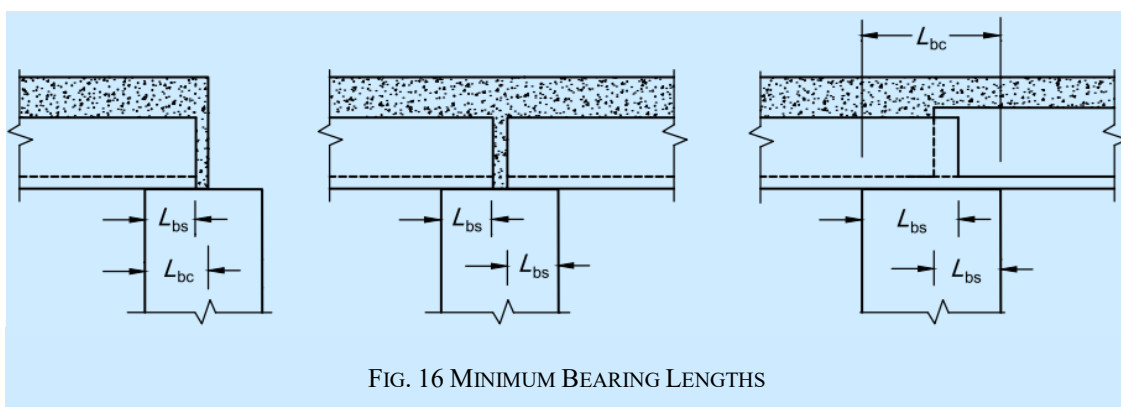


FIG. 16 MINIMUM BEARING LENGTHS

12.2.3 Reinforcement for Shrinkage and Temperature Stresses

The effect of temperature stress in composite slabs in buildings is normally neglected unless required in special conditions. The effect of shrinkage is to be considered. The total shrinkage strain for design may be taken as 0.0003 in the absence of test data.

12.2.4 Minimum Reinforcement

The minimum reinforcement in either direction should be according to [C-6B(2)]. The spacing of the reinforcement bar should be in accordance with as per good practice [C-6B(2)].

12.2.5 Size of Aggregates

In addition to the above, the largest nominal aggregate size should be according to as per good practice [C-6B(2)].

12.3 Analysis for Internal Forces and Moments

12.3.1 Analysis of Profile Steel Sheeting as Shuttering

- The elastic analysis shall be used where sheeting is considered. The design based on the elastic distribution of bending moment is conservative, as it does not take into account of redistribution of moments that can occur between support and mid span sections;
- In the analysis of propped profile steel sheeting, the magnitude of bending moments and shear forces could be conservatively calculated by using the coefficients of bending moment and shear force as per good practice [C-6B(2)] ;
- More accurate analysis and design of profile steel sheeting as a shuttering should be in accordance with as per good practice [C-6B(18)] ;
- If the deflection at the center of the shuttering due to its own weight and the wet weight of concrete is less than (1/10) of the depth of the composite slab, the effect of 'ponding' due to shuttering deflection can be disregarded. Otherwise, that effect of additional load shall be evaluated by taking the additional uniform depth of concrete over the span as 0.7 times the central deflection. Ponding is explained in C-3'
- The effect of ponding, especially in longer span, can be reduced by propping the shuttering during concreting until the concrete hardens. Plastic redistribution of moments in the propped shuttering is not allowed in evaluating stresses due to self-weight and wet concrete weight.

12.3.2 Analysis of Composite Slab

- The application of linear methods of analysis is suitable for the serviceability limit states as well as for the ultimate limit states. Plastic methods shall be used only in the ultimate limit state;

- b) The following method of analysis may be used for ultimate limit states:
- 1) Linear analysis without redistribution (serviceability limit state), and with redistribution (Ultimate limit state). If the effects of cracking of concrete are neglected in the analysis for ultimate limit states, the hogging bending moments at internal supports may be optionally reduced by up to 30 percent and corresponding increases are made to the sagging bending moments in the adjacent spans;
 - 2) Rigid-plastic global analysis may be used provided it is shown that sections where plastic rotations are required have sufficient rotation capacity. Plastic analysis without any direct check on rotation capacity may be used for the ultimate limit state if reinforcing steel satisfying the ductility requirement in [6.2.3](#) is used and the span length is not greater than 3.0 m. (Ultimate limit state); and
 - 3) Elastic-plastic analysis accounting for the non-linear material behavior. (Ultimate limit states).
- c) A continuous slab may be designed as a series of simply supported spans. Nominal reinforcement should be provided in accordance with as per good practice [C-6B(2)], over the intermediate supports to manage crack size; and
- d) Concentrated loads may be assumed to be distributed over an effective width of the slab, assuming the load is distributed at an angle of 45° to the horizontal up to the top surface of the decking.

12.4 Design of Composite Slabs

Proprietary data backed by analysis and tests may be used in the design. Otherwise, the procedure given below may be used to evaluate the strength of composite slabs.

12.4.1 Effective Span

- a) When a continuous composite slab is designed as a series of simply supported spans, for simplicity, the effective span can be taken as the lesser of the following:
- 1) Distance between centres of the supports; and
 - 2) The clear span plus the effective depth of the slab.
- b) Where the composite slab is designed as continuous, it is permitted to use an equivalent isostatic span for the determination of the resistance. The span length should be taken as:
- 1) $0.8 L$ for internal span (effective span L); and
 - 2) $0.9 L$ for external span (effective span L).

12.4.2 Design of Profile Steel Sheeting as Shuttering During Construction

The stress analysis of the profiled steel sheeting should be evaluated using the design equations or tests. For the ultimate limit state, the resistance of the sheet to sagging and hogging bending, together with the effects of combined bending and web crushing, are normally critical:

- a) For the serviceability limit state, the limiting value of deflection $\delta_{\max}$ of steel sheeting under its own weight plus the weight of wet concrete may be considered as $L/180$ (where L is the effective span between supports) as per good practice [C-6B(3)];
- b) Profiled decking acting alone resists construction, dead, and imposed load until the concrete hardens and gains strength. 'Ponding' effect, especially in long spans due to excessive deflection of decking, has to be accounted while evaluating design forces and design strength unless sufficient propping is provided during the construction stage until the concrete hardens; and
- c) The longer spans will require propping to eliminate substantial deflection. Otherwise, a significant increase in concrete weight due to ponding has to be accounted.

12.4.3 Design of Composite Slab for the Ultimate States

The bending moment and shear force distribution in continuous beams for secondary, as well as primary moment

resistant frames/beams, have to be determined by structural analysis. The design values of load effects shall not exceed the design values of resistance for the relevant ultimate limit states.

12.4.3.1 Design flexural resistance

The design flexural resistance calculation shall take the following provisions into consideration:

- The design bending resistance, M_d of any cross section should be in accordance with Annex C, where the yield stress of the steel sheeting should be taken as f_{yp} ;
- The effect of local buckling of compressed parts of the sheeting should be considered by using effective widths not exceeding twice the limiting values of Class-1 steel webs as discussed in as per good practice [C-6B(3)];
- For the effective area A_p of the steel sheeting, the width of embossments and indentations in the sheet should be neglected unless tests infer that a larger area is effective;
- The sagging bending resistance of a cross-section should be calculated from the equilibrium of stress distribution as shown in [C-4](#);
- In hogging bending regions, the contribution of steel sheeting shall be taken into account only where the sheet is continuous. The redistribution of moments that can occur between support and mid-span sections shall not be considered. If the contribution of the steel sheeting is neglected, the hogging bending resistance of a cross-section should be calculated as reinforced concrete for the stress distribution, as shown in [C-4.3](#); and
- In the case of partial shear connection, the sagging design bending resistance, M_d , shall be calculated as per [C-5](#).

12.5 Shear Resistance of Composite Slab

12.5.1 Punching Shear

The punching shear resistance of a composite slab should be calculated as per good practice [C-6B(2)], where the critical perimeter is as shown in [Fig. 17](#).

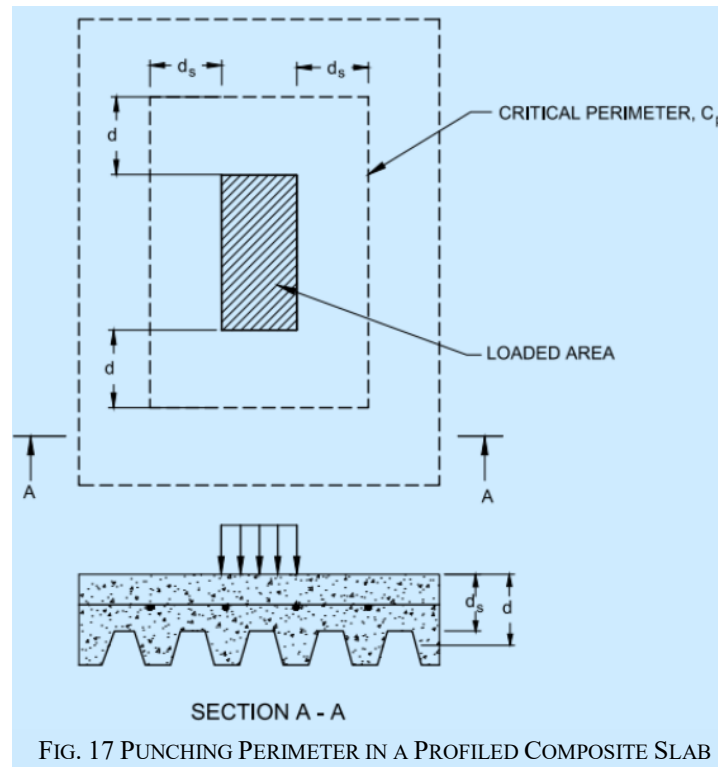


FIG. 17 PUNCHING PERIMETER IN A PROFILED COMPOSITE SLAB

12.5.2 Resistance to Shear

The vertical shear resistance V_{dvd} of composite slab over a width equal to the distance between centres of ribs and the effective depth, which depends on the effective depth of the cross section to the centroid of the tensile reinforcement, should be in accordance with as per good practice [C-6B(2)].

The sheeting may be considered as the tensile reinforcement, provided that it is fully anchored beyond the section considered. For heavily loaded slabs, additional reinforcement may be required at the support when the profiled steel sheeting is discontinuous and has only limited anchorage.

In the case of partial shear connection, the shear capacity may be governed by the interface shear strength between steel decking and concrete, as presented in [Annex C](#).

12.6 Serviceability Limit State

12.6.1 Design against Cracking

The crack width is calculated at the top surface in the hogging moment region using standard methods prescribed for reinforced concrete in as per good practice [C-6B(2)]. Crack width should not exceed 0.3 mm. The provision of 0.4 percent steel at the top of the slab will normally avoid cracking problems in propped construction. The provision of 0.2 percent of steel is normally sufficient for the same in un-propped construction.

If the environment is corrosive, it is advisable to design the slab as continuous at supports and take advantage of steel provided for negative bending moment resistance and for minimizing cracking during service loads.

12.6.2 Deflection Limits

As per good practice [C-6B(2)] gives a stringent deflection limitation of span/350 or 20 mm whichever is less, which may be unrealistic for un-propped construction. The span to depth ratio in the range of 25 to 35 for the composite condition is recommended for the simply supported slabs and the continuous slabs. The deflection of the composite slabs is influenced by the slip between sheeting and concrete. Testing is the best method to estimate the actual deflection for the conditions adopted.

12.6.3 Fire Resistance

The fire resistance is based on (a) thermal insulation criterion concerned with limiting the transmission of heat by conduction and (b) integrity criteria concerned with preventing the spread of flames and hot gases to nearby compartments. These are met by specifying the adequate thickness of insulation below the metal decking. When the metal decking is used as a permanent shuttering, the fire resistance of the slab shall be considered as per as per good practice [C-6B(2)].

13 COMPOSITE COLUMNS

13.1 General

13.1.1 This section applies to various forms of steel-concrete composite columns, including fully or partly encased steel columns and concrete in-filled rectangular or circular steel tubes, provided:

- The columns or compression members consist of structural steel with grade conforming to the accepted standard [C-6B(4)] and normal weight concrete of strength M20 to M60.
- This clause applies to isolated columns or compression members in framed structures where the other structural members are either composite or steel members.
- The steel contribution ratio δ ([Table 12](#)) should fulfill the criteria:

$$0.2 \leq \delta \leq 0.9$$

- Further:

- The influence of local buckling of the structural steel section on the resistance of the composite section as a whole shall be considered in the design; and

- 2) The effects of local buckling may be neglected for a steel section fully encased in accordance with [13.2](#), and for other types of cross-section, the maximum width to thickness ratio given in as per good practice [C-6B(3)] shall not be exceeded.

13.1.2 Types of Composite Columns

Composite columns can be of two types:

- a) Encased where concrete encases the steel section (see [Fig. 18](#)); and
- b) In-filled where concrete fills rectangular or circular steel tube (see [Fig. 19](#)).

13.2 Details for Composite Action

In composite columns with fully encased steel sections, concrete cover to structural steel sections shall be at least 40 mm or one-sixth of the breadth b of the flange, over steel section. The concrete shall be adequately held by steel reinforcements and stirrups all around. The steel section shall be unpainted to ensure bond and friction between steel and concrete, but cleaned to ensure protection against corrosion and spalling of concrete. The cover to steel reinforcement should be in accordance with as per good practice [C-6B(2)].

Shear transfer between steel-concrete interfaces is ensured basically through bond for which calculated shear stress at interface shall be limited in accordance with [Table 15](#), beyond which mechanical shear connectors are to be provided.

13.3 Members Under Axial Compression

Standard composite sections used as columns are as shown at [Fig. 18](#) and [Fig. 19](#).

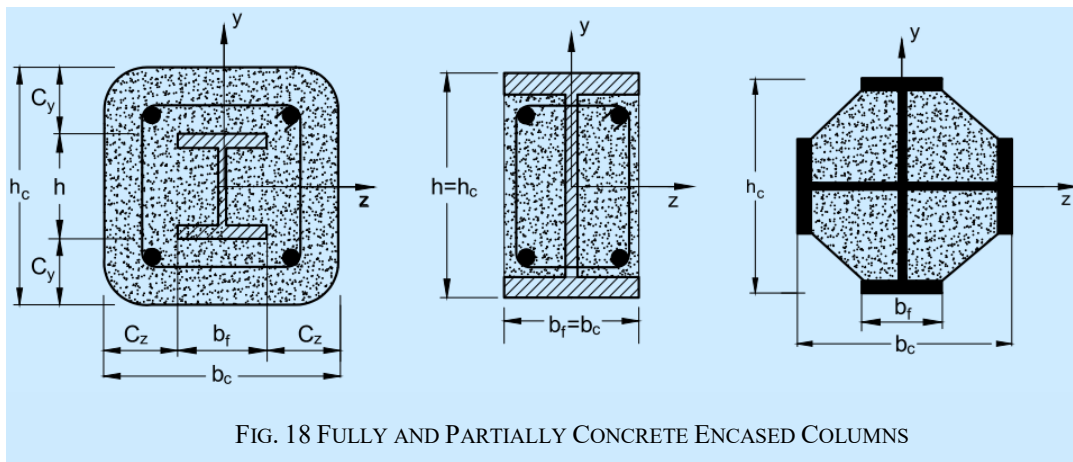


FIG. 18 FULLY AND PARTIALLY CONCRETE ENCASED COLUMNS

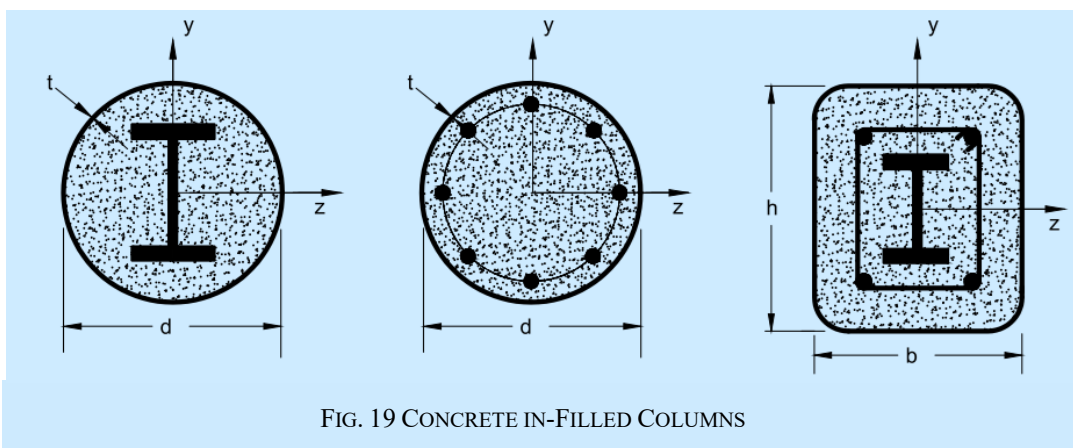


FIG. 19 CONCRETE IN-FILLED COLUMNS

13.3.1 General Design Philosophy

To ensure structural stability of a compression member, second-order effects like residual stresses, geometrical imperfections, local instability, cracking of concrete, creep/shrinkage of concrete, yielding of structural steel and reinforcement etc. shall be considered. While designing the compression members, including design for the above effects, the following shall be taken into account:

- The above effects shall be considered in any direction in which failure might occur if they affect the structural stability significantly;
- Internal forces shall be determined by any of the analysis methods discussed in [6](#);
- Full composite action between steel and concrete shall be considered up to failure; and
- Effects of creep and shrinkage shall be considered if they are likely to reduce the structural stability significantly. For simplification, creep and shrinkage effects may be ignored if the increase in the first-order bending moments due to creep deformations from permanent loads is not greater than 10 percent of first-order bending moments due to total design loads.

13.3.2 Design Guidelines

The simplified design provisions given in the following sections is applicable for composite members that are doubly symmetrical and uniform in cross-section throughout the length of the member. This method is not applicable to members consisting of two or more unconnected sections. Further, the composite member should conform to the following conditions.

- The non-dimensional slenderness λ shall not be greater than 2.
- For a fully encased steel section ([Fig. 18](#)), the maximum thickness of concrete cover that may be used in the strength calculation is limited to:

$$c_z (Max) = 0.4 b_f$$

$$c_y (Max) = 0.3 h$$
- The longitudinal steel reinforcement that may be used in the calculation should not exceed 6 percent of the concrete area; and
- The ratio of the cross-section's depth h_c to width b_c , (see [Fig. 18](#)), should be within limits ($0.2 \leq h_c/b_c \leq 5.0$)

13.4 Local Buckling of Steel Sections

13.4.1 To prevent premature local buckling of structural steel components in partly encased steel section and concrete filled steel sections ([Fig. 18](#) and [Fig. 19](#)), the width to thickness ratio of individual elements of the steel sections in compression must satisfy the following limits:

$$\frac{d}{t} \leq 88\epsilon^2 \quad \text{for concrete-filled circular tubular sections}$$

$$\frac{h}{t} \leq 50\epsilon \quad \text{for concrete-filled rectangular tubular sections}$$

$$b_f/t_f \leq 43\epsilon \quad \text{for partially encased I sections}$$

where

$$\epsilon = \sqrt{\frac{250}{f_y}} \text{ and } f_y \text{ is the yield strength of the steel section, in MPa.}$$

13.4.2 For fully encased steel sections, the above local buckling check is not required. However, the concrete cover to the flange of a fully encased steel section should not be less than 40 mm, nor less than one-sixth of the breadth, b_f , of the flange.

13.4.3 Design of concrete-filled rectangular tubular sections where h/t ratios exceed the local buckling limits for semi-compact sections should be verified by tests.

13.4.4 The cover to reinforcement should be in accordance with as per good practice [C-6B(2)].

13.4.5 The steel section shall not be painted to ensure friction between steel and concrete but cleaned at the abutting surface to ensure protection against corrosion and spalling of concrete.

13.5 Design Compressive Resistance of Short Composite Columns

13.5.1 A column is termed as a 'short column' when one of the following conditions is satisfied.

- its non-dimensional slenderness ratio $\lambda \leq 0.2$; and
- If the actual design compressive force on the composite column, P , is less than $0.1 P_{cr}$.

where

$$\lambda = \sqrt{\frac{P_n}{P_{cr}}}$$

P_{cr} = the elastic buckling load of the column $= \frac{\pi^2 (EI)_e}{L^2}$;

$(EI)_e$ = effective elastic flexural stiffness of the composite column (13.5.2); and

L = the effective length of the column, which may be conservatively taken as the overall length L for an isolated non-sway composite column.

The design compressive strength P_n of short columns of plastic and compact sections may be evaluated using the design strength of the material with appropriate partial safety factors. The most common column sections are bi-symmetric. The design compressive strength of short bi-symmetric columns is evaluated using equations given in D-2.

NOTE — While providing spiral ties it must be designed as per of [C-6B(2)], and the ratio of the volume of helical reinforcement to the volume of the core is not less than $0.36 (A_g/A_{co}-1)f_{ck}/f_{yk}$.

(Where, A_g = Gross area of section; A_{co} = Area of core of the confined core of column measured to the outer diameter of the helix).

13.5.2 Effective Elastic Flexural Stiffness:

- Short Term Loading** — The effective elastic flexural stiffness, $(EI)_e$, is obtained by adding up the flexural stiffness of the individual components of the cross-section:

$$(EI)_e = E_s I_s + 0.6 E_{cm} I_c + E_{st} I_{st}$$

where

E_s and E_{st} = modulus of elasticity of the structural steel section and the reinforcement, respectively; and

E_{cm} = secant modulus of the concrete.

- Long Term Loading** — In slender columns ($\lambda > 0.2$), the effect of long-term loading should be considered.

$$(EI)_e = E_s I_s + 0.6 E_{ceff} I_c + E_{st} I_{st}$$

$$E_{ceff} = E_{cm} \frac{1}{1 + \left(\frac{P_{G,Ed}}{P_{Ed}} \right) \varphi_t}$$

where

φ_t = creep coefficient according to [C-6B(2)]

P_{Ed} = Total design axial load

$P_{G,Ed}$ = Part of the axial load P_{Ed} that is caused due to the permanent gravity loads

If the eccentricity 'e' of loading is more than twice the cross-section dimension 'D' ($e > 2D$), the effect on the bending moment distribution caused by increased deflections due to creep and shrinkage of concrete will be very small and may be neglected. The effect of long-term loading need not be considered if the non-dimensional slenderness (λ) of the composite column is less than the limiting values given in [Table 12](#).

Table 12 Limiting Values of λ or Long Term Loading

(Clause [13.1](#) and [13.5.2](#))

Sl No.	Cross-sections	Braced Non-sway Systems	Un-braced and/or Sway System
(1)	(2)	(3)	(4)
i)	Concrete encased cross-section	0.8	0.5
ii)	Concrete filled cross-section	$\frac{0.8}{1 - \delta}$	$\frac{0.5}{1 - \delta}$
NOTE — δ is the steel contribution ratio defined as $\delta = \frac{A_s f_y}{P_d \gamma_{mo}}$. P_d is defined in D-2.			

When λ exceeds the limits prescribed above and $e/D < 2$, the effect of creep and shrinkage of concrete should be considered by adopting the modulus of elasticity of concrete E_{cs} instead of E_{cm} where E_{cs} is defined as follows:

$$E_{cs} = 0.75E_{cm} \left[1 - \frac{0.5P_{dd}}{P} \right]$$

where

P = the applied factored axial load; and

P_{dd} = the part of the factored load permanently acting on the column.

The effect of long-term loading may be ignored for concrete filled tubular sections with $\lambda \leq 2.0$ provided that $\delta > 0.6$ for braced (non-sway) columns and $\delta > 0.75$ for unbraced (sway) columns.

13.6 Long Members Subjected to Axial Compression

The non-sway composite columns need not be checked for buckling if any one of the following conditions is satisfied:

- $P < 0.1P_{cr}$ where P is the axial force in the column, and P_{cr} is the elastic buckling load of the column, defined in [13.5.1](#).
- Non-dimensional slenderness $\lambda < 0.2$.

For columns not satisfying the above conditions, safety against buckling strength shall be checked about the corresponding axis. The following equation needs to be satisfied for buckling load:

$$P \leq \chi \cdot P_d$$

where

P = the applied factored axial load;

P_d = is the design plastic resistance to compression ([D-2](#));

$\chi = \frac{1}{\phi + \sqrt{\phi^2 - \lambda^2}}$ = is the stability reduction factor;

$\phi = 0.5[1 + \alpha(\lambda - 0.2) + \lambda^2]$;

λ = Non-dimensional slenderness ratio;

α = A value that represents initial imperfections and residual stresses which is based on the buckling curve to be adopted, which is further dependent on the type of section and the axis of bending as given below (see Table 13):

Curve a = for concrete-filled tubular sections with reinforcement percentage less than 3 percent of gross cross section area and buckling about both axes.

Curve b = for fully or partially concrete-encased I-sections buckling about the strong axis (z-z axis) of the steel sections and for concrete-filled tubular sections with steel reinforcement percentage more than 3 percent of gross cross section area and buckling about both axes or concrete filled tubular section with the addition of I section at the core and buckling about both axes.

Curve c = for fully and partially concrete-encased I-sections buckling about the weak axis (y-y axis) of the steel sections.

Table 13 Imperfection Factor α for the Buckling Curves

(Clause 13.6)

Sl No.	Buckling Class	a_0	a	b	c	d
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	α	0.13	0.21	0.34	0.49	0.76

13.7 Members Subjected to Combined Compression and Bending

When the bending moment in the section is zero (that is, $M = 0$), the design compressive strength is as given in 13.5. Similarly, the plastic moment of resistance of a column at zero compression load is as given D-3. The equations for evaluating the design bending resistance of symmetric column sections are given in D-4.

13.7.1 Bending Resistance under High Shear

While determining the plastic resistance when the simultaneous shear force is high at the section as given below, the bending resistance shall be reduced as given below:

If the shear force V on the steel section exceeds 60 percent of the design shear resistance V_d of the steel (8.3.6.1), the influence of the transverse shear on the resistance in combined bending and compression shall be taken into account by a reduced design steel strength $(1 - \beta)f_y/Y_{m0}$ in the shear area A_v (β is determined as per 8.3.6.2).

- a) Unless a more accurate analysis is used, the design shear action, V_d , may be distributed into (V_s , acting on the structural steel) and (V_c , acting on the reinforced concrete section) as:

$$V_s = V \frac{M_{ds}}{M_d}$$

$$V_c = V - V_s$$

where

M_{ds} = design moment of resistance of steel section alone;

M = design moment of resistance of the entire composite section; and

V_d = may be assumed to be resisted by steel sections alone as a simplifying approximation.

13.7.2 Second Order Effects on Bending Moment

If neither of the two conditions (a) and (b) for short compression members in 13.5.1 is satisfied, then the second-order effects on first-order analysis moment shall be considered by modifying the maximum first-order bending

moment (moment obtained linear elastic analysis), $M_{\max}$, with an amplification factor k , which is defined as:

$$k = \frac{C_m}{1 - \frac{P}{P_{cr}}} \geq 1.0$$

where

- P = the applied factored axial load;
- P_{cr} = elastic critical load for the relevant axis and corresponding to a modified effective flexural stiffness given by $(EI)_e$ as given in [13.5.2](#) and the effective length of the column is taken as per [C-6B(3)];
- C_m = equivalent moment factor given in [Table 14](#) to account for non-uniform bending moment over the length of the member; and
- ψ = the ratio of moments at the ends of the laterally unsupported members.

13.8 Members Subjected to Axial Force and Uniaxial Bending

13.8.1 While checking a section for combined axial force and bending moment, first, it should be ensured that the section is safe against an axial force acting alone, considering buckling along each principal axis. The resistance of the section shall then be checked for combined axial compression and uniaxial bending moment, as described below:

The design against combined bending and axial compression are adequate when the following condition is satisfied:

$$M'_d = \alpha_m \mu_d M_d$$

where

- α_m = reduction coefficient based on steel yield strength;
= 0.9 for $250 \leq f_y \leq 400$ MPa;
= 0.8 for $f_y > 400$ MPa;
- M'_d = design bending moment, which is to be amplified to account for the second-order effects as discussed in [13.7.2](#), when the column is not short; and
- M_d = Design bending resistance of the section about the bending axis evaluated as in [D-3](#).

The moment resistance reduction ratio μ_d for a long composite column under combined compression and uniaxial bending shall be evaluated as follows:

$$\mu_d = \frac{(\chi - \chi_d)}{(1 - \chi_c)\chi} \quad \text{when } \chi_d \geq \chi_c \text{ and}$$

$$\mu_d = 1 - \frac{(1 - \chi)\chi_d}{(1 - \chi_c)\chi} \quad \text{when } \chi_d < \chi_c$$

- χ_c = Axial resistance ratio to the concrete P_c/P_d ;
- χ_d = Design axial resistance ratio to the concrete P/P_d ;
- χ = reduction factor due to column buckling as per good practice [C-6B(3)];
- P_c = axial compressive strength of concrete;
= $A_c \frac{\alpha_{cc}}{\gamma_c} \cdot \eta \cdot \lambda \cdot f_{ck}$; and
- P_d = see [D-2](#).

13.8.2 Members Subjected to Axial Force and Bi-axial Bending

Members subjected to combined axial compression and biaxial bending shall satisfy the following interaction relationships equations:

$$\frac{M_z}{\mu_{dz}M_{dz}} + \frac{M_y}{\mu_{dy}M_{dy}} \leq 1$$

$$\frac{M_z}{\mu_{dz}M_{dz}} \leq \alpha_{mz}$$

$$\frac{M_y}{\mu_{dy}M_{dy}} \leq \alpha_{my}$$

where

μ_{dz} and μ_{dy} = moment resistance reduction factors in z and y directions, respectively. (see [13.8](#))

$\alpha_{mz} = \alpha_{my} = \alpha_m$ (see [13.8](#))

Table 14 Equivalent Uniform Moment Factor

(Clause [13.7.2](#))

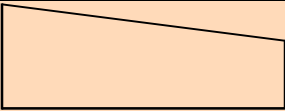
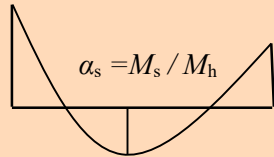
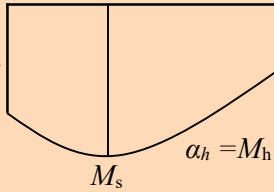
Bending Moment Diagram (1)	Range (2)		C_{my}, C_{mz}, C_{mLT}	
			Uniform Loading (3)	Concentrated Load (4)
	$-1 \leq \psi \leq 1$		$0.6 + 0.4 \psi \geq 0.4$	
	$0 \leq \alpha_s \leq 1$	$-1 \leq \psi \leq 1$	$0.2 + 0.8 \alpha_s \geq 0.4$	$0.2 + 0.8 \alpha_s \geq 0.4$
	$-1 \leq \alpha_s \leq 0$	$0 \leq \psi \leq 1$	$0.1 - 0.8 \alpha_s \geq 0.4$	$-0.8 \alpha_s \geq 0.4$
		$-1 \leq \psi \leq 0$	$0.1(1-\psi) - 0.8 \alpha_s \geq 0.4$	$0.2(-\psi) - 0.8 \alpha_s \geq 0.4$
	$0 \leq \alpha_h \leq 1$	$-1 \leq \psi \leq 1$	$0.95 - 0.05 \alpha_h$	$0.90 + 0.10 \alpha_h$
	$-1 \leq \alpha_h \leq 0$	$0 \leq \psi \leq 1$	$0.95 + 0.05 \alpha_h$	$0.90 + 0.10 \alpha_h$
		$-1 \leq \psi \leq 0$	$0.95 + 0.05 \alpha_h (1 + 2 \psi)$	$0.90 + 0.1 \alpha_h (1 + 2 \psi)$
For members with sway buckling mode, the equivalent uniform moment factor $C_{my} = C_{mz} = 0.9$.				

Table 14 (Concluded)

Bending Moment Diagram (1)	Range (2)	C_{my}, C_{mz}, C_{mLT}	
		Uniform Loading (3)	Concentrated Load (4)
C_{my}, C_{mz}, C_{mLT} shall be obtained according to the bending moment diagram between the relevant braced points			
Moment factor	Bending axis	Points braced in direction	
C_{my}	y-y	z-z	
C_{mz}	z-z	y-y	
C_{mLT}	z-z	z-z	

The diagram illustrates a beam with four braced points, represented by circles with diagonal lines. Below the beam, two bending moment diagrams are shown. The first diagram, labeled M_y for C_{my} , shows a linear variation of moment between braced points. The second diagram, labeled M_z for C_{mz} and C_{mLT} , shows a parabolic variation of moment between braced points.

13.9 Mechanical Shear Connection and Load Introduction

Proper sharing of loads between the steel section and concrete of composite columns should be ensured at points of load introduction due to load and moment reactions coming from members connected to the ends of the column and also for axial loads applied anywhere within the length of the column, considering the shear resistance at the interface between steel and concrete.

Where composite columns and compression members are subjected to significant transverse shear, for example, by local transverse loads and by end moments, the provision shall be made for the transfer of the corresponding local longitudinal shear stress at the interface between steel and concrete. For axially loaded columns and compression members, longitudinal shear outside the region of load introduction need not be considered.

13.9.1 Load Introduction

Shear connectors should be provided at regions of load introduction and at regions with change in cross section, if the design shear resistance at the concrete-steel interface, τ , ([Table 15](#)) is exceeded. The shear forces should be determined from the change of sectional forces of the steel or reinforced concrete section within the introduction length. If the loads are introduced into the concrete cross section only, the values resulting from an elastic analysis considering creep and shrinkage should be taken into account. Otherwise, the forces at the interface should be determined by elastic theory or plastic theory to determine the more severe case. In the absence of any accurate method, the introduction length should not exceed $2d$ or $L/3$, where d is the minimum transverse dimension of the column and L is the column length.

Table 15 Design Shear Strength (τ)

(Clauses [13.2](#), [13.9.1](#) and [13.9.3](#))

Sl No.	Type of Cross-section	Design Shear Strength τ (N/mm ²)
(1)	(2)	(3)
i)	Completely concrete-encased steel sections	0.30
ii)	Concrete filled circular hollow sections	0.55
iii)	Concrete filled rectangular hollow sections	0.40
iv)	Flanges of partially encased sections	0.20
v)	Webs of partially encased columns	0.00

Due to the action of creep and shrinkage, no shear connection is required for composite columns or compression members if the load application is by endplate, where the full interface between steel and concrete is permanently under compression. Otherwise, the load application/introduction should be verified as given below:

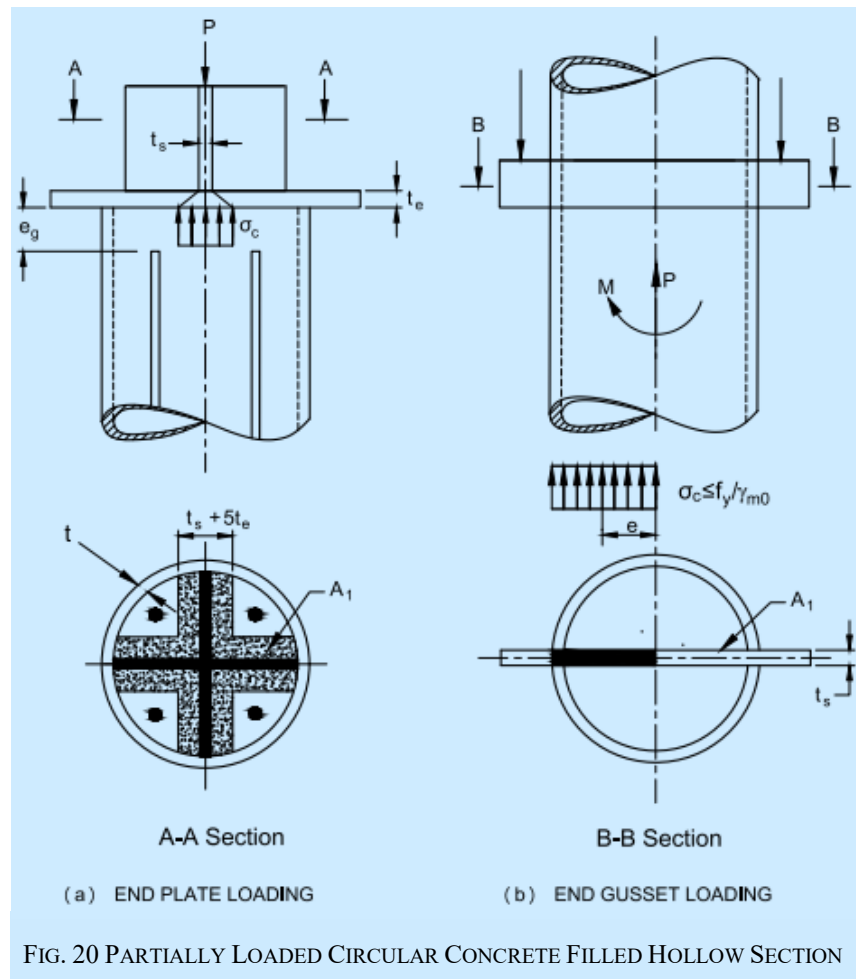
- If the cross-section is partially loaded, as shown in [Fig. 20 \(a\)](#), the loads may be distributed with a ratio of 1 : 2.5 over the thickness, t_e of the endplate. The concrete stresses should then be limited in the area of effective load introduction.
- For concrete-filled circular hollow section or square hollow section, under partial loading as shown in [Fig. 20\(b\)](#), for example, by gusset plates or by stiffeners, the local design strength of concrete σ_c under the gusset or stiffener resulting from the sectional forces of the concrete section shall be determined as:

$$\sigma_c = \frac{\alpha_{cc} f_{ck}}{\gamma_c} \left[1 + \eta_0 \frac{t}{d} \frac{f_y}{\alpha_{cc} f_{ck}} \right] \sqrt{\frac{A_c}{A_1}} \leq \frac{A_c \cdot \alpha_{cc} f_{ck}}{A_1 \cdot \gamma_c}; \leq \frac{f_y}{\gamma_{m0}}$$

where

$$A_c/A_1 \leq 20$$

- t = is the wall thickness of the steel tube;
- d = diameter of the tube or width of the square section;
- A_c = is the cross sectional area of the concrete section of the column;
- A_1 = is the loaded area under the gusset plate (*see* [Fig. 20](#)); and
- η_0 = 4.9 for circular steel tubes and 3.5 for square sections.



- c) For concrete-filled circular hollow sections, longitudinal reinforcements may be taken into account while determining the resistance of the composite column, even if the reinforcement is not directly connected to the endplate, provided that the gap [Fig. 20(a)] between the end of reinforcement and the surface of the endplate does not exceed 30mm.

13.9.2 Shear connection

When a mechanical connection is introduced in the form of stud connectors to the web of a fully or partially concrete-encased steel I-section, an account may be taken of the frictional forces that develop due to the restraint in lateral expansion of the concrete by the adjacent steel flanges. This resistance is assumed to be equal to $\mu \cdot Q_u/2$ on each flange and each horizontal row of studs, as shown in Fig. 21 and may be added to the calculated resistance of the shear connectors. μ is the relevant coefficient of friction and be taken as 0.5. Q_u is the resistance of a single stud as per 11.2.

The clear distance between the flanges should not exceed the values given in Fig. 21 to ensure the development of the frictional forces between concrete and steel flanges.

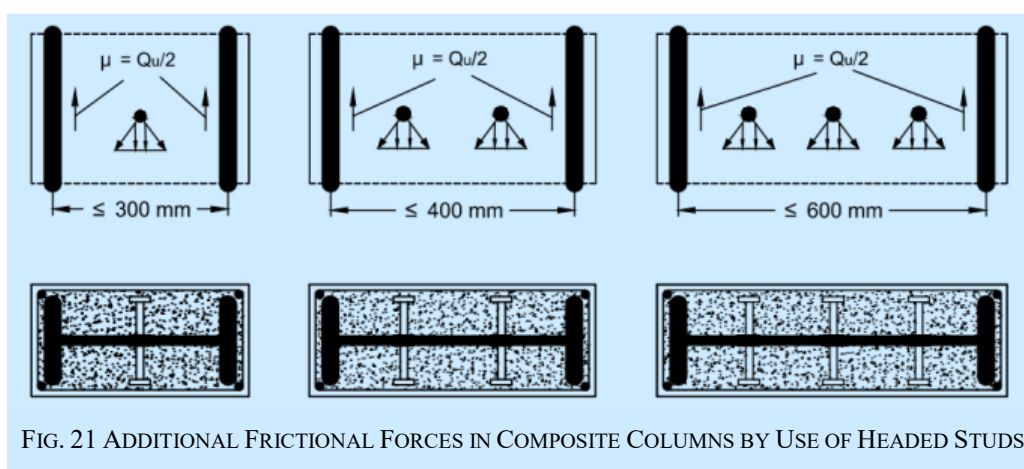


FIG. 21 ADDITIONAL FRICTIONAL FORCES IN COMPOSITE COLUMNS BY USE OF HEADED STUDS

13.9.3 Longitudinal Shear Outside Area of Load Introduction

- a) Outside the area of load introduction, longitudinal shear at the interface between concrete and steel should be verified where it is caused by transverse loads and /or end moments. Shear connectors should be provided based on the distribution of the design value of longitudinal shear, where this exceeds the design shear resistance τ . In the absence of a more accurate method, elastic analysis, considering long-term effects and cracking of the concrete, may be used to determine the longitudinal shear at the interface. Provided that the surface of the steel section, in contact with the concrete, is unpainted and free from oil, grease, and loose scale or rust, the values given in Table 15 may be assumed for τ . The value of τ given in Table 15 for fully concrete-encased steel sections applies to sections with a minimum concrete cover of 40 mm. For greater concrete cover and adequate reinforcement, higher values of τ may be used. Unless verified by tests, for completely encased sections, the increased value $\beta_c \cdot \tau$ may be used, with β_c given by:

$$\beta_c = 1 + 0.02 c_z \left[1 - \frac{c_{z,min}}{c_z} \right] \leq 2.5$$

where

c_z = is the wall thickness of the steel tube; and

$c_{z,min}$ = diameter of the tube or width of the square section.

- b) Unless otherwise verified, for partially encased I-sections with transverse shear due to bending about the weak axis caused by lateral loading or end moments, shear connectors should always be provided. If the resistance of the structural steel section alone against transverse shear is not sufficient to take care of the total transverse shear on the composite section, then the required transverse reinforcement for the shear force, V_c according to 12.5.2, should be welded to the web of the steel section or should pass through the web of the steel section.

13.10 Shear Check

The factored shear force in the compression members should be less than the design shear strength of the member, which is the sum total of the shear resistance given by the concrete section along with steel reinforcements as per good practice [C-6B(2)] and the shear resistance given by the steel section as per [8.3.6](#). The shear force shall be distributed between the steel section and the concrete section in accordance with [13.9](#).

14 CONSTRUCTION AND ERECTION

14.1 Fabrication and Inspection Procedure for Structural Steel

Fabrication and erection of steel sections and components, as and where situated in a composite structure would include fabrication procedures, both shop and site fabrications, along with fabrication tolerances, inspection, testing, handling, transportation, site storage, erection along with erection tolerances, etc.

Fabrication and erection specifications of all steel components of a composite structure shall refer to stipulations laid down in good practice [C-6B(3)].

15 FIRE RESISTANCE AND FIRE DESIGN

15.1 In closed structures, an accidental fire may lead to a rise in temperature under which failure of the material may take place unless a proper design and construction against fire is carried out. Open structures such as bridges are not generally vulnerable to failure under fire since the temperature does not go up to the level which may cause material damage. Also, in an open structure, the fire can be extinguished easily and quickly. Fire tests on open structures such as elevated parking lots and bridges have shown that the structure does not undergo any material damage due to reasons indicated above. However, all structures, including both closed and open ones, shall be protected from all possible accidental fires caused by different kinds of hazards.

Fire-resistant designs for open and closed structures at specialized locations, such as those in proximity to oil installations or pipelines carrying inflammable materials, etc as in the case of industrial buildings and structures, shall be done based on recommendations given in specialized literature. Also, adequate provisions may be made for firefighting equipment to access all parts of the structure as per Part F 'Fire and Life Safety' of this NBCS. After the occurrence of fire in a structure, it should be mandatory for the concerned authorities to have the structure inspected by competent experts in order to ascertain the condition of the structure before it can be declared safe for re-use.

In addition to the above, locations in any structural system that may be prone to accidental occurrence of fire, shall be adequately provided with basic fire protection methods as per specialist literature. These will include both active as well as passive fire protection.

15.2 Response to Fire

Steel is a good conductor of heat that experiences an almost uniform temperature increase over its entire volume. Concrete is a poor conductor of heat has a large temperature gradient within its volume. Due to this, the entire steel member gains the heat faster, whereas the surfaces of concrete exposed directly to heat experience high temperature, with the temperature dropping off drastically in the interior of the concrete mass.

Both steel and concrete exhibit a drastic reduction in strength with an increase in temperature above 400 °C. Due to this, the resistance of steel members tends to decrease drastically above 400 °C, whereas concrete at the surfaces exposed to high temperature tends to spall off due to high temperature gradient. The steel structure, if it does not fail under fire, regains its strength after the fire, the damage at the surface of the concrete is permanent due to chemical changes. The steel reinforcement in concrete is protected from loss of strength due to an increase in temperature by providing adequate concrete cover as per good practice [C-6B(2)]. Composite floor system deck sheeting is to be provided with fire protection.

However, literature shows that the minimum necessary fire resistance can be achieved by using the composite floor systems and the composite columns, without applying fire protection. Interaction of the steel beam and the concrete slab in the composite flooring leads to a higher critical temperature and higher fire resistance.

The requirements for fire resistance shall apply to steel elements of steel-concrete composite structure designed to exhibit a required fire resistant level as per the relevant specifications.

15.2.1 Design Fire

The design of a structure against fire load is dependent on the required fire resistant level (FRL), which is dependent on the function of the structure itself and the period of structural adequacy (PSA). These shall be calculated based on the stipulation laid down in as per good practice [C-6B(3)].

The response of steel elements of a composite structure against fire as laid down in as per good practice [C-6B(3)] shall be binding on all steel elements of the composite structures which are susceptible to rise in temperature during its design life. For all general buildings, an adequate fire protection methodology shall be adopted.

15.2.2 Fire Protection Methodology

Apart from the direct design of steel components against fire as required in as per good practice [C-6B(3)] (protected as well as unprotected section), other protective measures, (both active and passive), also may be adopted as recommended in literature as fire-resistant procedures.

For active fire resistance, provisions of fire locating and fighting measures like the smoke detectors, fire extinguishers inside a building along with accessory fire water supply, sprinkler system, etc, shall be made available at vantage points. The planning of the structure shall be made in such a way that it is accessible from all sides to fire extinguishing vehicles.

For passive fire resistance, protective paints and materials like intumescent paints, vermiculite boards etc, may be used on exposed steel surfaces, and their provision shall be made as per the required fire-resistant level and as per their properties and specification provided by the manufacturers. The design of protected sections shall be done as per the stipulations laid down in the relevant section of as per good practice [C-6B(3)]. Detailed design against the fire may be based on provisions of other composite codes and specialized literature.

15.2.3 Fire Resistance of Composite Slabs

The fire resistance is assumed based on the following two criteria:

- a) Thermal insulation criterion, concerned with limiting the transmission of heat by conduction; and
- b) Integrity criterion is concerned with preventing the flames and hot gases from nearby compartments. It is met by specifying an adequate thickness of insulation.

16 TESTING METHODS

For testing methods, [Annex E](#) may be referred.

ANNEX A

(Clause 4)

SYMBOLS

A-1 The letter symbols and notations given below apply to the provision of this section:

a	Span length of the slab transverse to the steel beams
A_s	Area of structural steel cross section
A_{sl}	Area of structural steel cross section in tension
A_c	Gross area of concrete
A_{co}	Equivalent area of the cracked composite beam section
A_{ec}	Area of concrete effective in compression
A_p	Effective cross-sectional area of profile steel sheeting
A_{st}	Area of steel reinforcements
A_f	Area of each flange of steel section
A_e	Effective cross sectional area
A_{sv}	Area of shear resistance
B	Centre to centre distance between beams and is equal to the transverse span of inner beam
b_c	Width of concrete encasement
b_p	Distance between the centres of adjacent ribs or one wave length of a profile deck
b_e	Effective width of the flange between pair of bolts
b_{eff}	Effective width of concrete flange
b_f	Width of the structural steel flange
D	Overall depth of girder/diameter of the steel cross section.
d	Depth of the web, Nominal diameter of bolts/rivets/studs.
d_2	Twice the clear distance from the compression flange of angles, plates, or tongue plates to the neutral axis
d_c	Vertical distance between the centroid of concrete slab and centroid of steel beam
d_s	Overall depth of the concrete slab
d_o	Nominal diameter of the pipe column or the dimensions of the column in the direction of depth of the base plate
d_p	Depth of profiled sheet deck
d_{sp}	Overall depth of the profiled sheet deck and concrete slab
e	Distance between the C.G. axis of the profiled steel sheeting and the extreme tension fiber of the composite slab
e_p	Distance between the plastic neutral axis and the extreme tension fiber of the composite slab
$E_{c,l}$	Modulus of elasticity of lightweight concrete
E_s	Modulus of elasticity of structural steel
E_{cm}	Secant Modulus of elasticity of concrete
E_{st}	Modulus of elasticity of steel reinforcements
F_f	Axial capacity of a single flange
F_{sr}	Axial capacity of reinforcing steel

F_{cc}	Compressive force in the concrete above the neutral axis
F_{sc}	Compressive force in the steel section above the neutral axis
F_{st}	Tensile force in the steel section below the neutral axis
F_w	Design capacity of the web in bearing
f	Actual normal stress range for the detail category
f_c	Actual axial compressive stress in concrete at service load
$f_{ctm, l}$	Tensile strength of lightweight concrete
f_{ck}	Characteristic compressive cube strength of concrete at 28 days
$f_{ctk(0.05)}$	Characteristic axial tensile strength of concrete based on 5 percent fractile
f_{ctm}	Mean axial tensile strength of concrete
f_f	Fatigue stress range corresponding to 5×10^6 cycles of loading
f_{fd}	Design fatigue normal stress range
f_{feq}	Equivalent constant amplitude stress
$f_{f \max}$	Highest normal stress range
f_{fn}	Normal fatigue stress range
f_o	Proof stress
f_{yk}	Characteristic yield strength of reinforcement
f_{yf}, f_{yw}	Yield strength of flange and web, respectively
f_u	Characteristic ultimate tensile stress
f_{up}	Characteristic ultimate tensile stress of the profiled deck
f_y	Characteristic yield stress of structural steel
f_{yp}	Characteristic yield stress of profiled deck
h	Depth of the section/total depth of steel beam
H	Distance between top of concrete and bottom of bottom flange of steel beam
h_s	Distance between centroids of top and bottom flanges /nominal height of stud
h_y	Distance between shear centre of the two flanges of the cross section
I	Moment of inertia of the member about an axis perpendicular to the plane of the frame
I_c	Moment of inertia of concrete (assumed uncracked) about the axis of bending for column
I_{co}	Moment of inertia of the composite section
$I_{fc,y}$	Moment of Inertia of the compression flange about the minor axis of the steel beam
$I_{ft,y}$	Moment of Inertia of the tension flange about the minor axis of the steel beam
I_s	Moment of inertia of the steel section about the axis of bending for column
I_{st}	Moment of inertia of reinforcement about the axis of bending for column
I_z	Moment of inertia about the major axis
I_{ys}, I_{zs}	Second moment of area of steel section about the minor and major axis respectively
I_{ybf}	Moment of inertia of the bottom flange about the minor axis of the steel section
I_{zco}	Second moment of the equivalent area of the cracked composite beam section for major axis bending
k_1	Flexural stiffness of the steel web per unit length along the beam
k_2	Flexural stiffness of the cracked concrete or composite slab transverse to the spanning direction of steel beam

KL	Effective length of the member
KL/r	Appropriate, effective slenderness ratio of the section
KL/r_y	Effective slenderness ratio of the section about the minor axis
KL/r_z	Effective slenderness ratio of the section about the major axis
L	Actual span of composite beams
L_{bs}	Minimum bearing lengths of steel decking on the support
L_{bc}	Composite slab including the cast in place concrete
L_c	Effective span of the cantilever for overhang
L_o	Length between points of zero moments (inflection) in the span
L_s	Shear span
L_x	Distance of the maximum moment cross section under consideration to the support
M	Bending moment
M_v	Reduced bending moment due to effect of shear force
M_{cr}	Elastic critical moment corresponding to lateral torsional buckling
M_d	Design bending strength/design bending resistance under only bending moment
M'_d	Design bending resistance under combined bending and compression
M_{dp}	Design bending resistance of profiled steel sheeting
M_e	Elastic moment capacity of the section
M_f	Design plastic resistance of the flange alone for steel section
M_p	Plastic moment capacity of the section
M_y	Factored applied moments about the minor axis of the cross section
M_z	Factored applied moments about the major axis of the cross section
m	Modular ratio
m_{dl}	Modular ratio (long term)
m_{ll}	Modular ratio (short term)
N_{SC}	Number of stress cycles
P	Design axial force
P_{cr}	Elastic critical buckling load
P_p	Plastic resistance of encased steel column section or concrete filled rectangular or square column section
R_h	Flange stress reduction factor for hybrid section
r	Appropriate radius of gyration
r_y	Radius of gyration about the minor axis
r_z	Radius of gyration about the major axis
S	Spacing
S_l	Spacing of shear connectors for longitudinal shear due to flexural force
S_r	Spacing of shear connectors due to bending moment
t	Thickness of element/angle, time in minutes
t_f	Thickness of flange of steel section
t_p	Thickness of profiled deck sheet

t_w	Thickness of web of steel section,
V, V_v, V_L	Factored applied shear force
V_d	Design shear resistance
V_{dvd}	Vertical shear resistance
V_p	Plastic shear resistance under pure shear
V_{vd}	Design vertical shear resistance
W	Total equivalent load
X	Distance from the centre line of edge beam to the edge of slab
x_e	Depth of elastic neutral axis of the composite section from the centroid of steel section
x_u	Depth of neutral axis at limit state of flexure from top of the concrete
Z_e	Elastic section modulus
Z_p	Plastic section modulus
Z_{pc}, Z_{pcn}	Plastic section modulus of concrete about its own centroid and about the neutral axis of the composite section, respectively
Z_{pr}, Z_{prn}	Plastic section modulus of reinforcement about its own centroid and about the neutral axis of the composite section, respectively
Z_{ps}, Z_{psn}	Plastic section modulus of structural steel section about its own centroid and about the neutral axis of the composite section, respectively
y_g	Distance between the point of application of the load and shear centre of the cross section
α	Imperfection factor
α_c	Strength coefficient of concrete
δ	Steel contribution ratio
χ	Stress reduction factor due to buckling under compression
χ_m	Stress reduction factor, χ at f_{ym}
χ_m	Stress reduction factor to account for lateral torsional buckling of a beam
γ	Unit weight of steel
γ_c	Partial safety factor for material (concrete)
γ_f	Partial safety factor for load
γ_{m0}	Partial safety factor against yield stress and buckling (structural steel)
γ_{m1}	Partial safety factor against ultimate stress (structural steel)
γ_{mf}	Partial safety factor for the strength of bolt
γ_{ft}	Partial safety factor for fatigue load
γ_{mft}	Partial safety factor for fatigue strength
γ_{mv}	Partial safety factor against shear failure
γ_{mw}	Partial safety factor for the strength of the weld
γ_k	Partial safety factor for material (reinforcements)
ε	Yield stress ratio, $(250/f_y)^{1/2}$
λ, λ_r	Non dimensional slenderness ratio $= \sqrt{f_y(KL/r)^2/\pi^2 E} = \sqrt{f_y/f_{cr}} = \sqrt{P_y/P_{cr}}$
λ_e	Equivalent slenderness ratio
μ_s	Poisson's ratio of structural steel

μ_c	Correction factor
η_1, η_2	Strength coefficients
η_{10}, η_{20}	Strength coefficients
τ	Actual shear stress range for the detail category
τ_f	Fatigue shear stress range
τ_{fd}	Design fatigue shear stress range
$\tau_{f,max}$	Highest shear stress range
τ_{fn}	Fatigue shear stress range at N_{SC} cycle for the detail category

ANNEX B

(Clauses [8.1.1](#), [8.1.2](#), [8.3.5.2](#), [8.3.5.3](#) and [8.3.7](#))

BENDING RESISTANCE OF COMPOSITE BEAMS

The stress distribution and the methods of evaluation of the sagging and hogging bending resistances under full and partial shear interaction of composite beams are discussed in this Annex.

B-1 SAGGING MOMENT RESISTANCE OF COMPOSITE SECTION WITH PLASTIC OR COMPACT STRUCTURAL STEEL BEAMS

B-1.1 Sagging Bending Resistance with Full Shear Interaction

The equations for calculating the bending resistance of composite beams are presented in this section. It is assumed that sufficient shear connectors are provided between the structural steel and concrete to develop full shear with no-slip during the service load.

Initially, evaluating the resistance is based on parabolic stress block as per good practice [C-6B(2)], over the entire depth of concrete in compression is presented. This is theoretically more accurate than the rectangular stress block (maximum of 2-3 percent difference), but computationally more tedious, particularly when it comes to sections subjected to combined bending and compression.

Subsequently, equations based on rectangular stress block for concrete in compression is presented. This is assumed so that the calculations are simplified. This method is easy to use, particularly when the neutral axis is in or closer to compression flange of the steel section.

In both the above methods, the stress in the reinforcing steel in the concrete slab under compression is disregarded.

B-1.1.1 Based on Parabolic Stress Block Satisfying Stress Strain Relationship

Calculation of bending resistance is based on parabolic stress block, which is compatible with strain diagram. The design bending resistance depends on the location of the plastic neutral axis. [Table 16](#) gives the various cases of design bending moments of a composite section depending upon the location of the plastic neutral axis as shown in [Fig. 22](#), [Fig. 23](#) and [Fig. 24](#). For hybrid sections, appropriate yield strength in flanges and web shall be considered for calculation of the plastic moments.

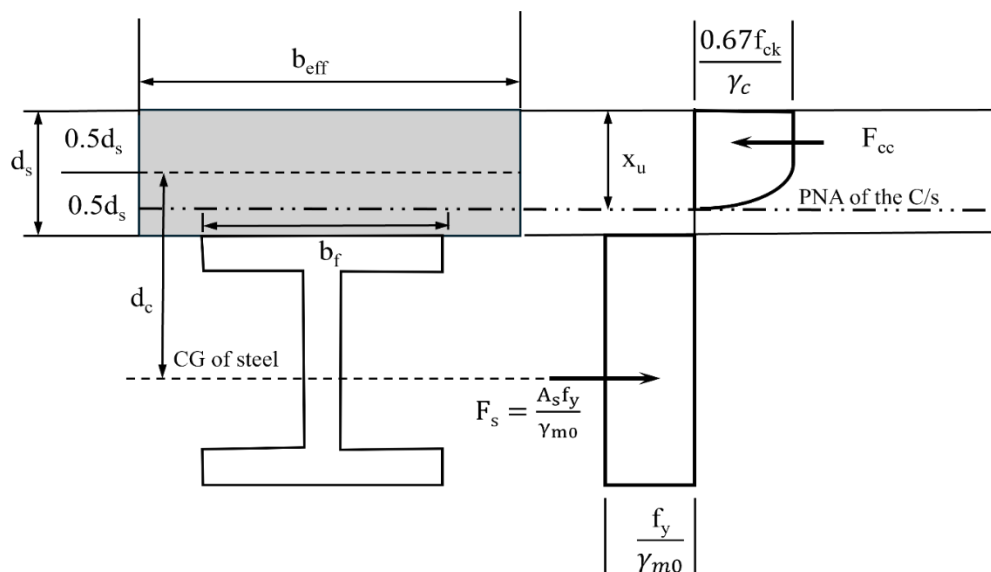


FIG. 22 STRESS DISTRIBUTION IN A COMPOSITE BEAM WITH PLASTIC NEUTRAL AXIS WITHIN THE CONCRETE SLAB AT ULTIMATE MOMENT

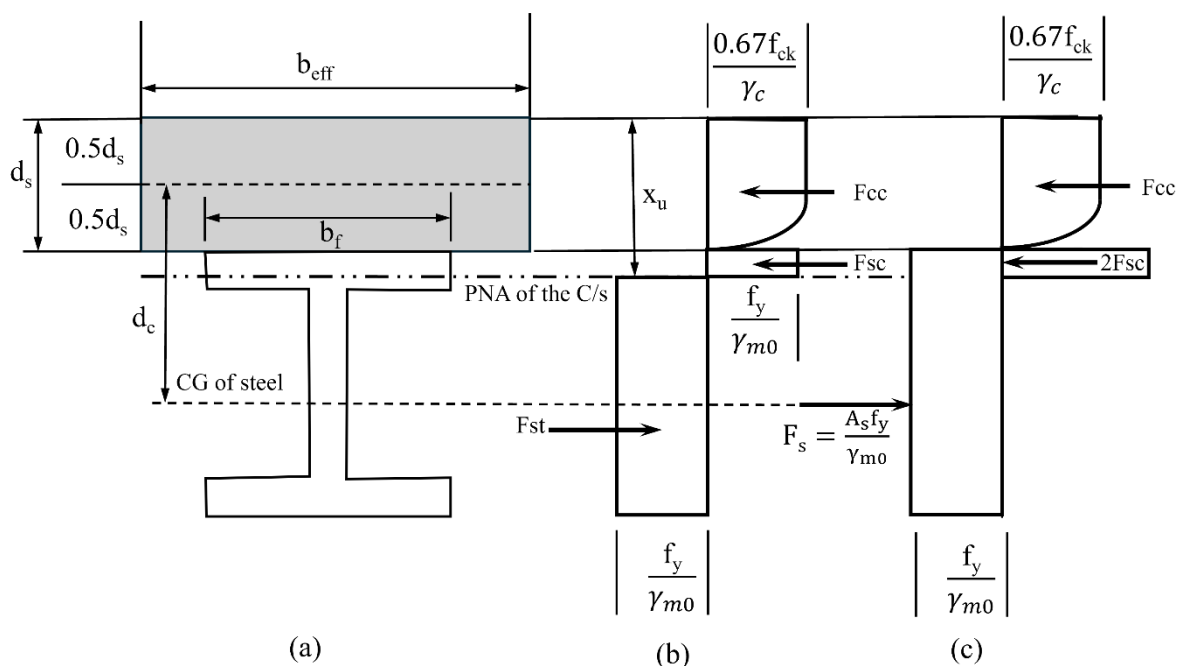


FIG. 23 STRESS DISTRIBUTION IN A COMPOSITE BEAM WITH PLASTIC NEUTRAL AXIS WITHIN THE FLANGE OF STEEL BEAM AT ULTIMATE MOMENT

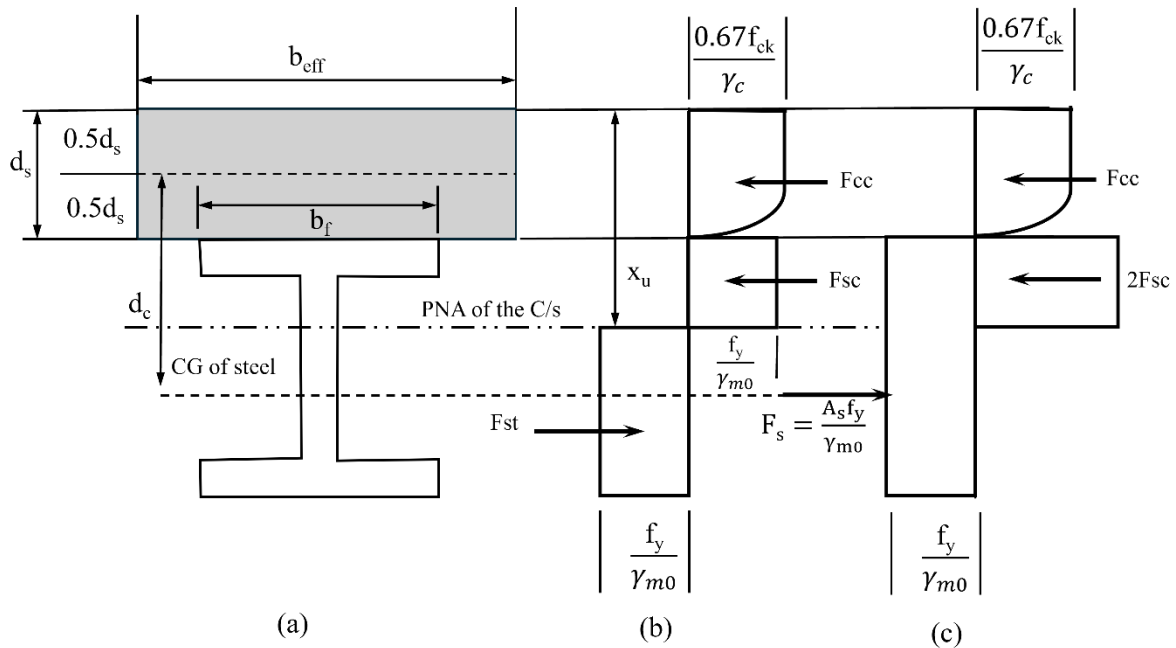


FIG. 24 STRESS DISTRIBUTION IN COMPOSITE BEAM WITH PLASTIC NEUTRAL AXIS WITHIN THE WEB OF STEEL BEAM AT ULTIMATE MOMENT

Table 16 Sagging Moment Resistance of Composite Section with full Shear Interaction (Parabolic Stress Block)

(Clause B-1.1.1)

Case	Position of Plastic Neutral Axis	Location of PNA x_u	Design Moment Capacity M_d
(1)	(2)	(3)	(4)
1	Within slab (Fig. 22) $b_{eff} d_s > a A_s$	$x_u = a A_s / b_{eff}$	$M_d = A_s (f_y / \gamma_{m0}) (d_c + 0.5 d_s - 0.42 x_u)$
2	In steel flange (Fig. 23) $b_{eff} d_s < a A_s < (b_{eff} d_s + 2a A_f)$	$x_u = d_s + \frac{(a A_s - b_{eff} d_s)}{2 b_f a}$	$M_d = (f_y / \gamma_{m0}) [\{ A_s (d_c + 0.08 d_s) \} - \{ b_f (x_u - d_s) (x_u + 0.16 d_s) \}]$
3	In web (Fig. 24) $a A_s > (b_{eff} d_s + 2a A_f)$	$x_u = d_s + t_f + \frac{a(A_s - 2A_f) - b_{eff} d_s}{2 a t_w}$	$M_d = (f_y / \gamma_{m0}) [\{ A_s (d_c + 0.08 d_s) \} - \{ 2 A_f (0.5 t_f + 0.58 d_s) \} - \{ 2 t_w (x_u - d_s - t_f) (0.5 x_u + 0.08 d_s + 0.5 t_f) \}]$

where $a = \frac{f_y}{0.36 f_{ck} \gamma_{m0}}$ and d_s is taken as per 8.1.2.1

B-1.1.2 Based on Rectangular Stress Block

The design bending resistance depends on the location of the plastic neutral axis, as shown in Fig. 25, Fig. 26 and Fig. 27. Table 17 gives the equations for evaluating the bending resistance of a composite section, depending upon the location of the plastic neutral axis. For hybrid sections appropriate yield strength in flanges and web shall be considered for calculation of the plastic moments.

Fig. 26 (b) and 27 (b), show the actual stress distribution and Fig. 26 (c) and 27 (c), show the equivalent stress distribution to simplify the equations.

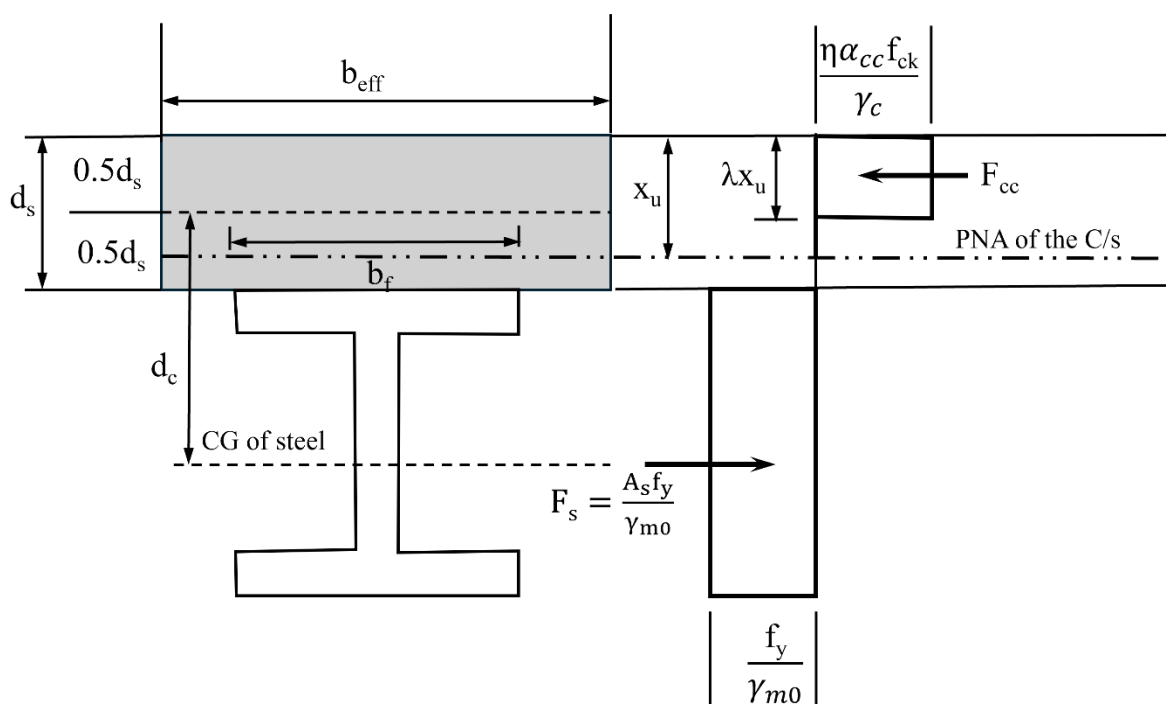


FIG. 25 STRESS DISTRIBUTION IN A COMPOSITE BEAM WITH PLASTIC NEUTRAL AXIS WITHIN CONCRETE SLAB AT ULTIMATE MOMENT

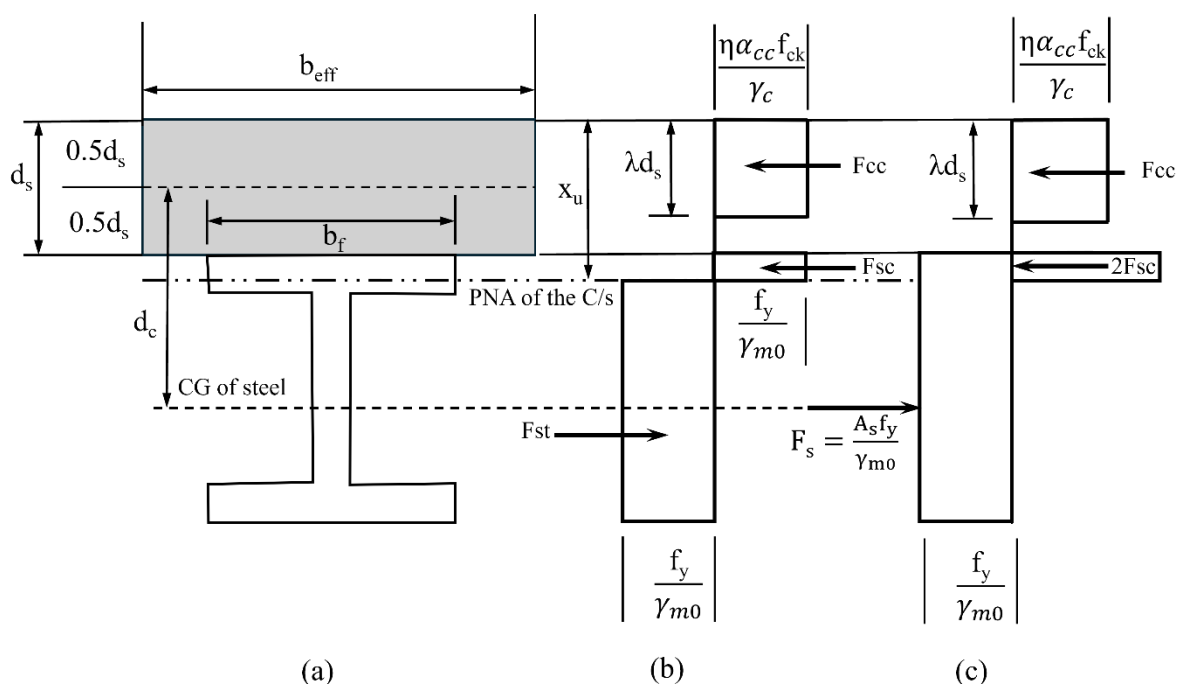


FIG. 26 STRESS DISTRIBUTION IN A COMPOSITE BEAM WITH PLASTIC NEUTRAL AXIS WITHIN THE FLANGE OF THE STEEL BEAM AT ULTIMATE MOMENT

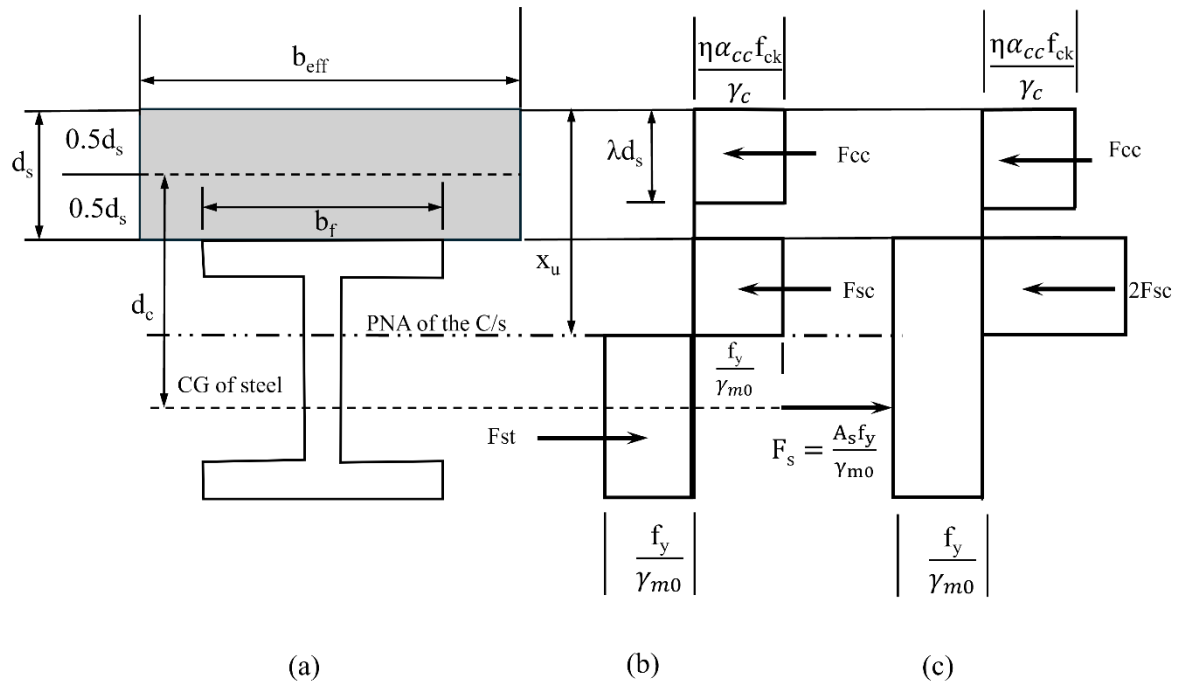


FIG. 27 STRESS DISTRIBUTION IN A COMPOSITE BEAM WITH PLASTIC NEUTRAL AXIS WITHIN THE WEB OF THE STEEL BEAM AT ULTIMATE MOMENT

Table 17 Sagging Moment Resistance of Composite Section with Full Shear Interaction (Rectangular Stress Block)

(Clause [B-1.1.2](#))

Case	Position of Plastic Neutral Axis	Location of PNA x_u	Design Moment Capacity, M_d
(1)	(2)	(3)	(4)
1	Within slab (Fig. 25) $b_{eff} d_s > a A_s$	$x_u = a A_s / b_{eff}$	$M_d = A_s f_y / \gamma_{m0} (d_c + 0.5 d_s - \lambda x_u / 2)$
2	Plastic neutral axis in steel flange (Fig. 26) $b_{eff} d_s < a A_s < (b_{eff} d_s + 2 a A_f)$	$x_u = d_s + \frac{(a A_s - b_{eff} d_s)}{2 b_f a}$	$M_d = f_y / \gamma_{m0} [A_s \{d_c + 0.5 d_s (1 - \lambda)\} - b_f (x_u - d_s) \{x_u + (1 - \lambda) d_s\}]$
3	Plastic neutral axis in web (Fig. 27) $b_{eff} d_s + 2 a A_f < a A_s$	$x_u = d_s + t_f + \frac{a(A_s - 2A_f) - b_{eff}d_s}{2at_w}$	$M_d = f_y / \gamma_{m0} [A_s \{d_c + 0.5 d_s (1 - \lambda)\} - 2A_f \{0.5 t_f + (1 - \lambda/2) d_s\} - t_w (x_u - d_s - t_f) \{x_u + (1 - \lambda) d_s + t_f\}]$

where

$$a = \frac{f_y}{\frac{\alpha_{cc}}{\gamma_c} \eta \lambda (f_{ck})} \text{ and } d_s \text{ is taken as per } \text{8.1.2.1}.$$

Table 17 (Concluded)

Case	Position of Plastic Neutral Axis	Location of PNA x_u	Design Moment Capacity, M_d
(1)	(2)	(3)	(4)
NOTE — Notations used in the determination of plastic bending resistance A_f = area of the top flange of steel beam of a composite section. A_s = cross sectional area of structural steel beam of a composite section. b_{eff} = effective width of concrete slab. b_f = width of top flange of steel section. d_s = overall depth of concrete slab. d_c = vertical distance between centroids of concrete slabs and steel beam in a composite section. t_f = average thickness of the top flange of the steel section. t_w = thickness of the web of the steel section. x_u = depth of neutral axis at ultimate limit state of flexure from top of concrete. M_d = design bending resistance. α_{cc} = 0.67. γ_c = material safety factor for concrete . = 1.50 (for basic and seismic combinations). = 1.0 (for accidental combinations). γ_{m0} = material safety factor for structural steel = 1.10 γ_k = material safety factor for reinforcing steel = 1.15 η = 1.0 [for $f_{ck} \leq 60$ MPa] = $1.0 - (f_{ck} - 60) / 250$ [for $60 < f_{ck} \leq 110$ MPa] λ = 0.8 [for $f_{ck} \leq 60$ MPa] = $0.8 - (f_{ck} - 60) / 500$ [for $60 < f_{ck} \leq 110$ MPa]			

B-1.2 Bending Moment with Partial Shear Interaction

B-1.2.1 Partial shear connection is applicable only in the case of plastic resistance of sections with plastic and compact classifications. Provisions for partial shear connection is applicable either for attaining economy without losing much in moment capacity of the composite section or in conditions where the number of shear connectors required for full shear interactions cannot be provided due to lack of space or increased spacing of shear connectors due to provision of lesser numbers of shear connectors than that needed for full composite action.

Degree of shear connection S_c is given as

$$S_c = \frac{n_p}{n_f} = \frac{F_{cp}}{F_{cf}} = \frac{M - M_{ds}}{M_d - M_{ds}}$$

where

- n_p = Number of shear connectors provided for partial shear connection;
- n_f = Number of shear connectors required for full shear connection;
- F_{cp} = Capacity of shear connectors in partial shear connection with n_p numbers of connectors;
- F_{cf} = Capacity of shear connectors in full shear connection with n_f numbers of connectors;
- M = Required bending resistance of the section;
- M_d = Design bending resistance of the composite section; and
- M_{ds} = Design bending resistance of steel section alone.

B-1.2.2 To obtain required bending resistance M , the number of shear connectors required (assuming all connectors have equal capacity) is given as:

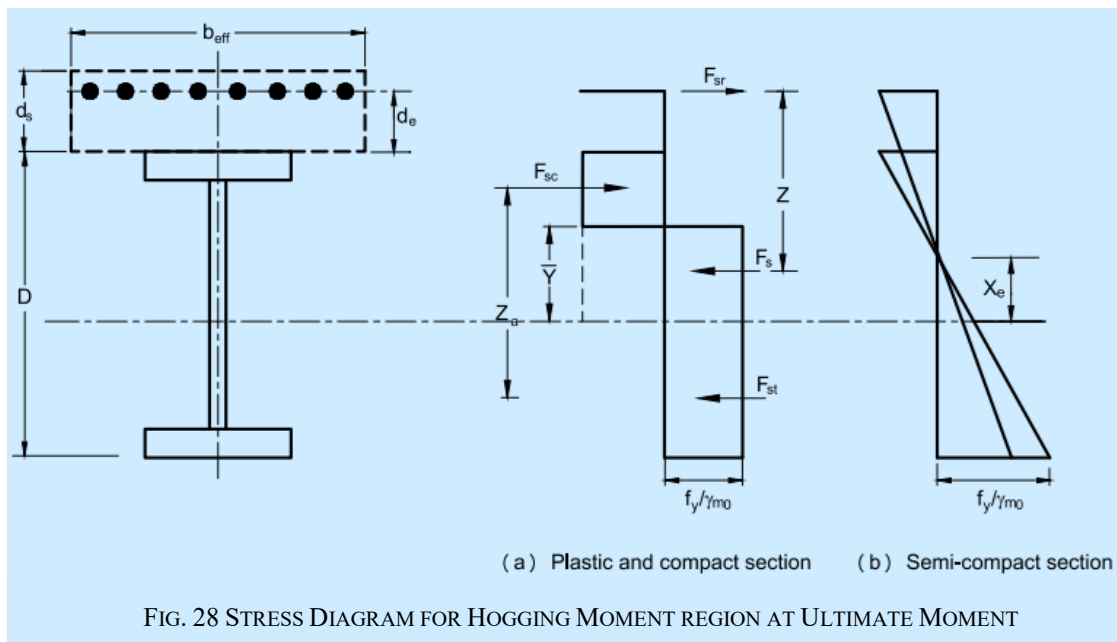
$$n_p = n_f \{(M - M_{ds}) / (M_d - M_{ds})\}$$

B-1.3 Sagging Bending Resistance of Composite Section with Non-Compact Structural Steel Beams

Since the compression flange of semi-compact steel sections buckle locally under compression on reaching yield stress f_y , the design resistance of the composite section consisting of semi-compact sections is evaluated using equations of compact sections as per [B-1](#), wherein the effective width of elements of section like the compression flange is restricted to limiting value of that of the compact section.

B-2 MOMENT OF RESISTANCE OF COMPOSITE SECTION (NEGATIVE MOMENT-CONTINUOUS BEAMS)

[Figure 28](#) shows stress distribution across a composite beam section subjected to hogging bending moment. Since the steel bottom flange is in compression section classification shall be done as per [C-6B(3)]. For classification of the web, the distance, $\bar{y}$, of the plastic neutral axis above the centroid of the area of the steel section shall first be found.



B-2.1 Hogging Bending Resistance for Plastic and Compact Structural Steel Sections

The design tensile force in reinforcement is given as:

$$F_{sr} = f_{yk} \cdot A_{st} / \gamma_k$$

where

- γ_k = partial safety factor for reinforcing steel = 1.15;
- f_{yk} = Characteristic yield strength of the reinforcing steel; and
- A_{st} = the effective area of longitudinal reinforcement within the effective width b_{eff} of the beam

The plastic moment of resistance for plastic and compact structural steel section is given as:

$$M_{ds} = \frac{Z_p \cdot f_y}{\gamma_{m0}}$$

where

Z_p is the plastic section modulus of the structural steel section and γ_{m0} is the material safety factor to be taken as 1.10. In the absence of any tensile reinforcements, the bending resistance of the section would be that of the structural steel section as given by M_{ds} above. To allow for reinforcements, it is assumed that the stress in a depth $\bar{y}$ changes from tension to compression for Plastic and compact section. The corresponding depth for a non-compact section is x_e . For plastic and compact sections, stress distribution $\bar{y}$ may be determined from

$$\bar{y} t_w \frac{2f_y}{\gamma_{m0}} = F_s$$

The locations of the neutral axis and the bending resistance for the plastic and compact section are given in [Table 18](#).

Table 18 Hogging Bending Resistance of Plastic and Compact Composite Section

(Clause [B-2.1](#))

Case	Position of Plastic Neutral Axis	Value of $\bar{y}$	Moment Capacity M_{dh}
(1)	(2)	(3)	(4)
1	In Web	$\bar{y} \leq \frac{D}{2} - t_f$	$M_{dh} = M_{ds} + F_{sr} Z$
2	In Flange	$\frac{D}{2} \geq \bar{y} > \frac{D}{2} - t_f$	$M_{dh} = F_b \frac{D}{2} + F_{sr} d_s - \frac{(F_b - F_{sr})^2 t_f}{F_f 4}$

where

$$Z = \frac{D}{2} + d_s - \frac{\bar{y}}{2}$$

$$F_f = \text{Axial capacity of a single flange} = \frac{f_y \cdot A_f}{\gamma_{m0}}$$

$$F_b = \text{Axial capacity of steel section} = \frac{f_y \cdot A_s}{\gamma_{m0}}$$

$$F_{sr} = \text{Axial capacity of reinforcing steel} = \frac{f_{yk} \cdot A_{st}}{\gamma_k}$$

A_s = area of structural steel cross section

A_f = area of top flange of structural steel cross section

A_{st} = area of reinforcing steel within the effective width of concrete flange

d_s – overall depth of the concrete slab

d_e = Effective depth of the slab

t_f = Average thickness of the top flange of the steel section

t_w = Thickness of the web of the steel section

NOTE — The web shall be classified as being in compression throughout.

B-2.2 Hogging Bending Resistance for Non-Compact Section

Where elastic analysis is used, creep is considered in the choice of modular ratio $m (= E_s/E_{cm})$. Here, at the section considered, the loading causes hogging bending moment $M_{e(s)}$ in the steel member alone and $[M_{e(c)}]$ in the composite member.

The location of the elastic neutral axis of the composite semi-compact section [see Fig. 28(b)] is given as:

$$x_e(A_s + A_{st}) = A_{st} \left(\frac{D}{2} + d_s \right)$$

and the second moment of area of the composite section is:

$$I_{co} = I_s + A_s \cdot x_e^2 + A_{st} \left(\frac{D}{2} + d_s - x_e \right)^2$$

Where, I_s is the second moment of area of the steel section alone.

The yield moment is mostly governed by the total stress in the steel bottom flange. The locations of the neutral axis and the moment of resistance for non-compact section is given in Table 19.

The bending moment $[M_{e(s)}]$ causes no stress in the slab reinforcements. In propped construction, the tensile stress in the reinforcement may govern the design. It is given in the following equation:

$$\sigma_{sr} = \frac{\left(\frac{f_y}{\gamma_{m0}} - f_s \right) \left(\frac{D}{2} + d_s - x_e \right)}{\left(\frac{D}{2} + x_e \right)} \leq \frac{f_{yk}}{\gamma_k}$$

Table 19 Hogging Bending Resistance of Non-compact Composite Section
(Clause B-2.2)

Location of Neutral Axis	Moment Capacity of Steel Section Alone $[f_s] \bar{y}$	Moment Capacity $M_{d(c)}$
(1)	(2)	(3)
$x_e(A_s + A_{st}) = A_{st} \left(\frac{D}{2} + d_s \right)$	$f_s = M_{e(s)} \cdot \frac{D/2}{I_s}$	$M_{d(c)} = M_{e(s)} + \frac{\left(\frac{f_y}{\gamma_{m0}} - f_s \right) I_{co}}{\left(\frac{D}{2} + x_e \right)}$

NOTE — The web shall be classified as being in compression throughout.

where

- A_{st} = cross sectional area of reinforcements within the effective width of the concrete flange;
- A_s = cross sectional area of steel beam of a composite section;
- d_s = overall depth of the slab;
- d_e = effective depth of the slab;
- $M_{e(s)}$ = hogging moment in the steel section alone;
- f_s = compressive stress in steel flange due to moment $M_{e(s)}$;
- I_s = moment of inertia of the steel beam alone;
- I_{co} = moment of inertia of the composite section;
- t_f = average thickness of the top flange of the steel section; and
- t_w = thickness of the web of steel section.

B-3 FLANGE STRESS REDUCTION FACTOR (R_h) FOR HYBRID SECTIONS

Flange stress reduction factor is applicable for hybrid sections using higher-grade steel flanges where the section is non-compact, or in other words, where the section reaches the plastic moment capacity. In such cases, design limiting stress for both compression and tension shall be modified by the reduction factor R_h and shall be taken as:

$$f_n = R_h \cdot f_{yf} / \gamma_{m0}$$

where

- R_h = $(D \cdot f_{yw}) / (d \cdot f_{yf}) \leq 1.0$;
- f_{yf}, f_{yw} = the yield strength of the flange and web respectively; and
- D, d = the distance of the extreme compression/tension flange and the corresponding extreme fiber of the web from the neutral axis of the composite section respectively.

For homogeneous grade steel sections, R_h shall be 1.0. The reduction factor shall not apply to compact or plastic sections because the effect of lower strength material in the web is accounted for in calculating the plastic moment as specified in [B-1.1](#).

B-4 MOMENT OF RESISTANCE AGAINST LATERAL-TORSIONAL BUCKLING (CONSTRUCTION STAGE)

At the construction stage, before the concrete hardens, the structural steel beam strength may be dictated by the lateral buckling strength. This may be evaluated as per good practice [C-6B(3)].

Effect of lateral-torsional buckling on flexural strength need not be considered if $\lambda_{LT} \leq 0.4$.

B-4.1 Sagging Lateral Buckling Resistance of Structural Steel Beams

The design buckling resistance moment of a laterally unrestrained beam under un-propped condition during the construction stage shall be taken as:

$$M_d = \beta_b Z_p \left(\frac{\chi_{LT} f_y}{\gamma_{m0}} \right)$$

where

$$\chi_{LT} = \frac{1}{[\varphi_{LT} + (\varphi_{LT}^2 - \lambda_{LT}^2)^{\frac{1}{2}}]} \leq 1.0 \text{ and}$$

$$\varphi_{LT} = 0.5 [1 + \alpha_{LT}(\lambda - 0.2) + \lambda_{LT}^2] \text{ and}$$

$$\alpha_{LT} = \text{As per Table 13}$$

The non-dimensional slenderness ratio, λ_{LT} , is given by:

$$\lambda_{LT} = \sqrt{\frac{\beta_b Z_p f_y}{M_{cr}}} \leq \sqrt{\frac{1.2 Z_e f_y}{M_{cr}}}$$

$$\lambda_{LT} = \sqrt{\frac{f_y}{f_{crb}}}$$

where

- β_b = 1.0 for Plastic and Compact sections;
- = Z_e / Z_p for semi-compact sections; and
- M_{cr} = the elastic critical moment corresponding to lateral torsional buckling.

In the case of simply supported prismatic members with symmetric cross-section, the elastic critical moment, M_{cr} , can be determined as:

$$M_{cr} = \sqrt{\left\{ \left(\frac{\pi^2 E I_y}{(L_{LT})^2} \right) \left[G I_t + \frac{\pi^2 E I_w}{(L_{LT})^2} \right] \right\}} = \beta_b Z_p f_{crb}$$

The maximum bending compressive stress corresponding to lateral buckling, $f_{cr,b}$ of non-slender rolled I-sections may be approximately calculated using the following equation:

$$f_{crb} = \frac{1.1 \pi^2 E}{(L_{LT}/r_y)^2} \left[1 + \frac{1}{20} \left(\frac{L_{LT}/r_y}{h_f/t_f} \right)^2 \right]^{0.5}$$

The following simplified conservative equation may be used in the case of prismatic members made of standard rolled I-sections and welded doubly symmetric I-sections for calculating the elastic critical lateral buckling moment.

$$M_{cr} = \frac{\pi^2 E I_y h_f}{2 L_{LT}^2} \left[1 + \frac{1}{20} \left(\frac{L_{LT}/r_y}{h_f/t_f} \right)^2 \right]^{0.5} = \beta_b Z_p f_{cr,b}$$

where

- $f_{cr,b}$ = lateral torsional buckling stress;
- I_w = warping constant;
- I_t = torsional constant = $\sum b_i t_i^3 / 3$ for open sections;
- I_y = moment of inertia about the weak axis;
- r_y = radius of gyration of the section about the weak axis];
- L_{LT} = effective length for lateral torsional buckling;
- h_f = Center to center distance between flanges; and
- t_f = thickness of the flange.

M_{cr} for different beam sections, considering loading, support condition, and non-symmetric cross-section, shall be more accurately calculated using the method given in as per good practice [C-6B(3)].

B-4.2 Hogging Lateral Buckling Resistance of Composite Beams

The effective unsupported length, L_{LT} , in case of bare steel section may be taken as recommended in as per good practice [C-6B(3)]. When a slab connects in the hogging moment region the top flange of two or more beams running parallel, then the effect of U-Frame action in restraining the lateral buckling of bottom flange under compression (Fig. 29) may be considered as given below:

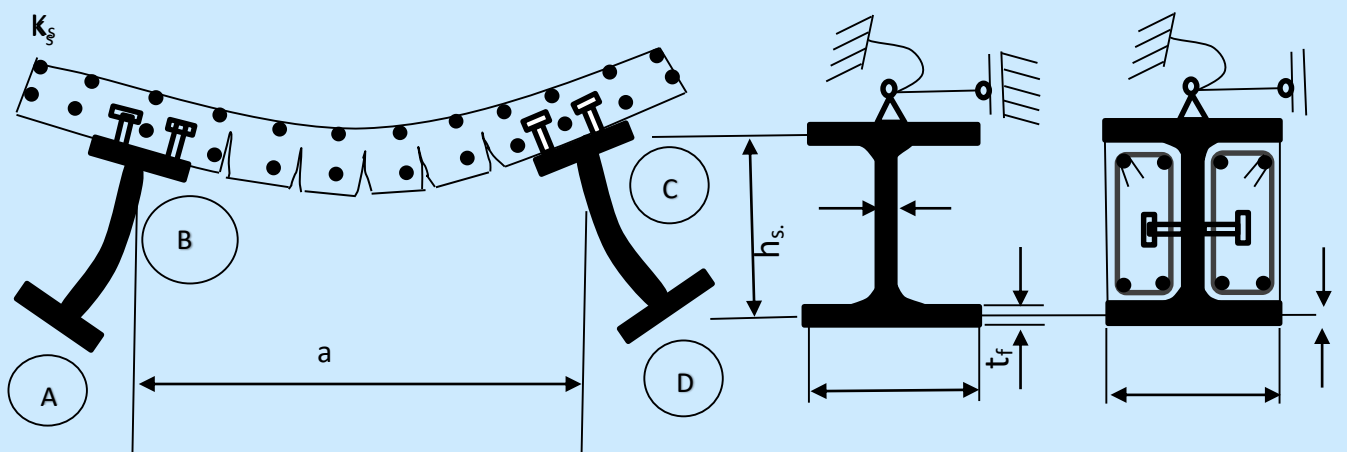


FIG. 29 INVERTED U-FRAME ACTION TO RESTRAIN LATERAL BUCKLING OF BOTTOM FLANGE

B-4.2.1 The rotational spring restraint stiffness of the concrete slab, k_s , per unit length of steel beam is given by

$$k_s = k_1 k_2 / (k_1 + k_2)$$

where

k_1 = Flexural stiffness of the cracked concrete or composite slab transverse to the spanning direction of steel beam = $\alpha (EI)_s / a$

α = 2 for an edge beam,

= 3 for inner beam and

= 4 for inner beam with four or more such beams running parallel and supporting the continuous slab;

a = span length of the slab transverse to the steel beams

$(EI)_s$ = cracked flexural stiffness of the concrete or composite slab per unit width parallel to the steel beam, taken as the lower of the sagging bending slab stiffness of the slab in between the steel beams and hogging bending stiffness of the slab at the supporting steel section;

k_2 = flexural stiffness of the steel web per unit length along the beam,

= $E_s t_w^3 / [4(1 - \nu_s^2) h_s]$ in un-encased steel section

= $E_s t_w b_c^2 / [16 h_s (1 + 4 m t_w / b_c)]$ in encased steel section as per [8.3.8](#)

E_s = modulus of elasticity of structural steel;

ν_s = Poisson's ratio of structural steel;

t_w = thickness of the steel section web;

b_c = width of concrete encasement ([Fig. 27](#));

h_s = distance between centroids of top and bottom flanges

m = long term modular ratio of encasing concrete as per good practice [C-6B(2)].

B-4.2.2 The torsional rigidity of the steel section may be taken as $G_s I_{st}$, where G_s is the shear modulus of steel, I_{st} is the St. Venant's torsion constant of the steel section. The torsional rigidity stiffness contribution of encasing concrete may be taken as being equal to $G_c I_{ct} / 10$, where G_c may be taken as equal to $0.3 E_s / m$ and I_{ct} is the St. Venant's torsion constant of encasing concrete of rectangular shape and m is the modular ratio for long term effects.

B-4.2.3 The elastic hogging critical buckling moment of the composite beam at an internal support of a continuous beam is given by:

$$M_{cr} = (k_c C_4 / L) [(G_s I_{st} + k_s L^2 / \pi^2) E_s I_{ybf}]^2$$

where

I_{ybf} = moment of inertia of the bottom flange about the minor axis of the steel section;

I_{zco} = second moment of the equivalent area, A_{co} , of the cracked composite beam section for major axis bending;

K_c = composite beam section property = $(h_s I_{zco} / I_{ys}) / [(h_s^2 / 4 + (I_{ys} + I_{zs}) / A_s) / e + h_s]$;

e = $A_{co} I_{zs} / [A_s Z_c (A_{co} - A_s)]$;

h_s = distance between the centers of the flanges of the steel section;

I_{ys}, I_{zs} = second moment of area of steel section about the minor and major axis, respectively;

Z_c = distance between the centroid of steel section and the mid depth of concrete slab; and

C_4 = constant depending upon the bending stress distribution over the span length as given in [Table 20](#) and [Table 21](#).

For equilibrium, $F_{st} = F_{sc} + F_{cc}$

where

F_{st} = Tensile force in the steel section below the neutral axis;

F_{sc} = Compressive force the steel section above the neutral axis; and

F_{cc} = Compressive force in the concrete above the neutral axis.

The depth of the neutral axis is given as,

$$x_u = H - x_g$$

where

$$x_g = \frac{\frac{\alpha_{cc}}{\gamma_c} \eta f_{ck} [BH - b_f t_f - t_w(h - t_f)] + t_w h \frac{f_y}{\gamma_{m0}}}{\frac{\alpha_{cc}}{\gamma_c} \eta f_{ck} (B - t_w) + 2 t_w \frac{f_y}{\gamma_{m0}}}$$

Bending resistance,

$$M_p = F_{sc} \cdot X_{sc} + F_{cc} \cdot X_{cc} + F_{st} \cdot X_{st}$$

where

X_{sc} , X_{cc} and X_{st} are respectively the distance between the neutral axis of the composite beam and the individual centre of gravities of the corresponding forces].

B = center-to-center distance between two filler beams;

b_{eff} = effective width of concrete for one filler beam;

b_f = width of the top flange of the steel section;

t_w = thickness of the web of steel section;

H = distance between top of concrete and bottom of bottom flange of steel beam;

h = total depth of steel girder;

x_g = distance of neutral axis from bottom of bottom flange of steel beam;

A_{st} = cross sectional area of reinforcements within the effective width of the concrete flange;

A_s = cross sectional area of steel beam of a composite section;

d_s = overall depth of the slab;

d_c = effective depth of the slab; and

M_p = ultimate bending moment.

ANNEX C

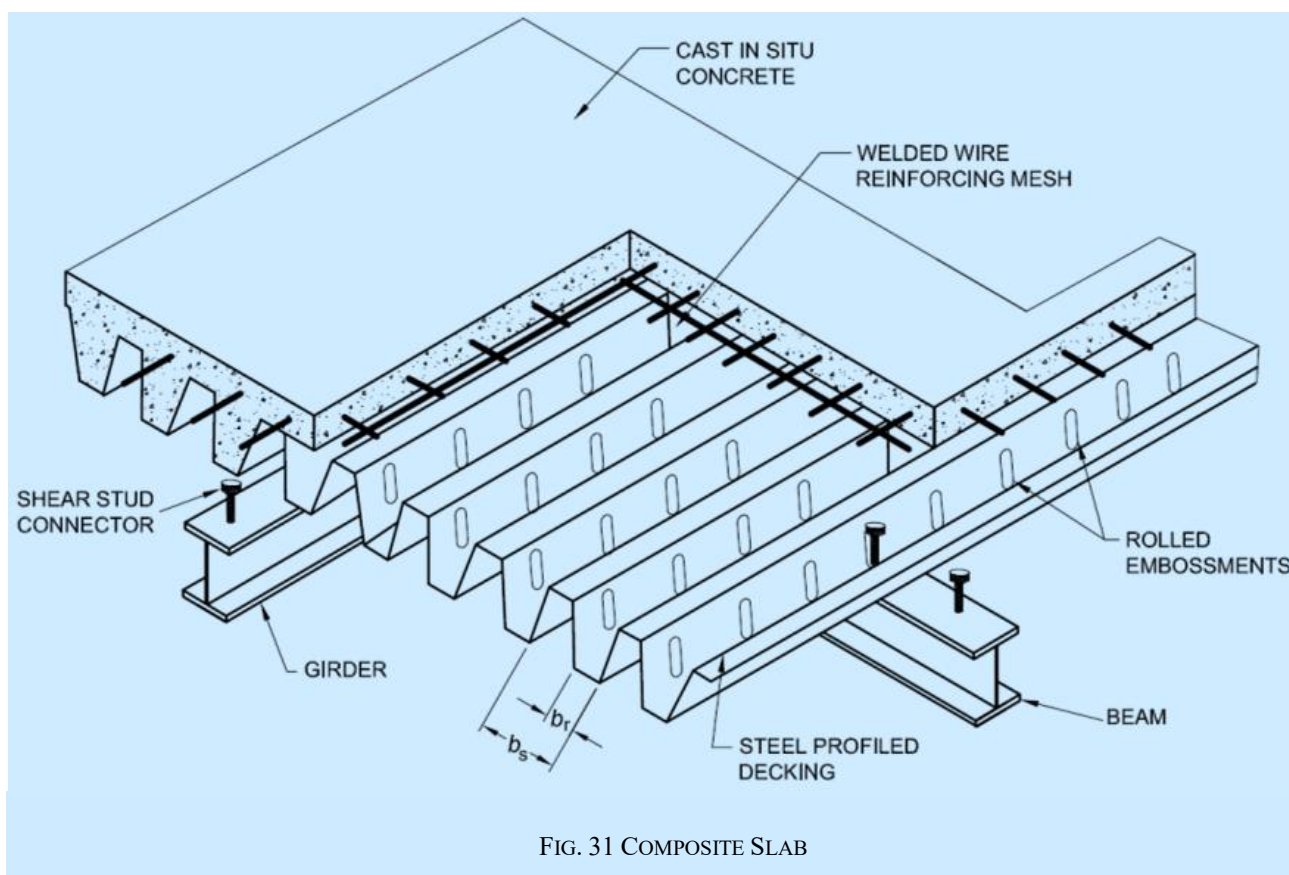
(Clauses [12.4.3.1](#) and [12.5.2](#))

COMPOSITE SLABS

C-1 GENERAL

Composite slabs consist of metal decking spanning between the supports and acting integrally with the concrete after the concrete hardens ([Fig. 31](#)). The steel decking performs several roles as given below:

- It supports loads during construction and acts as a working platform;
- It develops adequate composite action with concrete to resist the imposed loading;
- It transfers in-plane loading by diaphragm action to vertical bracing or shear walls;
- It may stabilize the compression flanges of the beams against lateral buckling until the concrete hardens;
- It reduces the volume of concrete in the tension zone of the slab; and
- It distributes shrinkage strains, thus preventing serious cracking of concrete.



C-2 STRUCTURAL ELEMENTS

C-2.1 Composite floors with profiled decking consist of the following structural elements along with in-situ concrete and steel beams:

- Profiled decking;
- Shear connectors; and
- Reinforcement for shrinkage and temperature stresses.

Connections between the structural steel beam and decking elements are generally designed as 'simple', that is, not moment-resisting. Stud shear connectors are welded through the sheeting onto the top flange of the beam to facilitate the shear transfer between concrete and steel decking, serving as end anchorage, as well as the shear transfer between the slab and steel beam for mobilising composite action along the beam span.

C-2.2 Profile Decking

The steel deck is normally rolled into the desired profile from 0.9 mm to 1.5 mm galvanized coil sheets. It is profiled such that the profile heights are usually in the range of 38 mm to 75 mm, and the pitch of corrugations is between 150 mm and 350 mm. Generally, composite slab spans of the order of 2.5 m to 3.5 m between the beams are chosen, and the beams are designed to span between 6 m to 12 m. There are two well-known generic types of profiles as given below (Fig. 32):

- a) Dovetail profile with flange indentations; and
- b) Trapezoidal profile with web indentations.

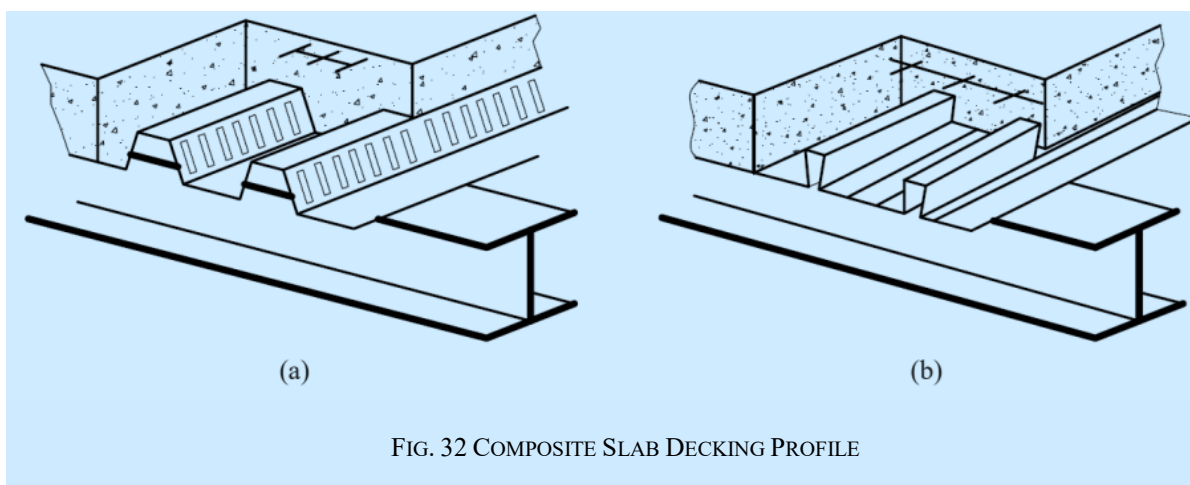
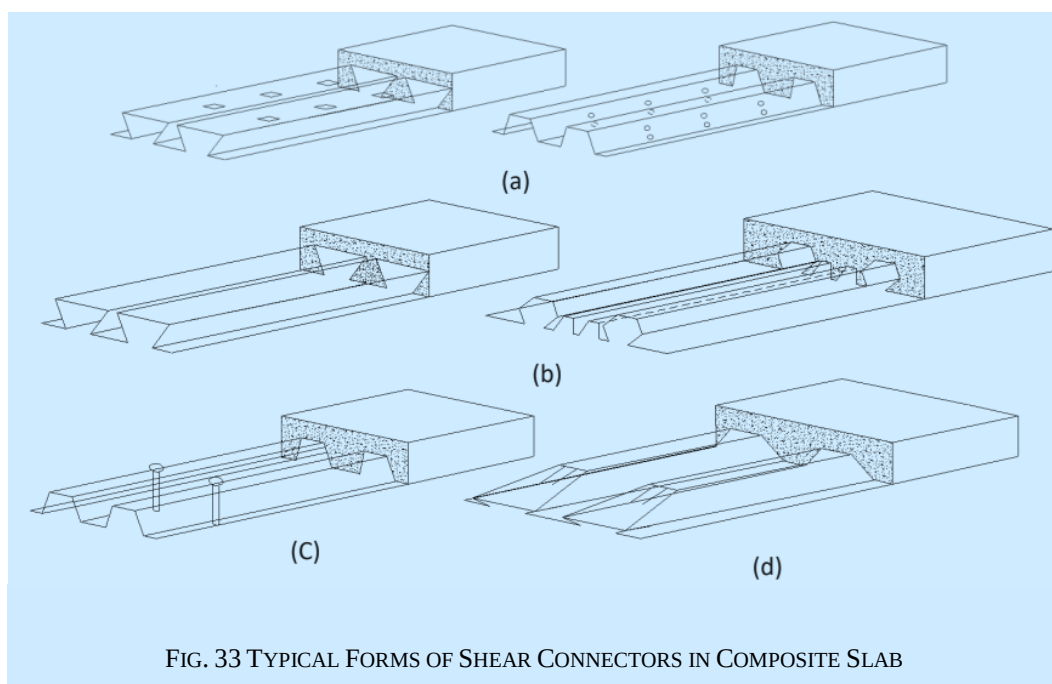


FIG. 32 COMPOSITE SLAB DECKING PROFILE

C-2.3 Shear Connectors

To enable composite action to be assumed between the profiled steel sheet and the concrete, the longitudinal shear force be transferred by the sheet by the above following form of connection (Fig. 33):

- a) Mechanical interlock through the provision of indentations or embossments rolled into the profile;
- b) Frictional interlock of re-entrant profile;
- c) Through-deck welded stud connectors or any other local connection between the steel and the concrete; and
- d) Deformation of the ends of the ribs at the ends of the sheeting.

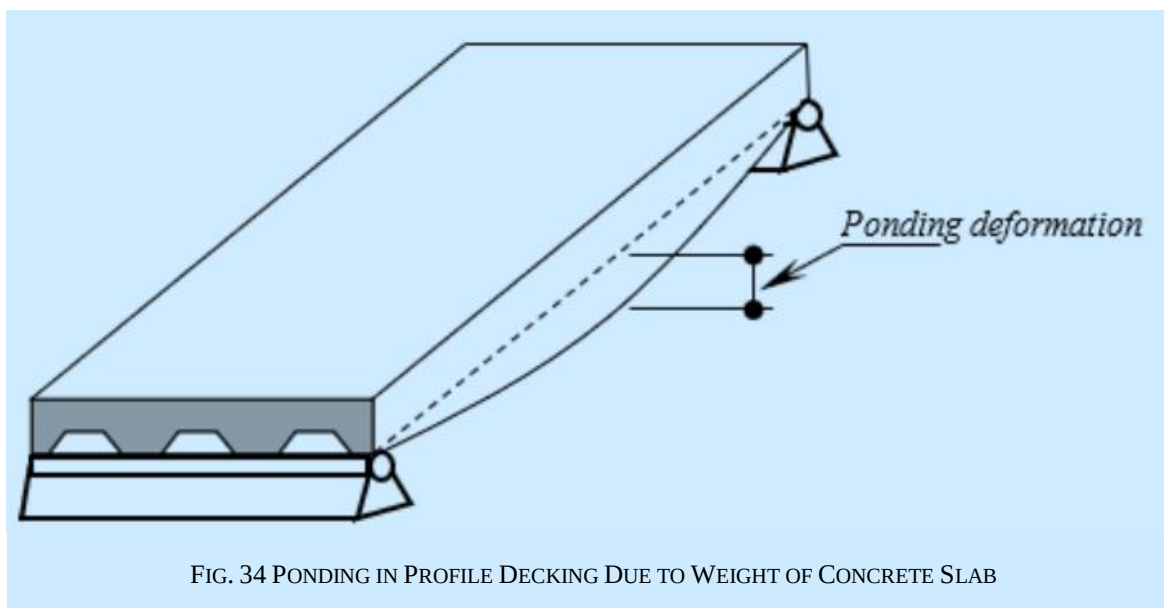


C-2.4 Reinforcement for Shrinkage and Temperature Stresses

In buildings, the temperature difference in the slabs is negligible; thus, there is no need to provide reinforcement to account for temperature stresses. The effect of shrinkage is to be considered, and the total shrinkage strain for design may be taken as 0.003 in the absence of test data.

C-3 PONDING OF PROFILED DECKING

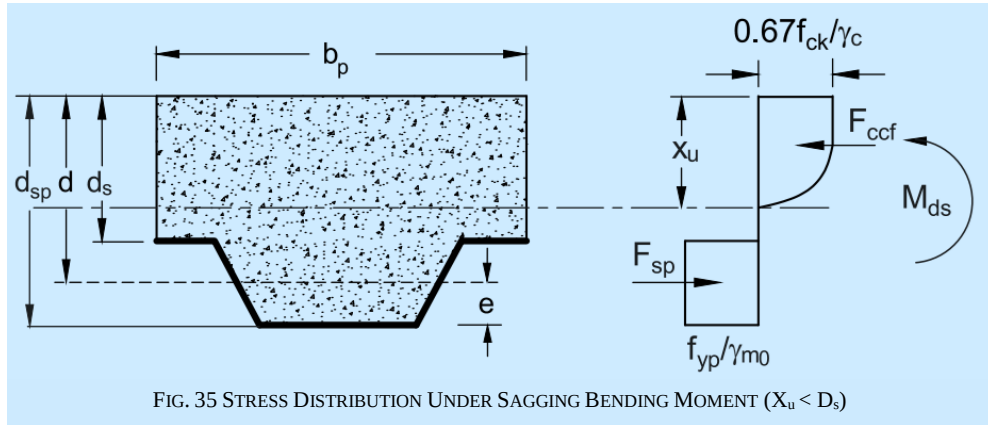
[Fig. 34](#) shows ponding of the profiled deck. In unpropped construction, the profiled decking acting as a formwork to support wet concrete and construction imposed load. Due to this, the deck sheeting may deflect and cause an increase in the thickness of concrete at or closer to mid-span. This is referred to as 'Ponding.' Ponding increases the volume of concrete consumed as well as dead load. These effects should be accounted for, particularly in long spans.



C-4 BENDING RESISTANCE OF COMPOSITE SLAB WITH FULL SHEAR CONNECTION

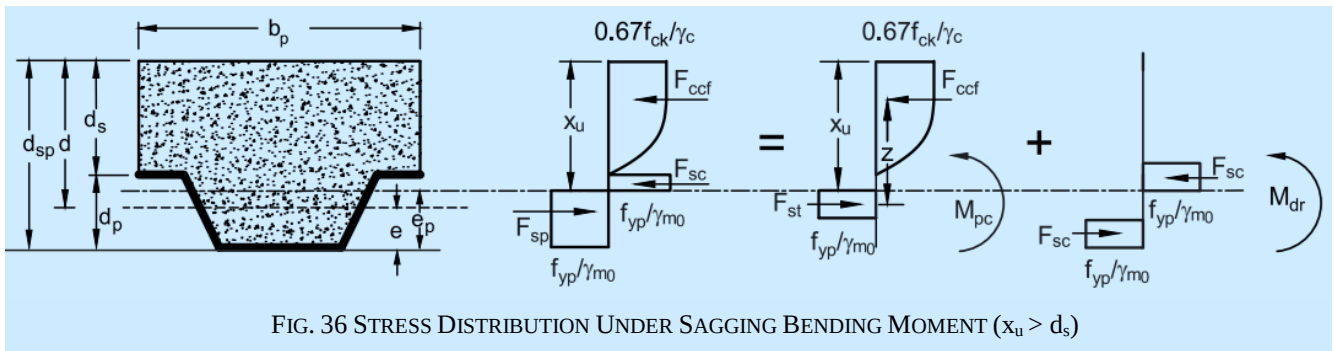
NOTE — In [C-4](#), instead of the usual F_{cc} , the concrete compressive force is denoted as F_{ccf} to represent the concrete compressive force with a full shear connection.

C-4.1 Sagging Bending Resistance [Neutral Axis is above the Steel Decking ([Fig. 35](#))]



The design stress distribution is shown in [Fig. 35](#). This is valid when $x_u \leq d_s$, that is, when the neutral axis lies above steel decking. Centroid of concrete force lies at $0.42x_u$ from top concrete surface. The equations for the design resistance are given in [Table 22](#).

C-4.2 Sagging Bending Resistance [Neutral Axis is within the Steel Decking ([Fig. 36](#))]



The design stress distribution is shown in [Fig. 36](#). This is valid when $x_u > d_s$, that is, when the plastic neutral axis lies in the steel decking. The depth of the parabolic compressive stress block in concrete is between the top of the concrete slab to the top fibre of the deck sheet as shown in [Fig. 36](#). Also, e_p is the distance between the plastic neutral axis of the composite slab to the extreme fibre of the deck sheet in tension and e is the distance between the centroidal axis of the profiled steel sheeting to the extreme fibre of the deck sheet in tension. The centroid of concrete force F_{ccf} lies at $0.42d_s$ from top of the concrete surface. The equations for the design resistance are given in [Table 22](#). The value M_{dp} used in [Table 22](#) is plastic moment capacity of the deck sheet which may be provided by the manufacturer or calculated independently.

C-4.3 Hogging Bending Resistance of Composite Deck

The contribution of steel decking is neglected. The design stress distribution is shown in [Fig. 37](#). This is valid when x_u is in the range of ($d_s < x_u < d_{cp}$), that is., when the neutral axis lies in the steel decking. The centroid of concrete force lies at $0.42x_u$ from the bottom concrete surface b_0 the concrete width at the neutral axis is to be evaluated by trial and error. The equations for the design resistance are given in [Table 22](#).

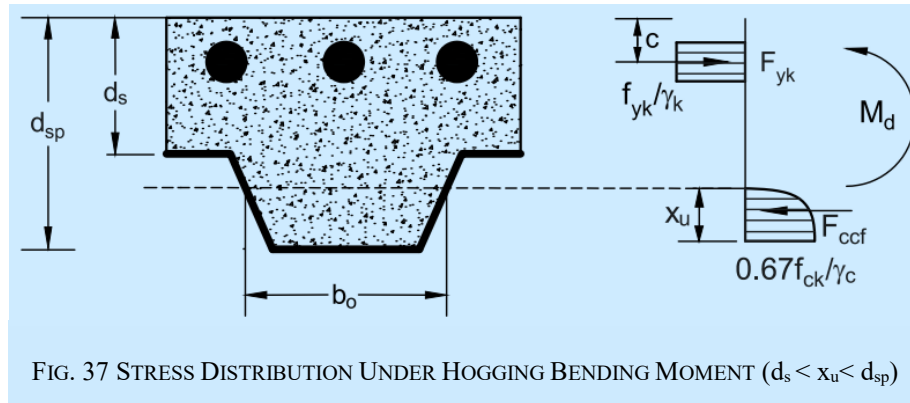


FIG. 37 STRESS DISTRIBUTION UNDER HOGGING BENDING MOMENT ($d_s < x_u < d_{sp}$)

Table 22 Design Moment Capacity, M_{ds} , of Composite Slabs with Full Shear Connection

(Clauses [C-4.1](#), [C-4.2](#) and [C-4.3](#))

Case	Position of Plastic Neutral Axis	Value of x_u	Design Moment Capacity, M_{ds}
(1)	(2)	(3)	(4)
1	Sagging moment, NA above sheeting material (Fig. 35)	$x_u = \frac{A_p f_{yp}}{0.36 f_{ck} b_p} \leq d_s$	$M_{ds} = \frac{A_p f_{yp}}{\gamma_{mo}} (d - 0.42 x_u)$
2	Sagging moment NA within the steel decking (Fig. 36)	$x_u > d_s$ A simplified stress block is shown in Fig. 36 . Using which M_{ds} is calculated without the need for x_u	$M_{ds} = F_{ccf} z + M_{dr}$ $M_{dr} = 1.25 M_{dp} \left[1 - \left(\frac{F_{ccf}}{A_p f_{yp} / \gamma_{mo}} \right) \right] < M_{dp}$ $z = d_{cp} - 0.42 d_s - e_p + \frac{F_{ccf}}{F_{sp}} (e_p - e)$ $F_{ccf} = 0.36 f_{ck} b_p d_s$ $F_{sp} = \frac{A_p f_{yp}}{\gamma_{mo}}$
3	Hogging moment (Fig. 37)	$x_u = \frac{A_{st} f_{yk}}{0.36 f_{ck} b_o}$	$M_d = \frac{A_{st} f_{yk}}{\gamma_k} \times (d_{sp} - c - 0.42 x_u)$

Notations:

A_{st} – area of the reinforcing bar within one wave length b_s of decking sheet

d – distance between the C.G axis of the profiled steel sheeting and the extreme fibre of the composite slab in compression ($d_{sp} - e$)

e – distance between the C.G axis of the profiled steel sheeting and the extreme fibre of the composite slab in tension

e_p – distance between the plastic neutral axis and the extreme tension fibre of the composite slab.

b_p – distance between the centres of adjacent ribs or one wave length of a profile deck

A_p – effective cross-sectional area of profile decking sheeting

M_{dr} – design bending resistance of profile decking sheeting

M_{ps} – Plastic moment of resistance of the profile decking sheeting

f_{ck} – characteristic compressive strength of concrete

f_{yp} – yield strength of profile decking sheeting

d_{sp} – total depth of the composite slab ($d_{sp} = d_s + d_p$)

C-5 BENDING RESISTANCE OF COMPOSITE SLAB WITH PARTIAL SHEAR CONNECTION

NOTE — In [C-5](#), instead of the usual F_{cc} , the concrete compressive force is denoted as F_{ccp} to represent the concrete compressive force with a partial shear connection.

C-5.1 The stress distribution in slabs with partial shear connection is similar to [Fig. 37](#). However, the compressive force in the concrete above the neutral axis ($F_{ccp} < F_{ccf}$) is controlled by the interface shear resistance due to the partial shear connection. The interface shear resistance is obtained from tests. The details of tests to obtain the shear strength between the interface of decking and concrete is given in [C-5.2](#).

C-5.2 Shear Resistance of Composite Slab

The longitudinal shear behaviour may be considered as ductile if the failure load exceeds the load causing a recorded end slip of 0.1 mm by more than 10 percent. If the maximum load is reached at a mid-span deflection exceeding $L/50$, the failure load shall be taken as the load at the mid-span deflection of $L/50$. Otherwise, the behaviour is classified as brittle.

The shear resistance due to the chemical bond shall not be considered. This clause is applicable to the composite slabs with the mechanical or frictional interlock. Literature gives two main methods for determining the shear transfer for partial shear connection from test results. They are (i) the m-k method and (ii) the partial shear connection method. The m-k method is valid for both ductile and brittle failure cases but disregards the effect of overhang beyond the support, whereas the partial shear connection method is valid only for ductile behaviour, but accounts for the effect of the support reaction and the overhang beyond the support.

C-5.2.1 m-k Method for Composite Slabs without End Anchorage

The m-k method is based on establishing the gradient and intercept of an assumed linear relationship between two parameters of the slab, obtained from two groups of composite slab tests. The evaluation of m-k values is illustrated in [Fig. 38](#).

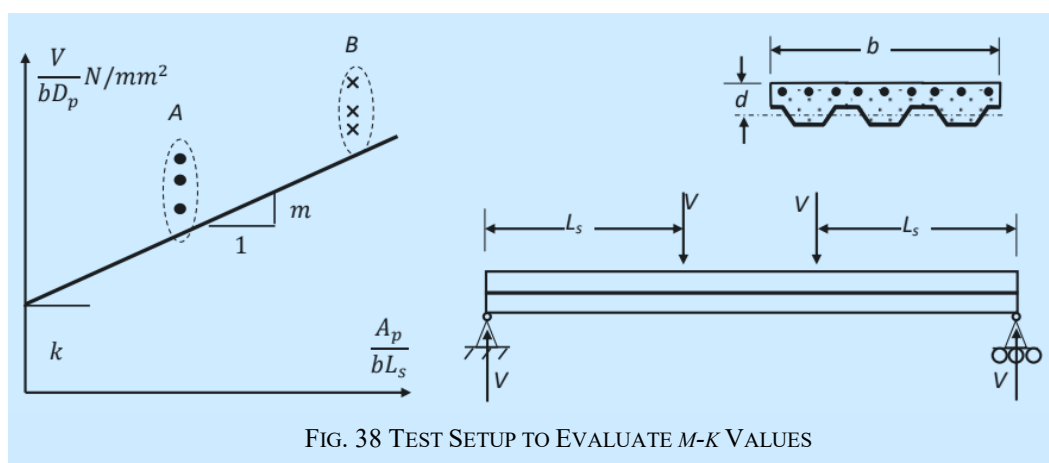


FIG. 38 TEST SETUP TO EVALUATE M-K VALUES

For cases when the longitudinal shear behaviour may be considered ductile, V is taken as the value of the support reaction at the failure load. However, if the behavior is brittle, the value should be reduced using a factor 0.8. By plotting the results from the composite slab tests in terms of the vertical shear parameter $[V/(b D_p)]$ against the shear bond parameter $[A_p/(b L_s)]$ for two groups of data corresponding to the long specimens (Group-A) and the short specimens (Group-B). The relationship between vertical shear and shear bond capacity is approximated by constructing a straight line through the two groups of data ([Fig. 38](#)). The effect of any overhang in the test specimen is neglected in the m-k method, unlike in the partial shear connection approach.

From all the values of V , 5 percent fractile of the characteristic shear strength linear regression line should be calculated to define the characteristic m and k values ([Fig. 38](#)). The minimum value of each group is further reduced by 10 percent for design consideration if two groups of three tests are used and the deviation of any individual test result in a group does not differ by more than 10 percent from the mean.

If the m-k method is used, it should be shown that the maximum design vertical shear V for a width of slab 'b' does not exceed the design shear resistance V_{ld} specified in [Table 23](#).

C-5.2.2 Partial Interaction Method

The degree of shear connection η is defined as,

$$\eta = \frac{F_{ccp}}{F_{ccf}}$$

where

F_{ccp} = compressive force in the concrete as governed by partial shear transfer, as obtained from a standard tests; and

F_{ccf} = compressive force in the concrete for full shear connection as given in [C-4](#).

The variation in bending resistance with the degree of shear connection is shown in [Fig. 39](#). In case $\eta = 0$ composite action between the steel sheet and the concrete does not exist, and it is assumed that the bending resistance is provided by the bending resistance, M_{dp} , of profiled steel sheet alone. For the case $\eta = 1$, in the full shear connection, the full tensile resistance of the sheet is developed, or the full compressive resistance of concrete above the ribs of the sheet is attained. For intermediate cases such that $0 < \eta < 1$, the partial shear connection exists. Typical behaviour of open trough profile steel sheeting with partial shear connection is illustrated in [Fig. 39](#).

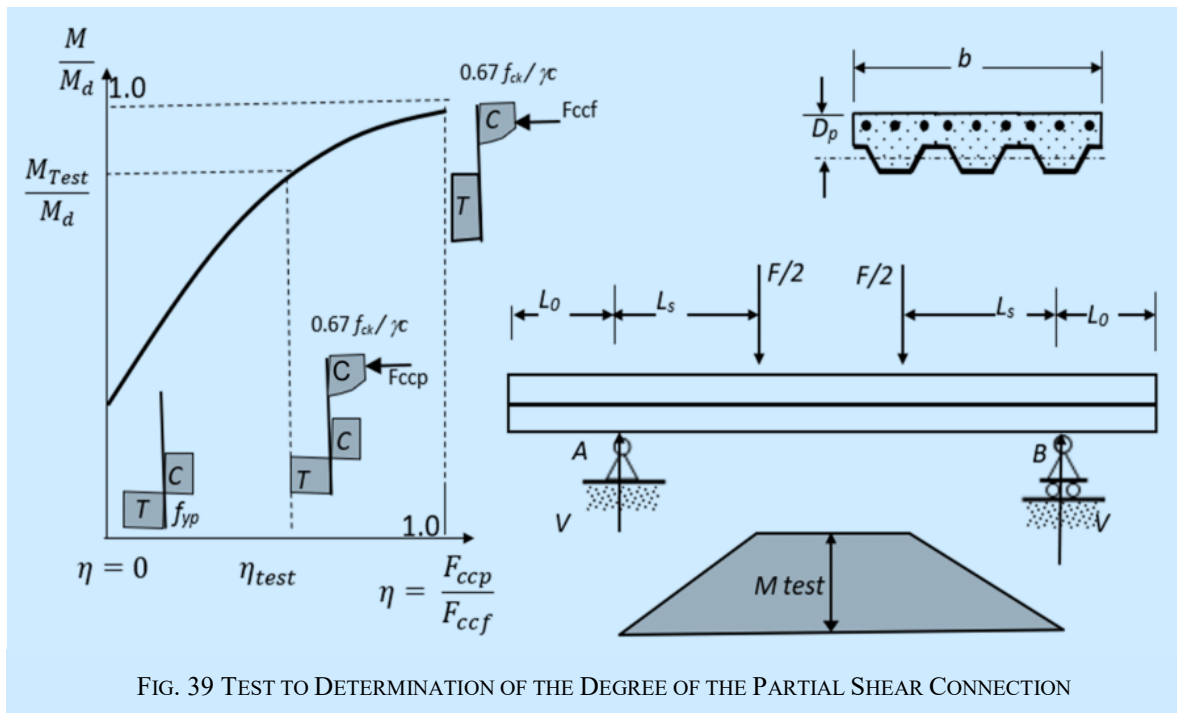


FIG. 39 TEST TO DETERMINATION OF THE DEGREE OF THE PARTIAL SHEAR CONNECTION

For a given bending resistance, the degree of shear connection provided in the test η_{test} can be assessed from the points on the partial shear interaction curve in [Fig. 39](#). By neglecting the effect of the support reaction, the longitudinal shear strength τ_u can be obtained from

$$\tau_u = \frac{\eta_{test} F_{ccf}}{b (L_s + L_o)}$$

where

F_{ccf} = Compressive normal force in the concrete flange with full shear connection;

b = Width of the slab;

L_s = Shear span; and

L_o = Length of the overhang ([Fig. 39](#)).

If the additional longitudinal shear, the additional resistance caused by the friction due to the support reaction is taken into account, the equation becomes

$$\tau_u = \frac{\eta_{\text{test}} F_{\text{ccf}} - \mu V}{b (L_s + L_o)}$$

where

μ = frictional coefficient taken as 0.5; and

V = support reaction under the test load.

The characteristic shear strength $\tau_{u, \text{fk}}$ should be calculated from the test values as the minimum value of stress below which not more than a specified percentage (usually 5 percent) of the corresponding stresses of samples tested are expected to occur. This is divided by the partial safety factor governed by the ultimate state to obtain the design value, $\tau_{u, \text{d}}$.

The determination of the design moment resistance of the composite slab with the partial shear connection is similar to the case ([Fig. 39](#)). The design bending resistance of the partial shear connection slab is evaluated using equations given in [Table 23](#).

Table 23 Design Moment Capacity, M_{dp} , of Composite Slabs with Partial Shear Connection

(Clauses [C-5.2.1](#), [C-5.2.2](#) and [D-2](#))

Case	Position of Plastic Neutral Axis	Methods/ Value of x_u	Design Moment Capacity, M_{dp}
(1)	(2)	(3)	(4)
1	Sagging moment (Fig. 38)	m - k method	$M_{dp} = V_{lp} L_s$ $V_{lp} = \frac{bd_p}{\gamma_{m1}} \left(\frac{mA_p}{bL_s} + k \right)$
2	Sagging moment, NA above profile decking sheet (Fig. 39)	Partial shear connection method $x_u = \frac{F_{ccp}}{0.36f_{ck}b_p} \leq d_s$	$M_{dp} = \frac{A_p f_{yp}}{\gamma_{mo}} (d - 0.42x_u)$ $F_{ccp} = \tau_{ud} b_p L_x \leq F_{ccf}$
3	Sagging moment, NA within the profile decking sheet (Fig. 39)	Partial shear connection method A simplified stress block is shown in Fig. 39 . Using which M_{dp} is calculated with $x_u = \frac{F_{ccp}}{0.36f_{ck}b_p}$	$M_{dp} = F_{ccp}z + M_{pr}$ $F_{ccp} = \tau_{ud} b_p L_x \leq F_{ccf}$ $z = d_{cp} - 0.42x_u - e_p + \frac{F_{ccp}}{A_p f_{yp} / \gamma_{mo}} (e_p - e)$ $M_{dr} = 1.25 M_{dp} \left[1 - \left(\frac{F_{ccp}}{A_p f_{yp} / \gamma_{mo}} \right) \right] < M_{dp}$

Where L_x is the distance of the cross section being considered to the nearest support.

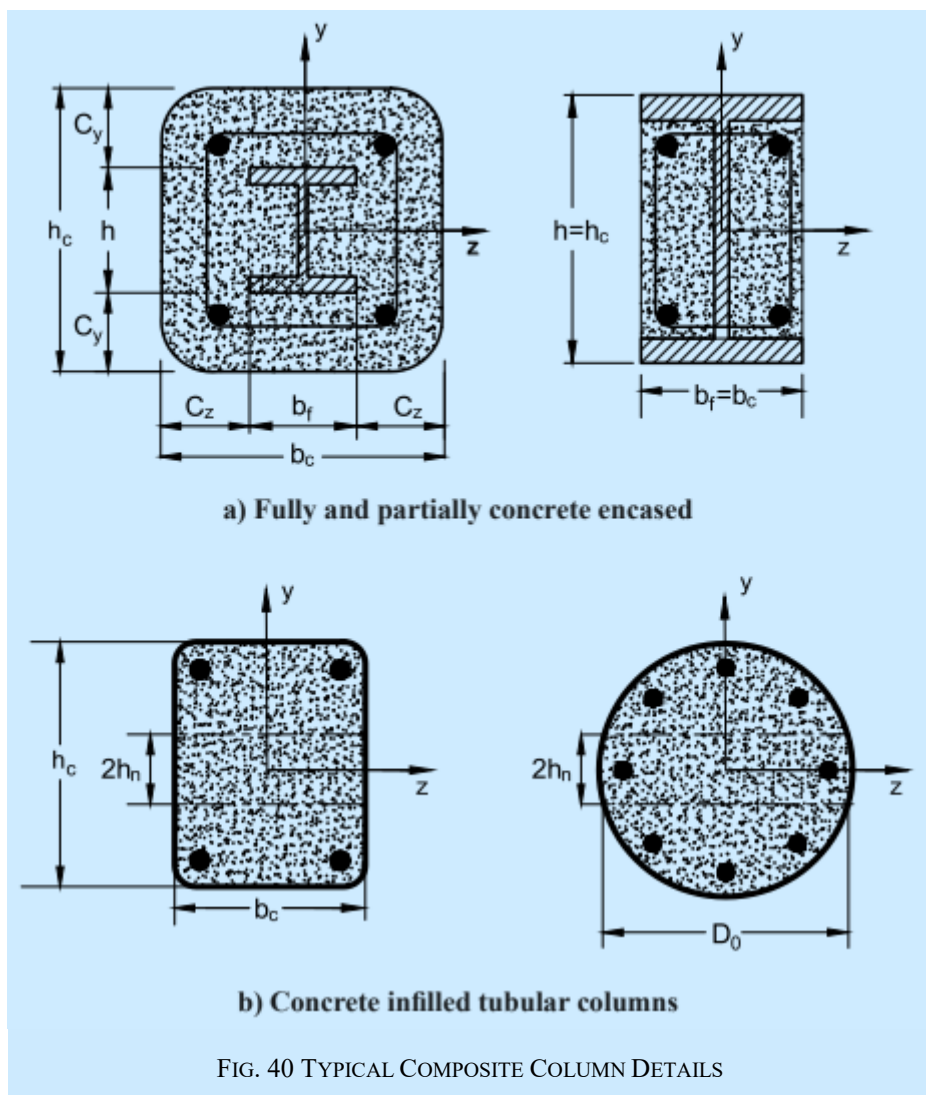
ANNEX D

(Clause 13.8.1)

PLASTIC DESIGN RESISTANCE OF SYMMETRIC COMPOSITE COLUMNS

D-1 GENERAL

This section deals with the calculation of the design resistance of the doubly symmetric composite columns. The resistance equations are derived based on the rectangular stress block in concrete. The general dimensional view of the fully or partially concrete-encased single I-section is as given in Fig. 40 (a). The general dimensional view of the concrete infilled tubular composite column section is given in Fig. 40 (b).



The following are the notations used in this section, in addition to the dimensions shown in Fig. 40.

- A_c = cross sectional area of concrete;
- A_s = cross sectional area of structural steel;
- A_{st} = cross sectional area of total steel reinforcement;
- A_{stc} = cross sectional area of steel reinforcement in the compression zone;
- f_{ck} = characteristic compressive cube strength of concrete;
- f_y = minimum guaranteed yield strength of structural steel section;

- f_{yk} = minimum Guaranteed yield strength of steel reinforcement;
 M_d = design bending resistance under only bending moment;
 M_n = ultimate bending resistance under only bending moment;
 P_d = design compressive resistance under compression only;
 P_n = ultimate compressive resistance under compression only;
 M'_d = design bending resistance under combined bending and compression;
 M'_n = ultimate bending resistance under combined bending and compression;
 P'_d = design compressive resistance under combined bending and compression;
 P'_n = ultimate compressive resistance under combined bending and compression;
 r = inner radius of the rounded corners in RHS;
 γ_c = partial safety factor for the compressive strength of concrete = 1.5;
 γ_{m0} = partial safety factor for the yield strength of structural steel = 1.1; and
 γ_k = partial safety factor for the yield strength of steel reinforcement = 1.15.

D-2 DESIGN COMPRESSIVE RESISTANCE OF SHORT COMPOSITE COLUMN (Point A)

The stresses on different components of the composite columns at the ultimate stage under only axial force are as shown in Fig. 41. The ultimate design compressive resistance, P_n , of the plastic, compact and semi-compact sections are given by:

$$P_n = A_s f_y + A_{st} f_{yk} + \alpha_c A_c (\alpha_{cc} f_{ck})$$

where

$\alpha_c = 0.85$, for partially or fully encased column; and

$\alpha_c = 1.0$, for infilled columns.

$\alpha_{cc} = 0.67$

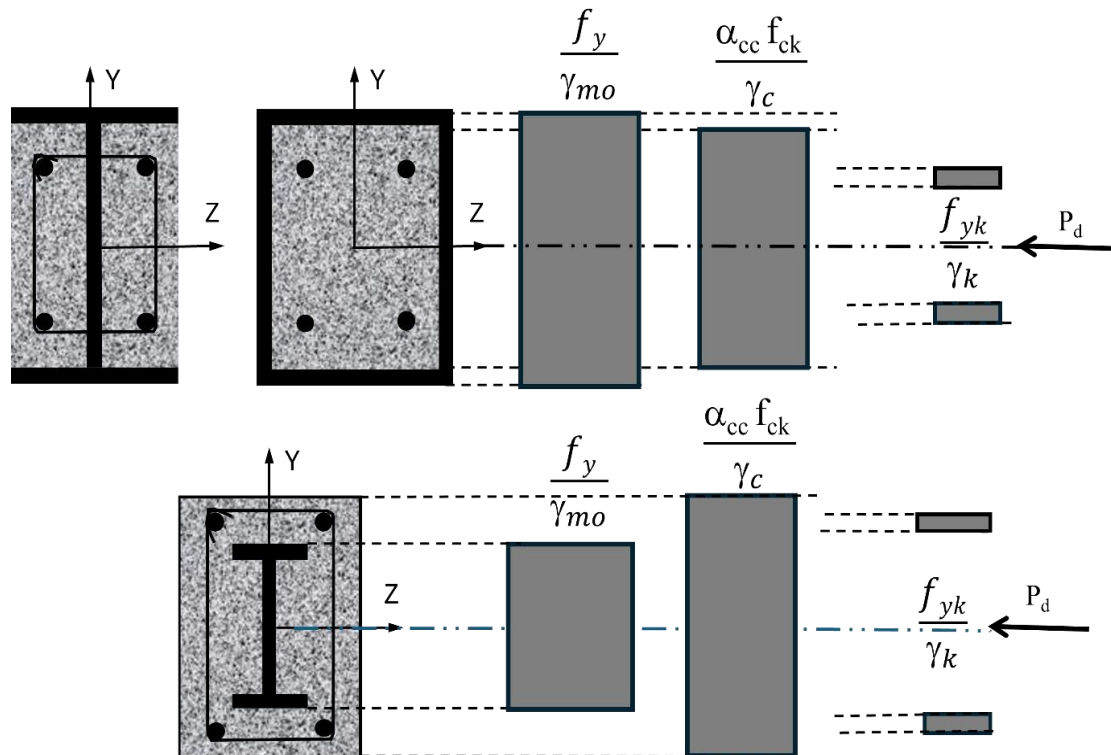


FIG. 41 STRESS DISTRIBUTION AT ULTIMATE STAGE UNDER ONLY AXIAL FORCE

The axial design resistances of different types of the composite column are given in [Table 24](#). For concrete filled circular tubular sections in [Table 24](#), the increase in concrete strength due to confinement may be taken, provided $\lambda \leq 0.5$ calculated using [13.5.1](#) and $e/d < 0.1$ where e is the eccentricity of the loading, given by the ratio of the design moment to design axial load and d is the external diameter of the circular column. If anyone of the conditions is not satisfied then $\eta_1 = 0$ and $\eta_2 = 1$.

To account for minimum eccentricity and grade of reinforcing steel, the design axial capacity can be estimated using the following expression,

$$P_d = A_s \frac{f_y}{\gamma_{m0}} + \alpha_k A_{st} \frac{f_{yk}}{\gamma_k} + 0.6 \alpha_c A_c \frac{f_{ck}}{\gamma_c}$$

α_k values can be taken as

Grade of Reinforcing Steel	Value of α_k
Fe250	0.90
Fe415	0.82
Fe500	0.77

Table 24 Design Compressive Resistance of Short Composite Compression Members

(Clause [D-2](#))

Type of Section	Design Compressive Resistance
(1)	(2)
Partially or fully encased composite section	$P_d = A_s \frac{f_y}{\gamma_{m0}} + A_{st} \frac{f_{yk}}{\gamma_k} + 0.57 A_c \frac{f_{ck}}{\gamma_c}$
Concrete filled rectangular tubular section	$P_d = A_s \frac{f_y}{\gamma_{m0}} + A_{st} \frac{f_{yk}}{\gamma_k} + 0.67 A_c \frac{f_{ck}}{\gamma_c}$
Concrete filled circular tubular section	$P_d = A_s \eta_2 \frac{f_y}{\gamma_{m0}} + A_{st} \frac{f_{yk}}{\gamma_k} + A_c \frac{(\alpha_{cc} f_{ck})}{\gamma_c} \left[1 + \eta_1 \frac{t}{d} \frac{f_y}{(\alpha_{cc} f_{ck})} \right]$ $\eta_1 = \eta_{10} \left[1 - \frac{10e}{d} \right]$ $\eta_2 = \eta_{20} + (1 - \eta_{20}) \frac{10e}{d}$ $\eta_{10} = 4.9 - 18.5 \lambda + 17 \lambda^2 \geq 0.0$ $\eta_{20} = 0.25(3 + 2 \lambda) \leq 1.0$

D-3 DESIGN BENDING RESISTANCE OF COMPOSITE COLUMN

The stresses on the different components of the composite columns at the ultimate stage under only the bending moment about the z-z axis are as shown in Fig. 42.

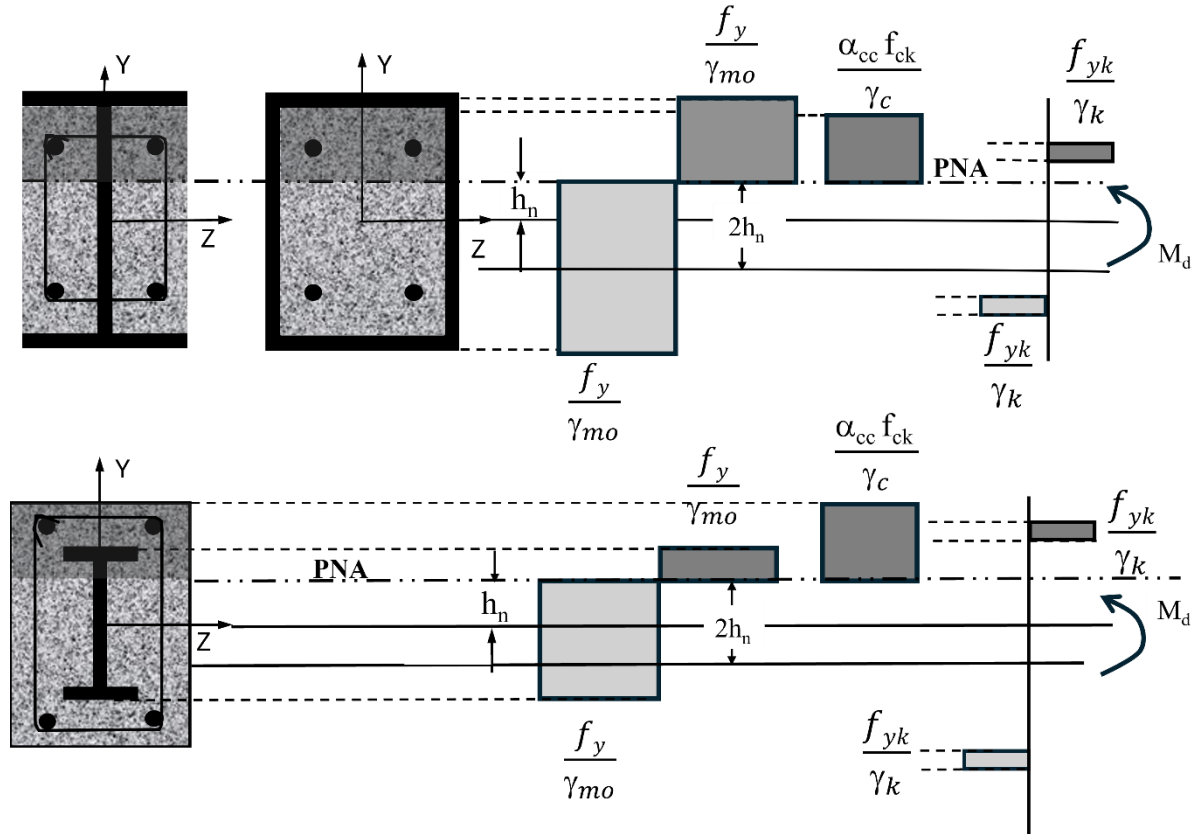


FIG. 42 STRESS DISTRIBUTION AT ULTIMATE STAGE UNDER ONLY BENDING MOMENT

The design bending resistance is given by:

$$M_d = (Z_{ps} - Z_{psn}) f_y / \gamma_{mo} + (Z_{pr} - Z_{prn}) f_{yk} / \gamma_k + 0.5 \alpha_c (Z_{pc} - Z_{pcn}) (\alpha_{cc} f_{ck}) / \gamma_c$$

where

Z_{ps} , Z_{pr} , and Z_{pc} = Plastic section moduli of the steel section, reinforcement and concrete about their own centroidal axis, respectively (the concrete in tension region is assumed to be uncracked for this calculation).

Z_{psn} , Z_{prn} and Z_{pcn} = Plastic section moduli of the steel section, reinforcement and concrete, all within the depth h_n on both sides of the neutral axis of the gross section, about the neutral axis of gross section, respectively. (Table 25 gives equations for evaluating the value of h_n)

- α_c = the strength coefficient of concrete,
- = 0.85 for fully and partially encased composite sections,
- = 1.0 for concrete filled hollow sections and
- = 0.89 for fully encased or concrete filled hollow sections with spiral ties

Table 25 Location of Neutral Axis in Composite Column at Ultimate Moment

(Clause [D-3](#))

Section	Case	Depth, h_n
(1)	(2)	(3)
$f_{cd} = \frac{(\alpha_{cc} f_{ck})}{\gamma_c},$ $A_{sn} = \text{Area of structural steel within } 2h_n \text{ distance}$ $A_{stn} = \text{Area of reinforcing steel within } 2h_n \text{ distance}$		
a) <i>Major axis Bending</i>		
Concrete encased sections	$h_n \leq [h/2 - t_f]$	$\frac{A_c a_c f_{cd} - A_{stn} \left(\frac{2f_{yk}}{\gamma_k} - a_c f_{cd} \right)}{2b_c a_c f_{cd} + 2t_w \left(2\frac{f_y}{\gamma_{m0}} - a_c f_{cd} \right)}$
	$[h/2 - t_f] \leq h_n \leq h/2$	$\frac{a_c A_c f_{cd} - A_{stn} \left(2\frac{f_{yk}}{\gamma_k} - a_c f_{cd} \right) + (b - t_w)(h - 2t_f) \left(2\frac{f_y}{\gamma_{m0}} - a_c f_{cd} \right)}{2b_c a_c f_{cd} + 2b \left(2\frac{f_y}{\gamma_{m0}} - a_c f_{cd} \right)}$
	$h/2 \leq h_n \leq h/2$	$\frac{A_c a_c f_{cd} - A_{stn} \left(2\frac{f_{yk}}{\gamma_k} - a_c f_{cd} \right) - A_{sn} \left(2\frac{f_y}{\gamma_{m0}} - a_c f_{cd} \right)}{2b_c a_c f_{cd}}$

Table 25 (Continued)

Section	Case	Depth, h_n
(1) Concrete filled sections	(2) <i>RHS</i>	(3) $\frac{A_c a_c f_{cd} - A_{stn} \left(2 \frac{f_{yk}}{\gamma_k} - a_c f_{cd} \right)}{2 b_c a_c f_{cd} + 4 t \left(2 \frac{f_y}{\gamma_{m0}} - a_c f_{cd} \right)}$
	CHS	In RHS equation substitute D for b_c as an approximation
b)	<i>Minor axis Bending</i>	
Concrete encased sections	$h_n \leq t_w/2$	$\frac{A_c a_c f_{cd} - A_{stn} \left(2 \frac{f_{yk}}{\gamma_k} - a_c f_{cd} \right)}{2 h_c a_c f_{cd} + 2 h \left(2 \frac{f_y}{\gamma_{m0}} - a_c f_{cd} \right)}$
	$t_w/2 < h_n < b/2$	$\frac{A_c a_c f_{cd} - A_{stn} \left(2 \frac{f_{yk}}{\gamma_k} - a_c f_{cd} \right) - t_w (h - 2t_f) \left(2 \frac{f_y}{\gamma_{m0}} - a_c f_{cd} \right)}{2 h_c a_c f_{cd} + 4 t_f \left(2 \frac{f_y}{\gamma_{m0}} - a_c f_{cd} \right)}$

Table 25 (Concluded)

Section	Case	Depth, h_n
(1)	(2)	(3)
	$b_f/2 \leq h_n \leq b_c/2$	$\frac{A_c a_c f_{cd} - A_{stn} \left(2 \frac{f_{yk}}{\gamma_k} - a_c f_{cd} \right) - A_{sn} \left(2 \frac{f_y}{\gamma_{m0}} - a_c f_{cd} \right)}{2 h_c a_c f_{cd}}$
Concrete filled sections	RHS	$\frac{A_c a_c f_{cd} - A_{stn} \left(2 \frac{f_{yk}}{\gamma_k} - a_c f_{cd} \right)}{2 h_c a_c f_{cd} + 4t \left(2 \frac{f_y}{\gamma_{m0}} - a_c f_{cd} \right)}$
	CHS	In RHS equation substitute D for h_c as an approximation

D-4 DESIGN RESISTANCE OF COMPOSITE SHORT COLUMN UNDER COMBINED BENDING AND COMPRESSION

The stresses on the different components of the composite columns at the ultimate stage under only the bending moment about the z-z axis are as shown in Fig. 43.

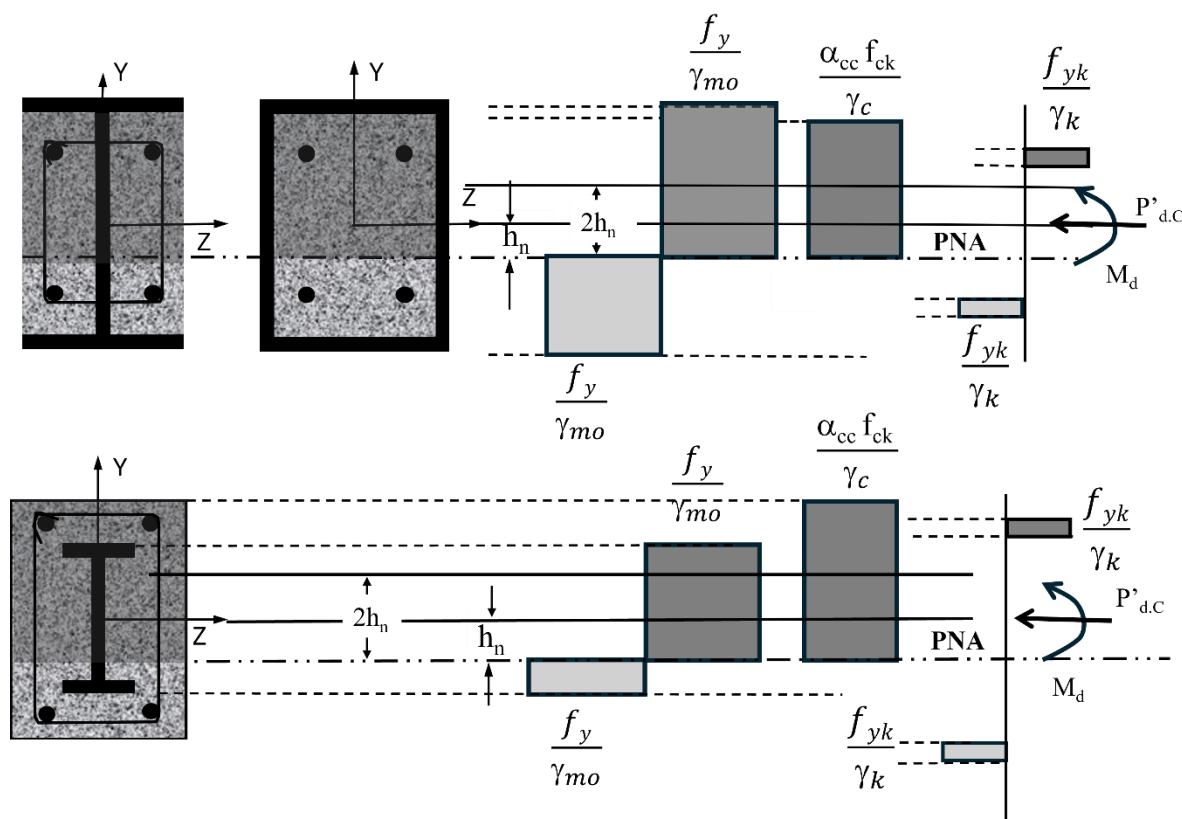


FIG. 43 SHORT COLUMN COMPRESSION AND BENDING INTERACTION

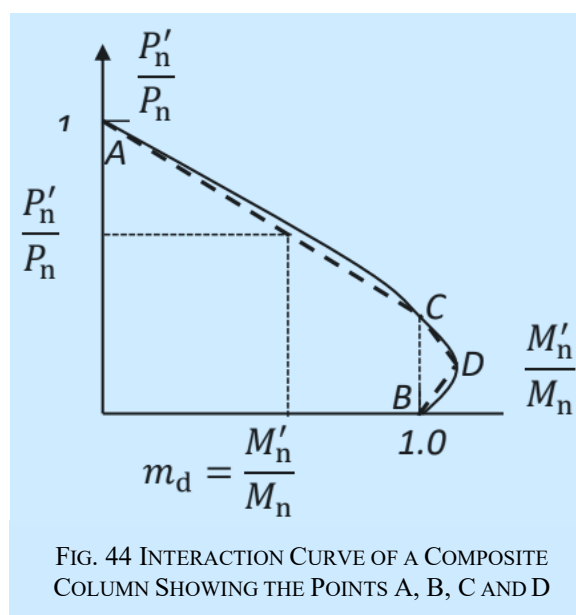


FIG. 44 INTERACTION CURVE OF A COMPOSITE COLUMN SHOWING THE POINTS A, B, C AND D

If the concrete section in compression is below the neutral axis of the gross section, h_n is assumed. The integration of the stresses over the cross section gives the ultimate nominal compressive resistance, P'_n , for the cross section. The integration of the moment of the stresses over the cross section about the neutral axis of the gross section gives the corresponding ultimate bending resistance, M'_n . For various values of h_n , the corresponding simultaneous compressive and bending resistance of the cross section can be calculated to obtain an interaction curve as shown in Fig. 44.

It is seen that the resistance under only compression (Table 23) is a point, (A), on the curve. The resistance under only bending is point, (B), on the curve. Further, from the case of pure bending, as the axial load is increased from zero value, the bending resistance also increases until point D. Similarly, the point C corresponds to the case, where in spite of the axial compression, the ultimate plastic bending resistance, M_n , is reached.

It is tedious and time-consuming to develop the full interaction curve for every section designed. In order to simplify, the values corresponding to the points A, B, C, and D are computed and in between a straight-line variation is assumed in the design. The simplified procedure and the equations are discussed in the following sections.

The values of P_d and M_d corresponding to plastic section under pure compression (point A, Fig. 41) and pure bending (Point B, Fig. 42) are obtained as in D-2 and D-3.

D-4.1 The Design Resistance at Point C (Fig. 45)

The design bending resistance corresponding to point C is the same as pure bending resistance.

$$M'_{d,C} = M_d$$

The design compressive resistance corresponding to point C on the interaction diagram is obtained as given below. The typical position of the neutral axis for points B is shown in Fig. 43 and at the point C in the beam-column interaction curve in Fig. 44. The location of the neutral axis from the CG of section, h_n , can be determined from

Table 21. The integration of the stresses shown in Fig. 46 is equal to $P'_{pd,C}$ at point C. The axial force corresponding to point C in the interaction curve is given by the equation in Table 26.

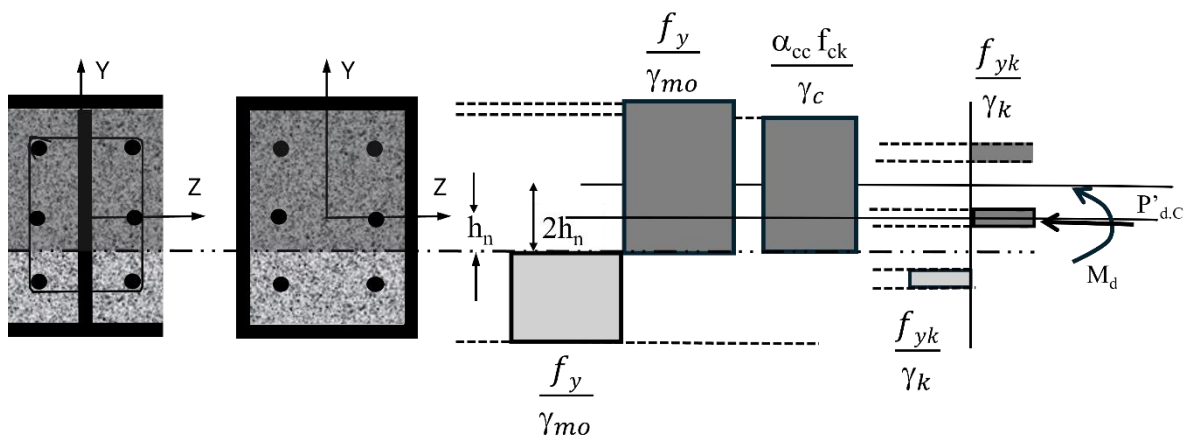


FIG. 45 STRESS DISTRIBUTIONS FOR THE POINT C OF THE INTERACTION CURVE FOR CONCRETE FILLED RECTANGULAR TUBULAR SECTIONS

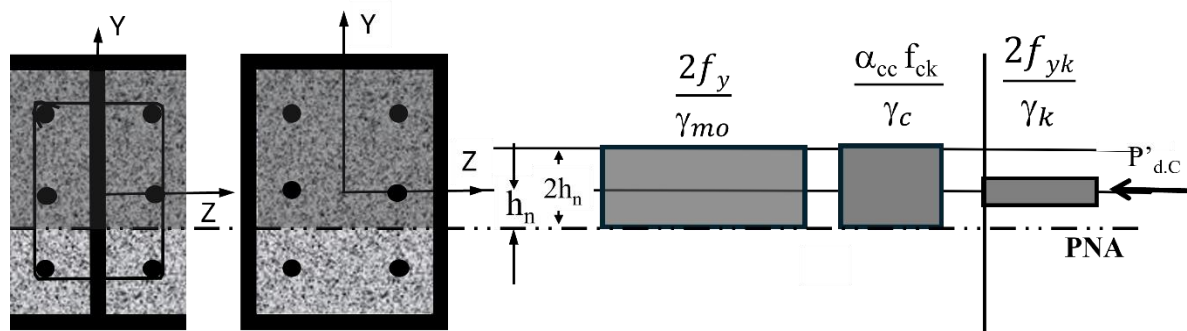


FIG. 46 STRESS CAUSING RESISTING COMPRESSION AT POINT C

Table 26 Design Compressive Resistance at Point C

(Clause [D-4.1](#))

Section	Case	$P'_{d,c}$
(1)	(2)	(3)
a) Major Axis Bending		
Concrete encased column	$h_n \leq [h/2 - t_f]$	$P'_{d,c} = 2 h_n b_c a_c f_{cd} + 2 h_n t_w \left(2 \frac{f_y}{\gamma_{m0}} - a_c f_{cd} \right) + A_{stn} \left(2 \frac{f_{yk}}{\gamma_k} - a_c f_{cd} \right)$
	$[h/2 - t_f] \leq h_n \leq h/2$	$P'_{d,c} = 2 h_n b_c a_c f_{cd} + (A_s - (h - 2 h_n) b_f) \left(2 \frac{f_y}{\gamma_{m0}} - a_c f_{cd} \right) + A_{stn} \left(2 \frac{f_{yk}}{\gamma_k} - a_c f_{cd} \right)$
	$h/2 \leq h_n \leq h_c/2$	$P'_{d,c} = 2 h_n b_c a_c f_{cd} + (A_{sn}) \left(2 \frac{f_y}{\gamma_{m0}} - a_c f_{cd} \right) + A_{stn} \left(2 \frac{f_{yk}}{\gamma_k} - a_c f_{cd} \right)$
	RHS	$P'_{d,c} = 2 h_n b_c f_{cd} + 4 h_n t \left(2 \frac{f_y}{\gamma_m} - a_c f_{cd} \right) + A_{stn} \left(2 \frac{f_{yk}}{\gamma_k} - a_c f_{cd} \right)$
Concrete infilled sections	CHS	In RHS equation substitute D for b_c
b) Minor Axis Bending		
Concrete encased column	$h_n \leq t_w/2$	$P'_{d,c} = 2 h_n h_c a_c f_{cd} + 2 h_n h \left(2 \frac{f_y}{\gamma_{m0}} - a_c f_{cd} \right) + A_{stn} \left(2 \frac{f_{yk}}{\gamma_k} - a_c f_{cd} \right)$
	$t_w/2 < h_n < b/2$	$P'_{d,c} = 2 h_n h_c a_c f_{cd} + (4 h_n t_f + t_w (h - t_f)) \left(2 \frac{f_y}{\gamma_{m0}} - a_c f_{cd} \right) + A_{stn} \left(2 \frac{f_{yk}}{\gamma_k} - a_c f_{cd} \right)$
	$b/2 \leq h_n \leq b_c/2$	$P'_{d,c} = 2 h_n h_c a_c f_{cd} + A_{sn} \left(2 \frac{f_y}{\gamma_{m0}} - a_c f_{cd} \right) + A_{stn} \left(2 \frac{f_{yk}}{\gamma_k} - a_c f_{cd} \right)$
Concrete infilled sections	RHS	$P'_{d,c} = 2 h_n h_c a_c f_{cd} + 4 h_n t_f \left(2 \frac{f_y}{\gamma_{m0}} - a_c f_{cd} \right) + A_{stn} \left(2 \frac{f_{yk}}{\gamma_k} - a_c f_{cd} \right)$

D-4.2 The Design Resistance at Point D (Fig. 44)

The stress diagram on the section corresponding to point D on the interaction curve is shown in Fig. 47.

$$P'_{d,D} = 0.5 P'_{d,C}$$

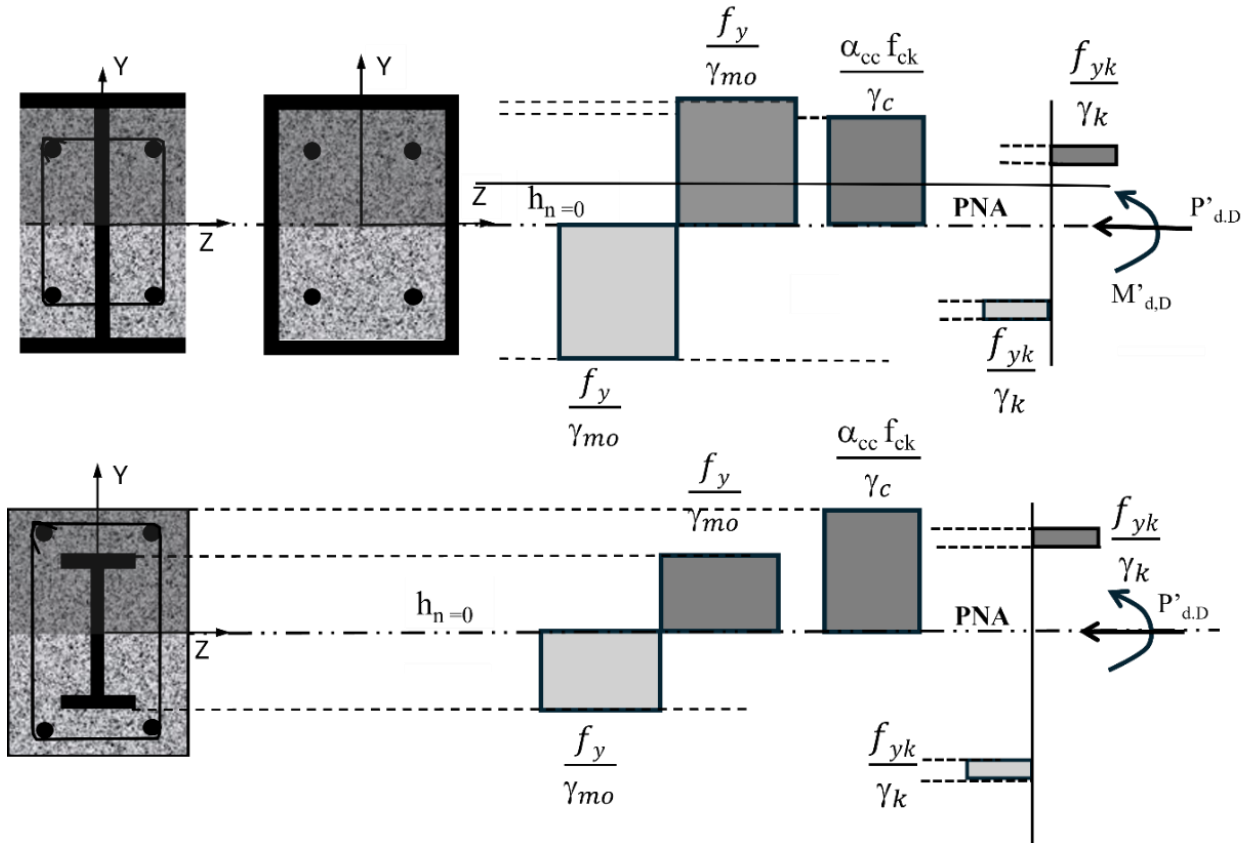


FIG. 47 STRESS DISTRIBUTION AT POINT D AT ULTIMATE STAGE UNDER COMBINED COMPRESSION AND BENDING

$$M'_{d,D} = M_{\max} = Z_{ps} \frac{f_y}{\gamma_{mo}} + Z_{pr} \frac{f_{yk}}{\gamma_k} + 0.5 a_c Z_{pc} \frac{\alpha_{cc} f_{ck}}{\gamma_c}$$

Knowing the values of the compressive and bending design resistances at the points A, B, C and D the values of the resistance at any other point may be obtained approximately by linear interpolation, as shown by dashed lines in Fig. 44, assuming a linear variation.

ANNEX E

(Clause 16)

TESTING

E-1 TESTING OF MATERIALS

Testing of materials shall be done as per standard laid down norms given in [E-1.1](#) to [E-1.3](#).

E-1.1 Concrete

For testing of concrete, reference shall be made to as per good practice [C-6B(2)].

E-1.2 Steel Sections

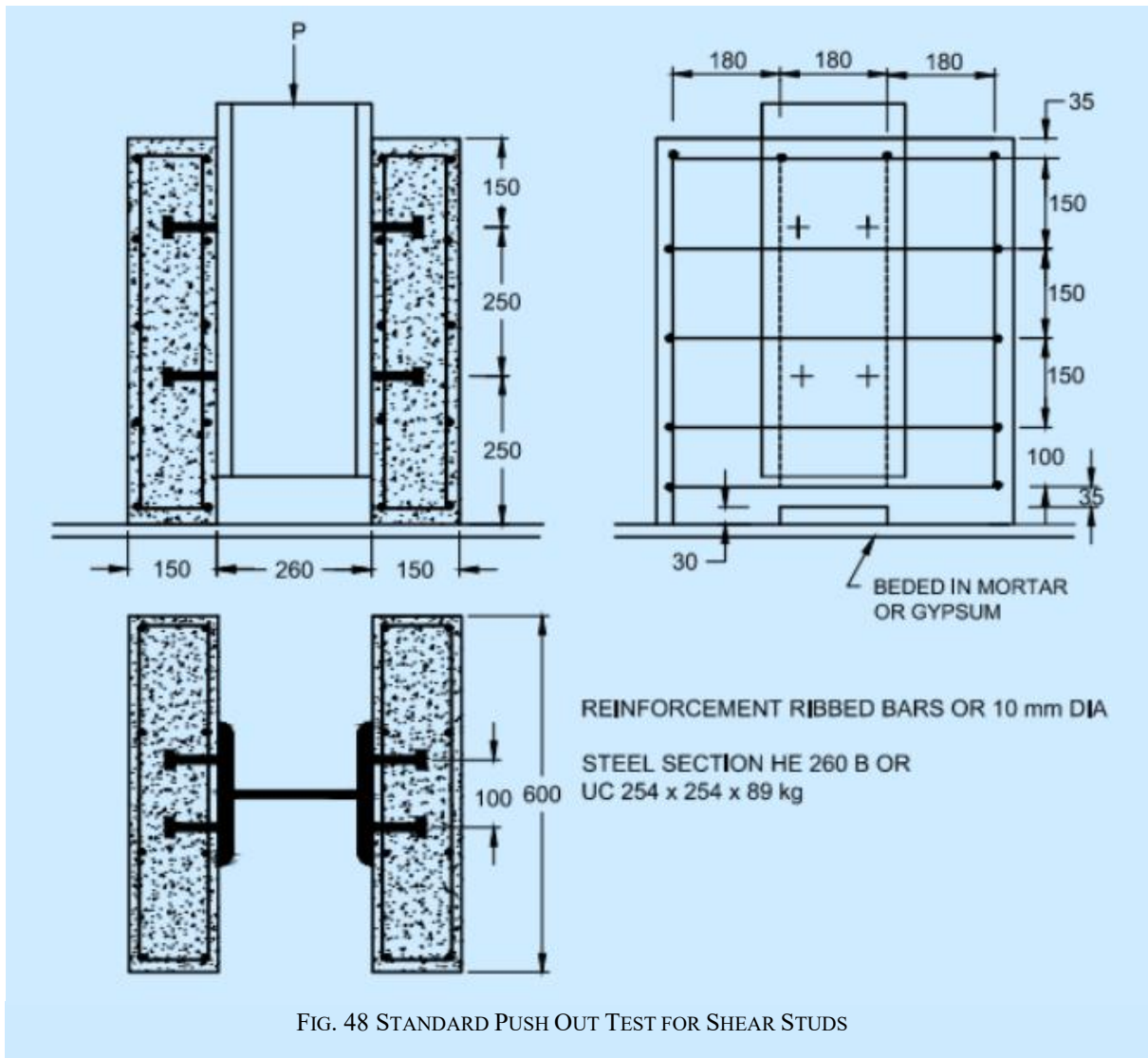
All the structural steel and accessories like rivets, bolts, nuts, washers, welding consumables, steel forging, casting, etc., shall be tested for mechanical and chemical properties as applicable and shall conform to requirements of appropriate Indian standards referred to in as per good practice [C-6B(3)]. Steel shall conform to the requirements of the accepted standard [C-6B(4)] and the accepted standard [C-6B(19)]. Bolts and bolted connection joints with high strength bolts shall be inspected according to as per good practice [C-6B(20)].

For testing of the strength, flexibility, and other relevant properties of the shear connectors, the test procedures as indicated in [E-1.3](#) shall be adopted.

E-1.3 Testing of Shear Studs for Steel-Concrete Interface Shear

The nominal static strength of a shear connector may be determined by push-out tests. The conditions and procedures which shall be followed while performing the tests are as indicated below:

- The dimensions of the standard test are as shown in [Fig. 48](#);
- The bond at the steel-concrete interface shall be prevented by greasing the flanges or by any other suitable method;
- The rate of application of load should be uniform and such that the failure load is reached in not less than 10 minutes;
- The strength of the concrete f_c , at the time of testing, should not differ from the specified cube strength f_{ck} of the concrete by more than ± 20 percent;
- Not less than three tests shall be done, and the nominal static strength P_u shall be taken as the lowest value of $f_{ck}P / f_c$ for any of the tests, where P is the failure loads of the connectors at concrete strength f_c and f_{ck} is the characteristic cube strength of the concrete; and
- The load-deformation curve, the ultimate resistance of the shear per shear connector, relative slip at ultimate load shall be recorded.



ANNEX F

(Clauses [5.3.2](#), [6.1.2](#), [8.1.2.1](#), [9.3](#) and [11.2.1](#))

STRENGTH DETAILS AND ELASTIC MODULUS OF NORMAL CONCRETE

<i>Sl No.</i>	<i>Symbol</i>	<i>Grade of Concrete</i>	M20	M25	M30	M35	M40	M45	M50	M55	M60	M65	M70	M75
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
i)	$f_{ck}(MPa)$	Characteristic cube compressive strength	20	25	30	35	40	45	50	55	60	65	70	75
ii)	$f_{cm}(MPa)$	Mean concrete compressive strength	30	35	40	45	50	55	60	65	70	75	80	85
iii)	$f_{ctm}(MPa)$	Mean tensile strength of concrete	1.91	2.21	2.50	2.77	3.03	3.28	3.52	3.75	3.97	4.42	4.54	4.66
iv)	$f_{ctk}(0.05)$ (MPa)	Characteristic tensile strength at 5 percent fractile of concrete	1.34	1.55	1.75	1.94	2.12	2.29	2.46	2.62	2.78	3.09	3.18	3.26
v)	$E_{cm}(MPa)$	Elastic modulus	28 608	29 962	31 187	32 308	33 346	34 313	35 220	36 076	36 887	37 659	38 395	39 100

LIST OF STANDARDS

The following list records those standards which are acceptable as ‘good practice’ and ‘accepted standards’ in the fulfilment of the requirements of the Code. The latest version of a standard shall be adopted at the time of enforcement of the Code. The standards listed may be used for conformance with the requirements of the referred clauses in the Code.

In the following list, the number appearing in the first column within parentheses indicates the number of the reference in this Part/Section.

<i>Sl No.</i>	<i>IS No.</i>	<i>Title</i>
(1)	IS 456 : 2000	Plain and reinforced concrete — Code of practice (<i>fourth revision</i>)
	IS 800 : 2026	General construction in steel — Code of practice (<i>fourth revision</i>)
(2)	IS 456 : 2000	Plain and reinforced concrete — Code of practice (<i>fourth revision</i>)
(3)	IS 800 : 2026	General construction in steel — Code of practice (<i>fourth revision</i>)
(4)	IS 2062 (Part 1) : 2025	Structural steel: Part 1 Hot rolled medium and high tensile steel (<i>eighth revision</i>)
(5)	IS 1608 (Part 1) : 2018	Metallic materials tensile testing: Part 1 Method of test at room temperature (<i>fourth revision</i>)
(6)	IS 808 : 2021	Dimensions for hot rolled steel beam, column, channel and angle sections (<i>fourth revision</i>)
	IS 1161 : 2014	Steel tubes for structural purposes — Specification (<i>fourth revision</i>)
	IS 1239 (Part 1) : 2004	Steel tubes, tubulars and other wrought steel fittings — Specification: Steel tubes (<i>sixth revision</i>)
	IS 1239 (Part 2) : 2011	Steel tubes, tubulars and other wrought steel fittings — Specification: Steel pipe fittings (<i>fifth revision</i>)
	IS 1730 : 1989	Dimensions for steel plates, sheets, strips and flats for general engineering purposes (<i>second revision</i>)
	IS 1732 : 1989	Dimensions for steel bars, round and square for structural and general engineering purposes (<i>second revision</i>)
	IS 1852 : 1985	Specification for rolling and cutting tolerances for hot rolled steel products (<i>fourth revision</i>)
	IS 2062 (Part 1) : 2025	Structural steel: Part 1 Hot rolled medium and high tensile steel (<i>eighth revision</i>)
	IS 4923 : 2017	Hollow steel sections for structural use — Specification (<i>third revision</i>)
	IS 11587 : 1986	Specification for structural weather resistant steels
	IS 12779 : 1989	Rolling and cutting tolerances for hot rolled parallel flange beam and column sections
	IS 1875 : 1992	Carbon steel billets, blooms, slabs and bars for forgings (<i>fifth revision</i>)
(7)	IS 6911 : 2017	Stainless steel plate, sheet and strip — Specification (<i>first revision</i>)
	IS 1030 : 1998	Carbon steel castings for general engineering purposes — Specification (<i>fifth revision</i>)
(8)	IS 1875 : 1992	Carbon steel billets, blooms, slabs and bars for forgings (<i>fifth revision</i>)
	IS 2004 : 1991	Carbon steel forgings for general engineering purposes — Specification (<i>third revision</i>)

<i>Sl No.</i>	<i>IS No.</i>	<i>Title</i>
	IS 2644 : 1994	High strength steel castings for general engineering and structural purposes — Specification (<i>fourth revision</i>)
	IS 4367 : 1991	Alloy steel forgings for general industrial use — Specification (<i>first revision</i>)
(9)	IS 1148 : 2009	Steel rivet bars (medium and high tensile) for structural purposes (<i>fourth revision</i>)
	IS 1363	Hexagon head bolts, screws and nuts of product grade C:
	Part 1 : 2019	Part 1 Hexagon head bolts (size range M 5 to M 64) (<i>fifth revision</i>)
	Part 2 : 2018	Hexagon Head Screws (Size Range M 5 to M 64) (<i>fifth revision</i>)
	Part 3 : 2018	(Style 1) Hexagon nuts (size range M 5 to M 64) (<i>fifth revision</i>)
	IS 1364 (Part 1) : 2018	Hexagon head bolts, screw and nuts products grade A and B: Part 1 Hexagon head bolts (size range M 1.6 to M 64) (<i>fifth revision</i>)
	IS 1367	Technical supply conditions for threaded steel fasteners:
	Part 1 : 2014	General requirements for bolts, screws, studs and nuts (<i>fourth revision</i>)
	Part 2 : 2002	Tolerances for fasteners — Bolts, screws, studs and nuts — Product grades a, b and c (third revision)
	Part 3 : 2017	Mechanical properties of fasteners made of carbon steel and bolts, screws and studs (fifth revision)
	Part 5 : 2018	Mechanical properties of fasteners made of carbon steel and alloy steel — Set screws and similar threaded fasteners with specified hardness classes — Coarse thread and fine pitch thread (fourth revision)
	Part C : 2018	Mechanical properties of fasteners made of carbon steel and alloy steel — Nuts with specified property classes — Coarse thread and fine pitch thread (fourth revision)
	Part 7 : 1980	Mechanical properties and test methods for nuts without specified proof loads (<i>second revision</i>)
	Part 8 : 2020	Prevailing torque type steel nuts — Functional properties (<i>fifth revision</i>)
	Part 9/Sec 1 : 1993	Surface discontinuities, Section 1 bolts, screws and studs for general applications (<i>third revision</i>)
	Part 9/Sec 2 : 1993	Surface discontinuities, Section 2 bolts, screws and studs for special applications (<i>third revision</i>)
	Part 10 : 2002	Surface discontinuities — Nuts (<i>third revision</i>)
	Part 11 : 2020	Electroplated coatings systems (<i>fourth revision</i>)
	Part 12 : 1981	Phosphate coatings on threaded fasteners (<i>second revision</i>)
	Part 13 : 2020	Hot — Dip galvanized coatings on threaded fasteners (<i>third revision</i>)
	Part 14/Sec 1: 2018	Mechanical properties of corrosion — Resistant stainless — Steel fasteners, Section 1 Bolts, screws and studs (<i>fourth revision</i>)
	Part 14/Sec 2 : 2018	Mechanical properties of corrosion-resistant stainless steel fasteners, Section 2 Nuts (<i>fourth revision</i>)
	Part 16 : 2002	Designation system for fasteners (third revision)
	Part 17 : 2005	Inspections, sampling and acceptance procedure (<i>fourth revision</i>)

<i>Sl No.</i>	<i>IS No.</i>	<i>Title</i>
	Part 18 : 1996	Packaging (<i>third revision</i>)
	IS 1929 : 1982	Specification for hot forged steel rivets for hot closing (12 to 36 mm diameter) (<i>first revision</i>)
	IS 2155 : 1982	Specification for cold forged solid steel rivets for hot closing (6 to 16 mm diameter) (<i>first revision</i>)
	IS 3640 : 1982	Specification for hexagon fit bolts (<i>first revision</i>)
	IS 3757 : 1985	Specification for high strength structural bolts (<i>second revision</i>)
	IS 4000 : 1992	High strength bolts in steel structures — Code of practice (<i>first revision</i>)
	IS 5369 : 1975	General requirements for plain washers and lock washers (<i>first revision</i>)
	IS 5370 : 1969	Specification for plain washers with outside diameter 3 x inside diameter
	IS 5372 : 1975	Specification for taper washers for channels (ISMC) (<i>first revision</i>)
	IS 5374 : 1975	Specification for taper washer for I-beams (ISMB) (<i>first revision</i>)
	IS 5624 : 1993	Foundation bolts — Specification (<i>first revision</i>)
	IS 6610 : 1972	Specification for heavy washers for steel structures
	IS 6623 : 2004	High strength structural nuts (<i>second revision</i>)
	IS 6649 : 1985	Specification for hardened and tempered washers for high strength structural bolts and nuts (<i>first revision</i>)
	IS 7002 : 2018	Prevailing torque type hexagon nuts (with non-metallic insert), style — Property classes 5, 8 and 10 (<i>third revision</i>)
(10)	IS 813 (Part 1) : 2018	Welding and allied processes symbolic representation on drawings-
	ISO 2553:2013	welded joints (<i>second revision</i>)
	IS 814 : 2004	Covered electrodes for manual metal arc welding of carbon and carbon manganese steel — Specification (<i>sixth revision</i>)
	IS IS 3613 : 1974	Acceptance tests for wire-flux combinations for submerged-arc welding of structural steels
	IS 1395 : 1982	Specification for low and medium alloy steel covered electrodes for manual metal arc welding (<i>third revision</i>)
	IS 6419 : 1996	Welding rods and bare electrodes for gas shielded arc welding of structural steel — Specification (<i>first revision</i>)
	IS 6560 : 2017	Welding consumables — wire electrodes, wires rods and deposits for gas shielded arc welding of creep resisting steel — Classification (<i>second revision</i>)
	IS 7280 : 1974	Specifications for bare wire electrodes for submerged arc welding of structural steel
	IS 812 : 1957	Glossary of terms relating to welding and cutting of metals
	IS 816 : 1969	Code of practice for use of metal arc welding for general construction in mild steel (<i>first revision</i>)
	IS 822 : 1970	Code of procedure for inspection of welds
	IS 1024 : 1999	Use of welding in bridges and structures subject to dynamic loading — Code of practice (<i>second revision</i>)
	IS 1182 : 1983	Recommended practice for radiographic examination of fusion welded butt joints in steel plates (<i>second revision</i>)

<i>Sl No.</i>	<i>IS No.</i>	<i>Title</i>
	IS 4853 : 1982	Recommended practice for radiographic inspection of fusion welded butt joints in steel pipes (<i>first revision</i>)
	IS 5334 : 2003	Code of practice for magnetic particle flaw detection of welds (<i>third revision</i>)
	IS 7307 (Part 1) : 1974	Approval tests for welding procedures: Part 1 Fusion welding of steel
	IS 7310 (Part 1) : 2019	Approval tests for welders procedures: Part 1 Fusion welding of steel (<i>first revision</i>)
	IS 7318 (Part 1) : 1974	Approval tests for welders when welding procedure approval is not required: Part 1 Fusion welding of steel
	IS 9595 : 1996	Metal — Arc welding of carbon and carbon manganese steels — Recommendations (<i>first revision</i>)
	IS 15977 : 2013	Classification and acceptance tests for bare solid wire electrodes and wire flux combination for submerged arc welding of structural steel — Specification
(11)	IS 1785 (Part 1)	Specification for plain hard-drawn steel wire for pre-stressed concrete: Part 1 Cold drawn stress relieved wire (<i>second revision</i>)
	IS 1785 (Part 2)	Specification for plain hard-drawn steel wire for pre-stressed concrete: Part 2 As-drawn wire (<i>first revision</i>)
	IS 2266 : 2019	Steel wire ropes for general engineering purposes – Specification (<i>fifth revision</i>)
	IS 2315 : 1978	Specification for thimbles for wire ropes (<i>first revision</i>)
(12)	IS 432 (Part 1) : 1982	Specification for mild steel and medium tensile steel bars and hard-drawn steel wire for concrete reinforcement: Part 1 Mild steel and medium tensile steel bars (<i>third revision</i>)
(13)	IS 1786 : 2008	High strength deformed steel bars and wires for concrete reinforcement — Specification (<i>fourth revision</i>)
(14)	IS 13620 : 1993	Fusion bonded epoxy coated reinforcing bars — Specification
(15)	IS 16651 : 2017	High strength deformed stainless steel bars and wires for concrete reinforcement — Specification
(16)	IS 1343 : 2012	Prestressed concrete — Code of practice (<i>second revision</i>)
(17)	IS 808 : 1989	Dimensions for hot rolled steel beam, column, channel and angle sections (<i>third revision</i>)
(18)	IS 801 : 1975	Code of practice for use of cold — Formed light gauge steel structural members in general building construction (<i>first revision</i>)
(19)	IS 11587 : 1986	Specification for structural weather resistant steels
(20)	IS 4000 : 1992	High strength bolts in steel structures — Code of practice (<i>first revision</i>)

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